Cadence IP for Gigabit Ethernet MAC with DMA, 1588, and TSN/AVB for Automotive

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User Guide

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About this Document

This User Guide describes the interface and operational details for integration of the Cadence IP for Gigabit Ethernet MAC with DMA, 1588, and TSN/AVB for Automotive (IP7014A) into a target design. The intended audience for this document includes design, verification, and software engineers.

This guide is organized as follows:

References and Related Documents

This guide references the following documents:

Table 1: References and Related Documents

Reference	Version/Description
IEEE 802.3 standard	2015 base standard
IEEE 802.3br	Frame Pre-Emption (or Interspersing Express Traffic)
IEEE 802.1AS	Timing and Synchronization for Time-Sensitive Applications
IEEE 802.1CB	Frame Replication and Elimination for Reliability
IEEE 802.1Qav	Credit-Based Shaper
IEEE 802.1Qaz	Enhanced Transmission Selection
IEEE 802.1Qbv	Enhancements for Scheduled Traffic
IEEE 802.1Qci	Pre-Stream Filtering and Policing
IEEE 1588	Precision Clock Synchronization Protocol
Cadence IPs for 2.5G and Gigabit Ethernet MAC Integration Guide	Revision 1.0, 2017
Cadence IP for Gigabit Ethernet MAC Demo Testbench User Guide	Revision 0.8, 2017

Acronyms and Abbreviations

The following acronyms and abbreviations are used in this guide:

Table 2: Acronyms and Abbreviations

Term	Definition
ADAS	Advanced Driver Assistance Systems
AHB	Advanced High-Performance Bus
AMBA®	Advanced Microcontroller Bus Architecture
APB	Advanced Peripheral Bus
AR	AXI Read
ARP	Address Resolution Protocol
ASIL	Automotive Safety Integrity Level
ASF	Automotive Safety Features
AVB	Audio Video Bridging
AW	AXI Write
AXI	Advanced eXtensible Interface
BD	Buffer Descriptor
CBS	Credit-Based Shaper
CRC	Cyclic Redundancy Check
CFI	Canonical Format Indicator
CSMA/CD	Carrier-Sense Multiple Access with Collision Detection
DMA	Direct Memory Access
DPRAM	Dual-Port Random Access Memory
DS	Differentiated Services field
DUT	Device Under Test
DWRR	Deficit Weighted Round Robin
ECC	Error Correcting Code
EEE	Energy-Efficient Ethernet
EnST	Enhanced for Scheduled Traffic
ETS	Enhanced Transmission Selection
FCS	Frame Check Sequence
FF	Flip-Flop
FIFO	First In First Out
FMEDA	Failure Modes, Effects, and Diagnostic Analysis
FRER	Frame Replication and Elimination for Reliability
GEM	Short-form name for Cadence IP for Gigabit Ethernet MAC with DMA, 1588, and TSN/AVB for Automotive (IP7014A)
GMII	Gigabit Media-Independent Interface
HSTL	High-Speed Transceiver Logic
IPG	Inter Packet Gap
LLDP	Link Layer Discovery Protocol
LFSM	Link Fault State Machine
LPI	Low Power Idle
LSB	Least Significant Bit
LSO	Large Send Offload
MAC	Medium Access Controller
MDC	Management Data Clock
MDIO	Management Data Input Output
MFS	Maximum Frame Size
MIB	Management Information Base
MII	Media-Independent Interface
MMSL	MAC Merge Sublayer
MSB	Most Significant Bit
MSS	Maximum Segment Size
PCB	Printed Circuit Board

Table 2: Acronyms and Abbreviations (Continued)

Term	Definition
PCS	Physical Coding Sublayer
PFC	Priority-Based Flow Control
PHY	Physical layer
PMA	Physical Medium Attachment
PPS	Pulse per Second
QoS	Quality of Service
PTP	Precision Time Protocol
QoS	Quality of Service
RGMII	Reduced Gigabit Media-Independent Interface
RGMII-ID	RGMII-Internal Delay
RMII	Reduced Media-Independent Interface
RMON	Remote network MONitoring
RSC	Receive Side Coalescing
RX	Receiver
SerDes	Serializer/Deserializer
SFD	Start Frame Delimiter
SGMII	Serial Gigabit Media-Independent Interface
SoC	System on Chip
SOP	Start of Packet
SMD	Start mPacket Delimiter
SNAP	SubNetwork Access Protocol
SPRAM	Single-Port Random Access Memory
SRAM	Static Random Access Memory
TBI	Ten-Bit Interface
TCP	Transmission Control Protocol
TSN	Time-Sensitive Networking
TSO	TCP Segmentation Offload
TSU	Timestamp Unit
TX	Transmitter
UDP	User Datagram Protocol
UFO	UDP Fragmentation Offload
UTLT	Use Transmit Launch Time
UVM	Universal Verification Methodology
VIP	Verification IP
VLAN	Virtual LAN

Typographical Conventions

The following conventions are used in this guide:

Table 3: Typographical Conventions

Convention	Description
`courier font`	Indicates code
Memory endianess	The little-endian format is used.

Terminology

The following terminologies are used in this guide:

Table 4: Terminology

Term	Definition
aclk	ARM® AMBA AXI clock used with DMA for both transmit and receive DMA operation
col	Collision
crs	Carrier sense
eMAC	express MAC
hclk	ARM AHB bus clock
pclk	AMBA APB peripheral bus clock
pMAC	pre-emptable MAC
rbc0, rbc1	Clocks used in the PCS receive channel

1 Introduction

The Cadence IP for Gigabit Ethernet MAC with DMA, 1588, and TSN/AVB for Automotive (IP7014A) − subsequently also referred to in this document as GEM − is a highly customizable soft controller IP compliant with IEEE 802.3 standard.

To support a range of Ethernet applications, GEM features an integrated 1000BASE-X Physical Coding Sublayer (PCS), a high-performance DMA with advanced AXI offloading capabilities and descriptor caching, 1588, and TSN/AVB. Furthermore, it supports a host of other features including IEEE 802.3az Energy-Efficient Ethernet (EEE), QoS, VLAN, TCP/IP offload, and remote network monitoring (RMON).

GEM is ISO 26262 ASIL-B Ready, accelerating functional safety assessments at system level, helping the designers and integrators of this IP to reach the required ASIL levels. It is delivered with a safety manual, intended to support safety system developers in building their safety-related systems using defined safety mechanisms. This extra document describes any additional system-level hardware or software safety mechanisms that should be implemented to achieve the desired system-level Safety Integrity Level.

As shown in *Figure 1*, GEM connects to a PHY through standard media independent interfaces such as MII, RMII, GMII, RGMII, or SGMI (SGMII operation requires a 1.25G SerDes to be present in the SoC). Host layer access to GEM is through industry-standard AXI or AHB interfaces when DMA is being used or through an external FIFO interface. A separate APB interface allows host applications to configure GEM.

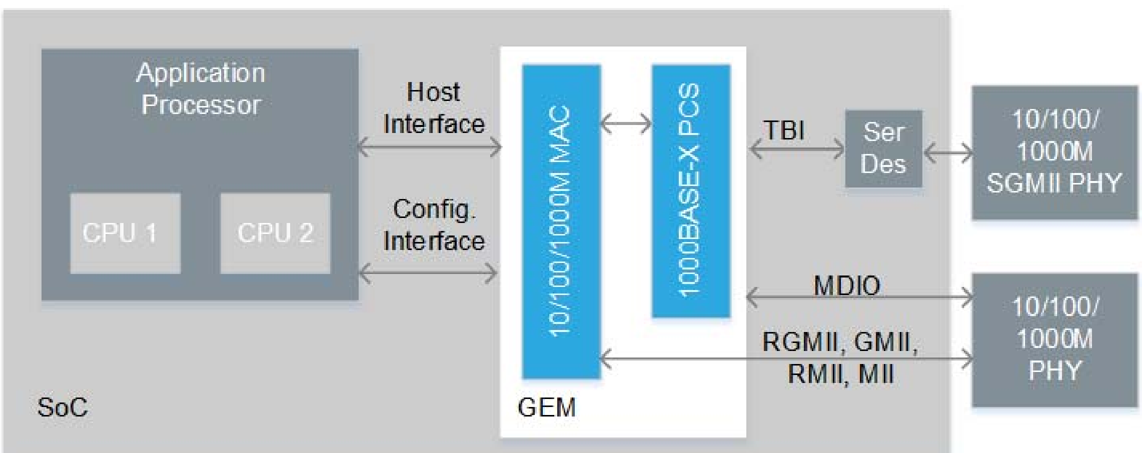

Figure 1: System-Level Block Diagram

2 Architecture Overview

2.1 Architectural Block Diagram

Figure 2 depicts the GEM's top-level block diagram.

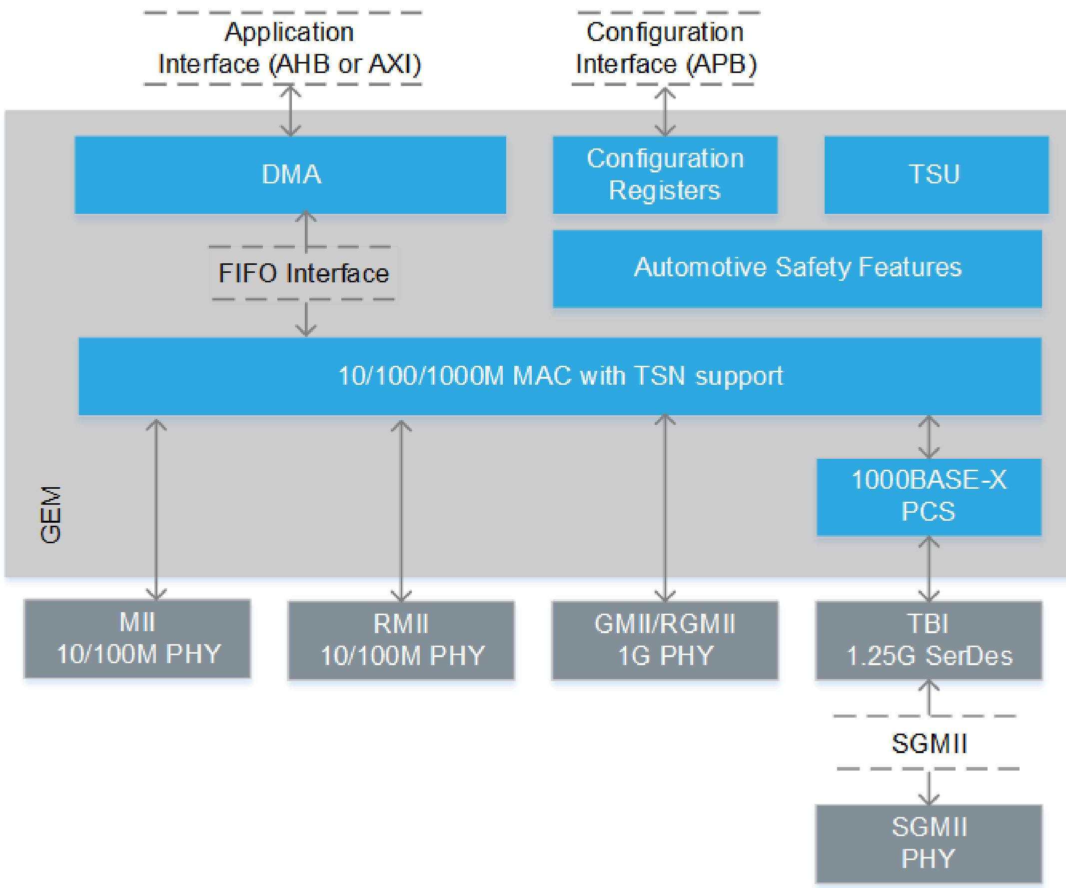

Figure 2: Top-Level Block Diagram

GEM implements a 10/100/1000Mbps Ethernet MAC compatible with the IEEE 802.3 standard. GEM can operate in either half or full duplex. The network configuration register is used to select the speed, duplex mode, and interface type (MII, RMII, GMII, RGMII or SGMII).

GEM comprises the following constituent components:

- MAC controlling transmit, receive, address checking and loopback
- Configuration registers (REG_TOP) providing control and status registers, statistics registers and synchronization logic
- Physical Coding Sublayer (PCS) controlling 8B/10B encode/decode, PCS transmit, PCS receive, and PCS auto-negotiation
- Direct Memory Access (DMA_TOP) controlling DMA transmit, DMA receive and AHB/AXI arbitration logic
- Timestamp Unit (TSU) to calculate the IEEE 1588 timer values and containing the real time clock
- Automotive Safety Features (ASF)

In system applications where no DMA operation is required, the DMA_TOP module can be removed using a configuration option and an external FIFO interface used to incorporate GEM into a SoC environment. GEM can also be configured to expose the FIFO interface when DMA_TOP is present so that either the FIFO or the DMA interface of the GEM instance may be used. GEM's PCS module may also be removed using a configuration option.

2.2 Top-Level Blocks

2.2.1 MAC

The MAC transmit block takes data from the external FIFO interface, adds preamble and, if necessary, pad and frame check sequence (FCS). Both half-duplex and full-duplex Ethernet modes of operation are supported. When operating in half-duplex mode, the MAC transmit block generates data according to the carrier sense multiple access with collision detect (CSMA/CD) protocol. The start of transmission is deferred if carrier sense (crs) is active. If collision (col) becomes active during transmission, a jam sequence is asserted and the transmission is re-tried after a random back off. The crs and col signals have no effect in full-duplex mode. Refer to *Section 6.1.6 MAC Transmit Block* on page 89 for more details.

The MAC receive block checks for valid preamble, FCS, alignment and length, and presents received frames to the MAC address checking block and external FIFO interface. Software can configure GEM to receive jumbo frames up to 16,383 bytes. It can optionally strip FCS from the received frame prior to transfer to the external FIFO interface. Refer to *Section 6.1.8 MAC Receive Block* on page 92 for more details.

The address checker recognizes a configurable number of maskable source or destination specific 48-bit addresses, can recognize four different type ID values, and contains a 64-bit hash register for matching multicast and unicast addresses as required. It can recognize the broadcast address of all ones, copy all frames and act on external address matching signals. The MAC can also reject all frames that are not VLAN tagged and recognize Wake-on-LAN events. Address comparison against individual bits of Specific Address Register 1 can be masked by means of the Specific Address Mask Register. All other specific address filters are byte maskable. Refer to *Section 6.1.9 MAC Filtering Block* on page 93 for more details.

The MAC receive block supports offloading of IP, TCP, and UDP checksum calculations (both IPv4 and IPv6 packet types supported), and can automatically discard frames with bad checksum.

The MAC receive block can be set up to identify 802.1CB streams and automatically eliminate duplicate frames. Statistics are provided to report counts of rogue and out-of-order frames, latent errors, and timer reset events.

2.2.2 Physical Coding Sublayer (PCS)

A PCS is incorporated for 1000BASE-X operation and provides a configurable interface to the PMA layer. A PCS transmit module and 8B/10B encoder form the transmit path, and a PCS receiver and two 8B/10B decoders form the receive path, which is expanded into 16 bits to simplify clocking. Auto-negotiation is also provided for network configuration. The PMA interface may be configured to be 10 or 20-bits and appropriate gearboxes for data width conversion will be automatically instantiated.

The PCS can be configured through the network configuration register to support an SGMII interface between the MAC and an external PHY. For a complete SGMII solution, a SerDes hard macro is also required.

2.2.3 Configuration Registers (REG_TOP)

Control registers drive the management data input/output (MDIO) interface, set up DMA activity, start frame transmission and select modes of operation such as full-duplex, half-duplex, and 10/100/1000Mbps operation. The register interface is compatible with AMBA APB interface standard, version 2.0.

The statistics register block contains registers for counting various types of event associated with transmit and receive operation. These registers, along with the status words stored in the receive buffer list, enable software to generate network management statistics compatible with IEEE 802.3 Clause 30.

2.2.4 Direct Memory Access (DMA_TOP)

The DMA may be configured in either a low-latency buffering mode using internal transmit and receive FIFOs or a packet buffering mode using external memories.

Datapath bus widths of 32, 64, and 128 bit are supported at all data rates (Note that 128-bit datapath is not available when an AHB interface is selected) although system designers should carefully consider the bandwidth requirements of the external datapath being used — either external FIFO interface or AMBA AHB/AXI interface. The DMA block connects to external memory through its AMBA AHB or AXI bus interface. The DMA gathers data to be transmitted from transmit data buffers in system memory and scatters received data to receive data buffers in system memory. Receive data is not sent to memory until the address checking logic has determined that the frame should be copied.

Receive or transmit frames are stored in one or more DMA buffers. The receive buffer size is programmable between 64 bytes and 16KB. Transmit buffers range in length between 1 and 16,383 bytes, and up to 128 buffers are permitted per frame. The DMA block manages the transmit and receive frame buffer queues.

For chip architecture where DMA is not required, GEM is supplied with an external transmit and receive external FIFO interface, thus simplifying design complexity and reducing gate count.

Note the DMA block does not use AMBA AHB split and retry operations.

2.2.5 Timestamp Unit (TSU)

The IEEE 1588 timestamp unit (TSU) is a timer implemented as a 102-bit register, with the bits designated as follows:

- The 48 upper bits count seconds
- The next 30 lower bits count nanoseconds
- The lowest 24 bits count sub nanoseconds
- The 54 lower bits roll over when they have counted to one second

The timer increments by a programmable period (to approximately 59.6 attosecond resolution – 5.96E-17) with each PCLK or TSU_CLK period and can also be adjusted in 1ns resolution (incremented or decremented) through APB register accesses.

An optional external TSU port can be built into the design using a configuration option. When the external TSU port is available, the selection between internal or external TSU is performed using a programmable register. The default selection is internal TSU. The external port is 94-bits wide, mapping to the most significant bits of the internal TSU count as defined above.

2.2.6 Automotive Safety Features

The following automotive safety components are included:

- Fault Logging and Reporting module
- SRAM protection wrappers
- Parity generators and checkers
- Generic transaction timeout monitor

2.2.6.1 Functional Safety

A functional safety manual is provided as one of the deliverables. A representative configuration of the GEM core has undergone Failure Modes, Effects, and Diagnostic Analysis (FMEDA) and achieved ASIL-B ready certification from SGS-TUV.

The following configuration was used for the FMEDA analysis:

- No PCS
- No RGMII

- Internal loopback
- Statistics registers included
- Four specific address filters
- 64-bit data bus width
- 32-bit addressing
- AXI host interface
- Dual-port SRAM (16KB each for transmit and receive)
- Three priority queues
- No cut-through
- No LSO
- Four type 1 screener registers
- Four type 2 screener registers
- 802.1Qav included
- 802.1Qbv included
- Twelve screener 2 compare registers
- TSU included
- tsu_clk included

The provided safety manual should be referenced and the suggested external safety mechanisms should be, where possible, implemented in order to meet the required ASIL safety level.

3 User Defined Options

3.1 Parameterization File

GEM has many parameterization options to configure the IP during compilation stage. This is achieved using Verilog `` `define `` compiler directives in an include file called gem_gxl_defs.v. To aid in debug, these configurations are readable through the design configuration registers starting from address 0x280 – For further details, refer to1‘ *Section 11 Registers*.

You should create your own gem_gxl_defs.v configuration file and place it in the appropriate include path for your own simulation tool.

The following sections detail the parameterization options that are available by modifying the gem_gxl_defs.v file.

Note: A configuration generator tool that has been created to help you create a configuration file. Offering detailed descriptions on the options available, this GUI-based tool can:

- Automatically generate the configuration file (gem_gxl_defs.v) and source modules list file (gem_gxl.f) needed in order to simulate and synthesize the design with the appropriate options
- Generate example test cases that you can simulate specific to your configuration (these are in addition to the example test cases already provided with the delivery that are targeted to example reference design configurations).
- Also generate synthesis constraints specific to the configuration and simulation scripts

To run the tool, please refer to the *Creating a Design Configuration section* in the Quick Start Guide.

3.2 Configuration ‘defines

The gem_gxl_defs.v file describes GEM configuration that has been defined, e.g. for RGMII − select whether to expose a GMII/MII or an RGMII interface: comment the following if a GMII/MII interface is require; uncomment for RGMI:

```
//`define gem_use_rgmii
```

3.2.1 Common Configuration

GEM has a set of common configuration as listed in *Table 5*.

Table 5: Common Configuration ‘defines

Name of ‘define	Description
gem_use_rgmii	Selects whether to expose a GMII/MII or an RGMII interface. Set this define to select an RGMII interface. If not defined, the interface will be GMII/MII.
gem_include_rmii	Selects whether to include an additional RMII interface. Set this define to include a separate RMII port.
gem_no_pcs	Selects the optional PCS module. Set this define to omit the PCS module from the design.
gem_pcs_legacy_if gem_pcs_10b_if gem_pcs_20b_if	If using the PCS, defines the interface option. The interface can be configured to be 'legacy' which is the interface defined in the IEEE 802.3 Clause 36 specifications, or a generic 10 or 20-bit interface. For the generic interface options, the PCS will also contain comma and code group alignment functions. Only one of these options should be defined.

Table 5: Common Configuration ‘defines (Continued)

Name of ‘define	Description
gem_int_loopback	Selects the internal loop back functionality. Set this define to include the internal loop back functionality. The internal loop back functionality requires gating of the tx_clk and rx_clk. This can be a problem for FPGA prototyping. If you are compiling for FPGA operation, you may find it convenient to remove the internal loop back functionality. **Note:** When using internal loopback, tx_clk and rx_clk must be provided using the same loopback reference clock, and n_tx_clk must be provided with the inverted version of the loopback reference clock. An extra output ‘loopback_local’ provides indication to the external system clock generator, and bit 1 in the network control register is provided for software to enable the mode. Removing the loop back module will also remove the top-level pins associated with it.
Note: Only one of the following defines can be active at one time: gem_ext_fifo_interface gem_tx_add_fifo_if gem_host_if_soft_select	
gem_ext_fifo_interface	Activates the exposed the FIFO interface and removes DMA module. When using the external FIFO interface, the DMA module is automatically omitted from the design.
gem_tx_add_fifo_if	Activates the Low-Latency TX interface. **Note:** The low_latency_tx interface is mutually exclusive with the exposed FIFO interface i.e. only one can be compiled at a time. This is because the low-latency TX interface uses the same ports as the external FIFO transmit interface. When using the low_latency_tx interface the DMA is also in the design. If gem_tx_add_fifo_if is configured, the DMA is used as the primary interface, but a secondary TX FIFO interface will be present for specific fixed latency transmit traffic. In this mode, the traffic sent to the TX FIFO interface will be prioritized. Unless the user has a requirement for FIFO interfaces, the recommendation would be to not include this option.
gem_host_if_soft_select	Selects External FIFO Interface OR DMA interface using a software control register. This define will build the design with both external FIFO interface port and DMA port. You can then select which port to use by programming the enable_external_fifo_interface register. **Note:** This enable is not intended to be a dynamic selection and should be performed after the device has had a hard reset but before the datapath is enabled using transmit enable and receive enable bits in the network control register. Unless the user has a requirement for FIFO interfaces, the recommendation would be to not include this option.
gem_user_io gem_user_out_width 8 gem_user_in_width 8	If gem_user_io is defined, the number of user inputs and user outputs are parameterized between 1 and 16 using gem_user_out_width and gem_user_in_width. A value of zero is not allowed. Default is set to 8 for both inputs and outputs with the following statements in the gem_gxl_defs.v file: • `` `define gem_user_io `` • `` `define gem_user_out_width 8 `` • `` `define gem_user_in_width 8 ``

Table 5: Common Configuration ‘defines (Continued)

Name of ‘define	Description
gem_no_stats	Removes the statistics registers. Set this define to remove GEM’s optional statistics registers.
gem_no_snapshot	Excludes the snapshot function in the statistics registers. To include this function, this define should not be set. **Note:** If this function is required, approximately 984 flip-flops and 10K gates are added to the implementation.
gem_irq_read_clear	The `gem_irq_read_clear define can be removed to cause the interrupt status register to use a write-one-to-clear mechanism rather than being cleared when read. To allow the interrupt status register to be cleared only by writing a '1' to the appropriate bit, do not set this define.
gem_jumbo_max_length	Controls the default size of accepted jumbo frames. In DMA mode, this shouldn’t be set any larger than 16383 due to the fact the descriptor writeback only has enough bits to register 16k in the length field. Below is an example of the define: `‘define gem_jumbo_max_length 16’d10240` This is the default value of the max_frame_length APB programmable register and so can be overwritten by writing to this register.
gem_pfc_multi_quantum	Instantiates three additional registers (tx_pause_quantum1, tx_pause_quantum2 and tx_pause_quantum3) to store separate values for the transmitted PFC pause quantum fields. This allows eight separate values to be set. Not setting this define saves gates if only a single value is required.

3.2.2 Filter Configuration

Table 6 lists GEM filter configuration.

Table 6: Filter Configuration ‘defines

Name of ‘define	Description
num_spec_add_filters	Configures the number of specific address match registers. At least one specific address match register should be defined because the value of specific address 1 is used in pause frame generation. Setting this define in pbuf DMA mode to a number greater than 4 will remove external address match from receive buffer descriptor entry word 1 and bit 28 of word 1 will be set if there is an address match in specific address register 1 to 8. Bits 27 to 25 then indicate which specific address register matched. The first four filters are located at address offset 0x88 to 0xA4. Filters 5 to 36 are located between address offset 0x300 through 0x3fc. The following is the default configuration: `` `define num_spec_add_filters 4 ``

Table 6: Filter Configuration ‘defines (Continued)

Name of ‘define	Description
gem_rx_pipeline_delay	Configures the depth of the MAC receive pipeline~~.~~ It is recommended that this value is set to 10. This is because the design has been thoroughly verified with a value of 10. The only purpose of the receive pipeline is to delay frame reception until the frame filtering process completes. If an external filter that looks deep into a frame is used, a deeper pipeline may be required. If very low latency and only simple frame filtering is required, you can consider using a shallower pipeline. When using internal address matching and DMA, this must be a minimum of seven. Frames that are not recognized in time will be dropped. The following is the default configuration: `` `define gem_rx_pipeline_delay 10 ``

3.2.3 IEEE 802.1CB Configuration

Table 7 lists GEM configuration for IEEE 802.1CB.

Table 7: IEEE 802.1CB Configuration ‘defines

Name of ‘define	Description
gem_no_of_cb_streams	If support for IEEE 802.1CB frame elimination is required, then set this define to the number of stream functions required, up to a maximum of 16. The following is an example where n is the number of stream functions required and should not exceed the value defined for num_type2_screeners. `` `define gem_no_of_cb_streams n ``
gem_seq_history_len	For the Vector Recovery algorithm, set to the history depth required for each stream function up to a maximum of 64. The following is an example where each stream function keeps track of m previous frames: `` `define gem_seq_history_len m ``

3.2.4 Datapath Widths

Configure the MAC bus width (Used in conjunction with DMA configuration in *Table 8*)

1. For non DMA modes of operation, the gem_emac_bus_width parameter defines the width of the tx_r_data and rx_w_data buses - i.e. the FIFO interface buses.
2. For Packet Buffer DMA operation, the gem_emac_bus_width paramter should be set to the same size as the gem_dma_bus_width parameter, unless gem_dma_bus_width is set to 128. If gem_dma_bus_width is set to 128, then gem_emac_bus_width should be set to 64.
3. For Non Packet Buffer DMA operation, gem_emac_bus_width should be set to gem_dma_bus_width.

Table 8: Standard Configurations of Datapath Widths

Configuration	gem_dma_bus_width	gem_rx_pbuf_data/ gem_tx_pbuf_data	gem_emac_bus_width	Notes
Packet buffer DMA (use ..._tx_pkt_buffer & ..._rx_pkt_buffer)	32	32	32	
	64	64	64	
	128	128	64	
Packet buffer DMA with SPRAM(use gem_spram)	32	128	32	
	64	128	64	
	128	Not applicable	Not applicable	Not supported

Table 8: Standard Configurations of Datapath Widths (Continued)

Configuration	gem_dma_bus_width	gem_rx_pbuf_data/ gem_tx_pbuf_data	gem_emac_bus_width	Notes
Non packet buffer DMA	32	32	32	
	64	64	64	
	128	128	128	
External FIFO interface only	Not applicable	Not applicable	32	
			64	
			128	
Packet buffer DMA with additional external FIFO IF	64	128	32	More complex configuration with external FIFO IF width as 32b and AHB/AXI width as 64b

Note: When using AXI mode with a single-port RAM (gem_spram == 1) mode and a 32b DMA bus width (gem_dma_bus_width == 32 or bits 22:21 of the network_config register are set to 0), the following hidden registers need to be updated (these registers are used for fine-tuning AXI DMA bursts and normally should not be touched):

- Write 0x4f8 to register 0x002A010A
- Write 0x4fc to register 0x0010001C

3.2.5 Priority Queues and DMA Configuration

The defines listed in *Table 9* are general DMA configuration.

Table 9: DMA Configuration ‘defines

Name of ‘define	Description
gem_axi	Determines whether to use AXI or AHB host interface. Set this define for the AXI interface. If an AXI DMA is selected, you may choose the level of pipelining in the DMA.
AXI Pipelining	
Buffering is required to improve AXI performance, allowing a greater degree of AXI pipelining (where pipelining here refers to the number of AR or AW/W requests issued before the accompanying R or B response is returned from the AXI fabric)	
gem_axi_access_pipeline_bits	Sets the number of bits used to describe the depth of the AXI pipelining allowed. A value of 1 means the maximum level of AXI pipelining is 2. A value of 4 means the maximum level of AXI pipelining is 16. Maximum value is 4. The degree of pipelining can also be fine-tuned using a programmable register in the GEM core. The following is an example configuration: `` `define gem_axi_access_pipeline_bits 4'h4 ``
Depth of the FIFO	
gem_axi_tx_descr_rd_buff_bits	Sets the number of bits used to describe the depth of the FIFO needed to buffer TX descriptor read responses The minimum value is 1. The maximum tested value is 4. A higher value improves the ability of the DMA and MAC to maintain maximum performance in high latency systems. The following is an example configuration: `` `define gem_axi_tx_descr_rd_buff_bits 4'h4 ``

Table 9: DMA Configuration ‘defines (Continued)

Name of ‘define	Description
gem_axi_rx_descr_rd_buff_bits	Sets the number of bits used to describe the depth of the FIFO needed to buffer RX descriptor read responses. The minimum value is 1. The maximum tested value is 4. A higher value improves the ability of the DMA and MAC to maintain maximum performance. The following is an example configuration: `` `define gem_axi_rx_descr_rd_buff_bits 4'h4 ``
gem_axi_tx_descr_wr_buff_bits	Sets the number of bits used to describe the depth of the FIFO needed to buffer TX descriptor write requests. This buffer is needed to avoid the underlying TX DMA from being blocked from continuing transmission as a result of a large RX data write in progress. The minimum value is 1. The maximum tested value is 4. A higher value improves the ability of the DMA and MAC to maintain maximum performance. The following is an example configuration: `` `define gem_axi_tx_descr_wr_buff_bits 4'h4 ``
gem_axi_rx_descr_wr_buff_bits	Sets the number of bits used to describe the depth of the FIFO needed to buffer RX descriptor write requests. It is needed primarily for high-latency systems. The DMA has the ability to issue multiple write requests and data before the first response (on B channel) has returned. Since the descriptor write must not be issued until the fabric identifies the last data write of a packet has been successfully written, and the underlying DMA issues these at request phase time, the DMA must buffer these descriptor writes. The minimum value is 1. The maximum tested value is 4. A higher value improves the ability of the DMA and MAC to maintain maximum performance in high latency systems. The following is an example configuration: `` `define gem_axi_rx_descr_wr_buff_bits 4'h4 ``
Width of DMA AHB/AXI Bus	
gem_dma_bus_width	Configures the maximum width of DMA AHB/AXI bus. Valid settings are 32, 64, or 128. Reducing the bus width will save area resources. The current bus width can be programmed through the network configuration register but will be forced to a value no greater than the configured width.The following is an example configuration: `` `define gem_dma_bus_width        128 ``
gem_dma_addr_width	Configures the maximum width of DMA AHB/AXI address bus. Valid settings are 32 and 64.
AHB and AXI Outputs	

Table 9: DMA Configuration ‘defines (Continued)

<table>
<tr><th>Name of ‘define</th><th>Description</th></tr>
<tr><td>gem_hprot_value</td><td>Controls the value for hprot AHB output.
Individual bits are assigned as follows:
<table>
<tr><th>Bit</th><th>Set to 1</th><th>Set to 0</th></tr>
<tr><td>hprot[0]</td><td>data access</td><td>opcode access</td></tr>
<tr><td>hprot[1]</td><td>privileged</td><td>user</td></tr>
<tr><td>hprot[2]</td><td>bufferable</td><td>not bufferable</td></tr>
<tr><td>hprot[3]</td><td>cacheable</td><td>not cacheable</td></tr>
</table>
The following is an example configuration:
<code>`define gem_hprot_value 4'b0001</code></td></tr>
<tr><td>gem_axi_prot_value</td><td>Controls the value for arprot and awprot AXI output.
Defaulted to nonsecure and data access.:
Individual bits are assigned as follows:
<table>
<tr><th>Bit</th><th>Set to 1</th><th>Set to 0</th></tr>
<tr><td>hprot[0]</td><td>privileged</td><td>normal</td></tr>
<tr><td>hprot[1]</td><td>nonsecure</td><td>secure</td></tr>
<tr><td>hprot[2]</td><td>instruction</td><td>data access</td></tr>
</table>
The following is an example configuration:
<code>`define gem_axi_prot_value 3'b010</code></td></tr>
<tr><td>gem_axi_awcache_value
gem_axi_arcache_value</td><td>Sets the value for arcache and awcache AXI outputs.
Defaulted to non-cacheable and non-bufferable.
The following are example configurations:
<code>`define gem_axi_awcache_value 4'b0000</code><br><code>`define gem_axi_arcache_value 4'b0000</code></td></tr>
<tr><td colspan="2">Packet Buffer DMA Implementation</td></tr>
<tr><td>gem_rx_pkt_buffer
gem_tx_pkt_buffer</td><td>Sets the use packet buffer configuration.
The DMA can be configured to use low-latency internal FIFOs implemented using synthesizable flip-flops or to utilize the packet buffer with external DPRAM.
Set these defines to use the packet buffer configuration.
The packet buffer configuration should not be set if using the exposed FIFO interface.</td></tr>
<tr><td>gem_spram</td><td>Sets the option of using a SPRAM, rather than the default DPRAM, when in packet buffer mode.
If this option is used, there are frequency limitations where the AHB/AXI clock must be going 2X faster than the MAC data rate.</td></tr>
</table>

Table 9: DMA Configuration ‘defines (Continued)

Name of ‘define	Description

gem_rx_pbuf_addr
gem_tx_pbuf_addr
	Sets the maximum size of external DPRAM if the DMA is configured to use packet buffers.
	Maximum number allowed is 16 (represents 256KB with 32-bit datapath).
	The following are example configurations:
	`` `define gem_rx_pbuf_addr  11 // RX buffer depth = 2 ^ gem_rx_pbuf_addr ``
	`` `define gem_tx_pbuf_addr  10 // TX buffer depth = 2 ^ gem_tx_p-buf_addr ``

gem_tx_pbuf_data	Defines the size of the TX packet buffer SRAM data width if gem_tx_pkt_buffer is defined. Valid settings are 32, 64, and 128. **Note:** The SRAM data width cannot be less than `gem_dma_bus_width and when using SPRAM mode (`gem_spram is set), this MUST be set to 128 (i.e. you need a 128b SRAM for this mode). The following is an example configuration: `` `define gem_tx_pbuf_data            128 ``
gem_rx_pbuf_data	Defines the size of the RX packet buffer SRAM data width if gem_rx_pkt_buffer is defined. Valid settings are 32, 64, and 128. **Note:** The SRAM data width cannot be less than `gem_dma_bus_width and when using SPRAM mode (`gem_spram is set), this MUST be set to 128 (i.e. you need a 128b SRAM for this mode). The following is an example configuration: `` `define gem_rx_pbuf_data            128. ``

3.2.5.1 Partial Store and Forward Mode

The DMA can support partial store and forward mode.

Table 10: Partial Store and Forward ‘define

Name of ‘define	Description
gem_pbuf_cutthru	Sets the partial store and forward mode operation. The DMA can support partial store and forward mode, which facilitates lower latency through the packet buffer and also allows frames larger than the buffer size to be supported. Partial store and forward mode needs additional FIFOs and if there is no plan to use partial store and forward mode then by not setting this define there is an area saving of around 2k gates. If this define is set then partial store and forward mode additionally needs a software enable by setting the appropriate enables in the cutthru registers.

3.2.5.2 Priority Queues and Transmit Queues

The DMA can support priority queues.

Table 11: Priority Queue ‘defines

Name of ‘define	Description
dma_priority_queue1 dma_priority_queue2 dma_priority_queue3 dma_priority_queue4 dma_priority_queue5 dma_priority_queue6 dma_priority_queue7 dma_priority_queue8 dma_priority_queue9 dma_priority_queue10 dma_priority_queue11 dma_priority_queue12 dma_priority_queue13 dma_priority_queue14 dma_priority_queue15	Configures the number of queues to support. To disable priority queuing, do not set these defines.

The lowest priority queue will be queue0. For each transmit queue (starting from queue0), there is an associated transmit DPRAM region and the address space allocated to it will be defined as shown in *Table 12*.

Table 12: Transmit Queue ‘defines

Name of ‘define	Description
gem_tx_pbuf_addr	Configures the total DPRAM space. To support priority queues, this space will be split into equal sized segments.
gem_tx_pbuf_queue_segment_size	Configures the size of the segments. This define only needs to be set if priority queuing is enabled and defines how many of the DPRAM address bits specified in `gem_tx_pbuf_addr will be used for the number of segments. Eg. If `gem_tx_pbuf_queue_segment_size = 4, then there are 16 segments each holding 1/16 of the total DPRAM space. If `gem_tx_pbuf_queue_segment_size = 1, then there are 2 segments each holding 1/2 of the total DPRAM space.

Table 12: Transmit Queue ‘defines (Continued)

Name of ‘define	Description

gem_tx_pbuf_num_segments_qN	Allocates the number of segments (N) required by each active queue.
	This is achieved by defining the number of bits to ensure a power of two is always maintained.
	Note: If only 1 segment is required for a queue, then the define associated with that queue should not be set.
	Eg. For 4 queues, with queue0 using a quarter of the TX DPRAM space, queue1 and queue2 each using an eighth and queue3 using half, the defines would be setup as follows:
	`` `define dma_priority_queue1 ``
	`` `define dma_priority_queue2 ``
	`` `define dma_priority_queue3 ``
	`` `define gem_tx_pbuf_queue_segment_size 3 // Eight segments ``
	`` `define gem_tx_pbuf_num_segments_q0 1 // Queue 0 uses 2 segments ``
	`` `define gem_tx_pbuf_num_segments_q3 2 // Queue 3 uses 4 segments ``
	`gem_tx_pbuf_num_segments_q1 and `gem_tx_pbuf_num_segments_q2 are not defined as they only use 1 segment.
	The defines should be set so that at least 512 bytes is available for each transmit queue.

gem_exclude_cbs	When multiple transmit queues are present, choose whether to include the credit based shaper algorithm for the top 2 queues. A credit-based shaping algorithm is available on the two highest priority queues and is defined in 802.1Qav: Forwarding and Queuing Enhancements for Time-Sensitive Streams. This allows traffic on these queues to be limited and to allow other queues to transmit. This define is negative. So only set this define if you do NOT want the CBS logic.

3.2.5.3 Receive Priority Queues

Table 13 lists GEM’s receive priority queue configuration.

Table 13: RX Priority Queue ‘defines

Name of ‘define	Description

num_type1_screeners
num_type2_screeners
	Enables and configures the RX priority queue screening algorithm in the receive direction used to identify receive queue information.
	Do not define either of the following if you do not wish to include any screeners, otherwise compilation errors will occur.
	`` `define num_type1_screeners 8'd16   // Number of screener type 1 registers ``
	`` `define num_type2_screeners 8'd16   // Number of screener type 2 registers ``

num_scr2_ethtype_regs	If the num_type2_screeners define has been set, the number of ethertype match registers will need to be defined up to a maximum of 8. For zero ethertype match registers (to minimize area), then do NOT define num_scr2_ethtype_regs at all. `` `define num_scr2_ethtype_regs 8'd8   // Number of ethertype match registers ``

Table 13: RX Priority Queue ‘defines (Continued)

Name of ‘define	Description
num_scr2_compare_regs	If the num_type2_screeners define has been set, the number of field compare match registers will need to be defined up to a maximum of 32. For zero field compare match registers (to minimize area), then do NOT define num_scr2_compare_regs at all. `` `define num_scr2_compare_regs 8'd16  // Number of field compare match registers ``

3.2.5.4 Enhancement for Scheduled Traffic (EnST) Module

Table 14 lists the Verilog defines for the IEEE 802.1Qbv functionality, which is Enhancement for Scheduled Traffic (EnST).

Table 14: EnST Configuration ‘defines

Name of ‘define	Description
gem_exclude_qbv	Excludes the EnST module. To exclude the EnST module, set this define.

3.2.5.5 IEEE 802.3br Pre-Emption Support

Table 15 lists the Verilog defines for the IEEE 802.3br functionality, which is Frame Pre-Emption (or Interspersing Express Traffic).

Table 15: IEEE 802.3br Pre-Emption Support ‘defines

Name of ‘define	Description
gem_has_802p3br	When defined, the core will support IEEE 802.3br standard. If choosing to include this functionality the AHB interface will not be supported and two instances of the MAC and DMA will be instantiated internally.
gem_emac_rx_pbuf_addr	Define the size of the RX packet buffer SRAM for the eMAC (express MAC), if gem_rx_pkt_buffer is defined.
gem_emac_tx_pbuf_addr	Define the size of the TX packet buffer SRAM for the eMAC, if gem_rx_pkt_buffer is defined.

3.2.5.6 FIFO-Based DMA Implementation

Table 16 lists the Verilog defines specific to the FIFO-based DMA implementation. These defines are ignored if the `gem_rx_pkt_buffer is defined.

Table 16: FIFO-Based DMA Implementation Configuration ‘defines

Name of ‘define	Description

gem_rx_fifo_size
gem_rx_base2_fifo_size
gem_rx_fifo_cnt_width
 	
gem_tx_fifo_size
gem_tx_base2_fifo_size
gem_tx_fifo_cnt_width
	Sets the size of the FIFOs.
	If the DMA is configure to used internal FIFOs, the following statements set the size of the FIFOs:
	`` `define gem_rx_fifo_size 10 ``
	`` `define gem_rx_base2_fifo_size 4’b1010 ``
	`` `define gem_rx_fifo_cnt_width 4 ``
	
	`` `define gem_tx_fifo_size 10 ``
	`` `define gem_tx_base2_fifo_size 4’b1010 ``
	`` `define gem_tx_fifo_cnt_width 4 ``
	This will set the receive FIFO and transmit FIFO word depths to 10. GEM has been verified with these parameters set to 10.
	The use of any other sizes would require verification and possible consequential code changes.

3.2.6 Timestamp Unit Inclusion

Table 17 lists the configuration for TSU inclusion.

Table 17: TSU Inclusion Configuration ‘defines

Name of ‘define	Description
gem_tsu	If TSU enhancements for 1588 support are not required, do not set this define
gem_ext_tsu_timer	If External TSU port is not used, do not set this define
gem_tsu_clk	Selects TSU clock from tsu_clk rather than pclk

3.2.7 Transmit and Receive Offload Inclusion

Table 18 lists the configuration for transmit and receive offload inclusion.

Table 18: Transmit and Receive Offload Configuration ‘defines

Name of ‘define	Description
gem_pbuf_lso	If Large Send Offload is not required, do not set this define
gem_pbuf_rsc	If Receive Side Coalescing is not required, do not set this define

3.2.8 Automotive Safety Features (ASF)

Table 19 lists the configuration for the GEM's ASF capabilities.

Table 19: ASF Configuration ‘defines

Name of ‘define	Description
gem_asf_enable	Activates the following ASF options under the hood. These are used in other places throughout this document but will be set by this global define to reduce the number of available customer-driven options: gem_asf_dap_prot gem_asf_csr_prot gem_asf_trans_to_prot gem_asf_integrity_prot
gem_asf_ecc_sram	Enables SECDED ECC protection of SRAM. Only valid if above gem_asf_enable is also defined.
gem_asf_prot_tsu	If defined, the TSU will be duplicated. Refer to Section 5 of the Integration Guide for area impact. Only valid if gem_asf_enable is also defined.
gem_asf_prot_tx_sched	If defined, the transmit scheduler will be duplicated. Refer to Section 5 of the Integration Guide for area impact. Only valid if gem_asf_enable is also defined.
gem_asf_host_par	Adds parity to host side interfaces. Only valid if gem_asf_enable is also defined.
The following Verilog defines are automatically set by selecting *gem_asf_enable.* This is to allow possible future configuration of specific features.	
gem_asf_dap_prot	Enables Data and Address Path parity protection
gem_asf_csr_prot	Enables Configuration/control and Status Registers protection
gem_asf_trans_to_prot	Enables transactions timeout monitoring on host-side interface
gem_asf_integrity_prot	Enables internal integrity checks

3.2.9 Revision Register Values

GEM contains readable bit fields for part number and revision number. The top 4 bits are for the fix number, the next 12 bits the part number, and the bottom 16 bits are module revision.

The part number is fixed as follows:

- 0x009 for IP7010
- 0x107 for IP7014
- 0x207 for IP7014A
- 0x10a for IP7016
- 0x00a for IP7017
- 0x20a for IP7017A

The revision register is incremented after each major release of the IP.

For example, the revision register value for IP7014 release 1p11f1 is set as follows:

```
`define gem_revision_reg_value 32’h1107010b
```

The hardwired value of the PCS PHY identification registers can be configured using the following statements. If the gem_phy_id_top define is commented out, then these registers do not appear in the address map:

```
`define gem_phy_id_top 16’h0002
```

```
`define gem_phy_id_bot 16’h010b
```

3.2.10 Default Register Values

The default values for some programmable registers can be defined at compile time, though they are still over-writeable by software.

Table 20 lists GEM’s filter configuration.

Table 20: Default Register Values Configuration ‘defines

Name of ‘define	Description

gem_dma_bus_width_def	Sets the default DMA bus width in the network configuration register.
	The default DMA bus width can be set to 32, 64, or 128 bits with the following define:
	`// 2'b00 - 32 bits`
	`// 2'b01 - 64 bits`
	`// 2'b1x - 128 bits`
	`` `define gem_dma_bus_width_def 2'b00 ``

gem_mdc_clock_div	Sets the default amount PCLK is divided to generate MDC.
	This can be set in the network configuration register as follows:-
	`// 3'b000 - divide pclk by 8`
	`// 3'b001 - divide pclk by 16`
	`// 3'b010 - divide pclk by 32`
	`// 3'b011 - divide pclk by 48`
	`// 3'b100 - divide pclk by 64`
	`// 3'b101 - divide pclk by 96`
	`// 3'b110 - divide pclk by 128`
	`// 3'b111 - divide pclk by 224`
	`` `define gem_mdc_clock_div 3'b010 ``

gem_endian_swap_def	Sets the default DMA endianism value in DMA control register.
	This can be set as follows:-
	`//             packet data            management descriptors`
	`// 2'b00 -    little endian              little endian`
	`// 2'b01 -    little endian                 big endian`
	`// 2'b10 -       big endian              little endian`
	`// 2'b11 -       big endian                 big endian`
	`` `define gem_endian_swap_def 2'b00 ``

gem_rx_pbuf_size_def	Sets the default RX DMA packet buffer memory size in DMA configuration register.
	This can be set as follows:-
	`// 2'b11 - use full configured memory size`
	`// 2'b10 - use half of configured memory size`
	`// 2'b01 - use quarter of configured memory size`
	`// 2'b00 - use eighth of configured memory size`
	`` `define gem_rx_pbuf_size_def 2'b11 ``

gem_tx_pbuf_size_def	Sets the default TX DMA packet buffer memory size in DMA configuration register.
	This can be set as follows:-
	`// 1'b1 - use full configured memory size`
	`// 1'b0 - use half of configured memory size`
	`` `define gem_tx_pbuf_size_def 1'b1 ``

gem_rx_buffer_length_def	Sets the default value of the external DMA receiver buffers. This is set to 128 byte with the following code and must be a multiple of 64 bytes up to 16320 bytes: `‘define gem_rx_buffer_length_def 8’d2`

4 Pin List

4.1 I/O Signals List

Figure 3 shows GEM's signal interfaces.

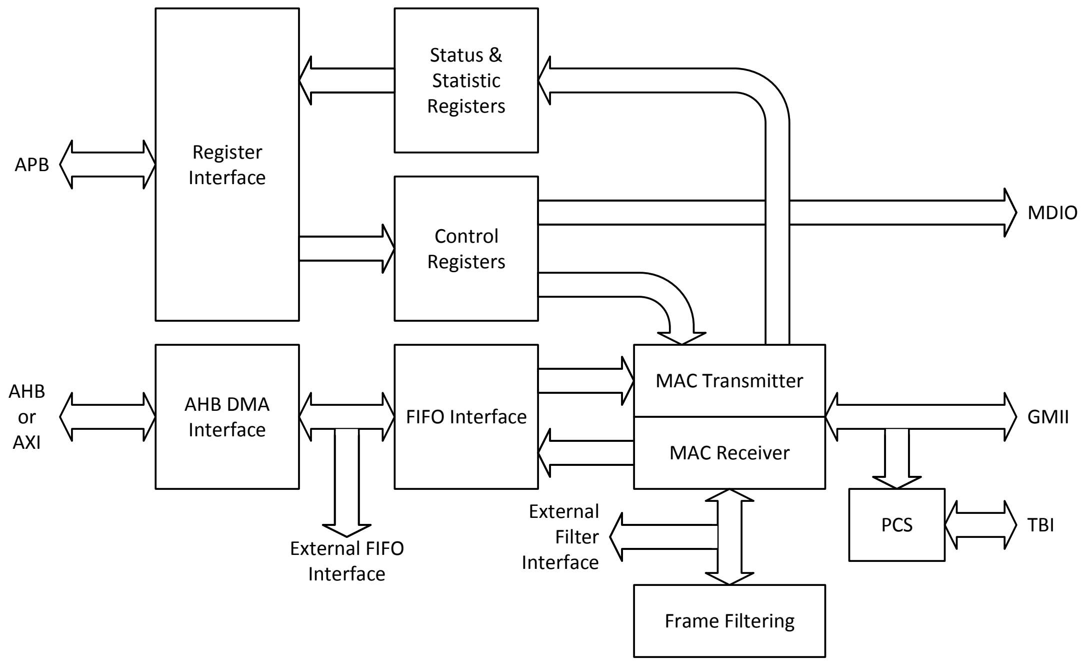

Figure 3: Signal Interfaces

GEM includes the following signal interfaces:

- GMII, RGMII, MII, RMII to an external PHY
- 10 or 20-bit interface to an internal SerDes
- MDIO interface for external PHY management
- AMBA Advanced Peripheral Bus (APB version 2) slave interface for accessing the GEM's registers
- AMBA Advanced High Speed Bus (AHB versions 2, 3, or 4; or AXI4) master interface for memory access
- An optional FIFO interface in applications where DMA functionality is not required
- An optional timestamp interface

The AMBA interfaces fully conform to the ARM AMBA Revision 2.0 specification.

Table 21 lists the controller ports and their descriptions.

Table 21: Signal List

Signal Name	Direction	Clock Domain	Polarity / Bus Size	Description
Ethernet Interface (GMII/MII)				
col	Input	Asynchronous	High	Collision detect from the PHY
crs	Input	Asynchronous	High	Carrier sense from the PHY

Table 21: Signal List (Continued)

Signal Name	Direction	Clock Domain	Polarity / Bus Size	Description
tx_er	Output	tx_clk	High	Transmit error signal to the PHY. Asserted if the DMA block fails to fetch data from memory during frame transmission.
txd[7:0]	Output	tx_clk	8	Transmit data to the PHY. 10/100 mode: txd[3:0] used, txd[7:4] tied to logic 0. 1000 mode: txd[7:0] used.
tx_en	Output	tx_clk	High	Transmit enable to the PHY
tx_clk	Input			Transmit clock from the system clock generator. • 125MHz for gigabit operation • 25MHz for 100M operation • 2.5MHz for 10M operation • 12.5MHz for 100M SGMII operation • 1.25MHz for 10M SGMII operation
rxd[7:0]	Input	rx_clk	8	Receive data from the PHY: • 10/100 mode: rxd[3:0] used, rxd[7:4] not used. • 1000 mode: rxd[7:0] used.
rx_er	Input	rx_clk	High	Receive error signal from the PHY.
rx_clk	Input			Receive clock from the system clock generator. • 125MHz for gigabit GMII operation • 25MHz for 100M operation • 2.5MHz for 10M operation • 62.5MHz for gigabit SGMII/TBI operation • 12.5MHz for 100M SGMII operation • 1.25MHz for 10M SGMII operation
rx_dv	Input	rx_clk	High	Receive data valid signal from the PHY.
Ethernet Interface (RGMII) (Optional)				
Note: From release 1p09 onwards rx_er, tx_er, col, and crs are brought out at the top-level when RGMII is configured in the defines file. This is to allow reconfiguration of the RGMII interface by firmware to support MII. If MII support is not required, it is important to tie the rx_er, col and crs inputs low.				
rgmii_tx_clk_sig	Input	tx_clk	High	Slightly delayed version of tx_clk used as the signal to control the multiplexer for rgmii_txd output data. Can be set high for 10/100M operation to prevent rgmii_txd driving zero on the falling edge of tx_clk.
rgmii_txd[3:0]	Output	tx_clk	4	Transmit data signal to the PHY.
rgmii_tx_ctl	Output	tx_clk		Transmit control signal to the PHY.
rgmii_rxd[3:0]	Input	rx_clk	4	Receive data signal from the PHY.
rgmii_rx_ctl	Input	rx_clk		Receive control signal from the PHY.
tx_clk	Input			Transmit clock from the system clock generator. • 125MHz for gigabit GMII operation • 25MHz for 100M operation • 2.5MHz for 10M operation This clock must also be output from the SoC to support source synchronous clocking to the PHY.
n_tx_clk	Input			Inverted tx_clk
rx_clk	Input			Receive clock from the PHY. • 125MHz for gigabit GMII operation • 25MHz for 100M operation • 2.5MHz for 10M operation
n_rx_clk	Input			Inverted rx_clk
rgmii_speed[1:0]	Output	rx_clk	2	RGMII extracted signal indicating speed of operation

Table 21: Signal List (Continued)

Signal Name	Direction	Clock Domain	Polarity / Bus Size	Description
rgmii_link_status	Output	rx_clk	High	RGMII extracted link status signal
rgmii_duplex_out	Output	rx_clk	High	RGMII extracted signal indicating duplex mode
Ethernet Interface (RMII) (Optional)				
mii_select	Input	Static		Drive high to select standard GMII interface, also drive high in gigabit mode. Drive low for RMII operation.
ref_clk	Input			50MHz reference clock for RMII interface module.
n_ref_reset	Input	ref_clk	Low	Active low asynchronous reset for the ref_clk domain
rmii_tx_clk	Output	ref_clk		ref_clk timed signal to be used as a source for tx_clk.
rmii_rx_clk	Output	ref_clk		ref_clk timed signal to be used as a source for rx_clk.
rmii_crs_dv	Input	ref_clk		Receive data valid signal from the PHY
rmii_txd[1:0]	Output	ref_clk	2	Transmit data to the PHY
rmii_tx_en	Output	ref_clk	High	Transmit enable to the PHY
rmii_rxd[1:0]	Input	ref_clk	2	Receive data from the PHY
rmii_rx_er	Input	ref_clk	High	Receive error signal from the PHY
MDIO Interface				
mdc	Output	pclk		Management data clock to pin
mdio_in	Input	Asynchronous		Management data input from MDIO pin
mdio_out	Output	pclk		Management data output to MDIO pin

mdio_en	Output	pclk		Management data output enable to MDIO pin.
				At the top-level the three mdio output pins can be used to drive a single tri-state pin.
				Alternatively, MDIO may be implemented as a pull-down. In this case the enable to the pull-down should be driven with `~mdio_out & mdio_en.`
				Example top-level RTL:
				`assign mdio_in = mdio;`
				`assign mdio = (mdio_en) ? mdio_out : 1'bz;`
				or if an external pull-up is used:
				`assign mdio = (mdio_en & ~mdio_out) ? 1'b0 : 1'bz;`

Control/Status Interface				
loopback	Output	pclk	High	Selects external loopback — not used by most PHYs. This signal does not need to be bonded out, but is used by the testbench.
loopback_local	Output	pclk	High	Internal loopback indication to the system clock generator
half_duplex	Output	pclk	High	Selects half duplex. This signal does not need to be bonded out, but is used by the testbench. It is also used by the RGMII core, half_duplex is inverted and connected to gmii_duplex_en of the RGMII core.

Table 21: Signal List (Continued)

Signal Name	Direction	Clock Domain	Polarity / Bus Size	Description
speed_mode[3:0]	Output	pclk	4	Indicates speed and external interface that GEM is currently configured to use to the system clock generator: • 0000 - 10 Mbps Ethernet operation using MII interface • 0001 - 100 Mbps operation using MII interface • 001x - 1000 Mbps operation using GMII interface • 0100 - 10 Mbps operation using SGMII interface • 0101 - 100 Mbps operation using SGMII interface • 011x - 1000 Mbps operation using TBI or SGMII interface speed_mode[3] indicates the value of the two_pt_five_-gig bit 29 of the network control register speed_mode[2:0] indicates the value of the tbi, gigabit and speed bits 11, 10 and 0 of the network configuration register.
tx_pause	Input	Asynchronous		Toggle input for pause frame transmission. When tx_pfc_sel is LOW, a classic 802.3 pause frame will be transmitted. The pause quantum value to be transmitted depends on the state of the tx_pause_zero pin. When tx_pfc_sel is HIGH, an 802.1Qbb, or PFC priority based pause frame will be transmitted. The pause quantum value to be transmitted depends on the state of the tx_pfc_priority bus. Tie this input low if hardware initiated flow control is not being used. This asynchronous input is synchronized into the tx_clk domain.
tx_pfc_sel	Input	Asynchronous		Indicates whether the pause frame to be transmitted should be classic 802.3 (when set to logic 0) or PFC priority based pause frame (when set to logic 1). This signal should be valid when tx_pause toggles and remain valid for at least two tx_clk cycles afterwards. Tie this low if hardware initiated flow control is not being used.
tx_pause_zero	Input	Asynchronous		Indicates whether the classic 802.3 pause frame to be transmitted is to have zero pause quantum (when set to logic 1) or the value of the transmit pause quantum register (when set to logic 0). This signal should be valid when tx_pause toggles and remain valid for at least two tx_clk cycles afterwards. Tie this low if hardware initiated flow control is not being used.
tx_pfc_pause[7:0]	Input	Asynchronous	8	Indicates the value to be used in the 8bit priority enable vector of the PFC priority based pause frame. This signal should be valid when tx_pause toggles and remain valid for at least two tx_clk cycles afterwards. Tie this low if hardware initiated flow control is not being used.

Table 21: Signal List (Continued)

Signal Name	Direction	Clock Domain	Polarity / Bus Size	Description
tx_pfc_pause_zero [7:0]	Input	Asynchronous	8	Indicates whether each priority entry in the PFC priority based pause frame is to have zero pause quantum (when set to logic 1) or the value of the transmit pause quantum register (when set to logic 0). This signal should be valid when tx_pause toggles and remain valid for at least two tx_clk cycles afterwards. Tie this low if hardware initiated flow control is not being used.
trigger_dma_tx_start	Input	Asynchronous		Toggle input for starting transmit DMA. Only present when using the DMA configured for internal packet buffering. This signal is used as a hardware alternative to setting the TX START bit of the network control register. If you only wish to use software to initiate transmission, then this input can be tied low. This asynchronous input is synchronized into the hclk domain.
rx_pfc_paused[7:0]	Output	rx_clk	8	Each bit corresponds to a priority indicated within the PFC priority based pause frame. Each bit is set when a PFC priority based pause frame has been received, and the associated priority pause time quantum is non-zero. Each bit is cleared when the associated pause time identified by the received pause time quantum has elapsed.
pfc_negotiate	Output	rx_clk		Identifies that PFC priority based pause flow control has been negotiated. • No PFC priority based pause frames have yet been received, flow control is being handled using classic 802.3 pause frames. • At least one PFC priority based pause frames has been received. All subsequent 802.3 pause frames will be dropped.
rx_databuf_wr_q0	Output	aclk/hclk		When priority queuing is disabled, this output is toggled on a DMA write to the first word of each DMA data buffer. When priority queuing is enabled, this output is toggled on a DMA write to the first word of each DMA data buffer associated with queue 0. This output can be left unconnected.
rx_databuf_wr_q1	Output	aclk/hclk		Only present when priority queuing support is enabled. This output is toggled on a DMA write to the first word of each DMA data buffer associated with queue 1.
rx_databuf_wr_q2	Output	aclk/hclk		Only present when priority queuing support is enabled. This output is toggled on a DMA write to the first word of each DMA data buffer associated with queue 2.
rx_databuf_wr_q3	Output	aclk/hclk		Only present when priority queuing support is enabled. This output is toggled on a DMA write to the first word of each DMA data buffer associated with queue 3.

Table 21: Signal List (Continued)

Signal Name	Direction	Clock Domain	Polarity / Bus Size	Description
rx_databuf_wr_q4	Output	aclk/hclk		Only present when priority queuing support is enabled. This output is toggled on a DMA write to the first word of each DMA data buffer associated with queue 4.
rx_databuf_wr_q5	Output	aclk/hclk		Only present when priority queuing support is enabled. This output is toggled on a DMA write to the first word of each DMA data buffer associated with queue 5.
rx_databuf_wr_q6	Output	aclk/hclk		Only present when priority queuing support is enabled. This output is toggled on a DMA write to the first word of each DMA data buffer associated with queue 6.
rx_databuf_wr_q7	Output	aclk/hclk		Only present when priority queuing support is enabled. This output is toggled on a DMA write to the first word of each DMA data buffer associated with queue 7.
halfduplex_flow_control_en	Input	Asynchronous		This input is ignored when GEM is configured for full duplex or gigabit modes. Setting this bit high will enable the half duplex flow control mechanism. The transmit block will transmit 64 bits of data, whenever it sees an incoming frame in order to force a collision. Tie this signal low if not being used.
dma_bus_width[1:0]	Output	pclk	2	Indicates width of DMA or external FIFO data bus as follows: • 00 - 32 bit data bus • 01 - 64 bit data bus • 10 or 11 - 128 bit data bus This signal does not need to be bonded out, but is used by the testbench.
External Filter Interface				
ext_match1 ext_match2 ext_match3 ext_match4	Input	rx_clk	High	Optional external match 1, 2, 3 and 4. Any one of these input signals from the external filter logic can be asserted indicating a match was found. Filter matches can take place on either ext_sa, ext_da, ext_type or ext_vid. If bit 26 ext_rxq_sel_en is set in the network control register then these inputs select which queue a matched receive frame is sent to.
ext_sa[47:0]	Output	rx_clk	48	Source address for comparison, extracted from packet being received
ext_sa_stb	Output	rx_clk	High	Validates source address
ext_da[47:0]	Output	rx_clk	48	Destination address for comparison, extracted from packet being received
ext_da_stb	Output	rx_clk	High	Validates destination address
ext_type[15:0]	Output	rx_clk	16	Length/type field for comparison, extracted from packet being received
ext_type_stb	Output	rx_clk	High	Validates length/type field
ext_vlan_tag1[31:0]	Output	rx_clk	32	VLAN identifier and VLAN ID field for comparison, extracted from packet being received. For packets incorporating two VLAN tags (stacked VLAN), this represents the first VLAN tag
ext_vlan_tag1_stb	Output	rx_clk	High	Validates VLAN ID field

Table 21: Signal List (Continued)

Signal Name	Direction	Clock Domain	Polarity / Bus Size	Description
ext_vlan_tag2[31:0]	Output	rx_clk	32	VLAN identifier and VLAN ID field for comparison, extracted from packet being received. For packets incorporating two VLAN tags (stacked VLAN), this represents the second VLAN tag. These bits are only valid for packets incorporating the stacked VLAN processing feature.
ext_vlan_tag2_stb	Output	rx_clk	High	Validates VLAN ID field
ext_ip_sa[127:0]	Output	rx_clk	128	IP source address for comparison, extracted from the packet being received. For IPv4 packets, only bits [31:0] are valid. For Ipv6 packets, bits [127:0] are valid.
ext_ip_sa_stb	Output	rx_clk	High	Validates IP source address field
ext_ip_da[127:0]	Output	rx_clk	128	IP destination address for comparison, extracted from the packet being received. For IPv4 packets, only bits [31:0] are valid. For Ipv6 packets, bits [127:0] are valid.
ext_ip_da_stb	Output	rx_clk	High	Validates IP destination address field
ext_source_port[15:0]	Output	rx_clk	16	TCP/UDP source port number for comparison
ext_sp_stb	Output	rx_clk	High	Validates ext_source_port
ext_dest_port[15:0]	Output	rx_clk	16	TCP/UDP destination port number for comparison
ext_dp_stb	Output	rx_clk	High	Validates ext_dest_port
ext_ipv6	Output	rx_clk	High	Asserted high for ipv6 frames
wol	Output	rx_clk	High	Asserted for 64 rx_clk cycles if a magic packet, ARP request, specific address 1 or multicast hash event is decoded and is enabled through the Wake-on-LAN register. Only bond this pin out if Wake on LAN functionality is required.
IEEE 1588 PTP Frame Recognition and Timestamp Unit Interface				
sof_tx	Output	tx_clk	High	Asserted high synchronous to tx_clk when the SFD is detected on a transmit frame, deasserted at end of frame.
sync_frame_tx	Output	tx_clk	High	Asserted high synchronous to tx_clk if PTP sync frame is detected on transmit.
delay_req_tx	Output	tx_clk	High	Asserted high synchronous to tx_clk if PTP delay request frame is detected on transmit.
pdelay_req_tx	Output	tx_clk	High	Asserted high synchronous to tx_clk if PTP peer delay request frame is detected on transmit.
pdelay_resp_tx	Output	tx_clk	High	Asserted high synchronous to tx_clk if PTP peer delay response frame is detected on transmit.
sof_rx	Output	rx_clk	High	Asserted high synchronous to rx_clk when the SFD is detected on a receive frame
sync_frame_rx	Output	rx_clk	High	Asserted high synchronous to rx_clk if PTP sync frame is detected on receive.
delay_req_rx	Output	rx_clk	High	Asserted high synchronous to rx_clk if PTP delay request frame is detected on receive.
pdelay_req_rx	Output	rx_clk	High	Asserted high synchronous to rx_clk if PTP peer delay request frame is detected on receive.

Table 21: Signal List (Continued)

Signal Name	Direction	Clock Domain	Polarity / Bus Size	Description
pdelay_resp_rx	Output	rx_clk	High	Asserted high synchronous to rx_clk if PTP peer delay response frame is detected on receive.
tsu_clk	Input			Alternative clock source for the time stamp unit. If gem_tsu_clk is defined in the gem_gxl_defs.v file then the TSU is clocked by tsu_clk rather than pclk. This clock must have a frequency greater than $1/8^{th}$ the frequency of tx_clk or rx_clk. Timestamp accuracy improves with higher frequencies.
gem_tsu_ms	Input	tsu_clk		TSU master/slave. Used with gem_tsu_inc_ctrl to control incrementing of the TSU and loading the sync strobe register.
gem_tsu_inc_ctrl[1:0]	Input	tsu_clk	2	Used to control incrementing of the TSU. Synchronous to tsu_clk or pclk. Drive high when not being used.
tsu_timer_cnt[93:0]	Output	tsu_clk	94	TSU timer count value. Synchronized to tsu_clk or pclk. The upper 48 bits indicate the seconds value and the lower 46 bits the nano-seconds / sub-nanoseconds. Bit 46 toggles every second, i.e. 1pps (also bit 45's falling edge is co-incident with this toggle).
tsu_timer_cmp_val	Output	tsu_clk		TSU timer comparison valid, synchronized to tsu_clk or pclk. Asserted high when upper 70 bits of TSU timer count value are equal to programmed comparison value.
ext_tsu_timer[93:0]	Input	tsu_clk	94	A 94-bit external timestamp interface enabled using the Network_Control register.
SerDes Interface (Optional)				
gtx_clk	Input			PCS transmit clock running at 125MHz for gigabit operation. This clock uses the same clock source as tx_clk. When operating in SGMII applications tx_clk is derived from this clock and divided down appropriately for 10 and 100Mbps speeds
gtx20_clk	Input			Optional interface clock used when the gem_pcs_20b_if define is used to select a 20-bit PHY interface. This clock is half the frequency of gtx_clk and both clocks must be phase aligned.
pcs_rx_clk	Input			Optional interface clock for when either gem_pcs_10b_if or gem_pcs_20b_if is defined. This clock is 62.5MHz for gigabit operation. This clock needs to be balanced with rx_clk. When operating in SGMII applications rx_clk is derived from either this clock or rbc1 and divided down appropriately for 10 and 100Mb/s speeds.
pcs_rx10_clk	Input			Optional interface clock used when the gem_pcs_10b_if define is used to select a 10-bit PHY interface. This clock is twice the frequency of pcs_rx_clk and both clocks must be phase aligned. This clock needs to be balanced with rx_clk.

Table 21: Signal List (Continued)

Signal Name	Direction	Clock Domain	Polarity / Bus Size	Description
rbc0 rbc1	Input			These optional clocks are used if gem_pcs_legacy_if is defined to select a TBI interface as defined in IEEE 802.3 Clause 36. The clocks will be 62.5MHz and rbc1 will be 180 degrees phase shifted with respect to rbc0.
tx_group[n:0]	Output	gtx_clk or gtx20_clk	10 or 20	8B/10B encoded transmit data to PHY transceiver, synchronized to either gtx_clk or gtx20_clk depending on interface configuration.
rx_group[n:0]	Input	rbc0/rbc1 or pcs_rx_clk or pcs_rx10_clk	10 or 20	8B/10B encoded receive data from PHY transceiver. The data should be synchronous to rbc0/rbc1 clocks if gem_pcs_legacy_if is defined otherwise depending on interface width will be synchronous to either pcs_rx_clk or pcs_rx10_clk.
pcs_cal_bypass	Input	Static		Optional static control to enable bypass of comma alignment module when gem_pcs_legacy_if is not used. Unless an external comma alignment function is used, this signal should always be tied to 1'b0.
pcs_cgalign_bypass	Input	Static		Optional static control to enable bypass of codegroup alignment module when gem_pcs_legacy_if is not used. If a 10-bit interface is used, it is recommended that this signal is always tied to 1'b0.
ewrap	Output			Control for loopback operation in the PHY transceiver.
en_cdet	Output			Control for comma alignment in the PHY transceiver (see IEEE 802.3 subclause 36.3.2.4). Leave unconnected if an on-chip SerDes is being used.
signal_detect	Input	Asynchronous		Valid signal detected from PMA. This signal must be driven high for PCS synchronization to occur. You are advised to check that this signal is driven high by their SerDes or to provide a control signal to over-ride the signal from the SerDes.
Interrupt Controller Interface				
ethernet_int	Output	pclk	High	Ethernet interrupt signal synchronous to pclk. When priority queuing is enabled, this interrupt represents the interrupt dedicated to Q0. When priority queuing is disabled, this interrupt represents the only interrupt output.
ethernet_int_q1 to ethernet_int_q15	Output	pclk	High	Ethernet interrupt signals for queues 1 to 15
emac_ethernet_int	Output	pclk	High	Ethernet interrupt signal synchronous to pclk for the eMAC. Only present for 802.3br configurations.
mmsl_int	Output	pclk	High	Interrupt signal synchronous to pclk for the MAC merge sublayer. Only present for 802.3br configurations.
ext_interrupt_in	Input	pclk	High	External interrupt input pin. The core can be programmed to generate an interrupt on the rising edge of this pin. Tie this signal low if not being used. This asynchronous input is synchronized into the pclk domain.

Table 21: Signal List (Continued)

Signal Name	Direction	Clock Domain	Polarity / Bus Size	Description
External Transmit FIFO Interface (Optional)				
Note: The transmit FIFO interface inputs must be pre-synchronized to the tx_clk clock domain.				
tx_r_data_rdy	Input	tx_clk		When set to logic 1, indicates enough data is present in the external FIFO for Ethernet frame transmission to commence on the current packet. If multiple queues are supported this signal becomes a vector with a bit for each queue configured.
tx_r_rd	Output	tx_clk		Single tx_clk clock cycle wide active high output requesting a word (either 32-bits, 64-bits or 128-bits) of information from the external FIFO interface. If multiple queues are supported this signal becomes a vector with a bit for each queue configured.
tx_r_valid	Input	tx_clk		Single tx_clk clock cycle wide active high input indicating requested FIFO data is now valid. Validates the following inputs: tx_r_data[127:0], tx_r_sop, tx_r_eop, tx_r_err and tx_r_mod[3:0].
tx_r_data[127:0]	Input	tx_clk	128	FIFO data for transmission, either 32-bit, 64-bit or 128-bit depending on dma_bus_width[2:0] which is configured through the register interface. This input is only valid whilst tx_r_valid is high.
tx_r_sop	Input	tx_clk		Start of packet, indicates the word received from the external FIFO interface is the first in a packet. This input is only valid whilst tx_r_valid is high.
tx_r_eop	Input	tx_clk		End of packet, indicates the word received from the external FIFO interface is the last in a packet. This input is only valid whilst tx_r_valid is high.
tx_r_mod[3:0]	Input	tx_clk	4	Read module, indicates how many bytes are valid on tx_r_data during the last transfer of the packet: • 0000- all bytes are valid • 0001 - tx_r_data[7:0] is valid • 0010 - tx_r_data[15:0] is valid • 0011 - tx_r_data[23:0] is valid, and so on until • 1111 - tx_r_data[119:0] is valid. This input is only valid when both tx_r_eop and tx_r_valid are high.
tx_r_err	Input	tx_clk		Error, active high input indicating the current packet contains an error. This signal is only valid whilst tx_r_valid is high and may be set at any time during the packet transfer. If tx_r_err is set concurrently with tx_r_sop then no frame will be transmitted and dma_tx_end_tog will not be toggled.
tx_r_underflow	Input	tx_clk		FIFO under flow, indicating the transmit FIFO was empty when a read was attempted. This signal is only valid when GEM has attempted a read by asserting tx_r_rd and the tx_r_valid signal has not yet been indicated. tx_r_flushed should be asserted following this event to indicate to GEM when it is safe to resume reading.

Table 21: Signal List (Continued)

Signal Name	Direction	Clock Domain	Polarity / Bus Size	Description
tx_r_flushed	Input	tx_clk		This signal must be driven high and then low after a major error event to indicate to GEM that the external FIFO has been flushed. This will enable GEM to resume reading data. Events that require this to be set are indicated by any bit of tx_r_status being asserted
tx_r_control	Input	tx_clk		tx_no_crc, set active high at start of packet to indicate current frame is to be transmitted without crc being appended. This input is only valid whilst both tx_r_valid and tx_r_sop are high.
tx_r_frame_size[13:0]	Input	tx_clk	14	If DWRR/ETS scheduling is not used, then these signals should be tied to 0. If DWRR/ETS scheduling is used, then this indicates the length of the frame to be transmitted.
tx_r_frame_size_vld	Input	tx_clk		Validates the tx_r_frame_size signal. If multiple queues are supported, this signal becomes a vector with a bit for each queue configured. If DWRR/ ETS scheduling is not used, then this signal should be tied to 0.
tx_r_queue[3:0]	Output	tx_clk		This vector is only present if priority queueing is enabled and is asserted with tx_r_rd to indicate which transmit queue is being read.
External Transmit FIFO Interface Status (Optional)				
dma_tx_end_tog	Output	tx_clk		Toggled to indicate that a frame has been completed and status is now valid on the tx_r_status and tx_r_time-stamp outputs. **Note:** This signal is not activated when a frame is being retried due to a collision.
dma_tx_status_tog	Input	Asynchronous		This signal must be toggled each time either dma_tx_end_tog or collision_occured are activated, to indicate that the status has been acknowledged.
tx_r_timestamp[77:0]	Output	tx_clk	78	Timestamp value at sop for transmit frame 48b seconds, 30b nanoseconds.

Table 21: Signal List (Continued)

Signal Name	Direction	Clock Domain	Polarity / Bus Size	Description
tx_r_status[3:0]	Output	tx_clk		[3]: fifo_underrun - status output indicating that the MAC transmitter has under run due to one of the following conditions. Data under run indicated by tx_r_underflow input from the external FIFO interface, status write back error due to status write back not completing when another status write back is attempted or a tx_r_err was driven on the external FIFO interface during the last frame transfer. Reset once dma_tx_status_tog changes logic state. [2]: collision_occured – HALF DUPLEX status output indicating that the frame in progress has suffered a collision and that re-transmission of the frame should take place. Set when the collision occurs and reset once dma_tx_status_tog changes logic state. Note that dma_tx_end_tog is not activated. (**Note**: This bit is not driven when the external TX FIFO interface is used in conjunction with the DMA (when `gem_tx_add_fifo_if verilog define is set) [1]: late_coll_occured – HALF DUPLEX status output indicating that the frame in progress suffered a late collision and was aborted. Valid when dma_tx_end_tog changes logic state. Cleared once dma_tx_status_tog changes logic state. (**Note**: This bit is not driven when the external TX FIFO interface is used in conjunction with the DMA (when `gem_tx_add_fifo_if verilog define is set) [0]: too_many_retries – HALF DUPLEX status output indicating that the frame in progress experienced excess collisions and was aborted. Valid when dma_tx_end_tog changes logic state. Cleared once dma_tx_status_tog changes logic state. (**Note**: This bit is not driven when the external TX FIFO interface is used in conjunction with the DMA (when `gem_tx_add_fifo_if verilog define is set)
External Receive FIFO Interface (Optional)				
Note: Receive FIFO interface outputs are synchronized to the rx_clk clock domain.				
rx_w_wr	Output	rx_clk		Single rx_clk clock cycle wide active high output indicating a write to the external FIFO interface.
rx_w_data[127:0]	Output	rx_clk	128	Received data for output to the external FIFO interface, either 32, 64, or 128-bit depending on dma_bus_width[2:0] which is configured through the register interface. This output is only valid when rx_w_wr is high.
rx_w_sop	Output	rx_clk		Start of packet, indicates the word output to the external FIFO interface is the first in a packet. This output is only valid when rx_w_wr is high.

Table 21: Signal List (Continued)

Signal Name	Direction	Clock Domain	Polarity / Bus Size	Description
rx_w_eop	Output	rx_clk		End of packet, indicates the word output to the external FIFO interface is the last in a packet. This output is only valid when rx_w_wr is high. **Note**: This signal will not be asserted if the receive path is disabled during frame reception however rx_w_flush will be asserted.

Table 21: Signal List (Continued)

Signal Name	Direction	Clock Domain	Polarity / Bus Size	Description
rx_w_status[44:0]	Output	rx_clk	45	Status signals, valid with rx_w_eop (with the exception of bits 20:15, which are valid on both rx_w_sop and rx_w_eop), definitions as follows: [44] - rx_w_code_error indicates a code error. [43] - rx_w_too_long indicates the frame was too long. [42] - rx_w_too_short indicates the frame was too short. [41] - rx_w_crc_error indicates the frame had a bad crc. [40] - rx_w_length_error indicates the length field was checked and was incorrect. [39] - rx_w_snap_match indicates the frame was SNAP encoded and had either no VLAN tag or a VLAN tag with the CFI bit not set. [38] - rx_w_checksumu indicates the UDP checksum was checked and was correct. [37] - rx_w_checksumt indicates the TCP checksum was checked and was correct. [36] - rx_w_checksumi indicates the IP checksum was checked and was correct. [35] - rx_w_type_match4, indicates the received frame was matched on type ID register 4. [34] - rx_w_type_match3, indicates the received frame was matched on type ID register 3. [33] - rx_w_type_match2, indicates the received frame was matched on type ID register 2. [32] - rx_w_type_match1, indicates the received frame was matched on type ID register 1. [31] - rx_w_add_match4, indicates the received frame was matched on specific address register 4. [30] - rx_w_add_match3, indicates the received frame was matched on specific address register 3. [29] - rx_w_add_match2, indicates the received frame was matched on specific address register 2. [28] - rx_w_add_match1, indicates the received frame was matched on specific address register 1. [27] - rx_w_ext_match4, indicates the received frame was matched externally by the ext_match4 input pin. [26] - rx_w_ext_match3, indicates the received frame was matched externally by the ext_match3 input pin. [25] - rx_w_ext_match2, indicates the received frame was matched externally by the ext_match2 input pin. [24] - rx_w_ext_match1, indicates the received frame was matched externally by the ext_match1 input pin. [23] - rx_w_uni_hash_match, indicates the received frame was matched as a unicast hash frame. [22] - rx_w_mult_hash_match, indicates the received frame was matched as a multicast hash frame. [21] - rx_w_broadcast_frame, indicates the received frame is a broadcast frame. [20] - rx_w_prty_tagged, indicates a VLAN priority tag detected with received packet. [19:16] - rx_w_tci[3:0], indicates VLAN priority of received packet (not supported for the corner case of a 802.1CB frame with both a redundancy tag and a second stacked VLAN tag). [15] - rx_w_vlan_tagged, indicates VLAN tag detected with received packet. [14] - rx_w_bad_frame, indicates received packet is bad. [13:0] - rx_w_frame_length, indicates number of bytes in received packet.
add_match_vec[X:0]	Output	rx_clk		These outputs indicate matches for all configured specific address registers. 'X' indicates the number of configured specific address filters. Matches for registers 1 to 4 are also indicated in the rx_w_status vector described above.

Table 21: Signal List (Continued)

Signal Name	Direction	Clock Domain	Polarity / Bus Size	Description
rx_w_queue[3:0]	Output	rx_clk		If priority queuing is enabled this vector indicates the priority queue selected by the screener registers. Valid with rx_w_eop.
rx_w_mod[3:0]	Output	rx_clk		Write modulus, indicates how many bytes are valid on rx_w_data during the last transfer of the packet: • 0000- all bytes are valid • 0001 - rx_r_data[7:0] is valid • 0010 - rx_r_data[15:0] is valid • 0011 - rx_r_data[23:0] is valid, and so on until • 1111 - rx_r_data[119:0] is valid This output is only valid when coincident with both rx_w_wr and rx_w_eop being high.
rx_w_err	Output	rx_clk		Error, active high output indicating the current packet contains an error. This signal is only valid when rx_w_wr is active high and may be set at any time during packet transfer. This signal will also be asserted high to indicate the current packet should be discarded due to IEEE 802.1CB elimination rules.
rx_w_overflow	Input	rx_clk		FIFO overflow, indicates to the MAC that the external RX FIFO has overflowed. The MAC uses this signal for status reporting at the end of frame.
rx_w_flush	Output	rx_clk		FIFO flush, active high output indicating that the external RX FIFO should be cleared. This signal is set when the receive path is disabled.
rx_w_timestamp[77:0]	Output	rx_clk	78	Timestamp value at sop for receive frame 48b seconds, 30b nanoseconds. Valid at rx_w_eop.
External Receive FIFO Interface for eMAC (Optional)				
Note: If 802.3br frame pre-emption is configured along with an external FIFO interface then the pMAC (pre-emptable MAC) will have the receive and transit signal names as defined above. There will be a separate external FIFO interface exposed for the eMAC which will have the same signals and the same signal names with the exception that the eMAC's signal names will be pre-pended with the four characters 'emac_'				
AMBA (APB) Interface				
n_preset	Input	pclk		Active low AMBA reset. This signal must be asserted low asynchronously, and deasserted high synchronously with pclk. Resets all APB registers and the pclk_syncs block.
pclk	Input			Peripheral bus clock.
psel	Input	pclk		Peripheral select. Active high select signal to indicate that a valid access is being made to one of the GEM's registers.
penable	Input	pclk		Peripheral enable. This indicates the second clock cycle of an access and indicates that the write data may be strobed into a register on the next rising edge of pclk, or that the read data is expected to be valid at the next rising edge of pclk.

Table 21: Signal List (Continued)

Signal Name	Direction	Clock Domain	Polarity / Bus Size	Description
pwrite	Input	pclk		Peripheral write strobe. This indicates that a write access is taking place (if psel is active).
paddr[11:2]	Input	pclk		Address bus from selected master (PADDR). Indicates which register is being accessed. This address is word-aligned within GEM and therefore should be connected to bits [11:2] of the APB address bus. If 802.3br is configured the address width is expanded to [12:2] and if APB parity protection is configured it is expanded to [15:0] and the host drives the two LSB to support parity protection on the additional paddr_par[1:0] inputs.
pwdata[31:0]	Input	pclk		Write data. Data to be written into the addressed register.
prdata[31:0]	Output	pclk		Read data. Data read from the addressed register. For APB Rev 2.0, it is driven to logic 0 when not addressed.
perr	Output	pclk		Indicates an APB access to an invalid address. Should be sampled when penable is high. This signal is optional and not a standard AMBA signal.
AMBA (AHB) Interface (Optional)				
n_hreset	Input	hclk	Low	Active low asynchronous reset for the DMA. This signal must be asserted low asynchronously and deasserted high synchronously with hclk.
hclk	Input			AHB bus clock
hready	Input	hclk		Slave device drives this low to extend data phase
hresp[1:0]	Input	hclk	2	Slave response. 00 for OK, any other response is not OK and triggers an interrupt. AMBA split and retry operations are not supported.
hgrant	Input	hclk		AHB bus grant
haddr[63:0]	Output	hclk		AHB address. Configurable for 32 and 64-bit bus widths.
htrans[1:0]	Output	hclk		AHB transfer type as follows: • 00 - Idle • 01 - Busy (not used by GEM) • 10 - Non-sequential • 11 - Sequential
hwrite	Output	hclk		High for AHB write, low for AHB read
hrdata[127:0]	Input	hclk		Data read from memory, configurable for 32, 64, or 128-bit bus widths. Unused bits should be tied to logic 0 or logic 1.
hsize[2:0]	Output	hclk		Transfer size, configured according to AMBA AHB data bus width as follows: • 010 - 32 bit transfers • 011 - 64 bit transfers • 100 - 128 bit transfers

Table 21: Signal List (Continued)

Signal Name	Direction	Clock Domain	Polarity / Bus Size	Description
hburst[2:0]	Output	hclk		Burst type. • 000 for single transfer • 001 for incrementing burst • 011 for four beat incrementing burst
hprot[3:0]	Output	hclk		Protection type. Parameterizable via `gem_hprot_value.
hwdata[127:0]	Output	hclk		Data written to memory. Configurable for 32, 64, or 128-bit bus widths.
hbusreq	Output	hclk		AHB bus request.
hlock	Output	hclk		Always asserted with hbusreq to lock the grant. This signal may be left unconnected if required (refer to *Section 6.1.1.6 Locked Accesses When Using AHB and AHB Bus Ownership* on page 61).
AMBA (AXI) Interface (Optional)				
n_areset	Input	aclk		Active low asynchronous reset for the DMA. This signal must be asserted low asynchronously and deasserted high synchronously with aclk.
aclk	Input			AXI bus clock
awaddr[63:0]	Output	aclk		AXI write channel address. Configurable for 32 and 64-bit bus widths.
awlen[7:0]	Output	aclk		AXI write channel burst length. Maximum value is 255.
awsize[2:0]	Output	aclk		Transfer size. Configured according to AXI data bus width as follows: • 010 - 32-bit transfers (i.e. 4 byte) • 011 - 64-bit transfers (i.e. 8 byte) • 100 - 128-bit transfers (i.e. 16 byte)
awburst[1:0]	Output	aclk		All bursts generated by the AXI DMA are incrementing, so this output is tied to 2'b01.
awvalid	Output	aclk		AXI write request available from DMA
awready	Input	aclk		AXI write request accepted by slave
awid[3:0]	Output	aclk		Not currently used by the GEM DMA. This output is tied low.
awcache[3:0]	Output	aclk		Cache support. Parameterizable via `gem_axi_awcache_value.
awlock[1:0]	Output	aclk		Not currently used by the GEM DMA. This output is tied low.
awprot[2:0]	Output	aclk		Protection unit support. Parameterizable via `gem_axi_prot_value.
awqos[3:0]	Output	aclk		Driven on descriptor and data write operations with the appropriate value from the axi_qos_cfg register for the active queue
wdata[127:0]	Output	aclk		Data write to memory. Configurable for 32, 64, or 128-bit bus widths. Unused bits are tied to logic 0.

Table 21: Signal List (Continued)

Signal Name	Direction	Clock Domain	Polarity / Bus Size	Description
wstrb[15:0]	Output	aclk		Byte write strobes to memory. Configurable for 32, 64, or 128-bit data bus widths. Unused bits are tied to logic 0.
wlast	Output	aclk		Identifies the last data of a burst
wid[3:0]	Output	Static		Not currently used by the GEM DMA. This output is tied low.
wvalid	Output	aclk		AXI write data valid
wready	Input	aclk		AXI write data accepted by slave
bresp[1:0]	Input	aclk		Slave response. 00 for OK, any other response is not OK and triggers an interrupt.
bid[3:0]	Input	aclk		Not currently used by the GEM DMA.
bvalid	Input	aclk		AXI write response bus valid.
bready	Output	aclk		AXI write response accepted.
araddr[63:0]	Output	aclk		AXI read channel address. Configurable for 32 and 64-bit bus widths.
arlen[7:0]	Output	aclk		AXI read channel burst length. Maximum value is 255.
arsize[2:0]	Output	aclk		Transfer size. Configured according to AXI data bus width as follows: • 010 - 32 bit transfers (i.e. 4 byte) • 011 - 64 bit transfers (i.e. 8 byte) • 100 - 128 bit transfers (i.e. 16 byte).
arburst[1:0]	Output	aclk		All bursts generated by the AXI DMA are incrementing, so this output is tied to 2'b01.
arvalid	Output			AXI read request available from DMA
arready	Input			AXI read request accepted by slave
arid[3:0]	Output	aclk		Not currently used by the GEM DMA. This output is tied low.
arcache[3:0]	Output	aclk		Cache support. Parameterizable via `gem_axi_arcache_value.
arlock[1:0]	Output	aclk		Not currently used by the GEM DMA. This output is tied low.
arprot[2:0]	Output	aclk		Protection unit support. Parameterizable via `gem_axi_prot_value.
arqos[3:0]	Output	aclk		Driven on descriptor and data read operations with the appropriate value from the axi_qos_cfg register for the active queue
rdata[127:0]	Input	aclk		Data read from memory. Configurable for 32, 64, or 128-bit bus widths. Unused bits should be tied to logic 0 or logic 1.
rlast	Input	aclk		Identifies the last data of a read burst
rresp[1:0]	Input	aclk		Slave response. 00 for OK, any other response is not OK and triggers an interrupt.
rid[3:0]	Input	aclk		Not currently used by the GEM DMA

Table 21: Signal List (Continued)

Signal Name	Direction	Clock Domain	Polarity / Bus Size	Description
rvalid	Input	aclk		AXI read data valid
rready	Output	aclk		AXI read data accepted
Transmit Packet Buffer Memory Interface (Dual-Port SRAM Configuration)				
Port A of the transmit packet memory should be clocked on hclk/aclk and port B should be clocked on tx_clk. **Note**: If the design is configured to support ECC, then the SRAM data ports are increased to support the additional protection bits such that *m* would be 39, 72 or 144-bits depending on whether the packet buffer width is configured to be 32, 64 or 128-bits respectively. If the design is configured to NOT support ECC but datapath parity protection is enabled, then *m* would be 36, 72 or 144-bits respectively.				
txdpram_ena	Output	hclk/aclk		Transmit packet buffer memory Port A chip select
txdpram_wea	Output	hclk/aclk		Transmit packet buffer memory Port A write enable
txdpram_addra[N:0]	Output	hclk/aclk		Transmit packet buffer memory Port A word address bus. Address width configurable.
txdpram_dia[m-1:0]	Output	hclk/aclk		Transmit packet buffer memory Port A write data bus. Either 32, 64, or 128 bit depending on dma_bus_width[1:0]
txdpram_enb	Output	tx_clk		Transmit packet buffer memory Port B chip select
txdpram_web	Output	Static	Tied low	Transmit packet buffer memory Port B write enable
txdpram_addrb[N:0]	Output	tx_clk		Transmit packet buffer memory Port B word address bus. Address width configurable.
txdpram_dob[m-1:0]	Input	tx_clk		Transmit packet buffer memory Port B read data bus. Either 32, 64, or 128 bit depending on dma_bus_width[1:0].
Transmit Packet Buffer Memory Interface (Single-Port SRAM Configuration)				
The transmit packet buffer memory should be clocked from hclk/aclk.				
txspram_we	Output	hclk/aclk		Transmit packet buffer memory write enable
txspram_en	Output	hclk/aclk		Transmit packet buffer memory chip select
txspram_addr[N:0]	Output	hclk/aclk		Transmit packet buffer memory word address bus. Address width configurable.
txspram_do[m-1:0]	Input	hclk/aclk	128	Transmit packet buffer memory read data bus. If the design is configured to support ECC or datapath parity protection, this port is increased to 144 bits.
txspram_di[m-1:0]	Output	hclk/aclk	128	Transmit packet buffer memory write data bus. If the design is configured to support ECC or datapath parity protection, this port is increased to 144 bits.
Receive Packet Buffer Memory Interface (Dual-Port SRAM Configuration)				
Port A of the receive packet memory should be clocked on rx_clk and port B should be clocked on hclk/aclk. **Note**: If the design is configured to support ECC, then the SRAM data ports are increased to support the additional protection bits such that *m* would be 39, 72, or 144 bits depending on whether the packet buffer width is configured to be 32, 64, or 128-bits respectively. If the design is configured to NOT support ECC but datapath parity protection is enabled, then *m* would be 36, 72, or 144 bits respectively.				
rxdpram_ena	Output	rx_clk		Receive packet buffer memory Port A chip select
rxdpram_wea	Output	rx_clk		Receive packet buffer memory Port A write enable
rxdpram_addra[N:0]	Output	rx_clk		Receive packet buffer memory Port A word address bus. Address width configurable.

Table 21: Signal List (Continued)

Signal Name	Direction	Clock Domain	Polarity / Bus Size	Description
rxdpram_dia[m-1:0]	Output	hclk/aclk		Receive packet buffer memory Port A write data bus. Either 32, 64, or 128 bit depending on dma_bus_width[1:0].
rxdpram_enb	Output	hclk/aclk		Receive packet buffer memory Port B chip select
rxdpram_web	Output	hclk/aclk	Tied low	Receive packet buffer memory Port B write enable
rxdpram_addrb[N:0]	Output	hclk/aclk		Receive packet buffer memory Port B word address bus. Address width configurable.
rxdpram_dob[m-1:0]	Input	hclk/aclk		Receive packet buffer memory Port B read data bus. Either 32 bit, 64 bit, or 128 bit depending on dma_bus_width[1:0]
Receive Packet Buffer Memory Interface (Single-Port SRAM Configuration)				
The receive packet buffer memory should be clocked from hclk/aclk.				
rxspram_we	Output	hclk/aclk		Receive packet buffer memory write enable
rxspram_en	Output	hclk/aclk		Receive packet buffer memory chip select
rxspram_addr[N:0]	Output	hclk/aclk		Receive packet buffer memory word address bus. Address width configurable.
rxspram_do[m-1:0]	Input	hclk/aclk	128	Receive packet buffer memory read data bus. If the design is configured to support ECC or datapath parity protection, this port is increased to 144 bits.
rxspram_di[m-1:0]	Output	hclk/aclk	128	Receive packet buffer memory write data bus. If the design is configured to support ECC or datapath parity protection, this port is increased to 144 bits.
Packet Buffer Memory Interface for eMAC				
Note:If IEEE 802.3br frame pre-emption is configured along with packet buffer DMA then the pMAC will have the SRAM signal names as defined above. Two additional SRAM instances will be required for the eMAC which will have the same signals and the same signal names with the exception that the eMAC’s signal names will be pre-pended with the four characters ‘emac_’.				
TSU I/O Requirements with Datapath (Optional)				
If gem_tsu is defined along with gem_asf_dap_prot then the following additional I/O will be present. There is no change to the existing timing requirements of this interface, the I/O will be timed to either tsu_clk or pclk.				
tsu_timer_cnt_par[11:0]	Output	tsu_clk		Parity bus for tsu_timer_cnt output.
ext_tsu_timer_par[11:0]	Input	tsu_clk		Parity bus for ext_tsu_timer of gem_ext_tsu_timer is defined.
User Input/Output Interface				
user_in[x:0]	Input	pclk		User inputs. These are read from the user_io_register which has a configurable width of 1 to 16 inputs.
user_out[y:0]	Output	pclk		User outputs. These are controlled by writing to the user_io_register which has a configurable width of 1 to 16 outputs.
Reset Interface				
n_txreset	Input	tx_clk		Active low tx_clk domain reset. This signal should be asserted low asynchronously, and deasserted high synchronously with tx_clk.
n_rxreset	Input	rx_clk		Active low rx_clk domain reset. This signal should be asserted low asynchronously, and deasserted high synchronously with rx_clk.

Table 21: Signal List (Continued)

Signal Name	Direction	Clock Domain	Polarity / Bus Size	Description
n_gtxreset	Input	gtx_clk		Active low gtx_clk domain reset. This signal should be asserted low asynchronously, and deasserted high synchronously with gtx_clk.
n_gtx20reset	Input	gtx20_clk		Active low gtx20_clk domain reset. This signal should be asserted low asynchronously, and deasserted high synchronously with gtx20_clk.
n_pcs_rxreset	Input	pcs_rx_clk		Active low pcs_rx_clk domain reset. This signal should be asserted low asynchronously, and deasserted high synchronously with pcs_rx_clk.
n_pcs_rx10reset	Input	pcs_rx10_clk		Active low pcs_rx10_clk domain reset. This signal should be asserted low asynchronously, and deasserted high synchronously with pcs_rx10_clk.
n_rbc0reset	Input	rbc0		Active low rbc0 domain reset. This signal should be asserted low asynchronously, and deasserted high synchronously with rbc0.
n_rbc1reset	Input	rbc1		Active low rbc1 domain reset. This signal should be asserted low asynchronously, and deasserted high synchronously with rbc1.
n_ntxreset	Input	n_tx_clk		Active low n_tx_clk domain reset. This signal should be asserted low asynchronously, and deasserted high synchronously with n_tx_clk.
n_nrxreset	Input	n_rx_clk		Active low n_rx_clk domain reset. This signal should be asserted low asynchronously, and deasserted high synchronously with n_rx_clk.
n_ref_reset	Input	ref_clk		Active low ref_clk domain reset. This signal should be asserted low asynchronously, and deasserted high synchronously with ref_clk.
n_tsureset	Input	tsu_clk		Active low tsu_clk domain reset. This signal should be asserted low asynchronously, and deasserted high synchronously with tsu_clk.

4.2 Signal List (ASF)

Table 22 lists controller signals for ASF capabilities.

Table 22: ASF Signal List

Signal Name	Direction	Clock Domain	Polarity / Bus Size	Description
ASF Fault Logging Outputs				
asf_int_nonfatal	Output	pclk	High	Non-fatal interrupt pin
asf_int_fatal	Output	pclk	High	Fatal interrupt pin
asf_sram_corr_err	Output	pclk	High	Status flag indicating an error has occurred in the SRAM and it was corrected using ECC. Driven directly by the relevant raw interrupt status bit.

Table 22: ASF Signal List (Continued)

Signal Name	Direction	Clock Domain	Polarity / Bus Size	Description
asf_sram_uncorr_err	Output	pclk	High	Status flag indicating an error has occurred in the SRAM and it was not corrected. Driven directly by the relevant raw interrupt status bit.
asf_dap_err	Output	pclk	High	Status flag indicating an error has been detected within the internal parity protected address or datapath logic. Driven directly by the relevant raw interrupt status bit.
asf_csr_err	Output	pclk	High	Status flag indicating an error has been detected within the configuration/control/status register space for the IP. Driven directly by the relevant raw interrupt status bit.
asf_trans_to_err	Output	pclk	High	Status flag indicating that a transaction timeout has occurred on one of the host interface buses. Driven directly by the relevant raw interrupt status bit.
asf_protocol_err	Output	pclk	High	Status flag indicating that an IP protocol specific error has occurred. Driven directly by the relevant raw interrupt status bit.
asf_integrity_err	Output	pclk	High	Status flag indicating that an IP specific internal control path error has occurred. Driven directly by the relevant raw interrupt status bit.
ASF Fault Logging Outputs for eMAC				
Note: If IEEE 802.3br frame pre-emption is configured along with ASF functionality then the pMAC will have the ASF signal names as defined above. The eMAC which will have the same signals and the same signal names with the exception that the eMAC’s signal names will be pre-pended with the four characters ‘emac_’.				
Host Interface Parity Requirements for ASF				
If *gem_asf_host_par* is defined, then the following new I/O will be added: N=gem_dma_bus_width/8 M=edma_addr_width/8 There is no change to the existing timing requirements for this interface. The AXI parity bus width is dependent on the configured data width of the IP and is 1-bit per byte even parity.				
paddr_par[1:0]	Input	pclk	2	Parity from host associated with paddr[15:0]
pwdata_par[3:0]	Input	pclk	4	Parity from host associated with pwdata[31:0]
prdata_par[3:0]	Output	pclk	4	Parity to host associated with prdata[31:0]
wdata_par[N:0]	Output	pclk	N+1	Parity bus for wdata if configured to use AXI interface
rdata_par[N:0]	Input	pclk	N+1	Parity bus for rdata if configured to use AXI interface
awaddr_par[M:0]	Output	pclk	M+1	Parity bus for awaddr if configured to use AXI interface
araddr_par[M:0]	Output	pclk	M+1	Parity bus for araddr if configured to use AXI interface

4.2.1 SRAM I/O Requirements with Datapath Protection

If the Data/Address Path protection ASF feature is enabled, the data width for the SRAM with ECC protection will increase as shown in *Table 23*, where Addr_P is the address parity which is one bit for every 8 bits of address.

Table 23: SRAM I/O Width Increase for ECC Protection

Original Width	Width to Protect	Parity Bits	SRAM Width
32	32 + Addr_P	7	39 + Addr_P
64	64 + Addr_P	8	72 + Addr_P
128	128 + Addr_P	16	144 + Addr_P

The data width for the SRAM without ECC protection (but with parity protection) will increase as shown in *Table 24*.

Table 24: SRAM I/O Width Increase without ECC Protection

Original Width	Width to Protect	Parity Bits	SRAM Width
32	32 + Addr_P	4	36 + Addr_P
64	64 + Addr_P	8	72 + Addr_P
128	128 + Addr_P	16	144 + Addr_P

There is no change to the existing timing requirements for this interface.

5 Register Address Map

Table 25 shows the address space allocation of control and status registers within the GEM module. The full register list and register details are provided in *Section 11 Registers* on page 146.

Table 25: Memory Map

Address Offset (hex)	Register Type	Present in IP7014A?	Width (bits)
MAC registers or pre-emptable MAC (pMAC)* registers			
0x0000	Control and Status	Y	32
0x0100	Statistics	Y	32
0x01BC	Timestamp Unit (TSU)	Y	32
0x0200	Physical Coding Sublayer (PCS)	Y	32
0x0260	Miscellaneous	Y	32
0x0300	Extended Filter	Y	32
0x0400	Priority Queue and Screening	Y	32
0x0800	Time-Sensitive Networking (TSN)	Y	32
0x0E00	Automotive Safety Feature (ASF)	Y	32
0x0F00	MAC Merge Sublayer (MMSL)	Y	32
Express MAC (eMAC) registers			
0x1000 to 0x1FFF	eMAC	Y	32

*If 802.3br is configured

6 Functional Description

6.1 Block-Level Description

Figure 4 shows an overview of GEM and its interfaces.

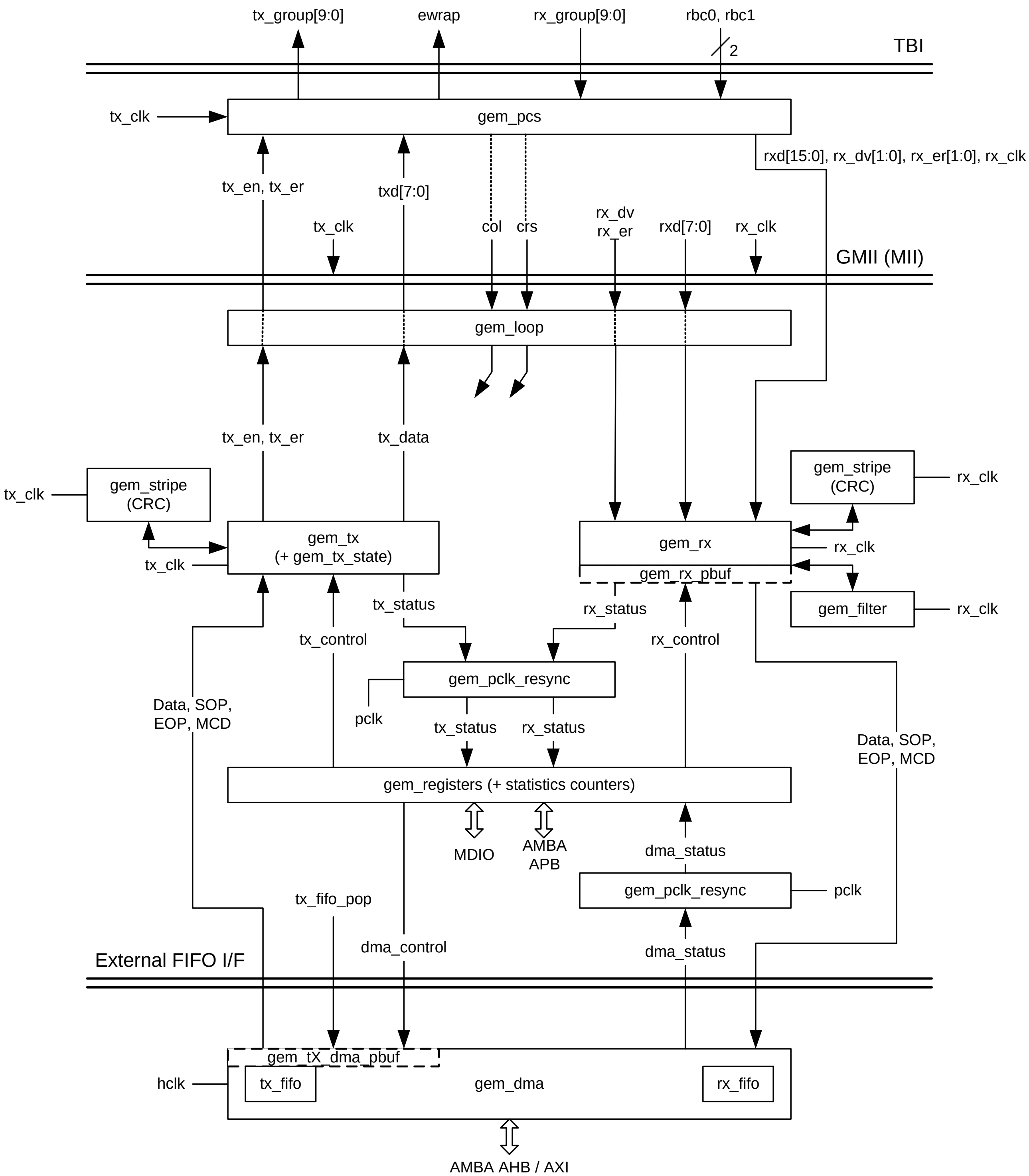

Figure 4: MAC Signal Interfaces

The top-level module name is gem_gxl. The block diagram above shows a GMII interface.

GEM is also capable of supporting RMII and RGMII interfaces (selected as a configuration option, refer to *Section 3 User Defined Options* on page 17 for details):

- If RMII is configured, an extra RMII interface is available at the top-level as shown in *Figure 5*. Use of this interface is done by driving the top-level input signal mii_select low. Driving mii_select low selects the RMII interface rather than the RGMII/GMII interface which will also be present at the top-level.
- If RGMII is configured, then the 8-bit GMII interface is replaced by a 4-bit RGMII interface. From release 1p09 onwards, it is possible to use software to reconfigure the RGMII interface for 4-bit MII operation using bit 28 "sel_mii_on_rgmii" in the network_control register.
- It is permissible to configure GEM to support both RMII and RGMII simultaneously. However, it is only worthwhile configuring for RMII if there is a need to support PHYs that only support 10/100 speeds (that is no gigabit support).

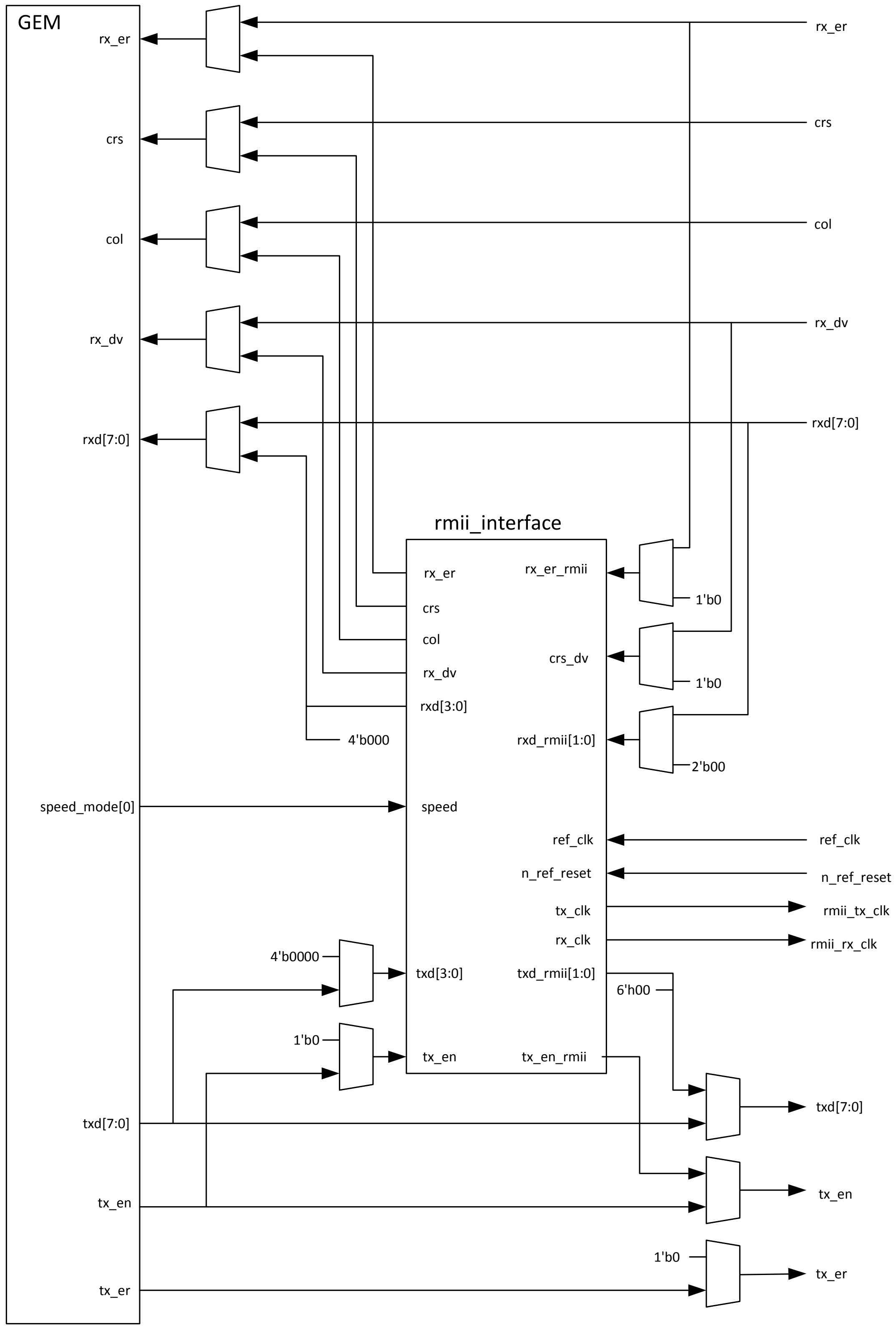

Figure 5: RMII Interface Included

When the RGMII interface is configured (refer to the configuration options in *Section 3 User Defined Options* on page 17 for details), it is implemented as a bridge function (GMII to RGMII) and replaces the standard GMII interface as shown in *Figure 6*.

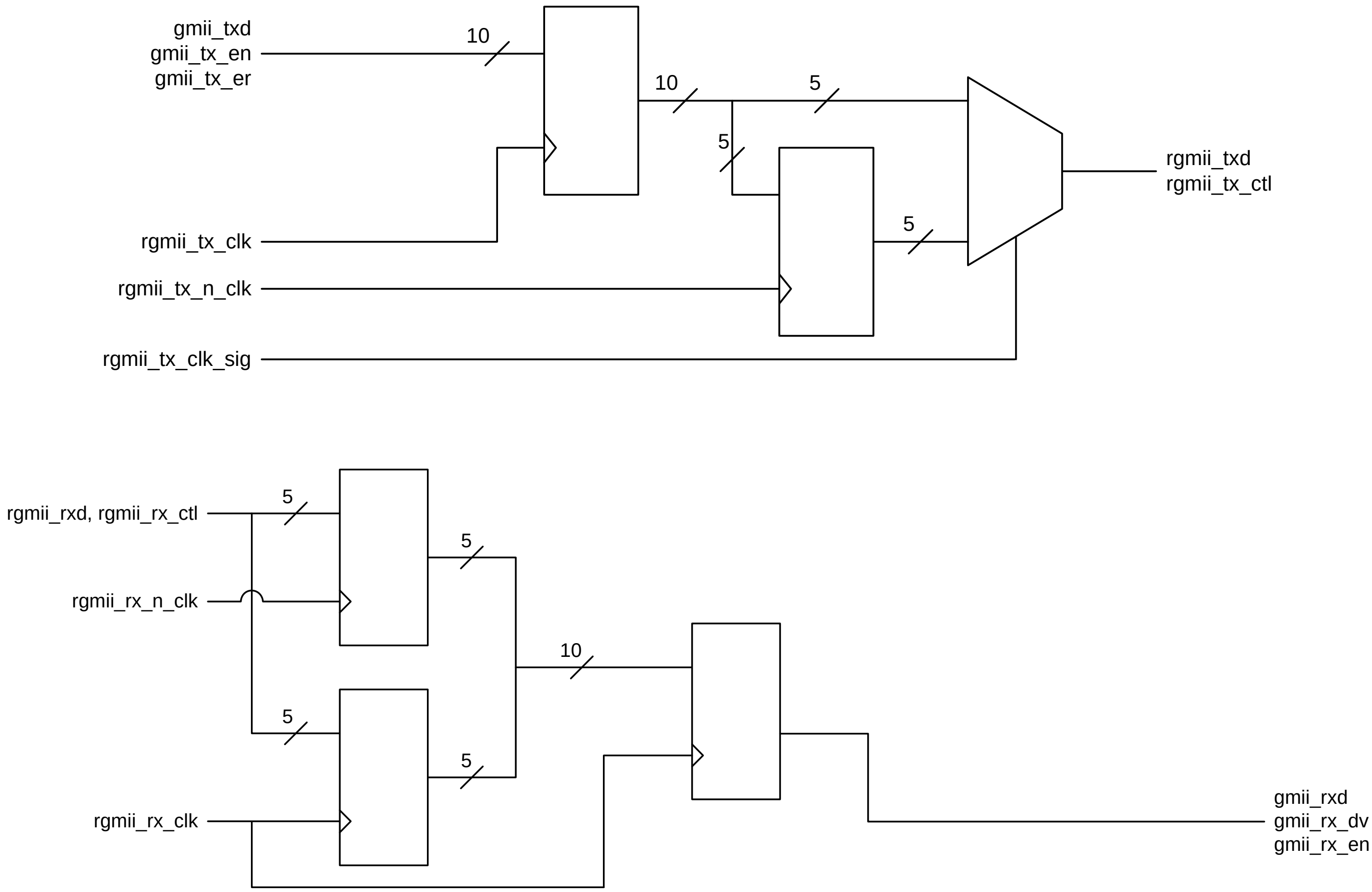

Figure 6: RGMII Interface Include

6.1.1 Direct Memory Access Interface

GEM is supplied with an optional DMA interface. When GEM is configured to use the DMA, it is attached to the MAC module's external FIFO interfaces to provide a scatter gather type capability for packet data storage in an embedded processor system or System on Chip (SoC). If GEM is not configured to use the DMA, then the external FIFO interface is left exposed for connecting to external memory controllers and the DMA is removed to save area.

When included in GEM, the DMA can be configured in two possible modes:

- The first using internal transmit and receive FIFOs implemented as synthesizable flip-flops
- The second is a packet buffering mode, where Single-Port or Dual-Port memories are used to buffer multiple frames.

Both have identical software and hardware interfaces (although the packet buffer mode reduces software overhead in certain circumstances).

6.1.1.1 DMA with Internal Transmit and Receive FIFOs

The key features of this legacy mode of operation are:

- Internal transmit/receive FIFO DMA
- AHB support only
- Lowest possible latency solution

- 32 and 64-bit databus width support

6.1.1.2 Packet Buffer DMA

- 32, 64, and 128-bit databus width support
- 32 or 64-bit address bus width support
- Easier to guarantee maximum line rate due to the ability to store multiple frames in the packet buffer
- Makes efficient use of the AHB / AXI interface
- Support for up to 16 outstanding AXI transactions in flight. These transactions can cross multiple frame transfers providing excellent support for systems with high latency.
- Optional local descriptor caching to improve performance for AXI systems with high latency.
- Full store and forward, or partial store and forward programmable options (partial will cater for shorter latency requirements). **Note:** Partial store and forward is not supported for an AHB configuration with more than one priority queue. Partial store and forward mode is however supported for AXI configurations with priority queues.
- Support for Transmit TCP/IP checksum offload
- Support for priority queuing
- Support for TCP/IP advanced offloads to reduce CPU overhead
- When a collision on the line occurs during transmission, the packet will be automatically replayed directly from the packet buffer memory rather than having to re-fetch through the AHB (full store and forward ONLY)
- Received errored packets are automatically dropped before any of the packet is presented to external memory (full store and forward ONLY), thus reducing AHB or AXI activity
- Supports manual RX packet flush capabilities
- Optional RX packet flush when there is lack of resource
- Optional burst padding at end of packet and end of buffer to maximize AHB / AXI efficiency
- Optional 64-bit addressing for Data Buffer start address within Buffer Descriptor Entry
- Optional TX / RX Timestamp capture to Buffer Descriptor Entry
- Dual-port or single-port SRAM support

6.1.1.3 DMA Transactions

The DMA uses separate transmit and receive lists of buffer descriptors, with each descriptor describing a buffer area in memory. This can allow Ethernet packets to be broken up and scattered around the system memory (multi buffer operation), although one buffer per frame is also permitted.

The DMA controller performs four types of operation on the AMBA bus. When the GEM DMA is configured in internal FIFO mode, in order of priority these are:
1. Receive buffer manager write/read
2. Transmit data DMA read
3. Receive data DMA write
4. Transmit buffer manager write/read

When the GEM DMA is configured in AHB packet buffer mode, in order of priority these are:
1. Receive buffer manager write/read
2. Transmit buffer manager write/read
3. Receive data DMA write
4. Transmit data DMA read

When using AHB, the bus request signal hbusreq is asserted when the DMA block needs to perform any of these operations. The DMA block takes control of the bus when it sees both hgrant and hready asserted high.

When using AXI, all read operations are routed to the AXI read channel and all write operations to the write channel. Both read and write channels may operate simultaneously. Arbitration logic is required when multiple requests are active on the same channel (for example when the transmit DMA requests a transmit data read at the same time the receive DMA requests a receive descriptor read). In these cases, the receive DMA is granted the bus before the transmit DMA. However, the vast majority of requests are either receive data writes or transmit data reads both of which can operate in parallel and can execute simultaneously.

Transfer size can be programmed to be 32, 64, or 128-bit words using the DMA bus width select bits in the network configuration register, and burst size may be programmed to single access or bursts of 4, 8, 16, or 256 words using the DMA Configuration Register.

6.1.1.4 Using the AXI Interface

The highest data throughput and system performance will be met when GEM is configured to use the AXI interface. The GEM's AXI4 interface provides separate data and address connections for Reads and Writes, which allow simultaneous, bidirectional data transfers. GEM supports multiple outstanding transactions on both the Read and Write address channels, up to a programmable limit (see AXI MAX PIPELINE programmable register in *Section 11 Registers* on page 146). This means the issuing of Read and Write requests from the GEM DMA (on AXI AR and AW channels) are decoupled from the AXI slave responses (on R and B channels). The issuing of outstanding transactions are allowed to span multiple frames, maintaining high data transfer rates even with very high latency systems (when the transactions can take a very long time to complete).

To further provide support for high latency systems, the DMA will buffer the descriptor accesses locally to avoid the underlying DMA from pausing while descriptor transactions are completed by the system fabric.

- TX and RX descriptor reads are issued up-front and stored in a local buffer to feed the underlying DMA when required. This optimizes performance and avoids the need for the underlying DMA to pause while new descriptor fetches are sent to the system bus. The size of the descriptor read buffers are configurable. For low-latency systems, these buffers can be configured small. The buffer descriptor list should not contain fewer entries than the size of the descriptor read buffers.
- TX and RX descriptor writes issued by the underlying DMA are buffered locally to avoid holding up the underlying DMA if and when the system delays the completion of descriptor writes. The size of the descriptor write buffers are configurable. For low-latency systems, these buffers can be configured small. **Note:** A descriptor write transaction is not considered complete until the 'B' response associated with that transaction has been collected from the fabric.

By providing these descriptor buffers, the DMA does not need to use AXI ID's to support high latency systems. AXI ID's could have been used for example to differentiate between traffic types. This could theoretically ensure that transactions issued by the underlying DMA that require to be completed before normal data-flow can continue are given unique ID's that could be processed out-of-order by the system fabric (the traffic types that would require this would be the traffic associated with descriptor accesses). The use of AXI ID's requires parallel out-of-order completion support in the system fabric and slaves. Any fabric or system limitations with regards to out-of-order support would need to be carefully analyzed to ensure system throughput could be guaranteed. GEM does not provide AXI ID support and encompasses its own buffering to support high latency systems using a single ID.

The maximum burst lengths GEM uses are programmable. Single, bursts up to 4, 8, 16, or 256 can be selected (256 is supported by AXI4 only). With a 64-bit datapath and a burst length setting of 256, up to 2KB transfers can be made with a single request. The burst length is controlled via the DMA configuration register.

The DMA adheres to the AXI4 specification and will never issue transactions that break a 4KB boundary. For example, if the burst length is set to 256, a packet to be transmitted is 100 bytes long, the datapath is 64-bits wide, and the data is located in system memory 60 bytes from a 4KB boundary, the GEM DMA will issue two burst requests, one of length 7 and the other of length 5.

GEM provides an AXI4 interface by default, although it is possible to connect this to an AXI3 system as necessary. Refer to the AMBA specification for details on how to connect AXI3 slaves to an AXI4 master.

6.1.1.5 Partial Store and Forward Using Packet Buffer DMA

When the DMA is configured to use SRAM-based packet buffers, it can be programmed into a low-latency mode, known as partial store and forward. This allows for a reduced latency as the full packet is not buffered before forwarding. **Note**: This option is only available when the device is configured for full-duplex operation and when not using multi buffer frames.

This feature is enabled via the TX and RX partial store and forward programmable registers. When the transmit partial store and forward mode is activated, the transmitter will only begin to forward the packet to the MAC when there is enough packet data stored in the packet buffer. Likewise, when the receive partial store and forward mode is activated, the receiver will only begin to forward the packet to the external AHB or AXI slave when enough packet data is stored in the packet buffer.

The amount of packet data required to activate the forwarding process is programmable via watermark registers which are located at the same address as the partial store and forward enable bits. **Note**: The minimum operational value for the TX partial store and forward watermark is 20. There is no operational limit for the RX partial store and forward watermark. To reduce the bandwidth requirements of the receive buffer manager the receive buffer size can be increased above its default value of 128 bytes by writing to the DMA configuration register.

Enabling partial store and forward is a useful means to reduce latency, but there are performance implications. In essence, the packet buffer DMA will start behaving in a similar way to the internal FIFO DMA mode when partial store and forward is enabled.

6.1.1.6 Locked Accesses When Using AHB and AHB Bus Ownership

The GEM DMA has been designed to support the loss of hgrant at any time during transmit or receive AHB activities. However, there are certain implications tied to losing hgrant that as the system integrator, you should be aware of:

- During an AHB data burst, the GEM DMA will drive hburst appropriately. The loss of hgrant during a burst will cause all remaining AHB accesses of that burst to be cancelled thereby breaking the burst. Although the remaining accesses will be restarted when the grant is re-established, an AHB protocol error with respect to hburst will have occurred. If the system integrator requires the use of hburst and this behaviour is not acceptable, then the loss of grant during data bursts must be avoided.
- When the GEM DMA is configured to be in the internal FIFO mode, or in the packet buffer partial store and forward mode there is a finite time that the DMA has to offload data to the AHB in the receive direction or fetch data from the AHB in the transmit direction. If hgrant is lost during DMA data transfers in a very frequent and/or prolonged manner, there is potential for internal FIFO overflows or underflows causing occasional packet loss. The frequency of this scenario occurring can be minimized by increasing the size of the internal DMA buffers.
- When the GEM DMA is configured to be in the packet buffer mode (and in full store and forward mode), all packets are fully buffered before being passed between the DMA and MAC. Therefore, the loss of hgrant while in this mode has no effect on data integrity.
- When the GEM DMA is configured in internal FIFO mode the hlock output of the DMA is asserted together with hbusreq. By connecting this hlock output to the AHB system arbiter, the grant to the DMA should never be removed during a locked transfer and consequently any potential issues discussed above associated with losing the grant will never occur. If you wish to deassert hgrant during certain operating conditions, the hlock output of GEM should be left unconnected. When the GEM DMA is configured to be in the packet buffer mode, the hlock output of the DMA is tied low.

6.1.1.7 Receive DMA Buffers

Received frames, optionally including FCS, are written to receive buffers stored in AHB or AXI memory. The receive buffer depth is programmable in the range of 64 bytes to 16320 bytes. If received frames are being routed to different priority queues (via the packet inspection screeners – refer to *Section 6.1.1.16 Priority Queuing in the DMA* on page 79), it is possible to program different receive buffer depths for each queue:

- For queue 0, the receive buffer depth is programmed through the DMA Configuration register (offset 0x10)
- For the other queues, they are programmed in the independent queue configuration registers (starting from offset 0x4a0)
- The default is 128 bytes

The start location for each receive buffer is stored in memory in a list of receive buffer descriptors at an address location pointed to by the receive buffer queue pointer. The base address for the receive buffer queue pointer is configured in software using the receive buffer queue base address register(s). For 64-bit addressing mode, the upper receive queue base address register at 0x04d4 is used to set the upper 32 bits of the receive buffer descriptor queue base address, and the upper transmit queue base address register at 0x04c8 is used to set the upper 32 bits of the transmit buffer descriptor queue base address. With 64-bit addressing, there is a restriction that all the descriptors must be located within a region of memory that does not cross a 4GB region, in other words the upper 32 bits of the 64-bit address must be fixed. This is only true of the descriptors and not the packet data which can be anywhere in the 64-bit address space.

The number of words in each buffer descriptor (BD) is dependent on the operating mode. Each BD word is defined as 32-bits.

The first two words (Word 0 and Word 1) are used for all BD modes.

In receive Extended Buffer Descriptor Mode (DMA configuration register bit 28 = 1), two BD words are added for 64 bit addressing mode and two BD words are added for timestamp capture. There are therefore either two, four or six BD words in each BD entry depending on operating mode, and every BD entry will have the same number of words. To summarize:

- Every descriptor will be 64-bit wide when 64-bit addressing is disabled and extended buffer descriptor mode is disabled
- Every descriptor will be 128-bit wide when 64-bit addressing is enabled and extended buffer descriptor mode is disabled
- Every descriptor will be 128-bit wide when 64-bit addressing is disabled and extended buffer descriptor mode is enabled
- Every descriptor will be 192-bit wide when 64-bit addressing is enabled and extended buffer descriptor mode is enabled

The following description details the functionality of Word 0 and Word 1, but Word 2 must also be set when using 64-bit addressing mode. Each list entry consists of the same first two words. The first contains the start location of the receive buffer and the second the main receive status. If the length of a receive frame exceeds the DMA buffer length, the status word for the used buffer is written with zeroes except for the “start of frame” bit, which is always set for the first buffer in a frame. Bit zero of the address field is written to 1 to show the buffer has been used. The receive buffer manager then reads the location of the next receive buffer and fills that with the next part of the received frame data. Receive buffers are filled until the frame is complete and the final buffer descriptor status word contains the complete frame status. Refer to *Table 26*, *Table 27*, and *Table 28* for details of the receive buffer descriptor list.

When using receive descriptor timestamp capture, (DMA configuration register bit 28 = 1) bit 2 of Word 0 is used to indicate a valid timestamp has been captured in the BD. The use of bit 2 for this purpose also necessitates the Data Buffer being located on 64-bit address boundaries. Also note the timestamp can be considered status and will only be present for the final buffer descriptor of a frame.

Each receive buffer start location is a word address. The start of the first buffer in a frame can be offset by up to three bytes depending on the value written to bits 14 and 15 of the network configuration register. Note that when the `define gem_pbuf_rsc has been set then these bits are not used. For 64-bit datapath, the start of the frame can be offset by up to a further four bytes if bit 2 of the DMA receive buffer start location in the buffer descriptor is set (This is applicable to the packet buffer DMA only – for the FIFO based DMA configured with a 64bit datapath, bit 2 of the buffer start location is ignored). If the start location of the buffer is offset the available length of the first buffer is reduced by the corresponding number of bytes.

Table 26: Receive Buffer Descriptor Entry

Bit	Function
	Word 0
31:3	Address [31:3] of beginning of buffer
2	Address [2] of beginning of buffer. Or In Extended Buffer Descriptor Mode (DMA configuration register[28] = 1), indicates a valid timestamp in the BD entry

Table 26: Receive Buffer Descriptor Entry (Continued)

Bit	Function
1	Wrap - marks last descriptor in receive buffer descriptor list.
0	Ownership - needs to be zero for GEM to write data to the receive buffer. GEMsets this to 1 once it has successfully written a frame to memory. Software has to clear this bit before the buffer can be used again.
Word 1	
31	Global all ones broadcast address detected.
30	Multicast hash match.
29	Unicast hash match.
28	External address match. Note if the packet buffer mode and the number of configured specific address filters is greater than four in gem_gxl_defs.v then external address matching is not reported in this bit and instead it is set if there has been a match in the first eight specific address registers. Bit 27 is then used along with bits 26:25 to indicate which register matched.
27	Specific address register match found, bit 25 and bit 26 indicates which specific address register causes the match. See note for bit 28 above.
26:25	Specific address register match. Encoded as follows: • 00 - Specific address register 1 match • 01 - Specific address register 2 match • 10 - Specific address register 3 match • 11 - Specific address register 4 match If more than one specific address is matched only one is indicated with priority 4 down to 1.
24	This bit has a different meaning depending on whether RX checksum offloading is enabled. **With RX checksum offloading disabled:** (bit 24 clear in Network Configuration) Type ID register match found, bit 22 and bit 23 indicate which type ID register causes the match. **With RX checksum offloading enabled:** (bit 24 set in Network Configuration) 0 - the frame was not SNAP encoded and/or had a VLAN tag with the CFI bit set. 1 - the frame was SNAP encoded and had either no VLAN tag or a VLAN tag with the CFI bit not set.
23:22	This bit has a different meaning depending on whether RX checksum offloading is enabled. **With RX checksum offloading disabled:** (bit 24 clear in Network Configuration) Type ID register match. Encoded as follows: • 00 -Type ID register 1 match • 01 -Type ID register 2 match • 10 -Type ID register 3 match • 11 -Type ID register 4 match If more than one Type ID is matched only one is indicated with priority 4 down to 1. **With RX checksum offloading enabled:** (bit 24 set in Network Configuration) • 00 -Neither the IP header checksum nor the TCP/UDP checksum was checked. • 01 - The IP header checksum was checked and was correct. Neither the TCP nor UDP checksum was checked. • 10 -Both the IP header and TCP checksum were checked and were correct. • 11 -Both the IP header and UDP checksum were checked and were correct.
21	VLAN tag detected — type ID of 0x8100. For packets incorporating the stacked VLAN processing feature, this bit will be set if the second VLAN tag received has a type ID of 0x8100
20	Priority tag detected — type ID of 0x8100 and null VLAN identifier. For packets incorporating the stacked VLAN processing feature, this bit will be set if the second VLAN tag received has a type ID of 0x8100 and a null VLAN identifier.

Table 26: Receive Buffer Descriptor Entry (Continued)

Bit	Function
19:17	When bit 15 (End of frame) and bit 21 (VLAN tag) are set, these bits represent the VLAN priority. When header/data splitting is enabled (via bit 5 of the DMA configuration register, offset 0x10) bit 17 indicates this descriptor is pointing to the last buffer of the header.
16	This bit has a different meaning depending on the state of bit 13 (report bad FCS in bit 16 of word 1 of the receive buffer descriptor) and bit 5 (header/data splitting) of the DMA Configuration register (offset 0x10). When header/data splitting is enabled and this buffer descriptor (BD) is not the last BD of the frame (as indicated in bit 15 of this BD), this bit will indicate that the BD is pointing to a data buffer containing header bytes. When this BD is the last BD of the frame (as indicated in bit 15 of this BD), and bit 13 of the DMA configuration register is set, this bit represents FCS/CRC error. When this BD is the last BD of the frame (as indicated in bit 15 of this BD), and bit 13 of the DMA configuration register is clear, and the received frame is VLAN tagged, this bit represents the Canonical format indicator (CFI).
15	End of frame - when set the buffer contains the end of a frame. If end of frame is not set, then the only valid status bit (unless header/data splitting is enabled) is start of frame (bit 14). If header/data splitting is enabled, then bits 16 and 17 are also valid status bits when this bit is not set.
14	Start of frame - when set the buffer contains the start of a frame. If both bits 15 and 14 are set, the buffer contains a whole frame.
13	This bit has a different meaning depending on whether jumbo frames and ignore FCS mode are enabled. If neither mode is enabled this bit will be zero. **With jumbo frame mode enabled:** (bit 3 set in Network Configuration Register) Additional bit for length of frame (bit[13]), that is concatenated with bits[12:0] **With ignore FCS mode enabled and jumbo frames disabled:** (bit 26 set in Network Configuration Register and bit 3 clear in Network Configuration Register) This indicates per frame FCS status as follows: • 0 – Frame had good FCS • 1 – Frame had bad FCS, but was copied to memory as ignore FCS enabled
12:0	When header/data splitting enabled (via bit 5 of the DMA configuration register, offset 0x10) and bit 17 is set (last buffer of header), these bits represent the length of the header in bytes. When bit 15 (End of frame) is set, these bits represent the length of the received frame which may or may not include FCS depending on whether FCS discard mode is enabled. **With FCS discard mode disabled:** (bit 17 clear in Network Configuration Register) Least significant 12-bits for length of frame **including** FCS. If jumbo frames are enabled, these 12-bits are concatenated with bit[13] of the descriptor above. **With FCS discard mode enabled:** (bit 17 set in Network Configuration Register) Least significant 12-bits for length of frame **excluding** FCS. If jumbo frames are enabled, these 12-bits are concatenated with bit[13] of the descriptor above.

When 64-bit addressing mode is enabled, *Table 27* identifies the added descriptor words.

Table 27: Receive Buffer Descriptor Entry − 64-Bit Addressing

Bit	Function
Word 2 (64-bit addressing)	
31:0	Upper 32-bit address of Data Buffer
Word 3 (64-bit addressing)	
31:0	Unused

When Descriptor Timestamp Capture mode is enabled, *Table 28* identifies the added descriptor words:

Table 28: Receive Buffer Descriptor Entry − Timestamp Capture

Bit	Function
Word 2 (64-bit addressing) or Word 4 (64-bit addressing)	
31:30	Timestamp seconds[1:0] (See Note:1)
29:0	Timestamp nanosecs [29:0] (See Note:1)
Word 3 (64-bit addressing) or Word 5 (64-bit addressing)	
31:10	Unused
9:0	Timestamp seconds[11:2] (See Note:1) (prior to release 1p08f1 this was [5:2])
	Note1: The timestamp is mode is controlled using the rx_bd_control_register. The RX Descriptor Timestamp Insertion mode bits are defined as: • 00: TS insertion disable • 01: TS inserted for PTP Event Frames only • 10: TS inserted for All PTP Frames only • 11: TS insertion for All Frames The timestamp bits are written back to the last buffer descriptor of a frame only.

To receive frames, the receive buffer descriptors must be initialized by writing an appropriate address to bits 31:2 (or 31:3 for timestamp capture mode) in the first word of each list entry. Bit 0 must be written with zero. Bit 1 is the wrap bit and indicates the last entry in the buffer descriptor list.

The start location of the receive buffer descriptor list must be written with the receive buffer queue base address before reception is enabled (receive enable in the network control register). Once reception is enabled, any writes to the receive buffer queue base address register are ignored. When read, it will return the current pointer position in the descriptor list, though this is only valid and stable when receive is disabled.

If the filter block indicates that a frame should be copied to memory, the receive data DMA operation starts writing data into the receive buffer. If an error occurs, the buffer is recovered.

An internal counter within GEM represents the receive buffer queue pointer and it is not visible through the CPU interface. The receive buffer queue pointer increments by two, four, or six words after each buffer has been used, depending on the descriptor size. It re-initializes to the receive buffer queue base address if any descriptor has its wrap bit set. **Note**: This operation is different from Cadence’s Ethernet MAC 10/100 (MACB), which will also wrap after 1024 buffers have been used.

As receive buffers are used, the receive buffer manager sets bit zero of the first word of the descriptor to logic one indicating the buffer has been used.

Software should search through the “used” bits in the buffer descriptors to find out how many frames have been received, checking the start of frame and end of frame bits.

When the DMA is configured for internal FIFO mode, received frames are written out to the AHB buffers as soon as the frame is matched by the filtering logic even though there may still be more data to be received. Similarly, when the DMA is configured in the packet buffer partial store and forward mode, received frames are written out to the AHB/AXI buffers as soon as enough frame data exists in the packet buffer. For both cases, this may mean several full buffers are used before some error conditions can be detected. If a receive error is detected the receive buffer currently being written will be recovered. Previous buffers will not be recovered. As an example, when receiving frames with CRC errors or excessive length, it is possible that a frame fragment might be stored in a sequence of receive buffers. Software can detect this by looking for start of frame bit set in a buffer following a buffer with no end of frame bit set.

For a properly working 10/100/1000 Ethernet system, there should be no excessive length frames or frames greater than 128 bytes with CRC errors. Collision fragments will be less than 128-bytes long, therefore it will be a rare occurrence to find a frame fragment in a receive buffer, when using the default value of 128 bytes for the receive buffers size.

When in packet buffer full store and forward mode only good received frames are written out of the DMA, so no fragments will exist in the buffers due to MAC receiver errors. There is still the possibility of fragments due to DMA errors, for example used bit read on the second buffer of a multi-buffer frame.

If bit zero of the receive buffer descriptor is already set when the receive buffer manager reads the location of the receive buffer, then the buffer has been already used and cannot be used again until software has processed the frame and cleared bit zero. In this case, the “buffer not available” bit in the receive status register is set and an interrupt triggered. The receive resource error statistics register is also incremented.

When the DMA is configured in the packet buffer full store and forward mode, the user can optionally select whether received frames should be automatically discarded when no buffer resource is available. This feature is selected via bit 24 of the DMA Configuration register (by default, the received frames are not automatically discarded). If this feature is off, then received packets will remain to be stored in the SRAM based packet buffer until AHB or AXI buffer resource next becomes available. This may lead to an eventual packet buffer overflow if packets continue to be received when bit zero (used bit) of the receive buffer descriptor remains set. **Note**: After a used bit has been read, the receive buffer manager will reread the location of the receive buffer descriptor every time a new packet is received. When the DMA is not configured in the packet buffer full store and forward mode and a used bit is read, the frame currently being received will be automatically discarded.

When the DMA is configured in the packet buffer full store and forward mode, a receive overrun condition occurs when the receive SRAM based packet buffer is full, or because an AMBA error occurred (via hresp or bresp). In all other modes, a receive overrun condition occurs when either the AHB/AXI bus was not granted quickly enough, or because an AMBA AHB or AXI error occurred, or because a new frame has been detected by the receive block when the status update or write back for the previous frame has not yet finished. For a receive overrun condition, the receive overrun interrupt is asserted and the buffer currently being written is recovered. The next frame that is received whose address is recognized reuses the buffer.

When the DMA is configured for packet buffer mode, a write to bit 18 of the network control register will force a packet from the external SRAM based receive packet buffer to be flushed. This feature is only acted upon when the RX DMA is not currently writing packet data out to memory– i.e. it is in an IDLE state. If the RX DMA is active, a write to this bit is ignored.

When the DMA is configured for packet buffer mode, the upper bits of the data buffer address stored in bits 31:2 in the first word of each list entry can be dynamically altered in real-time without physically changing the external memory holding the list entry. This feature is useful if the user needs to select the destination based on CPU usage or other flow control hardware. It is achieved using a MUX structure whereby the user can define whether the upper 4 bits of the 32-bit data-buffer AHB or AXI address should come from the descriptor list entry or from a programmable register. For further details, refer to the “Receive Data Buffer Address Mask” programmable register in *Section 11 Registers* on page 146. **Note:** Any changes to this register will be ignored while the DMA is currently processing a receive packet. It will only affect the next full packet to be written to AHB or AXI memory.

6.1.1.8 Transmit Buffers

Frames to transmit are stored in one or more transmit buffers. Transmit frames can be between 14 and 16383 bytes long, so it is possible to transmit frames longer than the maximum length specified in the IEEE 802.3 standard. It should be noted that zero length buffers are allowed and that the maximum number of buffers permitted for each transmit frame is 128.

The start location for each transmit buffer is stored in AHB or AXI memory in a list of transmit buffer descriptors at a location pointed to by the transmit buffer queue pointer. The base address for this queue pointer is set in software using the transmit buffer queue base address register(s). For 64-bit addressing mode, the upper transmit queue base address register at 0x04c8 is used to set the upper 32 bits of the transmit buffer descriptor queue base address. With 64-bit addressing, there is a restriction that all the descriptors must be located within a region of memory that does not cross a 4GB region, in other words the upper 32 bits of the 64-bit address must be fixed. The actual 32 bits chosen for the upper bits are programmed in the upper receive queue base address register at 0x04d4. This is only true of the descriptors and not the packet data which can be anywhere in the 64-bit address space.

The number of words in each buffer descriptor (BD) is dependent on the operating mode.

The first two words (Word 0 and Word 1) are used for all BD modes.

In transmit Extended Buffer Descriptor Modes (bit 29 in the DMA configuration register), two BD words are added for 64 bit addressing mode and two BD words are added for timestamp capture. There are therefore either two, four or six BD words in each BD entry depending on operating mode, and every BD entry will have the same number of words. To summarize:

- Every descriptor will be 64-bit wide when 64-bit addressing is disabled and extended buffer descriptor mode is disabled
- Every descriptor will be 128-bit wide when 64-bit addressing is enabled and extended buffer descriptor mode is disabled
- Every descriptor will be 128-bit wide when 64-bit addressing is disabled and extended buffer descriptor mode is enabled
- Every descriptor will be 192-bit wide when 64-bit addressing is enabled and extended buffer descriptor mode is enabled

The following description details the functionality of Word 0 and Word 1, but Word 2 must also be set when using 64-bit addressing mode.

Each list entry consists of the same first two words. The first is the byte address of the transmit buffer and the second containing the transmit control and status. For the packet buffer DMA, the start location for each AHB or AXI buffer is a byte address, the bottom bits of the address being used to offset the start of the data from the data-word boundary (i.e. bits 2,1 and 0 are used to offset the address for 64-bit datapaths). For the FIFO based DMA configured with a 32-bit datapath the address of the buffer is also a byte address. For bus widths of 64 or 128-bits however, the address of the buffer must be aligned to the correct 64-bit or 128-bit boundary, plus an offset of less than 4 bytes (**Note**: This alignment restriction in FIFO based DMA mode only should be sufficient for applications as the main purpose is to allow alignment of the encapsulated IP packet - Given the 14 bytes of MAC encapsulation, an offset of 2 will always align the IP header to a 128-bit boundary).

Frames can be transmitted with or without automatic CRC generation. If CRC is automatically generated, pad will also be automatically generated to take frames to a minimum length of 64 bytes. When CRC is not automatically generated (as defined in word 1 of the transmit buffer descriptor or through the control bus of the external FIFO interface), the frame is assumed to be at least 64 bytes long and pad is not generated.

An entry in the transmit buffer descriptor list is described in *Table 29*.

To transmit frames, the buffer descriptors must be initialized by writing an appropriate byte address to bits 31:0 in the first word of each descriptor list entry to indicate the location of the data to be transmitted.

The second word of the transmit buffer descriptor is initialized with control information that indicates the length of the frame, whether or not the MAC is to append CRC and whether the buffer is the last buffer in the frame. It also contains the “used” and “wrap” bits. **It is very important that the transmit buffer descriptor list contains at least one entry with its “used” bit set.** This is because the transmit DMA can read the buffer descriptor list very fast and will loop round retransmitting data when it encounters the wrap bit. When initializing the descriptor list the user needs add an additional buffer descriptor with its “used” bit set after the buffer descriptors which describe the data to be transmitted.

Also if descriptor read buffering has been configured (with `define gem_axi_tx_descr_rd_buff_bits), the transmit buffer descriptor list should not contain fewer entries than the size of the descriptor read buffers.

After transmission the status bits are written back to the second word of the first buffer descriptor along with the used bit. Bit 31 is the used bit which must be zero when the control word is read if transmission is to take place. It is written to one once the frame has been transmitted. Bits[29:20] indicate various transmit error conditions. Bit 23 indicates a valid timestamp has been captured in the BD. Bit 30 is the wrap bit which can be set for any buffer within a frame. If no wrap bit is encountered the queue pointer continues to increment. This operation is different from Ethernet MAC 10/100 (MACB) from Cadence, which will wrap after 1024 buffers have been used.

The transmit buffer queue base address register can only be updated whilst transmission is disabled or halted; otherwise any attempted write will be ignored. When transmission is halted the transmit buffer queue pointer will maintain its value. Therefore, when transmission is restarted the next descriptor read from the queue will be from immediately after the last successfully transmitted frame. Whilst transmit is disabled (bit 3 of the network control set low), the transmit buffer queue pointer resets to point to the address indicated by the transmit buffer queue base address register (disabling transmit resets the pointer however reading the transmit queue pointer register through the APB interface may still return the old value until transmission is restarted). **Note**: Disabling receive does not have the same effect on the receive buffer queue pointer.

Once the transmit queue is initialized, transmit is activated either by writing to the transmit start bit (bit 9) of the network control register, or in hardware by toggling the trigger_dma_tx_start input. Transmit is halted when a buffer descriptor with its used bit set is read, a transmit error occurs, or by writing to the transmit halt bit of the network control register. Transmission is suspended if a pause frame is received while the pause enable bit is set in the network configuration register. Rewriting the start bit while transmission is active is allowed. This is implemented with a tx_go variable which is readable in the transmit status register at bit location 3. The tx_go variable is reset when:

- Transmit is disabled
- A buffer descriptor with its ownership bit set is read
- Bit 10, tx_halt, of the network control register is written
- There is a transmit error such as too many retries, late collision (gigabit mode only) or a transmit underrun

To set tx_go write to bit 9, tx_start, of the network control register. Transmit halt does not take effect until any ongoing transmit finishes.

If the DMA is configured for internal FIFO mode and a collision occurs during transmission of a multi-buffer frame transmission will automatically restart from the first buffer of the frame. For packet buffer mode, the entire contents of the frame are read into the transmit packet buffer memory, so the retry attempt will be replayed directly from the packet buffer memory rather than having to re-fetch through the AHB or AXI.

If the transmit buffer list is incorrectly set up in such a way that a used bit is read mid-way through a multi buffer frame, transmission will stop. If cut-through is in operation and the MAC has actually starting transmitting the frame which has its used bit set, the MAC treats it as a transmit error, and asserts tx_er truncates the frame and corrupts the FCS.

Table 29: Transmit Buffer Descriptor Entry − Non-LSO Frame

Bit	Function
Word 0	
31:0	Byte address of buffer
Word 1	
31	Used – must be zero for GEM to read data to the transmit buffer. GEM sets this to one for the first buffer of a frame once it has been successfully transmitted. Software must clear this bit before the buffer can be used again.
30	Wrap – marks last descriptor in transmit buffer descriptor list. This can be set for any buffer within the frame.
29	Retry limit exceeded, transmit error detected
28	Transmit underrun. Occurs when the start of packet data has been written into the FIFO and either hresp is not OK, or the transmit data could not be fetched in time, or when buffers are exhausted. This is not set when the DMA is configured for packet buffer mode.
27	Transmit frame corruption due to AHB or AXI error – set if an error occurs whilst midway through reading transmit frame from the AHB/AXI, including HRESP or RRESP/BRESP errors and buffers exhausted mid frame (if the buffers run out during transmission of a frame then transmission stops, FCS shall be bad and tx_er asserted). Also set in AHB (not AXI) DMA packet buffer mode if single frame is too large for configured packet buffer memory size.
26	Late collision, transmit error detected. Late collisions only force this status bit to be set in gigabit mode.
25:24	Reserved.
23	For Extended Buffer Descriptor Mode this bit Indicates a timestamp has been captured in the BD. Otherwise Reserved.

Table 29: Transmit Buffer Descriptor Entry − Non-LSO Frame (Continued)

Bit	Function
22:20	Transmit IP/TCP/UDP checksum generation offload errors: • 000 - No Error • 001 - The Packet was identified as a VLAN type, but the header was not fully complete, or had an error in it • 010 - The Packet was identified as a SNAP type, but the header was not fully complete, or had an error in it • 011 - The Packet was not of an IP type, or the IP packet was invalidly short, or the IP was not of type IPv4/ IPv6 • 100 - The Packet was not identified as VLAN, SNAP or IP • 101 - Non supported packet fragmentation occurred. For IPv4 packets, the IP checksum was generated and inserted • 110 - Packet type detected was not TCP or UDP. TCP/UDP checksum was therefore not generated. For IPv4 packets, the IP checksum was generated and inserted • 111- A premature end of packet was detected and the TCP/UDP checksum could not be generated
19:17	Reserved. Must be set to 3’b000 to disable TSO and UFO
16	No CRC to be appended by MAC. When set this implies that the data in the buffers already contains a valid CRC and hence no CRC or padding is to be appended to the current frame by the MAC. This control bit must be set for the first buffer in a frame and will be ignored for the subsequent buffers of a frame. This operation is different from Cadence’s Ethernet MAC 10/100 (Enhanced), which reads the no CRC bit from the final buffer descriptor in the frame. Note that this bit must be clear when using the transmit IP/TCP/UDP checksum generation offload, otherwise checksum generation and substitution will not occur. **Note**: This bit must also be cleared when TX Partial Store and Forward mode is active.
15	Last buffer. When set, this bit will indicate the last buffer in the current frame has been reached.
14	Reserved.
13:0	Length of buffer.
Word 3 (64-bit addressing)	
31:0	Unused
31:0	Unused
31:0	Unused

Table 30: Transmit Buffer Descriptor Entry − TSO Header Buffer

Bit	Function
Word 0	
31:0	Byte address of buffer
Word 1	
31	Used – must be zero for GEM to read data to the transmit buffer. GEM sets this to one for the first buffer of a frame once it has been successfully transmitted. Software must clear this bit before the buffer can be used again.
30	Wrap – marks last descriptor in transmit buffer descriptor list. This can be set for any buffer within the frame.
29	Retry limit exceeded, transmit error detected
28	Transmit under run – always 0 for TSO

Table 30: Transmit Buffer Descriptor Entry − TSO Header Buffer (Continued)

Bit	Function
27	Transmit frame corruption due to AHB or AXI error. Set if an error occurs whilst midway through reading transmit frame from the AHB/AXI, including HRESP or RRESP/BRESP errors and buffers exhausted mid frame (if the buffers run out during transmission of a frame then transmission stops, FCS shall be bad and tx_er asserted). Also set in DMA packet buffer mode if single frame is too large for configured packet buffer memory size.
26	Late collision, transmit error detected. Late collisions only force this status bit to be set in gigabit mode.
25:24	TCP Stream Identifier. Used to select the hardware counter that is used for TCP sequence number generation.
23	For Extended Buffer Descriptor Mode this bit Indicates a timestamp has been captured in the BD. Otherwise Reserved.
22:20	Transmit IP/TCP/UDP checksum generation offload errors: • 000 - No Error • 001 - The Packet was identified as a VLAN type, but the header was not fully complete, or had an error in it • 010 - The Packet was identified as a SNAP type, but the header was not fully complete, or had an error in it • 011 - The Packet was not of an IP type, or the IP packet was invalidly short, or the IP was not of type IPv4/IPv6 • 100 - The Packet was not identified as VLAN, SNAP or IP • 101 - Non supported packet fragmentation occurred. For IPv4 packets, the IP checksum was generated and inserted • 110Packet type detected was not TCP or UDP. TCP/UDP checksum was therefore not generated. For IPv4 packets, the IP checksum was generated and inserted • 111 - A premature end of packet was detected and the TCP/UDP checksum could not be generated
19	TCP Sequence Number Source Select • 0 – Use sequence number value from the header buffer for the first small TCP frame and hardware generated sequence number values for subsequent small TCP frames • 1 – Use hardware generated sequence number values for all small TCP frames
18:17	LSO Control. Set to 2’b10 or 2’b11 to enable TSO
16	No CRC to be appended by MAC. Must be clear for TSO operation
15	Last buffer. Must be clear as TSO requires at least one payload buffer
14	Reserved.
13:0	Length of buffer.

Table 31: Transmit Buffer Descriptor Entry − TSO Payload Buffer

Bit	Function
Word 0	
31:0	Byte address of buffer.
Word 1	
31	Used – must be zero for GEM to read data to the transmit buffer. GEM sets this to one for the first buffer of a frame once it has been successfully transmitted. Software must clear this bit before the buffer can be used again.

Table 31: Transmit Buffer Descriptor Entry − TSO Payload Buffer (Continued)

Bit	Function
30	Wrap – marks last descriptor in transmit buffer descriptor list. This can be set for any buffer within the frame.
29:16	TCP Maximum Segment Size value in bytes. TSO will use a default value of 536 bytes if the programmed value is zero.
15	Last buffer. When set, this bit will indicate the last buffer in the current frame has been reached.
14	Reserved.
13:0	Length of buffer.

Table 32: Transmit Buffer Descriptor Entry − UFO Header Buffer

Bit	Function
	Word 0
31:0	Byte address of buffer.
	Word 1
31	Used – must be zero for GEM to read data to the transmit buffer. GEM sets this to one for the first buffer of a frame once it has been successfully transmitted. Software must clear this bit before the buffer can be used again.
30	Wrap – marks last descriptor in transmit buffer descriptor list. This can be set for any buffer within the frame.
29	Retry limit exceeded, transmit error detected
28	Transmit underrun. Always 0 for UFO
27	Transmit frame corruption due to AHB or AXI error. Set if an error occurs whilst midway through reading transmit frame from the AHB/AXI, including HRESP or RRESP/BRESP errors and buffers exhausted mid frame (if the buffers run out during transmission of a frame then transmission stops, FCS shall be bad and tx_er asserted). Also set in DMA packet buffer mode if single frame is too large for configured packet buffer memory size.
26	Late collision, transmit error detected. Late collisions only force this status bit to be set in gigabit mode.
25:24	Reserved
23	For Extended Buffer Descriptor Mode this bit Indicates a timestamp has been captured in the BD. Otherwise Reserved.
22:20	Transmit IP/TCP/UDP checksum generation offload errors: • 000 - No Error • 001 - The Packet was identified as a VLAN type, but the header was not fully complete, or had an error in it • 010 - The Packet was identified as a SNAP type, but the header was not fully complete, or had an error in it • 011 - The Packet was not of an IP type, or the IP packet was invalidly short, or the IP was not of type IPv4/IPv6 • 100 - The Packet was not identified as VLAN, SNAP or IP • 101 - Non supported packet fragmentation occurred. For IPv4 packets, the IP checksum was generated and inserted • 110 - Packet type detected was not TCP or UDP. TCP/UDP checksum was therefore not generated. For IPv4 packets, the IP checksum was generated and inserted • 111 - A premature end of packet was detected and the TCP/UDP checksum could not be generated
19	Reserved

Table 32: Transmit Buffer Descriptor Entry − UFO Header Buffer (Continued)

Bit	Function
18:17	LSO Control. Set to 2’b01 to enable UFO
16	No CRC to be appended by MAC. Must be clear for UFO operation
15	Last buffer. Must be clear as TSO requires at least one payload buffer
14	Reserved.
13:0	Length of buffer.

Table 33: Transmit Buffer Descriptor Entry − UFO Payload Buffer

Bit	Function
Word 0	
31:0	Byte address of buffer.
Word 1	
31	Used – must be zero for GEM to read data to the transmit buffer. GEM sets this to one for the first buffer of a frame once it has been successfully transmitted. Software must clear this bit before the buffer can be used again.
30	Wrap – marks last descriptor in transmit buffer descriptor list. This can be set for any buffer within the frame.
29:16	Maximum Ethernet Frame Size value in bytes (including FCS). UFO will use a default value of 1518 bytes if the programmed value is zero.
15	Last buffer. When set, this bit will indicate the last buffer in the current frame has been reached.
14	Reserved.
13:0	Length of buffer.

When 64-bit addressing mode is enabled, *Table 34* identifies the added descriptor words.

Table 34: Transmit Buffer Descriptor Entry − 64-Bit Addressing

Bit	Function
Word 2 (64-Bit Addressing)	
31:0	Upper 32-bit address of Data Buffer.
Word 3 (64-Bit Addressing)	
31:10	Unused

When Descriptor Timestamp Capture mode is enabled, *Table 35* identifies the added descriptor words.

Table 35: Transmit Buffer Descriptor Entry − Timestamp Capture

Bit	Function
Word 2 (64-Bit Addressing) or Word 4 (64-Bit Addressing)	
31:30	Timestamp seconds[1:0] / launch time (See Note:1)
29:0	Timestamp nanosecs [29:0] / launch time (See Note:1)
Word 3 (64-Bit Addressing) or Word 5 (64-Bit Addressing)	
31:10	Unused
9:0	Timestamp seconds[11:2] (See Note:1) (prior to release 1p08f1 this was [5:2])

Table 35: Transmit Buffer Descriptor Entry − Timestamp Capture (Continued)

Bit	Function
	Note1: The timestamp is mode is controlled using the tx_bd_control_register. The TX Descriptor Time-stamp Insertion mode bits are defined as: • 00: TS insertion disable • 01: TS inserted for PTP Event Frames only • 10: TS inserted for All PTP Frames only • 11: TS insertion for All Frames After transmission the timestamp bits are written back only to the first buffer descriptor.

6.1.1.9 Time-Based Transmit Scheduling

The time-based shaper enables the user to specify on a per-frame basis the TSU time (launch time) at which a frame is to be transmitted.

If a transmit buffer descriptor has the UTLT (Use Transmit Launch Time) bit set, then the value of the timestamp stored in bits [31:0] in word 2 or 4 (depending on whether 64 bit addressing is enabled) of the buffer descriptor becomes the launch time, and is used to delay transmission until the TSU clock time reaches this time.

The UTLT bit maps to bit 31 of the final word (word 3 or 5) of the transmit buffer descriptor.

Programming the value of the launch time in the transmit descriptor should follow the flow below:
1. Read the current timer module value
2. Calculate the frame launch time and program this value in the launch time field of the transmit descriptor
3. Set the UTLT bit in the transmit descriptor to indicate the launch time field is significant

The time-based shaper enables the user to specify when a frame can be transmitted. It can be used in combination with other available scheduling options such as Fixed Priority, DWRR, ETS, 802.1Qav credit-based shaper and the IEEE 802.1Qbv EnST protocol. Use of the time-based shaper ensures that frames are transmitted in the correct time slot.

Time-Based Scheduling is only available if the TSU and AXI configurations have been defined, and transmit extended buffer descriptor mode has been enabled by writing bit 29 (tx_bd_extended_mode_en) in the DMA configuration register. Time-based scheduling only works for AXI configrations, AHB is not supported.

The launch time consists of 2 bits for a seconds value and 30 bits for a nanoseconds value after which transmission of the frame pointed to by the transmit descriptor can start. Note the launch time is only valid for the first descriptors for a frame – i.e. if a frame is described by multiple descriptor/data buffer pairs, then only the launch time in the first descriptor of that frame will be used.

The 32-bit launch time is compared against the equivalent 32 bits in the TSU timer when the descriptor is first read by the DMA. If the 2 seconds bits match then transmission will be delayed if the 30 bit TSU nanosecond value is less than that of the launch time nanosecond value. Transmission of the frame will be delayed until the nanoseconds value of the TSU reaches that of the launch time. Transmission will also be delayed if the 2 seconds bits of the launch time is one more than that of the TSU regardless of the nanosecond value. This will allow frame transmission to be delayed at least one second beyond the current TSU time. Transmission will then start when the TSU seconds value matches the launch time value and the TSU nanoseconds reaches that of the launch time. If the launch time seconds value is 2 or more above that of the TSU then frame transmission will not be delayed. The conditions in *Table 36* will cause frame transmission to be delayed.

Table 36: TSU States which Cause Frame Transmission to be Delayed

TSU Seconds	Launch Time Seconds
00	00 and TSU ns < launch time ns
00	01
01	01 and TSU ns < launch time ns
01	10
10	10 and TSU ns < launch time ns
10	11
11	11 and TSU ns < launch time ns

Table 36: TSU States which Cause Frame Transmission to be Delayed

TSU Seconds	Launch Time Seconds
11	00

In summary the transmit buffer descriptors are defined as follows:

Extended Descriptor (32-bit addressing):

UTLT	RSVD	TS	USED	WRP	STS1	LSO1	STS2	LSO2	CRC	LAST	RSVD	LEN	ADDL

- UTLT[127] = Use Transmit Launch Time
- RSVD[126:106] = Reserved
- TS[105:64] = Timestamp (seconds = 105:94, nanoseconds = 93:64). Written as STATUS.
- USED[63] = Used bit
- WRAP[62] = Wrap bit
- STS1[61:58] = Status bits (written during descriptor writeback only)
- LSO1[57:56] = Bits to control and enable LSO features
- STS2[55:52] = Status bits (written during descriptor writeback only)
- LSO2[51:49] = Bits to control and enable LSO features
- CRC[48] = Set if no CRC is to be appended to frame
- LAST[47] = Set if this is the last descriptor buffer of the transmit frame
- RSVD[46] = Reserved
- LEN[45:32] = Buffer length
- ADDL[31:0] = Data Buffer Address (lower 32 bits)

Extended Descriptor (64-bit addressing):

UTLT	RSVD	TS	ADDM	USED	WRAP	STS1	LSO1	STS2	LSO2	CRC	LAST	RSVD	LED	ADDL

The mappings are the same as 32 bit addressing with the following changes:

- UTLT[191] = Use Transmit Launch Time
- RSVD[190:170] = Reserved
- TS[169:128] = Timestamp (seconds = 169:158, nanoseconds = 157:128). Written as STATUS.
- RSVD[127:96] = Reserved

6.1.1.10 DMA Bursting

The AXI DMA will always use INCR type accesses. The AHB DMA will always use SINGLE, or INCR type AHB accesses. When performing data transfers, the burst length used can be programmed using bits [4:0] of the DMA Configuration Register. Either single or fixed length incrementing bursts up to a maximum of 256 (AXI4) or 16 (AHB) are used as appropriate.

When there is sufficient space and enough data to be transferred, the programmed fixed length bursts will be used. If there is not enough data or space available, for example when at the end of a buffer or packet, bursts of sizes less than the programmed burst length value will be issued. The same is true at 1024 byte boundaries for AHB or 4096 byte boundaries for AXI, so that the relevant boundaries are not burst over as per AMBA requirements. Where the number of accesses remaining is less than 4 for AHB, single accesses will be used.

When the DMA is configured for packet buffer mode, an option to force the GEM DMA to pad the remaining bursts at the end of a buffer or EOP to the programmed burst length value is available via bits 26 and 25 of the DMA Configuration Register. Bit 26 will control the TX and bit 25 the RX. For RX, the data to burst is padded with ‘0’s up to the burst boundary defined by burst length. For TX, the extra data that is read is ignored by the DMA. This feature has been included for performance reasons when external AHB or AXI slaves that are being accessed by GEM perform better when accessed using fixed maximum length bursts. **Note:** Enabling this feature will not break the AHB 1KB rule, or the AXI 4KB rule and the enable bits are ignored if the programmed burst length is 256.

The DMA will not terminate fixed length bursts early, unless an AHB HRESP error condition is detected (note AXI response errors will never cause a burst to be terminated early) or if receive/transmit operation is disabled by software by writing to the network control register.

6.1.1.11 DMA Endianism

The default configuration of the DMA is to use little endian format. In order to interface to different CPU systems with different endianism requirements, GEM DMA may be programmed to swap the endianism using bits 6 and 7 of the DMA Configuration Register. Bit 6 controls the endianism of the management operations and bit 7 controls the endianism of the data operations.

When either of these modes are enabled, the order of the bytes on the AHB or AXI are swapped for either or both of the buffer management and data operations, as illustrated in the diagram below for a 128-bit system for data operations (bit 7 set). The same swapping applies for management operations if bit 6 is set.

Figure 7 shows the endian swap.

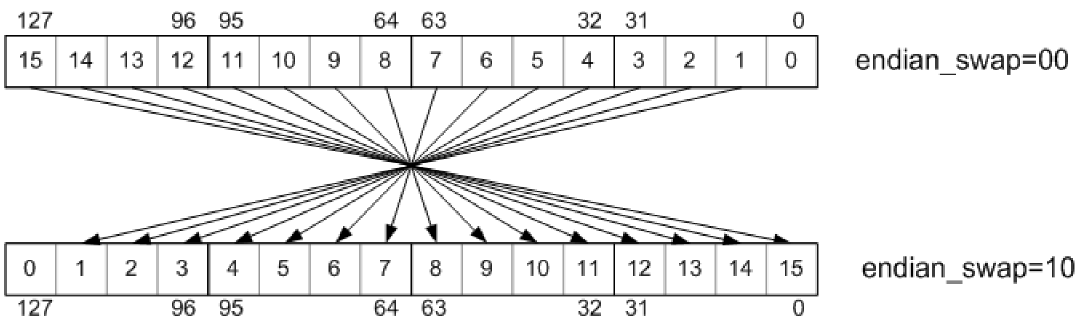

Figure 7: Endian Swap

6.1.1.12 DMA Transmit and Receive FIFOs

This section is not relevant if GEM is configured to use SRAM based packet buffer DMA mode.

The DMA can be configured to use a very low-latency mode by not including the packet buffer. In this mode, separate transmit and receive FIFOs are included within the DMA which are implemented using synthesizable flip-flop arrays. The FIFOs are used to smooth out data flow delays and latencies on the AHB fabric to ensure consistent service for the MAC pipelines and prevent underflows and overflows. The FIFOs are also used to transition data over the clock boundary between the DMA and the MAC rx_clk and tx_clk.

The transmitter and receiver FIFO depths are parameterizable. It is important to choose FIFO depths that are appropriate for the system clock speed, memory latency and network speed. In the default implementation of GEM, the receive and transmit FIFO depths are both set to ten locations each of 128 bits, giving a total of 160 bytes of storage per FIFO.

Data is typically transferred into and out of the FIFOs in bursts of the programmed burst length in the DMA configuration register.

For receive, an AHB bus request is asserted when the FIFO contains greater than or equal to the programmed AHB burst length. When the end of frame is reached, single word transfers may be used to complete the frame transfer if there is less data left in the frame to transfer than the programmed burst length. If the frame does not finish on the word boundary — depending on dma_bus_width, the frame length and the initial byte offset — the last word of the last transfer will contain redundant bytes, written as logic zero.

For transmit, an AHB bus request is generated whenever there is space for greater than or equal to the programmed AHB burst length in the transmit FIFO. This sequence is followed for every transfer in the buffer except the last. Once the last data transfer of a buffer is reached, the final burst may consist of single transfers if there is less data left in the frame to transfer than the programmed burst size. The last word of the last transfer contains redundant bytes if the frame does not finish on the word boundary. This depends upon dma_bus_width and the frame length.

Transmission of the frame will not begin until as much as the frame as possible has been read into the transmit FIFO. This is deemed to be when there is less space left in the FIFO than the programmed burst length, or if the end of the frame has already been written into the FIFO. Once either of these events occurs, a trigger is sent to the MAC transmitter, which will then begin reading out the frame from the FIFO. This way there is more of a buffer built up to allow for any delays in the AHB which may otherwise cause an underflow of the FIFO.

As the system designer, you should consider data bus width, transmit/receive clock rates, AMBA DMA bus latency, and the chosen AMBA AHB clock rate. These parameters will determine the FIFO depth.

The default FIFO depth of 10 means that the bus latency must be less than the time it takes to load the FIFO and transmit or receive six words — 24 bytes in 32-bit AHB mode — of data. At 1000Mbps, it takes 192ns to transmit or receive 24 bytes of data. In addition six hclk cycles should be allowed for data to be loaded from the bus and to propagate through the FIFOs. For an 80MHz hclk this takes 75ns, therefore making the bus latency requirement approximately 120ns.

Transmit buffers can be any length between 1 and 16383 bytes. For short length buffers, adequate bandwidth must be allowed for the buffer management overhead. It is recommended to perform simulations to ensure the system has adequate bandwidth if transmit frames are likely to contain a number of very short buffers. If the DMA block cannot access transmit data from memory in time, the “transmit underrun” status will be reported and an interrupt asserted.

Note: The FIFO depths must be chosen to allow for the desired burst length. For instance. if a 16-beat burst is desirable then the depths of both the receiver and transmitter FIFOs must be at least 16 words deep. The DMA Configuration register automatically prevents programming of a burst length that is greater than the FIFOs depth, by masking out the appropriate upper bits of the burst length field in the register.

AXI operation is not supported when the DMA is configured in this mode.

6.1.1.13 DMA Packet Buffer

This section is only relevant if GEM is configured to use SRAM based packet buffer DMA mode.

The DMA can be configured to use packet buffers for both transmit and receive paths. In this mode, the user has the option of an AXI4 master interface or an AHB master interface. This mode:

- Allows multiple packets to be buffered in both transmit and receive directions
- Allows the DMA to withstand variable levels of access latencies on the AHB or AXI fabric
- Offers the most efficient use of the system bandwidth

As described earlier, when the DMA is configured to use packet buffers, it can be programmed into a low latency mode known as partial store and forward. When this is enabled the packet buffer DMA will functionally behave in a similar way to the internal FIFO DMA mode. For further details of this mode, refer to *Section 6.1.1.5 Partial Store and Forward Using Packet Buffer DMA* on page 60.

When the DMA is programmed in full store and forward mode, full packets are buffered providing the opportunity to:

- Discard packets that are received with errors before they are partially written out of the DMA thus saving system bus bandwidth and driver processing overhead
- Retry collided transmit frames from the buffer, thus saving AHB or AXI bus bandwidth
- Implement transmit IP/TCP/UDP checksum generation offload

Figure shows the structure of the GEM datapaths when the packet buffers are included.

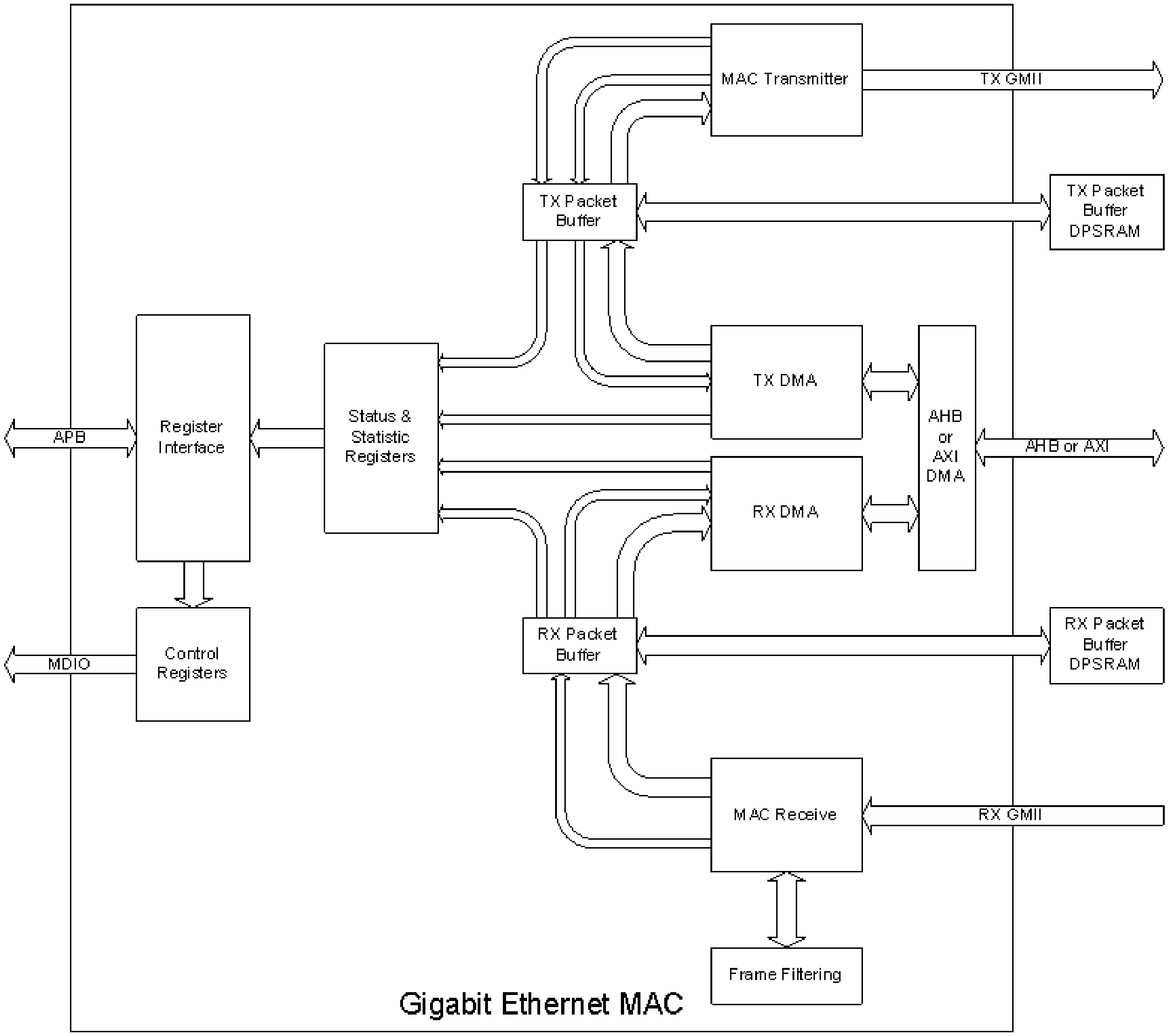

Figure 8: Datapath Structure with SRAM Packet Buffers

6.1.1.14 Transmit Packet Buffer

The transmitter packet buffer will continue attempting to fetch frame data from the AHB or AXI system memory until the packet buffer itself is full, at which point it will attempt to maintain its full level.

To accommodate the status and statistics associated with each frame, 3 words (or 2 if GEM is configured in 64 bit datapath mode) per packet are reserved at the end of the packet data. If the packet was bad and requires to be dropped, the status and statistics are the only information held on that packet. Storing the status in the DPRAM is required in order to decouple the DMA interface of the buffer from the MAC interface, to update the MAC status/stats and to generate interrupts in the order in which the packets that they represent were fetched from the external memory.

The transmit packet buffer will only attempt to read more frame data from the external memory when space is available in the internal packet buffer memory. If space is not available it must wait until a packet fetched by the MAC completes transmission and is subsequently removed from the packet buffer memory. Note that if full store and forward mode is active and if a single frame is fetched that is too large for the packet buffer memory, the frame is flushed and the DMA halted with an error status. This is because a complete frame must be written into the packet buffer before transmission can begin, and therefore the minimum packet buffer memory size should be chosen to satisfy the maximum frame to be transmitted in the application.

To ensure maximum throughput and full line-rate, it is recommended that the transmit packet buffer is greater than twice the size of the largest frame that will be transmitted for the intended application. This is to ensure the MAC pipeline can be kept full at all times. If only short frames will ever be transmitted, it is recommended that an SRAM size substantially more than twice the size of the largest frame is used – this is to offset and smooth potential random latency delays on the AHB or AXI data fetches.

In full store and forward mode, once the complete transmit frame is written into the packet buffer memory, a trigger is sent across to the MAC transmitter, which will then begin reading the frame from the packet buffer memory. Since the whole frame is present and stable in the packet buffer memory an underflow of the transmitter is not possible. The frame is kept in the packet buffer until notification is received from the MAC that the frame data has either been successfully transmitted or can no longer be re-transmitted (too many retries in half-duplex mode). When this notification is received, the frame is flushed from memory to make room for a new frame to be fetched from the AHB or AXI system memory.

In partial store and forward mode, a trigger is sent across to the MAC transmitter as soon as sufficient packet data is available in the internal buffer, which will then begin fetching the frame from the packet buffer memory. If, after this point, the MAC transmitter is able to fetch data from the packet buffer faster than the DMA can fill it, an underflow of the transmitter is possible. In this case, the transmission is terminated early, and the packet buffer is flushed. Transmission can only be restarted by writing to the transmit START bit, or by toggling the trigger_dma_tx_start input.

In half-duplex mode, the frame is kept in the packet buffer until notification is received from the MAC that the frame data has either been successfully transmitted or can no longer be retransmitted (too many retries in half-duplex mode). When this notification is received, the frame is flushed from memory to make room for a new frame to be fetched from external system memory.

In full-duplex mode, the frame is removed from the packet buffer on the fly.

Other than underflow, the only MAC related errors that can occur are due to collisions during half-duplex transmissions. When a collision occurs the frame still exists in the packet buffer memory so can be retried directly from there. Only once the MAC transmitter has failed to transmit after sixteen attempts is the frame finally flushed from the packet buffer.

6.1.1.15 Receive Packet Buffer

The receive packet buffer stores frames from the MAC receiver along with their status and statistics. Frames with errors are flushed from the packet buffer memory, whilst good frames are pushed onto the DMA AHB interface.

The receiver packet buffer monitors the external FIFO write interface from the MAC receiver and translates the FIFO pushes into packet buffer writes. At the end of the received frame, the status and statistics are buffered so that the information can be used when the frame is read out. When programmed in full store and forward mode, if the frame has an error, the frame data is immediately flushed from the packet buffer memory allowing subsequent frames to utilize the freed up space. The status and statistics for bad frames are still used to update the GEM registers.

To accommodate the status and statistics associated with each frame, up to 4 words (one being for descriptor timestamp capture when enabled) (or 2 if you are configured in 64-bit datapath mode) per packet are reserved at the end of the packet data. If the packet was bad and requires to be dropped, the status and statistics are the only information held on that packet.

The receiver packet buffer will also detect a full condition such that an overflow condition can be detected. If this occurs, subsequent packets will be dropped and an RX overflow interrupt is raised.

For full store and forward, the DMA will only begin packet fetches once the status and statistics for a frame are available. If the frame has a bad status due to a frame error, the status and statistics are passed onto the GEM registers. If the frame has a good status, the information is used to read the frame from the packet buffer memory and burst onto the AHB/AXI using the DMA buffer management protocol. Once the last frame data has been transferred to the FIFO, the status and statistics are updated to the GEM registers.

If partial store and forward mode is active, the DMA will begin fetching the packet data before the status is available. As soon as the status becomes available, the DMA will fetch this information before continuing to fetch the remainder of the frame. Once the last frame data has been transferred to the FIFO, the status and statistics are updated to the GEM registers.

6.1.1.16 Priority Queuing in the DMA

The DMA by default uses a single transmit and receive queue. This means the list of transmit/receive buffer descriptors point to data buffers associated with a single transmit/receive data stream. When the DMA is configured to use packet buffer memories, GEM can optionally select up to 16 priority queues. Each queue has an independent list of buffer descriptors pointing to separate data streams. The maximum number of queues is determined by setting appropriate defines in gem_gxl_defs.v (refer to *Section 3.2 Configuration ‘defines* on page 17). By default, each queue (up to the maximum configured in gem_gxl_defs.v) is active. Queues may be disabled by setting bit 0 of the Transmit or Receive Buffer Queue Base Address register, for which there is one per queue. Note that at least one queue must always remain enabled and only the top indexed queues may ever be disabled. For example for a system configured to have 12 queues, you may decide that only 7 of these need to be active. In this case, the user would disable queues 8 to 12 by setting bit 0 of the Transmit or Receive Buffer Queue Base Address registers specific to queues 8 to 12. (**Note**: IEEE 802.1Qav (CBS) works on the two highest enabled queues but for 802.1Qbv (EnST) if queues are disabled by software the control register mapping does not change. So EnST functionality is partially lost as transmit queues are disabled, for example with 12 queues and two disabled you would only have the ability to do EnST on 6 queues (rather than 8).)

In the transmit direction, higher priority queues are always serviced before lower priority queues. This strict priority scheme requires the user to ensure that high priority traffic is constrained such that lower priority traffic will have required bandwidth. The DMA will determine the next queue to service by initiating a sequence of buffer descriptor reads interrogating the ownership bits of each. The buffer descriptor corresponding to the highest priority queue is read first. If the ownership bit of this descriptor is set, then the DMA will progress to reading the 2nd highest priority queue's descriptor. If that ownership bit read of this lower priority queue is set, then the DMA will read the 3rd highest priority queue's descriptor, and so on. If all the descriptors return an ownership bit set, then a resource error has occurred, an interrupt is generated and transmission is automatically halted. Transmission can only be restarted by setting the START bit in the network control register or by toggling the trigger_dma_tx_start input. The DMA will need to identify the highest available queue to transmit from when the START bit in the network control register is written to and the TX is in a halted state, or when the last word of any packet has been fetched from external AHB or AXI memory.

The transmit DMA will maximize the effectiveness of priority queuing by ensuring that high priority traffic be transmitted as early as possible after being fetched from external memory. High priority traffic will be pushed to the MAC layer depending on the specific transmit scheduling algorithm employed (refer to *Section 6.1.7 Transmit Scheduling Algorithm* on page 91). In order to facilitate the scheduler, the SRAM based packet buffer used within the transmit DMA is split into regions, one region per queue. The size of each region determines the amount of SRAM space allocated per queue and will be configurable by setting relevant defines in gem_gxl_defs.v file. This is implemented by first splitting the transmit SRAM into a number of segments of equal size. This number must be a power of two. Then, each queue is granted a number of segments. These configuration options and how to set them are explained further in *Section 3.2 Configuration ‘defines* on page 17. Setting the configuration only sets the initial setting for SRAM queue allocation. You can change it by writing to the transmit segment allocation registers at offset 0x5a0 and 0x5a4. This allows you to tweak the SRAM allocation for each queue and is very useful if you wish to disable queues and reallocate the memory assigned to an active queue. There are three bits per queue. Segment allocation for the queue is done by writing a value of:

- 0 to allocate 1 segment
- 1 to allocate 2 segments
- 2 to allocate 4 segments
- 3 to allocate 8 segments
- 4 to allocate a maximum of 16 segments

Writing a value of 5, 6, or 7 is not permitted.

For each queue, there is an associated Transmit Buffer Queue Base Address register. For the lowest priority queue (or the only queue when only 1 queue is selected), the Transmit Buffer Queue Base Address is located at address 0x1C. For all other queues, the Transmit Buffer Queue Base Address registers are located at sequential addresses starting at address 0x440.

In the receive direction, each data packet is written to external AHB/AXI data buffers in the order that it is received. For each queue, there is an independent set of receive buffers for each queue. There is, therefore, a separate Receive Buffer Queue Base Address register for each queue. For the lowest priority queue (or the only queue when only one queue is selected), the Receive Buffer Queue Base Address is located at address 0x18. For all other queues, the Receive Buffer Queue Base Address registers are located at sequential addresses starting at address 0x480 for queues one to seven and from 0x5c0 for queues eight to fifteen. Every received packet will pass through a programmable screening algorithm which will allocate to that frame a particular queue to route it to. The user interface to the screeners is via two banks of programmable registers, screener type 1 match registers and screener type 2 match registers.

The operation of the screener registers is described in detail in *Section 6.1.9.8.1 Screener Type 1 and Type 2 Match Registers* on page 98.

6.1.2 Advanced DMA Features to Reduce CPU Overhead

The DMA design has support for various optional offload features to reduce CPU overhead.

6.1.2.1 Receive Header/Data Splitting

Receive header data splitting is a feature which when enabled will force the receive DMA to split a received frame into its header and payload constituent parts.

The header is not dropped, but instead separated from the payload and placed into its own DMA receive buffer. For example, a TCP/IPv4 frame will be split such that its Ethernet, IPv4 and TCP header is written into one or more buffers, and the TCP payload is written into one or more separate buffers.

A status bit in the receive descriptors will identify whether the descriptor is pointing to a header buffer or a payload buffer. Another status bit will identify whether the descriptor is pointing to the last buffer of a header (the header may be split over multiple buffers). The length of the header will also be written in the descriptor that points to the last buffer of the header.

When header/data splitting is enabled, ALL received frames will have their L2/L3/L4 headers separated. **Note**: Even standard Ethernet only frames containing just 14 bytes of header will be separated. Bit 5 of the DMA configuration register at offset 0x10 enables/disables header data splitting. When enabled there are specific bits of the DMA receive buffer descriptor (refer to the Receive Buffer Descriptor entry for details).

There are certain limitations to using this feature:

- Header Data Splitting applies to all received frames. Every frame received which includes a data payload will be split into at least 2 data buffers.
- Header Data Splitting is not currently available with the partial store and forward mode of operation. It should only be enabled when the full store and forward mode of operation is active.

6.1.2.2 Receive Side Coalescing

Receive segmentation coalescing (RS) is a mechanism to reduce CPU overhead.

This is done by coalescing received TCP message segments together into a single large message. This means that when the message is complete the CPU only has to process the single header and act upon the one data payload. This will only be effective when processing small non signaling frames. For RSC to work, the Header Data splitting feature must be enabled, and the priority queuing configuration options must be enabled (more details on this below).

The design will first inspect the low-level received frames and identify if they are part of a particular TCP stream or not. This is performed using the existing priority queuing feature (refer *Section 6.1.1.16 Priority Queuing in the DMA* on page 79). Each defined receive priority queue (other than queue 0) will map to a TCP stream capable of performing RSC. Therefore the priority queue screeners must be set up such that received frames can be inspected and routed to the appropriate queue. This can be achieved as follows:

Note: It is assumed the reader is already familiar with the priority queue features as defined earlier in this document.

When the ethertype = IPv4 (0800), then the subsequent TCP/IP streams (or flows) are identified by the 4-tuple of IP source and destination address and TCP source and destination port. This is done using the type 2 screeners. To provide sufficient compare functionality, all three A, B, and C compares are required and the full 32-bit comparison (i.e. mask functionality disabled) will be required for each comparison. *Figure* gives an example of the offsets required to identify the tuples.

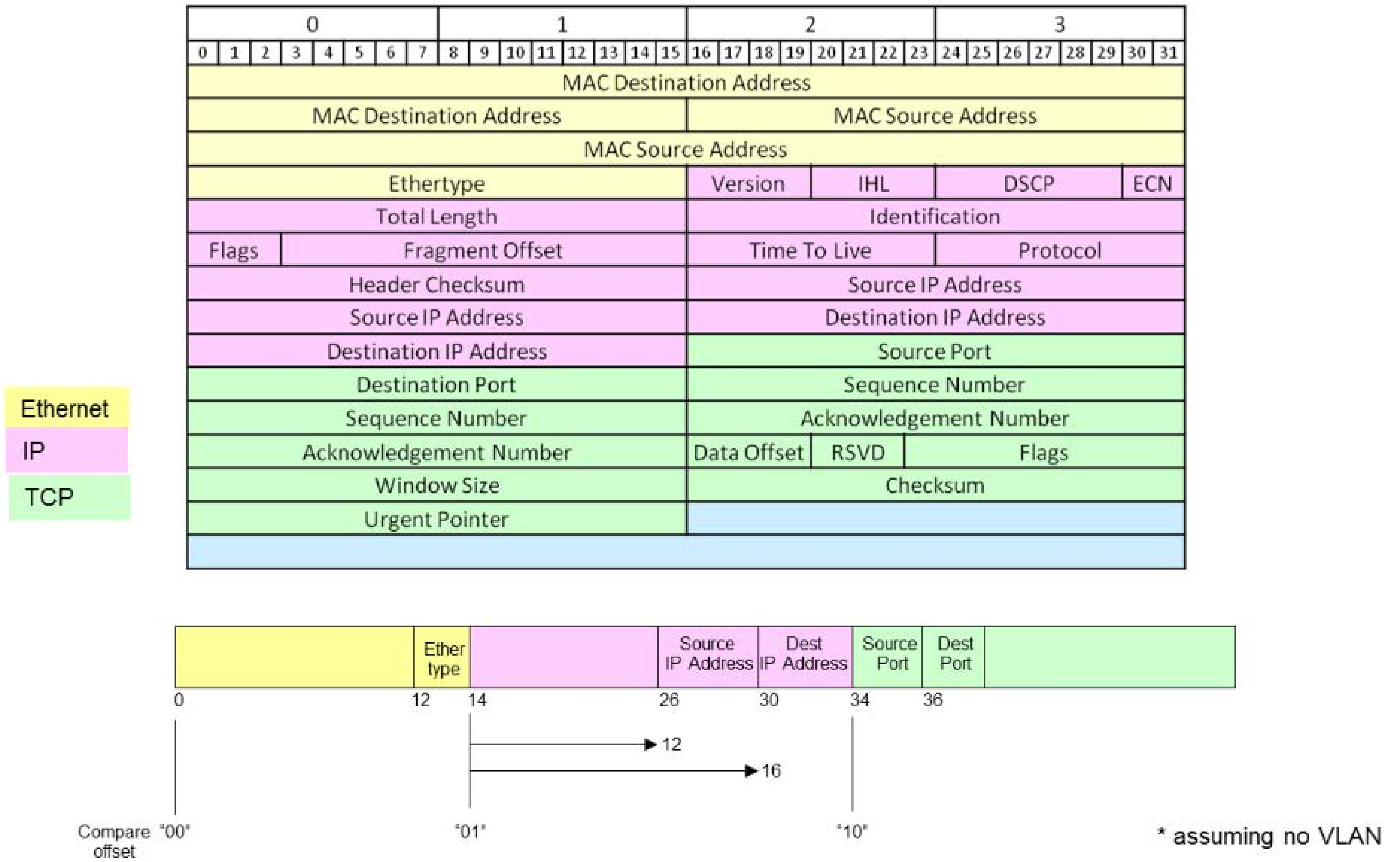

Figure 9: TCP/IP Encapsulation in an Ethernet Frame

The coalesced frames should be controlled using multiple queues therefore RSC requires the following to be set up:

- Hardware Configuration
- Memory Setup
- Enable RSC Control

6.1.2.2.1 Hardware Configuration Requirements

Table 37 lists the hardware configuration requirement for RSC.

Table 37: Hardware Configuration ‘defines

Name of Define	Description
gem_axi	RSC functionality is only supported when using an AXI interface. Set this define if RSC support is required.
dma_priority_queue1	RSC functionality is only available if the module is configured to have at least 2 queues. Multiple queues is a hardware configuration and is set using this `define prior to compile.
gem_pbuf_rsc	Enables RSC functionality. Set this define to enable RSC.

Table 37: Hardware Configuration 'defines (Continued)

Name of Define	Description
num_type2_screeners	Defines the number of type 2 screeners. RSC needs a number of ~~type2~~type 2 screeners defined. The number chosen should at minimum reflect the number of TCP streams required. Three compare registers per screener are also required. Below is an example define: `` `define num_type2_screeners 8'd3 ``

6.1.2.2.2 Memory Setup

Descriptor buffers point to the main system memory. Receive buffers should be configured to their maximum value of 16320 in the receive buffer size registers by setting dma_rx_q_buf_size to 0xFF. As header data splitting is enabled, software must provide at least two descriptors; the first being for the header followed by a second for the payload. This will result in a waste of memory space since the header will never be 16K. Therefore, it is suggested that software overlaps the second descriptor as shown in the *Figure 10*.

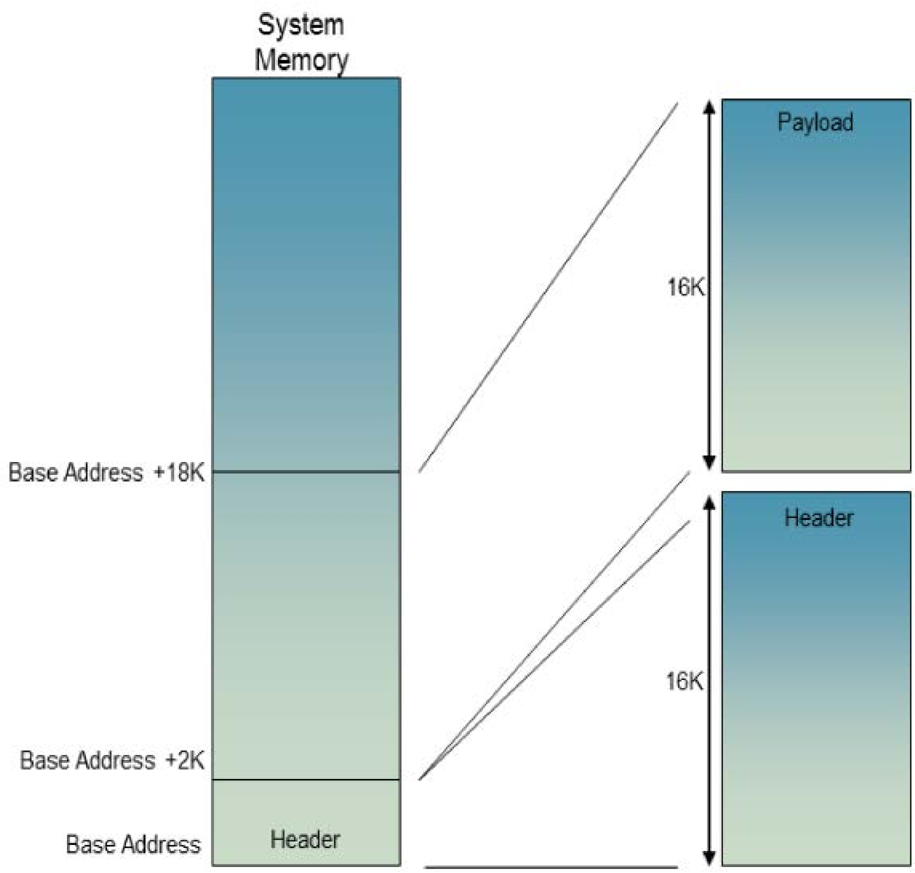

Figure 10: Memory Setup for RSC

The design will coalesce buffers up to the maximum size of 16K. If the next packet causes the coalesced buffer to exceed the maximum size, at this point it will close the buffer and restart a new buffer. Closing the buffer involves updating the IP length, checksum, and ACK fields in the header buffer and writing used bits for the header and payload buffers. The ACK field is written with the most recently received ACK number with the ACK flag set.

If the next packet has TCP Flags SYN, FIN, RST, or URG set, or a frame is received with an out-of-order TCP sequence number, these events are treated as "RSC STOP" events. In these cases, you can optionally select via the RSC control register offset 0x58 whether coalescing should cease (restarting only upon software setting up a new TCP stream for coalescing) or whether the design should simply restart coalescing into a new buffer on the frame following the “RSC STOP” event. The frame received that caused the “RSC STOP” event is always forwarded to main memory in isolation (non-coalesced).

If the next packet has the PSH TCP Flag set, this will also be treated as an “RSC STOP” event. However, coalescing will restart on the next frame received associated with the TCP stream.

The memory size for each priority queue is defined in the dma_rxbuf_size_q1 registers. For coalescing on a particular queue, these should be set to the maximum size of 16K.

6.1.2.2.3 RSC Control

Header Data splitting must be enabled prior to enabling RSC. RSC is enabled via the RSC Control register offset 0x58. Via this register, the number of TCP streams for coalescing is chosen (each priority queue will map to a particular TCP stream). There is also a control bit in this register to allow the user to select whether coalescing should cease after the “TCP STOP” event occurs.

6.1.2.2.4 RSC Limitations

- Little Endian only
- Header Data Splitting must be enabled to use RSC
- Buffer Sizes must be set to 16K
- RSC only works with IPv4 headers
- RSC should not be used if the GEM will receive jumbo frames
- Currently RSC cannot be used in conjunction with partial store and forward mode of operation
- When the gem_pbuf_rsc define has been set the receive buffer offset cannot be changed in the network configuration register

6.1.2.3 Large Send Offload

Large Send Offload functionality is included to reduce CPU overhead for transmit. Two mechanisms are included − TCP Segmentation Offload (TSO) and UDP Fragmentation Offload (UFO). CPU overhead is reduced by permitting software to hand off TCP and UDP frames which are larger than the current TCP Maximum Segment Size (MSS) and Ethernet Maximum Frame Size (MFS). Hardware splits the large software supplied frames into smaller frames for transmission and modifies the header fields as required. TCP offload is achieved by creating smaller segments at the TCP level. UDP offload is achieved by performing fragmentation at the IP level.

For correct TSO and UFO functionality, hardware checksum insertion must be enabled (DMA Configuration Register, bit 11).

For both TSO and UFO, software must supply the header and payload of the large frame in at least two buffers (one for header and one or more for payload). Interaction between hardware and software is similar to the use case where multiple buffers are used to supply a single Ethernet frame.

There are no control registers associated with TSO and UFO – the features are enabled and configured via fields in the descriptors for the header and payload buffers.

6.1.2.4 TSO Operation

Software supplies a large frame containing a large TCP segment via a header buffer and one or more payload buffers, and supplies the associated header and payload descriptor(s). The header buffer contains the Ethernet, IPv4 or IPv6, and TCP headers. The header descriptor has bits [18:17] set to 2’b10 or 2’b11 to enable TSO.

The header descriptor contains a sequence number select bit at bit 19. This is cleared if the hardware uses the sequence number from the header buffer for the first small segment (and generate appropriate sequence numbers for the other small segments). It is set if hardware generates sequence numbers for all of the small segments. In this case, the sequence number for the first small segment will be related to the most recent previously transmitted segment of the same stream that were transmitted with TSO enabled. The hardware supports four separate TCP streams per priority queue (and implements a sequence number counter for each). The TCP stream is identified by bits [25:24] of the header descriptor.

The payload descriptor contains a TCP Maximum Segment Size (MSS) field at bits [29:16] and hardware uses this MSS value to process the payload buffer. Each payload buffer is processed separately (i.e. an integer number of small segments is generated from each payload buffer) using the MSS value in the associated descriptor. Software may supply different MSS values in the payload descriptors of a large frame which has two or more payload buffers.

Hardware loads the header descriptor and stores it internally, and uses it to read the associated header buffer multiple times as the payload buffers are processed and small segments are transmitted. Hardware modifies the IPv4 Total Length, IPv4 Header Checksum, IPv6 Payload Length, TCP Sequence Number, and TCP Checksum fields. Software may modify the TCP Acknowledgement Number and TCP Window Size values in the header buffer during the hardware segmentation process – however as software has no notification of when hardware is reading the header buffer, care must be taken to avoid hardware reading a partially updated value. This can be achieved by locating the base address of the header buffer so that a software write of a field to be modified is achieved via a single AXI write access beat – i.e. the Acknowledgement Number field is on a 32-bit boundary in memory and the Window Size field is on a 16-bit boundary in memory. Placement of the header buffer base address to achieve this is dependent on the Ethernet header length.

Hardware discards the IPv4 Total Length and IPv6 Payload length values in the header buffer supplied by software and therefore it is possible for software to supply a very large TCP segment that exceeds the limitations imposed by IP encapsulation.

Hardware performs a write back to the header descriptor when the data contained in a payload buffer has been transmitted and the associated payload descriptor has the last bit set.

Software can make use of the sequence number generation functionality for all TCP segments regardless of size by:

- Delivering the frame in separate header and payload buffers and
- Enabling TSO in the header descriptor

It is not necessary for the supplied segment to exceed the TCP Maximum Segment Size in order to use hardware sequence number functionality.

6.1.2.4.1 TSO Limitations

- Ethernet Header extensions are limited to VLAN and Stacked VLAN
- Payload descriptors must not have a value of zero in the length field
- The Ethernet frame supplied by software must not contain IP fragments

6.1.2.5 UFO Operation

Software supplies a large frame containing a large UDP datagram via a header buffer and one or more payload buffers, and supplies the associated header and payload descriptor(s). The header buffer contains the Ethernet and IPv4 headers. The header descriptor has bits [18:17] set to 2’b01 to enable UFO. The payload buffer or buffers contain the IP payload, which consists of the UDP header and UDP payload.

The payload descriptor contains an Ethernet Maximum Frame Size (MFS) field at bits [29:16] and hardware uses this MFS value to process the payload buffer. Each payload buffer is processed separately (i.e. an integer number of fragments is generated from each payload buffer) using the MFS value in the associated descriptor. Software may supply different MFS values in the payload descriptors of a large frame which has two or more payload buffers. When software supplies two or more payload buffers, all payload buffers apart from the last must be a multiple of 8 bytes in size.

Hardware loads the header descriptor and stores it internally, and uses it to read the associated header buffer multiple times as the payload buffers are processed and fragments are transmitted. Hardware modifies the IPv4 Total Length, IPv4 More Fragments Flag, and IPv4 Fragment Offset fields. Hardware does not calculate the UDP checksum or modify the UDP checksum field. Therefore, software must set a value of zero in the checksum field in the UDP header (in the first payload buffer) to indicate to the receiver that the UDP datagram does not include a checksum.

Hardware performs a write back to the header descriptor when the data contained in a payload buffer has been transmitted and the associated payload descriptor has the last bit set.

6.1.2.5.1 UFO Limitations

- Ethernet Header extensions are limited to VLAN and Stacked VLAN
- IP encapsulation is limited to IPv4
- Payload descriptors must not have a value of zero in the length field
- Payload buffers, apart from those with the last bit set in the associated descriptor, must be a multiple of 8 bytes in length
- The Ethernet frame supplied by software must not contain IP fragments

6.1.3 External FIFO Interface

6.1.3.1 Description

In applications where the DMA operation is not required, an exposed FIFO interface is available for both the transmit and receive datapaths. Data bus widths in each direction can be configured to be either 32, 64, or 128 bit. The dma_data_width bits (bits 21 and 22 of the network configuration register) determine the bus width used within the MAC.

The tx_r_data_rdy signal indicates to the MAC that there is sufficient data in the FIFO for transmission to commence. Once this signal becomes active, the transmit module initiates a read cycle by asserting tx_r_rd for one tx_clk cycle. The FIFO should indicate valid data at the FIFO interface by asserting tx_r_valid for a single cycle. The latency between the read and valid data is controlled using the tx_r_valid response, which may be returned during the same cycle as the tx_r_rd request. Once a read has commenced, it must be terminated with tx_r_valid or tx_r_underflow even if tx_r_data_rdy is de-asserted.

The MAC transmitter searches for start of packet (SOP), indicated by tx_r_sop, and transmission commences once this input becomes valid coincident with tx_r_valid. The MAC will continuously search for tx_r_sop whilst tx_r_data_rdy is set. Once the SOP has been read, data is extracted from the FIFO on the tx_r_data input every time tx_r_valid is set. The MAC continues to read the frame from memory using tx_r_rd and transmission takes place.

The end of frame is indicated by tx_r_eop being set coincident with tx_r_valid. Once this condition occurs, the tx_r_mod input indicates how many data bytes are valid in the last transfer. tx_r_mod is only valid at the end of the frame, and cannot be used to insert an offset at the start of frame.

For frames smaller than the data width, both the SOP and end of packet (EOP) indicators must be set in the same data transfer.

If two SOPs occur with no intervening EOP, then there is an underrun and both frames are lost, unless the second SOP occurs on the same cycle as EOP in which case the second SOP is ignored. A properly configured system should not generate two SOPs with no intervening EOP.

The tx_r_err signal may be asserted at any stage in the frame, being driven coincident with tx_r_valid being set.

In applications where the external FIFO interface is required to operate in a half-duplex system, tx_r_status information is available indicating where collisions, excess collisions, late collisions, and underruns have occurred. Upon each of these conditions, it is necessary to flush the external FIFO and the MAC must wait for this operation to complete before commencing further frames. A falling edge on the tx_r_flushed input signal indicates when the flush is complete. If a collision occurs, it is necessary for the external FIFO interface to repeat the transfer of the current frame, so that the frame may be successfully re-transmitted.

tx_r_status information must be acknowledged by the external FIFO interface by toggling the tx_r_status_tog input to the MAC each time status is taken. This causes the tx_r_status bus to be cleared until the next end of frame or collision occurs.

For reception, once it has been determined that a frame should be written to memory, the MAC receiver writes data to the FIFO using rx_w_wr and rx_w_data. SOP is indicated by rx_w_sop and EOP using rx_w_eop. For the last transfer of a frame, the rx_w_mod is driven to indicate how many bytes are valid in this last transfer. rx_w_sop and rx_w_eop will not be asserted in the same cycle.

The rx_w_err signal is set when the MAC encounters a reception error such as frame too short or CRC error. An rx_w_status bus is available giving status about the frame being received, such as frame length, matched internally or externally, broadcast, multicast, etc. The rx_w_eop signal is always asserted in the same cycle as rx_w_err.

The rx_w_overflow input signal can be asserted in the rx_clk domain when an external FIFO fails to receive a frame from the GEM FIFO interface. If rx_w_overflow is asserted sometime between the SOP and EOP writes, the remainder of the packet will continue to be written out, but at the end of the packet rx_w_err will be asserted together with rx_w_eop. Additionally, if rx_w_overflow is asserted then the receive statistics registers will not count the frame as good. rx_w_overflow must be asserted one cycle after rx_w_eop or earlier.

rx_w_flush is asserted when the GEM receive path is disabled in the network control register. If you disable frame reception while a frame is being transferred on the FIFO interface, you will not see an rx_w_eop indication. rx_w_flush is an rx_clk timed signal that you can use to clear the external receive FIFO when receive is disabled.

6.1.3.2 Priority Queuing without the DMA

Priority Queuing is supported via the FIFO interface. By default, there is a single transmit and receive queue. GEM can optionally select up to 16 priority queues, determined by setting appropriate defines in gem_gxl_defs.v (For details, refer to *Section 3.2 Configuration ‘defines* on page 17). When there is more than one queue, the external FIFO interface is extended support queue ID signals. The width of tx_r_data_rdy and tx_r_rd is extended such that the width matches the number of configured queues. However, the overall functionality of these signals does not change. While multiple bits of tx_r_data_rdy may be set simultaneously to indicate there is sufficient data in the various external FIFOs (one per queue) for transmission to commence, GEM will only ever set one bit of tx_r_rd at any one time. As the integrator, it is for you to multiplex in the various external FIFO’s to interoperate with this signaling.

For the receiver, rx_w_queue[3:0] indicates the priority queue selected by the screener registers. The operation of the screener registers is described in *Section 6.1.9.8.1 Screener Type 1 and Type 2 Match Registers* **on page 98**.

6.1.3.3 External FIFO Interface Timing Criteria

Figure 11 shows detailed timing relationships for a frame on the external FIFO interface (MAC transmit) with one cycle between read request and data valid. Only one priority queue is shown here.

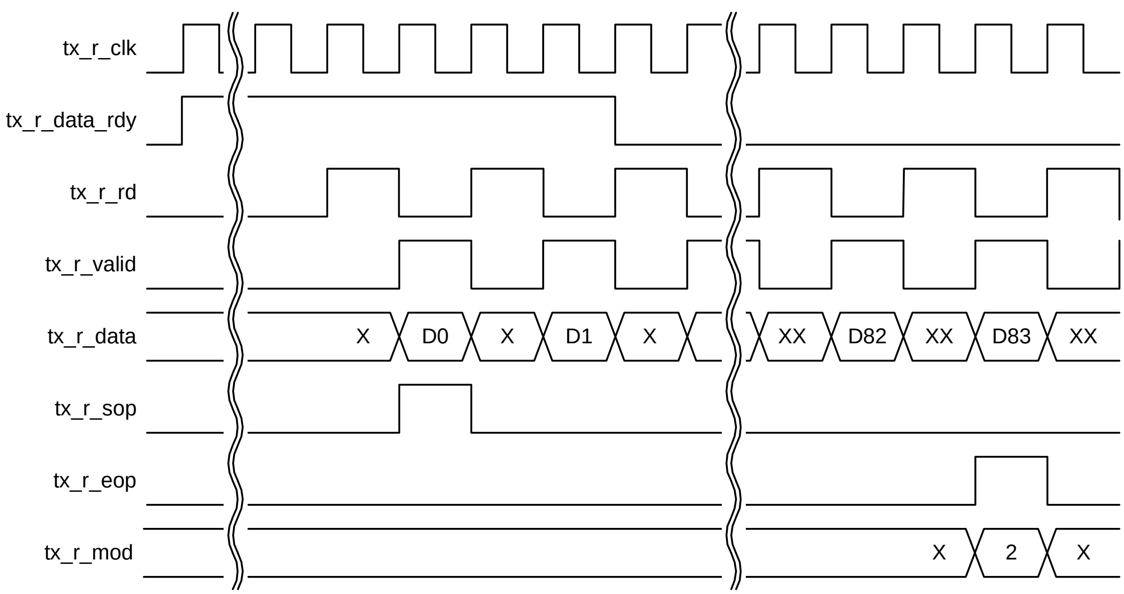

Figure 11: External FIFO Interface Timing (1)

Figure 12 is a similar diagram showing a three-queue system is shown below:

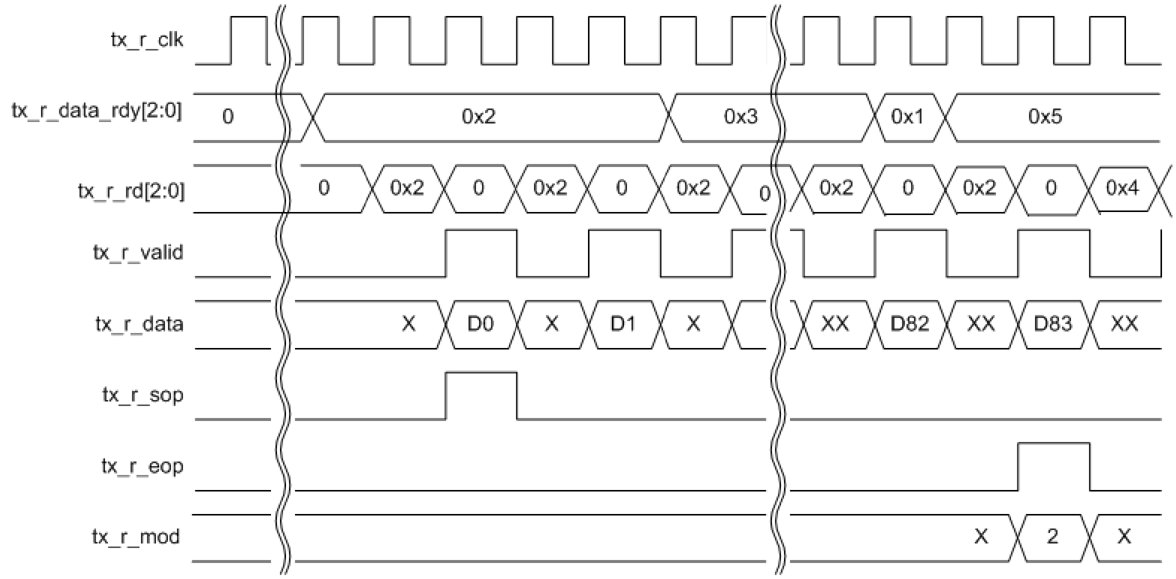

Figure 12: External FIFO Interface Timing (2)

Figure 13 shows detailed timing relationships for a frame on the external FIFO interface (MAC transmit) that incorporates a 2-byte frame with a SOP and EOP in the same transfer.

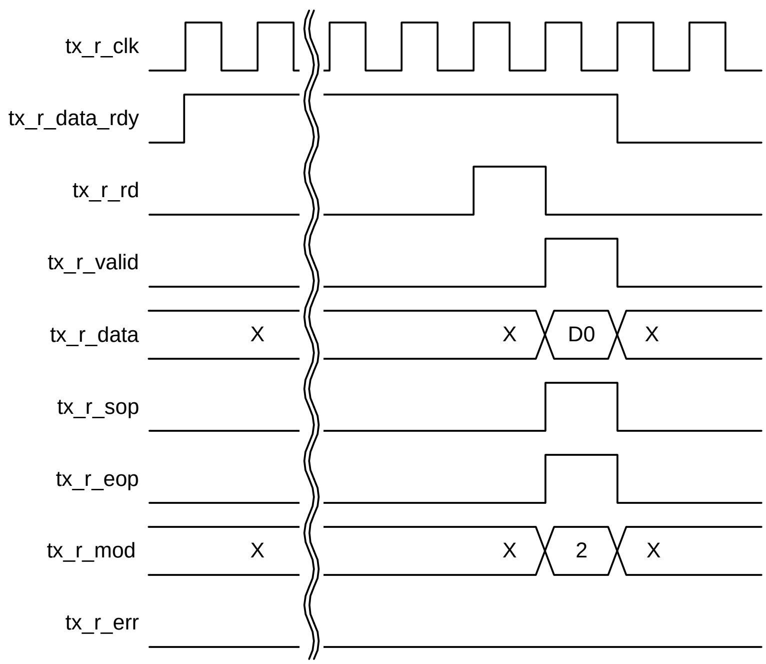

Figure 13: External FIFO Interface Timing (3)

Figure 14 shows detailed timing relationships for a frame on the external FIFO interface (MAC transmit) with frame error.

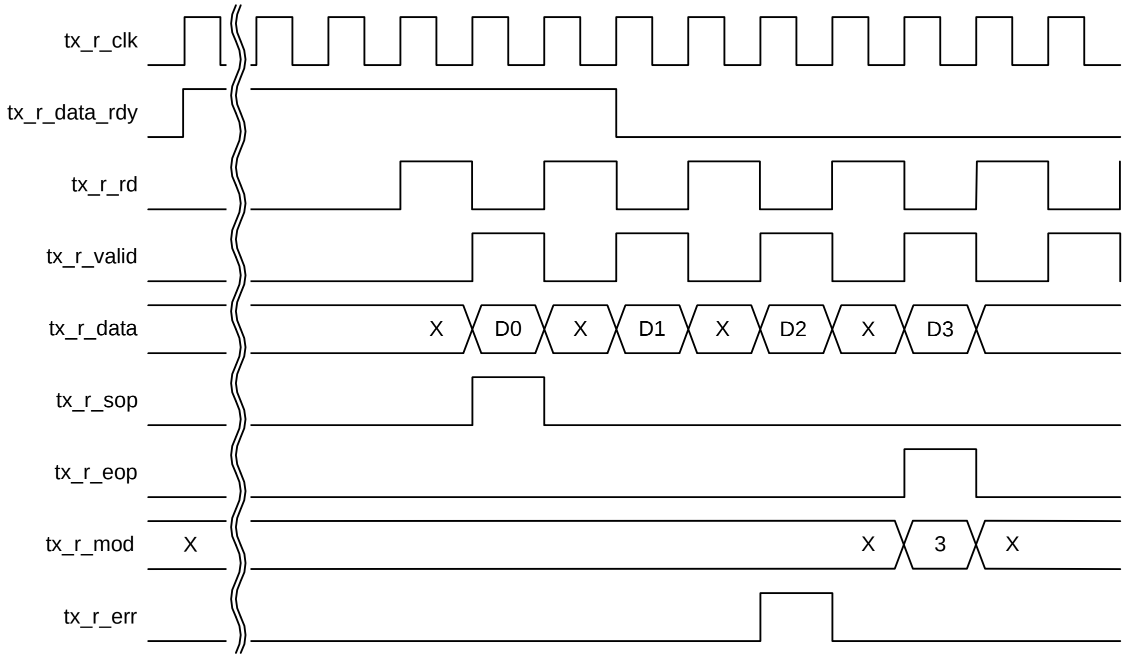

Figure 14: External FIFO Interface Timing (4)

Figure 15 shows a frame on the external FIFO interface (MAC receive).

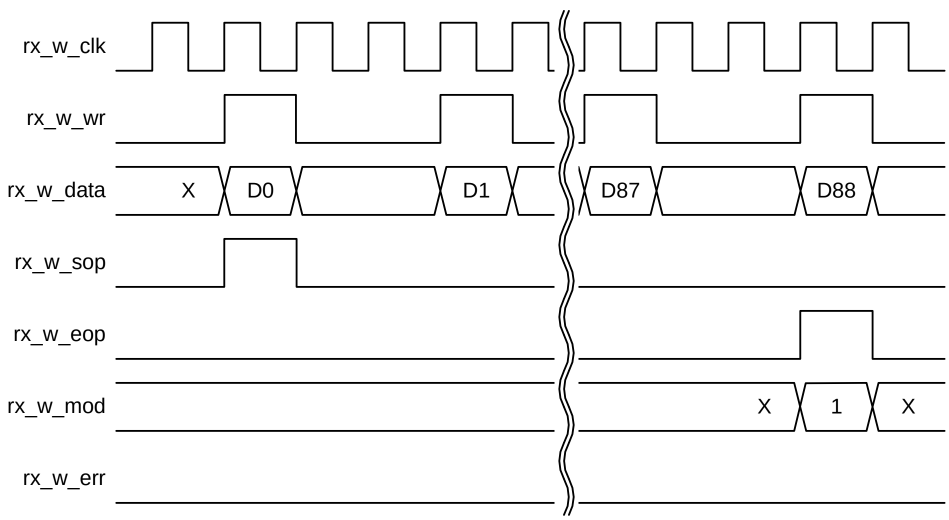

Figure 15: External FIFO Interface Timing (5)

Figure 16 shows a frame on the external FIFO interface (MAC receive) with frame error.

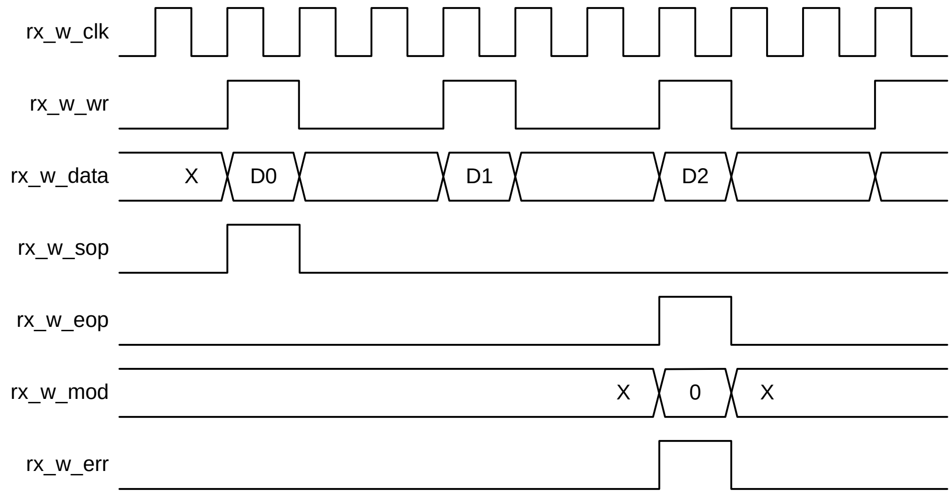

Figure 16: External FIFO Interface Timing (6)

6.1.4 Additional Low-Latency TX FIFO Interface for DMA Configurations

When the design is configured to include the SRAM-based packet buffer DMA, it is possible for you to additionally add an extra low latency FIFO TX host interface to inject high priority fixed latency traffic directly into the transmit path of the MAC bypassing the DMA altogether. This additional TX FIFO host interface is optionally included by defining `gem_tx_add_fifo_if verilog define. Packets that are injected into the MAC via the low latency FIFO interface have the same latency as the external fifo_interface mode, plus one tx_clk clock cycle, when no DMA frame is being sent or from the end of the current DMA frame ends, and is useful if particular packets needing a fixed transmit latency are required. DMA and low latency TX frames are dynamically arbitrated with low latency TX FIFO interface frames being highest priority. If a frame from the DMA is being transmitted to the MAC, it will complete before the Low Latency TX frame is sent but following DMA frames will be delayed if required.

As there is no RX FIFO interface in this mode, all received frames are passed to the RX DMA.

When using the additional low latency transmit fifo the following restrictions apply:

- Half duplex is not supported
- No register status updating for packet complete for fifo frames

Underflow errors that occur when packets are being fetched via the external FIFO interface are not reported in the status registers or interrupts. Underflow status will still get reported via the FIFO interface outputs.

6.1.5 Host Interface Soft Select Configuration External FIFO / DMA Interfaces

The design can be configured to have both external FIFO interface and the SRAM packet buffer DMA interfaces. This can optionally be built by defining `define gem_host_if_soft_select.

In this mode, either the external FIFO interface or the DMA interface can be selected. They cannot be used together. The interface selection is a static programming option that should be setup before enabling the RX and TX paths.

6.1.6 MAC Transmit Block

The MAC transmitter operates in either half duplex or full duplex, and transmits frames in accordance with the Ethernet IEEE 802.3 standard. In half-duplex mode, the CSMA/CD protocol of the IEEE 802.3 specification is followed.

A small input buffer receives data through the external FIFO interface (from either the DMA module or external to the IP Core) which, depending on the dma_bus_width control bits in the network configuration register, will extract data in either 32, 64, or 128-bit form. All subsequent processing prior to the final output is performed in bytes.

Transmit data can be output using the GMII/MII interface or through the TBI.

If the TBI is selected (gigabit and SGMII modes only), then the MAC transmitter passes 8-bit data to the PCS for further processing prior to output on the TBI. In 10 and 100Mbps SGMII mode, the MAC is clocked slower than the PCS which has the effect of the PCS sampling the same 8-bit data 10 or 100 times.

Frame assembly starts by adding preamble and the start frame delimiter. Data is taken from the transmit FIFO interface a word at a time. When GEM is configured for gigabit operation, the data output to the PHY uses all 8 bits of the txd[7:0] output. In 10/100M mode, transmit data to the PHY is nibble wide and least significant nibble first using txd[3:0] with txd[7:4] tied to logic 0.

If necessary, padding is added to take the frame length to 60 bytes. CRC is calculated using an order 32-bit polynomial. This is inverted and appended to the end of the frame taking the frame length to a minimum of 64 bytes. If the no CRC bit is set in the second word of the last buffer descriptor of a transmit frame, neither pad nor CRC are appended. The no CRC bit can also be set through the FIFO interface.

In full-duplex mode (at all data rates), frames are transmitted immediately. Back to back frames are transmitted at least 96 bit times apart to guarantee the interframe gap.

In half-duplex mode, the transmitter checks carrier sense. If asserted, the transmitter waits for the signal to become inactive, and then starts transmission after the interframe gap of 96 bit times. If the collision signal is asserted during transmission, the transmitter will transmit a jam sequence of 32 bits taken from the data register and then retry transmission after the back-off time has elapsed. If the collision occurs during either the preamble or start frame delimiter (SFD), then these fields will be completed prior to generation of the jam sequence.

The back-off time is based on an XOR of the 10 least significant bits of the data coming from the transmit FIFO interface and a 10-bit pseudo random number generator. The number of bits used depends on the number of collisions seen. After the first collision 1 bit is used, then the second 2 bits, and so on up to the maximum of 10 bits. All 10 bits are used above ten collisions. An error will be indicated and no further attempts will be made if 16 consecutive attempts cause collision. This operation is compliant with the description in Clause 4.2.3.2.5 of the IEEE 802.3 standard, which refers to the truncated binary exponential back-off algorithm.

In 10/100M mode, both collisions and late collisions are treated identically, and back off and retry will be performed up to 16 times. When operating in gigabit mode, late collisions are treated as an exception and transmission is aborted, without retry. This condition is reported in the transmit buffer descriptor word 1 (late collision, bit 26) and also in the transmit status register (late collision, bit 7). An interrupt can also be generated (if enabled) when this exception occurs, and bit 5 in the interrupt status register will be set.

In all modes of operation, if the transmit DMA underruns, a bad CRC is automatically appended using the same mechanism as jam insertion and the tx_er signal is asserted. For a properly configured system, this should never happen, and also it is impossible if configured to use the DMA with packet buffers as the complete frame is buffered in local packet buffer memory.

By setting when bit 28 is set in the network configuration register, the Inter Packet Gap (IPG) may be stretched beyond 96 bits depending on the length of the previously transmitted frame and the value written to the IPG_STRETCH register. The least significant 8 bits of the IPG_STRETCH register multiply the previous frame length (including preamble) the next significant 8 bits (+1 so as not to get a divide by zero) divide the frame length to generate the IPG. IPG stretch only works in full-duplex mode and when bit 28 is set in the network configuration register. The IPG_STRETCH register cannot be used to shrink the IPG below 96 bits.

If the back pressure bit is set in the network control register or if the half_duplex_flow_control_en input is set in 10/100M half-duplex mode, the transmit block transmits 64 bits of data, which can consist of 16 nibbles of 1011 or in bit rate mode 64 1s, whenever it sees an incoming frame to force a collision. This provides a way of implementing flow control in half-duplex mode. **Note**: This feature is not available in gigabit half-duplex mode.

6.1.7 Transmit Scheduling Algorithm

When multiple priority queues have been selected, the transmit scheduler is automatically included in the design and is responsible for selecting the next queue to be serviced either from the attached DMA or from the external FIFO interface when the DMA is not included. There are three scheduling algorithms available to the user and, with some exceptions detailed further below, you can select one of the four modes for each queue.

6.1.7.1 IEEE 802.1Qav Support – Credit Based Shaping

A credit-based shaping algorithm is available on the two highest priority active queues and is defined in IEEE 802.1Qav: Forwarding and Queuing Enhancements for Time-Sensitive Streams. Traffic shaping is enabled via the CBS (Credit Based Shaping) Control register (0x4bc) or the general transmit scheduling control register (offset 0x580). These two registers are aliased, so updating one register will automatically update the other. **Note**: It is permitted to enable CBS only on the second highest priority queue and not on the highest, in which case the highest priority queue would always take precedence. Also if the highest priority queues are disabled by writing 1 to the disable bit in the transmit queue pointer registers, then IEEE 802.1Qav (CBS) applies to the two highest enabled queues.

Enabling CBS on a queue will enable a counter which stores the amount of transmit ‘credit’, measured in bytes that a particular queue has. A queue may only transmit if it has non-negative credit. If a queue has data to send, but is held off from doing as another queue is transmitting, then credit will accumulate in the credit counter at the rate defined in the IdleSlope register for that queue. IdleSlope is the rate of change of credit when waiting to transmit and must be less than the value of the portTransmitRate. When this queue is transmitting the credit counter is decremented at the rate of sendSlope which is defined as the portTransmitRate – IdleSlope. A queue can accumulate negative credit when transmitting which will hold off any other transfers from that queue until credit returns to a non-negative value. No transfers are halted when a queue’s credit becomes negative, it will accumulate negative credit until the transfer completes.

To ensure that the CBS scheduling is completely accurate a single transmit buffer should be used per Ethernet frame (rather than multi-buffer transmit frames).

6.1.7.2 Fixed Priority

Any of the active queues can be selected as fixed priority and this is the default mode of operation for all queues. The queue index is used as the priority, where a higher index will have a higher priority than a lower index. The scheduler will always attempt to transmit from fixed priority queues with the highest priority. I.e. A fixed priority queue with a high queue index will always take precedence over a priority queue with a lower index.

6.1.7.3 Deficit Weighted Round Robin (DWRR)

Any of the active queues can be selected as DWRR. If DWRR is required, then at least two of the active queues should be selected as DWRR. It should not be used in conjunction with Enhanced Transmission Selection (ETS), as both algorithms operating together would make little practical sense. A DWRR enabled queue has lower priority than a CBS enabled queue or a fixed priority queue with a higher index.

The DWRR algorithm works by scanning all non-empty queues in sequence. Each queue is allocated a ‘deficit counter’ and an 8-bit weighting (or quantum) value. The value of the deficit counter is the maximum number of bytes that can be sent at the current time.

- If the deficit counter of the scanned queue is greater than the length of the packet waiting for transmission, then the packet will be transmitted and the value of the deficit counter is decremented by the packet size. If it is not greater, the scheduler will skip to the next DWRR enabled queue.
- If there is insufficient credit to transmit, the queue is simply skipped.
- If the queue is empty, the value of the deficit counter is reset to 0.
- If all queues have insufficient credit then each tx_clk cycle every queue’s deficit counter is incremented by its quantum value until a queue’s deficit counter obtains sufficient credit to transmit its first queued frame. The higher the quantum value chosen the quicker deficit counter will reach the required value.

- If all DWRR queues have the same weighting, then all queues will be granted the same overall bandwidth. The weighting value is stored in four programmable registers starting at offset 0x590. Since the design supports up to 16 priority queues, four physical registers are required. Refer to *Section 11 Registers* on page 146 for further details.

Note: If fixed priority queues are to be used in conjunction with DWRR, the fixed priority queues must be at a higher index value than the DWRR queues. A consequence of this is that the enabled DWRR queues shall form a contiguous set of queues starting from queue 0.

If CBS is also used in conjunction with DWRR, the DWRR queues will share the remaining bandwidth after the CBS allocation has been deducted.

Note: Transmit cut-thru should not be enabled if the transmit scheduler is used.

6.1.7.4 Enhanced Transmission Selection (ETS)

The ETS algorithm is defined in IEEE 802.1Qaz: Enhanced Transmission Selection for Bandwidth Sharing between Traffic and allows traffic on specific queues to be bandwidth-limited. Any of the active queues can be selected as ETS. If ETS is required, then at least two of the active queues should be selected as ETS. It should not be used in conjunction with DWRR as both algorithms operating together would make little practical sense. An ETS-enabled queue has lower priority than a CBS-enabled queue or a fixed priority queue with a higher index.

For each ETS-enabled queue, you should program the bandwidth requirement for each queue as a percentage of total bandwidth (an 8-bit register is used and the sum of values programmed should not exceed decimal 100). This will be the maximum bandwidth to be granted to that queue. The actual scheduling algorithm operates in a round-robin style from lowest indexed queues up to the highest indexed queue in sequence. The bandwidth allocation percentage is stored in four programmable registers starting at offset 0x590 – these are the same registers used for DWRR. Since the design supports up to 16 priority queues, four physical registers are required. Refer to the register descriptions for further details.

If CBS is also used in conjunction with ETS, the sum of the ETS queue percentages should equal the remaining bandwidth after the CBS allocation has been deducted.

Note: Transmit cut-thru should not be enabled if the transmit scheduler is used.

6.1.7.5 Enhancement for Scheduled Traffic (EnST)

IEEE 802.1Qbv is a TSN (Time Sensitive Networking) standard for “Enhancements for Scheduled Traffic" and specifies “time-aware queue-draining procedures” based on “timing derived from IEEE 802.1AS”. It adds transmission gates to the eight priority queues which allow low-priority queues to be shut down at specific times to allow higher-priority queues immediate access to the network at specific times.

EnST operation is explained in detail in *Section 6.1.16 IEEE 802.1Qbv: Enhancement for Scheduled Traffic (EnST)* on page 112.

6.1.8 MAC Receive Block

The control bits described in the following subclause are static and should not be updated without disabling the RX path by clearing bit[2], enable receive, of the network control register first.

All processing within the MAC receive block is implemented using 16-bit datapath. The MAC receive block checks for valid preamble, FCS, alignment and length, presents received frames to the external FIFO interface (to either the DMA module or external to the IP core) and stores the frames destination address for use by the address checking block.

When the TBI is selected, 16-bit words are passed directly from the PCS layer to the MAC receiver, except in 10 and 100Mpbs SGMII modes when 8 bits are read. In 10 and 100Mpbs SGMII modes, the MAC rx_clk runs more slowly than the PCS receive datapath clock and the same 8 bits of data is read 5 or 50 times.

If, during frame reception, the frame is found to be too long, a bad frame indication is sent to the external FIFO interface. The receiver logic ceases to send data to memory as soon as this condition occurs.

At end of frame reception, the receive block indicates to the DMA block whether the frame is good or bad. The DMA block will recover the current receive buffer if the frame was bad.

Ethernet frames are normally stored in DMA memory or to the external FIFO interface complete with the FCS. Setting the FCS remove bit in the network configuration (bit 17) causes frames to be stored without their corresponding FCS. The reported frame length field is reduced by four bytes to reflect this operation.

The receive block signals to the register block to increment the alignment, CRC (FCS), short frame, long frame, jabber or receive symbol errors when any of these exception conditions occur.

If bit 26 is set in the network configuration CRC errors is ignored and CRC errored frames are not discarded, the Frame Check Sequence Errors statistic register will still be incremented. Additionally, if configured to use the DMA and not enabled for jumbo frames mode, then bit[13] of the receiver descriptor word 1 will be updated to indicate the FCS validity for the particular frame. This is useful for applications such as EtherCAT whereby individual frames with FCS errors must be identified.

Received frames can be checked for length field error by setting the length field error frame discard bit of the network configuration register (bit-16). When this bit is set, the receiver compares a frame’s measured length with the length field (bytes 13 and 14) extracted from the frame. The frame is discarded if the measured length is shorter. This checking procedure is for received frames between 64 bytes and 1518 bytes in length.

Each discarded frame is counted in the 10 bit length field error statistics register. Frames where the length field is greater than or equal to 0x0600 hex will not be checked.

When operating in gigabit mode (half duplex), the receiver will discard frames which do not meet the minimal slot time of 512 bytes. If a burst is detected, the first frame is checked to ensure it meets the slot time, but all subsequent frames of the burst are checked to ensure they meet the minimum frame size of 64 bytes.

6.1.9 MAC Filtering Block

The filter block determines which frames should be written to the external FIFO interface and on to the optional DMA.

Whether a frame is passed depends on what is enabled in the network configuration register, the state of the external matching pins, the contents of the specific address, type and hash registers, and the frame’s destination address and type field.

If bit 25 of the network configuration register is not set, a frame will not be copied to memory if GEM is transmitting in half-duplex mode at the time a destination address is received.

Ethernet frames are transmitted a byte at a time, least significant bit first. The first six bytes (48 bits) of an Ethernet frame make up the destination address. The first bit of the destination address, which is the LSB of the first byte of the frame, is the group or individual bit. This is one for multicast addresses and zero for unicast. The *all ones* address is the broadcast address and a special case of multicast.

GEM supports recognition of specific source or destination addresses. The number of specific source or destination address filters is configurable and can range from zero to 36. Each specific address filter requires two registers:

- Specific address register bottom − Stores the first four bytes of the compared source or destination address.
- Specific address register top − Contains the last two bytes of this address, a control bit to select between source or destination address filtering and a 6-bit byte mask field to allow the user to mask bytes during the comparison.

The first filter (Filter 1) is slightly different to all other filters in that there is no byte mask. Instead address comparison against individual bits of specific address register 1 can be masked using the unique specific address mask register. The addresses stored in all filters can be specific (unicast), group (multicast), local or universal.

The destination or source address of received frames is compared against the data stored in the specific address registers once they have been activated. The addresses are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written. If a receive frame address matches an active address, the frame is written to the external FIFO interface and on to DMA memory if used.

Frames may be filtered using the type ID field for matching. Four type ID registers exist in the register address space and each can be enabled for matching by writing a one to the MSB (bit 31) of the respective register. When a frame is received, the matching is implemented as an OR function of the various types of match.

The contents of each type ID registers (when enabled) are compared against the length/type ID of the frame being received (e.g. bytes 13 and 14 in non-VLAN and non-SNAP encapsulated frames) and copied to memory if a match is found. The encoded type ID match bits (Word 0, Bit 22 and Bit 23) in the receive buffer descriptor status are set indicating which type ID register generated the match, if the receive checksum offload is disabled.

The reset state of the type ID registers is zero, hence each is initially disabled.

The following example in *Table 38* illustrates the use of the address and type ID match registers for a MAC address of 21:43:65:87:A9:CB:

Table 38: Address and Type ID Match Register (Example)

Field	Value Checked
Preamble	55
SFD	D5
DA (Octet 0 - LSB)	21
DA (Octet 1)	43
DA (Octet 2)	65
DA (Octet 3)	87
DA (Octet 4)	A9
DA (Octet 5 - MSB)	CB
SA (LSB)	00*
SA	00*
SA	00*
SA	00*
SA	00*
SA (MSB)	0
Type ID (MSB)	43
Type ID (LSB)	21

Note:* **-** contains the address of the transmitting device.

The sequence above shows the beginning of an Ethernet frame. Byte order of transmission is from top to bottom as shown. For a successful match to specific address 1, the following address matching registers must be set up:

- Specific address 1 bottom (Address 0x088)0x87654321
- Specific address 1 top (Address 0x08C)0x0000CBA9

And for a successful match to the type ID, the following type ID match 1 register must be set up:

- Type ID Match 1 (Address 0x0A8)0x80004321

6.1.9.1 Broadcast Address

Frames with the broadcast address of 0xFFFFFFFFFFFF are stored to memory only if the ‘no broadcast’ bit in the network configuration register is set to zero.

6.1.9.2 Hash Addressing

The hash address register is 64-bit long and takes up two locations in the memory map. The least significant bits are stored in hash register bottom and the most significant bits in hash register top.

The unicast hash enable and the multicast hash enable bits in the network configuration register enable the reception of hash matched frames. The destination address is reduced to a 6-bit index into the 6- bit hash register using the following hash function. The hash function is an XOR of every sixth bit of the destination address.

hash_index[05] = da[05] ^ da[11] ^ da[17] ^ da[23] ^ da[29] ^ da[35] ^ da[41] ^ da[47]

hash_index[04] = da[04] ^ da[10] ^ da[16] ^ da[22] ^ da[28] ^ da[34] ^ da[40] ^ da[46]
hash_index[03] = da[03] ^ da[09] ^ da[15] ^ da[21] ^ da[27] ^ da[33] ^ da[39] ^ da[45]
hash_index[02] = da[02] ^ da[08] ^ da[14] ^ da[20] ^ da[26] ^ da[32] ^ da[38] ^ da[44]
hash_index[01] = da[01] ^ da[07] ^ da[13] ^ da[19] ^ da[25] ^ da[31] ^ da[37] ^ da[43]
hash_index[00] = da[00] ^ da[06] ^ da[12] ^ da[18] ^ da[24] ^ da[30] ^ da[36] ^ da[42]

da[0] represents the least significant bit of the first byte received, that is, the multicast/unicast indicator, and da[47] represents the most significant bit of the last byte received.

If the hash index points to a bit that is set in the hash register, then the frame will be matched according to whether the frame is multicast or unicast.

A multicast match will be signaled if the multicast hash enable bit is set, da[0] is logic 1 and the hash index points to a bit set in the hash register.

A unicast match will be signaled if the unicast hash enable bit is set, da[0] is logic 0 and the hash index points to a bit set in the hash register.

To receive all multicast frames, the hash register should be set with all ones and the multicast hash enable bit should be set in the network configuration register.

6.1.9.3 External Address Matching

In applications where the internal address matching circuitry of GEM is not sufficient, an external address matching interface is provided. Matches can be performed on source address, destination address, type or VLAN ID. Each field is provided as an output (ext_sa, ext_da, ext_type and ext_vid output pins) with a validating strobe signal (ext_sa_stb, ext_da_stb, ext_type_stb and ext_vid_stb output pins). The strobes for each field are asserted high for one cycle at the point the field becomes valid within the frame reception.

IP source and destination addresses along with UDP/TCP port numbers are also available for external address matching. These fields are provided as the ext_ip_sa, ext_ip_da, ext_source_port and ext_dest_port outputs, and are validated by the ext_ip_sa_stb, ext_ip_da_stb, ext_sp_stb and ext_dp_stb outputs.

The external filter interface is enabled by setting bit 9 in the network configuration register.

External logic connected to the filter interface should return the ext_match1, ext_match2, ext_match3, or ext_match4 inputs when a match is found. Any one of the four match inputs can indicate a filter match as the inputs are logically ORed within the MAC.

The time at which the output strobes for the various fields are valid is fixed relative to the time of frame reception and depends on the type of frame being decoded. Destination address, source address, and VLAN ID are always fixed relative to the start of the frame. Type ID and the IP address, however, depend on whether the frame has a VLAN tag and/or a valid SNAP encapsulation.

The number of rx_clk cycles available to make the comparison is design specific, depending on the parameterizable depth of the MAC receive pipeline. The ‘gem_rx_pipeline_delay define in gem_gxl_defs.v defines the depth of the receive pipeline. A decision as to whether the frame should be stored to memory is made when the destination address reaches the end of the pipeline.

Timings from each strobe becoming valid until the internal match takes place are shown below for the default receive pipeline of ten.

These timings also demonstrate the worst case of TBI gigabit mode, where 16 bits of data are processed every rx_clk cycle. For demonstration (see *Table 39*), the frame received has a VLAN tag. The type ID field, therefore, presented to the external filter interface is delayed until after the VLAN ID is presented so that the real type ID field is captured.

Table 39: External Address Match Timing

Strobe Signal	No. of rx_clk for a Match
ext_da_stb (destination address)	7

Table 39: External Address Match Timing (Continued)

Strobe Signal	No. of rx_clk for a Match
ext_sa_stb (source address)	4
ext_type_stb (type field)	1
ext_vid_stb (VLAN ID)	2

The interface assumes the connecting signals are synchronous to rx_clk as shown *Figure 17*.

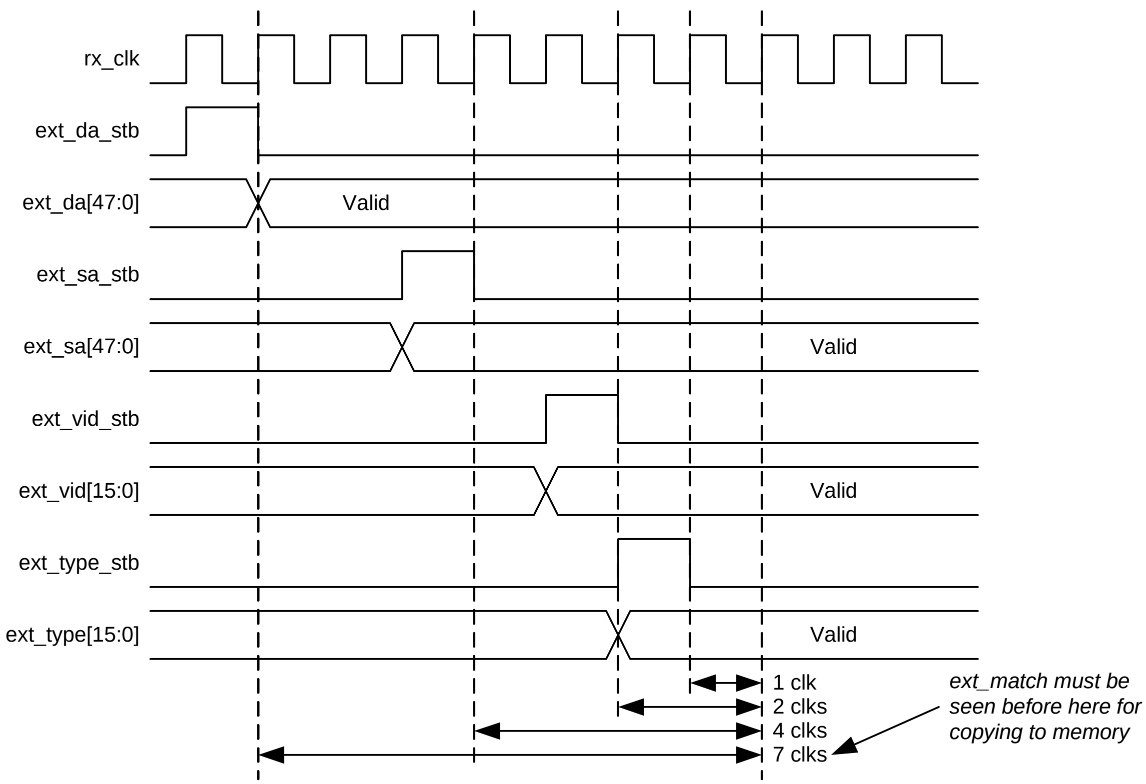

Figure 17: External Address Matching Timing

The depth of the default receive pipeline is insufficient for external matching of the extracted IP addresses. To cope with both VLAN and SNAP frames, the pipeline must be at least 25 deep when matching on the extracted IP destination address. SNAP and IP frames are only identified if the receive checksum offload is enable in the network configuration register.

6.1.9.4 Copy All Frames (or Promiscuous Mode)

If the copy all frames bit is set in the network configuration register, then all frames (except those that are too long, too short, have FCS errors or have rx_er asserted during reception) will be copied to memory. Frames with FCS errors will be copied if bit 26 is set in the network configuration register.

6.1.9.5 Disable Copy of Pause Frames

Pause frames can be prevented from being written to memory by setting the disable copying of pause frames control bit 23 in the network configuration register. When set, pause frames are not copied to memory regardless of the copy all frames bit, whether a hash match is found, a type ID match is identified, or a destination address match is found.

6.1.9.6 VLAN Support

Table 40 depicts an Ethernet encoded 802.1Q VLAN tag.

Table 40: VLAN Tag

TPID (Tag Protocol Identifier) 16 Bits	TCI (Tag Control Information) 16 Bits
0x8100	First 3 bits priority, then CFI bit, last 12 bits VID

The VLAN tag is inserted at the 13th byte of the frame, adding an extra four bytes to the frame. To support these extra four bytes, GEM can accept frame lengths up to 1536 bytes by setting bit 8 in the network configuration register.

If the VID (VLAN identifier) is null (0x000), this indicates a priority-tagged frame.

The following bits in the receive buffer descriptor status word give information about VLAN tagged frames:

- Bit 21 set if receive frame is VLAN tagged (i.e. type id of 0x8100).
- Bit 20 set if receive frame is priority tagged (i.e. type id of 0x8100 and null VID). (If bit 20 is set bit 21 will be set also.).
- Bit 19, 18 and 17 set to priority if bit 21 is set.
- Bit 16 set to CFI if bit 21 is set.

GEM can be configured to reject all frames except VLAN tagged frames by setting the discard non-VLAN frames bit in the network configuration register.

6.1.9.7 Wake-on-LAN Support

The receive block supports Wake on LAN by detecting the following events on incoming receive frames:

- Magic packet
- ARP request to the device IP address
- Specific address 1 filter match
- Multicast hash filter match

If one of these events occurs, Wake-on-LAN detection is indicated by asserting the wol output pin for 64 rx_clk cycles. These events can be individually enabled through bits[19:16] of the Wake on LAN register. Also, for Wake-on-LAN detection to occur receive enable must be set in the network control register, however a receive buffer does not have to be available. In addition to the wol output pin, an optional wol interrupt is generated for software purposes. If only the MAC receive clock is running at the time of the Wake-on-LAN event (and not the APB clock), the wol interrupt will not occur.

wol assertion due to ARP request, specific address 1 or multicast filter events will occur even if the frame is errored. For magic packet events, the frame must be correctly formed and error free.

A magic packet event is detected if all of the following are true:

- Magic packet events are enabled through bit 16 of the Wake-on-LAN register
- The frame's destination address matches specific address 1
- The frame is correctly formed with no errors
- The frame contains at least 6 bytes of 0xFF for synchronization
- There are 16 repetitions of the contents of specific address 1 register immediately following the synchronization

An ARP request event is detected if all of the following are true:

- ARP request events are enabled through bit 17 of the Wake-on-LAN register
- Broadcasts are allowed by bit 5 in the network configuration register
- The frame has a broadcast destination address (bytes 1 to 6)
- The frame has a typeID field of 0x0806 (bytes 13 and 14)
- The frame has an ARP operation field of 0x0001 (bytes 21 and 22)

- The least significant 16 bits of the frame's ARP target protocol address (bytes 41 and 42) match the value programmed in bits[15:0] of the Wake-on-LAN register

The decoding of the ARP fields adjusts automatically if a VLAN tag is detected within the frame. The reserved value of 0x0000 for the Wake-on-LAN target address value will not cause an ARP request event, even if matched by the frame.

A specific address 1 filter match event will occur if all of the following are true:

- Specific address 1 events are enabled through bit 18 of the Wake on LAN register
- The frame's destination address matches the value programmed in the specific address 1 registers

A multicast filter match event will occur if all of the following are true:

- Multicast hash events are enabled through bit 19 of the Wake on LAN register
- Multicast hash filtering is enabled through bit 6 of the network configuration register
- The frame destination address matches against the multicast hash filter
- The frame destination address is not a broadcast

6.1.9.8 Frame Screening

Frames matched by the frame filtering functionality can undergo further screening and be sent to individual receive queues. This screening functionality has the ability to inspect deep into a receive frame and from release 1p11 has the ability to discard frames in addition to directing them to a specified receive queue.

For configurations using the FIFO interface, rx_w_queue[3:0] indicates the priority queue selected by the screener registers.

For configurations using the DMA interface, there is an independent set of receive buffers for each queue. Every received packet will pass through a programmable screening algorithm which will allocate to that frame a particular queue to route it to. The user interface to the screeners is via two banks of programmable registers, screener type 1 match registers and screener type 2 match registers.

6.1.9.8.1 Screener Type 1 and Type 2 Match Registers

Screener type 1 match registers allow the user to route received frames based on particular IP and UDP fields extracted from the received frames. Specifically, these fields are DS (Differentiated Services field of IPv4 frames), TC (Traffic class field of IPv6 frames), and/or the UDP destination port. These fields are compared against the values stored in each of the screener type 1 match registers. If the result of this comparison is positive, then the received frame is routed to the priority queue specified in that screener type 1 match register. The number of type 1 screeners is determined by a define in the GEM defines file. Up to 16 screener type 1 registers are allocated address space in the default GEM address map, although this could be increased if necessary by redistributing other GEM registers. For further programming details, refer to the type 1 screening register descriptions in *Section 11 Registers*.

Screener type 2 match registers operate independently of screener type 1 match registers and offer additional match capabilities, extending the capabilities into vendor-specific protocols. The type 2 screening allows a screen to be configured that is the combination of all or any of the following comparisons:

1. An Enabled VLAN Priority. A VLAN Priority match will be performed if the VLAN priority enable is set. The extracted priority field in the VLAN header is compared against 3 bits within the screener type 2 register itself.
2. An Enabled EtherType. The ethertype field inside the screener type 2 register maps to one of 8 ethertype match registers. The extracted ethertype is compared against the ethertype register designated by this ethertype field. The number of ethertype registers is configurable (up to 8) via the GEM defines file.
3. An Enabled Field Compare A
4. An Enabled Field Compare B
5. An Enabled Field Compare C

Compare A, B, and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled, the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16-bit comparison or a 32-bit comparison is done. This selection is made via a control bit in the associated compare word1:

- If a 16-bit comparison is selected, then a 16-bit mask is also available to you for selecting which bits should be compared
- If the 32-bit compare option is selected, then no mask is available

The byte at the OFFSET number of bytes from the index start is compared through bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared through bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the ethertype field, the byte following the end of the IP header (IPv4 or IPv6) or the byte following the end of the TCP/UDP header. **Note**: The logic to decode the IP header or the TCP/UDP header is reused from the TCP/UDP/IP checksum offload logic and therefore has the same restrictions on use (the main limitation is that IP fragmentation is not supported). For further details, refer to *Section 6.1.10 Checksum Offload for IP, TCP, and UDP* on page 100. The Compare Register field points to a single pool of 32 compare Registers. Compare A, Compare B, and Compare C use a common set of compare registers. The number of compare registers is configurable (up to 32) via the in gem_gxl_defs.v file.

Note: Compare A, B and C together allow matching against an arbitrary 48 bits of data and so can be used to match against a MAC address.

All enabled comparisons are ANDed together to form the overall type 2 screener match.

For programming details, refer to the type 2 screening register descriptions in *Section 11 Registers*.

The number of type 2 screeners is determined by a define in the in gem_gxl_defs.v file. Up to 16 screener type 2 match registers are allocated address space in the default GEM address map, although this could be increased if necessary by redistributing other GEM registers.

The following summarizes the available configurations that the GEM core can support with regards to priority queuing:

- There can be between one and 16 priority queues configured
- When the number of priority queues is greater than 1, the user has the option of including up to 16 type 1 screeners, which allow the user to select the queue a receive packet will be routed to based on its UDP and/or TOS/TC fields
- When the number of priority queues is greater than 1, the user has the option of including up to 16 type 2 screeners, which allow the user to select the queue a receive packet will be routed to based on its VLAN, ethertype compare, or user-programmed compare fields
- When the number of type 2 screeners defined is greater than 0, the user has the option of including from zero up to 8 ethertype compare registers
- When the number of type 2 screeners defined is greater than 0, the user has the option of including from zero up to 32 user programmable field compare registers

Each screener register is programmable via the APB interface. Although this is not recommended, it is possible that more than one screener register will be programmed to match against a single frame. If this happens there are 2 cases to consider:

1. If a received frame matches against multiple screeners of the same type, then the frame will route to the queue mapped by the screener located at the lowest numeric APB address. E.g. if screener type 2 #0 and screener type 2 #1 both match, then the frame will route to the queue identified in bits [3:0] of the screener type 2 #0 register.
2. If a received frame matches against a type 2 screener and a type 1 screener, then the type 1 screener will take precedence

When a screener is matched, the received frame will be routed to a queue defined inside bits 3:0 of the screener register. Unmatched frames are routed to queue 0.

When the priority queuing feature is enabled, the number of interrupt outputs from the GEM core is increased to match the number of supported queues. The number of interrupt status registers are increased by the same number. Only DMA related events are reported using the individual interrupt outputs, as GEM can relate these events to specific queues. All other events generated within GEM are reported in the interrupt associated with the lowest priority queue. For the lowest priority queue (or the only queue when only 1 queue is selected), the interrupt status register is located at address 0x24. For all other queues, the interrupt status register is located at sequential addresses starting at address 0x400.

6.1.10 Checksum Offload for IP, TCP, and UDP

GEM can be programmed to perform IP, TCP, and UDP checksum offloading in both the receive and transmit direction which is enabled by setting:

- Bit 24 in the network configuration register for receive and
- Bit 11 in the DMA configuration register for transmit.

IPv4 packets contain a 16-bit checksum field, which is the 16-bit 1's complement of the 1's complement sum of all 16-bit words in the header. TCP and UDP packets contain a 16-bit checksum field, which is the 16-bit 1's complement of the 1's complement sum of all 16-bit words in the header, the data and a conceptual IP pseudo header.

To calculate these checksums in software requires each byte of the packet to be processed. For TCP and UDP, this can use a large amount of processing power. Offloading the checksum calculation to hardware can result in significant performance improvements.

For IP, TCP, or UDP checksum offload to be useful, the operating system containing the protocol stack must be aware that this offload is available so that it can make use of the fact that the hardware can either generate or verify the checksum.

6.1.10.1 Receiver Checksum Offload

When receive checksum offloading is enabled in GEM, the IPv4 header checksum is checked as per RFC 791, where the packet meets the following criteria:

- If present, the VLAN header must be four octets long and the CFI bit must not be set. (Also for receive one stacked VLAN is supported.)
- Encapsulation must be RFC 894 Ethernet Type Encoding or RFC 1042 SNAP Encoding or PPPoE Encoding
- IPv4 packet
- IP header is of a valid length
- IP options are supported

GEM also checks the TCP checksum as per RFC 793, or the UDP checksum as per RFC 768, if the following criteria are met:

- IPv4 or IPv6 packet
- IP options and all IPv6 extension headers (i.e. hop-by-hop, routing and destination) are supported (except for fragmentation headers)
- Good IP header checksum (if IPv4)
- IP fragmentation is not supported (If a packet is fragmented, then the checksum will not be checked)
- Reserved bit must not be set in the IPv4 header flags field
- TCP or UDP packet

When an IP, TCP, or UDP frame is received, the receive buffer descriptor gives an indication if GEM was able to verify the checksums. There is also an indication if the frame had SNAP encapsulation. These indication bits will replace the type ID match indication bits when the receive checksum offload is enabled. For details of these indication bits, please refer to the table *Table 26 Receive Buffer Descriptor Entry* on page 62.

If any of the checksums are verified incorrect by GEM, the packet is discarded and the appropriate statistics counter incremented.

6.1.10.2 Transmitter Checksum Offload

The transmitter checksum offload is only available if GEM is configured to use the DMA in packet buffer mode and full store and forward mode is enabled. This is because the complete frame to be transmitted must be read into the packet buffer memory before the checksum can be calculated and written back into the headers at the beginning of the frame.

Transmitter checksum offload is enabled by setting bit [11] in the DMA Configuration Register. When enabled, it will monitor the frame as it is written into the transmitter packet buffer memory to automatically detect the protocol of the frame. Protocol support is identical to the receiver checksum offload.

For transmit checksum generation and substitution to occur, the protocol of the frame must be recognized and the frame must be provided without the FCS field, by making sure that bit [16] of the transmit descriptor word 1 is clear. If the frame data already had the FCS field, this would be corrupted by the substitution of the new checksum fields. Stacked VLANs are supported as long as the VLAN type field of the stacked VLAN is set to 0x88A8. This differs from receive where the stacked VLAN type field is a programmable value. 0x88A8 is the ethertype for the S-TAG defined in the IEEE 802.1ad QinQ standard.

If these conditions are met, the transmit checksum offload engine will calculate the IP, TCP, and UDP checksums as appropriate. Once the full packet is completely written into packet buffer memory, the checksums will be valid and the relevant DPRAM locations will be updated for the new checksum fields as per standard IP/TCP and UDP packet structures.

If the transmitter checksum engine is prevented from generating the relevant checksums, bits [22:20] of the transmitter DMA writeback status will be updated to identify the reason for the error. **Note**: The frame will still be transmitted but without the checksum substitution, as typically the reason that the substitution did not occur was that the protocol was not recognized.

6.1.11 MAC IEEE 802.3 Pause Frame Support

Note: See Clause 31 and Annex 31A and 31B of the IEEE 802.3 standard for a full description of pause operation.

Table 41 shows the start of an 802.3 pause frame.

Table 41: IEEE 802.3 Pause Frame (Start)

Destination Address	Source Address	Type (MAC Control Frame)	Pause Opcode	Pause Time
0x0180C2000001	6 bytes	0x8808	0x0001	2 bytes

GEM supports both hardware controlled pause of the transmitter upon reception of a pause frame and hardware generated pause frame transmission.

6.1.11.1 IEEE 802.3 Pause Frame Reception

Bit 13 of the network configuration register is the pause enable control for reception. When this bit is set, transmission will pause if a non zero pause quantum frame is received.

If a valid pause frame is received, then the pause time register is updated with the new frame’s pause time regardless of whether a previous pause frame is active or not. An interrupt (either bit 12 or bit 13 of the interrupt status register) is triggered when a pause frame is received, but only if the interrupt has been enabled (bit 12 and bit 13 of the interrupt mask register). Pause frames received with non-zero quantum are indicated through the interrupt bit 12 of the interrupt status register. Pause frames received with zero quantum are indicated on bit 13 of the interrupt status register.

Once the pause time register is loaded and the frame currently being transmitted has been sent, no new frames are transmitted until the pause time reaches zero. The loading of a new pause time, and hence the pausing of transmission, only occurs when GEM is configured for full-duplex operation. If GEM is configured for half duplex, there will be no transmission pause, but the pause frame received interrupt will still be triggered. A valid pause frame is defined as having a destination address that matches either the address stored in specific address register 1 or if it matches the reserved address of 0x0180C2000001. It must also have the MAC control frame type ID of 0x8808 and have the pause opcode of 0x0001.

Pause frames that have FCS or other errors will be treated as invalid and will be discarded. IEEE 802.3 Pause frames that are received after Priority based flow control (PFC) has been negotiated will also be discarded. Valid pause frames received will increment the pause frames received statistic register.

The pause time register decrements every 512 bit times once transmission has stopped. For test purposes, the retry test bit (bit 12 in the network configuration register) can be set, which causes the pause time register to decrement every tx_clk cycle once transmission has stopped.

The interrupt (bit 13 in the interrupt status register) is asserted whenever the pause time register decrements to zero (assuming it has been enabled by bit 13 in the interrupt mask register). This interrupt is also set when a zero quantum pause frame is received.

6.1.11.2 IEEE 802.3 Pause Frame Transmission

Automatic transmission of pause frames is supported through the transmit pause frame bits of the network control register and from the external input pins tx_pause, tx_pause_zero and tx_pfc_sel. If either bit 11 or bit 12 of the network control register is written with logic 1, or if the input signal tx_pause is toggled when tx_pfc_sel is low, an 802.3 pause frame will be transmitted providing full duplex is selected in the network configuration register and the transmit block is enabled in the network control register.

Pause frame transmission will happen immediately if transmit is inactive or if transmit is active between the current frame and the next frame due to be transmitted.

Transmitted pause frames comprise of the following:

- A destination address of 01-80-C2-00-00-01
- A source address taken from specific address register 1
- A type ID of 88-08 (MAC control frame)
- A pause opcode of 00-01
- A pause quantum register
- Fill of 00 to take the frame to minimum frame length
- Valid FCS

The pause quantum used in the generated frame will depend on the trigger source for the frame as follows:

If bit 11 is written with a one, the pause quantum will be taken from the transmit pause quantum register. The transmit pause quantum register resets to a value of 0xFFFF giving maximum pause quantum as default.

If bit 12 is written with a one, the pause quantum will be zero.

If the tx_pause input is toggled, tx_pfc_sel is low and the tx_pause_zero input is held low until the next toggle, the pause quantum will be taken from the transmit pause quantum register.

If the tx_pause input is toggled, tx_pfc_sel is low and the tx_pause_zero input is held high until the next toggle, the pause quantum will be zero.

After transmission, a pause frame transmitted interrupt will be generated (bit 14 of the interrupt status register) and the only statistics register that will be incremented will be the pause frames transmitted register.

Pause frames can also be transmitted by the MAC using normal frame transmission methods.

6.1.12 MAC PFC Priority Based Pause Frame Support

Note: Refer to the IEEE 802.1Qbb standard for a full description of priority based pause operation.

GEM supports PFC Priority Based Pause transmission and reception. Before PFC pause frames can be received, bit 16 of the network control register must be set.

Table 42 shows the start of a PFC pause frame.

Table 42: PFC Pause Frame (Start)

Destination Address	Source Address	Type (MAC Control Frame)	Pause Opcode	Priority Enable Vector	Pause Time
0x0180C2000001	6 bytes	0x8808	0x0101	2 bytes	8 * 2 bytes

6.1.12.1 PFC Pause Frame Reception

The ability to receive and decode priority based pause frames is enabled by setting bit 16 of the network control register. When this bit is set, GEM will match either classic IEEE 802.3 pause frames or PFC priority based pause frames. Once a priority based pause frame has been received and matched, then from that moment on GEM will only match on priority based pause frames (this is an IEEE 802.1Qbb requirement, known as PFC negotiation). Once priority based pause has been negotiated, any received 802.3x format pause frames will not be acted upon. The state of PFC negotiation is identified via the output pfc_negotiate.

If a valid priority based pause frame is received then GEM will decode the frame and determine which, if any, of the 8 priorities require to be paused. Up to 8 pause time registers are then updated with the 8 pause times extracted from the frame regardless of whether a previous pause operation is active or not. An interrupt (bit 12 of the interrupt status register) is triggered when a PFC pause frame is received, but only if the interrupt has been enabled (bit 12 of the interrupt mask register). Pause frames received with non-zero and zero quanta are indicated through the interrupt bit 12 of the interrupt status register (bit 13 is never set for PFC pause frames). The output vector rx_pfc_paused is used to indicate when the pause time for a particular priority reaches zero. The state of the 8 pause time counters are indicated through the outputs rx_pfc_paused. These outputs will remain high for the duration of the pause time quanta. The loading of a new pause time only occurs when GEM is configured for full-duplex operation. If GEM is configured for half duplex, the pause time counters will not be loaded, but the pause frame received interrupt will still be triggered. A valid pause frame is defined as having a destination address that matches either the address stored in specific address register 1 or if it matches the reserved address of 0x0180C2000001. It must also have the MAC control frame type ID of 0x8808 and have the pause opcode of 0x0101.

Pause frames that have FCS or other errors will be treated as invalid and will be discarded. Valid pause frames received will increment the pause frames received statistic register.

The pause time registers decrement every 512 bit times immediately following the PFC frame reception. For test purposes, the retry test bit (bit 12 in the network configuration register) can be set, which causes the pause time register to decrement every rx_clk cycle once transmission has stopped.

6.1.12.2 PFC Pause Frame Transmission

Automatic transmission of pause frames is supported through the transmit priority based pause frame bit of the network control register and from the external input pins tx_pause, tx_pfc_pause[7:0], tx_pfc_pause_zero[7:0] and tx_pfc_sel. If bit 17 of the network control register is written with logic 1, or if the input signal tx_pause is toggled when tx_pfc_sel is high, a PFC pause frame will be transmitted providing full duplex is selected in the network configuration register and the transmit block is enabled in the network control register. When bit 17 of the network control register is set, the fields of the priority based pause frame will be built using the values stored in the transmit pfc pause register.

Pause frame transmission will happen immediately if transmit is inactive or if transmit is active between the current frame and the next frame due to be transmitted.

Transmitted pause frames comprise of the following:

- A destination address of 01-80-C2-00-00-01
- A source address taken from specific address register 1
- A type ID of 88-08 (MAC control frame)
- A pause opcode of 01-01
- A priority enable vector taken from tx_pfc_pause or the transmit pfc pause register

- 8 pause quantum registers
- Fill of 00 to take the frame to minimum frame length
- Valid FCS

The pause quantum registers used in the generated frame will depend on the trigger source for the frame as follows:

If bit 17 of the network control register is written with a one then the priority enable vector of the priority based pause frame will be set equal to the value stored in the transmit pfc pause quantum register [7:0]. For each entry equal to zero in the transmit pfc pause quantum register[15:8], the pause quantum field of the pause frame associated with that entry will be taken from the transmit pause quantum register. For each entry equal to one in the transmit PFC pause quantum register[15:8], the pause quantum associated with that entry will be zero.

The transmit pause quantum register resets to a value of 0xFFFF giving maximum pause quantum as default.

The pause quantum registers are classed as static and these registers should be updated only when no PFC frame is being transmitted.

A configuration option to use 8 different pause quantums, one for each pause priority, is available by defining gem_pfc_multi_quantum in the gem_defs_*.v file.

When this define is active, the default operation is to use bits[15:0] of the original transmit pfc pause quantum register for every pause priority.

To use the eight pause priority quantum register, set Network Control Register bit[24] = ‘1’. This will cause tx_pause_quantum/1/2/3 register to be used, (note that two priority quantums are stored per register, one in bits [31:16] and one in bits [15:0]).

If the tx_pause input is toggled and tx_pfc_sel is high, then the priority enable vector of the priority based pause frame will be set equal to the value in tx_pfc_pause [7:0]. For each entry equal to zero in tx_pfc_pause_zero[7:0], the pause quantum field of the pause frame associated with that entry will be taken from the transmit pause quantum register. For each entry equal to one in tx_pfc_pause_zero [7:0], the pause quantum associated with that entry will be zero.

After transmission, a pause frame transmitted interrupt will be generated (bit 14 of the interrupt status register) and the only statistics register that will be incremented will be the pause frames transmitted register.

PFC Pause frames can also be transmitted by the MAC using normal frame transmission methods.

6.1.13 IEEE 1588 and IEEE 802.1AS Support

IEEE 1588 is a standard for precision time synchronization in local area networks. It works with the exchange of special Precision Time Protocol (PTP) frames. The PTP messages can be transported over IEEE 802.3/Ethernet, over Internet Protocol Version 4 or over Internet Protocol Version 6 as described in the annex of IEEE Std 1588-2008.

Most IEEE 1588 functionality can be implemented in software but for greatest accuracy, hardware assist is required to detect when PTP event messages pass the GMII interface (clock timestamp point).

GEM detects when the PTP event messages: sync, delay_req, pdelay_req and pdelay_resp are transmitted and received.

GEM output pins indicate the message timestamp point (asserted on the start packet delimiter and deasserted at end of frame) for all frames and the passage of PTP event frames (asserted when a PTP event frame is detected and deasserted at end of frame).

Synchronization between master and slave clocks is a two stage process:

- First, the offset between the master and slave clocks is corrected by the master sending a sync frame to the slave with a follow-up frame containing the exact time the sync frame was sent. Hardware assist modules at the master and slave side detect exactly when the sync frame was sent by the master and received by the slave. The slave then corrects its clock to match the master clock.

- Second, the transmission delay between the master and slave is corrected. The slave sends a delay request frame to the master which sends a delay response frame in reply. Hardware assist modules at the master and slave side detect exactly when the delay request frame was sent by the slave and received by the master. The slave will now have enough information to adjust its clock to account for delay. For example, if the slave was assuming zero delay the actual delay will be half the difference between the transmit and receive time of the delay request frame (assuming equal transmit and receive times) because the slave clock will be lagging the master clock by the delay time already.

For hardware assist, it is necessary to time-stamp when sync and delay_req messages are sent and received. The timestamp is taken when the message timestamp point passes the clock timestamp point. For Ethernet, the message time-stamp point is the SFD and the clock timestamp point is the MII interface. (The 1588 spec refers to sync and delay_req messages as event messages as these require time-stamping. Follow up, delay response and management messages do not require time-stamping and are referred to as general messages.)

IEEE 1588 version 2 defines two additional PTP event messages. These are the peer delay request (Pdelay_Req) and peer delay response (Pdelay_Resp) messages. These messages are used to calculate the delay on a link. Nodes at both ends of a link send both types of frames (regardless of whether they contain a master or slave clock). The Pdelay_Resp message contains the time at which a Pdelay_Req was received and is itself an event message. The time at which a Pdelay_Resp message is received is returned in a Pdelay_Resp_Follow_Up message.

IEEE 1588 version 2 introduces two kinds of transparent clocks: peer-to-peer (P2P) and end-to-end (E2E). Transparent clocks measure the transit time of event messages through a bridge and amend a correction field within the message to allow for the transit time. P2P transparent clocks additionally correct for the delay in the receive path of the link using the information gathered from the peer delay frames. With P2P transparent clocks delay_req messages are not used to measure link delay. This simplifies the protocol and makes larger systems more stable.

The sof_tx and sof_rx signals are provided to indicate the message timestamp point and follow up signals are provided to indicate the presence of an event frame. With IEEE 1588 version 1, for a given data-rate, the assertion of the event frame signals will be a fixed delay after the SOF signals so taking the timestamp could be delayed until the event signals are asserted and suitable compensation made.

GEM recognizes ten different encapsulations for PTP event messages:
1. IEEE 1588 version 1 (UDP/IPv4 multicast)
2. IEEE 1588 version 1 (UDP/IPv4 multicast with VLAN)
3. IEEE 1588 version 2 (UDP/IPv4 multicast)
4. IEEE 1588 version 2 (UDP/IPv4 multicast with VLAN)
5. IEEE 1588 version 2 (UDP/IPv4 unicast)
6. IEEE 1588 version 2 (UDP/IPv4 unicast with VLAN)
7. IEEE 1588 version 2 (UDP/IPv6 multicast)
8. IEEE 1588 version 2 (UDP/IPv6 multicast with VLAN)
9. IEEE 1588 version 2 (Ethernet multicast)
10. IEEE 1588 version 2 (Ethernet multicast with VLAN)

Unicast PTP frame recognition is enabled via bit 20 of the network control register. The unicast addresses themselves are user programmable via the unicast PTP address register for which there are two (0x0D4 and 0x0D8). The first holds the RX unicast IP destination address and the other the TX unicast IP destination address. The unicast PTP address register should only be changed when the unicast PTP frame recognition is disabled.

Table 43 and *Table 44* show examples of these frames for IEEE 1588 version 1 format.

Table 43: Example of a Sync Frame − IEEE 1588 version 1

Field	Value Checked
Preamble/SFD	55555555555555D5
DA (Octets 0 - 5)	
SA (Octets 6 - 11)	
Type (Octets 12-13)	0800
IP header (Octets 14-22)	
IP header - UDP identifier (Octet 23)	11
IP header (Octets 24-29)	

Table 43: Example of a Sync Frame − IEEE 1588 version 1 (Continued)

Field	Value Checked
IP header - DA (Octets 30-32)	E00001
IP header - DA (Octet 33)	81 or 82 or 83 or 84
UDP header - source port (Octets 34-35)	
UDP header - dest port (Octets 36-37)	013F
UDP header – length and checksum (Octets 38-41)	
PTP header (Octet 42)	
PTP header - version PTP (Octet 43)	01
PTP header (Octets 44-73)	
PTP header - control (Octet 74)	00
PTP header (Octets 75-168)	

Table 44: Example of a Delay Request Frame − IEEE 1588 version 1

Field	Value Checked
Preamble/SFD	55555555555555D5
DA (Octets 0 - 5)	
SA (Octets 6 - 11)	
Type (Octets 12-13)	0800
IP header (Octets 14-22)	
IP header - UDP identifier (Octet 23)	11
IP header (Octets 24-29)	
IP header - DA (Octets 30-32)	E00001
IP header - DA (Octet 33)	81 or 82 or 83 or 84
UDP header - source port (Octets 34-35)	
UDP header - dest port (Octets 36-37)	013F
UDP header – length and checksum (Octets 38-41)	
PTP header (Octet 42)	
PTP header - version PTP (Octet 43)	01
PTP header (Octets 44-73)	
PTP header - control (Octet 74)	01
PTP header (Octets 75-168)	

For 1588 version 1 messages, sync and delay request frames are indicated by GEM if:

- The frames type field indicates TCP/IP, UDP protocol is indicated
- The destination IP address is 224.0.1.129/130/131 or 132
- The destination UDP port is 319 and the control field is correct

The control field is 0x00 for sync frames and 0x01 for delay request frames.

For 1588 version 2 messages, the type of frame is determined by looking at the message type field in the first byte of the PTP frame. Whether a frame is version 1 or version 2 can be determined by looking at the version PTP field in the second byte of both version 1 and version 2 PTP frames.

In version 2 messages, sync frames have a message type value of 0x0, delay_req have 0x1, pdelay_req have 0x2 and pdelay_resp have 0x3.

Table 45 to *Table 48* show examples of these frames for IEEE 1588 version 2 format.

Table 45: Example of a Sync Frame − IEEE 1588 version 2 (UDP/IPv4)

Field	Value Checked
Preamble/SFD	55555555555555D5
DA (Octets 0 - 5)	

Table 45: Example of a Sync Frame − IEEE 1588 version 2 (UDP/IPv4) (Continued)

Field	Value Checked
SA (Octets 6 - 11)	
Type (Octets 12-13)	0800
IP header (Octets 14-22)	
IP header - UDP identifier (Octet 23)	11
IP header (Octets 24-29)	
IP header - DA (Octets 30-33)	E0000181
UDP header - source port (Octets 34-35)	
UDP header - dest port (Octets 36-37)	013F
UDP header – length and checksum (Octets 38-41)	
PTP header – messagetype (Octet 42[3:0])	0
PTP header – transport specific (Octet 42[7:4])	not checked
PTP header - version PTP (Octet 43)	02

Table 46: Example of a pdelay_req − IEEE 1588 version 2 (UDP/IPv4)

Field	Value Checked
Preamble/SFD	55555555555555D5
DA (Octets 0 - 5)	
SA (Octets 6 - 11)	
Type (Octets 12-13)	0800
IP header (Octets 14-22)	
IP header - UDP identifier (Octet 23)	11
IP header (Octets 24-29)	
IP header - DA (Octets 30-33)	E000006B
UDP header - source port (Octets 34-35)	
UDP header - dest port (Octets 36-37)	013F
UDP header – other stuff (Octets 38-41)	
PTP header – messagetype (Octet 42[3:0])	2
PTP header – transport specific (Octet 42[7:4])	not checked
PTP header - version PTP (Octet 43)	02

Table 47: Example of a Sync Frame − IEEE 1588 version 2 (UDP/IPv6)

Field	Value Checked
Preamble/SFD	55555555555555D5
DA (Octets 0 - 5)	
SA (Octets 6 - 11)	
Type (Octets 12-13)	86dd
IP header (Octets 14-19)	
IP header - UDP identifier (Octet 20)	11
IP header (Octets 21-37)	
IP header - DA (Octets 38-53)	FF0X000000000181
UDP header - source port (Octets 54-55)	
UDP header - dest port (Octets 56-57)	013F
UDP header – (Octets 58-61)	
PTP header – messagetype (Octet 62[3:0])	0
PTP header – transport specific (Octet 62[7:4])	not checked
PTP header - version PTP (Octet 63)	02

Table 48: Example of a pdelay_req − IEEE 1588 version 2 (UDP/IPv6)

Field	Value Checked
Preamble/SFD	55555555555555D5
DA (Octets 0 - 5)	
SA (Octets 6 - 11)	
Type (Octets 12-13)	86dd
IP header (Octets 14-19)	
IP header - UDP identifier (Octet 20)	11
IP header (Octets 21-37)	
IP header - DA (Octets 38-53)	FF0200000000006B
UDP header – source port (Octets 54-55)	
UDP header – dest port (Octets 56-57)	013F
UDP header – (Octets 58-61)	
PTP header – messagetype (Octet 62[3:0])	0
PTP header – transport specific (Octet 62[7:4])	not checked
PTP header - version PTP (Octet 63)	02

Table 49 is an example of a sync frame in the 1588 version 2 (Ethernet multicast) format. For the multicast address 011B19000000 sync and delay request frames are recognized depending on the messagetype field, 00 for sync and 01 for delay request.

Table 49: Example of a Sync Frame − IEEE 1588 version 2 (Ethernet Multicast)

Field	Value Checked
Preamble/SFD	55555555555555D5
DA (Octets 0 - 5)	011B19000000
SA (Octets 6 - 11)	
Type (Octets 12-13)	88F7
messagetype (Octet 14[3:0])	0
transport specific (Octet 14[7:4])	not checked
version PTP (Octet 15)	02

Table 50 is an example of a pdelay_req frame in the 1588 version 2 (Ethernet multicast) format which uses the “nearest bridge group address” of 0180C200000E as defined in 802.1Q. 0180C200000E is a special multicast address that is not forwarded by bridges. Also IEEE 802.1AS requires the use of this address with Ethernet encapsulation without VLAN tagging (see 11.3.3 and 11.3.4 in the IEEE 802.1AS spec), so sync frames are recognized with this address in addition to pdelay request and pdelay response frames depending on the messagetype field, 00 for sync, 02 for pdelay request and 03 for pdelay response. However sync frames will not be indicated with this address if they are VLAN tagged.

Table 50: Example of a pdelay_req frame − IEEE 1588 version 2 (Ethernet Multicast)

Field	Value Checked
Preamble/SFD	55555555555555D5
DA (Octets 0 - 5)	0180C200000E
SA (Octets 6 - 11)	
Type (Octets 12-13)	88F7
messagetype (Octet 14[3:0])	0
transport specific (Octet 14[7:4])	not checked
version PTP (Octet 15)	02

GEM contains an optional timestamp unit (TSU) selectable with a tick define. The TSU consists of a timer and registers to capture the time at which PTP event frames cross the message time-stamp point. These are accessible through the GEM APB interface. An interrupt is issued when a capture register is updated.

6.1.13.1 Support for Timestamping and Timestamp Accuracy

The MAC has the responsibility of sampling the TSU timer value when the TX or RX SOF event of the frame passes the MII/GMII boundary. This event is an existing signal synchronous to MAC TX/RX clock domains. The MAC uses the sampled timestamp to insert the timestamp into transmitted PTP sync frames (if one step sync feature is enabled), or to pass to the register block to capture the timestamp in APB accessible registers, or to pass to the DMA to insert into TX or RX descriptors. For each of these, the SOF event, which is captured in the tx_clk and rx_clk domains respectively, is synchronized to the tsu_clk domain and the resulting signal is used to sample the TSU count value. This value will be kept stable for an entire frame, or specifically at least 64 TX/RX clock cycles, since the minimum frame size in Ethernet is 64bytes and worst case is a transfer rate of 1byte per cycle. It is used as the source for all the various components within GEM that require the timestamp value.

Since the SOF event had to pass a clock boundary, there is a degree of inaccuracy in the captured timestamp. The level of inaccuracy depends on the frequency of tsu_clk. There will be no more than 1 cycle of inaccuracy as in *Figure 18*.

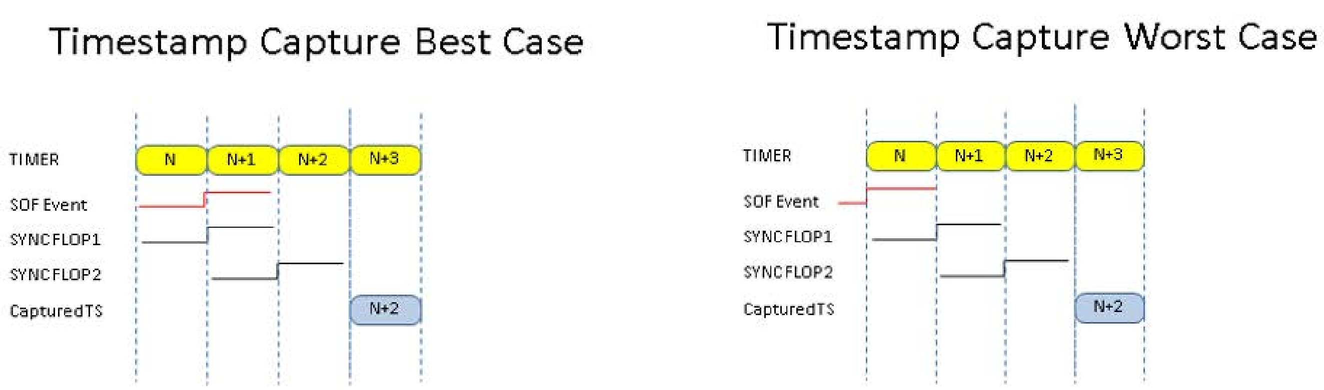

Figure 18: Timestamp Capture

In the best case, the SOF event (which is in the rx/tx clk domain) just meets the setup time of the tsu_clk domain at the input to the first sync flop. The captured TS is N+2, but it really should be N+1. In the worst case, the captured TS is also N+2, but it really should be N. There is 1 cycle of uncertainty.

6.1.14 Single-Step Timestamping

Support of one step clock for TX Sync frames can be enabled by setting a bit in the network control register. In this mode the timestamp field, within the 1588 (ver 2) sync frame, is replaced by the TSU timestamp value at the time the Sync frame sof passes the MII interface. To use single step timestamping, the sampled timestamp must be stable before the point at which GEM requires to insert the timestamp. This can be guaranteed by enforcing a rule that TSU clock (tsu_clk) is greater than 1/8th the frequency of tx_clk or rx_clk. The UDP checksum offload functionality cannot be used with single step timestamping.

6.1.14.1 Single-Step Update of Correction Field in PTP Sync Frames

Version 2 PTP frames have a common message header of 34 octets defined in *Table 18 of the IEEE 1588-2008 spec*. Sync frames contain the origin Timestamp immediately after the header. The correction field begins at the ninth octet in the header.

The correction field is defined in 13.3.2.7 of IEEE 1588-2008. The correction field is measured in nanoseconds and is multiplied by 2^16. For example, 2.5ns is represented by the 64-bit value 0x0000000000028000 and is transmitted most significant byte first. A value of one in all bits, except the most significant, indicates that the correction is too big to be represented.

From release 1p09 onwards, if bit 27 “oss_correction_field” is set in the network_control register then the correction field will be updated.

Section 11.5 of the IEEE 1588 spec describes updating of the correction field. *Sections 13.3 and 13.6* describe the fields of the sync frame. At offset 8 in the PTP sync frame there is an 8 byte correction field. The requirement is to add the sync frame residence time to this field in a single step process.

The residence time is the difference between the ingress and egress timestamps of the sync frame. The ingress time stamp will be available to firmware as is the correction field which can be extracted from the received sync frame.

The TSU value is stored as nanoseconds and seconds, while the correction field is stored as nanoseconds (48 bits for ns and 16 bits for subns). The subns bits of the correction field can be used by firmware to signal to the gem_tx block what action it should take as timestamps cannot be taken with sub-nanosecond precision. Firmware can also update the correction field to take account of the ingress timestamp.

For single step updating of the correction field to work firmware has to do the following:
1. Clear the 16 least significant bits of the correction field except any bits that need to be set as below
2. If the correction field is at its maximum value, set bit[15] of the correction field to 1 and take no further action
3. Set bit[8] of the correction field to the value of the LSB of the captured ingress seconds TSU value. The captured value refers to the time the sync frame entered the SoC.
4. Subtract the nanosecond value of the ingress time-stamp from the correction field (ignoring the subns bits of the correction field). If the resulting value is negative, set bit[14] of the correction field.
5. Write the absolute result (i.e. positive integer nanosecond number) to the correction field

Because firmware will have initialized the correction field by subtracting the nanosecond value of the ingress timestamp from it, adding the nanosecond egress time to the correction field results in it being incremented by the residence time of the sync frame in device.

GEM adds the egress nanosecond value to the correction field and clears the least significant 16 bits. If the egress least significant bit of the seconds value is different from the value stored in bit [8] then an extra billion nanoseconds are added to the correction field.

If there is an overflow or bit[15] is set, then the correction field is set to the maximum value of 0x7FFFFFFFFFFFFFFF.

The least significant bits of the correction field are subnanoseconds and GEM cannot take timestamps with this precision, so it is allowed to reuse the subnanosecond field of the correction field to carry seconds and other data. It is also assumed that the residence time of the sync frame will be less than a second, so only a single second bit needs to be carried.

For reference for IEEE 802.1AS, the origin Timestamp field is reserved and the correction field is zero in sync frames because IEEE 802.1AS does not support one-step (single step) time stamping (see 7.5 f, 10.5.2.2.7 and 11.4.3 in the IEEE 802.1AS spec).

6.1.14.2 Timestamp Capture in APB registers

There are eight 80-bit status registers that capture the time at which PTP event frames are transmitted and received. An interrupt is issued when these registers are updated.

6.1.14.3 Timestamp Capture in DMA descriptors

The TX and/or RX Timestamp can optionally be captured in an extended buffer descriptor when configured using bits in the DMA configuration register. The timestamp can be captured for a number of frame types (ptp event or ptp general, or all frames, or none as defined in tx/rx_bd_control registers) and a bit within BD word 0/1 is used to indicate that the timestamp is present.

6.1.15 Controlling the GEM TSU

The timer is implemented as a 102-bit register with the bits designated as follows:

- The upper 48 bits counting seconds
- The next 30 bits counting nanoseconds
- The lowest 24 bits counting sub-nanoseconds
- The lower 54 bits roll over when they have counted to one second

An interrupt is generated when the seconds increment. The timer value can be read, written, and adjusted through the APB interface.

The initial value of the timer is written through the tsu_timer_msb_sec, tsu_timer_sec, and tsu_timer_nsec registers. The timer can be adjusted by adding or subtracting an integral number of nanoseconds in a one-off write to the tsu_timer_adjust register. The amount the timer increments by each clock cycle is set by the tsu_timer_incr and tsu_timer_incr_sub_nsec registers.

The timer is clocked by either pclk or by tsu_clk if gem_tsu_clk is defined the gem_gxl_defs.v file.

There are three modes of operation to control the way the timer varies over time. These are:

1. Increment mode: Increment timer by a fixed value every input clock (tsu_clk or pclk)
2. Alternative increment mode: Increment timer by a fixed value for a fixed number of input clocks, followed by an alternative increment value for a single clock. This is a legacy mode of operation and it is not recommended to use the alternative increment mode of operation.
3. Timer adjust mode: Adjust the timer by adding or subtracting 1ns from the timer. This mode is controlled by gem_tsu_inc_ctrl and gem_tsu_ms inputs.

These modes are described in *Table 51*.

Table 51: Timer Modes of Operations

Mode	Description
Increment Mode	The amount by which the timer increments each clock cycle is controlled by the timer increment register. Bits 7:0 are the default increment value in nanoseconds and an additional 24 bits of sub-nsec resolution are available using the timer increment sub_nsec register. If the rest of the timer increment register is written with zero the timer increments by the value in 7:0 plus sub-nsec register, each clock cycle. The sub-nsec register allows a resolution of approximately 59.6 attoseconds (5.96E-17). The sub-nsec register is split into two fields with the MSBs at 15:0 and the LSBs at 31:24. This is because when the number of significant bits was increased from 16 to 24, the location of the MSBs was left unchanged and the LSBs were added to the top of the register. To calculate the value to write to the sub-ns register, you take the decimal value of the sub-nanosecond value, then multiply it by 2 to the power of 24 (16777216) and convert the result to hexadecimal. For example, for a sub-nanosecond value of 0.33333333 you would write 0x55005555.
Alternative Increment Mode	It is NOT recommended to use this legacy mode of operation. Bits 15:8 of the increment register are the alternative increment value in nanoseconds and bits 23:16 are the number of increments after which the alternative increment value is used. If 23:16 are zero then the alternative increment value will never be used. Taking the example of 10.2MHz, you have 102 cycles every ten microseconds or 51 every five microseconds. So a timer with a 10.2MHz clock source is constructed by incrementing by 98ns for fifty cycles and then incrementing by 100ns (98*50+100=5000). This is programmed by setting the IEEE 1588 timer increment register to 0x00326462. For a 49.8MHz clock source, it would be 20ns for 248 cycles followed by an increment of 40ns (20*248+40=5000) programmed as 0x00F82814. Having eight bits for the ‘number of increments’ field allows frequencies up to 50MHz to be supported with 200kHz resolution.

Table 51: Timer Modes of Operations (Continued)

Mode	Description
Timer Adjust Mode	An alternative way of controlling the way the timer increment register is to use the gem_tsu_inc_ctrl inputs. When gem_tsu_inc_ctrl [1:0] = • 2b’11 − timer register increments as normal. • 2b’01 − timer register increments by an additional nanosecond. • 2b’10 − timer increments by a nanosecond less. • 2b’00 − • When gem_tsu_ms = 1b’1, the “nanoseconds” timer register is cleared and the “seconds” timer register is incremented with each clock cycle. • When gem_tsu_ms = 1b’0, the timer register increments as normal but the timer value is copied to the sync strobe register. The timer sync strobe register is loaded with the value of the timer when the input signals gem_tsu_ms xgm_tsu_ms and gem_tsu_inc_ctrl are set to zero.

The TSU timer count value bits are passed out of the core so that it can optionally be used directly by external hardware.

The TSU timer count value can be compared to a programmable comparison value. For the comparison, the 48 bits of the seconds value and the upper 22 bits of the nanoseconds value are used. A signal is output from the core to indicate when the TSU timer count value is equal to the comparison value stored in the TSU timer comparison value registers (0x0DC, 0x0E0 and 0x0E4). An interrupt can also be generated (if enabled) when the TSU timer count value and comparison value are equal, mapped to bit 29 of the interrupt status register.

IEEE 802.1AS is mostly a subset of IEEE 1588. There is one difference in that IEEE 802.1AS uses the Ethernet multicast address 0180C200000E for sync frame recognition whereas 1588 does not. GEM is designed to recognize sync frames with both 802.1AS and 1588 addresses and so can support both 1588 and 802.1AS frame recognition simultaneously.

An optional external timestamp port can be used instead of the internal timer. The external port is 94 bits wide, conforming to the 94 most significant bits of the internal TSU count as defined above.

The external port must be synchronous to the supplied tsu_clk.

The external port is enabled using a bit in the network control register.

When using external timestamp port the registers which modify the timestamp in internal mode, will have no effect, but the timer read and compare registers operate as described above.

6.1.16 IEEE 802.1Qbv: Enhancement for Scheduled Traffic (EnST)

IEEE 802.1Qbv is a TSN (Time Sensitive Networking) standard for “Enhancements for Scheduled Traffic" and specifies “time-aware queue-draining procedures” based on “timing derived from IEEE 802.1AS”. It adds transmission gates to the eight priority queues which allow low priority queues to be shut down at specific times to allow higher priority queues immediate access to the network at specific times. There are two main use cases for this:

- Firstly to allow guaranteed access for high priority low latency control frames (this is similar to Time Triggered Ethernet previously specified by the SAE in 2011 - AS6802)
- Secondly to allow periodic transmission of AVB traffic such as 1722 talker class A streams which need frames to be transmitted every 125 microseconds

GEM supports IEEE 802.1Qbv by allowing time-aware control of individual transmit queues. GEM has the ability to enable and disable transmission on a particular queue on a periodic basis with the on/off cycling starting at a specified TSU clock time.

For each transmit queue, configured up to a maximum of eight, there is a gate on timer and a gate off timer.

The on and off timer values are set in two 17-bit registers. These registers determine the “on” and “off” times for each queue in bytes. The actual “on” and “off” time will be a function of the number of bytes programmed in these registers and the speed of operation. 17 bits allow for a maximum value of about 1.05 milliseconds at 1G speed of operation (there is an exception for 2.5G operation where the on-time register counts 20-bit quantities, 2.5 bytes – so 17 bits is also equivalent to 1.05 milliseconds).

For each transmit queue, configured up to a maximum of eight, there is a 32-bit register to control the start time for IEEE 802.1Qbv queuing. The bottom 30 bits define the nanosecond value (as matched against the TSU timer) at which the queue will start and the top two bits the second value.

There is a 16-bit register that enables and disables IEEE 802.1Qbv traffic scheduling on each queue. The bottom 8 bits are for enabling each queue and the top 8 bits are for disabling. The disabling bits start at bit location 16. You should not set the enable and disable bit simultaneously for a particular queue, but if this is done, the disable bit has priority. The disable bits are write-only. The enable bits are read/write with a read returning the queue's status.

When enabled traffic scheduling starts for a particular queue at the time the TSU rolls over to the start time value in the relevant 32-bit register, the queue will then be on for the time in the "on" time register and off for the time in the "off" time register and the sequence will then repeat ignoring the time on the start time register. The functionality is similar to but not identical to the "Cycle Timer state machine" specified in Figure 8-14 of the IEEE 802.1Qbv standard. The state machine in the IEEE 802.1Qbv standard allows complete freedom in setting a queue's "on" and "off" times. The GEM implementation assumes that the "on" and "off" times will occur with a fixed period. If it is necessary to change the on/off sequencing, you need to write to the enst_control register to disable the queue and reprogram the "on", "off", and start times before restarting EnST on the queue.

When traffic scheduling is disabled, the queue will no longer be gated and will be scheduled as normal.

If scheduling is enabled for a queue, transmission will not start if the frame to be transmitted will not complete in the remaining on-time available, that is, it is too long. If the frame is too long to ever be transmitted, then the frame will be dropped and an error indicated.

IEEE 802.1Qbv scheduling works in addition to other queue arbitration schemes such as CBS and priority queuing.

The EnST registers should be set so that the queues are not overlapping when they are open. For example, for a three-queue configuration, the following is a correct setting (depicted in *Figure 19*):

- If more than eight queues exist in the hardware configuration, then EnST applies only to the eight highest queues.
- If the highest priority queues are disabled by writing 1 to the disable bit in the transmit queue pointer registers, the IEEE 802.1Qbv (EnST) control register mapping does not change and EnST functionality is partially lost as queues are disabled. For example, with 12 queues and 2 disabled, you would only have the ability to do EnST on six queues (rather than eight).

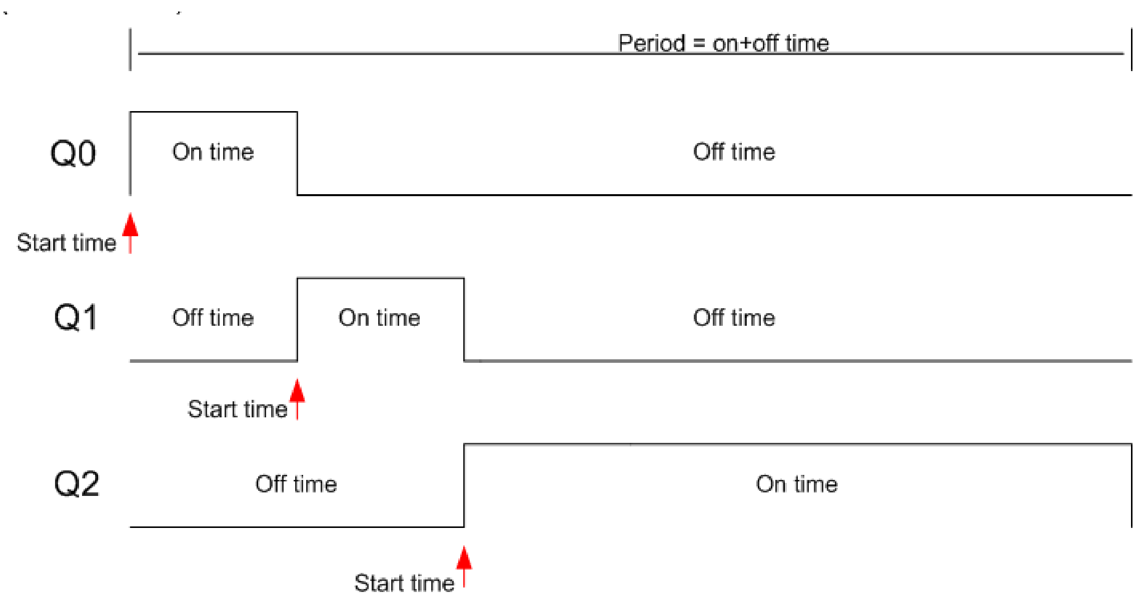

Figure 19: 802.1Qbv Transmit Scheduling

Table 52 lists the programmable registers of the EnST module.

Table 52: EnST Registers

Register	Type	Description
start_time[31:0]	R/W	The TSU Time at which to begin the “on” time. The 2 MSB are seconds, the 30 LSB are expressing nanoseconds.
on_time[16:0]	R/W	The time in which the gate has to be open, **expressed in bytes.**
off_time[16:0]	R/W	The time in which the gate has to be closed, **expressed in bytes.**
enst_en	R/W	EnST function enabled

To simplify implementation the on_time and the off_time are specified in bytes rather than nanoseconds.

The relation between the on_time (nanoseconds) and the on_time (bytes, the register of the module), that you should use for 1Gbps, 100Mbps, and 10Mbps, is the following:

$$on\_time_{bytes} = \frac{on\_time_{nanoseconds}}{X}$$

Where the value of X is given in table below for various operating speeds:

Table 53: 802/1Qbv “On” Time Calculation

Speed	Value of X
2.5Gbps	8
1Gbps	8
100Mbps	80
10Mbps	800

So, for example a gate open time of 800 ns at 10Mbps is equivalent to one byte:

$$on_{time_{bytes}} = \frac{on_{time_{nanoseconds}}}{800} = \frac{800}{8} = 1$$

So, 1 byte represents 800ns at 10Mbps, 80ns at 100Mbps and 8ns at 1Gbps.

The maximum value of $on\_time_{nanoseconds}$ and $off\_time_{nanoseconds}$ that can be set is 1,048,568ns.

6.1.17 IEEE 802.1CB Support

IEEE 802.1CB “Frame Replication and Elimination for Reliability” is one of the TSN (Time Sensitive Networking) standards developed by the 802.1 working group. Using Frame Replication and Elimination for Reliability (FRER) within a network increases the probability that a given packet will be delivered using multipath paths through the network.

The MAC supports a subset of this standard and provides the capability for stream identification and frame elimination but does not provide support for the replication of frames.

6.1.17.1 Configuration

The IEEE 802.1CB FRER functionality is a configuration option. When gem_no_of_cb_streams is zero or undefined 802.1CB functionality will not be present.

Table 54 lists the configuration options for this feature.

Table 54: FRER Configuration ‘define

‘define Name	Description
gem_no_of_cb_streams	When defined or non-zero defines the number of compound CB streams supported. This number cannot exceed the number of type 2 screener registers (as defined by num_type2_screeners) (valid range 0..16).
gem_seq_history_len	Indicates the length of the 2-bit vector used by the vector recovery algorithm for each 802.1CB stream. The maximum value allowed for the history length is 64 (valid range 1..64).

Note: The identification of IEEE 802.1CB streams relies on the use of the ‘Type 2 Screener’ functionality, therefore there must be at least `gem_no_of_cb_streams number of num_type2_screeners defined.

6.1.17.2 Stream Identification

The identification of IEEE 802.1CB streams relies on the use of the existing type 2 screener functionality. Each type 2 screener may be configured to inspect various fields of the receive frame and compared against values programmed into the type 2 screeners and compare registers.

For each implemented IEEE 802.1CB stream function, its associated control register contains the two fields ‘member_stream_1’ and ‘member_stream_2’:

- Each contains a value between 0x0 and 0xf representing the type 2 screener associated with this stream
- Each may point to the same or different type 2 screeners providing the option to eliminate packets which may have been sent across two different paths using different VLAN ID for example.

6.1.17.3 Support for FRER Redundancy Tag and Sequence Number Identification

In order to be able to identify duplicate frames, a sequence number must be extracted from the incoming frame.

IEEE 802.1CB defines a new optional Redundancy Tag field that will immediately follow either the MAC source address field or the VLAN tag fields of the frame. This Redundancy Tag is identified by a new Ethertype with a value of 0xF1C1.

Draft 2.5 of the IEEE 802.1CB standard (see subclause 7.8) increased the length of the redundancy tag to 6 bytes from 4 bytes:

- The first two bytes are for the Ethertype (F1C1)
- The second two bytes are reserved and are set to zero
- The final two bytes are the 16-bit sequence number.

Releases 1p10 and 1p11 of GEM implemented 802.1CB in accordance with draft 2.4 of the 802.1CB standard which defined the redundancy tag to be four bytes in length without the second two bytes of zero. Release 1p12 of the GEM added support for 6 byte redundancy tags.

Control of Redundancy Tag detection is done through the frer_red_tag register. This selects between 4 and 6 byte operation and controls the ability to automatically strip out the tag from the received frame such that it will be transparent to the software stack.

Alternatively to the use of the redundancy tag, an offset from the start of the frame may be used to identify the location of the sequence number.

For each IEEE 802.1CB stream function, its associated control register should be set up to determine how the sequence number is to be extracted. If the use_r_tag field is set, then the above extracted sequence number from the Redundancy Tag will be used, otherwise the offset_value fields should be programmed to point to the byte offset of where the sequence number is, this may be a TCP sequence number or any other unique number that may be injected by software.

6.1.17.4 Supported Recovery Algorithms

The IEEE 802.1CB specification specifies two different algorithms for processing the received frames and determining whether the frame should be discarded. Both of these algorithms are supported and each stream function should be set up to use the appropriate algorithm via the en_vector_rec_alg field.

6.1.17.4.1 Match Recovery Algorithm

For each CB stream configured and enabled, the first packet matched by the relevant screeners is accepted as valid and its sequence number is stored in the RecovSeqNum register (this is a hidden register not accessible by firmware).

Subsequent packets matched by the screeners have their sequence number checked against the RecovSeqNum register and if there is a match they are dropped and the duplicate bit is set.

If there is no match the frame is accepted

- And if the duplicate bit is not set, then the latent error counter is incremented
- Then the RecovSeqNum register is updated and the duplicate bit is reset
- And if the received sequence number does not directly follow the previous sequence number, the out of order statistics counter is incremented
- The sequence recovery reset timer is reset to the value in the FRER timeout register

If the sequence recovery reset timer recovery timer decrements to zero, the RecovSeqNum register and the duplicate bit are reset, the sequence recovery reset field in the statistics register is incremented, and the next packet matched by the relevant screeners is accepted.

6.1.17.4.2 Vector Recovery Algorithm

The vector recovery algorithm uses the history vector which is as long as the gem_seq_history_len configured. This indicates which sequence numbers have been seen relative to the maximum sequence number seen as indicated by the value in the RecovSeqNum register.

For each CB stream configured and enabled, the first packet matched by the relevant screeners for a CB stream or after a sequence recovery reset event is accepted as valid and its sequence number is stored in the RecovSeqNum register.

Subsequent packets matched by the screeners have their sequence number checked against the RecovSeqNum register and if it is equal to it, then it is a duplicate of the last frame and is discarded. If it differs by more than the history depth, it is considered a rogue frame and is rejected, and the rogue frames statistics is incremented.

If the sequence number is greater than the value stored in the RecovSeqNum register, the frame is accepted and the RecovSeqNum register is updated with the new sequence number. The history vector is updated to take account of the new max sequence number. If the sequence number did not immediately follow RecovSeqNum, the out of order statistics is incremented. The latent error statistics counter is updated to take account of any non-duplicates as the history vector is updated.

If the sequence number is less than the value stored in the RecovSeqNum register the frame is accepted as long as the frame is not a duplicate. If the frame is accepted, the appropriate bit is set in the history vector and the out of order statistics register updated.

6.1.18 IEEE 802.3br Support

IEEE 802.3br “Interspering Express Traffic” is one of the TSN (Time Sensitive Networking) standards developed by the IEEE 802.3 working group, which defines a mechanism to allow an express frame to be transmitted with minimum delay at the expense of delaying completion of normal priority frames.

This standard has been implemented by instantiating two separate MAC modules with related DMA, an MMSL (MAC Merge Sub Layer) and an AXI arbiter. One MAC is termed the express or eMAC and the other is a pre-emptable or pMAC. The eMAC is designed to carry time sensitive traffic which must be delivered within a known time.

A conceptual view of this structure is shown in *Figure 20*.

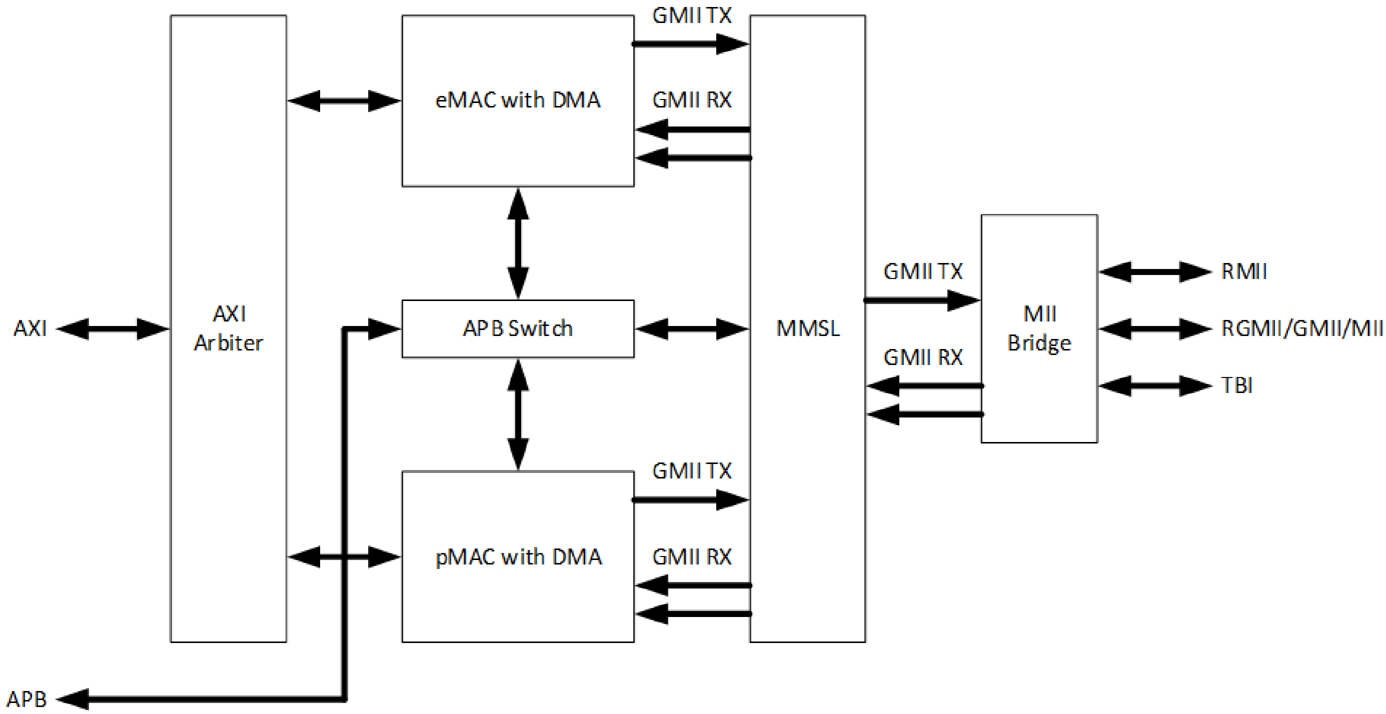

Figure 20: IEEE 802.3br Support Block Diagram

Note: The addition of the AXI Arbiter and APB Switch means that the main SoC interfaces to the core are not duplicated per MAC instance. Instead, a unified interface is provided to ease integration.

6.1.18.1 Frame Types

The use of this standard introduces a new format of frame, called an M-frame. M-frames are defined in the IEEE 802.3br standard as frames that have a special type of Start Frame Delimiter, called a Start mPacket Delimiter (SMD) and have a special type of CRC, called MCRC.

This new frame type is used by the MMSL to differentiate between the different types of traffic to determine whether:

- The frame is for the pMAC or eMAC
- An M-frame may consist of a full Ethernet frame or a fragment of an Ethernet frame

Frames to/from the eMAC are the same as regular Ethernet frames and when received will always contain a complete frame.

Frames to/from the pMAC uses a different SFD and may be fragmented if a frame was required to be sent from the eMAC, in which case the pMAC frame is transmitted in between at least two pMAC frame fragments.

6.1.18.2 Configuration

Refer to *Table 15 IEEE 802.3br Pre-Emption Support ‘defines* on page 27 for the defines available for this standard.

Note: In order to reduce the size of the eMAC, a number of configuration parameters of the eMAC are fixed and do not share the same configuration as the pMAC. These are listed below:

- The number of queues is fixed to 1
- If 802.1CB is supported, then the number of stream functions is fixed to 2
- The number of type 2 screeners and compare registers is fixed at 2 and 6 respectively if IEEEv802.1CB is supported; otherwise these are set to 0 since only 1 queue is supported
- The eMAC will share the TSU from the pMAC and is configured to use an external TSU.
- RSC is not supported in the eMAC

If the IEEE 802.3br functionality is to be used, then cut-thru and half-duplex and pause modes should not be enabled.

6.1.18.3 Memory Map

Table 55 shows the memory map defined for when 802.3br is used.

Table 55: Memory Map for IEEE 802.3br Support

Register Space	Slave
0x1000 – 0x1FFC	eMAC
0x0F00 – 0x0FFC	MMSL
0x0000 – 0x0EFC	pMAC

6.1.18.4 TX Interspersing Capability

On the transmit side, the interspersing capability is achieved by pre-empting M-frames from the pMAC in order to give the priority to Express frames from the eMAC.

If the pMAC is in the process of transmitting a frame and the eMAC requests to transmit, then the M-frame from the pMAC is halted with the M-frame transmission stopped and appended with a 32-bit MCRC before transmission of the express frame commences.

The additional support for IEEE 802.1Qbv also provides the capability to send time-scheduled express frames. This is supported by a set of programmable registers to define when the frame should be transmitted.

There are certain circumstances where a pMAC transmission will not be pre-empted, this is handled in accordance to the IEEE 802.3br standard as defined below:

- There are less than 64 M-frame bytes left to transmit
- The M-frame transmission started at the moment when an Express packet requested the channel (he minimum number of bytes to be transmitted before pre-emption can occur is programmable using the register add_frag_size register). In this case pre-emption will start after the minimum number of bytes of the M-frame have been transmitted.
- If the pre-emption function is not enabled
- If the pre-emption function is enabled, the verification function is not disabled and the link partner couldn’t complete the handshake protocol indicating that it can’t support the IEEE 802.3br standard.

6.1.18.5 RX Interspersing Capability

The MMSL receiver processes the received M-frames to determine whether the frame is destined for the eMAC or the pMAC and routes them appropriately.

Frames destined for the eMAC look like standard Ethernet frames and are passed un-modified to the eMAC.

Frames destined for the pMAC contain a modified SFD and this new SFD is also used to identify frame fragments in order to reconstruct a complete Ethernet frame for the pMAC.

If a frame for the pMAC is fragmented due to an express frame, the MMSL receive state machine and the pMAC receive path are halted while the express frame is routed to the eMAC. Once a continuation fragment is received, the MMSL receive state machine and pMAC receive path resumes operation.

6.1.18.6 Verification Process

The verification process is defined by the IEEE 802.3br as a mechanism to validate the support of the 802.3br standard by both ends of the link.

If the pre-emption capability and the verification process are enabled, after reset a “verify frame” which consists of a special type of M-frame is transmitted. This frame, when received by the link partner should cause the link partner to transmit a ‘Respond Frame’.

The proper reception of this ‘Respond Frame’ completes the verification process and enables the pre-emption capability.

If no ‘Respond Frame’ is received within 10ms, the process will automatically be repeated another two times before the pre-emption capability is disabled. If this happens, the verification process may be restarted through the programming registers.

During the time in which the core is waiting for the Respond packet, the receive traffic will be routed to either the pMAC or the eMAC depending on the value of the 802.3br configuration registers.

6.1.19 IEEE 802.1Qci: Receive (Ingress) Traffic Policing

The rx_q_flush registers starting at 0x0b00 allow frames to be discarded on a per-queue basis when the following events occur:

- All traffic received to a particular queue
- When a receive queue’s buffer descriptor is read with the used bit set
- When a queue’s SRAM utilization reaches a programmable depth (specified in 128-byte chunks)
- When a frame is received above a programmable length (specified in bytes)

The scr2_reg_rate_limit registers starting at 0x0b40 allow frames to be discarded that exceed a certain flow rate.

The TSN standards enable provisioning the resources in a network in such a way that high priority traffic is guaranteed to get through as long as it does not exceed its frame length and flow rate allocation.

802.1Qci is a policing mechanism that discards frames on ingress (ie received frames) if they exceed their allocated frame length or flow rate. The rx_q_flush and scr2_reg_rate_limit registers are provided for this purpose.

6.1.20 Physical Coding Sublayer

Figure 21 shows the PCS signal block diagram.

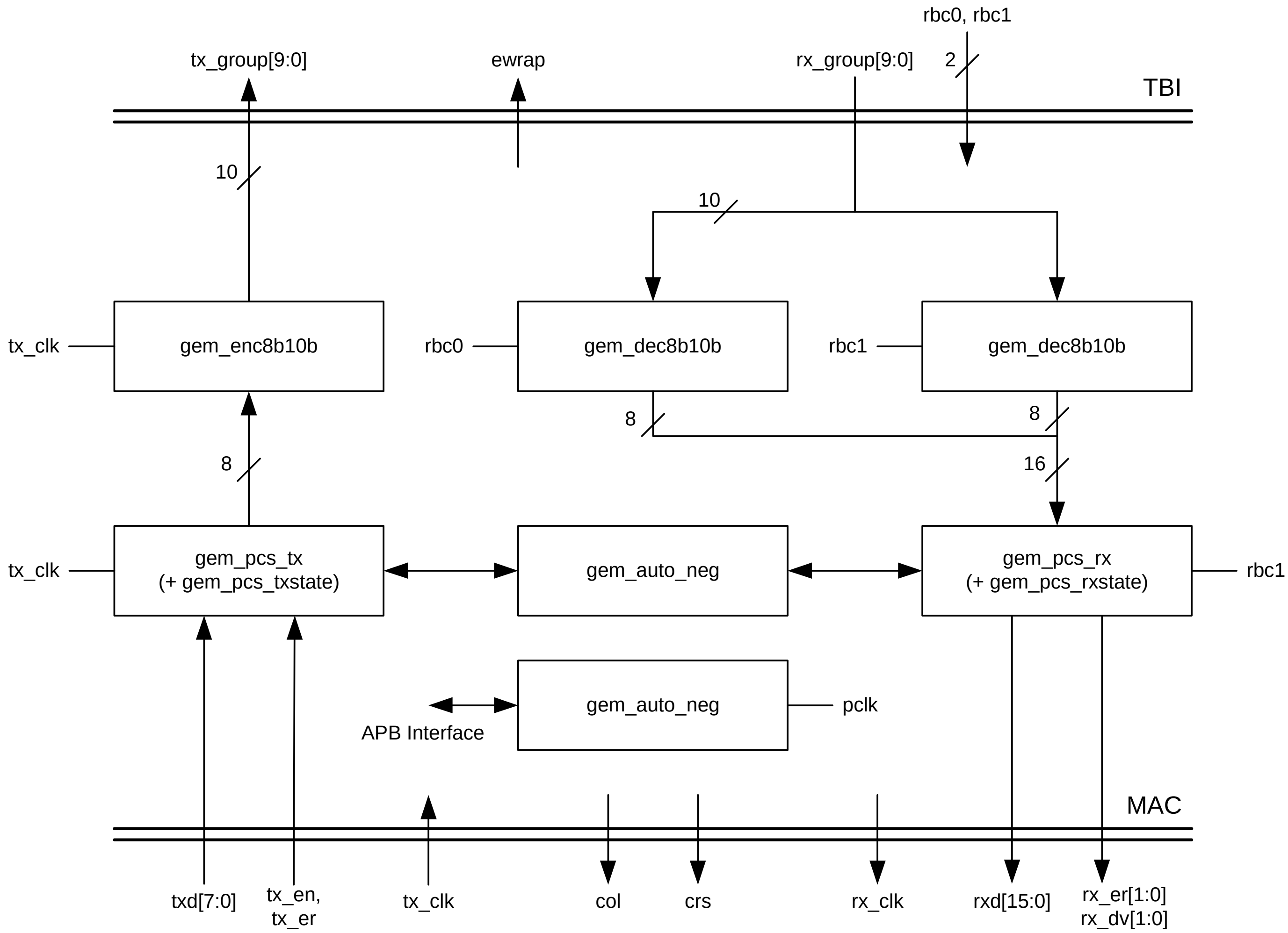

Figure 21: PCS Block Diagram

The PCS provides a TBI for connection to a PHY transceiver. It is operational when the PCS interface is selected through the PCS select bit in the network configuration register. Components of the PCS sub-system include a PCS transmitter, 8B/10B encoder, PCS receiver, 8B/10B decoder, and auto-negotiation.

6.1.20.1 PCS Transmit

Ethernet transmit data is passed to the PCS transmitter in byte form from the MAC transmitter and encoded into 10-bit words by the 8B/10B encoder for transmission. The 8B/10B encoder module is controlled by the transmit state machine and generates both data code groups and special code groups as per the IEEE 802.3 standard.

The transmitter generates ordered sets consisting of both special code groups and data code groups. Each Ethernet frame is encapsulated between a start of frame and end of frame. If auto-negotiation is requested, then the transmitter generates ordered sets to reconfigure the link.

When the PCS is not performing auto-negotiation and the MAC has no frame to transmit, the PCS transmitter generates idle code groups. Idle code groups will also be transmitted if the receive link is down unless bit 31 is set in the network configuration register.

Each MAC frame to be transmitted is encoded a byte at a time, with each 10-bit output word being based on the Ethernet octet for transmission and the running disparity from the previously transmitted word.

The PCS transmits its 10-bit symbols as alternate even and odd numbered code groups. The start packet delimiter has to be transmitted as an even numbered code group. To accomplish this, GEM may need to delete a byte of preamble, reducing the transmitted preamble from seven bytes to six. This is accordance with the IEEE 802.3 spec. Other implementations may not delete preamble and instead adjust the IPG so that it falls below 12 bytes.

For reference, see the following sections of the IEEE 802.3 standard:

- "36.2.4.14 Start_of_Packet (SPD) delimiter. A Start_of_Packet delimiter (SPD) is used to delineate the starting boundary of a data transmission sequence and to authenticate carrier events. Upon each fresh assertion of TX_EN by the GMII, and subsequent to the completion of PCS transmission of the current ordered_set, the PCS replaces the current octet of the MAC preamble with SPD."

If the PCS is configured to use a 20-bit interface to the PHY, an optional 10 to 20-bit gearbox is instantiated to perform the data width conversion and additional clock inputs will be present.

6.1.20.2 PCS Receiver

The receive interface between the PHY and the PCS is configurable and different clocks are present depending on the options chosen.

If a 10-bit PHY interface is chosen, then an optional 10 to 20-bit gearbox is automatically instantiated to perform the data width conversion with additional clock inputs added.

If the interface selection is not set to gem_pcs_legacy_if, then additional comma and code group alignment functions are instantiated to ensure that the receive data is correctly aligned ready for decoding.

The internal datapath of the PCS is 16 bits for easier timing closure.

Two 8B/10B decoder blocks perform the decoding function for the data received via the TBI. Running disparity is performed on the received data, with disparity indicators being passed between the two decoders.

Each 8B/10B decoder decodes special code groups and data code groups as per IEEE 802.3 standard. A receive state machine monitors special code groups and determines whether the TBI contains Ethernet frames, auto-negotiation link configuration information or if the link is idle.

Decoded 8-bit data, along with rx_dv and rx_er, is generated and passed to the MAC receiver in 16-bit form.

For reference see the following sections of the IEEE 802.3 standard:

- "35.2.3.2.2 Receive case. The conditions for assertion of RX_DV are defined in 35.2.2.7. The operation of 1000 Mb/s PHYs can result in shrinkage of the preamble between transmission at the source GMII and reception at the destination GMII. Table 35–3 depicts the case where no preamble bytes are conveyed across the GMII. This case may not be possible with a specific PHY, but illustrates the minimum preamble with which MAC shall be able to operate."

6.1.20.3 PCS Auto-Negotiation

An auto-negotiation block provides a means for the PCS to establish automatic link configuration. It is performed at power-up or during normal operation if requested by a link partner or through the restart auto-negotiation bit in the PCS control register.

By default, GEM has auto-negotiation enabled in the PCS control register and full and half-duplex capability enabled in the PCS auto-negotiation advertisement register. Pause capability is disabled in the advertisement register by default. If auto-negotiation is not required then bit 12 of the PCS control register needs to be set low.

When a new base or next page is received from the link partner, a PCS link partner page received interrupt is set (bit 17 of the interrupt status register). The first time this interrupt is received, it will indicate a base page received and subsequent reads will indicate next pages.

For next page exchange to work, the next page register (0x21c) must be written within 10 ms of receiving a new page from the link partner. If the link partner is requesting next pages and GEM has none to send then the next page register should be written with the null message (0x2001). The value 0x0000 must not be written to the next page register.

GEM signals completion of auto-negotiation through the PCS auto-negotiation complete interrupt, on bit 16 of the interrupt status register. Auto-negotiation completion is also indicated by bit 5 of the PCS status register.

The PCS resolves the GEM and link partner’s abilities and reports the result in the network status register. Pause transmit and receive resolution are reported in accordance with *Table 37-4 in the IEEE 802.3 specification*. If full-duplex capability is resolved the duplex resolution bit is set high in the network status register. If half-duplex capability is resolved the duplex resolution bit will be low. If GEM and its link partner cannot resolve a common duplex capability, the duplex resolution bit is not set and link will be indicated as being down (bit 2 in the PCS status register and bit 0 in the network status register will both be zero) when auto-negotiation completes. Although GEM reports the auto-negotiation resolution, it does not automatically reconfigure its duplex and pause states. So it is necessary for management software to set the duplex bit in the network configuration register if it reads the duplex resolution bit as being set in the network status register.

6.1.20.4 PCS Collision Detect and Carrier Sense

The PCS provides both the carrier sense and collision signals for use by the MAC sub-layer when the TBI is active.

CRS (Carrier Sense) is generated by the following conditions:

- The receiver has decoded a start of packet/end of packet or receive carrier extension is active. This state is indicated internally to the PCS by CRS receive.
- tx_en is active, or carrier extension is active for transmit.

The collision signal is generated whenever the PCS is requested to transmit an Ethernet frame at a time coincident with the crs receive signal being active. The col signal remains active for the duration of the collision. Both crs and col are asserted, regardless of whether the PCS is operating in half-duplex or full-duplex mode.

6.1.20.5 Link Status

The PCS link status is indicated on: [introduced bullets]

- Bit 2 of the PCS status register
- Bit 0 of the network status register and
- Bit 9 of the interrupt status register

An interrupt is generated each time the PCS link status changes (i.e. link good or link bad).

When auto-negotiation is disabled, the link status value is determined by whether or not the PCS is in a synchronized state. When auto-negotiation is enabled, the link status value is determined by successful completion of auto-negotiation.

6.2 Low-Power Features

GEM provides Energy-Efficient Ethernet (EEE) support.

IEEE 802.3az adds support for energy efficiency to Ethernet. These are the key features of 802.3az:

- Allows a system’s transmit path to enter a low power mode if there is nothing to transmit
- Allows a PHY to detect whether its link partner’s transmit path is in low-power mode, therefore allowing the system’s receive path to enter low-power mode
- Link remains up during lower power mode and no frames are dropped
- Asymmetric, one direction can be in low-power mode while the other is transmitting normally
- LPI (Low Power Idle) signaling is used to control entry and exit to and from low power modes
- LPI signaling can only take place if both sides have indicated support for it through auto-negotiation

802.3az operation:

- Low-power control is done at the MII (reconciliation sublayer).

- As an architectural convenience in writing the IEEE 802.3az, it is assumed that transmission is deferred by asserting carrier sense; in practice it is not done this way. This system will know when it has nothing to transmit and only enter low-power mode when it is not transmitting.
- LPI should not be requested unless the link has been up for at least one second
- LPI is signaled on the GMII transmit path by asserting 0x01 on txd with tx_en low and tx_er high
- A PHY, on seeing LPI requested on the MII, sends the sleep signal before going quiet. After going quiet, it periodically emits refresh signals.
- The sleep, quiet, and refresh periods are defined in Table 78-2 of IEEE 802.3az. For 1000BASE-X, the sleep period is 20 microseconds, the quiet period 2.5 milliseconds, and the refresh period 20 microseconds.
- 1000BASE-X is required to go quiet after sleep is signaled. The easiest way to do this is to write to a control register to disable transmit in the SerDes.
- SGMII is not part of IEEE 802.3az and should not go quiet after sleep is signaled.
- LPI mode ends by transmitting normal idle for the wake time. There is a default time for this but it can be adjusted in software using the Link Layer Discovery Protocol (LLDP) described in Clause 79 of 802.3az.
- LPI is indicated at the receive side when sleep and refresh signaling has been detected.

6.2.1 LPI Operation in Cadence IP

It is best to use firmware to control LPI. LPI operation is something that happens at the system level. Firmware gives maximum control and flexibility of operation. LPI operation is straightforward and firmware should be capable of responding within the required timeframes.

Auto-negotiation

- Indicate EEE capability using next page auto-negotiation

For the transmit path:

- If the link has been up for 1 second and there is nothing being transmitted, write to the LPI bit in network control register
- If connected to 1000BASE-T PHY using SGMII or RGMII, there is nothing more to do.
- If connected to a backplane using a 1000BASE-KX PHY, use firmware to periodically disable the SerDes transmit path. (Write to bit 1.160.0 for 1000BASE-KX.)
- Wake up by clearing the LPI bit in the network control register

From GEM 1p07 release onwards, the LPI bit is ORed into the transmit pause functionality so transmission will pause as long as tx_lpi_en is active. This allows queuing of new transmit frames while the LPI signal is active. The 16-bit Tw_sys_tx time register at 0x060 holds the system wake time. After tx_lpi_en is de-asserted transmission will continue to be paused while the value in the tw_sys_tx_time counts done fixed time periods. The time period counted down is:

- 64ns for gigabit operation
- 320ns for 100M operation
- 3200ns for 10M operation

In other words 8 tx_clk periods. So for example, if tw_sys_tx_time is set to 8 in 100M mode, transmission will be paused for 8 * 320 ns = 25.6 microseconds after deassertion of tx_lpi_en.

For the receive path:

- Wait for an interrupt to indicate that LPI has been received
- Disable relevant parts of the receive path if desired but keep the PCS and SerDes active
- Wait for an interrupt to indicate that regular idle has been received and then re-enable the receive path

6.3 Design For Test (DFT) Features

Setting the retry_text bit in the network configuration register aids directed functional testing by speeding up the collision back-off algorithm, the pause counter, and the frer timeout.

6.4 Automotive Safety Features (ASF)

In order to improve reliability of the GEM, a number of internal detection mechanisms are implemented to detect when there is abnormal behavior.

6.4.1 IP Fault Logging and Reporting

The IP Fault Logging and Reporting module provides a user interface for accessing from memory location 0x0e00, Active Safety fault information.

The fault detection capability covers the following:

- Lockup detection
- Mapping of existing internally detected faults to fault logging
- Mapping of protocol error detection to fault logging, e.g. various RX errors which would normally cause packet to be dropped with only update to statistics, these are also be reported as protocol errors that the user can mask.
- Parity errors on the datapath
- Parity errors with the control registers
- Error detection and correction on the SRAMs

The IP Fault Logging and Reporting module presents a consistent view of the ASF status.

Figure 22 shows a basic block diagram for this module.

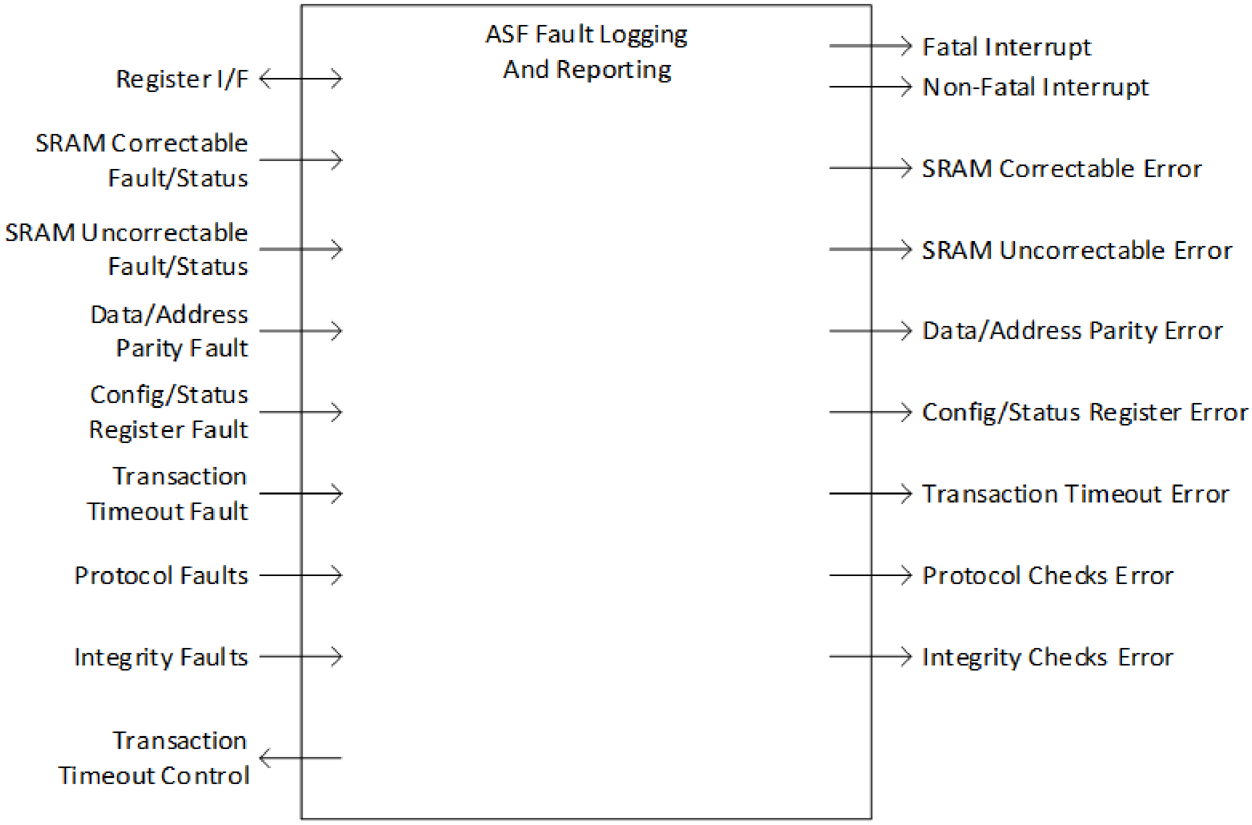

Figure 22: IP Fault Logging and Reporting Block Diagram

The module defines seven different ASF functions, each of which has its own set of fault and status inputs. If a specific ASF function is not implemented within the IP, the inputs related to that function should be tied to 0 and the outputs left unconnected. Parameterization of this module will ensure that the relevant interrupt bits do not exist and access to the associated status registers will result in *asf_err* being asserted which you may map to host error signaling where appropriate (e.g. Perr).

6.4.2 Configuration/Control and Status Registers Protection

When ASF is configured, all configuration, control, and status registers may be protected either by parity or duplication. The only registers that are not protected are statistics registers, i.e. everything in gem_reg_stats.

Parity should be either sourced from the host as the primary input pwdata_par if the host has parity generation capabilities, otherwise a single parity generator will be instanced to generate parity based on pwdata. This parity is stored alongside all registers that are to be parity protected.

The stored parity is checked alongside the stored data every clock cycle for errors.

The following registers are protected by duplication as they involve interaction with the hardware or complex manipulation preventing parity alone from being a comprehensive fault protection method

- Network Control
- Transmit Status
- Receive Status
- All interrupt registers
- PHY Management registers and state machine
- Part of transmit scheduler registers

Each sub-module also generates a fault signal associated with that module to indicate when a fault has been detected. These signals are combined with ‘OR’ to generate overall fault indication to the Fault Logging and Reporting module [01].

Fault detection is active every clock cycle.

If the host interface requires parity, the prdata_par is generated using the parity generator modules directly from prdata which should have sufficiently protected source registers.

Parity generation and checking is expected to use the new reuse modules developed as part of [01] and will need to be checked out through an SVN extern definition to an appropriate directory.

7 How to Set Up the IP

7.1 SRAM Protection

The IP can be configured to add Error Correcting Code (ECC) protection to SRAM data such that corruption within the SRAM may be detected and possibly corrected. The protection code used is capable of correcting 1-bit and detecting 2-bit errors.

When this configuration option is selected, the dataports of the SRAM are widened to include the additional correction codes. The data width is extended by 7, 8, or 16 bits depending on whether the datapath is configured to be 32, 64, or 128 bits respectively.

For uncorrectable errors, the address associated with the error is logged in either the txdpram_non_crr_err_addr or rxdpram_non_crr_err_addr register and is not updated again until the software reads this register.

The SRAM protection component is instantiated if the Data/Address Path protection feature is enabled. Otherwise, the existing SRAM interface signals go direct to the top level I/O.

If Data/Address Path protection is enabled, the SRAM protection can be configured to support either parity or ECC protection.

If ECC protection is selected, the parity will be checked and terminated at the SRAM ECC boundary and parity will be regenerated on the read side after the SRAM ECC boundary.

7.2 Transaction Timeout Monitoring

Host side interfaces have the option of transaction timeout monitoring to detect potential lockup conditions in the host.

There is one transaction timeout monitor instantiated on each AXI channel. The timeout value is controlled by the Fault Logging and Reporting module and each fault condition is routed to a separate input of the asf_trans_to_fault bus of that module.

7.3 Datapath and Address Path Parity Protection

When the design is configured for datapath parity protection, additional parity generators and checkers are used to ensure integrity of the main datapaths through the design. If this option is selected but the above ECC protection (*Section 7.1 SRAM Protection*) is not, then the SRAM I/O are parity protected with no correction capability. This causes the SRAM datawidth to widen by 4, 8, or 16 bits depending on whether the datapath is configured to be 32, 64, or 128 bits respectively.

When this configuration option is enabled, the main datapaths of GEM have parity protection added and wherever possible this is maintained for end to end protection.

Note: Similar to the CSR protection scheme, parity is generated/checked at the host interface depending on the option of gem_asf_host_par.

The following dataflow points will need protection:

7.3.1 Transmit Datapath

Parity for transmit data originates from either the host if gem_asf_dma_par is defined, otherwise a parity generator is instanced to generate an internal parity from the DMA read data. As the entire DMA datapath operates on a multiple of bytes, this parity can be easily maintained and should not need to be regenerated at any point except for TCP checksum generationand timestamp insertion.

In the MAC, the parity is maintained up to the point of transmission on the GMII where it is checked. If the interface is using nibbles, then the parity will be checked every second clock cycle.

7.3.2 Transmit Descriptor Writeback Data

The writeback information, including the address for the writeback has parity protection tagged along to ensure this data is properly protected.

7.3.3 Receive Datapath

Parity is generated in the MAC RX to match the receive data. This is done at the point where the receive data has been gathered to 16-bit data words.

This parity is maintained all way to the DMA write port.

7.3.4 Receive Descriptor Writeback Data

The writeback information including the address for the writeback has parity protection tagged along to ensure this data is properly protected.

7.3.5 Address Path Protection

For both transmit and receive, the starting buffer address originates from a descriptor read which already has parity data. This parity is maintained along with the address information and output to the host for the first access of the data buffer. Subsequent accesses will be based on an increment of this starting address and parity will need to be regenerated for these following addresses.

7.4 IP Integrity Protection

7.4.1 TSU Protection Enhancement

The timestamp unit is responsible for generating the time value used for the IEEE 802.1AS protocol. The value of this is critical to ensure that packets are timestamped correctly. To protect against any faults within this timer, the TSU module is duplicated with the Verilog define gem_protect_tsu is defined. The timer output of these two modules are compared every cycle to ensure the values are identical. If any fault is detected, this is signaled as an interrupt.

If the datapath parity protection configuration option is selected, the timestamp value generated by the TSU unit is also parity protected and checked wherever it is used to ensure that the value used and logged is valid. If a fault occurs in any of the timestamp bits, then an interrupt is signaled.

TSU duplication protection has the following features:

- ALL outputs are compared for mismatches
- Parity generation is added to the TSU module and this parity is maintained and stored along with the timer value. The parity does not require regenerating. Parity checks are performed at the following points:
  - TX datapath endpoint at GMII
  - Registers. There are several paths that eventually end at the registers, the parity should be stored along with the timer value and will be checked when the host reads the relevant status registers. If CSR protection is not configured, then the parity is not stored in the registers but it will be checked at the point where the timer value is stored.
  - FIFO interface, this will be a check point
  - DMA. The parity is maintained as far as possible into the DMA, this should be only used for the writeback data so should form part of the writeback data parity which should be checked at the hand-off interface to the SRAM wrappers.
- Supports external timer input mode and assumes that if datapath parity protection is enabled then this external TSU source will also supply parity input

7.4.2 Optional Duplication Protection of Transmit Scheduling

The user may optionally duplicate all the transmit scheduling functions:

- Transmit Scheduler (including CBS traffic shaper)
- EnST function

Both these features are instantiated within gem_tx_fifo_if so new option is to allow duplication of this entire module and sub-hierarchy.

This feature adds significant area and is expected to only be used by customers who require guaranteed bandwidth allocation characteristics.

For protection against total failure of these functions resulting in no packets being transmitted, this would be detected by the lockup detection features.

The fault detection will operate every cycle comparing all the outputs of the two modules.

7.5 Lockup Detection

A simple lockup detection mechanism is included in all configurations of the controller IP. This mechanism may be used to detect abnormal behaviour of the controller where packets do not pass through the datapath as expected within a programmable time frame. The lockup time is controlled by a number of registers including a ‘prescaler’ register which is common for all the lockup mechanisms and allows the actual lockup timers to be smaller in size.

There are four separate lockup detection points defined in the following sections:

7.5.1 DMA Transmit Lockup Detection

A DMA transmit lockup detection module is instantiated once for each configured queue in the design and covers two detection points:

1. Timeout detection from point of ‘start’ trigger (hardware or software start) until a complete packet has been written into the SRAM and cleared for any of the following conditions:
   - Packet fully written into SRAM for this queue.
   - A ‘used bit read’ detected for this queue.
   - A flush condition has occurred in the DMA.
2. Count number of packets in the SRAM, this is incremented each time a full packet has been written into the SRAM and decremented each time a full packet has been transferred to the MAC. A timer is started whenever this count is non-zero and is reset each time the count is decremented. A separate software controlled enable is provided for each queue such that the timer is not started unless the queue specific timer is enabled. This allows for cases where a fixed priority scheduling scheme is used whereby the software knows a large number of high priority packets will be transmitted which would prevent lower priority queues from transmitting.

This lockup mechanism does not support the following features and should not be enabled if any of these are enabled:

- Cut-thru mode
- Half-duplex mode
- Large Send Offload

There is a single register for this timeout value and takes into account the prescaler value and is shared by all DMA lockup detection mechanisms (both TX and RX).

7.5.2 MAC Transmit Lockup Detection

This lockup detection mechanism continues on from the end point of the DMA Transmit Lockup Detection and counts the time from the end of packet on the MAC internal FIFO interface to the end of packet output from the MII bridge after the optional IEEE 802.3br MMSL. The result of this is that for the case of the pMAC, the programmed lockup detection time should be set high enough to account for potential pre-emption by the eMAC. There is a single dedicated register for this timeout value and takes into account the prescaler value.

7.5.3 MAC Receive Lockup Detection

A generic lockup detection mechanism is available for the MAC RX and relies on the system being set up to have a known maximum packet interval. A timeout is triggered if a good (valid packet passing all filtering checks) is not received periodically within the programmed timeout time. There is a single dedicated register for this timeout value and takes into account the prescaler value.

7.5.4 DMA Receive Lockup Detection

A single DMA receive lockup detection module is instantiated (this is not per queue) and counts the number of packets in the receive SRAM. The packet count is incremented when a good ‘end of packet’ is written by the MAC RX and overflow is not signaled. The packet count is decremented on either DMA packet complete indication or a packet has been flushed from the SRAM due to resource error. A timer is started whenever this count is non-zero and is reset each time the count is decremented.

This lockup mechanism does not support the following features and should not be enabled if any of these are enabled:

- Cut-thru mode
- Receive Side Coalescing

7.5.5 Lockup Recovery Mechanism

GEM can be configured to automatically disable the transmit and receive datapath circuits in the event that a lockup is detected. This will ensure that the design is quickly forced into a quiescent state to prevent possible further corruption of the main interfaces.

As the system integrator, it is your responsibility to ensure that the overall system where this IP is used is also transitioned into the appropriate fault-detected state and/or appropriate recovery procedures are taken.

7.6 Error Handling

7.6.1 IP Protocol Checking

GEM contains existing checks for the Ethernet protocol and these normally result in increment of various statistics registers.

These are also routed to the IP Fault Logging and Reporting module to allow the user to be optionally notified and take action on these events.

The following are items that can be logged:

- Collision
- Too many retries
- Underflow
- RX length error
- RX too long
- RX jabber error
- RX CRC error
- RX too short
- RX symbol error
- RX align error

- RX checksum error

Default for the protocol masking is that these are all masked as the IP is designed to be able to deal with these errors. You may optionally want to unmask these to be notified of these events happening and possibly take action if too many such errors occur.

7.6.2 Fatal and Non-Fatal Interrupts Classification

When the design is configured to include the ASF features, two new interrupt pins called asf_int_fatal and asf_int_nonfatal provide interrupt event notification specific to ASF events and this is configurable through the asf_int_mask register.

In addition, a new register asf_fatal_nonfatal_select is provided to allow routing a number of different interrupt sources to either of these interrupts.

8 How to Program the IP

All register addresses are 32-bit word aligned and all read/writes occur over 32-bit data buses. Byte accesses are not supported. Write only (WO) bits return zero if read.

8.1 Initialization

8.1.1 Configuration

Initialization of the GEM configuration must be done while the transmit and receive functionality is disabled. See the description of the network control register and network configuration register *Section 11 Registers*.

To change loopback mode, the following sequence of operations **must** be followed:
1. Write to network control register to disable transmit and receive circuits
2. Write to network control register to change loop back mode
3. Write to network control register to re-enable transmit or receive circuits

Note: These writes to network control register cannot be combined in any way.

8.1.2 Receive Buffer List

Receive data is written to areas of data (i.e. buffers) in system memory. These buffers are listed in another data structure that also resides in main memory. This data structure (receive buffer queue) is a sequence of descriptor entries as defined in the *Table 26 Receive Buffer Descriptor Entry* on page 62.

The receive buffer queue pointer register points to this data structure as shown in *Figure 23*.

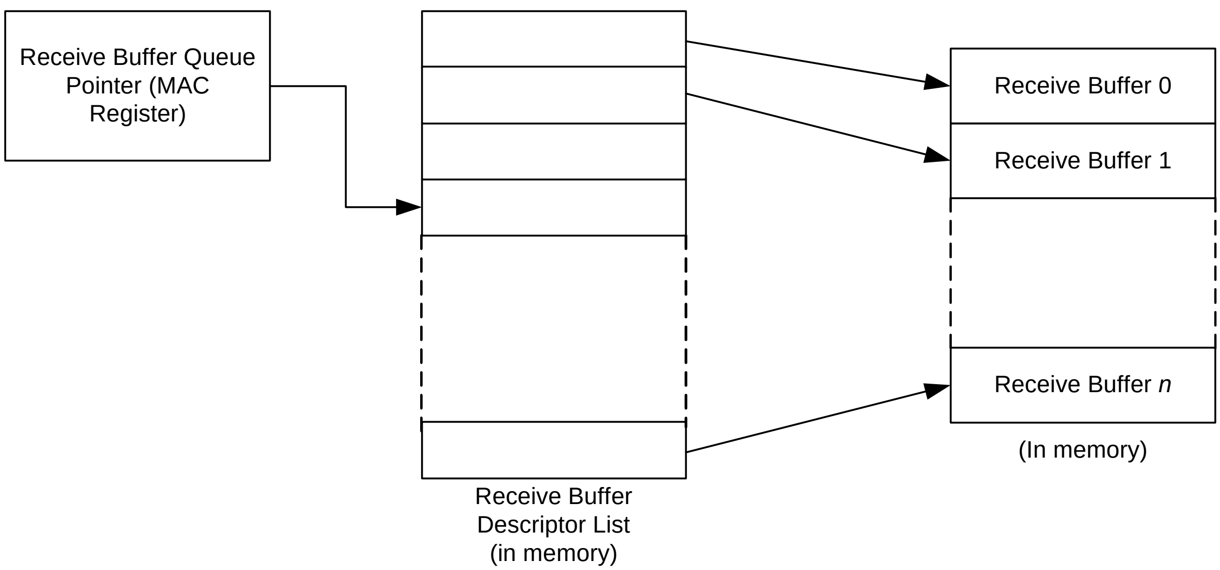

Figure 23: Receive Buffer List

To create this list of buffers: (**Note**: M is number of Buffer Descriptor Words / 2, and will be 1 or 2 or 3 depending on 64 bit addressing and timestamp insertion modes being enabled)
1. Allocate a number (*N*) of buffers of X bytes in system memory, where X is the DMA buffer length programmed in the DMA Configuration register.
2. Allocate an area 8NxM bytes for the receive buffer descriptor list in system memory and create N entries in this list. Mark all entries in this list as owned by GEM, i.e. bit 0 of word 0 set to 0.
3. Add an extra descriptor to the end of queue with its used bit set (bit 0 in word 0 set to 1). This last descriptor in the queue may also have its wrap bit set (bit 1 in word 0 set to 1) in addition to its used bit. When receive is enabled at least one

entry in the buffer descriptor ring needs its used bit set so it is not sufficient to set the wrap bit of the last buffer in the queue without also setting its used bit. GEM can now prefetch receive descriptors and the used bit will be used as an indication to the hardware that all available descriptors have been prefetched.
4. Write address of receive buffer descriptor list and control information to GEM register receive buffer queue pointer.
5. The receive circuits can then be enabled by writing to the address recognition registers and the network control register.

8.1.3 Transmit Buffer List

Transmit data is read from areas of data (the buffers) in system memory. These buffers are listed in another data structure that also resides in main memory. This data structure (Transmit Buffer Queue) is a sequence of descriptor entries as defined in the *Table 29 Transmit Buffer Descriptor Entry − Non-LSO Frame* on page 68.

The transmit buffer queue pointer register points to this data structure as shown in *Figure 24*.

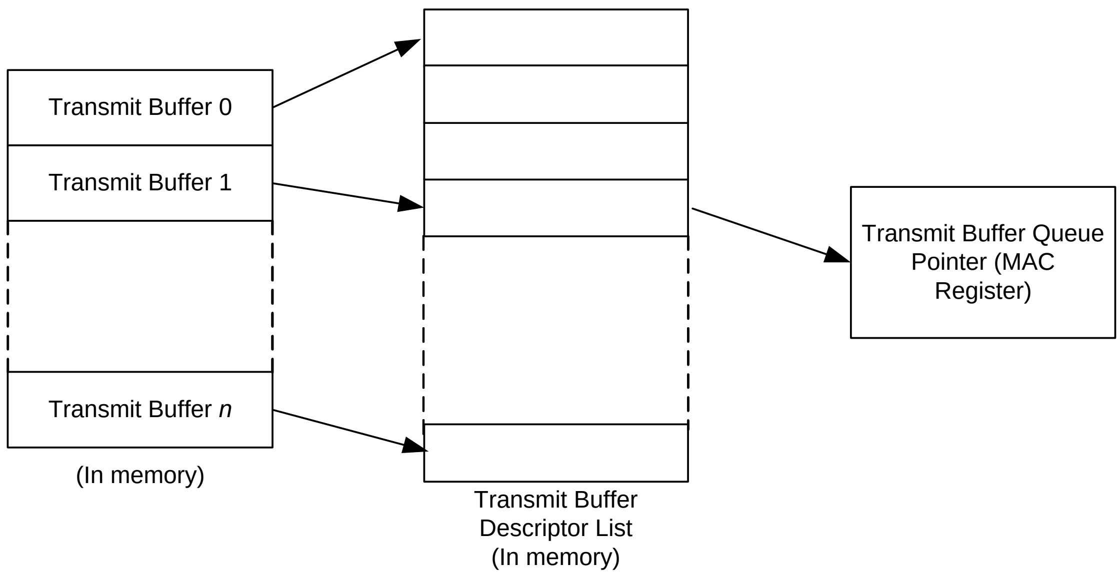

Figure 24: Transmit Buffer List

To create this list of buffers: (**Note**: M is number of Buffer Descriptor Words / 2, and will be 1 or 2 or 3 depending on 64 bit addressing and timestamp insertion modes being enabled)

1. Allocate a number (N) of buffers of between 1 and 16383 bytes of data to be transmitted in system memory. Up to 128 buffers per frame are allowed.
2. Allocate an area 8NxM bytes for the transmit buffer descriptor list in system memory and create N entries in this list. Mark all entries in this list as owned by GEM, i.e. bit 31 of word 1 set to 0.
3. Add an extra descriptor to the end of queue with its used bit set (bit 31 in word 1 set to 1). This last descriptor in the queue may also have its wrap bit set (bit 1 in word 0 set to 1) in addition to its used bit. When transmit is enabled at least one entry in the buffer descriptor ring must have its used bit set. When the buffer descriptor ring is initialized for the first time there must be a used bit set before or at the buffer descriptor with the wrap bit. Once transmission halts due to reading a used bit firmware can reuse the transmit buffers and clear the used bits before and including the one with the wrap bit and restart transmission by writing the tx_start bit.
4. GEM can now read transmit data so fast that all data may be read in before it sets the used bit of the first buffer descriptor in the queue.
5. Write address of transmit buffer descriptor list and control information to GEM register transmit buffer queue pointer.
6. The transmit circuits can then be enabled by writing to the network control register.

8.1.4 Address Matching

The GEM register pair hash address and the specific address register-pairs must be written with the required values. Each register pair comprises of a bottom register and top register with the bottom register being written first. The address matching is:

- Disabled for a particular register pair after the bottom register has been written and

- Re-enabled when the top register is written.

Each register pair may be written at any time, regardless of whether the receive circuits are enabled or disabled.

As an example, to set specific address register 1 to recognize destination address 21:43:65:87:A9:CB, the following values would be written to specific address register 1 bottom and specific address register 1 top:

- Specific address register 1 bottom bits 31:0 (0x98): 8765_4321 hex.
- Specific address register 1 top bits 15:0 (0x9C): cba9 hex.

8.1.5 PHY Management

An MDIO interface is provided to allow GEM to access the PHY's management registers. This interface is controlled by the PHY management register. Writing to this register causes a PHY management frame to be sent to the PHY over the MDIO interface. PHY management frames are used to either write or read the PHY's control and status registers.

The PHY management register is implemented as a shift register. Writing to the register starts a shift operation which is signaled as complete when bit two is set in the network status register (about 2000 pclk cycles later when bits [18:16] are set to 010 in the network configuration register). An interrupt is generated as this bit is set.

During this time, the MSB of the register is output on the mdio_in pin and the LSB updated from the mdio_in pin with each MDC cycle. This causes the transmission of a PHY management frame on MDIO. See *Section 22.2.4.5 of the IEEE 802.3 standard*.

Reading during the shift, operation will return the current contents of the shift register. At the end of the management operation the bits will have shifted back to their original locations. For a read operation the data bits will be updated with data read from the PHY. It is important to write the correct values to the register to ensure a valid PHY management frame is produced.

MDC should not toggle faster than 2.5 MHz (minimum period of 400ns), as defined by the IEEE 802.3 standard. mdc is generated by dividing down pclk. Three bits in the network configuration register determine by how much pclk should be divided to produce MDC. The default is 32 which is acceptable for pclk running up to 80 MHz.

8.2 Interrupts

There are multiple interrupt conditions that are detected within GEM. These are ORed to make a single interrupt (or multiple interrupts if priority queuing is enabled). Depending on the overall system design this may be passed through a further level of interrupt collection (interrupt controller). On receipt of the interrupt signal, the CPU shall enter the interrupt handler. Refer to your documentation for the interrupt controller to identify that it is GEM that is generating the interrupt. To ascertain which interrupt, read the interrupt status register. **Note**: In the default configuration, this register will clear itself after being read, though this may be configured to be write-one-to-clear if desired.

At reset all interrupts are disabled. To enable an interrupt, write to interrupt enable register with the pertinent interrupt bit set to 1. To disable an interrupt, write to interrupt disable register with the pertinent interrupt bit set to 1. To check whether an interrupt is enabled or disabled, read interrupt mask register: if the bit is set to 1, the interrupt is disabled.

8.3 Transmitting Frames

To set up a frame for transmission:

1. Allocate an area of system memory for transmit data. This does not have to be contiguous, varying byte lengths can be used if they conclude on byte borders.
2. Write the address of the first buffer descriptor to transmit buffer descriptor queue pointer. When priority queues are used, each queue's buffer descriptor queue pointer should be updated and Q0's pointer should be updated last of all.
3. Enable transmit in the network control register.
4. Set-up the transmit buffer list by writing buffer addresses to word zero of the transmit buffer descriptor entries and control and length to word one. Make sure there is an extra dummy descriptor at the end of the list with its "used bit" set.
5. Write data for transmission into the buffers pointed to by the descriptors

6. Enable appropriate interrupts
7. Write to the transmit start bit in the network control register, or toggle the trigger_dma_tx_start pin

8.4 Receiving Frames

When a frame is received and the receive circuits are enabled, GEM checks the address and, in the following cases, the frame is written to system memory if:

- It matches one of the four specific address registers
- It matches one of the four type ID registers
- It matches the hash address function
- It is a broadcast address (0xFFFFFFFFFFFF) and broadcasts are allowed
- GEM is configured to “copy all frames”
- A match is found in the external address filtering interface

The register receive buffer queue pointer points to the next entry in the receive buffer descriptor list and GEM uses this as the address in system memory to write the frame to.

Once the frame has been completely and successfully received and written to system memory, GEM then updates the receive buffer descriptor entry (*Table 26 Receive Buffer Descriptor Entry* on page 62) with the reason for the address match and marks the area as being owned by software. Once this is complete, a receive complete interrupt is set. Software is then responsible for copying the data to the application area and releasing the buffer (by writing the ownership bit back to 0).

If GEM is unable to write the data at a rate to match the incoming frame, then a receive overrun interrupt is set. If there is no receive buffer available, i.e. the next buffer is still owned by software, a receive buffer not available interrupt is set. If the frame is not successfully received, the receive overrun statistic register is incremented and the frame is discarded without informing software.

9 Clocks and Reset

9.1 Clock Domains

Depending on the configuration selected, GEM has up to nine clock domains used internally within the design. These are described in *Table 56*.

Table 56: Clock Domains

Clock	Speed (MHz)	Description
hclk	10 to 500	AMBA AHB clock used by the DMA block. (the MAX clock speed is reduced to 250MHz if SRAM error correction is enabled)
aclk	10 to 500	AMBA AXI clock used by the DMA block. (the MAX clock speed is reduced to 400MHz if SRAM error correction is enabled)
pclk	10 to 500	AMBA APB clock used by the MAC register block and PCS register block.
tsu_clk	5 to 400	Alternate clock source for the timestamp unit. Timestamp accuracy improves with higher frequencies. To support single step timestamping, tsu_clk frequency must be greater than 1/8th the frequency of tx_clk or rx_clk.
tx_clk	1.25, 2.5,12.5, 25, 125	MAC transmit clock, used by the MAC transmit block (gem_tx, gem_dma_tx_pbuf_rd, and gem_tx_state blocks). In 10/100 GMII mode, tx_clk will run at either 2.5 MHz or 25 MHz as determined by the external PHY MII clock input. When using gigabit mode, the transmit clock must be sourced from a 125 MHz reference clock. Depending upon system architecture, this reference clock may be sourced from an on-chip clock multiplier, generated directly from an off-chip oscillator, or taken from the PHY rx_clk. In RMII mode, tx_clk should be connected to the rmii_tx_clk outputs. In SGMII mode this clock is sourced from gtx_clk. In 100Mbps SGMII mode, nine out of ten gtx_clk pulses need to be deleted to bring the effective frequency down to 12.5MHz and in 10Mbps SGMII down to 1.25MHz by deleting 99 out of 100 pulses.
n_tx_clk	2.5, 25, or 125	Inverted tx_clk used for RGMII and also for the loopback module. For loopback, tx_clk domain signals are re-timed to this clock before being passed to the rx_clk domain receiver inputs
gtx_clk	125	PCS transmit clock. In SGMII applications, this is sourced from the SerDes and needs to be balanced with tx_clk.
gtx20_clk	62.5	PCS transmit clock used at the PCS to PHY interface to support a 20-bit PHY interface. This must be phase aligned with gtx_clk.

Table 56: Clock Domains (Continued)

Clock	Speed (MHz)	Description
rx_clk	1.25, 2.5, 12.5, 25, 62.5, 125	Clock used by the MAC receive synchronization (gem_rx, gem_rx_decode, gem_dma_pbuf_rx_wr, and gem_filter) blocks. In 10/100 and gigabit mode using the GMII/MII interface, this clock is sourced from the rx_clk input of the external PHY and will be either 2.5 MHz, 25 MHz, or 125 MHz. In RMII modes, rx_clk should be connected to the rmii_rx_clk outputs. When the PCS is selected in gigabit mode, the clock must be sourced from either the rbc1 or pcs_rx_clk and will be 62.5MHz for Gigabit PHYs. In 10/100Mbps SGMII mode, the clock needs to be divided down to bring the effective frequency down to 12.5MHz in 100Mbps mode and in 10Mbps SGMII down to 1.25MHz. In 10/100 SGMII, 8 bits of data are transferred from the PCS to the MAC each clock cycle rather than 16.
rbc0 rbc1	Two 62.5MHz clocks 180° out of phase	Clocks used in the PCS receive channel when gem_pcs_legacy_if is defined.
pcs_rx_clk	62.5	Clocks used in the PCS receive channel when gem_pcs_legacy_if is not defined. This clock needs to be balanced with rx_clk.
pcs_rx10_clk	125	Clocks used at the PCS to PHY interface (when gem_pcs_legacy_if is not defined) to support a 10-bit PHY interface. This clock needs to be balanced with rx_clk.
ref_clk	50	50MHz reference clock present only when the RMII configuration option is selected. Both the tx_clk and rx_clk are internally generated from ref_clk within the RMII interface. In order to provide maximum flexibility for you, these generated clocks are fed out and then back into the design. This allows you to determine your own scan insertion technique and helps with clock tree insertion and balancing. These signals can optionally be multiplexed with an external clock to provide clocking in internal loopback mode. gem_clk_ctrl.v (in the delivery package) gives an example of how this can be done.
n_rx_clk	2.5, 25, or 125	Inverted rx_clk used for RGMII

GEM contains handshaking logic and synchronizers that ensure reliable propagation of signals across clock boundaries. Most clocks run asynchronously with respect to each other with data transfers implemented using handshaking logic. In tbi mode rbc1 (or pcs_rx_clk) and rx_clk need to run synchronously and in loopback mode the transmit and receive path need to be synchronous to each other.

As far as GEM is concerned, management data clock (MDC) is not a clock, but a pclk timed output from the registers block.

The maximum frequency of hclk, aclk, and pclk is determined by the speed of the technology library.

The hclk and aclk clock frequency must be chosen such that the bandwidth requirements of the chosen transmit and receive Ethernet data rates are met.

When using internal loopback mode, tx_clk and rx_clk must be provided using the same loopback reference clock, and n_tx_clk must be provided with the inverted version of the loopback reference clock.

It is important that receive and transmit are disabled when making the switch into and out of internal loopback, as the clocks provided may glitch whilst switching to the loopback reference clock. Also TBI mode must be disabled for internal loopback. This is because TBI mode configures the receive path to be 16 bits and the transmit path to 8 bits wide.

When operating at gigabit speed using the GMII interface, the system must provide a 125MHz reference clock to the PHY, normally termed g_tx_clk. This clock must be the same clock used for the tx_clk input of GEM to maintain correct relationship between the reference clock and data on the GMII.

9.2 Clock Requirements

9.2.1 RMII Clocking

When operating in RMII mode, both the tx_clk and rx_clk are internally generated from ref_clk. In order to provide maximum flexibility for you, these generated clocks are fed out and then back into the design. This allows you to determine your own scan insertion technique and helps with clock tree insertion and balancing. These signals can optionally be multiplexed with an external clock to provide clocking in internal loopback mode. gem_clk_ctrl.v (in the delivery package) gives an example of how this can be done, but it is also described in *Figure 25*.

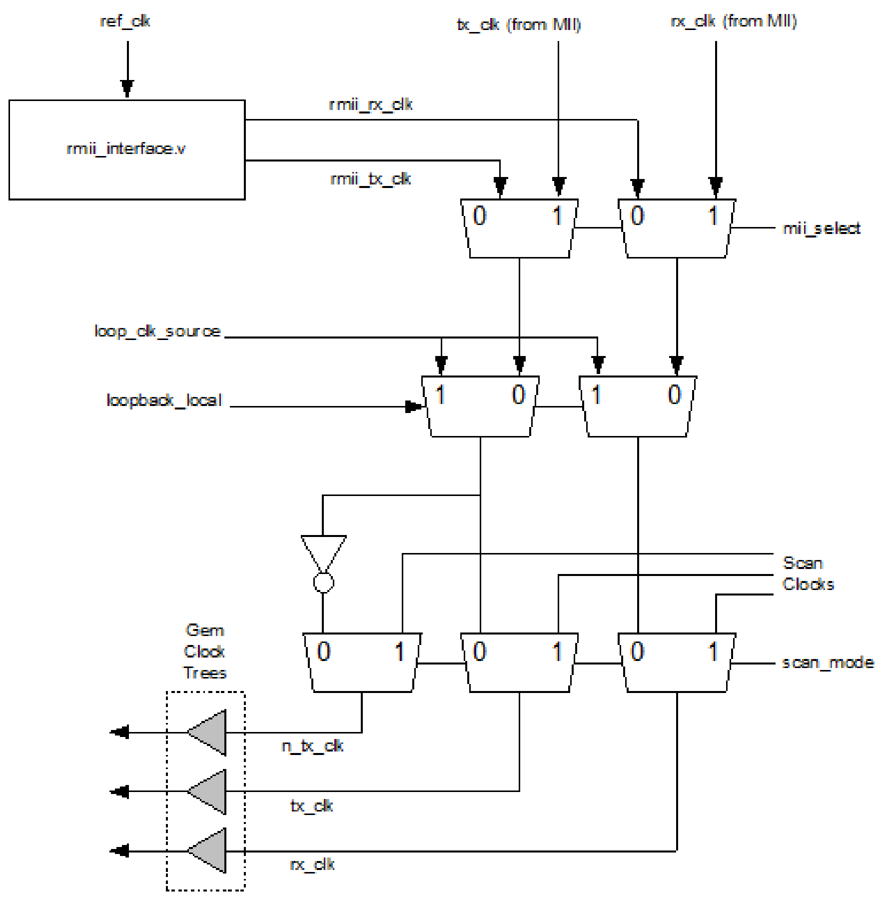

Figure 25: RMII Clocking

9.2.2 RGMII Clocking

For 1Gbps operation, clocks operate at 125MHz. For 100Mbps and 10Mbps operation, clocks operate at 25MHz and 2.5MHz respectively.

The RGMII standard specifies a source synchronous clock with the data. It relies on the clock having a longer path delay than the data so that the data is resampled using the same edge of the clock on which it was generated.

In version 1.3 of the RGMII specification, a 1.5 to 2ns clock delay is achieved through a PCB trace delay. In version 2.0, there is the option of introducing the delay on-chip at the source. Devices which support the internal delay are referred to as RGMII-ID.

Whether to support RGMII-ID is an implementation choice. The Cadence IP supports both versions of the specification, however implementing the on-chip internal delay (ID) is a physical implementation task that requires instancing a delay buffer to achieve the required 2ns delay on rgmii_tx_clk relative to rgmii_txd.

The RGMII module does not generate the rgmii_tx_clk signal. This must be generated by the system clock controller, and provided to both the RGMII module and to the PHY. To ensure that the skew requirements of the RGMII are met, for RGMII version 1.3, the delays on this external clock output should be matched to the RGMII transmit data and control outputs, for RGMII-ID the transmit clock output should be delayed 2ns relative to these outputs.

The RGMII uses the transmit clock to control the multiplexer logic in addition to its use as a real clock. When this clock is being used as a data signal, it is sourced from rgmii_tx_clk_sig. rgmii_tx_clk_sig should be a slightly delayed version of rgmii_tx_clk.

In *Figure 26* showing RGMII transmit operation the gmii_txd[7:0], data signals are sampled on the rising edge of rgmii_tx_clk. The lower nibble of the data byte rgmii_txd[3:0] is available on the rising edge of rgmii_tx_clk and the upper nibble is available on the following falling edge of rgmii_tx_clk (if sel_mii_on_rgmii is active to select MII operation then gmii_txd is replaced with {txd[3:0], txd[3:0]} in *Figure 26* to make the upper and lower nibbles identical and a delayed version of tx_er is output at the top-level). The signal rgmii_tx_clk_sig selects the lower nibble to be output on rgmii_txd when high and the upper nibble when low. The rgmii_tx_clk_sig signal needs to be a delayed version of rgmii_tx_clk so that the relevant input to the mux driving rgmii_txd is stable when rgmii_tx_clk_sig changes.

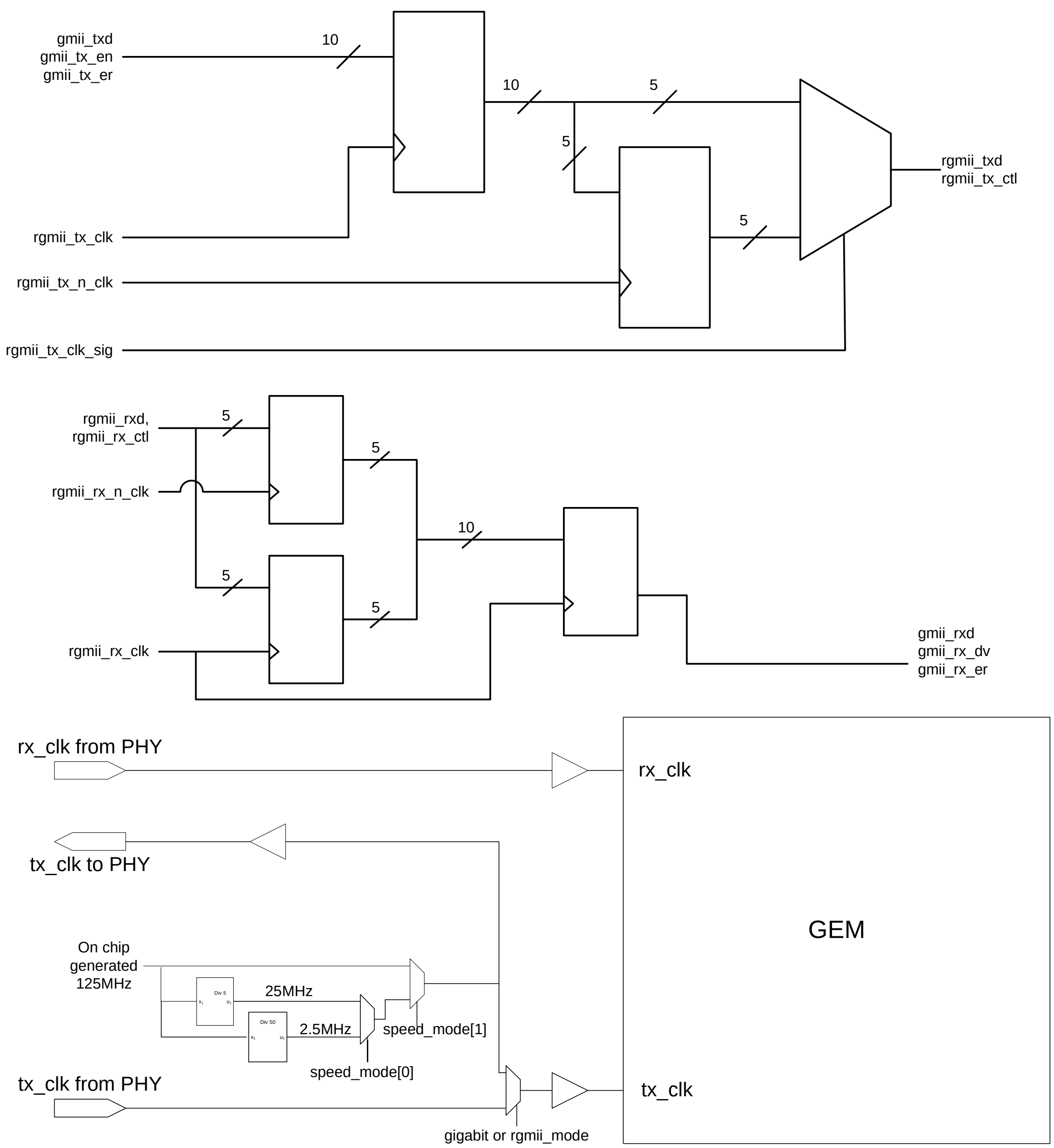

Figure 26: RGMII Clocking

9.2.3 PCS Clocking

In *Table 57* – tbi, gigabit, and speed reflect the status of bits 11, 10, and 0 in the network configuration register and are output at the top level as speed_mode bits 2, 1, and 0.

Table 57: Speed Modes

Speed mode bits 2 1 0	Speed (Mbps)	Interface
1 1 x	1000	Using TBI
0 1 x	1000	Using GMII
0 0 1	100	Using MII
0 0 0	10	Using MII
1 0 1	100	Using SGMII
1 0 0	10	Using SGMII

Figure 27 shows a typical clock tree for configurations using a legacy TBI interface with rbc0 and rbc1 clocks.

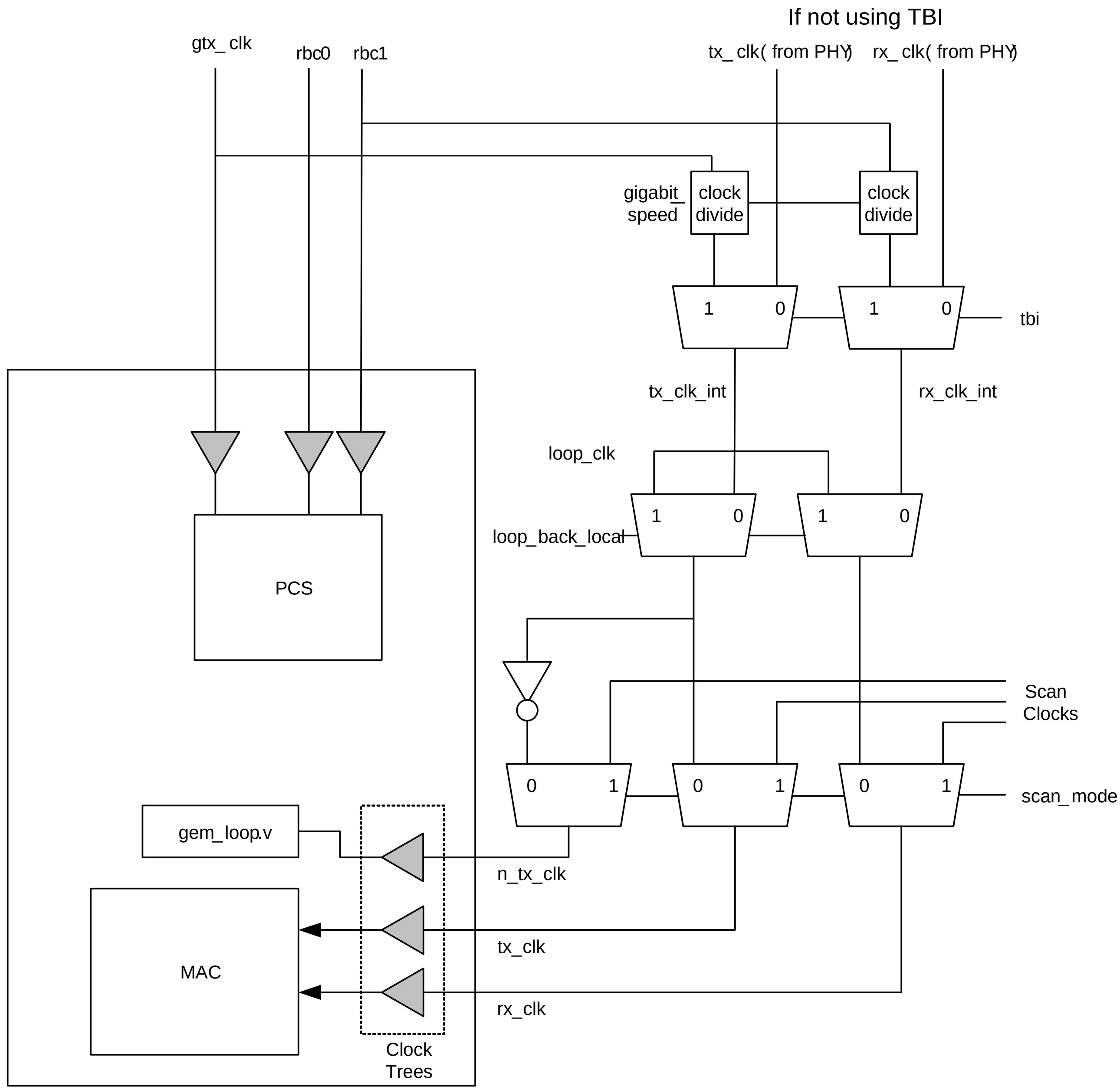

Figure 27: Example of a Clock Tree for TBI

Figure 28 shows the clock tree for using 10 and 20-bit PCS configurations, gtx20_clk is only used if the 20-bit interface is configured and pcs_rx10_clk is only used if the 10-bit interface is configured. The transmit and receive clocks require a common source. For a SGMII interface, the receive PCS is clocked at 62.5MHz and the transmit PCS at 125MHz (with a wrapper clocked at 62.5MHz to provide a 20-bit interface to the SerDes transmit path):

- gtx_clk is 125MHz
- gtx20_clk is 62.5MHz – only used in 20-bit mode
- tx_clk is 125MHz for 1Gbps operation, 12.5MHz for 100Mbps, and 1.25MHz for 10Mbps
- pcs_rx_clk is 62.5MHz
- pcs_rx10_clk is 125MHz – not used in 20-bit mode
- rx_clk is 62.5MHz for 1Gbps operation, 12.5MHz for 100Mbps, and 1.25MHz for 10Mbps

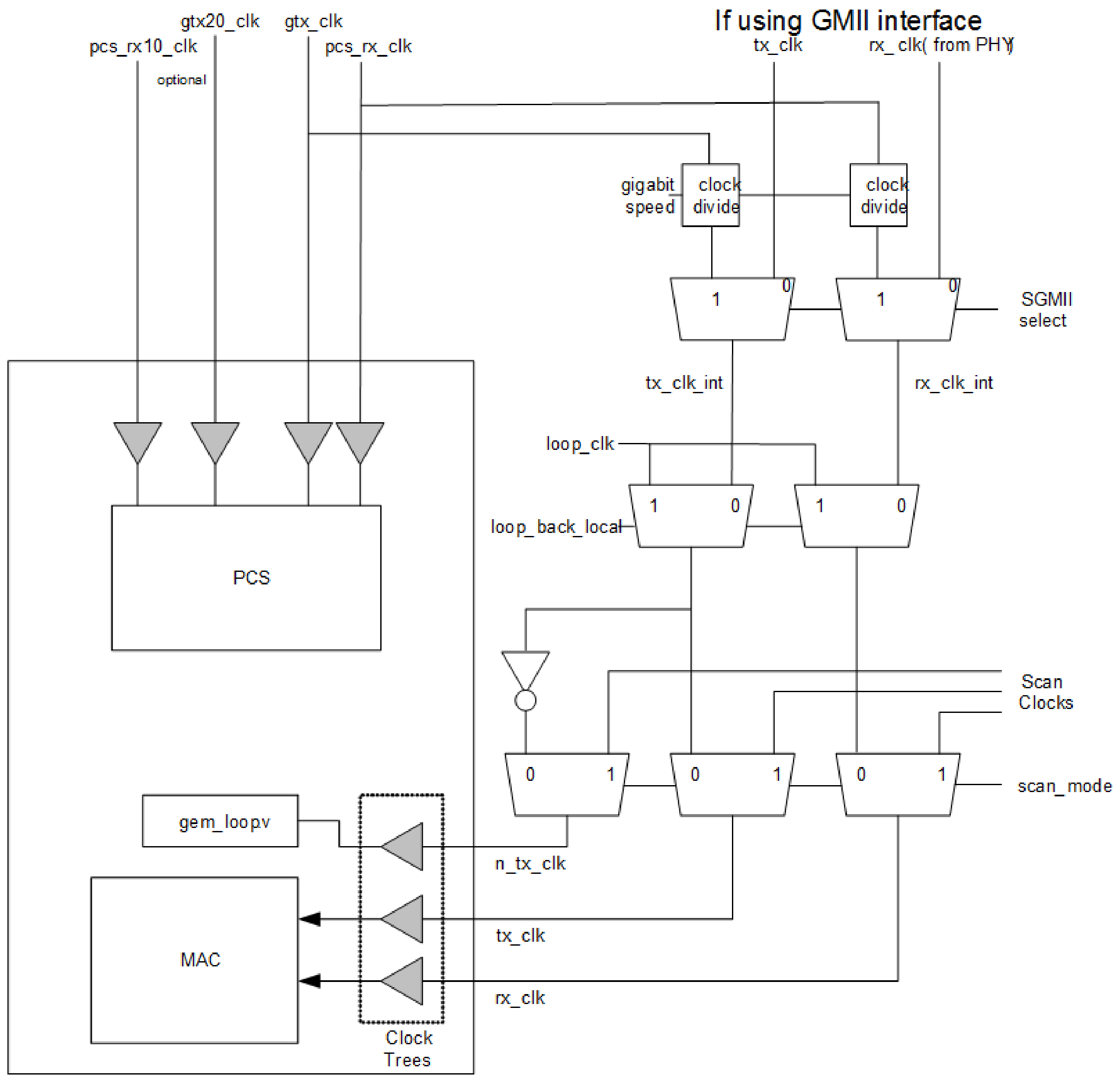

Figure 28: Example of a Clock Tree Using PCS Configuration

9.2.4 Resets

Each clock domain requires an active low reset to be provided. These resets are used directly to reset the synchronous parts of the design. The reset signals may be set low asynchronously with respect to the clock, but must be set high synchronously.

9.2.5 Metastability

To address the problem of metastability, all critical signals passing between clock domains are synchronized by passing through at least two flip-flops as shown in *Figure 29*. Because the clocks can all be asynchronous with respect to each other, each data transfer across a clock domain is implemented using handshaking logic.

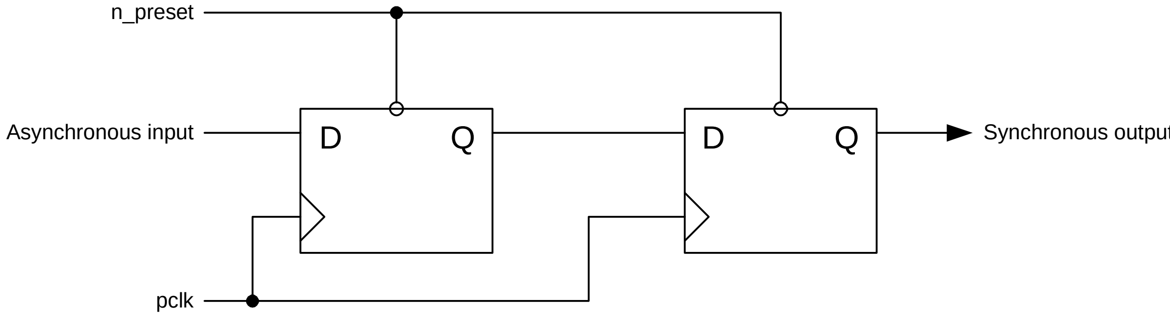

Figure 29: Metastability

10 Verification

10.1 Verification Environment

GEM is delivered with two verification environments in the func_ver/sanity directory. The vlog_tb directory contains a testbench utilizing Perl and Verilog constructs.

The demo_tb directory contains a UVM simulation environment which demonstrates the use of VIP to exercise GEM.

10.2 Verilog TB Simulation Scripts

There are two supplied testbench in the func_ver/sanity directory. The vlog_tb testbench has been developed around the RTL description and has been used with a synthesized netlist (not provided). Numerous tests are provided with the testbench.

Simulation should be performed in the work directory which contains a Makefile to run the supplied tests. This is configured to use the Cadence Incisive® Simulator.

Test scenarios are stored in the func_ver/sanity/vlog_tb/tests/tests_<configuration> directories. These test scenarios are translated using a Perl program called trans.pl, stored in the func_ver/sanity/vlog_tb/scripts directory. The outputs are test stimuli and comparison data stored in the work/files directory.

To simulate a test scenario, the following commands should be invoked from the root directory of the IP:

```
cd work

make run_sim TB=directed CFG=<config_name> TST=<test_name>
```

10.2.1 Example Test Scenario

`rx_filter` is an example test scenario.

This will run the specified test from the /func_ver/sanity/vlog_tb/tests/common/filter directory.

If a simulator other than ncsim is used, take the following steps:
1. Compile and elaborate the RTL and testbench. The RTL source files are present in the hdl/hdl_src directory and the testbench source files are listed in the directory func_ver/sanity/vlog_tb/src/common. The top-level module for elaboration is called tb_gem.
2. Translate the test scenario using the trans.pl Perl program.
3. Invoke the simulator on the elaborated tb_gem module. The testbench will automatically stop at the end of the test scenario.

10.3 Demo Testbench

The GEM Demo Testbench provides a starting point to the integration of related Cadence VIPs and has been developed to demonstrate base features of the GEM IP through simple test cases. This testbench also contains support to interface C functions and models to demonstrate basic health-checking of the DUT using a more system-oriented approach.

Please refer to the *GEM Demo TestBench User Guide* for details on verification procedures and running simulation.

10.3.1 Running the Demo TB

This simulation environment demonstrates the use of VIP and UVM to test the IP component.

Change to directory gem_gxl/work

The Demo Testbench is supplied with example test scenarios written in C and system Verilog.

To see what test scenarios are available, use the following command:

- make show_tests TB=uvm_demotb

The source for the example C tests is in the directory:

- func_ver/sanity/demo_tb/verif/uvc_lib/cdn_gem_demo/sve/tests_c/

The source for the example UVM tests is in the directory:

- func_ver/sanity/demo_tb/verif/uvc_lib/cdn_gem_demo/sve/tests/

11 Registers

This section contains the full register list and register details. The user's actual memory map will depend on the configuration selected. Once you have chosen a configuration, Cadence can generate an IP-XACT description of the user registers.

Note: The Offset Address in the following table is byte-aligned as required by the CPU. It is therefore four times that supplied on paddr[11:2]

11.1 Register Map of IP7014A

Table 11-1: GEM_IP7014A_UG Register Map

Register Name	Type	Width	Reset Value	Address Offset
network_control	RW	32	0x0000 0000	0x0000
network_config	RW	32	0x0008 0000	0x0004
network_status	RO	32	0x0000 0004	0x0008
user_io_register	RW	32	0x0000 0000	0x000C
dma_config	RW	32	0x0002 07C4	0x0010
transmit_status	RW	32	0x0000 0000	0x0014
receive_q_ptr	RW	32	0x0000 0000	0x0018
transmit_q_ptr	RW	32	0x0000 0000	0x001C
receive_status	RW	32	0x0000 0000	0x0020
int_status	RW	32	0x0000 0000	0x0024
int_enable	RW	32	0x0000 0000	0x0028
int_disable	RW	32	0x0000 0000	0x002C
int_mask	RO	32	0xFFFF FEFF	0x0030
phy_management	RW	32	0x0000 0000	0x0034
pause_time	RO	32	0x0000 0000	0x0038
tx_pause_quantum	RW	32	0xFFFF FFFF	0x003C
pbuf_txcutthru	RW	32	0x0000 3FFF	0x0040
pbuf_rxcutthru	RW	32	0x0000 07FF	0x0044
jumbo_max_length	RW	32	0x0000 2800	0x0048
external_fifo_interface	RW	32	0x0000 0000	0x004C
axi_max_pipeline	RW	32	0x0000 0101	0x0054
rsc_control	RW	32	0x0000 0000	0x0058
int_moderation	RW	32	0x0000 0000	0x005C
sys_wake_time	RW	32	0x0000 0000	0x0060
lockup_config	RW	32	0x07FF FFFF	0x0068
mac_lockup_time	RW	32	0x07FF FFFF	0x006C
lockup_config3	RW	32	0x0000 0000	0x0070
rx_water_mark	RW	32	0x0000 0000	0x007C
hash_bottom	RW	32	0x0000 0000	0x0080
hash_top	RW	32	0x0000 0000	0x0084
spec_add1_bottom	RW	32	0x0000 0000	0x0088
spec_add1_top	RW	32	0x0000 0000	0x008C
spec_add2_bottom	RW	32	0x0000 0000	0x0090
spec_add2_top	RW	32	0x0000 0000	0x0094
spec_add3_bottom	RW	32	0x0000 0000	0x0098
spec_add3_top	RW	32	0x0000 0000	0x009C
spec_add4_bottom	RW	32	0x0000 0000	0x00A0
spec_add4_top	RW	32	0x0000 0000	0x00A4
spec_type1	RW	32	0x0000 0000	0x00A8
spec_type2	RW	32	0x0000 0000	0x00AC
spec_type3	RW	32	0x0000 0000	0x00B0
spec_type4	RW	32	0x0000 0000	0x00B4
wol_register	RW	32	0x0000 0000	0x00B8
stretch_ratio	RW	32	0x0000 0000	0x00BC
stacked_vlan	RW	32	0x0000 0000	0x00C0
tx_pfc_pause	RW	32	0x0000 0000	0x00C4
mask_add1_bottom	RW	32	0x0000 0000	0x00C8
mask_add1_top	RW	32	0x0000 0000	0x00CC
dma_addr_or_mask	RW	32	0x0000 0000	0x00D0
rx_ptp_unicast	RW	32	0x0000 0000	0x00D4
tx_ptp_unicast	RW	32	0x0000 0000	0x00D8

Register Name	Type	Width	Reset Value	Address Offset
tsu_nsec_cmp	RW	32	0x0000 0000	0x00DC
tsu_sec_cmp	RW	32	0x0000 0000	0x00E0
tsu_msb_sec_cmp	RW	32	0x0000 0000	0x00E4
tsu_ptp_tx_msb_sec	RO	32	0x0000 0000	0x00E8
tsu_ptp_rx_msb_sec	RO	32	0x0000 0000	0x00EC
tsu_peer_tx_msb_sec	RO	32	0x0000 0000	0x00F0
tsu_peer_rx_msb_sec	RO	32	0x0000 0000	0x00F4
dpram_fill_dbg	RW	32	0x0000 0000	0x00F8
revision_reg	RO	32	0x0007 0200	0x00FC
octets_txed_bottom	RO	32	0x0000 0000	0x0100
octets_txed_top	RO	32	0x0000 0000	0x0104
frames_txed_ok	RO	32	0x0000 0000	0x0108
broadcast_txed	RO	32	0x0000 0000	0x010C
multicast_txed	RO	32	0x0000 0000	0x0110
pause_frames_txed	RO	32	0x0000 0000	0x0114
frames_txed_64	RO	32	0x0000 0000	0x0118
frames_txed_65	RO	32	0x0000 0000	0x011C
frames_txed_128	RO	32	0x0000 0000	0x0120
frames_txed_256	RO	32	0x0000 0000	0x0124
frames_txed_512	RO	32	0x0000 0000	0x0128
frames_txed_1024	RO	32	0x0000 0000	0x012C
frames_txed_1519	RO	32	0x0000 0000	0x0130
tx_underruns	RO	32	0x0000 0000	0x0134
single_collisions	RO	32	0x0000 0000	0x0138
multiple_collisions	RO	32	0x0000 0000	0x013C
excessive_collisions	RO	32	0x0000 0000	0x0140
late_collisions	RO	32	0x0000 0000	0x0144
deferred_frames	RO	32	0x0000 0000	0x0148
crs_errors	RO	32	0x0000 0000	0x014C
octets_rxed_bottom	RO	32	0x0000 0000	0x0150
octets_rxed_top	RO	32	0x0000 0000	0x0154
frames_rxed_ok	RO	32	0x0000 0000	0x0158
broadcast_rxed	RO	32	0x0000 0000	0x015C
multicast_rxed	RO	32	0x0000 0000	0x0160
pause_frames_rxed	RO	32	0x0000 0000	0x0164
frames_rxed_64	RO	32	0x0000 0000	0x0168
frames_rxed_65	RO	32	0x0000 0000	0x016C
frames_rxed_128	RO	32	0x0000 0000	0x0170
frames_rxed_256	RO	32	0x0000 0000	0x0174
frames_rxed_512	RO	32	0x0000 0000	0x0178
frames_rxed_1024	RO	32	0x0000 0000	0x017C
frames_rxed_1519	RO	32	0x0000 0000	0x0180
undersize_frames	RO	32	0x0000 0000	0x0184
excessive_rx_length	RO	32	0x0000 0000	0x0188
rx_jabbers	RO	32	0x0000 0000	0x018C
fcs_errors	RO	32	0x0000 0000	0x0190
rx_length_errors	RO	32	0x0000 0000	0x0194
rx_symbol_errors	RO	32	0x0000 0000	0x0198
alignment_errors	RO	32	0x0000 0000	0x019C
rx_resource_errors	RO	32	0x0000 0000	0x01A0
rx_overruns	RO	32	0x0000 0000	0x01A4
rx_ip_ck_errors	RO	32	0x0000 0000	0x01A8
rx_tcp_ck_errors	RO	32	0x0000 0000	0x01AC
rx_udp_ck_errors	RO	32	0x0000 0000	0x01B0
auto_flushed_pkts	RO	32	0x0000 0000	0x01B4
tsu_timer_incr_sub_nsec	RW	32	0x0000 0000	0x01BC
tsu_timer_msb_sec	RW	32	0x0000 0000	0x01C0

Register Name	Type	Width	Reset Value	Address Offset
tsu_strobe_msb_sec	RO	32	0x0000 0000	0x01C4
tsu_strobe_sec	RO	32	0x0000 0000	0x01C8
tsu_strobe_nsec	RO	32	0x0000 0000	0x01CC
tsu_timer_sec	RW	32	0x0000 0000	0x01D0
tsu_timer_nsec	RW	32	0x0000 0000	0x01D4
tsu_timer_adjust	RW	32	0x0000 0000	0x01D8
tsu_timer_incr	RW	32	0x0000 0000	0x01DC
tsu_ptp_tx_sec	RO	32	0x0000 0000	0x01E0
tsu_ptp_tx_nsec	RO	32	0x0000 0000	0x01E4
tsu_ptp_rx_sec	RO	32	0x0000 0000	0x01E8
tsu_ptp_rx_nsec	RO	32	0x0000 0000	0x01EC
tsu_peer_tx_sec	RO	32	0x0000 0000	0x01F0
tsu_peer_tx_nsec	RO	32	0x0000 0000	0x01F4
tsu_peer_rx_sec	RO	32	0x0000 0000	0x01F8
tsu_peer_rx_nsec	RO	32	0x0000 0000	0x01FC
pcs_control	RW	32	0x0000 9040	0x0200
pcs_status	RO	32	0x0000 0109	0x0204
pcs_phy_top_id	RO	32	0x0000 0007	0x0208
pcs_phy_bot_id	RO	32	0x0000 0200	0x020C
pcs_an_adv	RW	32	0x0000 0020	0x0210
pcs_an_lp_base	RO	32	0x0000 0000	0x0214
pcs_an_exp	RO	32	0x0000 0004	0x0218
pcs_an_np_tx	RW	32	0x0000 0000	0x021C
pcs_an_lp_np	RO	32	0x0000 0000	0x0220
pcs_an_ext_status	RO	32	0x0000 8000	0x023C
tx_pause_quantum1	RW	32	0xFFFF FFFF	0x0260
tx_pause_quantum2	RW	32	0xFFFF FFFF	0x0264
tx_pause_quantum3	RW	32	0xFFFF FFFF	0x0268
pfc_status	RO	32	0x0000 0000	0x026C
rx_lpi	RO	32	0x0000 0000	0x0270
rx_lpi_time	RO	32	0x0000 0000	0x0274
tx_lpi	RO	32	0x0000 0000	0x0278
tx_lpi_time	RO	32	0x0000 0000	0x027C
designcfg_debug1	RO	32	0x0498 4310	0x0280
designcfg_debug2	RO	32	0x7AF1 2800	0x0284
designcfg_debug3	RO	32	0x2400 0000	0x0288
designcfg_debug4	RO	32	0x0000 0000	0x028C
designcfg_debug5	RO	32	0x502F A345	0x0290
designcfg_debug6	RO	32	0x07D4 FFFE	0x0294
designcfg_debug7	RO	32	0x0000 0000	0x0298
designcfg_debug8	RO	32	0x1010 0820	0x029C
designcfg_debug9	RO	32	0x0000 0000	0x02A0
designcfg_debug10	RO	32	0x2224 4444	0x02A4
designcfg_debug11	RO	32	0x0000 00FF	0x02A8
designcfg_debug12	RO	32	0x0377 4010	0x02AC
axi_qos_cfg_0	RW	32	0x0000 0000	0x02E0
axi_qos_cfg_1	RW	32	0x0000 0000	0x02E4
axi_qos_cfg_2	RW	32	0x0000 0000	0x02E8
axi_qos_cfg_3	RW	32	0x0000 0000	0x02EC
spec_add5_bottom	RW	32	0x0000 0000	0x0300
spec_add5_top	RW	32	0x0000 0000	0x0304
spec_add6_bottom	RW	32	0x0000 0000	0x0308
spec_add6_top	RW	32	0x0000 0000	0x030C
spec_add7_bottom	RW	32	0x0000 0000	0x0310
spec_add7_top	RW	32	0x0000 0000	0x0314
spec_add8_bottom	RW	32	0x0000 0000	0x0318
spec_add8_top	RW	32	0x0000 0000	0x031C

Register Name	Type	Width	Reset Value	Address Offset
spec_add9_bottom	RW	32	0x0000 0000	0x0320
spec_add9_top	RW	32	0x0000 0000	0x0324
spec_add10_bottom	RW	32	0x0000 0000	0x0328
spec_add10_top	RW	32	0x0000 0000	0x032C
spec_add11_bottom	RW	32	0x0000 0000	0x0330
spec_add11_top	RW	32	0x0000 0000	0x0334
spec_add12_bottom	RW	32	0x0000 0000	0x0338
spec_add12_top	RW	32	0x0000 0000	0x033C
spec_add13_bottom	RW	32	0x0000 0000	0x0340
spec_add13_top	RW	32	0x0000 0000	0x0344
spec_add14_bottom	RW	32	0x0000 0000	0x0348
spec_add14_top	RW	32	0x0000 0000	0x034C
spec_add15_bottom	RW	32	0x0000 0000	0x0350
spec_add15_top	RW	32	0x0000 0000	0x0354
spec_add16_bottom	RW	32	0x0000 0000	0x0358
spec_add16_top	RW	32	0x0000 0000	0x035C
spec_add17_bottom	RW	32	0x0000 0000	0x0360
spec_add17_top	RW	32	0x0000 0000	0x0364
spec_add18_bottom	RW	32	0x0000 0000	0x0368
spec_add18_top	RW	32	0x0000 0000	0x036C
spec_add19_bottom	RW	32	0x0000 0000	0x0370
spec_add19_top	RW	32	0x0000 0000	0x0374
spec_add20_bottom	RW	32	0x0000 0000	0x0378
spec_add20_top	RW	32	0x0000 0000	0x037C
spec_add21_bottom	RW	32	0x0000 0000	0x0380
spec_add21_top	RW	32	0x0000 0000	0x0384
spec_add22_bottom	RW	32	0x0000 0000	0x0388
spec_add22_top	RW	32	0x0000 0000	0x038C
spec_add23_bottom	RW	32	0x0000 0000	0x0390
spec_add23_top	RW	32	0x0000 0000	0x0394
spec_add24_bottom	RW	32	0x0000 0000	0x0398
spec_add24_top	RW	32	0x0000 0000	0x039C
spec_add25_bottom	RW	32	0x0000 0000	0x03A0
spec_add25_top	RW	32	0x0000 0000	0x03A4
spec_add26_bottom	RW	32	0x0000 0000	0x03A8
spec_add26_top	RW	32	0x0000 0000	0x03AC
spec_add27_bottom	RW	32	0x0000 0000	0x03B0
spec_add27_top	RW	32	0x0000 0000	0x03B4
spec_add28_bottom	RW	32	0x0000 0000	0x03B8
spec_add28_top	RW	32	0x0000 0000	0x03BC
spec_add29_bottom	RW	32	0x0000 0000	0x03C0
spec_add29_top	RW	32	0x0000 0000	0x03C4
spec_add30_bottom	RW	32	0x0000 0000	0x03C8
spec_add30_top	RW	32	0x0000 0000	0x03CC
spec_add31_bottom	RW	32	0x0000 0000	0x03D0
spec_add31_top	RW	32	0x0000 0000	0x03D4
spec_add32_bottom	RW	32	0x0000 0000	0x03D8
spec_add32_top	RW	32	0x0000 0000	0x03DC
spec_add33_bottom	RW	32	0x0000 0000	0x03E0
spec_add33_top	RW	32	0x0000 0000	0x03E4
spec_add34_bottom	RW	32	0x0000 0000	0x03E8
spec_add34_top	RW	32	0x0000 0000	0x03EC
spec_add35_bottom	RW	32	0x0000 0000	0x03F0
spec_add35_top	RW	32	0x0000 0000	0x03F4
spec_add36_bottom	RW	32	0x0000 0000	0x03F8
spec_add36_top	RW	32	0x0000 0000	0x03FC
int_q1_status	RO	32	0x0000 0000	0x0400

Register Name	Type	Width	Reset Value	Address Offset
int_q2_status	RO	32	0x0000 0000	0x0404
int_q3_status	RO	32	0x0000 0000	0x0408
int_q4_status	RO	32	0x0000 0000	0x040C
int_q5_status	RO	32	0x0000 0000	0x0410
int_q6_status	RO	32	0x0000 0000	0x0414
int_q7_status	RO	32	0x0000 0000	0x0418
int_q8_status	RO	32	0x0000 0000	0x041C
int_q9_status	RO	32	0x0000 0000	0x0420
int_q10_status	RO	32	0x0000 0000	0x0424
int_q11_status	RO	32	0x0000 0000	0x0428
int_q12_status	RO	32	0x0000 0000	0x042C
int_q13_status	RO	32	0x0000 0000	0x0430
int_q14_status	RO	32	0x0000 0000	0x0434
int_q15_status	RO	32	0x0000 0000	0x0438
transmit_q1_ptr	RW	32	0x0000 0000	0x0440
transmit_q2_ptr	RW	32	0x0000 0000	0x0444
transmit_q3_ptr	RW	32	0x0000 0000	0x0448
transmit_q4_ptr	RW	32	0x0000 0000	0x044C
transmit_q5_ptr	RW	32	0x0000 0000	0x0450
transmit_q6_ptr	RW	32	0x0000 0000	0x0454
transmit_q7_ptr	RW	32	0x0000 0000	0x0458
transmit_q8_ptr	RW	32	0x0000 0000	0x045C
transmit_q9_ptr	RW	32	0x0000 0000	0x0460
transmit_q10_ptr	RW	32	0x0000 0000	0x0464
transmit_q11_ptr	RW	32	0x0000 0000	0x0468
transmit_q12_ptr	RW	32	0x0000 0000	0x046C
transmit_q13_ptr	RW	32	0x0000 0000	0x0470
transmit_q14_ptr	RW	32	0x0000 0000	0x0474
transmit_q15_ptr	RW	32	0x0000 0000	0x0478
receive_q1_ptr	RW	32	0x0000 0000	0x0480
receive_q2_ptr	RW	32	0x0000 0000	0x0484
receive_q3_ptr	RW	32	0x0000 0000	0x0488
receive_q4_ptr	RW	32	0x0000 0000	0x048C
receive_q5_ptr	RW	32	0x0000 0000	0x0490
receive_q6_ptr	RW	32	0x0000 0000	0x0494
receive_q7_ptr	RW	32	0x0000 0000	0x0498
dma_rxbuf_size_q1	RW	32	0x0000 0002	0x04A0
dma_rxbuf_size_q2	RW	32	0x0000 0002	0x04A4
dma_rxbuf_size_q3	RW	32	0x0000 0002	0x04A8
dma_rxbuf_size_q4	RW	32	0x0000 0002	0x04AC
dma_rxbuf_size_q5	RW	32	0x0000 0002	0x04B0
dma_rxbuf_size_q6	RW	32	0x0000 0002	0x04B4
dma_rxbuf_size_q7	RW	32	0x0000 0002	0x04B8
cbs_control	RW	32	0x0000 0000	0x04BC
cbs_idleslope_q_a	RW	32	0x0000 0000	0x04C0
cbs_idleslope_q_b	RW	32	0x0000 0000	0x04C4
upper_tx_q_base_addr	RW	32	0x0000 0000	0x04C8
tx_bd_control	RW	32	0x0000 0000	0x04CC
rx_bd_control	RW	32	0x0000 0000	0x04D0
upper_rx_q_base_addr	RW	32	0x0000 0000	0x04D4
wd_counter	RW	32	0x0000 0007	0x04EC
axi_tx_full_thresh0	RW	32	0x0006 0008	0x04F8
axi_tx_full_thresh1	RW	32	0x0000 0000	0x04FC
screening_type_1_register_0	RW	32	0x0000 0000	0x0500
screening_type_1_register_1	RW	32	0x0000 0000	0x0504
screening_type_1_register_2	RW	32	0x0000 0000	0x0508
screening_type_1_register_3	RW	32	0x0000 0000	0x050C

Register Name	Type	Width	Reset Value	Address Offset
screening_type_1_register_4	RW	32	0x0000 0000	0x0510
screening_type_1_register_5	RW	32	0x0000 0000	0x0514
screening_type_1_register_6	RW	32	0x0000 0000	0x0518
screening_type_1_register_7	RW	32	0x0000 0000	0x051C
screening_type_1_register_8	RW	32	0x0000 0000	0x0520
screening_type_1_register_9	RW	32	0x0000 0000	0x0524
screening_type_1_register_10	RW	32	0x0000 0000	0x0528
screening_type_1_register_11	RW	32	0x0000 0000	0x052C
screening_type_1_register_12	RW	32	0x0000 0000	0x0530
screening_type_1_register_13	RW	32	0x0000 0000	0x0534
screening_type_1_register_14	RW	32	0x0000 0000	0x0538
screening_type_1_register_15	RW	32	0x0000 0000	0x053C
screening_type_2_register_0	RW	32	0x0000 0000	0x0540
screening_type_2_register_1	RW	32	0x0000 0000	0x0544
screening_type_2_register_2	RW	32	0x0000 0000	0x0548
screening_type_2_register_3	RW	32	0x0000 0000	0x054C
screening_type_2_register_4	RW	32	0x0000 0000	0x0550
screening_type_2_register_5	RW	32	0x0000 0000	0x0554
screening_type_2_register_6	RW	32	0x0000 0000	0x0558
screening_type_2_register_7	RW	32	0x0000 0000	0x055C
screening_type_2_register_8	RW	32	0x0000 0000	0x0560
screening_type_2_register_9	RW	32	0x0000 0000	0x0564
screening_type_2_register_10	RW	32	0x0000 0000	0x0568
screening_type_2_register_11	RW	32	0x0000 0000	0x056C
screening_type_2_register_12	RW	32	0x0000 0000	0x0570
screening_type_2_register_13	RW	32	0x0000 0000	0x0574
screening_type_2_register_14	RW	32	0x0000 0000	0x0578
screening_type_2_register_15	RW	32	0x0000 0000	0x057C
tx_sched_ctrl	RW	32	0x0000 0000	0x0580
bw_rate_limit_q0to3	RW	32	0x0000 0000	0x0590
bw_rate_limit_q4to7	RW	32	0x0000 0000	0x0594
bw_rate_limit_q8to11	RW	32	0x0000 0000	0x0598
bw_rate_limit_q12to15	RW	32	0x0000 0000	0x059C
tx_q_seg_alloc_q_lower	RW	32	0x0000 0000	0x05A0
tx_q_seg_alloc_q_upper	RW	32	0x0000 0000	0x05A4
receive_q8_ptr	RW	32	0x0000 0000	0x05C0
receive_q9_ptr	RW	32	0x0000 0000	0x05C4
receive_q10_ptr	RW	32	0x0000 0000	0x05C8
receive_q11_ptr	RW	32	0x0000 0000	0x05CC
receive_q12_ptr	RW	32	0x0000 0000	0x05D0
receive_q13_ptr	RW	32	0x0000 0000	0x05D4
receive_q14_ptr	RW	32	0x0000 0000	0x05D8
receive_q15_ptr	RW	32	0x0000 0000	0x05DC
dma_rxbuf_size_q8	RW	32	0x0000 0002	0x05E0
dma_rxbuf_size_q9	RW	32	0x0000 0002	0x05E4
dma_rxbuf_size_q10	RW	32	0x0000 0002	0x05E8
dma_rxbuf_size_q11	RW	32	0x0000 0002	0x05EC
dma_rxbuf_size_q12	RW	32	0x0000 0002	0x05F0
dma_rxbuf_size_q13	RW	32	0x0000 0002	0x05F4
dma_rxbuf_size_q14	RW	32	0x0000 0002	0x05F8
dma_rxbuf_size_q15	RW	32	0x0000 0002	0x05FC
int_q1_enable	RW	32	0x0000 0000	0x0600
int_q2_enable	RW	32	0x0000 0000	0x0604
int_q3_enable	RW	32	0x0000 0000	0x0608
int_q4_enable	RW	32	0x0000 0000	0x060C
int_q5_enable	RW	32	0x0000 0000	0x0610
int_q6_enable	RW	32	0x0000 0000	0x0614

Register Name	Type	Width	Reset Value	Address Offset
int_q7_enable	RW	32	0x0000 0000	0x0618
int_q1_disable	RW	32	0x0000 0000	0x0620
int_q2_disable	RW	32	0x0000 0000	0x0624
int_q3_disable	RW	32	0x0000 0000	0x0628
int_q4_disable	RW	32	0x0000 0000	0x062C
int_q5_disable	RW	32	0x0000 0000	0x0630
int_q6_disable	RW	32	0x0000 0000	0x0634
int_q7_disable	RW	32	0x0000 0000	0x0638
int_q1_mask	RO	32	0x0000 08E6	0x0640
int_q2_mask	RO	32	0x0000 08E6	0x0644
int_q3_mask	RO	32	0x0000 08E6	0x0648
int_q4_mask	RO	32	0x0000 08E6	0x064C
int_q5_mask	RO	32	0x0000 08E6	0x0650
int_q6_mask	RO	32	0x0000 08E6	0x0654
int_q7_mask	RO	32	0x0000 08E6	0x0658
int_q8_enable	RW	32	0x0000 0000	0x0660
int_q9_enable	RW	32	0x0000 0000	0x0664
int_q10_enable	RW	32	0x0000 0000	0x0668
int_q11_enable	RW	32	0x0000 0000	0x066C
int_q12_enable	RW	32	0x0000 0000	0x0670
int_q13_enable	RW	32	0x0000 0000	0x0674
int_q14_enable	RW	32	0x0000 0000	0x0678
int_q15_enable	RW	32	0x0000 0000	0x067C
int_q8_disable	RW	32	0x0000 0000	0x0680
int_q9_disable	RW	32	0x0000 0000	0x0684
int_q10_disable	RW	32	0x0000 0000	0x0688
int_q11_disable	RW	32	0x0000 0000	0x068C
int_q12_disable	RW	32	0x0000 0000	0x0690
int_q13_disable	RW	32	0x0000 0000	0x0694
int_q14_disable	RW	32	0x0000 0000	0x0698
int_q15_disable	RW	32	0x0000 0000	0x069C
int_q8_mask	RO	32	0x0000 08E6	0x06A0
int_q9_mask	RO	32	0x0000 08E6	0x06A4
int_q10_mask	RO	32	0x0000 08E6	0x06A8
int_q11_mask	RO	32	0x0000 08E6	0x06AC
int_q12_mask	RO	32	0x0000 08E6	0x06B0
int_q13_mask	RO	32	0x0000 08E6	0x06B4
int_q14_mask	RO	32	0x0000 08E6	0x06B8
int_q15_mask	RO	32	0x0000 08E6	0x06BC
screening_type_2_ethertype_reg_0	RW	32	0x0000 0000	0x06E0
screening_type_2_ethertype_reg_1	RW	32	0x0000 0000	0x06E4
screening_type_2_ethertype_reg_2	RW	32	0x0000 0000	0x06E8
screening_type_2_ethertype_reg_3	RW	32	0x0000 0000	0x06EC
screening_type_2_ethertype_reg_4	RW	32	0x0000 0000	0x06F0
screening_type_2_ethertype_reg_5	RW	32	0x0000 0000	0x06F4
screening_type_2_ethertype_reg_6	RW	32	0x0000 0000	0x06F8
screening_type_2_ethertype_reg_7	RW	32	0x0000 0000	0x06FC
type2_compare_0_word_0	RW	32	0x0000 0000	0x0700
type2_compare_0_word_1	RW	32	0x0000 0000	0x0704
type2_compare_1_word_0	RW	32	0x0000 0000	0x0708
type2_compare_1_word_1	RW	32	0x0000 0000	0x070C
type2_compare_2_word_0	RW	32	0x0000 0000	0x0710
type2_compare_2_word_1	RW	32	0x0000 0000	0x0714
type2_compare_3_word_0	RW	32	0x0000 0000	0x0718
type2_compare_3_word_1	RW	32	0x0000 0000	0x071C
type2_compare_4_word_0	RW	32	0x0000 0000	0x0720
type2_compare_4_word_1	RW	32	0x0000 0000	0x0724

Register Name	Type	Width	Reset Value	Address Offset
type2_compare_5_word_0	RW	32	0x0000 0000	0x0728
type2_compare_5_word_1	RW	32	0x0000 0000	0x072C
type2_compare_6_word_0	RW	32	0x0000 0000	0x0730
type2_compare_6_word_1	RW	32	0x0000 0000	0x0734
type2_compare_7_word_0	RW	32	0x0000 0000	0x0738
type2_compare_7_word_1	RW	32	0x0000 0000	0x073C
type2_compare_8_word_0	RW	32	0x0000 0000	0x0740
type2_compare_8_word_1	RW	32	0x0000 0000	0x0744
type2_compare_9_word_0	RW	32	0x0000 0000	0x0748
type2_compare_9_word_1	RW	32	0x0000 0000	0x074C
type2_compare_10_word_0	RW	32	0x0000 0000	0x0750
type2_compare_10_word_1	RW	32	0x0000 0000	0x0754
type2_compare_11_word_0	RW	32	0x0000 0000	0x0758
type2_compare_11_word_1	RW	32	0x0000 0000	0x075C
type2_compare_12_word_0	RW	32	0x0000 0000	0x0760
type2_compare_12_word_1	RW	32	0x0000 0000	0x0764
type2_compare_13_word_0	RW	32	0x0000 0000	0x0768
type2_compare_13_word_1	RW	32	0x0000 0000	0x076C
type2_compare_14_word_0	RW	32	0x0000 0000	0x0770
type2_compare_14_word_1	RW	32	0x0000 0000	0x0774
type2_compare_15_word_0	RW	32	0x0000 0000	0x0778
type2_compare_15_word_1	RW	32	0x0000 0000	0x077C
type2_compare_16_word_0	RW	32	0x0000 0000	0x0780
type2_compare_16_word_1	RW	32	0x0000 0000	0x0784
type2_compare_17_word_0	RW	32	0x0000 0000	0x0788
type2_compare_17_word_1	RW	32	0x0000 0000	0x078C
type2_compare_18_word_0	RW	32	0x0000 0000	0x0790
type2_compare_18_word_1	RW	32	0x0000 0000	0x0794
type2_compare_19_word_0	RW	32	0x0000 0000	0x0798
type2_compare_19_word_1	RW	32	0x0000 0000	0x079C
type2_compare_20_word_0	RW	32	0x0000 0000	0x07A0
type2_compare_20_word_1	RW	32	0x0000 0000	0x07A4
type2_compare_21_word_0	RW	32	0x0000 0000	0x07A8
type2_compare_21_word_1	RW	32	0x0000 0000	0x07AC
type2_compare_22_word_0	RW	32	0x0000 0000	0x07B0
type2_compare_22_word_1	RW	32	0x0000 0000	0x07B4
type2_compare_23_word_0	RW	32	0x0000 0000	0x07B8
type2_compare_23_word_1	RW	32	0x0000 0000	0x07BC
type2_compare_24_word_0	RW	32	0x0000 0000	0x07C0
type2_compare_24_word_1	RW	32	0x0000 0000	0x07C4
type2_compare_25_word_0	RW	32	0x0000 0000	0x07C8
type2_compare_25_word_1	RW	32	0x0000 0000	0x07CC
type2_compare_26_word_0	RW	32	0x0000 0000	0x07D0
type2_compare_26_word_1	RW	32	0x0000 0000	0x07D4
type2_compare_27_word_0	RW	32	0x0000 0000	0x07D8
type2_compare_27_word_1	RW	32	0x0000 0000	0x07DC
type2_compare_28_word_0	RW	32	0x0000 0000	0x07E0
type2_compare_28_word_1	RW	32	0x0000 0000	0x07E4
type2_compare_29_word_0	RW	32	0x0000 0000	0x07E8
type2_compare_29_word_1	RW	32	0x0000 0000	0x07EC
type2_compare_30_word_0	RW	32	0x0000 0000	0x07F0
type2_compare_30_word_1	RW	32	0x0000 0000	0x07F4
type2_compare_31_word_0	RW	32	0x0000 0000	0x07F8
type2_compare_31_word_1	RW	32	0x0000 0000	0x07FC
enst_start_time_q8	RW	32	0x0000 0000	0x0800
enst_start_time_q9	RW	32	0x0000 0000	0x0804
enst_start_time_q10	RW	32	0x0000 0000	0x0808

Register Name	Type	Width	Reset Value	Address Offset
enst_start_time_q11	RW	32	0x0000 0000	0x080C
enst_start_time_q12	RW	32	0x0000 0000	0x0810
enst_start_time_q13	RW	32	0x0000 0000	0x0814
enst_start_time_q14	RW	32	0x0000 0000	0x0818
enst_start_time_q15	RW	32	0x0000 0000	0x081C
enst_on_time_q8	RW	32	0x0001 FFFF	0x0820
enst_on_time_q9	RW	32	0x0001 FFFF	0x0824
enst_on_time_q10	RW	32	0x0001 FFFF	0x0828
enst_on_time_q11	RW	32	0x0001 FFFF	0x082C
enst_on_time_q12	RW	32	0x0001 FFFF	0x0830
enst_on_time_q13	RW	32	0x0001 FFFF	0x0834
enst_on_time_q14	RW	32	0x0001 FFFF	0x0838
enst_on_time_q15	RW	32	0x0001 FFFF	0x083C
enst_off_time_q8	RW	32	0x0000 0000	0x0840
enst_off_time_q9	RW	32	0x0000 0000	0x0844
enst_off_time_q10	RW	32	0x0000 0000	0x0848
enst_off_time_q11	RW	32	0x0000 0000	0x084C
enst_off_time_q12	RW	32	0x0000 0000	0x0850
enst_off_time_q13	RW	32	0x0000 0000	0x0854
enst_off_time_q14	RW	32	0x0000 0000	0x0858
enst_off_time_q15	RW	32	0x0000 0000	0x085C
enst_control	RW	32	0x0000 0000	0x0880
frer_timeout	RW	32	0x0000 0000	0x08A0
frer_red_tag	RW	32	0x4000 F1C1	0x08A4
frer_control_1_a	RW	32	0x0000 0000	0x08C0
frer_control_1_b	RW	32	0x0000 0000	0x08C4
frer_statistics_1_a	RO	32	0x0000 0000	0x08C8
frer_statistics_1_b	RO	32	0x0000 0000	0x08CC
frer_control_2_a	RW	32	0x0000 0000	0x08D0
frer_control_2_b	RW	32	0x0000 0000	0x08D4
frer_statistics_2_a	RO	32	0x0000 0000	0x08D8
frer_statistics_2_b	RO	32	0x0000 0000	0x08DC
frer_control_3_a	RW	32	0x0000 0000	0x08E0
frer_control_3_b	RW	32	0x0000 0000	0x08E4
frer_statistics_3_a	RO	32	0x0000 0000	0x08E8
frer_statistics_3_b	RO	32	0x0000 0000	0x08EC
frer_control_4_a	RW	32	0x0000 0000	0x08F0
frer_control_4_b	RW	32	0x0000 0000	0x08F4
frer_statistics_4_a	RO	32	0x0000 0000	0x08F8
frer_statistics_4_b	RO	32	0x0000 0000	0x08FC
frer_control_5_a	RW	32	0x0000 0000	0x0900
frer_control_5_b	RW	32	0x0000 0000	0x0904
frer_statistics_5_a	RO	32	0x0000 0000	0x0908
frer_statistics_5_b	RO	32	0x0000 0000	0x090C
frer_control_6_a	RW	32	0x0000 0000	0x0910
frer_control_6_b	RW	32	0x0000 0000	0x0914
frer_statistics_6_a	RO	32	0x0000 0000	0x0918
frer_statistics_6_b	RO	32	0x0000 0000	0x091C
frer_control_7_a	RW	32	0x0000 0000	0x0920
frer_control_7_b	RW	32	0x0000 0000	0x0924
frer_statistics_7_a	RO	32	0x0000 0000	0x0928
frer_statistics_7_b	RO	32	0x0000 0000	0x092C
frer_control_8_a	RW	32	0x0000 0000	0x0930
frer_control_8_b	RW	32	0x0000 0000	0x0934
frer_statistics_8_a	RO	32	0x0000 0000	0x0938
frer_statistics_8_b	RO	32	0x0000 0000	0x093C
frer_control_9_a	RW	32	0x0000 0000	0x0940

Register Name	Type	Width	Reset Value	Address Offset
frer_control_9_b	RW	32	0x0000 0000	0x0944
frer_statistics_9_a	RO	32	0x0000 0000	0x0948
frer_statistics_9_b	RO	32	0x0000 0000	0x094C
frer_control_10_a	RW	32	0x0000 0000	0x0950
frer_control_10_b	RW	32	0x0000 0000	0x0954
frer_statistics_10_a	RO	32	0x0000 0000	0x0958
frer_statistics_10_b	RO	32	0x0000 0000	0x095C
frer_control_11_a	RW	32	0x0000 0000	0x0960
frer_control_11_b	RW	32	0x0000 0000	0x0964
frer_statistics_11_a	RO	32	0x0000 0000	0x0968
frer_statistics_11_b	RO	32	0x0000 0000	0x096C
frer_control_12_a	RW	32	0x0000 0000	0x0970
frer_control_12_b	RW	32	0x0000 0000	0x0974
frer_statistics_12_a	RO	32	0x0000 0000	0x0978
frer_statistics_12_b	RO	32	0x0000 0000	0x097C
frer_control_13_a	RW	32	0x0000 0000	0x0980
frer_control_13_b	RW	32	0x0000 0000	0x0984
frer_statistics_13_a	RO	32	0x0000 0000	0x0988
frer_statistics_13_b	RO	32	0x0000 0000	0x098C
frer_control_14_a	RW	32	0x0000 0000	0x0990
frer_control_14_b	RW	32	0x0000 0000	0x0994
frer_statistics_14_a	RO	32	0x0000 0000	0x0998
frer_statistics_14_b	RO	32	0x0000 0000	0x099C
frer_control_15_a	RW	32	0x0000 0000	0x09A0
frer_control_15_b	RW	32	0x0000 0000	0x09A4
frer_statistics_15_a	RO	32	0x0000 0000	0x09A8
frer_statistics_15_b	RO	32	0x0000 0000	0x09AC
frer_control_16_a	RW	32	0x0000 0000	0x09B0
frer_control_16_b	RW	32	0x0000 0000	0x09B4
frer_statistics_16_a	RO	32	0x0000 0000	0x09B8
frer_statistics_16_b	RO	32	0x0000 0000	0x09BC
rx_q0_flush	RW	32	0x0000 0000	0x0B00
rx_q1_flush	RW	32	0x0000 0000	0x0B04
rx_q2_flush	RW	32	0x0000 0000	0x0B08
rx_q3_flush	RW	32	0x0000 0000	0x0B0C
rx_q4_flush	RW	32	0x0000 0000	0x0B10
rx_q5_flush	RW	32	0x0000 0000	0x0B14
rx_q6_flush	RW	32	0x0000 0000	0x0B18
rx_q7_flush	RW	32	0x0000 0000	0x0B1C
rx_q8_flush	RW	32	0x0000 0000	0x0B20
rx_q9_flush	RW	32	0x0000 0000	0x0B24
rx_q10_flush	RW	32	0x0000 0000	0x0B28
rx_q11_flush	RW	32	0x0000 0000	0x0B2C
rx_q12_flush	RW	32	0x0000 0000	0x0B30
rx_q13_flush	RW	32	0x0000 0000	0x0B34
rx_q14_flush	RW	32	0x0000 0000	0x0B38
rx_q15_flush	RW	32	0x0000 0000	0x0B3C
scr2_reg0_rate_limit	RW	32	0x0000 0000	0x0B40
scr2_reg1_rate_limit	RW	32	0x0000 0000	0x0B44
scr2_reg2_rate_limit	RW	32	0x0000 0000	0x0B48
scr2_reg3_rate_limit	RW	32	0x0000 0000	0x0B4C
scr2_reg4_rate_limit	RW	32	0x0000 0000	0x0B50
scr2_reg5_rate_limit	RW	32	0x0000 0000	0x0B54
scr2_reg6_rate_limit	RW	32	0x0000 0000	0x0B58
scr2_reg7_rate_limit	RW	32	0x0000 0000	0x0B5C
scr2_reg8_rate_limit	RW	32	0x0000 0000	0x0B60
scr2_reg9_rate_limit	RW	32	0x0000 0000	0x0B64

Register Name	Type	Width	Reset Value	Address Offset
scr2_reg10_rate_limit	RW	32	0x0000 0000	0x0B68
scr2_reg11_rate_limit	RW	32	0x0000 0000	0x0B6C
scr2_reg12_rate_limit	RW	32	0x0000 0000	0x0B70
scr2_reg13_rate_limit	RW	32	0x0000 0000	0x0B74
scr2_reg14_rate_limit	RW	32	0x0000 0000	0x0B78
scr2_reg15_rate_limit	RW	32	0x0000 0000	0x0B7C
scr2_rate_status	RO	32	0x0000 0000	0x0B80
asf_int_status	RW	32	0x0000 0000	0x0E00
asf_int_raw_status	RW	32	0x0000 0000	0x0E04
asf_int_mask	RW	32	0x0000 007F	0x0E08
asf_int_test	RW	32	0x0000 0000	0x0E0C
asf_fatal_nonfatal_select	RW	32	0x0000 007F	0x0E10
asf_sram_corr_fault_status	RO	32	0x0000 0000	0x0E20
asf_sram_uncorr_fault_status	RO	32	0x0000 0000	0x0E24
asf_sram_fault_stats	RW	32	0x0000 0000	0x0E28
asf_trans_to_ctrl	RW	32	0x0000 0000	0x0E30
asf_trans_to_fault_mask	RW	32	0x0000 001F	0x0E34
asf_trans_to_fault_status	RW	32	0x0000 0000	0x0E38
asf_protocol_fault_mask	RW	32	0x003F 01FF	0x0E40
asf_protocol_fault_status	RW	32	0x0000 0000	0x0E44
mmsl_control	RW	32	0x0000 0020	0x0F00
mmsl_status	RO	32	0x0000 0000	0x0F04
mmsl_err_stats	RO	32	0x0000 0000	0x0F08
mmsl_ass_ok_count	RO	32	0x0000 0000	0x0F0C
mmsl_frag_count_rx	RO	32	0x0000 0000	0x0F10
mmsl_frag_count_tx	RO	32	0x0000 0000	0x0F14
mmsl_int_status	RW	32	0x0000 0000	0x0F18
mmsl_int_enable	RW	32	0x0000 0000	0x0F1C
mmsl_int_disable	RW	32	0x0000 0000	0x0F20
mmsl_int_mask	RO	32	0x0000 003F	0x0F24
emac_network_control	RW	32	0x0000 0000	0x1000
emac_network_config	RW	32	0x0008 0000	0x1004
emac_network_status	RO	32	0x0000 0004	0x1008
emac_dma_config	RW	32	0x0002 07C4	0x1010
emac_transmit_status	RW	32	0x0000 0000	0x1014
emac_receive_q_ptr	RW	32	0x0000 0000	0x1018
emac_transmit_q_ptr	RW	32	0x0000 0000	0x101C
emac_receive_status	RW	32	0x0000 0000	0x1020
emac_int_status	RW	32	0x0000 0000	0x1024
emac_int_enable	RW	32	0x0000 0000	0x1028
emac_int_disable	RW	32	0x0000 0000	0x102C
emac_int_mask	RO	32	0xFFFF FEFF	0x1030
emac_phy_management	RW	32	0x0000 0000	0x1034
emac_pause_time	RO	32	0x0000 0000	0x1038
emac_tx_pause_quantum	RW	32	0xFFFF FFFF	0x103C
emac_pbuf_txcutthru	RW	32	0x0000 07FF	0x1040
emac_pbuf_rxcutthru	RW	32	0x0000 07FF	0x1044
emac_jumbo_max_length	RW	32	0x0000 2800	0x1048
emac_external_fifo_interface	RW	32	0x0000 0000	0x104C
emac_axi_max_pipeline	RW	32	0x0000 0101	0x1054
emac_int_moderation	RW	32	0x0000 0000	0x105C
emac_sys_wake_time	RW	32	0x0000 0000	0x1060
emac_lockup_config	RW	32	0x07FF FFFF	0x1068
emac_mac_lockup_time	RW	32	0x07FF FFFF	0x106C
emac_lockup_config3	RW	32	0x0000 0000	0x1070
emac_rx_water_mark	RW	32	0x0000 0000	0x107C
emac_hash_bottom	RW	32	0x0000 0000	0x1080

Register Name	Type	Width	Reset Value	Address Offset
emac_hash_top	RW	32	0x0000 0000	0x1084
emac_spec_add1_bottom	RW	32	0x0000 0000	0x1088
emac_spec_add1_top	RW	32	0x0000 0000	0x108C
emac_spec_add2_bottom	RW	32	0x0000 0000	0x1090
emac_spec_add2_top	RW	32	0x0000 0000	0x1094
emac_spec_add3_bottom	RW	32	0x0000 0000	0x1098
emac_spec_add3_top	RW	32	0x0000 0000	0x109C
emac_spec_add4_bottom	RW	32	0x0000 0000	0x10A0
emac_spec_add4_top	RW	32	0x0000 0000	0x10A4
emac_spec_type1	RW	32	0x0000 0000	0x10A8
emac_spec_type2	RW	32	0x0000 0000	0x10AC
emac_spec_type3	RW	32	0x0000 0000	0x10B0
emac_spec_type4	RW	32	0x0000 0000	0x10B4
emac_wol_register	RW	32	0x0000 0000	0x10B8
emac_stretch_ratio	RW	32	0x0000 0000	0x10BC
emac_stacked_vlan	RW	32	0x0000 0000	0x10C0
emac_tx_pfc_pause	RW	32	0x0000 0000	0x10C4
emac_mask_add1_bottom	RW	32	0x0000 0000	0x10C8
emac_mask_add1_top	RW	32	0x0000 0000	0x10CC
emac_dma_addr_or_mask	RW	32	0x0000 0000	0x10D0
emac_rx_ptp_unicast	RW	32	0x0000 0000	0x10D4
emac_tx_ptp_unicast	RW	32	0x0000 0000	0x10D8
emac_tsu_nsec_cmp	RW	32	0x0000 0000	0x10DC
emac_tsu_sec_cmp	RW	32	0x0000 0000	0x10E0
emac_tsu_msb_sec_cmp	RW	32	0x0000 0000	0x10E4
emac_tsu_ptp_tx_msb_sec	RO	32	0x0000 0000	0x10E8
emac_tsu_ptp_rx_msb_sec	RO	32	0x0000 0000	0x10EC
emac_tsu_peer_tx_msb_sec	RO	32	0x0000 0000	0x10F0
emac_tsu_peer_rx_msb_sec	RO	32	0x0000 0000	0x10F4
emac_dpram_fill_dbg	RW	32	0x0000 0000	0x10F8
emac_revision_reg	RO	32	0x0007 0200	0x10FC
emac_octets_txed_bottom	RO	32	0x0000 0000	0x1100
emac_octets_txed_top	RO	32	0x0000 0000	0x1104
emac_frames_txed_ok	RO	32	0x0000 0000	0x1108
emac_broadcast_txed	RO	32	0x0000 0000	0x110C
emac_multicast_txed	RO	32	0x0000 0000	0x1110
emac_pause_frames_txed	RO	32	0x0000 0000	0x1114
emac_frames_txed_64	RO	32	0x0000 0000	0x1118
emac_frames_txed_65	RO	32	0x0000 0000	0x111C
emac_frames_txed_128	RO	32	0x0000 0000	0x1120
emac_frames_txed_256	RO	32	0x0000 0000	0x1124
emac_frames_txed_512	RO	32	0x0000 0000	0x1128
emac_frames_txed_1024	RO	32	0x0000 0000	0x112C
emac_frames_txed_1519	RO	32	0x0000 0000	0x1130
emac_tx_underruns	RO	32	0x0000 0000	0x1134
emac_single_collisions	RO	32	0x0000 0000	0x1138
emac_multiple_collisions	RO	32	0x0000 0000	0x113C
emac_excessive_collisions	RO	32	0x0000 0000	0x1140
emac_late_collisions	RO	32	0x0000 0000	0x1144
emac_deferred_frames	RO	32	0x0000 0000	0x1148
emac_crs_errors	RO	32	0x0000 0000	0x114C
emac_octets_rxed_bottom	RO	32	0x0000 0000	0x1150
emac_octets_rxed_top	RO	32	0x0000 0000	0x1154
emac_frames_rxed_ok	RO	32	0x0000 0000	0x1158
emac_broadcast_rxed	RO	32	0x0000 0000	0x115C
emac_multicast_rxed	RO	32	0x0000 0000	0x1160
emac_pause_frames_rxed	RO	32	0x0000 0000	0x1164

Register Name	Type	Width	Reset Value	Address Offset
emac_frames_rxed_64	RO	32	0x0000 0000	0x1168
emac_frames_rxed_65	RO	32	0x0000 0000	0x116C
emac_frames_rxed_128	RO	32	0x0000 0000	0x1170
emac_frames_rxed_256	RO	32	0x0000 0000	0x1174
emac_frames_rxed_512	RO	32	0x0000 0000	0x1178
emac_frames_rxed_1024	RO	32	0x0000 0000	0x117C
emac_frames_rxed_1519	RO	32	0x0000 0000	0x1180
emac_undersize_frames	RO	32	0x0000 0000	0x1184
emac_excessive_rx_length	RO	32	0x0000 0000	0x1188
emac_rx_jabbers	RO	32	0x0000 0000	0x118C
emac_fcs_errors	RO	32	0x0000 0000	0x1190
emac_rx_length_errors	RO	32	0x0000 0000	0x1194
emac_rx_symbol_errors	RO	32	0x0000 0000	0x1198
emac_alignment_errors	RO	32	0x0000 0000	0x119C
emac_rx_resource_errors	RO	32	0x0000 0000	0x11A0
emac_rx_overruns	RO	32	0x0000 0000	0x11A4
emac_rx_ip_ck_errors	RO	32	0x0000 0000	0x11A8
emac_rx_tcp_ck_errors	RO	32	0x0000 0000	0x11AC
emac_rx_udp_ck_errors	RO	32	0x0000 0000	0x11B0
emac_auto_flushed_pkts	RO	32	0x0000 0000	0x11B4
emac_tsu_timer_incr_sub_nsec	RW	32	0x0000 0000	0x11BC
emac_tsu_timer_msb_sec	RW	32	0x0000 0000	0x11C0
emac_tsu_strobe_msb_sec	RO	32	0x0000 0000	0x11C4
emac_tsu_strobe_sec	RO	32	0x0000 0000	0x11C8
emac_tsu_strobe_nsec	RO	32	0x0000 0000	0x11CC
emac_tsu_timer_sec	RW	32	0x0000 0000	0x11D0
emac_tsu_timer_nsec	RW	32	0x0000 0000	0x11D4
emac_tsu_timer_adjust	RW	32	0x0000 0000	0x11D8
emac_tsu_timer_incr	RW	32	0x0000 0000	0x11DC
emac_tsu_ptp_tx_sec	RO	32	0x0000 0000	0x11E0
emac_tsu_ptp_tx_nsec	RO	32	0x0000 0000	0x11E4
emac_tsu_ptp_rx_sec	RO	32	0x0000 0000	0x11E8
emac_tsu_ptp_rx_nsec	RO	32	0x0000 0000	0x11EC
emac_tsu_peer_tx_sec	RO	32	0x0000 0000	0x11F0
emac_tsu_peer_tx_nsec	RO	32	0x0000 0000	0x11F4
emac_tsu_peer_rx_sec	RO	32	0x0000 0000	0x11F8
emac_tsu_peer_rx_nsec	RO	32	0x0000 0000	0x11FC
emac_tx_pause_quantum1	RW	32	0xFFFF FFFF	0x1260
emac_tx_pause_quantum2	RW	32	0xFFFF FFFF	0x1264
emac_tx_pause_quantum3	RW	32	0xFFFF FFFF	0x1268
emac_pfc_status	RO	32	0x0000 0000	0x126C
emac_rx_lpi	RO	32	0x0000 0000	0x1270
emac_rx_lpi_time	RO	32	0x0000 0000	0x1274
emac_tx_lpi	RO	32	0x0000 0000	0x1278
emac_tx_lpi_time	RO	32	0x0000 0000	0x127C
emac_designcfg_debug1	RO	32	0x0490 8510	0x1280
emac_designcfg_debug2	RO	32	0x6EF1 2800	0x1284
emac_designcfg_debug3	RO	32	0x2400 0000	0x1288
emac_designcfg_debug4	RO	32	0x0000 0000	0x128C
emac_designcfg_debug5	RO	32	0x502F A345	0x1290
emac_designcfg_debug6	RO	32	0x03D4 0000	0x1294
emac_designcfg_debug7	RO	32	0x0000 0000	0x1298
emac_designcfg_debug8	RO	32	0x0102 0006	0x129C
emac_designcfg_debug9	RO	32	0x0000 0000	0x12A0
emac_designcfg_debug10	RO	32	0x2224 4444	0x12A4
emac_designcfg_debug11	RO	32	0x0000 00FF	0x12A8
emac_designcfg_debug12	RO	32	0x0377 4002	0x12AC

Register Name	Type	Width	Reset Value	Address Offset
emac_axi_qos_cfg	RW	32	0x0000 0000	0x12E0
emac_spec_add5_bottom	RW	32	0x0000 0000	0x1300
emac_spec_add5_top	RW	32	0x0000 0000	0x1304
emac_spec_add6_bottom	RW	32	0x0000 0000	0x1308
emac_spec_add6_top	RW	32	0x0000 0000	0x130C
emac_spec_add7_bottom	RW	32	0x0000 0000	0x1310
emac_spec_add7_top	RW	32	0x0000 0000	0x1314
emac_spec_add8_bottom	RW	32	0x0000 0000	0x1318
emac_spec_add8_top	RW	32	0x0000 0000	0x131C
emac_spec_add9_bottom	RW	32	0x0000 0000	0x1320
emac_spec_add9_top	RW	32	0x0000 0000	0x1324
emac_spec_add10_bottom	RW	32	0x0000 0000	0x1328
emac_spec_add10_top	RW	32	0x0000 0000	0x132C
emac_spec_add11_bottom	RW	32	0x0000 0000	0x1330
emac_spec_add11_top	RW	32	0x0000 0000	0x1334
emac_spec_add12_bottom	RW	32	0x0000 0000	0x1338
emac_spec_add12_top	RW	32	0x0000 0000	0x133C
emac_spec_add13_bottom	RW	32	0x0000 0000	0x1340
emac_spec_add13_top	RW	32	0x0000 0000	0x1344
emac_spec_add14_bottom	RW	32	0x0000 0000	0x1348
emac_spec_add14_top	RW	32	0x0000 0000	0x134C
emac_spec_add15_bottom	RW	32	0x0000 0000	0x1350
emac_spec_add15_top	RW	32	0x0000 0000	0x1354
emac_spec_add16_bottom	RW	32	0x0000 0000	0x1358
emac_spec_add16_top	RW	32	0x0000 0000	0x135C
emac_spec_add17_bottom	RW	32	0x0000 0000	0x1360
emac_spec_add17_top	RW	32	0x0000 0000	0x1364
emac_spec_add18_bottom	RW	32	0x0000 0000	0x1368
emac_spec_add18_top	RW	32	0x0000 0000	0x136C
emac_spec_add19_bottom	RW	32	0x0000 0000	0x1370
emac_spec_add19_top	RW	32	0x0000 0000	0x1374
emac_spec_add20_bottom	RW	32	0x0000 0000	0x1378
emac_spec_add20_top	RW	32	0x0000 0000	0x137C
emac_spec_add21_bottom	RW	32	0x0000 0000	0x1380
emac_spec_add21_top	RW	32	0x0000 0000	0x1384
emac_spec_add22_bottom	RW	32	0x0000 0000	0x1388
emac_spec_add22_top	RW	32	0x0000 0000	0x138C
emac_spec_add23_bottom	RW	32	0x0000 0000	0x1390
emac_spec_add23_top	RW	32	0x0000 0000	0x1394
emac_spec_add24_bottom	RW	32	0x0000 0000	0x1398
emac_spec_add24_top	RW	32	0x0000 0000	0x139C
emac_spec_add25_bottom	RW	32	0x0000 0000	0x13A0
emac_spec_add25_top	RW	32	0x0000 0000	0x13A4
emac_spec_add26_bottom	RW	32	0x0000 0000	0x13A8
emac_spec_add26_top	RW	32	0x0000 0000	0x13AC
emac_spec_add27_bottom	RW	32	0x0000 0000	0x13B0
emac_spec_add27_top	RW	32	0x0000 0000	0x13B4
emac_spec_add28_bottom	RW	32	0x0000 0000	0x13B8
emac_spec_add28_top	RW	32	0x0000 0000	0x13BC
emac_spec_add29_bottom	RW	32	0x0000 0000	0x13C0
emac_spec_add29_top	RW	32	0x0000 0000	0x13C4
emac_spec_add30_bottom	RW	32	0x0000 0000	0x13C8
emac_spec_add30_top	RW	32	0x0000 0000	0x13CC
emac_spec_add31_bottom	RW	32	0x0000 0000	0x13D0
emac_spec_add31_top	RW	32	0x0000 0000	0x13D4
emac_spec_add32_bottom	RW	32	0x0000 0000	0x13D8
emac_spec_add32_top	RW	32	0x0000 0000	0x13DC

Register Name	Type	Width	Reset Value	Address Offset
emac_spec_add33_bottom	RW	32	0x0000 0000	0x13E0
emac_spec_add33_top	RW	32	0x0000 0000	0x13E4
emac_spec_add34_bottom	RW	32	0x0000 0000	0x13E8
emac_spec_add34_top	RW	32	0x0000 0000	0x13EC
emac_spec_add35_bottom	RW	32	0x0000 0000	0x13F0
emac_spec_add35_top	RW	32	0x0000 0000	0x13F4
emac_spec_add36_bottom	RW	32	0x0000 0000	0x13F8
emac_spec_add36_top	RW	32	0x0000 0000	0x13FC
emac_cbs_control	RW	32	0x0000 0000	0x14BC
emac_cbs_idleslope_q_a	RW	32	0x0000 0000	0x14C0
emac_cbs_idleslope_q_b	RW	32	0x0000 0000	0x14C4
emac_upper_tx_q_base_addr	RW	32	0x0000 0000	0x14C8
emac_tx_bd_control	RW	32	0x0000 0000	0x14CC
emac_rx_bd_control	RW	32	0x0000 0000	0x14D0
emac_upper_rx_q_base_addr	RW	32	0x0000 0000	0x14D4
emac_wd_counter	RW	32	0x0000 0007	0x14EC
emac_wd_axi_tx_full_thresh0	RW	32	0x0006 0008	0x14F8
emac_wd_axi_tx_full_thresh1	RW	32	0x0000 0000	0x14FC
emac_screening_type_1_register	RW	32	0x0000 0000	0x1500
emac_screening_type_2_register_0	RW	32	0x0000 0000	0x1540
emac_screening_type_2_register_1	RW	32	0x0000 0000	0x1544
emac_tx_sched_ctrl	RW	32	0x0000 0000	0x1580
emac_bw_rate_limit	RW	32	0x0000 0000	0x1590
emac_tx_q_seg_alloc_q_lower	RW	32	0x0000 0000	0x15A0
emac_type2_compare_0_word_0	RW	32	0x0000 0000	0x1700
emac_type2_compare_0_word_1	RW	32	0x0000 0000	0x1704
emac_type2_compare_1_word_0	RW	32	0x0000 0000	0x1708
emac_type2_compare_1_word_1	RW	32	0x0000 0000	0x170C
emac_type2_compare_2_word_0	RW	32	0x0000 0000	0x1710
emac_type2_compare_2_word_1	RW	32	0x0000 0000	0x1714
emac_type2_compare_3_word_0	RW	32	0x0000 0000	0x1718
emac_type2_compare_3_word_1	RW	32	0x0000 0000	0x171C
emac_type2_compare_4_word_0	RW	32	0x0000 0000	0x1720
emac_type2_compare_4_word_1	RW	32	0x0000 0000	0x1724
emac_type2_compare_5_word_0	RW	32	0x0000 0000	0x1728
emac_type2_compare_5_word_1	RW	32	0x0000 0000	0x172C
emac_enst_start_time	RW	32	0x0000 0000	0x1800
emac_enst_on_time	RW	32	0x0001 FFFF	0x1820
emac_enst_off_time	RW	32	0x0000 0000	0x1840
emac_enst_control	RW	32	0x0000 0000	0x1880
emac_frer_timeout	RW	32	0x0000 0000	0x18A0
emac_frer_red_tag	RW	32	0x4000 F1C1	0x18A4
emac_frer_control_1_a	RW	32	0x0000 0000	0x18C0
emac_frer_control_1_b	RW	32	0x0000 0000	0x18C4
emac_frer_statistics_1_a	RO	32	0x0000 0000	0x18C8
emac_frer_statistics_1_b	RO	32	0x0000 0000	0x18CC
emac_frer_control_2_a	RW	32	0x0000 0000	0x18D0
emac_frer_control_2_b	RW	32	0x0000 0000	0x18D4
emac_frer_statistics_2_a	RO	32	0x0000 0000	0x18D8
emac_frer_statistics_2_b	RO	32	0x0000 0000	0x18DC
emac_rx_q0_flush	RW	32	0x0000 0000	0x1B00
emac_scr2_reg0_rate_limit	RW	32	0x0000 0000	0x1B40
emac_scr2_reg1_rate_limit	RW	32	0x0000 0000	0x1B44
emac_scr2_rate_status	RO	32	0x0000 0000	0x1B80
emac_asf_int_status	RW	32	0x0000 0000	0x1E00
emac_asf_int_raw_status	RW	32	0x0000 0000	0x1E04
emac_asf_int_mask	RW	32	0x0000 007F	0x1E08

Register Name	Type	Width	Reset Value	Address Offset
emac_asf_int_test	RW	32	0x0000 0000	0x1E0C
emac_asf_fatal_nonfatal_select	RW	32	0x0000 007F	0x1E10
emac_asf_sram_corr_fault_status	RO	32	0x0000 0000	0x1E20
emac_asf_sram_uncorr_fault_status	RO	32	0x0000 0000	0x1E24
emac_asf_sram_fault_stats	RW	32	0x0000 0000	0x1E28
emac_asf_trans_to_ctrl	RW	32	0x0000 0000	0x1E30
emac_asf_trans_to_fault_mask	RW	32	0x0000 001F	0x1E34
emac_asf_trans_to_fault_status	RW	32	0x0000 0000	0x1E38
emac_asf_protocol_fault_mask	RW	32	0x003F 01FF	0x1E40
emac_asf_protocol_fault_status	RW	32	0x0000 0000	0x1E44

11.2 Registers Descriptions

Table 11-2: network_control

Register Information				
Description	"The network control register contains general MAC control functions for both receiver and transmitter."			
Offset	0x0000	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	reserved_31	"Reserved, read as 0, ignored on write."	RO	0
30	ifg_eats_qav_credit	"Setting this bit high modifies the CBS algorithm so the IFG/IPG associated with a transmit frame counts towards its 802.1Qav credit."	RW	0
29	reserved_29	"Reserved"	RW	0
28	sel_mii_on_rgmii	"If the RGMII interface is being used, set this bit high to configure the RGMII interface for MII operation (in 802.3br configurations this bit has no effect for the eMAC)."	RW	0
27	oss_correction_field	"1588 One Step Correction Field Update. Set this bit high to enable updating the correction field of PTP 1588 version 2 sync frames by adding the current TSU timer value."	RW	0
26	ext_rxq_sel_en	"Enable external selection of receive queue. When this bit is high the ext_match1, ext_match2, ext_match3 and ext_match4 inputs will determine which receive queue a frame is routed to. This will be the case regardless of the state of the external address match enable bit 9 of the network config register. Note that receive frames will be dropped unless they are matched by the internal frame filtering functionality. If the external address match enable bit 9 in the network config register is set frames may be matched by an external address match filter as long as one of the ext_match1, ext_match2, ext_match3 and ext_match4 inputs is asserted early enough. When set ext_rxq_sel_en takes precedence over the existing screener functionality. This bit is only relevant if priority queuing is configured."	RW	0
25	pfc_ctrl	"Enable multiple PFC pause quantums, one per pause priority."	RW	0
24	one_step_sync_mode	"1588 One Step Sync Mode. Write 1 to enable. Replace timestamp field in the 1588 header for TX Sync Frames with current TSU timer value."	RW	0
23	ext_tsu_port_enable	"Set to one to use the ext_tsu_timer input vector as the value to use for time-stamping frames rather than the internal TSU value. If a SoC has multiple GEM instances one GEM may be used as the TSU timer source for the other GEM instances."	RW	0

22	store_udp_offset	"Store UDP / TCP offset to memory. Setting this bit to one will cause the upper 16-bits of the CRC of every received frame to be replaced with the offset from start of frame to the beginning of the UDP or TCP header. The lower 16-bits of the CRC are replaced with zero and reserved for future use. The offset is measured in units of 2 bytes. Set to zero for normal operation."	RW	0
21	alt_sgmii_mode	"Alternative sgmii mode. If asserted with sgmii_mode in the network control register the ACK bit is driven before ability detect during transfer of status information from the PHY to the MAC."	RW	0
20	ptp_unicast_ena	"Enable detection of unicast PTP unicast frames."	RW	0
19	tx_lpi_en	"Enable LPI transmission when set LPI (low power idle) is immediately transmitted. Depending on configuration LPI is indicated on the GMII, RGMII or MII interface and can be encoded by the PCS used for SGMII. LPI is transmitted even if bit 3 transmit enable is disabled. Setting this bit also sends a pause signal to the transmit datapath. In 802.3br configurations LPI can only be controlled from the pMAC."	RW	0
18	flush_rx_pkt_pclk	"Flush the next packet from the external RX DPRAM. This bit flushes the frame that is the next in line to be pushed out to the AXI interface. Writing one to this bit will only have an effect if the DMA is not currently writing a packet already stored in the DPRAM to memory."	WO	0
17	transmit_pfc_priority_based_pause_frame	"Write a one to transmit PFC priority based pause frame. Takes the values stored in the Transmit PFC Pause Register."	WO	0
16	pfc_enable	"Enable PFC Priority Based Pause Reception capabilities. Setting this bit will enable PFC negotiation and recognition of priority based pause frames."	RW	0
15	store_rx_ts	"Store receive time stamp to memory. Setting this bit to one will cause the CRC of every received frame to be replaced with the value of the nanoseconds field of the 1588 timer that was captured as the receive frame passed the message time stamp point (for releases prior to 1p10 bits 31 and 30 of the value written to the CRC field were hardwired to zero, from release 1p10 onwards these bits represent the two least significant seconds bit of the captured timer value). Set to zero for normal operation."	RW	0
14	stats_read_snap	"Read snapshot - writing a one means that the snapshot value of the statistics register will be read back, otherwise the raw statistic register will be read. The default GEM configuration does not support this function. See Parameterization section under Implementation Application Notes for more details."	RW	0
13	stats_take_snap	"Take snapshot - writing a one will record the current value of all statistics registers in the snapshot registers and clear the statistics registers. The default GEM configuration does not support this function."	WO	0
12	tx_pause_frame_zero	"Transmit zero quantum pause frame - writing one to this bit causes a pause frame with zero quantum to be transmitted."	WO	0
11	tx_pause_frame_req	"Transmit pause frame - writing one to this bit causes a pause frame to be transmitted."	WO	0

10	transmit_halt	"Transmit halt - writing one to this bit resets the tx_go variable and halts the dma from reading more transmit frames into the transmit SRAM buffer. Any frames already read into the transmit SRAM will still be transmitted."	WO	0
9	transmit_start	"Start transmission - writing one to this bit starts transmission."	WO	0
8	back_pressure	"Back pressure if set in 10M or 100M half-duplex mode will force collisions on all received frames. Ignored in gigabit half-duplex mode."	RW	0
7	stats_write_en	"Write enable for statistics registers - setting this bit to one means the statistics registers can be written for functional test purposes."	RW	0
6	inc_all_stats_regs	"Incremental statistics registers - this bit is write only. Writing a one increments all the statistics registers by one for test purposes."	WO	0
5	clear_all_stats_regs	"Clear statistics registers - this bit is write only. Writing a one clears the statistics registers."	WO	0
4	man_port_en	"Management port enable - set to one to enable the management port. When zero forces mdio to high impedance state and mdc low."	RW	0
3	enable_transmit	"Transmit enable - when set, it enables the GEM transmitter to send data. When reset transmission will stop immediately, the transmit pipeline and control registers will be cleared and the transmit queue pointer register will reset to point to the start of the transmit descriptor list. This bit needs to be set before bit 9 the start transmission bit. (Setting this bit low resets the transmit queue pointer however reading the transmit queue pointer register through the APB interface may still return an old value until transmission is restarted.)"	RW	0
2	enable_receive	"Receive enable - when set, it enables the GEM to receive data. When reset frame reception will stop immediately and the receive pipeline will be cleared. The receive queue pointer register is unaffected."	RW	0
1	loopback_local	"Loopback local - asserts the loopback_local signal to the system clock generator. Also connects txd to rxd, tx_en to rx_dv and forces full duplex mode. If the GEM configuration contains the PCS then bit 11 of the network configuration register must be set low to disable TBI mode when in internal loopback. rx_clk and tx_clk may malfunction as the GEM is switched into and out of internal loopback. It is important that receive and transmit circuits have already been disabled when making the switch into and out of internal loopback. Local loopback functionality is optional and may not be supported by some instantiations of the GEM."	RW	0
0	loopback	"Loopback - controls the loopback output pin."	RW	0

Table 11-3: network_config

Register Information			
Description	"The network configuration register contains functions for setting the mode of operation for the Gigabit Ethernet MAC."		
Offset	0x0004	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	uni_direction_enable	"Uni-direction-enable. When low the PCS will transmit idle symbols if the link goes down. When high the PCS can transmit frame data when the link is down."	RW	0
30	ignore_ipg_rx_er	"Ignore IPG rx_er. When set rx_er has no effect on the GEMs operation when rx_dv is low. Set this when using the RGMII wrapper in half-duplex mode."	RW	0
29	nsp_change	"Receive bad preamble. When set frames with non-standard preamble are not rejected (the first byte of preamble needs to be 55 and the last D5, bytes other than 55 or D5 may be inserted in between)."	RW	0
28	ipg_stretch_enable	"IPG stretch enable - when set the transmit IPG can be increased above 96 bit times depending on the previous frame length using the IPG stretch register."	RW	0
27	sgmii_mode_enable	"SGMII mode enable - changes behaviour of the auto-negotiation advertisement and link partner ability registers to meet the requirements of SGMII and reduces the duration of the link timer from 10 ms to 1.6 ms."	RW	0
26	ignore_rx_fcs	"Ignore RX FCS - when set frames with FCS/CRC errors will not be rejected. FCS error statistics will still be collected for frames with bad FCS and FCS status will be recorded in frame's DMA descriptor. For normal operation this bit must be set to zero."	RW	0
25	en_half_duplex_rx	"Enable frames to be received in half-duplex mode while transmitting."	RW	0
24	receive_checksum_offload_enable	"Receive checksum offload enable - when set, the receive checksum engine is enabled. Frames with bad IP, TCP or UDP checksums are discarded."	RW	0
23	disable_copy_of_pause_frames	"Disable copy of pause frames - set to one to prevent pause frames being copied to memory. When set, neither control frames with type id 8808, nor pause frames with destination address 010000c28001 are copied to memory, this functionality was enhanced in release 1p09. Note that valid pause frames received will still increment pause statistics and pause the transmission of frames as required."	RW	0

22:21	data_bus_width	"Data bus width - set according to AMBA (AHB/AXI) or external FIFO data bus width. The reset value for this can be changed by defining a new value for gem_dma_bus_width_def in gem_defs. Only valid bus widths may be written if the system is configured to a maximum width less than 128-bits. 00: 32 bit data bus width 01: 64 bit AMBA (AHB/AXI) data bus width 10: 128 bit AMBA (AHB/AXI) data bus width 11: invalid Note. The reset value of this field is equal to the gem_dma_bus_width_def define, which is user configurable."	RW	0x0
20:18	mdc_clock_division	"MDC clock division - set according to pclk speed. These three bits determine the number pclk will be divided by to generate MDC. For conformance with the 802.3 specification, MDC must not exceed 2.5 MHz (MDC is only active during MDIO read and write operations). The reset value for this can be changed by defining a new value for gem_mdc_clock_div in gem_defs.v 000: divide pclk by 8 (pclk up to 20 MHz) 001: divide pclk by 16 (pclk up to 40 MHz) 010: divide pclk by 32 (pclk up to 80 MHz) 011: divide pclk by 48 (pclk up to 120MHz) 100: divide pclk by 64 (pclk up to 160 MHz) 101: divide pclk by 96 (pclk up to 240 MHz) 110: divide pclk by 128 (pclk up to 320 MHz) 111: divide pclk by 224 (pclk up to 540 MHz). Note. The reset value of this field is equal to the gem_mdc_clock_div define, which is user configurable."	RW	0x2
17	fcs_remove	"FCS remove - setting this bit will cause received frames to be written to memory without their frame check sequence (last 4 bytes). The frame length indicated will be reduced by four bytes in this mode."	RW	0
16	length_field_error_frame_discard	"Length field error frame discard - setting this bit causes frames with a measured length shorter than the extracted length field (as indicated by bytes 13 and 14 in a non-VLAN tagged frame) to be discarded. This only applies to frames with a length field less than 0x0600."	RW	0
15:14	receive_buffer_offset	"Receive buffer offset - indicates the number of bytes by which the received data is offset from the start of the receive buffer. Note that when the define gem_pbuf_rsc has been set then these bits cannot be used."	RW	0x0
13	pause_enable	"Pause enable - when set, transmission will pause if a non-zero 802.3 classic pause frame is received and PFC has not been negotiated."	RW	0
12	retry_test	"Retry test - must be set to zero for normal operation. If set to one the backoff between collisions will always be one slot time. Setting this bit to one helps test the too many retries condition. Also used in the pause frame tests to reduce the pause counter's decrement time from 512 bit times, to every rx_clk cycle."	RW	0

11	pcs_select	"PCS select - selects between MII/GMII and TBI. Must be set for SGMII operation. 0: GMII/MII interface enabled, TBI disabled 1: TBI enabled, GMII/MII disabled (in 802.3br configurations this bit must be set identically in the eMAC and pMAC)"	RW	0
10	gigabit_mode_enable	"Gigabit mode enable - setting this bit configures the GEM for 1000 Mbps operation. 0: 10/100 operation using MII or TBI interface 1: Gigabit operation using GMII or TBI interface (in 802.3br configurations this bit must be set identically in the eMAC and pMAC)"	RW	0
9	external_address_match_enable	"External address match enable - when set the external address match interface can be used to copy frames to memory."	RW	0
8	receive_1536_byte_frames	"Receive 1536 byte frames - setting this bit means the GEM will accept frames up to 1536 bytes in length. Normally the GEM would reject any frame above 1518 bytes."	RW	0
7	unicast_hash_enable	"Unicast hash enable - when set, unicast frames will be accepted when the 6 bit hash function of the destination address points to a bit that is set in the hash register."	RW	0
6	multicast_hash_enable	"Multicast hash enable - when set, multicast frames will be accepted when the 6 bit hash function of the destination address points to a bit that is set in the hash register."	RW	0
5	no_broadcast	"No broadcast - when set to logic one, frames addressed to the broadcast address of all ones will not be accepted."	RW	0
4	copy_all_frames	"Copy all frames - when set to logic one, all valid frames will be accepted."	RW	0
3	jumbo_frames	"Jumbo frames - set to one to enable jumbo frames up to `gem_jumbo_max_length bytes to be accepted. The default length is 10,240 bytes."	RW	0
2	discard_non_vlan_frames	"Discard non-VLAN frames - when set only VLAN tagged frames will be passed to the address matching logic."	RW	0
1	full_duplex	"Full duplex - if set to logic one, the transmit block ignores the state of collision and carrier sense and allows receive while transmitting. Also controls the half_duplex pin."	RW	0
0	speed	"Speed - set to logic one to indicate 100Mbps operation, logic zero for 10Mbps. The value of this pin is reflected on the speed_mode[0] output pin (in 802.3br configurations this bit has no effect for the eMAC)."	RW	0

Table 11-4: network_status

Register Information			
Description	"The network status register returns status information with respect to the PHY management MDIO interface, the PCS, priority flow control, LPI and other status."		
Offset	0x0008	**Type:**	RO

Bits	Name	Description	Access	Reset
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10:9	reserved_10_9	"Reserved, read as 0, ignored on write."	RO	0x0
8	axi_xaction_outstanding	"Outstanding AXI Transactions - This status bit is set when one or more AXI read or write transactions have been issued by the DUT but the responses have not yet been fully collected."	RO	0
7	lpi_indicate_pclk	"LPI Indication - Low power idle has been detected on receive. This bit is set when LPI is detected and reset when normal idle is detected. An interrupt is generated when the state of this bit changes."	RO	0
6	pfc_negotiate_pclk	"Set when PFC Priority Based Pause has been negotiated."	RO	0
5	mac_pause_tx_en	"PCS auto-negotiation pause transmit resolution."	RO	0
4	mac_pause_rx_en	"PCS auto-negotiation pause receive resolution."	RO	0
3	mac_full_duplex	"PCS auto-negotiation duplex resolution. Set to one if the resolution function determines that both devices are capable of full duplex operation. If zero half-duplex operation is possible as long as bit 0 (PCS link state) is also one."	RO	0
2	man_done	"The PHY management logic is idle (i.e. has completed)."	RO	1
1	mdio_in	"Returns status of the mdio_in pin."	RO	0
0	pcs_link_state	"Returns status of PCS link state. If auto-negotiation is disabled this returns the synchronisation status. If auto-negotiation is enabled it is set in the LINK_OK state as long as a compatible duplex mode is resolved, it is always set in the LINK_OK state in SGMII mode."	RO	0

Table 11-5: user_io_register

Register Information			
Description	"The GEM design provides up to 16 inputs and 16 outputs so that the I/O may be read or set under the control of the processor interface. The number of inputs and outputs are set by the user_in_width and user_out_width verilog macro defines. If the user I/O is disabled as a configuration option, this address space is defined as reserved, and hence will be a read-only register of the value 0x0. If enabled, the number of inputs and outputs can be configured separately. The first output will be represented in bit 0 of the user I/O register, the second output will use bit 1 and so on. The first input will be represented in bit 16 of the user I/O register, the second input will use bit 17 and so on."		
Offset	0x000C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	user_programmable_inputs	"User programmable inputs - the upper 16 bits of this register are used to monitor the state of the user inputs. A logic one read from a bit in this range will correspond to the input being in a high state. A logic zero read from a bit in this range will correspond to the input being in a low state. Any unused bits will be read as 0. Writing to any bits in this range will have no functional effect."	RO	0x0000
15:0	user_programmable_outputs	"User programmable outputs - the lower 16 bits of this register are used to control the state of the user outputs. A logic one written to a bit in this range will cause the corresponding output to be set high. A logic zero written to a bit in this range shall cause the corresponding output to be forced low. Any unused bits will be read as logic zero. Writing to any unused bits in this range will have no functional effect."	RW	0x0000

Table 11-6: dma_config

Register Information				
Description	"DMA Configuration Register"			
Offset	0x0010	**Type:**	RW	

Bits	Name	Description	Access	Reset
31	reserved_31	"Reserved, read as 0, ignored on write."	RO	0
30	dma_addr_bus_width_1	"DMA address bus width. 0 = 32b, 1 = 64b."	RW	0
29	tx_bd_extended_mode_en	"Enable TX extended BD mode. See TX BD control register definition for description of feature."	RW	0
28	rx_bd_extended_mode_en	"Enable RX extended BD mode. See RX BD control register definition for description of feature."	RW	0
27	reserved_27	"Reserved, read as 0, ignored on write."	RO	0
26	force_max_amba_burst_tx	"Force max length bursts on TX. Force the TX DMA to always issue max length bursts on EOP (end of packet) or EOB (end of buffer) transfers as defined by bits 4:0 of this register, even when there is less than max burst data bytes to read. Residual data read is ignored. AHB only - does not apply on AXI bursts (prior to release 1p10) or bursts that break 1k boundary rule - supported on AXI from release 1p10."	RW	0
25	force_max_amba_burst_rx	"Force max length bursts on RX. Force the RX DMA to always issue max length bursts on EOP (end of packet) or EOB (end of buffer) transfers, even if there is less than max burst real packet data required to write. Any extra bytes of pad data is set to 0x00. AHB only - does not apply on AXI bursts (prior to release 1p10) or bursts that break 1k boundary rule - supported on AXI from release 1p10."	RW	0
24	force_discard_on_err	"Auto Discard RX frames during lack of resource. When set, the GEM DMA will automatically discard the next frame (that is the oldest frame) from the receiver packet buffer memory when a receive buffer descriptor is read with its used bit set. When low, then received frames will remain to be stored in the SRAM based packet buffer until AMBA (AHB/AXI) buffer resource next becomes available. In this case if the SRAM memory fills up then there will be a receive overflow condition and the most recently received frame (that is the newest) will be discarded. A write to this bit is ignored if the DMA is not configured in the packet buffer full store and forward mode."	RW	0

23:16	rx_buf_size	"DMA receive buffer size in external AMBA (AHB/AXI) system memory. The value defined by these bits determines the size of buffer to use in main system memory when writing received data. The value is defined in multiples of 64 bytes. 0x01 corresponds to buffers of 64 bytes 0x02 corresponds to 128 bytes etc. For example: 0x02: 128 byte. 0x18: 1536 byte (1*max length frame/buffer) 0xA0: 10240 byte (1*10K jumbo frame/buffer) 0xFF: 16320 byte Note that this value should never be written as zero. Note. The reset value of this field is equal to the gem_rx_buffer_length_def define, which is user configurable."	RW	0x02
15:14	reserved_15_14	"Reserved, read as 0, ignored on write."	RO	0x0
13	crc_error_report	"When the bit is set, bit 16 of the receive buffer descriptor will represent FCS/CRC error (only if frames with FCS are copied to memory as enabled by bit 26 in the network config register). When this bit is clear, bit 16 of the receive buffer descriptor will represent the Canonical format indicator (CFI) bit as extracted from the receive frame (if the receive buffer descriptor is pointing to the last data buffer of the receive frame and the received frame was VLAN tagged)."	RW	0
12	infinite_last_dbuf_size_en	"Forces the receive DMA to consider the data buffer pointed to by last descriptor in the descriptor list to be of elastic size (the last descriptor is the one with its wrap bit set). This means the first buffer pointed to in the list will always contain the beginning of a frame (this helps if there is a desire to build custom logic that interfaces with the receive buffer directly without software intervention). When set the rx_buf_size bits 23:16 in the dma configuration register are ignored for the last receive buffer in the descriptor list and data will be written into the buffer sequentially until the frame is completely received and the buffer descriptor status will be updated with the frame length as normal."	RW	0
11	tx_pbuf_tcp_en	"Transmitter IP, TCP and UDP checksum generation offload enable (not supported when in TX Partial Store and Forward mode). When set, the transmitter checksum generation engine is enabled, to calculate and substitute checksums for transmit frames. When clear, frame data is unaffected. If the GEM is not configured to use the DMA packet buffer, this bit is not implemented and will be treated as reserved, read as 0, ignored on write."	RW	0

10	tx_pbuf_size	"Transmitter packet buffer memory size select. Having this bit at zero halves the amount of memory used for the transmit packet buffer. This reduces the amount of memory used by the GEM. It is important to set this bit to one if the full configured physical memory is available. The value in brackets below represents the size that would result for the default maximum configured memory size of 4 Kbytes. 1: Use full configured addressable space (4 Kb) 0: Do not use top address bit (2 Kb) If the GEM is not configured to use the DMA packet buffer, this bit is not implemented and will be treated as reserved, read as 0, ignored on write. Note. The reset value of this field is equal to the gem_tx_pbuf_size_def define, which is user configurable."	RW	1
9:8	rx_pbuf_size	"Receiver packet buffer memory size select. Having these bits at less than 11 reduces the amount of memory used for the receive packet buffer. This reduces the amount of memory used by the GEM. It is important to set these bits both to one if the full configured physical memory is available. The value in brackets below represents the size that would result for the default maximum configured memory size of 8 Kbytes. 11: Use full configured addressable space (8 Kb) 10: Do not use top address bit (4 Kb) 01: Do not use top two address bits (2 Kb) 00: Do not use top three address bits (1 Kb) If the GEM is not configured to use the DMA packet buffer, these bits are not implemented and will be treated as reserved, read as 0, ignored on write. Note. The reset value of this field is equal to the gem_rx_pbuf_size_def define, which is user configurable."	RW	0x3
7	endian_swap_packet	"endian swap mode enable for packet data accesses. When set, selects swapped endianism for AMBA (AHB/AXI) transfers. When clear, selects little endian mode."	RW	1
6	endian_swap_management	"endian swap mode enable for management descriptor accesses. When set, selects swapped endianism for AMBA (AHB/AXI) transfers. When clear, selects little endian mode."	RW	1
5	hdr_data_splitting_en	"Enable header data Splitting. When set, receive frames will be forwarded to main memory using a minimum of two DMA data buffers. The first X data buffers will contain the frame header, consisting of the Ethernet,VLAN,(IPv4 or IPv6),(TCP or UDP). X= (frame header size divided by rx_buf_size as defined in bits 23:16 of this register). The last Y data buffers will contain the frame payload. Y= (frame payload size divided by rx_buf_size). Note that for non VLAN/IP/TCP/UDP frames, the header will always be 14 bytes. When this feature is disabled, the frame is forwarded to main memory in blocks of rx_buf_size."	RW	0

4:0	amba_burst_length	"Selects the burst length to use on the AMBA (AHB/AXI) when transferring frame data. Not used for DMA management operations and only used where space and data size allow and respecting AXI/AHB burst boundary rules. One-hot priority encoding enforced automatically on register writes as follows, where x represents don't care: 1xxxx: Attempt to use bursts of up to 16. 01xxx: Attempt to use bursts of up to 8. 001xx: Attempt to use bursts of up to 4. 0001x: Always use SINGLE bursts. 00001: Always use SINGLE bursts. 00000: Attempt to use bursts of up to 256."	RW	0x04

Table 11-7: transmit_status

Register Information			
Description	"This register, when read, provides details of the status of the transmit path. Once read, individual bits may be cleared by writing 1 to them. It is not possible to set a bit to 1 by writing to the register."		
Offset	0x0014	**Type:**	RW

Bits	Name	Description	Access	Reset
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	tx_dma_lockup_detected	"TX DMA lockup detected - set when lockup has been detected the transmit DMA lockup monitor."	RW W1toClr	0
9	tx_mac_lockup_detected	"TX MAC lockup detected - set when lockup has been detected by the lockup monitor on the MAC transmit path."	RW W1toClr	0
8	resp_not_ok	"bresp/hresp not OK - set when the DMA block sees bresp/hresp not OK. Cleared by writing a one to this bit."	RW W1toClr	0
7	late_collision_occurred	"Late collision occurred - only set if the condition occurs in gigabit mode, as retry is not attempted. Cleared by writing a one to this bit."	RW W1toClr	0
6	transmit_under_run	"Transmit under run - this bit is set if the transmitter was forced to terminate a frame that it had already began transmitting due to further data being unavailable. This bit is set if a transmitter status write back has not completed when another status write back is attempted. When using the DMA interface configured for internal FIFO mode, this bit is also set when the transmit DMA has written the SOP data into the FIFO and either the AHB bus was not granted in time for further data, or because an AHB not OK response was returned, or because a used bit was read. When using the DMA interface configured for packet buffer mode, this bit will never be set. When using the external FIFO interface, this bit is also set when the tx_r_underflow input is asserted during a frame transfer. Cleared by writing a 1."	RW W1toClr	0
5	transmit_complete	"Transmit complete - set when a frame has been transmitted. Cleared by writing a one to this bit."	RW W1toClr	0
4	amba_error	"Transmit frame corruption due to AMBA (AHB/AXI) errors. Set if an error occurs whilst midway through reading transmit frame from external memory including HRESP(AHB), RRESP or BRESP(AXI) errors and buffers exhausted mid frame (if the buffers run out during transmission of a frame then transmission stops, FCS shall be bad and tx_er asserted). Also set in DMA packet buffer mode if single frame is too large for configured packet buffer memory size. Some AXI errors cause immediate reset of the transmit MAC datapath, in these cases tx_er will not be asserted and this bit will not be set. Cleared by writing a one to this bit."	RW W1toClr	0

3	transmit_go	"Transmit go - if high transmit is active. When using the exposed FIFO interface, this bit represents the transmit enable bit (bit 3) of the network control register. When using the DMA interface this bit represents the tx_go variable as specified in the transmit buffer description."	RO	0
2	retry_limit_exceeded	"Retry limit exceeded - cleared by writing a one to this bit."	RW W1toClr	0
1	collision_occurred	"Collision occurred - set by the assertion of collision. Cleared by writing a one to this bit. When operating in 10/100 mode, this status indicates either a collision or a late collision. In gigabit mode, this status is not set for a late collision."	RW W1toClr	0
0	used_bit_read	"Used bit read - set when a transmit buffer descriptor is read with its used bit set. Cleared by writing a one to this bit."	RW W1toClr	0

Table 11-8: receive_q_ptr

Register Information			
Description	"This register holds the start address of the receive buffer queue (receive buffers descriptor list). The receive buffer queue base address must be initialized before receive is enabled through bit 2 of the network control register. Once reception is enabled, any write to the receive buffer queue base address register is ignored. Reading this register returns the location of the descriptor currently being accessed. This value increments as buffers are used. Software should not use this register for determining where to remove received frames from the queue as it constantly changes as new frames are received. Software should instead work its way through the buffer descriptor queue checking the used bits. In terms of AMBA (AHB/AXI) operation, the receive descriptors are read from memory using a single 32bit AHB/AXI access. When the datapath is configured at 64bit or 128bit, the receive descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is written to using a single 64bit AHB/AXI access. For 32bit datapaths, the receive descriptors should be aligned at 32-bit boundaries and are written to using two individual non sequential 32-bit accesses."		
Offset	0x0018	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:2	dma_rx_q_ptr	"Receive buffer queue base address - written with the address of the start of the receive queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_rx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while receive is not enabled."	RW	0

Table 11-9: transmit_q_ptr

Register Information			
Description	"This register holds the start address of the transmit buffer queue (transmit buffers descriptor list). The transmit buffer queue base address register must be initialized before transmit is started through bit 9 of the network control register. Once transmission has started, any write to the transmit buffer queue base address register is illegal and therefore ignored. Note that due to clock boundary synchronization, it takes a maximum of four pclk cycles from the writing of the transmit start bit before the transmitter is active. Writing to the transmit buffer queue base address register during this time may produce unpredictable results. Reading this register returns the location of the descriptor currently being accessed. Because the DMA can store data for multiple frames at once, this may not necessarily be pointing to the current frame being transmitted. In terms of AMBA AHB/AXI operation, the transmit descriptors are written to memory using a single 32bit AHB access. When the datapath is configured as 64bit or 128bit, the transmit descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is read from memory using a single AHB/AXI access. For 32bit datapaths, the descriptors should be aligned at 32-bit boundaries and the descriptors are read from memory using two individual 32-bit non sequential accesses."		
Offset	0x001C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:2	dma_tx_q_ptr	"Transmit buffer queue base address - written with the address of the start of the transmit queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_tx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while transmit is not enabled."	RW	0

Table 11-10: receive_status

Register Information			
Description	"This register, when read provides details of the status of the receive path. Once read, individual bits may be cleared by writing 1 to them. It is not possible to set a bit to 1 by writing to the register."		
Offset	0x0020	**Type:**	RW

Bits	Name	Description	Access	Reset
31:6	reserved_31_6	"Reserved, read as 0, ignored on write."	RO	0x000 0000
5	rx_dma_lockup_detected	"RX DMA lockup detected - set when lockup has been detected the receive DMA lockup monitor."	RW W1toClr	0
4	rx_mac_lockup_detected	"RX MAC lockup detected - set when lockup has been detected by the lockup monitor on the MAC receive path."	RW W1toClr	0
3	resp_not_ok	"bresp/hresp not OK - set when the DMA block sees bresp/hresp not OK. Cleared by writing a one to this bit."	RW W1toClr	0
2	receive_overrun	"Receive overrun - this bit is set if either the gem_dma RX FIFO or external RX FIFO were unable to store the receive frame due to a FIFO overflow, or if the receive status, reported by the gem_rx module to the gem_dma was not taken at end of frame. This bit is also set in DMA packet buffer mode if the packet buffer overflows. For DMA operation the buffer will be recovered if an overrun occurs. This bit is cleared by writing a one to it."	RW W1toClr	0
1	frame_received	"Frame received - one or more frames have been received and placed in memory. Cleared by writing a one to this bit."	RW W1toClr	0
0	buffer_not_available	"Buffer not available - an attempt was made to get a new buffer and the pointer indicated that it was owned by the processor. The DMA will reread the pointer each time an end of frame is received until a valid pointer is found. This bit is set following each descriptor read attempt that fails, even if consecutive pointers are unsuccessful and software has in the meantime cleared the status flag. Cleared by writing a one to this bit."	RW W1toClr	0

Table 11-11: int_status

Register Information			
Description	"If not configured for priority queuing, the GEM generates a single interrupt. This register indicates the source of this interrupt. The corresponding bit in the mask register must be clear for a bit to be set. If any bit is set in this register the ethernet_int signal will be asserted. For test purposes each bit can be set or reset by writing to the interrupt mask register. The default configuration is shown below whereby all bits are reset to zero on read. Changing the validity of the `gem_irq_read_clear define will instead require a one to be written to the appropriate bit in order to clear it. In this mode reading has no effect on the status of the bit."		
Offset	0x0024	**Type:**	RW

Bitfield Details

Bits	Name	Description	Access	Reset
31	tx_lockup_detected	"TX lockup detection - set when lockup has been detected by any of the lockup monitors on the transmit datapath."	RW RtoClr	0
30	rx_lockup_detected	"RX lockup detection - set when lockup has been detected by any of the lockup monitors on the receive datapath."	RW RtoClr	0
29	tsu_timer_comparison_interrupt	"TSU timer comparison interrupt. Indicates when TSU timer count value is equal to programmed value."	RW RtoClr	0
28	wol_interrupt	"WOL interrupt. Indicates a WOL event has been received."	RW RtoClr	0
27	receive_lpi_indication_status_bit_change	"Receive LPI indication status bit change"	RW RtoClr	0
26	tsu_seconds_register_increment	"TSU seconds register increment indicates the register has incremented. Cleared on read."	RW RtoClr	0
25	ptp_pdelay_resp_frame_transmitted	"PTP pdelay_resp frame transmitted indicates a PTP pdelay_resp frame has been transmitted. Cleared on read."	RW RtoClr	0
24	ptp_pdelay_req_frame_transmitted	"PTP pdelay_req frame transmitted indicates a PTP pdelay_req frame has been transmitted. Cleared on read."	RW RtoClr	0
23	ptp_pdelay_resp_frame_received	"PTP pdelay_resp frame received indicates a PTP pdelay_resp frame has been received. Cleared on read."	RW RtoClr	0
22	ptp_pdelay_req_frame_received	"PTP pdelay_req frame received indicates a PTP pdelay_req frame has been received. Cleared on read."	RW RtoClr	0
21	ptp_sync_frame_transmitted	"PTP sync frame transmitted indicates a PTP sync frame has been transmitted. Cleared on read."	RW RtoClr	0
20	ptp_delay_req_frame_transmitted	"PTP delay_req frame transmitted indicates a PTP delay_req frame has been transmitted. Cleared on read."	RW RtoClr	0
19	ptp_sync_frame_received	"PTP sync frame received indicates a PTP sync frame has been received. Cleared on read."	RW RtoClr	0
18	ptp_delay_req_frame_received	"PTP delay_req frame received indicates a PTP delay_req frame has been received. Cleared on read."	RW RtoClr	0

17	pcs_link_partner_page_received	"PCS link partner page received - set when a new base page or next page is received from the link partner. The first time this interrupt is received, it will indicate base page received and subsequent reads will indicate next pages. The next page and base page registers should only be read when this interrupt is signalled. For next pages, the link partner next page register should be read first to avoid the register being over written. This interrupt also indicates when the host should write a new page into the next page register. If further next page exchange is only required by the link partner, this register should be written with a null message page (0x2001). Cleared on read."	RW RtoClr	0
16	pcs_auto_negotiation_complete	"PCS auto-negotiation complete - set once the internal PCS layer has completed auto-negotiation. Cleared on read."	RW RtoClr	0
15	external_interrupt	"External interrupt - set when a rising edge has been detected on the ext_interrupt_in input pin. Cleared on read."	RW RtoClr	0
14	pause_frame_transmitted	"Pause frame transmitted - indicates a pause frame has been successfully transmitted after being initiated from the network control register or from the tx_pause control pin. Cleared on read."	RW RtoClr	0
13	pause_time_elapsed	"Pause Time elapsed - set when either the pause time register at address 0x38 decrements to zero, or when a valid pause frame is received with a zero pause quantum field. Not set for PFC pause frames. Cleared on read."	RW RtoClr	0
12	pause_frame_with_non_zero_pause_quantum_received	"Pause frame with non-zero pause quantum received - indicates a valid legacy pause frame with a non-zero pause quantum field has been received or any valid PFC pause frame has been received. Cleared on read."	RW RtoClr	0
11	resp_not_ok	"bresp/hresp not OK - set when the DMA block sees bresp/hresp not OK. Cleared on read."	RW RtoClr	0
10	receive_overrun	"Receive overrun - set when the receive overrun status bit gets set. Cleared on read."	RW RtoClr	0
9	link_change	"Link change - set when the PCS link status changes. If auto-negotiation is enabled then link status goes high when it completes. If AN is disabled link status goes high when synchronization is achieved. If 802.3cb link fault signalling is enabled a link change interrupt is triggered when the link fault indication as reported in the network status register changes. Cleared on read."	RW RtoClr	0
8	reserved_8	"Reserved, read as 0, ignored on write."	RO	0
7	transmit_complete	"Transmit complete - set when a frame has been transmitted. Cleared on read."	RW RtoClr	0
6	amba_error	"Transmit frame corruption due to AMBA (AHB/AXI) error. Set if an error occurs whilst midway through reading transmit frame from external system memory, including HRESP errors(AHB), RRESP or BRESP(AXI) errors and buffers exhausted mid frame (if the buffers run out during transmission of a frame then transmission stops, FCS shall be bad and tx_er asserted). Also set in DMA packet buffer mode if single frame is too large for configured packet buffer memory size. Cleared on a read."	RW RtoClr	0

5	retry_limit_exceeded_or_late_collision	"Retry limit exceeded or late collision - transmit error. Late collision will only cause this status bit to be set in gigabit mode (as a retry is not attempted). Cleared on read."	RW RtoClr	0
4	transmit_under_run	"Transmit under run - this interrupt is set if the transmitter was forced to terminate a frame that it has already began transmitting due to further data being unavailable. If an under run occurs, the transmitter will force bad crc and tx_er high. This interrupt is set if a transmitter status write back has not completed when another status write back is attempted. When using the DMA interface configured for internal FIFO mode, this interrupt is also set when the transmit DMA has written the SOP data into the FIFO and either the AHB bus was not granted in time for further data, or because an AHB/AXI error response was returned by the connected slave, or because a used bit was read. When using the DMA interface configured for packet buffer mode, this bit will never be set. When using the external FIFO interface, this interrupt is also set when the tx_r_underflow input was asserted during a frame transfer. Cleared on read."	RW RtoClr	0
3	tx_used_bit_read	"TX used bit read - set when a transmit buffer descriptor is read with its used bit set. Cleared on read."	RW RtoClr	0
2	rx_used_bit_read	"RX used bit read - set when a receive buffer descriptor is read with its used bit set. Cleared on read."	RW RtoClr	0
1	receive_complete	"Receive complete - a frame has been stored in memory. Cleared on read."	RW RtoClr	0
0	management_frame_sent	"Management frame sent - the PHY management register has completed its operation. Cleared on read."	RW RtoClr	0

Table 11-12: int_enable

Register Information			
Description	"At reset all interrupts are disabled. Writing a one to the relevant bit location enables the required interrupt. This register is write only and when read will return zero."		
Offset	0x0028	**Type:**	RW

Bits	Name	Description	Access	Reset
31	enable_tx_lockup_detected_interrupt	"Enable TX lockup detection interrupt."	WO	0
30	enable_rx_lockup_detected_interrupt	"Enable RX lockup detection interrupt."	WO	0
29	enable_tsu_timer_comparison_interrupt	"Enable TSU timer comparison interrupt."	WO	0
28	enable_wol_event_received_interrupt	"Enable WOL event received interrupt"	WO	0
27	enable_rx_lpi_indication_interrupt	"Enable RX LPI indication interrupt"	WO	0
26	enable_tsu_seconds_register_increment	"Enable TSU seconds register increment"	WO	0
25	enable_ptp_pdelay_resp_frame_transmitted	"Enable PTP pdelay_resp frame transmitted"	WO	0
24	enable_ptp_pdelay_req_frame_transmitted	"Enable PTP pdelay_req frame transmitted"	WO	0
23	enable_ptp_pdelay_resp_frame_received	"Enable PTP pdelay_resp frame received"	WO	0
22	enable_ptp_pdelay_req_frame_received	"Enable PTP pdelay_req frame received"	WO	0
21	enable_ptp_sync_frame_transmitted	"Enable PTP sync frame transmitted"	WO	0
20	enable_ptp_delay_req_frame_transmitted	"Enable PTP delay_req frame transmitted"	WO	0
19	enable_ptp_sync_frame_received	"Enable PTP sync frame received"	WO	0
18	enable_ptp_delay_req_frame_received	"Enable PTP delay_req frame received"	WO	0
17	enable_pcs_link_partner_page_received	"Enable PCS link partner page received"	WO	0
16	enable_pcs_auto_negotiation_complete_interrupt	"Enable PCS auto-negotiation complete interrupt"	WO	0
15	enable_external_interrupt	"Enable external interrupt"	WO	0

14	enable_pause_frame_transmitted_interrupt	"Enable pause frame transmitted interrupt"	WO	0
13	enable_pause_time_zero_interrupt	"Enable pause time zero interrupt"	WO	0
12	enable_pause_frame_with_non_zero_pause_quantum_interrupt	"Enable pause frame with non-zero pause quantum interrupt"	WO	0
11	enable_resp_not_ok_interrupt	"Enable bresp/hresp not OK interrupt"	WO	0
10	enable_receive_overrun_interrupt	"Enable receive overrun interrupt"	WO	0
9	enable_link_change_interrupt	"Enable link change interrupt"	WO	0
8	reserved_8	"Reserved, read as 0, ignored on write."	RO	0
7	enable_transmit_complete_interrupt	"Enable transmit complete interrupt"	WO	0
6	enable_transmit_frame_corruption_due_to_amba_error_interrupt	"Enable transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	enable_retry_limit_exceeded_or_late_collision_interrupt	"Enable retry limit exceeded or late collision interrupt"	WO	0
4	enable_transmit_buffer_under_run_interrupt	"Enable transmit buffer under run interrupt"	WO	0
3	enable_transmit_used_bit_read_interrupt	"Enable transmit used bit read interrupt"	WO	0
2	enable_receive_used_bit_read_interrupt	"Enable receive used bit read interrupt"	WO	0
1	enable_receive_complete_interrupt	"Enable receive complete interrupt"	WO	0
0	enable_management_done_interrupt	"Enable management done interrupt"	WO	0

Table 11-13: int_disable

Register Information			
Description	"Writing a 1 to the relevant bit location disables that particular interrupt. This register is write only and when read will return zero."		
Offset	0x002C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	disable_tx_lockup_detected_interrupt	"Disable TX lockup detection interrupt."	WO	0
30	disable_rx_lockup_detected_interrupt	"Disable RX lockup detection interrupt."	WO	0
29	disable_tsu_timer_comparison_interrupt	"Disable TSU timer comparison interrupt."	WO	0
28	disable_wol_event_received_interrupt	"Disable WOL event received interrupt"	WO	0
27	disable_rx_lpi_indication_interrupt	"Disable RX LPI indication interrupt"	WO	0
26	disable_tsu_seconds_register_increment	"Disable TSU seconds register increment"	WO	0
25	disable_ptp_pdelay_resp_frame_transmitted	"Disable PTP pdelay_resp frame transmitted"	WO	0
24	disable_ptp_pdelay_req_frame_transmitted	"Disable PTP pdelay_req frame transmitted"	WO	0
23	disable_ptp_pdelay_resp_frame_received	"Disable PTP pdelay_resp frame received"	WO	0
22	disable_ptp_pdelay_req_frame_received	"Disable PTP pdelay_req frame received"	WO	0
21	disable_ptp_sync_frame_transmitted	"Disable PTP sync frame transmitted"	WO	0
20	disable_ptp_delay_req_frame_transmitted	"Disable PTP delay_req frame transmitted"	WO	0
19	disable_ptp_sync_frame_received	"Disable PTP sync frame received"	WO	0
18	disable_ptp_delay_req_frame_received	"Disable PTP delay_req frame received"	WO	0
17	disable_pcs_link_partner_page_received	"Disable PCS link partner page received"	WO	0
16	disable_pcs_auto_negotiation_complete_interrupt	"Disable PCS auto-negotiation complete interrupt"	WO	0
15	disable_external_interrupt	"Disable external interrupt"	WO	0

14	disable_pause_frame_transmitted_interrupt	"Disable pause frame transmitted interrupt"	WO	0
13	disable_pause_time_zero_interrupt	"Disable pause time zero interrupt"	WO	0
12	disable_pause_frame_with_non_zero_pause_quantum_interrupt	"Disable pause frame with non-zero pause quantum interrupt"	WO	0
11	disable_resp_not_ok_interrupt	"Disable bresp/hresp not OK interrupt"	WO	0
10	disable_receive_overrun_interrupt	"Disable receive overrun interrupt"	WO	0
9	disable_link_change_interrupt	"Disable link change interrupt"	WO	0
8	reserved_8	"Reserved, read as 0, ignored on write."	RO	0
7	disable_transmit_complete_interrupt	"Disable transmit complete interrupt"	WO	0
6	disable_transmit_frame_corruption_due_to_amba_error_interrupt	"Disable transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	disable_retry_limit_exceeded_or_late_collision_interrupt	"Disable retry limit exceeded or late collision interrupt"	WO	0
4	disable_transmit_buffer_under_run_interrupt	"Disable transmit buffer under run interrupt"	WO	0
3	disable_transmit_used_bit_read_interrupt	"Disable transmit used bit read interrupt"	WO	0
2	disable_receive_used_bit_read_interrupt	"Disable receive used bit read interrupt"	WO	0
1	disable_receive_complete_interrupt	"Disable receive complete interrupt"	WO	0
0	disable_management_done_interrupt	"Disable management done interrupt"	WO	0

Table 11-14: int_mask

Register Information			
Description	"The interrupt mask register is a read only register indicating which interrupts are masked. All bits are set at reset and can be reset individually by writing to the interrupt enable register or set individually by writing to the interrupt disable register. Having separate address locations for enable and disable saves the need for performing a read modify write when updating the interrupt mask register. For test purposes there is a write-only function to this register that allows the bits in the interrupt status register to be set or cleared, regardless of the state of the mask register."		
Offset	0x0030	**Type:**	RO

Bits	Name	Description	Access	Reset
31	tx_lockup_detected_mask	"TX lockup interrupt mask."	RO	1
30	rx_lockup_detected_mask	"RX lockup interrupt mask."	RO	1
29	tsu_timer_comparison_mask	"Enable TSU timer comparison interrupt mask."	RO	1
28	wol_event_received_mask	"A read of this register returns the value of the WOL event received mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
27	rx_lpi_indication_mask	"A read of this register returns the value of the RX LPI indication mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written"	RO	1
26	tsu_seconds_register_increment_mask	"A read of this register returns the value of the TSU seconds register increment mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
25	ptp_pdelay_resp_frame_transmitted_mask	"A read of this register returns the value of the PTP pdelay_resp frame transmitted mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
24	ptp_pdelay_req_frame_transmitted_mask	"A read of this register returns the value of the PTP pdelay_req frame transmitted mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1

23	ptp_pdelay_resp_frame_received_mask	"A read of this register returns the value of the PTP pdelay_resp frame received mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
22	ptp_pdelay_req_frame_received_mask	"A read of this register returns the value of the PTP pdelay_req frame received mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
21	ptp_sync_frame_transmitted_mask	"A read of this register returns the value of the PTP sync frame transmitted mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
20	ptp_delay_req_frame_transmitted_mask	"A read of this register returns the value of the PTP delay_req frame transmitted mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
19	ptp_sync_frame_received_mask	"A read of this register returns the value of the PTP sync frame received mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
18	ptp_delay_req_frame_received_mask	"A read of this register returns the value of the PTP delay_req frame received mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
17	pcs_link_partner_page_mask	"A read of this register returns the value of the PCS link partner page mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
16	pcs_auto_negotiation_complete_interrupt_mask	"A read of this register returns the value of the PCS auto-negotiation complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1

15	external_interrupt_mask	"External interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
14	pause_frame_transmitted_interrupt_mask	"Pause frame transmitted interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
13	pause_time_zero_interrupt_mask	"Pause time zero interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
12	pause_frame_with_non_zero_pause_quantum_interrupt_mask	"Pause frame with non-zero pause quantum interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
11	resp_not_ok_interrupt_mask	"bresp/hresp not OK interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
10	receive_overrun_interrupt_mask	"Receive overrun interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
9	link_change_interrupt_mask	"A read of this register returns the value of the link change interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
8	reserved_8	"Reserved, read as 0, ignored on write."	RO	0
7	transmit_complete_interrupt_mask	"Transmit complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
6	amba_error_interrupt_mask	"Transmit frame corruption due to AMBA (AHB/AXI) error interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1

5	retry_limit_exceeded_or_late_collision_mask	"A read of this register returns the value of the retry limit exceeded or late collision (gigabit mode only) interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
4	transmit_buffer_under_run_interrupt_mask	"Transmit buffer under run interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
3	transmit_used_bit_read_interrupt_mask	"Transmit used bit read interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
2	receive_used_bit_read_interrupt_mask	"Receive used bit read interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
1	receive_complete_interrupt_mask	"Receive complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
0	management_done_interrupt_mask	"Management done interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1

Table 11-15: phy_management

Register Information			
Description	"The PHY management register is implemented as a shift register. Writing to the register starts a shift operation which is signalled as complete when bit-2 is set in the network status register. It takes about 2000 pclk cycles to complete, when MDC is set for pclk divide by 32 in the network configuration register. An interrupt is generated upon completion. During this time, the MSB of the register is output on the MDIO pin and the LSB updated from the MDIO pin with each MDC cycle. This causes transmission of a PHY management frame on MDIO. See Section 22.2.4.5 of the IEEE 802.3 standard. Reading during the shift operation will return the current contents of the shift register. At the end of management operation, the bits will have shifted back to their original locations. For a read operation, the data bits will be updated with data read from the PHY. It is important to write the correct values to the register to ensure a valid PHY management frame is produced. The MDIO interface can read IEEE 802.3 clause 45 PHYs as well as clause 22 PHYs. To read clause 45 PHYs, bit 30 should be written with a 0 rather than a 1. For a description of MDC generation, see Network Configuration Register."		
Offset	0x0034	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	write0	"Must be written with 0."	RW	0
30	write1	"Must be written to 1 for a valid Clause 22 frame and to 0 for a valid Clause 45 frame."	RW	0
29:28	operation	"Operation. For a Clause 45 frame: 00 is an addr, 01 is a write, 10 is a post read increment, 11 is a read frame. For a Clause 22 frame: 10 is a read, 01 is a write."	RW	0x0
27:23	phy_address	"PHY address."	RW	0x00
22:18	register_address	"Register address - specifies the register in the PHY to access."	RW	0x00
17:16	write10	"Must be written with 10."	RW	0x0
15:0	phy_write_read_data	"For a write operation this is written with the data to be written to the PHY. After a read operation this contains the data read from the PHY."	RW	0x0000

Table 11-16: pause_time

Register Information			
Description	"Received Pause Quantum Register"		
Offset	0x0038	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	quantum	"Received pause quantum - stores the current value of the received pause quantum register which is decremented every 512 bit times."	RO	0x0000

Table 11-17: tx_pause_quantum

Register Information				
Description	"Transmit Pause Quantum Register"			
Offset	0x003C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	quantum_p1	"Transmit pause quantum - written with the pause quantum value for pause frame transmission of priority 1."	RW	0xFFFF
15:0	quantum	"Transmit pause quantum - written with the pause quantum value for pause frame transmission."	RW	0xFFFF

Table 11-18: pbuf_txcutthru

Register Information				
Description	"Transmit Partial Store and Forward Register. Partial store and forward is only applicable when using the DMA configured in SRAM based packet buffer mode. It is also not available when using multi buffer frames. This register contains the enable bit and watermark value for transmit cut-through operation."			
Offset	0x0040	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	dma_tx_cutthru	"Enable TX partial store and forward operation"	RW	0
30:14	reserved	"Reserved, read as 0, ignored on write."	RO	0x0 0000
13:0	dma_tx_cutthru_threshold	"Watermark value corresponding to number of SRAM locations. This value must be >= 0x14. The actual number of bytes for the watermark is obtained by multiplying the watermark value by the value of the gem_tx_pbuf_data define divided by 8. The reset value depends on the value of the configuration option `gem_tx_pbuf_addr, which is defined in the verilog defs configuration file. The value chosen for the generation of the userguide was `gem_tx_pbuf_addr = 14"	RW	0x3FFF

Table 11-19: pbuf_rxcutthru

Register Information			
Description	"Receive Partial Store and Forward Register. Partial store and forward is only applicable when using the DMA configured in SRAM based packet buffer mode. It is also not available when using multi buffer frames. This register contains the enable bit and watermark value for receive cut-through operation."		
Offset	0x0044	**Type:**	RW

Bits	Name	Description	Access	Reset
31	dma_rx_cutthru	"Enable RX partial store and forward operation"	RW	0
30:11	reserved	"Reserved, read as 0, ignored on write."	RO	0x0 0000
10:0	dma_rx_cutthru_threshold	"Watermark value corresponding to number of SRAM locations. The actual number of bytes for the watermark is obtained by multiplying the watermark value by the value of the gem_rx_pbuf_data define divided by 8. The reset value depends on the value of the configuration option `gem_rx_pbuf_addr, which is defined in the verilog defs configuration file. The value chosen for the generation of the userguide was `gem_rx_pbuf_addr = 11"	RW	0x7FF

Table 11-20: jumbo_max_length

Register Information			
Description	"Maximum Jumbo Frame Size."		
Offset	0x0048	**Type:**	RW

Bits	Name	Description	Access	Reset
31:14	reserved_31_14	"Reserved, read as 0, ignored on write."	RO	0x0 0000
13:0	jumbo_max_length	"Maximum Jumbo Frame Size - resets to the gem_jumbo_max_length define value."	RW	0x2800

Table 11-21: external_fifo_interface

Register Information			
Description	"External FIFO Interface Enable (only valid when gem_host_if_soft_select is defined)"		
Offset	0x004C	**Type:**	RW

Bits	Name	Description	Access	Reset
31:1	reserved_31_1	"Reserved, read as 0, ignored on write."	RO	0x0000 0000
0	external_fifo_interface	"Enable external fifo interface. 1: Enabled. 0: Disabled."	RW	0

Table 11-22: axi_max_pipeline

Register Information			
Description	"Used to set the maximum amount of outstanding transactions on the AXI bus between AR / R channels and AW / W channels. Cannot be more than the depth of the configured AXI pipeline FIFO (defined in verilog defs.v)"		
Offset	0x0054	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:17	reserved	"Reserved, read as 0, ignored on write."	RO	0x0000
16	use_aw2b_fill	"For the write issuing capability as defined in bits 15:8 of this register, select whether the max number of transactions operates between the AW to W AXI channel or the AW to B channel. Set to 0 to operate between the AW and W channels. Set to 1 to operate between the AW and B channels."	RW	0
15:8	aw2w_max_pipelin e	"Defines the maximum number of outstanding AXI write requests that can be issued by the DMA via the AW channel. This is effectively the write issuing capability."	RW	0x01
7:0	ar2r_max_pipeline	"Defines the maximum number of outstanding AXI read requests that can be issued by the DMA via the AR channel. This is effectively the read issuing capability."	RW	0x01

Table 11-23: rsc_control

Register Information			
Description	"Used to enable Receive side coalescing on queues 1-15"		
Offset	0x0058	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:17	reserved_31_17	"Reserved, read as 0, ignored on write."	RO	0x0000
16	rsc_clr_mask	"Mask the clearing of the rsc_en bit. When set to 1 this bit will prevent the hardware from clearing the rsc_en bit when the state machines detect a Flag set during the coalescing function."	RW	0
15:1	rsc_control	"Enables Receive Side Coalescing. Bit 1 enables RSC on queue 1, Bit 2 on queue 2 etc. RSC on queue 0 is not permitted."	RW	0x0000
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-24: int_moderation

Register Information			
Description	"Used to moderate the number of transmit and receive complete interrupts issued. With interrupt moderation enabled receive and transmit interrupts are not generated immediately a frame is transmitted or received. Instead when a receive or transmit event occurs a timer is started and the interrupt is asserted after it times out. This limits the frequency with which the CPU receives interrupts. When interrupt moderation is enabled interrupt status bit one is always used for receive and bit 7 is always used for transmit even when priority queuing is enabled. With interrupt moderation 800ns periods are counted. GEM determines what constitutes an 800ns period by looking at the tbi (bit 11), gigabit bit (10) and speed (bit 0) bits in the network configuration register and counting tx_clk cycles. Bit 0 needs to be set to 1 for 100M operation. From release 1p11 onwards a frame threshold value may also be used to moderate interrupts. If both time based and frame threshold moderation is enabled the interrupt will be asserted as soon as the first moderation method expires."		
Offset	0x005C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	tx_int_mod_thresh	"Count of transmitted frames before bit 7 is set in the interrupt status register. A non-zero value indicates transmit interrupt moderation will be performed."	RW	0x00
23:16	tx_int_moderation	"Count of 800ns periods before bit 7 is set in the interrupt status register after a frame is transmitted. A non-zero value indicates transmit interrupt moderation will be performed."	RW	0x00
15:8	rx_int_mod_thresh	"Count of received frames before bit 1 is set in the interrupt status register. A non-zero value indicates receive interrupt moderation will be performed."	RW	0x00
7:0	rx_int_moderation	"Count of 800ns periods before bit 1 is set in the interrupt status register after a frame is received. A non-zero value indicates receive interrupt moderation will be performed."	RW	0x00

Table 11-25: sys_wake_time

Register Information			
Description	"Used to pause transmission after deassertion of tx_lpi_en. Each unit in this register corresponds to 25.6ns in 2.5G mode, 64ns in gigabit mode, 320ns in 100M mode and 3200ns at 10M. After tx_lpi_en is deasserted transmission will pause for the set time."		
Offset	0x0060	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	sys_wake_time	"Count of 25.6ns, 64ns, 320ns or 3200ns intervals before transmission starts after deassertion of tx_lpi_en (each interval is equivalent to eight tx_clk periods and so varies with data rate)."	RW	0x0000

Table 11-26: lockup_config

Register Information			
Description	"The lockup detection and recovery configuration register."		
Offset	0x0068	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	tx_dma_lockup_mon_en	"Enable the TX DMA lockup detector. Enable the monitor that detects lockups in the transmit DMA."	RW	0
30	tx_mac_lockup_mon_en	"Enable the TX MAC lockup detector. Enable the monitor that detects lockups in the MAC transmit path."	RW	0
29	rx_dma_lockup_mon_en	"Enable the RX DMA lockup detector. Enable the monitor that detects lockups in the receive DMA."	RW	0
28	rx_mac_lockup_mon_en	"Enable the RX MAC lockup detector. Enable the monitor that detects lockups in the MAC receive path."	RW	0
27	lockup_recovery_en	"Lockup recovery. If this bit is set then the module will go into a soft reset state if a lockup is detected on the transmit or receive data paths."	RW	0
26:16	dma_lockup_time	"Prior to release 1p11 this field was reserved. From release 1p11 onwards this field defines the timeout value for transmit and receive DMA lockup detection defined as a multiple of the prescaler value stored in bits 15:0 of this register."	RW	0x7FF
15:0	prescaler_value	"For release 1p10 this field defined the time (measured in units of 1024 tx_clk periods) after which the lockup detection monitors would trigger, except for the receive MAC lock up detector which had its lockup time register. From release 1p11 onwards this field defines a prescaler value which is the number of tx_clk periods used to scale the timeout registers."	RW	0xFFFF

Table 11-27: mac_lockup_time

Register Information			
Description	"MAC lockup detection time register. For receive this register specifies the maximum time between received frames. If no valid EOP is seen at the receive FIFO interface within the timer period then a lockup is considered to have occurred in the receive MAC. For transmit the MAC lockup time is simply the time it takes for data to be seen on the MII output pins after entering on the MAC FIFO interface."		
Offset	0x006C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:27	reserved_31_27	"Reserved, read as 0, ignored on write."	RO	0x00
26:16	tx_mac_lockup_time	"Prior to release 1p11 this field was reserved. From release 1p11 onwards this field defines the timeout value for transmit MAC lockup detection defined as a multiple of the prescaler value stored in bits 15:0 of the lockup config register."	RW	0x7FF
15:0	rx_mac_lockup_time	"The time after which the receive MAC lockup detection monitor will trigger. For release 1p10 this was measured in units of 1024 tx_clk periods. From release 1p11 onwards it is used in conjunction with the prescaler stored in bits 15:0 of the lockup config register."	RW	0xFFFF

Table 11-28: lockup_config3

Register Information			
Description	"DMA TX Lockup Enable Control Register. This register enables the lockup timer for each individual queue. If this is set to 0, the number of outstanding packets are still counted but the actual timer does not run."		
Offset	0x0070	**Type:**	RW

Bits	Name	Description	Access	Reset
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15	dma_tx_lockup_en_q_15	"Enable DMA TX lockup timer for queue 15."	RW	0
14	dma_tx_lockup_en_q_14	"Enable DMA TX lockup timer for queue 14."	RW	0
13	dma_tx_lockup_en_q_13	"Enable DMA TX lockup timer for queue 13."	RW	0
12	dma_tx_lockup_en_q_12	"Enable DMA TX lockup timer for queue 12."	RW	0
11	dma_tx_lockup_en_q_11	"Enable DMA TX lockup timer for queue 11."	RW	0
10	dma_tx_lockup_en_q_10	"Enable DMA TX lockup timer for queue 10."	RW	0
9	dma_tx_lockup_en_q_9	"Enable DMA TX lockup timer for queue 9."	RW	0
8	dma_tx_lockup_en_q_8	"Enable DMA TX lockup timer for queue 8."	RW	0
7	dma_tx_lockup_en_q_7	"Enable DMA TX lockup timer for queue 7."	RW	0
6	dma_tx_lockup_en_q_6	"Enable DMA TX lockup timer for queue 6."	RW	0
5	dma_tx_lockup_en_q_5	"Enable DMA TX lockup timer for queue 5."	RW	0
4	dma_tx_lockup_en_q_4	"Enable DMA TX lockup timer for queue 4."	RW	0
3	dma_tx_lockup_en_q_3	"Enable DMA TX lockup timer for queue 3."	RW	0
2	dma_tx_lockup_en_q_2	"Enable DMA TX lockup timer for queue 2."	RW	0
1	dma_tx_lockup_en_q_1	"Enable DMA TX lockup timer for queue 1."	RW	0
0	dma_tx_lockup_en_q_0	"Enable DMA TX lockup timer for queue 0."	RW	0

Table 11-29: rx_water_mark

Register Information				
Description	"rx_water_mark - Receive water mark register. This register contains the high and low water-marks for automatic transmission of pause frames. The water-marks are compared against the internal signal rx_dpram_fill_lvl which is readable in the dpram_fill_dbg register; rx_dpram_fill_lvl indicates the number of words used in the receive SRAM (word length is configuration dependent and may be 4, 8 and 16 bytes). A value of zero in a field disables the corresponding functionality."			
Offset	0x007C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	rx_low_watermark	"If this field is non-zero and the last pause frame transmitted was non-zero then a zero length pause frame is transmitted when the receive SRAM fill level falls below this value."	RW	0x0000
15:0	rx_high_watermark	"If this field is non-zero and the receive SRAM fill level exceeds this value then a pause frame is transmitted"	RW	0x0000

Table 11-30: hash_bottom

Register Information				
Description	"The unicast hash enable and the multicast hash enable bits in the network configuration register enable the reception of hash matched frames. Hash Register Bottom 31:0."			
Offset	0x0080	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"The first 32 bits of the hash address register."	RW	0x0000 0000

Table 11-31: hash_top

Register Information				
Description	"Hash Register Top 63:32"			
Offset	0x0084	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"The remaining 32 bits of the hash address register."	RW	0x0000 0000

Table 11-32: spec_add1_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x0088	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-33: spec_add1_top

Register Information				
Description	"Specific Address Top"			
Offset	0x008C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:17	reserved_31_17	"Reserved, read as 0, ignored on write."	RO	0x0000
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. Octets 5 and 4 of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-34: spec_add2_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x0090	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-35: spec_add2_top

Register Information				
Description	"Specific Address Top"			
Offset	0x0094	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-36: spec_add3_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x0098	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-37: spec_add3_top

Register Information				
Description	"Specific Address Top"			
Offset	0x009C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-38: spec_add4_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x00A0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-39: spec_add4_top

Register Information				
Description	"Specific Address Top"			
Offset	0x00A4	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-40: spec_type1

Register Information				
Description	"Type ID Match 1"			
Offset	0x00A8	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	enable_copy	"Enable copying of type ID match 1 matched frames."	RW	0
30:16	reserved_30_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	match	"Type ID match 1. For use in comparisons with received frames type ID/length field."	RW	0x0000

Table 11-41: spec_type2

Register Information				
Description	"Type ID Match 2"			
Offset	0x00AC	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	enable_copy	"Enable copying of type ID match 2 matched frames."	RW	0
30:16	reserved_30_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	match	"Type ID match 2. For use in comparisons with received frames type ID/length field."	RW	0x0000

Table 11-42: spec_type3

Register Information					
Description	"Type ID Match 3"				
Offset	0x00B0		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31	enable_copy	"Enable copying of type ID match 3 matched frames."		RW	0
30:16	reserved_30_16	"Reserved, read as 0, ignored on write."		RO	0x0000
15:0	match	"Type ID match 3. For use in comparisons with received frames type ID/length field."		RW	0x0000

Table 11-43: spec_type4

Register Information					
Description	"Type ID Match 4"				
Offset	0x00B4		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31	enable_copy	"Enable copying of type ID match 4 matched frames."		RW	0
30:16	reserved_30_16	"Reserved, read as 0, ignored on write."		RO	0x0000
15:0	match	"Type ID match 4. For use in comparisons with received frames type ID/length field."		RW	0x0000

Table 11-44: wol_register

Register Information				
Description	"Wake on LAN Register"			
Offset	0x00B8	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:20	reserved_31_20	"Reserved - read 0, ignored when written."	RO	0x000
19	wol_mask_3	"Wake on LAN multicast hash event enable. When set multicast hash events will cause the wol output to be asserted."	RW	0
18	wol_mask_2	"Wake on LAN specific address register 1 event enable. When set specific address 1 events will cause the wol output to be asserted."	RW	0
17	wol_mask_1	"Wake on LAN ARP request event enable. When set ARP request events will cause the wol output to be asserted."	RW	0
16	wol_mask_0	"Wake on LAN magic packet event enable. When set magic packet events will cause the wol output to be asserted."	RW	0
15:0	addr	"Wake on LAN ARP request IP address. Written to define the least significant 16 bits of the target IP address that is matched to generate a Wake on LAN event. A value of zero will not generate an event, even if this is matched by the received frame."	RW	0x0000

Table 11-45: stretch_ratio

Register Information				
Description	"IPG stretch register"			
Offset	0x00BC	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	ipg_stretch	"IPG Stretch. Bits 7:0 are multiplied with the previously transmitted frame length (including preamble) bits 15:8 +1 divide the frame length. If the resulting number is greater than 96 and bit 28 is set in the network configuration register then the resulting number is used for the transmit inter-packet-gap. 1 is added to bits 15:8 to prevent a divide by zero."	RW	0x0000

Table 11-46: stacked_vlan

Register Information					
Description	"Stacked VLAN Register"				
Offset	0x00C0		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31	enable_processing	"Enable stacked VLAN processing mode"		RW	0
30:16	reserved_30_16	"Reserved, read as 0, ignored on write."		RO	0x0000
15:0	match	"User defined VLAN_TYPE field. When Stacked VLAN is enabled, the first VLAN tag in a received frame will only be accepted if the VLAN type field is equal to this user defined VLAN_TYPE OR equal to the standard VLAN type (0x8100). Note that the second VLAN tag of a Stacked VLAN packet will only be matched correctly if its VLAN_TYPE field equals 0x8100."		RW	0x0000

Table 11-47: tx_pfc_pause

Register Information					
Description	"Transmit PFC Pause Register"				
Offset	0x00C4		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."		RO	0x0000
15:8	vector	"Priority Vector Pause Size. If bit 17 of the network control register is written with a one then for each entry equal to zero in the Transmit PFC Pause Register[15:8], the PFC pause frame's pause quantum field associated with that entry will be taken from the transmit pause quantum register. For each entry equal to one in the Transmit PFC Pause Register [15:8], the pause quantum associated with that entry will be zero."		RW	0x00
7:0	vector_enable	"Priority Vector Enable. If bit 17 of the network control register is written with a one then the priority enable vector of the PFC priority based pause frame will be set equal to the value stored in this register [7:0]."		RW	0x00

Table 11-48: mask_add1_bottom

Register Information					
Description	"Specific Address Mask 1 Bottom 31:0"				
Offset	0x00C8		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	address_mask	"Specific Address Mask. Setting a bit to one masks the corresponding bit in the specific address 1 register"		RW	0x0000 0000

Table 11-49: mask_add1_top

Register Information				
Description	"Specific Address Mask 1 Top 47:32"			
Offset	0x00CC	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	address_mask	"Specific Address Mask. Setting a bit to one masks the corresponding bit in the specific address 1 register"	RW	0x0000

Table 11-50: dma_addr_or_mask

Register Information				
Description	"Receive DMA Data Buffer Address Mask - only applies to AHB operation"			
Offset	0x00D0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:28	mask_value	"Data Buffer Address Mask Value. Values used to force bits 31:28 of the receive data buffer AHB address to a particular value when the associated enable bits stored in this register [3:0] are set. Any changes to this register will be ignored while the DMA is currently processing a receive packet. It will only affect the next full packet to be written to external system memory."	RW	0x0
27:4	reserved_27_4	"Reserved, read as 0, ignored on write."	RO	0x00 0000
3:0	mask_enable	"Data Buffer Address Mask Enable. These bits are associated directly with bits[31:28].When bit 0 is set, the AHB address bit 28 used for accessing the receive data buffers will be forced to the value stored in bit 28 of this register. When bit 1 is set, the AHB address bit 29 used for accessing the receive data buffers will be forced to the value stored in bit 29 of this register. When bit 2 is set, the AHB address bit 30 used for accessing the receive data buffers will be forced to the value stored in bit 30 of this register. When bit 3 is set, the AHB address bit 31 used for accessing the receive data buffers will be forced to the value stored in bit 31 of this register. When these bits are clear, the associated value stored in bits 31:28 have no effect on the AHB address used for receive data buffer accesses. Any changes to this register will be ignored while the DMA is currently processing a receive packet. It will only affect the next full packet to be written to external memory."	RW	0x0

Table 11-51: rx_ptp_unicast

Register Information					
Description	"PTP RX unicast IP destination address"				
Offset	0x00D4		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:0	address	"Unicast IP destination address. Used for detection of PTP frames on receive path."	RW	0x0000 0000	

Table 11-52: tx_ptp_unicast

Register Information					
Description	"PTP TX unicast IP destination address"				
Offset	0x00D8		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:0	address	"Unicast IP destination address. Used for detection of PTP frames on transmit path."	RW	0x0000 0000	

Table 11-53: tsu_nsec_cmp

Register Information					
Description	"TSU timer comparison value nanoseconds"				
Offset	0x00DC		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:22	reserved_31_22	"Reserved, read as 0, ignored on write."	RO	0x000	
21:0	comparison_value	"TSU timer comparison value (ns). Value is compared to the bits[45:24] of the TSU timer count value (upper 22 bits of nanosecond value). The output tsu_timer_cmp_val is driven high when the timer comparison values match the TSU count value. From release 1p11 onwards this comparison only occurs once the nanosecond compare register has been written and when the match occurs tsu_timer_cmp_val will only be driven high for a single tsu_clk period."	RW	0x00 0000	

Table 11-54: tsu_sec_cmp

Register Information			
Description	"TSU timer comparison value seconds 31:0"		
Offset	0x00E0	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	comparison_value	"TSU timer comparison value (s). Value is compared to seconds value bits [31:0] of the TSU timer count value."	RW	0x0000 0000

Table 11-55: tsu_msb_sec_cmp

Register Information			
Description	"TSU timer comparison value seconds 47:32"		
Offset	0x00E4	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	comparison_value	"TSU timer comparison value (s). Value is compared to the top 16 bits (most significant 16-bits [47:32] of seconds value) of the TSU timer count value."	RW	0x0000

Table 11-56: tsu_ptp_tx_msb_sec

Register Information			
Description	"PTP Event Frame Transmitted Seconds Register 47:32"		
Offset	0x00E8	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	timer_seconds	"PTP Event Frame TX Seconds. The register is updated with the value that the 1588 timer seconds register held when the SFD of a PTP transmit primary event crosses the MII interface. The actual update occurs when the GEM recognizes the frame as a PTP sync or delay_req frame. An interrupt is issued when the register is updated."	RO	0x0000

Table 11-57: tsu_ptp_rx_msb_sec

Register Information					
Description	"PTP Event Frame Received Seconds Register 47:32"				
Offset	0x00EC		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000	
15:0	timer_seconds	"PTP Event Frame TX Seconds. The register is updated with the value that the 1588 timer seconds register held when the SFD of a PTP receive primary event crosses the MII interface. The actual update occurs when the GEM recognizes the frame as a PTP sync or delay_req frame. An interrupt is issued when the register is updated."	RO	0x0000	

Table 11-58: tsu_peer_tx_msb_sec

Register Information					
Description	"PTP Peer Event Frame Transmitted Seconds Register 47:32"				
Offset	0x00F0		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000	
15:0	timer_seconds	"PTP Peer Event Frame TX Seconds. The register is updated with the value that the 1588 timer seconds register held when the SFD of a PTP transmit peer event crosses the MII interface. The actual update occurs when the GEM recognizes the frame as a PTP pdelay_req or pdelay_resp frame. An interrupt is issued when the register is updated."	RO	0x0000	

Table 11-59: tsu_peer_rx_msb_sec

Register Information					
Description	"PTP Peer Event Frame Received Seconds Register 47:32"				
Offset	0x00F4		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000	
15:0	timer_seconds	"PTP Peer Event Frame RX Seconds. The register is updated with the value that the 1588 timer seconds register held when the SFD of a PTP receive peer event crosses the MII interface. The actual update occurs when the GEM recognizes the frame as a PTP pdelay_req or pdelay_resp frame. An interrupt is issued when the register is updated."	RO	0x0000	

Table 11-60: dpram_fill_dbg

Register Information				
Description	"The fill levels for the TX and RX packet buffer SRAMs can be read using this register, including the fill level for each queue in the TX direction. The fill level is reported as the number of word locations used. The number of bytes will depend on the SRAM data width."			
Offset	0x00F8	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	dma_tx_rx_fill_level	"Fill Level - TX or RX packet buffer fill level, selected by the tx_q_fill_level_select and tx_rx_fill_level_select registers. Read this register to determine the fill level."	RO	0x0000
15:8	reserved_15_8	"Reserved, read as 0, ignored on write."	RO	0x00
7:4	dma_tx_q_fill_level_select	"TX queue fill level select - select what TX queue to report fill levels for."	RW	0x0
3:1	reserved_3_1	"Reserved, read as 0, ignored on write."	RO	0x0
0	dma_tx_rx_fill_level_select	"TX/RX Fill Level select - report the fill level for the TX or RX packet buffer. Set 1 for transmit and 0 for receive."	RW	0

Table 11-61: revision_reg

Register Information				
Description	"This register indicates a Cadence module identification number and module revision. The value of this register is read only as defined by `gem_revision_reg_value"			
Offset	0x00FC	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:28	fix_number	"Fix number - incremented for fix releases."	RO	0x0
27:16	module_identification_number	"Module identification number - for the GEM, this value is fixed."	RO	0x007
15:0	module_revision	"Module revision - fixed value specific to the revision of the design which is incremented for each non-fix release of the IP."	RO	0x0200

Table 11-62: octets_txed_bottom

Register Information			
Description	"These registers reset to zero on a read and stick at all ones when they count to their maximum value. They should be read frequently enough to prevent loss of data. In order to reduce overall design area, the statistics registers may be optionally removed in the configuration file if they are deemed unnecessary for a particular design. The receive statistics registers are only incremented when the receive enable bit is set in the network control register. The statistics registers optionally have a snapshot capability which, when exercised, will simultaneously store and clear the current values of all the statistics registers into a snapshot register set in order to allow a consistent set of statistics to be read by the processor. The snapshot is controlled using bit 13 of the network control register. The read snapshot control indicated by bit 14 of the network control register determines whether the processor reads the snapshot registers (logic 1) or the incrementing registers (logic 0). The default GEM configuration does not support the snapshot capability. See Parameterization section under Implementation Application Notes for an explanation of how to enable this function. All the statistics registers are read only. For test purposes they may be written by setting bit 7 (Write Enable) in the network control register. Setting bit 6 (increment statistics) in the network control register causes all the statistics registers to increment by one, again for test purposes. Once a statistics register has been read, it is automatically cleared. When reading the octets transmitted and octets received registers, bits 31:0 should be read prior to bits 47:32 to ensure reliable operation. The statistics register block contains the following registers. Octets Transmitted [31:0]"		
Offset	0x0100	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	count	"Transmitted octets in frame without errors [31:0]. The number of octets transmitted in valid frames of any type. This counter is 48-bits, and is read through two registers. This count does not include octets from automatically generated pause frames."	RO RtoClr	0x0000 0000

Table 11-63: octets_txed_top

Register Information			
Description	"Octets Transmitted 47:32"		
Offset	0x0104	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	count	"Transmitted octets in frame without errors [47:32]. The number of octets transmitted in valid frames of any type. This counter is 48-bits, and is read through two registers. This count does not include octets from automatically generated pause frames."	RO RtoClr	0x0000

Table 11-64: frames_txed_ok

Register Information					
Description	"Frames Transmitted"				
Offset	0x0108		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:0	count	"Frames transmitted without error. A 32 bit register counting the number of frames successfully transmitted, i.e. no under run and not too many retries. Excludes pause frames."	RO RtoClr	0x0000 0000	

Table 11-65: broadcast_txed

Register Information					
Description	"Broadcast Frames Transmitted"				
Offset	0x010C		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:0	count	"Broadcast frames transmitted without error. A 32 bit register counting the number of broadcast frames successfully transmitted without error, i.e. no under run and not too many retries. Excludes pause frames."	RO RtoClr	0x0000 0000	

Table 11-66: multicast_txed

Register Information					
Description	"Multicast Frames Transmitted"				
Offset	0x0110		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:0	count	"Multicast frames transmitted without error. A 32 bit register counting the number of multicast frames successfully transmitted without error, i.e. no under run and not too many retries. Excludes pause frames."	RO RtoClr	0x0000 0000	

Table 11-67: pause_frames_txed

Register Information					
Description	"Pause Frames Transmitted"				
Offset	0x0114		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."		RO	0x0000
15:0	count	"Transmitted pause frames - a 16 bit register counting the number of pause frames transmitted. Only pause frames triggered by the register interface or through the external pause pins or automatically when rx_water_mark is reached are counted as pause frames. Pause frames received through the external FIFO interface are counted in the frames transmitted counter."		RO RtoClr	0x0000

Table 11-68: frames_txed_64

Register Information					
Description	"64 Byte Frames Transmitted"				
Offset	0x0118		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	count	"64 byte frames transmitted without error. A 32 bit register counting the number of 64 byte frames successfully transmitted without error, i.e. no under run and not too many retries. Excludes pause frames."		RO RtoClr	0x0000 0000

Table 11-69: frames_txed_65

Register Information					
Description	"65 to 127 Byte Frames Transmitted"				
Offset	0x011C		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	count	"65 to127 byte frames transmitted without error. A 32 bit register counting the number of 65 to127 byte frames successfully transmitted without error, i.e. no under run and not too many retries."		RO RtoClr	0x0000 0000

Table 11-70: frames_txed_128

Register Information					
Description	"128 to 255 Byte Frames Transmitted"				
Offset	0x0120		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	count	"128 to 255 byte frames transmitted without error. A 32 bit register counting the number of 128 to 255 byte frames successfully transmitted without error, i.e. no under run and not too many retries."		RO RtoClr	0x0000 0000

Table 11-71: frames_txed_256

Register Information					
Description	"256 to 511 Byte Frames Transmitted"				
Offset	0x0124		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	count	"256 to 511 byte frames transmitted without error. A 32 bit register counting the number of 256 to 511 byte frames successfully transmitted without error, i.e. no under run and not too many retries."		RO RtoClr	0x0000 0000

Table 11-72: frames_txed_512

Register Information					
Description	"512 to 1023 Byte Frames Transmitted"				
Offset	0x0128		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	count	"512 to 1023 byte frames transmitted without error. A 32 bit register counting the number of 512 to 1023 byte frames successfully transmitted without error, i.e. no under run and not too many retries."		RO RtoClr	0x0000 0000

Table 11-73: frames_txed_1024

Register Information					
Description	"1024 to 1518 Byte Frames Transmitted"				
Offset	0x012C		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	count	"1024 to 1518 byte frames transmitted without error. A 32 bit register counting the number of 1024 to 1518 byte frames successfully transmitted without error, i.e. no under run and not too many retries."		RO RtoClr	0x0000 0000

Table 11-74: frames_txed_1519

Register Information					
Description	"Greater Than 1518 Byte Frames Transmitted"				
Offset	0x0130		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	count	"Greater than 1518 byte frames transmitted without error. A 32 bit register counting the number of 1518 or above byte frames successfully transmitted without error, i.e. no under run and not too many retries."		RO RtoClr	0x0000 0000

Table 11-75: tx_underruns

Register Information					
Description	"Transmit Under Runs"				
Offset	0x0134		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:10	reserved_31_10	"Reserved, read as 0, ignored on write."		RO	0x00 0000
9:0	count	"Transmit under runs - a 10 bit register counting the number of frames not transmitted due to a transmit under run. If this register is incremented then no other statistics register is incremented."		RO RtoClr	0x000

Table 11-76: single_collisions

Register Information					
Description	"Single Collision Frames"				
Offset	0x0138		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:18	reserved_31_18	"Reserved, read as 0, ignored on write."	RO	0x0000	
17:0	count	"Single collision frames - an 18 bit register counting the number of frames experiencing a single collision before being successfully transmitted, i.e. no under run."	RO RtoClr	0x0 0000	

Table 11-77: multiple_collisions

Register Information					
Description	"Multiple Collision Frames"				
Offset	0x013C		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:18	reserved_31_18	"Reserved, read as 0, ignored on write."	RO	0x0000	
17:0	count	"Multiple collision frames - an 18 bit register counting the number of frames experiencing between two and fifteen collisions prior to being successfully transmitted, i.e. no under run and not too many retries."	RO RtoClr	0x0 0000	

Table 11-78: excessive_collisions

Register Information					
Description	"Excessive Collisions"				
Offset	0x0140		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:10	reserved_31_10	"Reserved, read as 0, ignored on write."	RO	0x00 0000	
9:0	count	"Excessive collisions - a 10 bit register counting the number of frames that failed to be transmitted because they experienced 16 collisions."	RO RtoClr	0x000	

Table 11-79: late_collisions

Register Information				
Description	"Late Collisions"			
Offset	0x0144	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:10	reserved_31_10	"Reserved, read as 0, ignored on write."	RO	0x00 0000
9:0	count	"Late collisions - a 10 bit register counting the number of late collision occurring after the slot time (512 bits) has expired. In 10/100 mode, late collisions are counted twice i.e. both as a collision and a late collision. In gigabit mode, a late collision causes the transmission to be aborted, thus the single and multi collision registers are not updated."	RO RtoClr	0x000

Table 11-80: deferred_frames

Register Information				
Description	"Deferred Transmission Frames"			
Offset	0x0148	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:18	reserved_31_18	"Reserved, read as 0, ignored on write."	RO	0x0000
17:0	count	"Deferred transmission frames - an 18 bit register counting the number of frames experiencing deferral due to carrier sense being active on their first attempt at transmission. Frames involved in any collision are not counted nor are frames that experienced a transmit under run."	RO RtoClr	0x0 0000

Table 11-81: crs_errors

Register Information					
Description	"Carrier Sense Errors"				
Offset	0x014C		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:10	reserved_31_10	"Reserved, read as 0, ignored on write."		RO	0x00 0000
9:0	count	"Carrier sense errors - a 10 bit register counting the number of frames transmitted where carrier sense was not seen during transmission or where carrier sense was deasserted after being asserted in a transmit frame without collision (no under run). Only incremented in half-duplex mode. The only effect of a carrier sense error is to increment this register. The behaviour of the other statistics registers is unaffected by the detection of a carrier sense error."		RO RtoClr	0x000

Table 11-82: octets_rxed_bottom

Register Information					
Description	"Octets Received 31:0"				
Offset	0x0150		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	count	"Received octets in frame without errors [31:0]. The number of octets received in valid frames of any type. This counter is 48-bits and is read through two registers. This count does not include octets from pause frames, and is only incremented if the frame is successfully filtered."		RO RtoClr	0x0000 0000

Table 11-83: octets_rxed_top

Register Information					
Description	"Octets Received 47:32"				
Offset	0x0154		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."		RO	0x0000
15:0	count	"Received octets in frame without errors [47:32]. The number of octets received in valid frames of any type. This counter is 48-bits and is read through two registers. This count does not include octets from pause frames, and is only incremented if the frame is successfully filtered."		RO RtoClr	0x0000

Table 11-84: frames_rxed_ok

Register Information					
Description	"Frames Received"				
Offset	0x0158		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	count	"Frames received without error. A 32 bit register counting the number of frames successfully received. Excludes pause frames, and is only incremented if the frame is successfully filtered."		RO RtoClr	0x0000 0000

Table 11-85: broadcast_rxed

Register Information					
Description	"Broadcast Frames Received"				
Offset	0x015C		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	count	"Broadcast frames received without error. A 32 bit register counting the number of broadcast frames successfully received without error. Excludes pause frames, and is only incremented if the frame is successfully filtered."		RO RtoClr	0x0000 0000

Table 11-86: multicast_rxed

Register Information					
Description	"Multicast Frames Received"				
Offset	0x0160		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	count	"Multicast frames received without error. A 32 bit register counting the number of multicast frames successfully received without error. Excludes pause frames, and is only incremented if the frame is successfully filtered."		RO RtoClr	0x0000 0000

Table 11-87: pause_frames_rxed

Register Information			
Description	"Pause Frames Received"		
Offset	0x0164	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	count	"Received pause frames - a 16 bit register counting the number of pause frames received without error."	RO RtoClr	0x0000

Table 11-88: frames_rxed_64

Register Information			
Description	"64 Byte Frames Received"		
Offset	0x0168	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	count	"64 byte frames received without error. A 32 bit register counting the number of 64 byte frames successfully received without error. Excludes pause frames, and is only incremented if the frame is successfully filtered."	RO RtoClr	0x0000 0000

Table 11-89: frames_rxed_65

Register Information			
Description	"65 to 127 Byte Frames Received"		
Offset	0x016C	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	count	"65 to 127 byte frames received without error. A 32 bit register counting the number of 65 to 127 byte frames successfully received without error. Excludes pause frames, and is only incremented if the frame is successfully filtered."	RO RtoClr	0x0000 0000

Table 11-90: frames_rxed_128

Register Information				
Description	"128 to 255 Byte Frames Received"			
Offset	0x0170	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	count	"128 to 255 byte frames received without error. A 32 bit register counting the number of 128 to 255 byte frames successfully received without error. Excludes pause frames, and is only incremented if the frame is successfully filtered."	RO RtoClr	0x0000 0000

Table 11-91: frames_rxed_256

Register Information				
Description	"256 to 511 Byte Frames Received"			
Offset	0x0174	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	count	"256 to 511 byte frames received without error. A 32 bit register counting the number of 256 to 511 byte frames successfully transmitted without error. Excludes pause frames, and is only incremented if the frame is successfully filtered."	RO RtoClr	0x0000 0000

Table 11-92: frames_rxed_512

Register Information				
Description	"512 to 1023 Byte Frames Received"			
Offset	0x0178	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	count	"512 to 1023 byte frames received without error. A 32 bit register counting the number of 512 to 1023 byte frames successfully received without error. Excludes pause frames, and is only incremented if the frame is successfully filtered."	RO RtoClr	0x0000 0000

Table 11-93: frames_rxed_1024

Register Information					
Description	"1024 to 1518 Byte Frames Received"				
Offset	0x017C		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	count	"1024 to 1518 byte frames received without error. A 32 bit register counting the number of 1024 to 1518 byte frames successfully received without error. Excludes pause frames, and is only incremented if the frame is successfully filtered."		RO RtoClr	0x0000 0000

Table 11-94: frames_rxed_1519

Register Information					
Description	"1519 to maximum Byte Frames Received"				
Offset	0x0180		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	count	"1519 to maximum byte frames received without error. A 32 bit register counting the number of 1519 byte or above frames successfully received without error. Maximum frame size is determined by the network configuration register bit 8 (1536 maximum frame size) or bit 3 (jumbo frame size). Excludes pause frames, and is only incremented if the frame is successfully filtered."		RO RtoClr	0x0000 0000

Table 11-95: undersize_frames

Register Information					
Description	"Undersized Frames Received"				
Offset	0x0184		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:10	reserved_31_10	"Reserved, read as 0, ignored on write."		RO	0x00 0000
9:0	count	"Undersize frames received - a 10 bit register counting the number of frames received less than 64 bytes in length (10/100 mode or gigabit mode, full duplex) that do not have either a CRC error or an alignment error. In gigabit mode, half-duplex, this register counts either frames not conforming to the minimum slot time of 512 bytes or frames not conforming to the minimum frame size once bursting is active."		RO RtoClr	0x000

Table 11-96: excessive_rx_length

Register Information					
Description	"Oversize Frames Received"				
Offset	0x0188		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:10	reserved_31_10	"Reserved read 0, ignored on write."		RO	0x00 0000
9:0	count	"Oversize frames received - a 10 bit register counting the number of frames received exceeding 1518 bytes (1536 bytes if bit 8 is set in network configuration register, 10,240 bytes if bit 3 is set in the network configuration register) in length but do not have either a CRC error, an alignment error nor a receive symbol error."		RO RtoClr	0x000

Table 11-97: rx_jabbers

Register Information					
Description	"Jabbers Received"				
Offset	0x018C		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:10	reserved_31_10	"Reserved, read as 0, ignored on write."		RO	0x00 0000
9:0	count	"Jabbers received - a 10 bit register counting the number of frames received exceeding 1518 bytes (1536 if bit 8 set in network configuration register, 10,240 bytes if bit 3 is set in the network configuration register) in length and have either a CRC error, an alignment error or a receive symbol error."		RO RtoClr	0x000

Table 11-98: fcs_errors

Register Information					
Description	"Frame Check Sequence Errors"				
Offset	0x0190		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:10	reserved_31_10	"Reserved, read as 0, ignored on write."		RO	0x00 0000
9:0	count	"Frame check sequence errors - a 10 bit register counting frames that are an integral number of bytes, have bad CRC and are between 64 and 1518 bytes in length (1536 if bit 8 set in network configuration register, 10,240 bytes if bit 3 is set in the network configuration register). This register is also incremented if a symbol error is detected and the frame is of valid length and has an integral number of bytes. This register is incremented for a frame with bad FCS, regardless of whether it is copied to memory due to ignore FCS mode being enabled in bit 26 of the network configuration register."		RO RtoClr	0x000

Table 11-99: rx_length_errors

Register Information					
Description	"Length Field Frame Errors"				
Offset	0x0194		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:10	reserved_31_10	"Reserved, read as 0, ignored on write."		RO	0x00 0000
9:0	count	"Length field frame errors - this 10-bit register counts the number of frames received that have a measured length shorter than that extracted from the length field (bytes 13 and 14). This condition is only counted if the value of the length field is less than 0x0600, the frame is not of excessive length and checking is enabled through bit 16 of the network configuration register."		RO RtoClr	0x000

Table 11-100: rx_symbol_errors

Register Information					
Description	"Receive Symbol Errors"				
Offset	0x0198		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:10	reserved_31_10	"Reserved, read as 0, ignored on write."	RO	0x00 0000	
9:0	count	"Receive symbol errors - a 10-bit register counting the number of frames that had rx_er asserted during reception. For 10/100 mode symbol errors are counted regardless of frame length checks. For gigabit mode the frame must satisfy slot time requirements in order to count a symbol error. Additionally, in gigabit half-duplex mode, carrier extension errors are also recorded. Receive symbol errors will also be counted as an FCS or alignment error if the frame is between 64 and 1518 bytes (1536 bytes if bit 8 is set in the network configuration register, 10240 bytes if bit 3 is set in the network configuration register). If the frame is larger it will be recorded as a jabber error."	RO RtoClr	0x000	

Table 11-101: alignment_errors

Register Information					
Description	"Alignment Errors"				
Offset	0x019C		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:10	reserved_31_10	"Reserved, read as 0, ignored on write."	RO	0x00 0000	
9:0	count	"Alignment errors - a 10 bit register counting frames that are not an integral number of bytes long and have bad CRC when their length is truncated to an integral number of bytes and are between 64 and 1518 bytes in length (1536 if bit 8 set in network configuration register, 10,240 bytes if bit 3 is set in the network configuration register). This register is also incremented if a symbol error is detected and the frame is of valid length and does not have an integral number of bytes."	RO RtoClr	0x000	

Table 11-102: rx_resource_errors

Register Information					
Description	"Receive Resource Errors"				
Offset	0x01A0		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:18	reserved_31_18	"Reserved, read as 0, ignored on write."	RO	0x0000	
17:0	count	"Receive resource errors - an 18 bit register counting the number of times a receive buffer descriptor was read with its used bit set. This can be used as a means of checking the efficiency of the software in processing and releasing receive buffers. In an ideally configured system, this counter would not increase."	RO RtoClr	0x0 0000	

Table 11-103: rx_overruns

Register Information					
Description	"Receive Overruns"				
Offset	0x01A4		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:10	reserved_31_10	"Reserved, read as 0, ignored on write."	RO	0x00 0000	
9:0	count	"Receive overruns - a 10 bit register counting the number of frames that are address recognized but were not copied to memory due to a receive overrun."	RO RtoClr	0x000	

Table 11-104: rx_ip_ck_errors

Register Information					
Description	"IP Header Checksum Errors"				
Offset	0x01A8		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:8	reserved_31_8	"Reserved, read as 0, ignored on write."	RO	0x00 0000	
7:0	count	"IP header checksum errors - an 8-bit register counting the number of frames discarded due to an incorrect IP header checksum, but are between 64 and 1518 bytes (1536 bytes if bit 8 is set in the network configuration register or 10240 bytes if bit 3 is in the network configuration register) and do not have a CRC error, an alignment error, nor a symbol error."	RO RtoClr	0x00	

Table 11-105: rx_tcp_ck_errors

Register Information					
Description	"TCP Checksum Errors"				
Offset	0x01AC		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:8	reserved_31_8	"Reserved, read as 0, ignored on write."	RO	0x00 0000	
7:0	count	"TCP checksum errors - an 8-bit register counting the number of frames discarded due to an incorrect TCP checksum, but are between 64 and 1518 bytes (1536 bytes if bit 8 is set in the network configuration register or 10240 bytes if bit 3 is in the network configuration register) and do not have a CRC error, an alignment error, nor a symbol error."	RO RtoClr	0x00	

Table 11-106: rx_udp_ck_errors

Register Information					
Description	"UDP Checksum Errors"				
Offset	0x01B0		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:8	reserved_31_8	"Reserved, read as 0, ignored on write."	RO	0x00 0000	
7:0	count	"UDP checksum errors - an 8-bit register counting the number of frames discarded due to an incorrect UDP checksum, but are between 64 and 1518 bytes (1536 bytes if bit 8 is set in the network configuration register or 10240 bytes if bit 3 is in the network configuration register) and do not have a CRC error, an alignment error, nor a symbol error."	RO RtoClr	0x00	

Table 11-107: auto_flushed_pkts

Register Information					
Description	"Receive DMA Flushed Packets"				
Offset	0x01B4		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000	
15:0	count	"Flushed RX packets counter. A 16 bit register counting the number of frames that have been flushed from the receive SRAM based packet buffer due to one of the following reasons: 1. When either partial store and forward mode, bit 24 of the DMA configuration register, or the drop_on_resource_err mode of the traffic policing feature is active, and a packet is received while there is no AMBA (AHB/AXI) resource (no free descriptors for the DUT to use). 2. When partial store and forward mode is enabled and an AMBA (AHB/AXI) error is encountered while writing the packet data to external memory. 3. When bit 18 of the network control register (software action to flush a packet from the head of the PBUF queue) is pulsed and the GEM DMA is not currently busy. 4. When a frame is dropped due to the policing actions defined in the rx_qX_flush registers located at 0x0b00-0x0b3c."	RO RtoClr	0x0000	

Table 11-108: tsu_timer_incr_sub_nsec

Register Information					
Description	"1588 Timer Increment Register sub nsec. From release 1p08f1 onwards this register must be written before the tsu_timer_incr register and the value written will not take effect until the tsu_timer_incr register is written to."				
Offset	0x01BC		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:24	sub_ns_incr_lsb	"These are the least significant bits [7:0] of the sub-ns value by which the 1588 timer will be incremented each clock cycle."	RW	0x00	
23:16	reserved_23_16	"Reserved, read as 0, ignored on write."	RO	0x00	
15:0	sub_ns_incr	"These are the most significant bits [23:8] of the sub-ns value by which the 1588 timer will be incremented each clock cycle. 24 bits of sub nanosecond precision gives resolution of approximately 5.86E-17 seconds (16 bits gives 15.2 femtoseconds)."	RW	0x0000	

Table 11-109: tsu_timer_msb_sec

Register Information				
Description	"1588 Timer Seconds Register 47:32"			
Offset	0x01C0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	timer	"TSU timer value (s). Most significant 16 bits of seconds timer count. The register is writeable. The 48-bit counter increments by one when the 1588 nanoseconds counter counts to one second. It may also be incremented or decremented when the timer adjust register is written (if decremented from zero the 48-bit combined count would roll back to 0xFFFFFFFFFFFF). Note: The value of this register is used only when the lower 32 bit register is written to. This is to ensure a single update of the 48 bit seconds value"	RW	0x0000

Table 11-110: tsu_strobe_msb_sec

Register Information				
Description	"1588 Timer Sync Strobe Seconds Register 47:32"			
Offset	0x01C4	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	strobe	"1588 Timer Sync Strobe Seconds. The most significant 16-bit value of the Timer Seconds register captured when gem_tsu_ms and gem_tsu_inc_ctrl are zero."	RO	0x0000

Table 11-111: tsu_strobe_sec

Register Information				
Description	"1588 Timer Sync Strobe Seconds Register 31:0"			
Offset	0x01C8	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	strobe	"1588 Timer Sync Strobe Seconds. The lowest significant 32-bit value of the Timer Seconds register captured when gem_tsu_ms and gem_tsu_inc_ctrl are zero."	RO	0x0000 0000

Table 11-112: tsu_strobe_nsec

Register Information					
Description	"1588 Timer Sync Strobe Nanoseconds Register"				
Offset	0x01CC		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."		RO	0x0
29:0	strobe	"1588 Timer Sync Strobe Nanoseconds. The value of the Timer Nanoseconds register captured when gem_tsu_ms and gem_tsu_inc_ctrl are zero."		RO	0x0000 0000

Table 11-113: tsu_timer_sec

Register Information					
Description	"1588 Timer Seconds Register 31:0"				
Offset	0x01D0		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	timer	"1588 Timer Seconds Register. TSU timer value (s). Least significant 32 bits of seconds timer count. This register is writeable. The 48-bit counter increments by one when the 1588 nanoseconds counter counts to one second. It may also be incremented or decremented when the timer adjust register is written (if decremented from zero the 48-bit combined count would roll back to 0xFFFFFFFFFFFF)."		RW	0x0000 0000

Table 11-114: tsu_timer_nsec

Register Information					
Description	"1588 Timer Nanoseconds Register"				
Offset	0x01D4		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."		RO	0x0
29:0	timer	"Timer count in nanoseconds. This register is writeable. It can also be adjusted by writes to the 1588 timer adjust register. It increments by the value of the 1588 timer increment register each clock cycle (if this register is close to zero and a write to the timer adjust register causes a decrement the seconds register will be decremented if necessary and the nanoseconds register will roll back to 9999999xx (decimal))."		RW	0x0000 0000

Table 11-115: tsu_timer_adjust

Register Information				
Description	"This register is used to adjust the value of the timer in the TSU. It allows an integral number of nanoseconds to be added or subtracted from the timer in a one-off operation. This register returns all zeroes when read."			
Offset	0x01D8	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	add_subtract	"Write as one to subtract from the 1588 timer. Write as zero to add to it."	WO	0
30	reserved_30	"Reserved, read as 0, ignored on write."	RO	0
29:0	increment_value	"Timer increment value. The number of nanoseconds to increment or decrement the 1588 timer nanoseconds register. If necessary the 1588 seconds register will be incremented or decremented."	WO	0x0000 0000

Table 11-116: tsu_timer_incr

Register Information				
Description	"1588 Timer Increment Register. From release 1p08f1 onwards this register must be written after the tsu_timer_incr_sub_ns register and the write operation will cause the value written to the tsu_timer_incr_sub_ns register to take effect."			
Offset	0x01DC	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	reserved_31_24	"Reserved, read as 0, ignored on write."	RO	0x00
23:16	num_incs	"Number of incs before alt inc. The number of increments after which the alternative increment is used."	RW	0x00
15:8	alt_ns_incr	"Alternative nanoseconds count. Alternative count of nanoseconds by which the 1588 timer nanoseconds register will be incremented each clock cycle."	RW	0x00
7:0	ns_increment	"A count of nanoseconds by which the 1588 timer nanoseconds register will be incremented each clock cycle. These are the most significant 8 bits of the 32 bit timer_increment counter. The tsu_timer_incr_sub_nsec register holds the least significant 24 bits of the increment."	RW	0x00

Table 11-117: tsu_ptp_tx_sec

Register Information				
Description	"PTP Event Frame Transmitted Seconds Register 31:0"			
Offset	0x01E0	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	timer	"PTP Event Frame Transmitted Seconds. The register is updated with the value that the 1588 timer seconds register held when the SFD of a PTP transmit primary event crosses the MII interface. The actual update occurs when the GEM recognizes the frame as a PTP sync or delay_req frame. An interrupt is issued when the register is updated."	RO	0x0000 0000

Table 11-118: tsu_ptp_tx_nsec

Register Information				
Description	"PTP Event Frame Transmitted Nanoseconds Register"			
Offset	0x01E4	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:0	timer	"PTP Event Frame Transmitted Nanoseconds. The register is updated with the value that the 1588 timer nanoseconds register held when the SFD of a PTP transmit primary event crosses the MII interface. The actual update occurs when the GEM recognizes the frame as a PTP sync or delay_req frame. An interrupt is issued when the register is updated."	RO	0x0000 0000

Table 11-119: tsu_ptp_rx_sec

Register Information				
Description	"PTP Event Frame Received Seconds Register 31:0"			
Offset	0x01E8	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	timer	"PTP Event Frame Received Seconds. The register is updated with the value that the 1588 timer seconds register held when the SFD of a PTP receive primary event crosses the MII interface. The actual update occurs when the GEM recognizes the frame as a PTP sync or delay_req frame. An interrupt is issued when the register is updated."	RO	0x0000 0000

Table 11-120: tsu_ptp_rx_nsec

Register Information				
Description	"PTP Event Frame Received Nanoseconds Register"			
Offset	0x01EC	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:0	timer	"PTP Event Frame Received Nanoseconds. The register is updated with the value that the 1588 timer nanoseconds register held when the SFD of a PTP receive primary event crosses the MII interface. The actual update occurs when the GEM recognizes the frame as a PTP sync or delay_req frame. An interrupt is issued when the register is updated."	RO	0x0000 0000

Table 11-121: tsu_peer_tx_sec

Register Information				
Description	"PTP Peer Event Frame Transmitted Seconds Register 31:0"			
Offset	0x01F0	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	timer	"PTP Peer Event Frame Received Seconds. The register is updated with the value that the 1588 timer seconds register held when the SFD of a PTP transmit peer event crosses the MII interface. The actual update occurs when the GEM recognizes the frame as a PTP pdelay_req or pdelay_resp frame. An interrupt is issued when the register is updated."	RO	0x0000 0000

Table 11-122: tsu_peer_tx_nsec

Register Information				
Description	"PTP Peer Event Frame Transmitted Nanoseconds Register"			
Offset	0x01F4	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:0	timer	"PTP Peer Event Frame Transmitted Nanoseconds. The register is updated with the value that the 1588 timer nanoseconds register held when the SFD of a PTP transmit peer event crosses the MII interface. The actual update occurs when the GEM recognizes the frame as a PTP pdelay_req or pdelay_resp frame. An interrupt is issued when the register is updated."	RO	0x0000 0000

Table 11-123: tsu_peer_rx_sec

Register Information					
Description	"PTP Peer Event Frame Received Seconds Register 31:0"				
Offset	0x01F8		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	timer	"PTP Peer Event Frame Received Seconds. The register is updated with the value that the 1588 timer seconds register held when the SFD of a PTP receive peer event crosses the MII interface. The actual update occurs when the GEM recognizes the frame as a PTP pdelay_req or pdelay_resp frame. An interrupt is issued when the register is updated."		RO	0x0000 0000

Table 11-124: tsu_peer_rx_nsec

Register Information					
Description	"PTP Peer Event Frame Received Nanoseconds Register"				
Offset	0x01FC		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."		RO	0x0
29:0	timer	"PTP Peer Event Frame Received Nanoseconds. The register is updated with the value that the 1588 timer nanoseconds register held when the SFD of a PTP receive peer event crosses the MII interface. The actual update occurs when the GEM recognizes the frame as a PTP pdelay_req or pdelay_resp frame. An interrupt is issued when the register is updated."		RO	0x0000 0000

Table 11-125: pcs_control

Register Information			
Description	"Note: All PCS registers are defined in the IEEE 802.3 Standard. PCS Control Register. This register provides the main control functions with respect to the PCS."		
Offset	0x0200	**Type:**	RW

Bits	Name	Description	Access	Reset
31:16	reserved_31_16	"Reserved. Set to zero."	RO	0x0000
15	pcs_software_reset	"PCS software reset - written by software to force the hardware logic into a reset state. This bit is self-clearing. When reading this bit, logic 1 is returned until both the soft reset has completed and the PCS is enabled through the PCS select bit of the network configuration register. Writing logic 0 has no effect."	RW	1
14	loopback_mode	"Loopback mode - the ewrap output pin of the GEM reflects this control bit, and can be used to select loopback mode in the PHY transceiver. 0: Loopback mode disabled. 1: Loopback mode enabled."	RW	0
13	speed_select_bit_1	"Speed select bit 1 - combined with speed select [0] to indicate the speed of operation of the PCS. As the GEM PCS is only intended to operate at 1000 Mbps, this bit is hardwired to logic 0."	RO	0
12	enable_auto_neg	"Enable auto-negotiation - when set active high, auto-negotiation operation is enabled."	RW	1
11:10	reserved_11_10	"Reserved. Set to zero."	RO	0x0
9	restart_auto_neg	"Restart auto-negotiation - when set active high, the hardware restarts auto-negotiation. This bit is self-clearing, but once set shall remain in this state until auto-negotiation has restarted. Writing logic 0 has no effect."	RW	0
8	mac_duplex_state	"MAC Duplex state. This returns the value of the MAC's duplex state as indicated in bit 1 of the MAC's network configuration register."	RO	0
7	collision_test	"Collision test - when set active high, the PCS generates collisions on transmit. This bit should only be set for test purposes."	RW	0
6	speed_select_bit_0	"Speed select bit 0 - combined with speed select [1] to indicate the speed of operation of the PCS. As the GEM PCS is only intended to operate at 1000 Mbps, this bit is hardwired to logic 1."	RO	1
5:0	reserved_5_0	"Reserved. Set to zero."	RO	0x00

Table 11-126: pcs_status

Register Information			
Description	"This register indicates general status information concerning the PCS."		
Offset	0x0204	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved. Set to zero."	RO	0x0000
15	base_100_t4	"100 BASE-T4 - the GEM PCS does not support 100 BASE-T4. This bit is hardwired to logic 0."	RO	0
14	base_100_x_full_duplex	"100 BASE-X full duplex - the GEM PCS does not support 100 BASE-X. This bit is hardwired to logic 0."	RO	0
13	base_100_x_half_duplex	"100 BASE-X half-duplex - the GEM PCS does not support 100 BASE-X. This bit is hardwired to logic 0."	RO	0
12	mbps_10_full_duplex	"10 Mbps full duplex - the GEM PCS does not support this mode. This bit is hardwired to logic 0."	RO	0
11	mbps_10_half_duplex	"10 Mbps half-duplex - the GEM PCS does not support this mode. This bit is hardwired to logic 0."	RO	0
10	base_100_t2_full_duplex	"100 BASE-T2 full duplex - the GEM PCS does not support 100 BASE-T2. This bit is hardwired to logic 0."	RO	0
9	base_100_t2_half_duplex	"100 BASE-T2 half-duplex - the GEM PCS does not support 100 BASE-T2. This bit is hardwired to logic 0."	RO	0
8	extended_status	"Extended status - when set active high, indicates extended status information is present in the PCS auto-negotiation extended status register. This bit is hardwired to logic 1."	RO	1
7:6	reserved_7_6	"Reserved. Set to zero."	RO	0x0
5	auto_neg_complete	"Auto-negotiation complete - set active high by the PCS hardware to indicate auto-negotiation has completed."	RO	0
4	remote_fault	"Remote fault - set active high if the link partner remote fault bits in the PCS auto-negotiation link partner ability register, indicates an error. Resets low when read."	RO	0
3	auto_neg_ability	"Auto-negotiation ability - this bit indicates whether the PCS has auto-negotiation ability and reflects the value of the auto-negotiation enable bit in the PCS control register. 0: PCS is not able to perform auto-negotiation. 1: PCS is able to perform auto-negotiation."	RO	1
2	link_status	"Link status - indicates the status of the physical connection to the link partner. When set to logic 1 the link is up, and when set to logic 0, the link is down. If auto-negotiation is disabled this returns the synchronisation status. Held at logic 0 if the link goes down until this bit is read."	RO	0
1	reserved_1	"Reserved. Set to zero."	RO	0
0	extended_capabilities	"Extended register capabilities - when set active high, indicates the PCS supports extended register capabilities. This bit is hardwired to logic 1."	RO	1

Table 11-127: pcs_phy_top_id

Register Information					
Description	"The value of this register indicates the upper 16-bits of the PHY's identification code. This is a read-only register with a value defined by `gem_phy_id_top"				
Offset	0x0208		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:16	reserved_31_16	"Reserved. Set to zero."		RO	0x0000
15:0	id_code	"Upper 16-bits of the PHY identification code, corresponding to the fixed identification for the GEM IP."		RO	0x0007

Table 11-128: pcs_phy_bot_id

Register Information					
Description	"The value of this register indicates the lower 16-bits of the PHY's identification code. This is a read-only register with a value defined by `gem_phy_id_bot"				
Offset	0x020C		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:16	reserved_31_16	"Reserved. Set to zero."		RO	0x0000
15:0	id_code	"Lower 16-bits of the PHY identification code, corresponding to the revision of the GEM IP, which is incremented after each release of the IP."		RO	0x0200

Table 11-129: pcs_an_adv

Register Information			
Description	"The value of this register is used to transmit the base page of the GEM PCS advertised capabilities as long as SGMII mode is not enabled. If SGMII mode is enabled by setting bit 27 in the network configuration register then this register becomes read only with a fixed value of 0x00000001. (SGMII specifies that the transmit configuration information sent from the MAC to the PHY is fixed with bit 14 set to 1 to indicate acknowledge, bit 0 set to 1 to indicate SGMII and all other bits set to 0.)"		
Offset	0x0210	**Type:**	RW

Bitfield Details

Bits	Name	Description	Access	Reset
31:16	reserved_31_16	"Reserved. Set to zero."	RO	0x0000
15	next_page	"Next page. When set active high, this bit is used during auto-negotiation to indicate to the link partner that the PCS requires exchanging next pages."	RW	0
14	reserved_14	"Reserved. Set to zero."	RO	0
13:12	remote_fault	"Remote fault [1:0] - indicates and classifies a remote fault condition to the link partner: 00: No error, Link OK 01: Link Failure. 10: Off line. 11: Auto-negotiation error."	RW	0x0
11:9	reserved_11_9	"Reserved. Set to zero."	RO	0x0
8:7	pause	"Pause[1:0] - used to provide a pause capability mechanism as follows: 00: No pause. 01: Symmetric pause. 10: Asymmetric pause toward link partner. 11: Both symmetric pause and asymmetric pause toward link device."	RW	0x0
6	half_duplex	"half-duplex - this bit defines to the link partner whether the GEM is capable of supporting half-duplex operation. 0: The GEM cannot support half-duplex. 1: The GEM can support half-duplex."	RW	0
5	full_duplex	"Full duplex - this bit defines to the link partner whether the GEM is capable of supporting full duplex operation. 0: The GEM cannot support full duplex. 1: The GEM can support full duplex."	RW	1
4:0	reserved_4_0	"Reserved. Set to zero."	RO	0x00

Table 11-130: pcs_an_lp_base

Register Information			
Description	"When SGMII mode is not enabled, the value of this register contains the link partner's base page received information. This register is updated in the ABILITY_DETECT state of the PCS auto-negotiation state machine so bit 14 will only be set if the link partner is sending acknowledge while the PCS in this state. The register is not updated in the ACK_DETECT state. For SGMII mode, the contents of this register change to the one defined in the SGMII standard. The value of this register contains the link partner's base page received information. In this case the link partner is the PHY connected by the SGMII."		
Offset	0x0214	**Type:**	RO

Bits	Name	Description	Access	Reset
31:16	reserved_31_16	"Reserved. Set to zero."	RO	0x0000
15	link_partner_next_page_status	"The contents of this register change depending on SGMII or non SGMII mode. non SGMII Mode: Link partner next page - when set active high, this bit indicates the link partner's intention to exchange next pages. SGMII Mode: Link Status. 0 : Link Down. 1 : Link Up."	RO	0
14	link_partner_acknowledge	"Link partner acknowledge - indicates the link partner has successfully received the transmitted base page"	RO	0
13:12	link_partner_remote_fault_duplex_mode	"The contents of this register change depending on SGMII or non SGMII mode. non SGMII Mode: Link partner remote fault [1:0] - indicates and classifies a remote fault condition has been detected by the link partner as follows: 00: No error, Link OK 01: Link Failure. 10: Off line. 11: Auto-negotiation error. SGMII Mode: Bit 13: Reserved. read as 0. Bit 12 : 0 : half-duplex. 1: Full Duplex."	RO	0x0
11:9	speed_reserved	"The contents of this register change depending on SGMII or non SGMII mode. non SGMII Mode: Reserved. Set to zero. SGMII Mode : Bits 11:10 : Speed : 11 : Reserved 10 : 1000 Mbps 01 : 100Mbps 00 : 10 Mbps Bit 9 : Reserved. read as 0."	RO	0x0

8:7	pause	"The contents of this register change depending on SGMII or non SGMII mode. non SGMII Mode: Pause[1:0] - provides the link partner's pause frame capability as follows: 00: No pause. 01: Symmetric pause. 10: Asymmetric pause toward link partner. 11: Both symmetric pause and asymmetric pause toward link device. SGMII Mode: Reserved. read as 0."	RO	0x0
6	link_partner_half_duplex	"The contents of this register change depending on SGMII or non SGMII mode. non SGMII Mode: Link partner half-duplex - this bit indicates whether the link partner is capable of supporting half-duplex operation. 0: The link partner cannot support half-duplex. 1: The link partner can support half-duplex. SGMII Mode: Reserved. read as 0."	RO	0
5	link_partner_full_duplex	"The contents of this register change depending on SGMII or non SGMII mode. non SGMII Mode: Link partner full duplex - this bit indicates whether the link partner is capable of supporting full duplex operation. 0: The link partner cannot support full duplex. 1: The link partner can support full duplex. SGMII Mode: Reserved. read as 0."	RO	0
4:0	reserved_4_0	"Reserved. Set to zero."	RO	0x00

Table 11-131: pcs_an_exp

Register Information				
Description	"This register contains auto-negotiation next page ability and page received information."			
Offset	0x0218	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:3	reserved_31_3	"Reserved. Set to zero."	RO	0x0000 0000
2	next_page_capability	"Next page capability - hard wired to logic 1 to indicate that the GEM PCS supports next page operation."	RO	1
1	page_received	"Page received - this bit is set active high when a new page has been received from the link partner. It is cleared when the link partner next page register has been read."	RO	0
0	reserved_0	"Reserved. Set to zero."	RO	0

Table 11-132: pcs_an_np_tx

Register Information			
Description	"The value of this register is used to transmit the next page information for the GEM PCS. For next page exchange to work this register must be written within 10 ms of receiving a new page from the link partner. If the link partner is requesting next pages and the GEM has none or no more to send then this register should be written with the null message (0x2001). The value 0x0000 must not be written to this register."		
Offset	0x021C	**Type:**	RW

Bits	Name	Description	Access	Reset
31:16	reserved_31_16	"Reserved. Set to zero."	RO	0x0000
15	next_page_to_transmit	"Next page to transmit - this bit indicates whether this is the last next page to be transmitted:0: Last page. 1: Additional page(s) to follow."	RW	0
14	reserved_14	"Reserved. Set to zero."	RO	0
13	message_page_indicator	"Message page indicator - this bit identifies the message. 0: Unformatted page. 1: Message page."	RW	0
12	acknowledge_2	"Acknowledge 2 - when set active high, indicates that the GEM PCS has the ability to comply with the last received message."	RW	0
11	reserved_11	"Reserved. Set to zero."	RO	0
10:0	message	"Message - contains data as defined by the message page indicator bit."	RW	0x000

Table 11-133: pcs_an_lp_np

Register Information				
Description	"This value of this register contains the next page received information from the link partner."			
Offset	0x0220	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved. Set to zero."	RO	0x0000
15	next_page_to_receive	"Next page to receive - this bit indicates whether this is the last page to be received in the sequence by the GEM PCS: 0: Last page. 1: Additional page(s) to follow."	RO	0
14	acknowledge	"Acknowledge - this bit indicates whether as part of the next page function, the link partner has successfully received the last message transmitted."	RO	0
13	message_page_indicator	"Message page indicator - this bit identifies the message. 0: Unformatted page. 1: Message page."	RO	0
12	acknowledge_2	"Acknowledge 2 - set active high by the link partner to indicate when it has the ability to comply with the last message received."	RO	0
11	toggle	"Toggle - this bit toggles with every received page."	RO	0
10:0	message	"Message - contains data as defined by the message page indicator bit."	RO	0x000

Table 11-134: pcs_an_ext_status

Register Information				
Description	"This register contains PCS auto-negotiation extended status information."			
Offset	0x023C	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved. Set to zero."	RO	0x0000
15	full_duplex_1000base_x	"Full duplex 1000BASE-X - hardwired to logic 1, indicates the GEM PCS can support full duplex operation of 1000BASE-X."	RO	1
14	half_duplex_1000base_x	"half-duplex 1000BASE-X - hardwired to logic 0, indicates the GEM MAC cannot support half-duplex operation at gigabit speeds."	RO	0
13	full_duplex_1000base_t	"Full duplex 1000BASE-T - hardwired to logic 0, indicates the GEM PCS cannot support 1000BASE-T full duplex operation."	RO	0
12	half_duplex_1000base_t	"half-duplex 1000BASE-T - hardwired to logic 0, indicates the GEM PCS cannot support 1000BASE-T half-duplex operation."	RO	0
11:0	reserved_11_0	"Reserved. Set to zero."	RO	0x000

Table 11-135: tx_pause_quantum1

Register Information				
Description	"Transmit Pause Quantum Register 1"			
Offset	0x0260	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	quantum_p3	"Transmit pause quantum - written with the pause quantum value for pause frame transmission of priority 3."	RW	0xFFFF
15:0	quantum_p2	"Transmit pause quantum - written with the pause quantum value for pause frame transmission of priority 2."	RW	0xFFFF

Table 11-136: tx_pause_quantum2

Register Information				
Description	"Transmit Pause Quantum Register 2"			
Offset	0x0264	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	quantum_p5	"Transmit pause quantum - written with the pause quantum value for pause frame transmission of priority 5."	RW	0xFFFF
15:0	quantum_p4	"Transmit pause quantum - written with the pause quantum value for pause frame transmission of priority 4."	RW	0xFFFF

Table 11-137: tx_pause_quantum3

Register Information				
Description	"Transmit Pause Quantum Register 3"			
Offset	0x0268	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	quantum_p7	"Transmit pause quantum - written with the pause quantum value for pause frame transmission of priority 7."	RW	0xFFFF
15:0	quantum_p6	"Transmit pause quantum - written with the pause quantum value for pause frame transmission of priority 6."	RW	0xFFFF

Table 11-138: pfc_status

Register Information				
Description	"Priority Flow Control status register - indicates whether PFC has been negotiated and the current state of the PFC counters for each priority."			
Offset	0x026C	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:9	reserved_31_9	"Reserved, read as 0, ignored on write."	RO	0x00 0000
8	pfc_negotiate_pclk	"Set when PFC Priority Based Pause has been negotiated."	RO	0
7:0	rx_pfc_paused	"Reflects the state of the rx_pfc_paused signals. Each bit in the vector corresponds to a priority indicated within the received PFC priority based pause frame. A bit is set when a PFC priority based pause frame has been received, and the associated priority pause time quantum is non-zero. The bit is cleared when the associated pause time has elapsed."	RO	0x00

Table 11-139: rx_lpi

Register Information				
Description	"Received LPI transitions"			
Offset	0x0270	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Unused, read zero"	RO	0x0000
15:0	count	"Count of RX LPI transitions. A count of the number of times there is a transition from receiving normal idle to receiving low power idle. Cleared on read."	RO RtoClr	0x0000

Table 11-140: rx_lpi_time

Register Information				
Description	"Received LPI time"			
Offset	0x0274	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	reserved_31_24	"Unused, read zero"	RO	0x00
23:0	lpi_time	"Time in LPI. This register increments once every 16 pclk cycles when the LPI indication bit 7 is set in the network status register. Cleared on read."	RO RtoClr	0x00 0000

Table 11-141: tx_lpi

Register Information				
Description	"Transmit LPI transitions"			
Offset	0x0278	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Unused, read zero"	RO	0x0000
15:0	count	"Count of TX LPI transmissions. A count of the number of times the enable LPI transmission bit 19 goes from low to high in the network control register. Cleared on read."	RO RtoClr	0x0000

Table 11-142: tx_lpi_time

Register Information				
Description	"Transmit LPI time"			
Offset	0x027C	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	reserved_31_24	"Unused, read zero"	RO	0x00
23:0	lpi_time	"Time in LPI. This register increments once every 16 pclk cycles when the enable LPI transmission bit 19 is set in the network control register. Cleared on read."	RO RtoClr	0x00 0000

Table 11-143: designcfg_debug1

Register Information			
Description	"Design Configuration Register 1 - The GEM has many parameterisation options to configure the IP during compilation stage. This is achieved using Verilog define compiler directives in an include file called gem_defs.v. This configuration is readable through APB addressable designcfg_debug registers."		
Offset	0x0280	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:28	axi_cache_value	"Takes the value of the edma_axi_awcache_value parameter which is set by the `gem_axi_awcache_value DEFINE or will be zero if no packet buffer DMA is present"	RO	0x0
27:25	dma_bus_width	"Takes the value of bits 7:5 of the `gem_dma_bus_width DEFINE. So if the define is set to decimal 64 this will return binary 010."	RO	0x2
24	exclude_cbs	"Takes the value of the edma_exclude_cbs parameter which is set by the `gem_exclude_cbs DEFINE."	RO	0
23	irq_read_clear	"Takes the value of the edma_irq_read_clear parameter which is set by the `gem_irq_read_clear DEFINE"	RO	1
22	no_snapshot	"Takes the value of the edma_no_snapshot parameter which is set by the `gem_no_snapshot DEFINE"	RO	0
21	no_stats	"Takes the value of the edma_no_stats parameter which is set by the `gem_no_stats DEFINE"	RO	0
20	reserved_20	"Reserved, read as 1, ignored on write."	RO	1
19:15	user_in_width	"Takes the value of the `gem_user_in_width DEFINE or 1 if user_io not defined"	RO	0x10
14:10	user_out_width	"Takes the value of the `gem_user_out_width DEFINE or 1 if user_io not defined"	RO	0x10
9	user_io	"Takes the value of the `gem_user_io DEFINE"	RO	1
8	reserved_8	"Reserved, read as 1, ignored on write."	RO	1
7	reserved_7	"Reserved, read as 0, ignored on write."	RO	0
6	ext_fifo_interface	"Takes the value of the edma_ext_fifo_interface parameter which is set by the `gem_ext_fifo_interface DEFINE"	RO	0
5	reserved_5	"Reserved, read as 0, ignored on write."	RO	0
4	int_loopback	"Takes the value of the edma_int_loopback parameter which is set by the `gem_int_loopback DEFINE"	RO	1
3:2	reserved_3_2	"Reserved, read as 0, ignored on write."	RO	0x0
1	exclude_qbv	"Takes the value of the edma_exclude_qbv parameter which is set by the `gem_exclude_qbv DEFINE"	RO	0
0	no_pcs	"Takes the value of the edma_no_pcs parameter which is set by the `gem_no_pcs DEFINE"	RO	0

Table 11-144: designcfg_debug2

Register Information				
Description	"Design Configuration Register 2"			
Offset	0x0284	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	spram	"Takes the value of the edma_spram parameter which is set by the `gem_spram DEFINE"	RO	0
30	axi	"Takes the value of the edma_axi parameter which is set by the `gem_axi DEFINE but will be zero if the FIFO interface has been configured"	RO	1
29:26	tx_pbuf_addr	"Takes the value of the `gem_tx_pbuf_addr DEFINE or zero if pbuf_address is decimal 16"	RO	0xE
25:22	rx_pbuf_addr	"Takes the value of the `gem_rx_pbuf_addr DEFINE or zero if pbuf_address is decimal 16"	RO	0xB
21	tx_pkt_buffer	"Takes the value of the edma_tx_pkt_buffer parameter which is set by the `gem_tx_pkt_buffer DEFINE but will be zero if the FIFO interface has been configured"	RO	1
20	rx_pkt_buffer	"Takes the value of the edma_rx_pkt_buffer parameter which is set by the `gem_rx_pkt_buffer DEFINE but will be zero if the FIFO interface has been configured"	RO	1
19:16	hprot_value	"Takes the value of the edma_hprot parameter which is set by the `gem_hprot_value DEFINE"	RO	0x1
15:14	reserved_15_14	"Unused, read zero"	RO	0x0
13:0	jumbo_max_length	"Takes the value of the edma_jumbo_max_length parameter which is set by the `gem_jumbo_max_length DEFINE"	RO	0x2800

Table 11-145: designcfg_debug3

Register Information				
Description	"Design Configuration Register 3"			
Offset	0x0288	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Unused, read zero"	RO	0x0
29:24	num_spec_add_filters	"Takes the value of the `num_spec_add_filters DEFINE"	RO	0x24
23:21	reserved_23_21	"Unused, read zero"	RO	0x0
20:0	reserved_20_0	"Unused, read zero - reserved for rx_base2_fifo_size and rx_fifo_size defines in internal FIFO mode"	RO	0x00 0000

Table 11-146: designcfg_debug4

Register Information				
Description	"Design Configuration Register 4"			
Offset	0x028C	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:20	reserved_31_20	"Unused, read zero"	RO	0x000
19:0	reserved_19_0	"Unused, read zero - reserved for tx_base2_fifo_size and tx_fifo_size defines in internal FIFO mode"	RO	0x0 0000

Table 11-147: designcfg_debug5

Register Information				
Description	"Design Configuration Register 5"			
Offset	0x0290	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:29	axi_prot_value	"Takes the value of the edma_axi_prot_value parameter which is set by the `gem_axi_prot_value DEFINE but will be zero if the FIFO interface has been configured"	RO	0x2
28	tsu_clk	"Takes the value of the `gem_tsu_clk DEFINE"	RO	1
27:20	rx_buffer_length_def	"Takes the value of the `gem_rx_buffer_length_def DEFINE"	RO	0x02
19	tx_pbuf_size_def	"Takes the value of the edma_tx_pbuf_size_def parameter which is set by the `gem_tx_pbuf_size_def DEFINE but will be zero if the FIFO inteface has been configured"	RO	1
18:17	rx_pbuf_size_def	"Takes the value of the `gem_rx_pbuf_size_def DEFINE"	RO	0x3
16:15	endian_swap_def	"Takes the value of the `gem_endian_swap_def DEFINE"	RO	0x3
14:12	mdc_clock_div	"Takes the value of the `gem_mdc_clock_div DEFINE"	RO	0x2
11:10	dma_bus_width_def	"Takes the value of the `gem_dma_bus_width_def DEFINE"	RO	0x0
9	phy_ident	"Indicates whether the top and bottom PHY_ID registers are present in the address map"	RO	1
8	tsu	"Takes the value of the `gem_tsu DEFINE"	RO	1
7:4	tx_fifo_cnt_width	"Takes the value of the `gem_tx_fifo_cnt_width DEFINE"	RO	0x4
3:0	rx_fifo_cnt_width	"Takes the value of the `gem_rx_fifo_cnt_width DEFINE"	RO	0x5

Table 11-148: designcfg_debug6

Register Information			
Description	"Design Configuration Register 6"		
Offset	0x0294	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:28	reserved_31_28	"Reserved. Set to zero."	RO	0x0
27	pbuf_lso	"Takes the value of the edma_lso parameter which is set by the `gem_pbuf_lso DEFINE this is zero if the FIFO interface or AHB is configured."	RO	0
26	pbuf_rsc	"Takes the value of the edma_rsc parameter which is set by the `gem_pbuf_rsc DEFINE or is zero if the FIFO interface or AHB is configured."	RO	1
25	pbuf_cutthru	"Takes the value of the `gem_pbuf_cutthru DEFINE"	RO	1
24	pfc_multi_quantum	"Takes the value of the `gem_pfc_multi_quantum DEFINE"	RO	1
23	dma_addr_width_is_64b	"This is one if the `gem_dma_addr_width DEFINE is set to 64."	RO	1
22	host_if_soft_select	"Takes the value of the `gem_host_if_soft_select DEFINE"	RO	1
21	tx_add_fifo_if	"Takes the value of the `gem_tx_add_fifo_if DEFINE"	RO	0
20	ext_tsu_timer	"Takes the value of the `gem_ext_tsu_timer DEFINE"	RO	1
19:16	tx_pbuf_queue_segment_size	"Takes the value of the `gem_tx_pbuf_queue_segment_size DEFINE"	RO	0x4
15	dma_priority_queue15	"Takes the value of the `dma_priority_queue15 DEFINE"	RO	1
14	dma_priority_queue14	"Takes the value of the `dma_priority_queue14 DEFINE"	RO	1
13	dma_priority_queue13	"Takes the value of the `dma_priority_queue13 DEFINE"	RO	1
12	dma_priority_queue12	"Takes the value of the `dma_priority_queue12 DEFINE"	RO	1
11	dma_priority_queue11	"Takes the value of the `dma_priority_queue11 DEFINE"	RO	1
10	dma_priority_queue10	"Takes the value of the `dma_priority_queue10 DEFINE"	RO	1
9	dma_priority_queue9	"Takes the value of the `dma_priority_queue9 DEFINE"	RO	1
8	dma_priority_queue8	"Takes the value of the `dma_priority_queue8 DEFINE"	RO	1
7	dma_priority_queue7	"Takes the value of the `dma_priority_queue7 DEFINE"	RO	1
6	dma_priority_queue6	"Takes the value of the `dma_priority_queue6 DEFINE"	RO	1
5	dma_priority_queue5	"Takes the value of the `dma_priority_queue5 DEFINE"	RO	1
4	dma_priority_queue4	"Takes the value of the `dma_priority_queue4 DEFINE"	RO	1

3	dma_priority_queue3	"Takes the value of the `dma_priority_queue3 DEFINE"	RO	1
2	dma_priority_queue2	"Takes the value of the `dma_priority_queue2 DEFINE"	RO	1
1	dma_priority_queue1	"Takes the value of the `dma_priority_queue1 DEFINE"	RO	1
0	reserved_0	"Reserved. Set to zero."	RO	0

Table 11-149: designcfg_debug7

Register Information				
Description	"Design Configuration Register 7"			
Offset	0x0298	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:28	tx_pbuf_num_segments_q7	"Takes the value of the `gem_tx_pbuf_num_segments_q7 DEFINE"	RO	0x0
27:24	tx_pbuf_num_segments_q6	"Takes the value of the `gem_tx_pbuf_num_segments_q6 DEFINE"	RO	0x0
23:20	tx_pbuf_num_segments_q5	"Takes the value of the `gem_tx_pbuf_num_segments_q5 DEFINE"	RO	0x0
19:16	tx_pbuf_num_segments_q4	"Takes the value of the `gem_tx_pbuf_num_segments_q4 DEFINE"	RO	0x0
15:12	tx_pbuf_num_segments_q3	"Takes the value of the `gem_tx_pbuf_num_segments_q3 DEFINE"	RO	0x0
11:8	tx_pbuf_num_segments_q2	"Takes the value of the `gem_tx_pbuf_num_segments_q2 DEFINE"	RO	0x0
7:4	tx_pbuf_num_segments_q1	"Takes the value of the `gem_tx_pbuf_num_segments_q1 DEFINE"	RO	0x0
3:0	tx_pbuf_num_segments_q0	"Takes the value of the edma_tx_pbuf_num_segments_q0 parameter which is set by the `gem_tx_pbuf_num_segments_q0 DEFINE or is zero if priority queuing is not configured."	RO	0x0

Table 11-150: designcfg_debug8

Register Information					
Description	"Design Configuration Register 8"				
Offset	0x029C		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:24	num_type1_screeners	"Takes the value of the `num_type1_screeners DEFINE"		RO	0x10
23:16	num_type2_screeners	"Takes the value of the `num_type2_screeners DEFINE"		RO	0x10
15:8	num_scr2_ethtype_regs	"Takes the value of the `num_scr2_ethtype_regs DEFINE"		RO	0x08
7:0	num_scr2_compare_regs	"Takes the value of the `num_scr2_compare_regs DEFINE"		RO	0x20

Table 11-151: designcfg_debug9

Register Information					
Description	"Design Configuration Register 9"				
Offset	0x02A0		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:28	tx_pbuf_num_segments_q15	"Takes the value of the `gem_tx_pbuf_num_segments_q15 DEFINE"		RO	0x0
27:24	tx_pbuf_num_segments_q14	"Takes the value of the `gem_tx_pbuf_num_segments_q14 DEFINE"		RO	0x0
23:20	tx_pbuf_num_segments_q13	"Takes the value of the `gem_tx_pbuf_num_segments_q13 DEFINE"		RO	0x0
19:16	tx_pbuf_num_segments_q12	"Takes the value of the `gem_tx_pbuf_num_segments_q12 DEFINE"		RO	0x0
15:12	tx_pbuf_num_segments_q11	"Takes the value of the `gem_tx_pbuf_num_segments_q11 DEFINE"		RO	0x0
11:8	tx_pbuf_num_segments_q10	"Takes the value of the `gem_tx_pbuf_num_segments_q10 DEFINE"		RO	0x0
7:4	tx_pbuf_num_segments_q9	"Takes the value of the `gem_tx_pbuf_num_segments_q9 DEFINE"		RO	0x0
3:0	tx_pbuf_num_segments_q8	"Takes the value of the `gem_tx_pbuf_num_segments_q8 DEFINE"		RO	0x0

Table 11-152: designcfg_debug10

Register Information				
Description	"Design Configuration Register 10"			
Offset	0x02A4	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:28	emac_bus_width	"Takes the value of the `gem_emac_bus_width DEFINE. 1 - The MAC has a datawidth of 32bits. 2 - The MAC has a datawidth of 64bits. 4 - The MAC has a datawidth of 128bits"	RO	0x2
27:24	tx_pbuf_data	"Takes the value of the `gem_tx_pbuf_data DEFINE. 1 - The TX DPRAM has a datawidth of 32bits. 2 - The TX DPRAM has a datawidth of 64bits. 4 - The TX DPRAM has a datawidth of 128bits"	RO	0x2
23:20	rx_pbuf_data	"Takes the value of the `gem_rx_pbuf_data DEFINE. 1 - The RX DPRAM has a datawidth of 32bits. 2 - The RX DPRAM has a datawidth of 64bits. 4 - RX The DPRAM has a datawidth of 128bits"	RO	0x2
19:16	axi_access_pipeline_bits	"Takes the value of the edma_axi_access_pipeline_bits parameter which is set by the `gem_axi_access_pipeline_bits DEFINE or is zero if the FIFO interface is configured."	RO	0x4
15:12	axi_tx_descr_rd_buff_bits	"Takes the value of the edma_axi_tx_descr_rd_buff_bits parameter which is set by the `gem_axi_tx_descr_rd_buff_bits DEFINE or is zero if the FIFO interface is configured."	RO	0x4
11:8	axi_rx_descr_rd_buff_bits	"Takes the value of the edma_axi_rx_descr_rd_buff_bits parameter which is set by the `gem_axi_rx_descr_rd_buff_bits DEFINE or is zero if the FIFO interface is configured."	RO	0x4
7:4	axi_tx_descr_wr_buff_bits	"Takes the value of the edma_axi_tx_descr_wr_buff_bits parameter which is set by the `gem_axi_tx_descr_wr_buff_bits DEFINE or is zero if the FIFO interface is configured."	RO	0x4
3:0	axi_rx_descr_wr_buff_bits	"Takes the value of the edma_axi_rx_descr_wr_buff_bits parameter which is set by the `gem_axi_rx_descr_wr_buff_bits DEFINE or is zero if the FIFO interface is configured."	RO	0x4

Table 11-153: designcfg_debug11

Register Information				
Description	"Design Configuration Register 11"			
Offset	0x02A8	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:8	reserved_31_8	"Reserved, read as 0, ignored on write."	RO	0x00 0000
7	asf_prot_tx_sched	"Takes the value of the edma_asf_asf_prot_tx_sched parameter which is set by the `gem_asf_prot_tx_sched DEFINE."	RO	1
6	asf_host_par	"Takes the value of the edma_asf_host_par parameter which is set by the `gem_asf_host_par DEFINE."	RO	1
5	asf_trans_to_prot	"Takes the value of the edma_asf_trans_to_prot parameter which is set by the `gem_asf_enable DEFINE."	RO	1
4	asf_integrity_prot	"Takes the value of the edma_asf_integrity_prot parameter which is set by the `gem_asf_enable DEFINE."	RO	1
3	protect_tsu	"akes the value of the edma_asf_prot_tsu parameter which is set by the `gem_asf_prot_tsu DEFINE or is zero if no TSU is present."	RO	1
2	csr_protection	"Takes the value of the edma_asf_csr_prot parameter which is set by the `gem_asf_enable DEFINE."	RO	1
1	dap_protection	"Takes the value of the edma_asf_dap_prot parameter which is set by the `gem_asf_enable DEFINE."	RO	1
0	ecc_sram	"Takes the value of the edma_asf_ecc_sram parameter which is set by the `gem_asf_ecc_sram DEFINE."	RO	1

Table 11-154: designcfg_debug12

Register Information				
Description	"Design Configuration Register 12"			
Offset	0x02AC	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:26	reserved_31_26	"Reserved, read as 0, ignored on write."	RO	0x00
25	gem_has_802p3_br	"Takes the value of the edma_has_br parameter which is set by the gem_has_802p3_br tick define. If defined the configuration contains both an express MAC (eMAC) and a pre-emptable MAC (pMAC) to allow 802.3br frame pre-emption."	RO	1
24:21	emac_tx_pbuf_addr	"Takes the value of the `gem_emac_tx_pbuf_addr DEFINE - this defines the size of the transmit SRAM for express MAC (eMAC) when 802.3br is configured - will be zero if emac_tx_pbuf_address is decimal 16"	RO	0xB
20:17	emac_rx_pbuf_addr	"Takes the value of the `gem_emac_rx_pbuf_addr DEFINE - this defines the size of the receive SRAM for express MAC (eMAC) when 802.3br is configured - will be zero if emac_rx_pbuf_address is decimal 16"	RO	0xB
16	gem_has_cb	"Indicates whether gem has 802.1CB/FRER configured. This is zero if `gem_no_of_cb_streams is undefined or zero."	RO	1
15:8	gem_cb_history_len	"Takes the value of the `gem_seq_history_len DEFINE or 0x01 if undefined."	RO	0x40
7:0	gem_num_cb_streams	"Takes the value of the `gem_no_of_cb_streams DEFINE or 0x01 if undefined or the defined value is zero."	RO	0x10

Table 11-155: axi_qos_cfg_0

Register Information			
Description	"AXI Quality of Service register 0. This register is only present if AXI is configured. This register contains 8-bits per queue to control driving of the ARQOS and AWQOS outputs (4-bits each). For each queue the lower four bits are for data accesses and the upper four bits are for descriptor accesses."		
Offset	0x02E0	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:28	q_3_descr_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 3 descriptor accesses."	RW	0x0
27:24	q_3_data_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 3 data accesses."	RW	0x0
23:20	q_2_descr_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 2 descriptor accesses."	RW	0x0
19:16	q_2_data_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 2 data accesses."	RW	0x0
15:12	q_1_descr_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 1 descriptor accesses."	RW	0x0
11:8	q_1_data_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 1 data accesses."	RW	0x0
7:4	q_0_descr_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 0 descriptor accesses."	RW	0x0
3:0	q_0_data_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 0 data accesses."	RW	0x0

Table 11-156: axi_qos_cfg_1

Register Information			
Description	"AXI Quality of Service register 1. This register is only present if AXI is configured. This register contains 8-bits per queue to control driving of the ARQOS and AWQOS outputs (4-bits each). For each queue the lower four bits are for data accesses and the upper four bits are for descriptor accesses."		
Offset	0x02E4	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:28	q_7_descr_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 7 descriptor accesses."	RW	0x0
27:24	q_7_data_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 7 data accesses."	RW	0x0
23:20	q_6_descr_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 6 descriptor accesses."	RW	0x0
19:16	q_6_data_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 6 data accesses."	RW	0x0
15:12	q_5_descr_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 5 descriptor accesses."	RW	0x0
11:8	q_5_data_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 5 data accesses."	RW	0x0
7:4	q_4_descr_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 4 descriptor accesses."	RW	0x0
3:0	q_4_data_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 4 data accesses."	RW	0x0

Table 11-157: axi_qos_cfg_2

Register Information			
Description	"AXI Quality of Service register 2. This register is only present if AXI is configured. This register contains 8-bits per queue to control driving of the ARQOS and AWQOS outputs (4-bits each). For each queue the lower four bits are for data accesses and the upper four bits are for descriptor accesses."		
Offset	0x02E8	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:28	q_11_descr_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 11 descriptor accesses."	RW	0x0
27:24	q_11_data_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 11 data accesses."	RW	0x0
23:20	q_10_descr_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 10 descriptor accesses."	RW	0x0
19:16	q_10_data_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 10 data accesses."	RW	0x0
15:12	q_9_descr_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 9 descriptor accesses."	RW	0x0
11:8	q_9_data_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 9 data accesses."	RW	0x0
7:4	q_8_descr_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 8 descriptor accesses."	RW	0x0
3:0	q_8_data_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 8 data accesses."	RW	0x0

Table 11-158: axi_qos_cfg_3

Register Information				
Description	"AXI Quality of Service register 3. This register is only present if AXI is configured. This register contains 8-bits per queue to control driving of the ARQOS and AWQOS outputs (4-bits each). For each queue the lower four bits are for data accesses and the upper four bits are for descriptor accesses."			
Offset	0x02EC	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:28	q_15_descr_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 15 descriptor accesses."	RW	0x0
27:24	q_15_data_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 15 data accesses."	RW	0x0
23:20	q_14_descr_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 14 descriptor accesses."	RW	0x0
19:16	q_14_data_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 14 data accesses."	RW	0x0
15:12	q_13_descr_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 13 descriptor accesses."	RW	0x0
11:8	q_13_data_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 13 data accesses."	RW	0x0
7:4	q_12_descr_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 12 descriptor accesses."	RW	0x0
3:0	q_12_data_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for queue 12 data accesses."	RW	0x0

Table 11-159: spec_add5_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x0300	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-160: spec_add5_top

Register Information				
Description	"Specific Address Top"			
Offset	0x0304		**Type:**	RW
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-161: spec_add6_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x0308		**Type:**	RW
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-162: spec_add6_top

Register Information				
Description	"Specific Address Top"			
Offset	0x030C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-163: spec_add7_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x0310	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-164: spec_add7_top

Register Information			
Description	"Specific Address Top"		
Offset	0x0314	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-165: spec_add8_bottom

Register Information			
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."		
Offset	0x0318	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-166: spec_add8_top

Register Information				
Description	"Specific Address Top"			
Offset	0x031C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-167: spec_add9_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x0320	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-168: spec_add9_top

Register Information				
Description	"Specific Address Top"			
Offset	0x0324	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-169: spec_add10_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x0328	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-170: spec_add10_top

Register Information				
Description	"Specific Address Top"			
Offset	0x032C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-171: spec_add11_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x0330	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-172: spec_add11_top

Register Information				
Description	"Specific Address Top"			
Offset	0x0334	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-173: spec_add12_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x0338	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-174: spec_add12_top

Register Information				
Description	"Specific Address Top"			
Offset	0x033C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-175: spec_add13_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x0340	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-176: spec_add13_top

Register Information			
Description	"Specific Address Top"		
Offset	0x0344	**Type:**	RW

Bits	Name	Description	Access	Reset
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-177: spec_add14_bottom

Register Information			
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."		
Offset	0x0348	**Type:**	RW

Bits	Name	Description	Access	Reset
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-178: spec_add14_top

Register Information				
Description	"Specific Address Top"			
Offset	0x034C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-179: spec_add15_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x0350	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-180: spec_add15_top

Register Information				
Description	"Specific Address Top"			
Offset	0x0354	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-181: spec_add16_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x0358	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-182: spec_add16_top

Register Information			
Description	"Specific Address Top"		
Offset	0x035C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-183: spec_add17_bottom

Register Information			
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."		
Offset	0x0360	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-184: spec_add17_top

Register Information				
Description	"Specific Address Top"			
Offset	0x0364	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-185: spec_add18_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x0368	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-186: spec_add18_top

Register Information				
Description	"Specific Address Top"			
Offset	0x036C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-187: spec_add19_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x0370	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-188: spec_add19_top

Register Information			
Description	"Specific Address Top"		
Offset	0x0374	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-189: spec_add20_bottom

Register Information			
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."		
Offset	0x0378	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-190: spec_add20_top

Register Information				
Description	"Specific Address Top"			
Offset	0x037C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-191: spec_add21_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x0380	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-192: spec_add21_top

Register Information				
Description	"Specific Address Top"			
Offset	0x0384	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-193: spec_add22_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x0388	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-194: spec_add22_top

Register Information				
Description	"Specific Address Top"			
Offset	0x038C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-195: spec_add23_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x0390	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-196: spec_add23_top

Register Information				
Description	"Specific Address Top"			
Offset	0x0394	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-197: spec_add24_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x0398	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-198: spec_add24_top

Register Information				
Description	"Specific Address Top"			
Offset	0x039C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-199: spec_add25_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x03A0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-200: spec_add25_top

Register Information				
Description	"Specific Address Top"			
Offset	0x03A4	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-201: spec_add26_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x03A8	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-202: spec_add26_top

Register Information				
Description	"Specific Address Top"			
Offset	0x03AC	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-203: spec_add27_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x03B0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-204: spec_add27_top

Register Information				
Description	"Specific Address Top"			
Offset	0x03B4	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-205: spec_add28_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x03B8	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-206: spec_add28_top

Register Information				
Description	"Specific Address Top"			
Offset	0x03BC	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-207: spec_add29_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x03C0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-208: spec_add29_top

Register Information				
Description	"Specific Address Top"			
Offset	0x03C4	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-209: spec_add30_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x03C8	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-210: spec_add30_top

Register Information				
Description	"Specific Address Top"			
Offset	0x03CC	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-211: spec_add31_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x03D0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-212: spec_add31_top

Register Information			
Description	"Specific Address Top"		
Offset	0x03D4	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-213: spec_add32_bottom

Register Information			
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."		
Offset	0x03D8	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-214: spec_add32_top

Register Information				
Description	"Specific Address Top"			
Offset	0x03DC	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-215: spec_add33_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x03E0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-216: spec_add33_top

Register Information				
Description	"Specific Address Top"			
Offset	0x03E4	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-217: spec_add34_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x03E8	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-218: spec_add34_top

Register Information				
Description	"Specific Address Top"			
Offset	0x03EC	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-219: spec_add35_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x03F0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-220: spec_add35_top

Register Information				
Description	"Specific Address Top"			
Offset	0x03F4	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-221: spec_add36_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x03F8	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-222: spec_add36_top

Register Information				
Description	"Specific Address Top"			
Offset	0x03FC	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-223: int_q1_status

Register Information				
Description	"Priority Queue Interrupt Status Register - the bits in this register are cleared when read as gem_irq_read_clear is set for this configuration"			
Offset	0x0400	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok	"bresp/hresp not OK"	RO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete	"Transmit complete"	RO	0
6	amba_error	"Transmit frame corruption due to AMBA (AHB/AXI) error set if an error occurs whilst midway through reading transmit frame from the external memory, including HRESP errors(AHB), RRESP and BRESP errors (AXI) and buffers exhausted mid frame"	RO	0
5	retry_limit_exceeded_or_late_collision	"Retry limit exceeded or late collision"	RO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_bit_read	"RX used bit read"	RO	0
1	receive_complete	"Receive complete"	RO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-224: int_q2_status

Register Information			
Description	"Priority Queue Interrupt Status Register - the bits in this register are cleared when read as gem_irq_read_clear is set for this configuration"		
Offset	0x0404	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok	"bresp/hresp not OK"	RO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete	"Transmit complete"	RO	0
6	amba_error	"Transmit frame corruption due to AMBA (AHB/AXI) error set if an error occurs whilst midway through reading transmit frame from the external memory, including HRESP errors(AHB), RRESP and BRESP errors (AXI) and buffers exhausted mid frame"	RO	0
5	retry_limit_exceeded_or_late_collision	"Retry limit exceeded or late collision"	RO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_bit_read	"RX used bit read"	RO	0
1	receive_complete	"Receive complete"	RO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-225: int_q3_status

Register Information			
Description	"Priority Queue Interrupt Status Register - the bits in this register are cleared when read as gem_irq_read_clear is set for this configuration"		
Offset	0x0408	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok	"bresp/hresp not OK"	RO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete	"Transmit complete"	RO	0
6	amba_error	"Transmit frame corruption due to AMBA (AHB/AXI) error set if an error occurs whilst midway through reading transmit frame from the external memory, including HRESP errors(AHB), RRESP and BRESP errors (AXI) and buffers exhausted mid frame"	RO	0
5	retry_limit_exceeded_or_late_collision	"Retry limit exceeded or late collision"	RO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_bit_read	"RX used bit read"	RO	0
1	receive_complete	"Receive complete"	RO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-226: int_q4_status

Register Information				
Description	"Priority Queue Interrupt Status Register - the bits in this register are cleared when read as gem_irq_read_clear is set for this configuration"			
Offset	0x040C	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok	"bresp/hresp not OK"	RO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete	"Transmit complete"	RO	0
6	amba_error	"Transmit frame corruption due to AMBA (AHB/AXI) error set if an error occurs whilst midway through reading transmit frame from the external memory, including HRESP errors(AHB), RRESP and BRESP errors (AXI) and buffers exhausted mid frame"	RO	0
5	retry_limit_exceeded_or_late_collision	"Retry limit exceeded or late collision"	RO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_bit_read	"RX used bit read"	RO	0
1	receive_complete	"Receive complete"	RO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-227: int_q5_status

Register Information				
Description	"Priority Queue Interrupt Status Register - the bits in this register are cleared when read as gem_irq_read_clear is set for this configuration"			
Offset	0x0410	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok	"bresp/hresp not OK"	RO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete	"Transmit complete"	RO	0
6	amba_error	"Transmit frame corruption due to AMBA (AHB/AXI) error set if an error occurs whilst midway through reading transmit frame from the external memory, including HRESP errors(AHB), RRESP and BRESP errors (AXI) and buffers exhausted mid frame"	RO	0
5	retry_limit_exceeded_or_late_collision	"Retry limit exceeded or late collision"	RO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_bit_read	"RX used bit read"	RO	0
1	receive_complete	"Receive complete"	RO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-228: int_q6_status

Register Information				
Description	"Priority Queue Interrupt Status Register - the bits in this register are cleared when read as gem_irq_read_clear is set for this configuration"			
Offset	0x0414	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok	"bresp/hresp not OK"	RO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete	"Transmit complete"	RO	0
6	amba_error	"Transmit frame corruption due to AMBA (AHB/AXI) error set if an error occurs whilst midway through reading transmit frame from the external memory, including HRESP errors(AHB), RRESP and BRESP errors (AXI) and buffers exhausted mid frame"	RO	0
5	retry_limit_exceeded_or_late_collision	"Retry limit exceeded or late collision"	RO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_bit_read	"RX used bit read"	RO	0
1	receive_complete	"Receive complete"	RO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-229: int_q7_status

Register Information				
Description	"Priority Queue Interrupt Status Register - the bits in this register are cleared when read as gem_irq_read_clear is set for this configuration"			
Offset	0x0418	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok	"bresp/hresp not OK"	RO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete	"Transmit complete"	RO	0
6	amba_error	"Transmit frame corruption due to AMBA (AHB/AXI) error set if an error occurs whilst midway through reading transmit frame from the external memory, including HRESP errors(AHB), RRESP and BRESP errors (AXI) and buffers exhausted mid frame"	RO	0
5	retry_limit_exceeded_or_late_collision	"Retry limit exceeded or late collision"	RO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_bit_read	"RX used bit read"	RO	0
1	receive_complete	"Receive complete"	RO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-230: int_q8_status

Register Information			
Description	"Priority Queue Interrupt Status Register - the bits in this register are cleared when read as gem_irq_read_clear is set for this configuration"		
Offset	0x041C	**Type:**	RO

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok	"bresp/hresp not OK"	RO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete	"Transmit complete"	RO	0
6	amba_error	"Transmit frame corruption due to AMBA (AHB/AXI) error set if an error occurs whilst midway through reading transmit frame from the external memory, including HRESP errors(AHB), RRESP and BRESP errors (AXI) and buffers exhausted mid frame"	RO	0
5	retry_limit_exceeded_or_late_collision	"Retry limit exceeded or late collision"	RO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_bit_read	"RX used bit read"	RO	0
1	receive_complete	"Receive complete"	RO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-231: int_q9_status

Register Information			
Description	"Priority Queue Interrupt Status Register - the bits in this register are cleared when read as gem_irq_read_clear is set for this configuration"		
Offset	0x0420	**Type:**	RO

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok	"bresp/hresp not OK"	RO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete	"Transmit complete"	RO	0
6	amba_error	"Transmit frame corruption due to AMBA (AHB/AXI) error set if an error occurs whilst midway through reading transmit frame from the external memory, including HRESP errors(AHB), RRESP and BRESP errors (AXI) and buffers exhausted mid frame"	RO	0
5	retry_limit_exceeded_or_late_collision	"Retry limit exceeded or late collision"	RO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_bit_read	"RX used bit read"	RO	0
1	receive_complete	"Receive complete"	RO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-232: int_q10_status

Register Information				
Description	"Priority Queue Interrupt Status Register - the bits in this register are cleared when read as gem_irq_read_clear is set for this configuration"			
Offset	0x0424	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok	"bresp/hresp not OK"	RO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete	"Transmit complete"	RO	0
6	amba_error	"Transmit frame corruption due to AMBA (AHB/AXI) error set if an error occurs whilst midway through reading transmit frame from the external memory, including HRESP errors(AHB), RRESP and BRESP errors (AXI) and buffers exhausted mid frame"	RO	0
5	retry_limit_exceeded_or_late_collision	"Retry limit exceeded or late collision"	RO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_bit_read	"RX used bit read"	RO	0
1	receive_complete	"Receive complete"	RO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-233: int_q11_status

Register Information				
Description	"Priority Queue Interrupt Status Register - the bits in this register are cleared when read as gem_irq_read_clear is set for this configuration"			
Offset	0x0428	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok	"bresp/hresp not OK"	RO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete	"Transmit complete"	RO	0
6	amba_error	"Transmit frame corruption due to AMBA (AHB/AXI) error set if an error occurs whilst midway through reading transmit frame from the external memory, including HRESP errors(AHB), RRESP and BRESP errors (AXI) and buffers exhausted mid frame"	RO	0
5	retry_limit_exceeded_or_late_collision	"Retry limit exceeded or late collision"	RO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_bit_read	"RX used bit read"	RO	0
1	receive_complete	"Receive complete"	RO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-234: int_q12_status

Register Information			
Description	"Priority Queue Interrupt Status Register - the bits in this register are cleared when read as gem_irq_read_clear is set for this configuration"		
Offset	0x042C	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok	"bresp/hresp not OK"	RO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete	"Transmit complete"	RO	0
6	amba_error	"Transmit frame corruption due to AMBA (AHB/AXI) error set if an error occurs whilst midway through reading transmit frame from the external memory, including HRESP errors(AHB), RRESP and BRESP errors (AXI) and buffers exhausted mid frame"	RO	0
5	retry_limit_exceeded_or_late_collision	"Retry limit exceeded or late collision"	RO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_bit_read	"RX used bit read"	RO	0
1	receive_complete	"Receive complete"	RO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-235: int_q13_status

Register Information			
Description	"Priority Queue Interrupt Status Register - the bits in this register are cleared when read as gem_irq_read_clear is set for this configuration"		
Offset	0x0430	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok	"bresp/hresp not OK"	RO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete	"Transmit complete"	RO	0
6	amba_error	"Transmit frame corruption due to AMBA (AHB/AXI) error set if an error occurs whilst midway through reading transmit frame from the external memory, including HRESP errors(AHB), RRESP and BRESP errors (AXI) and buffers exhausted mid frame"	RO	0
5	retry_limit_exceeded_or_late_collision	"Retry limit exceeded or late collision"	RO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_bit_read	"RX used bit read"	RO	0
1	receive_complete	"Receive complete"	RO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-236: int_q14_status

Register Information			
Description	"Priority Queue Interrupt Status Register - the bits in this register are cleared when read as gem_irq_read_clear is set for this configuration"		
Offset	0x0434	**Type:**	RO

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok	"bresp/hresp not OK"	RO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete	"Transmit complete"	RO	0
6	amba_error	"Transmit frame corruption due to AMBA (AHB/AXI) error set if an error occurs whilst midway through reading transmit frame from the external memory, including HRESP errors(AHB), RRESP and BRESP errors (AXI) and buffers exhausted mid frame"	RO	0
5	retry_limit_exceeded_or_late_collision	"Retry limit exceeded or late collision"	RO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_bit_read	"RX used bit read"	RO	0
1	receive_complete	"Receive complete"	RO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-237: int_q15_status

Register Information			
Description	"Priority Queue Interrupt Status Register - the bits in this register are cleared when read as gem_irq_read_clear is set for this configuration"		
Offset	0x0438	**Type:**	RO

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok	"bresp/hresp not OK"	RO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete	"Transmit complete"	RO	0
6	amba_error	"Transmit frame corruption due to AMBA (AHB/AXI) error set if an error occurs whilst midway through reading transmit frame from the external memory, including HRESP errors(AHB), RRESP and BRESP errors (AXI) and buffers exhausted mid frame"	RO	0
5	retry_limit_exceeded_or_late_collision	"Retry limit exceeded or late collision"	RO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_bit_read	"RX used bit read"	RO	0
1	receive_complete	"Receive complete"	RO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-238: transmit_q1_ptr

Register Information			
Description	"This register holds the start address of the transmit buffer queue (transmit buffers descriptor list). The transmit buffer queue base address register must be initialized before transmit is started through bit 9 of the network control register. Once transmission has started, any write to the transmit buffer queue base address register is illegal and therefore ignored. Note that due to clock boundary synchronization, it takes a maximum of four pclk cycles from the writing of the transmit start bit before the transmitter is active. Writing to the transmit buffer queue base address register during this time may produce unpredictable results. Reading this register returns the location of the descriptor currently being accessed. Because the DMA can store data for multiple frames at once, this may not necessarily be pointing to the current frame being transmitted. In terms of AMBA AHB/AXI operation, the transmit descriptors are written to memory using a single 32bit AHB access. When the datapath is configured as 64bit or 128bit, the transmit descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is read from memory using a single AHB/AXI access. For 32bit datapaths, the descriptors should be aligned at 32-bit boundaries and the descriptors are read from memory using two individual 32-bit non sequential accesses."		
Offset	0x0440	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:2	dma_tx_q_ptr	"Transmit buffer queue base address - written with the address of the start of the transmit queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_tx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while transmit is not enabled."	RW	0

Table 11-239: transmit_q2_ptr

Register Information			
Description	"This register holds the start address of the transmit buffer queue (transmit buffers descriptor list). The transmit buffer queue base address register must be initialized before transmit is started through bit 9 of the network control register. Once transmission has started, any write to the transmit buffer queue base address register is illegal and therefore ignored. Note that due to clock boundary synchronization, it takes a maximum of four pclk cycles from the writing of the transmit start bit before the transmitter is active. Writing to the transmit buffer queue base address register during this time may produce unpredictable results. Reading this register returns the location of the descriptor currently being accessed. Because the DMA can store data for multiple frames at once, this may not necessarily be pointing to the current frame being transmitted. In terms of AMBA AHB/AXI operation, the transmit descriptors are written to memory using a single 32bit AHB access. When the datapath is configured as 64bit or 128bit, the transmit descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is read from memory using a single AHB/AXI access. For 32bit datapaths, the descriptors should be aligned at 32-bit boundaries and the descriptors are read from memory using two individual 32-bit non sequential accesses."		
Offset	0x0444	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:2	dma_tx_q_ptr	"Transmit buffer queue base address - written with the address of the start of the transmit queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_tx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while transmit is not enabled."	RW	0

Table 11-240: transmit_q3_ptr

Register Information			
Description	"This register holds the start address of the transmit buffer queue (transmit buffers descriptor list). The transmit buffer queue base address register must be initialized before transmit is started through bit 9 of the network control register. Once transmission has started, any write to the transmit buffer queue base address register is illegal and therefore ignored. Note that due to clock boundary synchronization, it takes a maximum of four pclk cycles from the writing of the transmit start bit before the transmitter is active. Writing to the transmit buffer queue base address register during this time may produce unpredictable results. Reading this register returns the location of the descriptor currently being accessed. Because the DMA can store data for multiple frames at once, this may not necessarily be pointing to the current frame being transmitted. In terms of AMBA AHB/AXI operation, the transmit descriptors are written to memory using a single 32bit AHB access. When the datapath is configured as 64bit or 128bit, the transmit descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is read from memory using a single AHB/AXI access. For 32bit datapaths, the descriptors should be aligned at 32-bit boundaries and the descriptors are read from memory using two individual 32-bit non sequential accesses."		
Offset	0x0448	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:2	dma_tx_q_ptr	"Transmit buffer queue base address - written with the address of the start of the transmit queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_tx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while transmit is not enabled."	RW	0

Table 11-241: transmit_q4_ptr

Register Information			
Description	"This register holds the start address of the transmit buffer queue (transmit buffers descriptor list). The transmit buffer queue base address register must be initialized before transmit is started through bit 9 of the network control register. Once transmission has started, any write to the transmit buffer queue base address register is illegal and therefore ignored. Note that due to clock boundary synchronization, it takes a maximum of four pclk cycles from the writing of the transmit start bit before the transmitter is active. Writing to the transmit buffer queue base address register during this time may produce unpredictable results. Reading this register returns the location of the descriptor currently being accessed. Because the DMA can store data for multiple frames at once, this may not necessarily be pointing to the current frame being transmitted. In terms of AMBA AHB/AXI operation, the transmit descriptors are written to memory using a single 32bit AHB access. When the datapath is configured as 64bit or 128bit, the transmit descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is read from memory using a single AHB/AXI access. For 32bit datapaths, the descriptors should be aligned at 32-bit boundaries and the descriptors are read from memory using two individual 32-bit non sequential accesses."		
Offset	0x044C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:2	dma_tx_q_ptr	"Transmit buffer queue base address - written with the address of the start of the transmit queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_tx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while transmit is not enabled."	RW	0

Table 11-242: transmit_q5_ptr

Register Information			
Description	"This register holds the start address of the transmit buffer queue (transmit buffers descriptor list). The transmit buffer queue base address register must be initialized before transmit is started through bit 9 of the network control register. Once transmission has started, any write to the transmit buffer queue base address register is illegal and therefore ignored. Note that due to clock boundary synchronization, it takes a maximum of four pclk cycles from the writing of the transmit start bit before the transmitter is active. Writing to the transmit buffer queue base address register during this time may produce unpredictable results. Reading this register returns the location of the descriptor currently being accessed. Because the DMA can store data for multiple frames at once, this may not necessarily be pointing to the current frame being transmitted. In terms of AMBA AHB/AXI operation, the transmit descriptors are written to memory using a single 32bit AHB access. When the datapath is configured as 64bit or 128bit, the transmit descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is read from memory using a single AHB/AXI access. For 32bit datapaths, the descriptors should be aligned at 32-bit boundaries and the descriptors are read from memory using two individual 32-bit non sequential accesses."		
Offset	0x0450	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:2	dma_tx_q_ptr	"Transmit buffer queue base address - written with the address of the start of the transmit queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_tx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while transmit is not enabled."	RW	0

Table 11-243: transmit_q6_ptr

Register Information			
Description	"This register holds the start address of the transmit buffer queue (transmit buffers descriptor list). The transmit buffer queue base address register must be initialized before transmit is started through bit 9 of the network control register. Once transmission has started, any write to the transmit buffer queue base address register is illegal and therefore ignored. Note that due to clock boundary synchronization, it takes a maximum of four pclk cycles from the writing of the transmit start bit before the transmitter is active. Writing to the transmit buffer queue base address register during this time may produce unpredictable results. Reading this register returns the location of the descriptor currently being accessed. Because the DMA can store data for multiple frames at once, this may not necessarily be pointing to the current frame being transmitted. In terms of AMBA AHB/AXI operation, the transmit descriptors are written to memory using a single 32bit AHB access. When the datapath is configured as 64bit or 128bit, the transmit descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is read from memory using a single AHB/AXI access. For 32bit datapaths, the descriptors should be aligned at 32-bit boundaries and the descriptors are read from memory using two individual 32-bit non sequential accesses."		
Offset	0x0454	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:2	dma_tx_q_ptr	"Transmit buffer queue base address - written with the address of the start of the transmit queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_tx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while transmit is not enabled."	RW	0

Table 11-244: transmit_q7_ptr

Register Information			
Description	"This register holds the start address of the transmit buffer queue (transmit buffers descriptor list). The transmit buffer queue base address register must be initialized before transmit is started through bit 9 of the network control register. Once transmission has started, any write to the transmit buffer queue base address register is illegal and therefore ignored. Note that due to clock boundary synchronization, it takes a maximum of four pclk cycles from the writing of the transmit start bit before the transmitter is active. Writing to the transmit buffer queue base address register during this time may produce unpredictable results. Reading this register returns the location of the descriptor currently being accessed. Because the DMA can store data for multiple frames at once, this may not necessarily be pointing to the current frame being transmitted. In terms of AMBA AHB/AXI operation, the transmit descriptors are written to memory using a single 32bit AHB access. When the datapath is configured as 64bit or 128bit, the transmit descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is read from memory using a single AHB/AXI access. For 32bit datapaths, the descriptors should be aligned at 32-bit boundaries and the descriptors are read from memory using two individual 32-bit non sequential accesses."		
Offset	0x0458	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:2	dma_tx_q_ptr	"Transmit buffer queue base address - written with the address of the start of the transmit queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_tx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while transmit is not enabled."	RW	0

Table 11-245: transmit_q8_ptr

Register Information			
Description	"This register holds the start address of the transmit buffer queue (transmit buffers descriptor list). The transmit buffer queue base address register must be initialized before transmit is started through bit 9 of the network control register. Once transmission has started, any write to the transmit buffer queue base address register is illegal and therefore ignored. Note that due to clock boundary synchronization, it takes a maximum of four pclk cycles from the writing of the transmit start bit before the transmitter is active. Writing to the transmit buffer queue base address register during this time may produce unpredictable results. Reading this register returns the location of the descriptor currently being accessed. Because the DMA can store data for multiple frames at once, this may not necessarily be pointing to the current frame being transmitted. In terms of AMBA AHB/AXI operation, the transmit descriptors are written to memory using a single 32bit AHB access. When the datapath is configured as 64bit or 128bit, the transmit descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is read from memory using a single AHB/AXI access. For 32bit datapaths, the descriptors should be aligned at 32-bit boundaries and the descriptors are read from memory using two individual 32-bit non sequential accesses."		
Offset	0x045C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:2	dma_tx_q_ptr	"Transmit buffer queue base address - written with the address of the start of the transmit queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_tx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while transmit is not enabled."	RW	0

Table 11-246: transmit_q9_ptr

Register Information			
Description	"This register holds the start address of the transmit buffer queue (transmit buffers descriptor list). The transmit buffer queue base address register must be initialized before transmit is started through bit 9 of the network control register. Once transmission has started, any write to the transmit buffer queue base address register is illegal and therefore ignored. Note that due to clock boundary synchronization, it takes a maximum of four pclk cycles from the writing of the transmit start bit before the transmitter is active. Writing to the transmit buffer queue base address register during this time may produce unpredictable results. Reading this register returns the location of the descriptor currently being accessed. Because the DMA can store data for multiple frames at once, this may not necessarily be pointing to the current frame being transmitted. In terms of AMBA AHB/AXI operation, the transmit descriptors are written to memory using a single 32bit AHB access. When the datapath is configured as 64bit or 128bit, the transmit descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is read from memory using a single AHB/AXI access. For 32bit datapaths, the descriptors should be aligned at 32-bit boundaries and the descriptors are read from memory using two individual 32-bit non sequential accesses."		
Offset	0x0460	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:2	dma_tx_q_ptr	"Transmit buffer queue base address - written with the address of the start of the transmit queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_tx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while transmit is not enabled."	RW	0

Table 11-247: transmit_q10_ptr

Register Information			
Description	"This register holds the start address of the transmit buffer queue (transmit buffers descriptor list). The transmit buffer queue base address register must be initialized before transmit is started through bit 9 of the network control register. Once transmission has started, any write to the transmit buffer queue base address register is illegal and therefore ignored. Note that due to clock boundary synchronization, it takes a maximum of four pclk cycles from the writing of the transmit start bit before the transmitter is active. Writing to the transmit buffer queue base address register during this time may produce unpredictable results. Reading this register returns the location of the descriptor currently being accessed. Because the DMA can store data for multiple frames at once, this may not necessarily be pointing to the current frame being transmitted. In terms of AMBA AHB/AXI operation, the transmit descriptors are written to memory using a single 32bit AHB access. When the datapath is configured as 64bit or 128bit, the transmit descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is read from memory using a single AHB/AXI access. For 32bit datapaths, the descriptors should be aligned at 32-bit boundaries and the descriptors are read from memory using two individual 32-bit non sequential accesses."		
Offset	0x0464	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:2	dma_tx_q_ptr	"Transmit buffer queue base address - written with the address of the start of the transmit queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_tx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while transmit is not enabled."	RW	0

Table 11-248: transmit_q11_ptr

Register Information			
Description	"This register holds the start address of the transmit buffer queue (transmit buffers descriptor list). The transmit buffer queue base address register must be initialized before transmit is started through bit 9 of the network control register. Once transmission has started, any write to the transmit buffer queue base address register is illegal and therefore ignored. Note that due to clock boundary synchronization, it takes a maximum of four pclk cycles from the writing of the transmit start bit before the transmitter is active. Writing to the transmit buffer queue base address register during this time may produce unpredictable results. Reading this register returns the location of the descriptor currently being accessed. Because the DMA can store data for multiple frames at once, this may not necessarily be pointing to the current frame being transmitted. In terms of AMBA AHB/AXI operation, the transmit descriptors are written to memory using a single 32bit AHB access. When the datapath is configured as 64bit or 128bit, the transmit descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is read from memory using a single AHB/AXI access. For 32bit datapaths, the descriptors should be aligned at 32-bit boundaries and the descriptors are read from memory using two individual 32-bit non sequential accesses."		
Offset	0x0468	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:2	dma_tx_q_ptr	"Transmit buffer queue base address - written with the address of the start of the transmit queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_tx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while transmit is not enabled."	RW	0

Table 11-249: transmit_q12_ptr

Register Information			
Description	"This register holds the start address of the transmit buffer queue (transmit buffers descriptor list). The transmit buffer queue base address register must be initialized before transmit is started through bit 9 of the network control register. Once transmission has started, any write to the transmit buffer queue base address register is illegal and therefore ignored. Note that due to clock boundary synchronization, it takes a maximum of four pclk cycles from the writing of the transmit start bit before the transmitter is active. Writing to the transmit buffer queue base address register during this time may produce unpredictable results. Reading this register returns the location of the descriptor currently being accessed. Because the DMA can store data for multiple frames at once, this may not necessarily be pointing to the current frame being transmitted. In terms of AMBA AHB/AXI operation, the transmit descriptors are written to memory using a single 32bit AHB access. When the datapath is configured as 64bit or 128bit, the transmit descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is read from memory using a single AHB/AXI access. For 32bit datapaths, the descriptors should be aligned at 32-bit boundaries and the descriptors are read from memory using two individual 32-bit non sequential accesses."		
Offset	0x046C	**Type:**	RW

Bits	Name	Description	Access	Reset
31:2	dma_tx_q_ptr	"Transmit buffer queue base address - written with the address of the start of the transmit queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_tx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while transmit is not enabled."	RW	0

Table 11-250: transmit_q13_ptr

Register Information			
Description	"This register holds the start address of the transmit buffer queue (transmit buffers descriptor list). The transmit buffer queue base address register must be initialized before transmit is started through bit 9 of the network control register. Once transmission has started, any write to the transmit buffer queue base address register is illegal and therefore ignored. Note that due to clock boundary synchronization, it takes a maximum of four pclk cycles from the writing of the transmit start bit before the transmitter is active. Writing to the transmit buffer queue base address register during this time may produce unpredictable results. Reading this register returns the location of the descriptor currently being accessed. Because the DMA can store data for multiple frames at once, this may not necessarily be pointing to the current frame being transmitted. In terms of AMBA AHB/AXI operation, the transmit descriptors are written to memory using a single 32bit AHB access. When the datapath is configured as 64bit or 128bit, the transmit descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is read from memory using a single AHB/AXI access. For 32bit datapaths, the descriptors should be aligned at 32-bit boundaries and the descriptors are read from memory using two individual 32-bit non sequential accesses."		
Offset	0x0470	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:2	dma_tx_q_ptr	"Transmit buffer queue base address - written with the address of the start of the transmit queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_tx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while transmit is not enabled."	RW	0

Table 11-251: transmit_q14_ptr

Register Information			
Description	"This register holds the start address of the transmit buffer queue (transmit buffers descriptor list). The transmit buffer queue base address register must be initialized before transmit is started through bit 9 of the network control register. Once transmission has started, any write to the transmit buffer queue base address register is illegal and therefore ignored. Note that due to clock boundary synchronization, it takes a maximum of four pclk cycles from the writing of the transmit start bit before the transmitter is active. Writing to the transmit buffer queue base address register during this time may produce unpredictable results. Reading this register returns the location of the descriptor currently being accessed. Because the DMA can store data for multiple frames at once, this may not necessarily be pointing to the current frame being transmitted. In terms of AMBA AHB/AXI operation, the transmit descriptors are written to memory using a single 32bit AHB access. When the datapath is configured as 64bit or 128bit, the transmit descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is read from memory using a single AHB/AXI access. For 32bit datapaths, the descriptors should be aligned at 32-bit boundaries and the descriptors are read from memory using two individual 32-bit non sequential accesses."		
Offset	0x0474	**Type:**	RW

Bitfield Details

Bits	Name	Description	Access	Reset
31:2	dma_tx_q_ptr	"Transmit buffer queue base address - written with the address of the start of the transmit queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_tx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while transmit is not enabled."	RW	0

Table 11-252: transmit_q15_ptr

Register Information			
Description	"This register holds the start address of the transmit buffer queue (transmit buffers descriptor list). The transmit buffer queue base address register must be initialized before transmit is started through bit 9 of the network control register. Once transmission has started, any write to the transmit buffer queue base address register is illegal and therefore ignored. Note that due to clock boundary synchronization, it takes a maximum of four pclk cycles from the writing of the transmit start bit before the transmitter is active. Writing to the transmit buffer queue base address register during this time may produce unpredictable results. Reading this register returns the location of the descriptor currently being accessed. Because the DMA can store data for multiple frames at once, this may not necessarily be pointing to the current frame being transmitted. In terms of AMBA AHB/AXI operation, the transmit descriptors are written to memory using a single 32bit AHB access. When the datapath is configured as 64bit or 128bit, the transmit descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is read from memory using a single AHB/AXI access. For 32bit datapaths, the descriptors should be aligned at 32-bit boundaries and the descriptors are read from memory using two individual 32-bit non sequential accesses."		
Offset	0x0478	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:2	dma_tx_q_ptr	"Transmit buffer queue base address - written with the address of the start of the transmit queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_tx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while transmit is not enabled."	RW	0

Table 11-253: receive_q1_ptr

Register Information			
Description	"This register holds the start address of the receive buffer queue (receive buffers descriptor list). The receive buffer queue base address must be initialized before receive is enabled through bit 2 of the network control register. Once reception is enabled, any write to the receive buffer queue base address register is ignored. Reading this register returns the location of the descriptor currently being accessed. This value increments as buffers are used. Software should not use this register for determining where to remove received frames from the queue as it constantly changes as new frames are received. Software should instead work its way through the buffer descriptor queue checking the used bits. In terms of AMBA (AHB/AXI) operation, the receive descriptors are read from memory using a single 32bit AHB/AXI access. When the datapath is configured at 64bit or 128bit, the receive descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is written to using a single 64bit AHB/AXI access. For 32bit datapaths, the receive descriptors should be aligned at 32-bit boundaries and are written to using two individual non sequential 32-bit accesses."		
Offset	0x0480	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:2	dma_rx_q_ptr	"Receive buffer queue base address - written with the address of the start of the receive queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_rx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while receive is not enabled."	RW	0

Table 11-254: receive_q2_ptr

Register Information			
Description	"This register holds the start address of the receive buffer queue (receive buffers descriptor list). The receive buffer queue base address must be initialized before receive is enabled through bit 2 of the network control register. Once reception is enabled, any write to the receive buffer queue base address register is ignored. Reading this register returns the location of the descriptor currently being accessed. This value increments as buffers are used. Software should not use this register for determining where to remove received frames from the queue as it constantly changes as new frames are received. Software should instead work its way through the buffer descriptor queue checking the used bits. In terms of AMBA (AHB/AXI) operation, the receive descriptors are read from memory using a single 32bit AHB/AXI access. When the datapath is configured at 64bit or 128bit, the receive descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is written to using a single 64bit AHB/AXI access. For 32bit datapaths, the receive descriptors should be aligned at 32-bit boundaries and are written to using two individual non sequential 32-bit accesses."		
Offset	0x0484	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:2	dma_rx_q_ptr	"Receive buffer queue base address - written with the address of the start of the receive queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_rx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while receive is not enabled."	RW	0

Table 11-255: receive_q3_ptr

Register Information					
Description	"This register holds the start address of the receive buffer queue (receive buffers descriptor list). The receive buffer queue base address must be initialized before receive is enabled through bit 2 of the network control register. Once reception is enabled, any write to the receive buffer queue base address register is ignored. Reading this register returns the location of the descriptor currently being accessed. This value increments as buffers are used. Software should not use this register for determining where to remove received frames from the queue as it constantly changes as new frames are received. Software should instead work its way through the buffer descriptor queue checking the used bits. In terms of AMBA (AHB/AXI) operation, the receive descriptors are read from memory using a single 32bit AHB/AXI access. When the datapath is configured at 64bit or 128bit, the receive descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is written to using a single 64bit AHB/AXI access. For 32bit datapaths, the receive descriptors should be aligned at 32-bit boundaries and are written to using two individual non sequential 32-bit accesses."				
Offset	0x0488		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:2	dma_rx_q_ptr	"Receive buffer queue base address - written with the address of the start of the receive queue."		RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."		RO	0
0	dma_rx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while receive is not enabled."		RW	0

Table 11-256: receive_q4_ptr

Register Information					
Description	"This register holds the start address of the receive buffer queue (receive buffers descriptor list). The receive buffer queue base address must be initialized before receive is enabled through bit 2 of the network control register. Once reception is enabled, any write to the receive buffer queue base address register is ignored. Reading this register returns the location of the descriptor currently being accessed. This value increments as buffers are used. Software should not use this register for determining where to remove received frames from the queue as it constantly changes as new frames are received. Software should instead work its way through the buffer descriptor queue checking the used bits. In terms of AMBA (AHB/AXI) operation, the receive descriptors are read from memory using a single 32bit AHB/AXI access. When the datapath is configured at 64bit or 128bit, the receive descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is written to using a single 64bit AHB/AXI access. For 32bit datapaths, the receive descriptors should be aligned at 32-bit boundaries and are written to using two individual non sequential 32-bit accesses."				
Offset	0x048C		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:2	dma_rx_q_ptr	"Receive buffer queue base address - written with the address of the start of the receive queue."		RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."		RO	0
0	dma_rx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while receive is not enabled."		RW	0

Table 11-257: receive_q5_ptr

Register Information			
Description	"This register holds the start address of the receive buffer queue (receive buffers descriptor list). The receive buffer queue base address must be initialized before receive is enabled through bit 2 of the network control register. Once reception is enabled, any write to the receive buffer queue base address register is ignored. Reading this register returns the location of the descriptor currently being accessed. This value increments as buffers are used. Software should not use this register for determining where to remove received frames from the queue as it constantly changes as new frames are received. Software should instead work its way through the buffer descriptor queue checking the used bits. In terms of AMBA (AHB/AXI) operation, the receive descriptors are read from memory using a single 32bit AHB/AXI access. When the datapath is configured at 64bit or 128bit, the receive descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is written to using a single 64bit AHB/AXI access. For 32bit datapaths, the receive descriptors should be aligned at 32-bit boundaries and are written to using two individual non sequential 32-bit accesses."		
Offset	0x0490	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:2	dma_rx_q_ptr	"Receive buffer queue base address - written with the address of the start of the receive queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_rx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while receive is not enabled."	RW	0

Table 11-258: receive_q6_ptr

Register Information			
Description	"This register holds the start address of the receive buffer queue (receive buffers descriptor list). The receive buffer queue base address must be initialized before receive is enabled through bit 2 of the network control register. Once reception is enabled, any write to the receive buffer queue base address register is ignored. Reading this register returns the location of the descriptor currently being accessed. This value increments as buffers are used. Software should not use this register for determining where to remove received frames from the queue as it constantly changes as new frames are received. Software should instead work its way through the buffer descriptor queue checking the used bits. In terms of AMBA (AHB/AXI) operation, the receive descriptors are read from memory using a single 32bit AHB/AXI access. When the datapath is configured at 64bit or 128bit, the receive descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is written to using a single 64bit AHB/AXI access. For 32bit datapaths, the receive descriptors should be aligned at 32-bit boundaries and are written to using two individual non sequential 32-bit accesses."		
Offset	0x0494	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:2	dma_rx_q_ptr	"Receive buffer queue base address - written with the address of the start of the receive queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_rx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while receive is not enabled."	RW	0

Table 11-259: receive_q7_ptr

Register Information			
Description	"This register holds the start address of the receive buffer queue (receive buffers descriptor list). The receive buffer queue base address must be initialized before receive is enabled through bit 2 of the network control register. Once reception is enabled, any write to the receive buffer queue base address register is ignored. Reading this register returns the location of the descriptor currently being accessed. This value increments as buffers are used. Software should not use this register for determining where to remove received frames from the queue as it constantly changes as new frames are received. Software should instead work its way through the buffer descriptor queue checking the used bits. In terms of AMBA (AHB/AXI) operation, the receive descriptors are read from memory using a single 32bit AHB/AXI access. When the datapath is configured at 64bit or 128bit, the receive descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is written to using a single 64bit AHB/AXI access. For 32bit datapaths, the receive descriptors should be aligned at 32-bit boundaries and are written to using two individual non sequential 32-bit accesses."		
Offset	0x0498	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:2	dma_rx_q_ptr	"Receive buffer queue base address - written with the address of the start of the receive queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_rx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while receive is not enabled."	RW	0

Table 11-260: dma_rxbuf_size_q1

Register Information			
Description	"Receive Buffer Size"		
Offset	0x04A0	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:8	reserved_31_8	"Reserved, read as 0, ignored on write."	RO	0x00 0000
7:0	dma_rx_q_buf_size	"DMA receive buffer size in system memory. The value defined by these bits determines the size of buffer to use in main system memory when writing received data. The value is defined in multiples of 64 bytes. 0x01 corresponds to buffers of 64 bytes. 0x02 corresponds to 128 bytes etc. For example: 0x02: 128 byte 0x18: 1536 byte (1*max length frame/buffer) 0xA0: 10240 byte (1*10K jumbo frame/buffer) 0xFF: 16320 byte Note that this value should never be written as zero. Note. The reset value of this field is equal to the gem_rx_buffer_length_def define, which is user configurable."	RW	0x02

Table 11-261: dma_rxbuf_size_q2

Register Information					
Description	"Receive Buffer Size"				
Offset	0x04A4		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:8	reserved_31_8	"Reserved, read as 0, ignored on write."		RO	0x00 0000
7:0	dma_rx_q_buf_size	"DMA receive buffer size in system memory. The value defined by these bits determines the size of buffer to use in main system memory when writing received data. The value is defined in multiples of 64 bytes. 0x01 corresponds to buffers of 64 bytes. 0x02 corresponds to 128 bytes etc. For example: 0x02: 128 byte 0x18: 1536 byte (1*max length frame/buffer) 0xA0: 10240 byte (1*10K jumbo frame/buffer) 0xFF: 16320 byte Note that this value should never be written as zero. Note. The reset value of this field is equal to the gem_rx_buffer_length_def define, which is user configurable."		RW	0x02

Table 11-262: dma_rxbuf_size_q3

Register Information					
Description	"Receive Buffer Size"				
Offset	0x04A8		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:8	reserved_31_8	"Reserved, read as 0, ignored on write."		RO	0x00 0000
7:0	dma_rx_q_buf_size	"DMA receive buffer size in system memory. The value defined by these bits determines the size of buffer to use in main system memory when writing received data. The value is defined in multiples of 64 bytes. 0x01 corresponds to buffers of 64 bytes. 0x02 corresponds to 128 bytes etc. For example: 0x02: 128 byte 0x18: 1536 byte (1*max length frame/buffer) 0xA0: 10240 byte (1*10K jumbo frame/buffer) 0xFF: 16320 byte Note that this value should never be written as zero. Note. The reset value of this field is equal to the gem_rx_buffer_length_def define, which is user configurable."		RW	0x02

Table 11-263: dma_rxbuf_size_q4

Register Information				
Description	"Receive Buffer Size"			
Offset	0x04AC	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:8	reserved_31_8	"Reserved, read as 0, ignored on write."	RO	0x00 0000
7:0	dma_rx_q_buf_size	"DMA receive buffer size in system memory. The value defined by these bits determines the size of buffer to use in main system memory when writing received data. The value is defined in multiples of 64 bytes. 0x01 corresponds to buffers of 64 bytes. 0x02 corresponds to 128 bytes etc. For example: 0x02: 128 byte 0x18: 1536 byte (1*max length frame/buffer) 0xA0: 10240 byte (1*10K jumbo frame/buffer) 0xFF: 16320 byte Note that this value should never be written as zero. Note. The reset value of this field is equal to the gem_rx_buffer_length_def define, which is user configurable."	RW	0x02

Table 11-264: dma_rxbuf_size_q5

Register Information				
Description	"Receive Buffer Size"			
Offset	0x04B0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:8	reserved_31_8	"Reserved, read as 0, ignored on write."	RO	0x00 0000
7:0	dma_rx_q_buf_size	"DMA receive buffer size in system memory. The value defined by these bits determines the size of buffer to use in main system memory when writing received data. The value is defined in multiples of 64 bytes. 0x01 corresponds to buffers of 64 bytes. 0x02 corresponds to 128 bytes etc. For example: 0x02: 128 byte 0x18: 1536 byte (1*max length frame/buffer) 0xA0: 10240 byte (1*10K jumbo frame/buffer) 0xFF: 16320 byte Note that this value should never be written as zero. Note. The reset value of this field is equal to the gem_rx_buffer_length_def define, which is user configurable."	RW	0x02

Table 11-265: dma_rxbuf_size_q6

Register Information			
Description	"Receive Buffer Size"		
Offset	0x04B4	**Type:**	RW

Bits	Name	Description	Access	Reset
31:8	reserved_31_8	"Reserved, read as 0, ignored on write."	RO	0x00 0000
7:0	dma_rx_q_buf_size	"DMA receive buffer size in system memory. The value defined by these bits determines the size of buffer to use in main system memory when writing received data. The value is defined in multiples of 64 bytes. 0x01 corresponds to buffers of 64 bytes. 0x02 corresponds to 128 bytes etc. For example: 0x02: 128 byte 0x18: 1536 byte (1*max length frame/buffer) 0xA0: 10240 byte (1*10K jumbo frame/buffer) 0xFF: 16320 byte Note that this value should never be written as zero. Note. The reset value of this field is equal to the gem_rx_buffer_length_def define, which is user configurable."	RW	0x02

Table 11-266: dma_rxbuf_size_q7

Register Information			
Description	"Receive Buffer Size"		
Offset	0x04B8	**Type:**	RW

Bits	Name	Description	Access	Reset
31:8	reserved_31_8	"Reserved, read as 0, ignored on write."	RO	0x00 0000
7:0	dma_rx_q_buf_size	"DMA receive buffer size in system memory. The value defined by these bits determines the size of buffer to use in main system memory when writing received data. The value is defined in multiples of 64 bytes. 0x01 corresponds to buffers of 64 bytes. 0x02 corresponds to 128 bytes etc. For example: 0x02: 128 byte 0x18: 1536 byte (1*max length frame/buffer) 0xA0: 10240 byte (1*10K jumbo frame/buffer) 0xFF: 16320 byte Note that this value should never be written as zero. Note. The reset value of this field is equal to the gem_rx_buffer_length_def define, which is user configurable."	RW	0x02

Table 11-267: cbs_control

Register Information					
Description	"The IdleSlope value is defined as the rate of change of credit when a packet is waiting to be sent. This must not exceed the portTransmitRate which is dependent on the speed of operation, eg, portTransmitRate: 1Gb/s = 32'h07735940 (125 Mbytes/s), 100Mb/sec = 32'h017D7840 (25 Mnibbles/s), 10Mb/sec = 32'h002625A0 (2.5 Mnibbles/s). If 50% of bandwidth was to be allocated to a particular queue in 1Gb/sec mode then the IdleSlope value for that queue would be calculated as 32'h07735940/2. Note: Credit-Based Shaping should be disabled prior to updating the IdleSlope values. As another example, for a 1722 audio packet with a payload of 6 samples per channel, the packet size would be: 7 (preamble) + 1 (SFD) + 50 (packet header) + 6x4x2(payload) + 4 (CRC) = 110 bytes. For a rate of 8000 packets per second, the desired rate would 110 x 8000 bytes per second, so the programmed idleSlope value would be 880000 for a 1Gb/s connection, or 1760000 for a 100Mb/s or 10Mbs connection. See Figure 6.3 in the IEEE 1722 standard. In practice, the actual transmission rate will be vary slightly from that calculated. In this case, the idleSlope value should be recalibrated based on the variance between the measured and expected rate, and in this case very accurate transmission rates can be achieved. (The idleslope value is scaled by 2.5 in 2.5G operation and so the port transmit rate is the same for 1G and 2.5G.)"				
Offset	0x04BC		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:2	reserved_31_2	"Reserved, read as 0, ignored on write."		RO	0x0000 0000
1	cbs_enable_queue_b	"Enable Credit-Based shaping on the 2nd highest priority queue (queue B). Write 1 to enable"		RW	0
0	cbs_enable_queue_a	"Enable Credit-Based Shaping on the highest priority queue (queue A). Write 1 to enable"		RW	0

Table 11-268: cbs_idleslope_q_a

Register Information					
Description	"Queue A is the highest priority queue. This is the highest indexed active queue, e.g. For a system with Q0 to Q7, this would be Q7 if all queues were active."				
Offset	0x04C0		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	idleslope_a	"IdleSlope value for queue A in bytes/sec for gigabit operation and nibbles/sec for 10/100 operation"		RW	0x0000 0000

Table 11-269: cbs_idleslope_q_b

Register Information					
Description	"Queue B is the 2nd highest priority queue. This is the second highest indexed active queue, e.g. For a system with Q0 to Q7, this would be Q6 if all queues were active."				
Offset	0x04C4		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	idleslope_b	"IdleSlope value for queue B in bytes/sec for gigabit operation and nibbles/sec for 10/100 operation"		RW	0x0000 0000

Table 11-270: upper_tx_q_base_addr

Register Information					
Description	"Upper 32 bits of transmit buffer descriptor queue base address."				
Offset	0x04C8		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:0	upper_tx_q_base_addr	"Upper 32 bits of transmit buffer descriptor queue base address. Used when 64 bit addressing is enabled. (In releases earlier to 1p06f2 this register also affected the receive descriptor queue.)"	RW	0x0000 0000	

Table 11-271: tx_bd_control

Register Information					
Description	"Transmit buffer descriptor control register - this register determines which transmit frames, with time stamps reported in the buffer descriptor status field in extended buffer descriptor mode, set bit 23 in transmit buffer descriptor word 1."				
Offset	0x04CC		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:6	reserved_31_6	"Reserved, read as 0, ignored on write."	RO	0x000 0000	
5:4	tx_bd_ts_mode	"Transmit Descriptor Timestamp Insertion mode - bit 23 in transmit buffer descriptor word 1 when extended buffer descriptor mode is enabled, 00: Bit 23 always zero, 01: Bit 23 high for PTP Event Frames only, 10: Bit 23 high for all PTP Frames only, 11: Bit 23 always high"	RW	0x0	
3:0	reserved_3_0	"Reserved, read as 0, ignored on write."	RO	0x0	

Table 11-272: rx_bd_control

Register Information					
Description	"Receive buffer descriptor control register - this register determines which receive frames have time stamps reported in the buffer descriptor status field"				
Offset	0x04D0		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:6	reserved_31_6	"Reserved, read as 0, ignored on write."	RO	0x000 0000	
5:4	rx_bd_ts_mode	"Receive Descriptor Timestamp Insertion mode, 00: TS insertion disable, 01: TS inserted for PTP Event Frames only, 10: TS inserted for All PTP Frames only, 11: TS insertion for All Frames"	RW	0x0	
3:0	reserved_3_0	"Reserved, read as 0, ignored on write."	RO	0x0	

Table 11-273: upper_rx_q_base_addr

Register Information				
Description	"Upper 32 bits of receive buffer descriptor queue base address."			
Offset	0x04D4	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	upper_rx_q_base_addr	"Upper 32 bits of receive buffer descriptor queue base address. Used when 64 bit addressing is enabled."	RW	0x0000 0000

Table 11-274: wd_counter

Register Information				
Description	"Hidden control register - do not alter"			
Offset	0x04EC	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:4	reserved_31_4	"Reserved, read as 0, ignored on write."	RO	0x000 0000
3:0	rx_bd_reread_timer	"Controls counter used to retry receive descriptor reads (retries may occur if there was a resource error and there are frames waiting in the internal SRAM buffer to be off-loaded to host memory)"	RW	0x7

Table 11-275: axi_tx_full_thresh0

Register Information				
Description	"Hidden control register - do not alter"			
Offset	0x04F8	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	reserved_31_24	"Reserved, read as 0, ignored on write."	RO	0x00
23:16	axi_tx_full_adj_0	"AXI single port SRAM fine tuning"	RW	0x06
15:8	reserved_15_8	"Reserved, read as 0, ignored on write."	RO	0x00
7:0	axi_tx_full_adj_1	"AXI single port SRAM fine tuning"	RW	0x08

Table 11-276: axi_tx_full_thresh1

Register Information				
Description	"Hidden control register - do not alter"			
Offset	0x04FC	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	reserved_31_24	"Reserved, read as 0, ignored on write."	RO	0x00
23:16	axi_tx_full_adj_2	"AXI single port SRAM fine tuning"	RW	0x00
15:8	reserved_15_8	"Reserved, read as 0, ignored on write."	RO	0x00
7:0	axi_tx_full_adj_3	"AXI single port SRAM fine tuning"	RW	0x00

Table 11-277: screening_type_1_register_0

Register Information				
Description	"Screening type 1 registers are used to allocate up to 16 priority queues to received frames based on certain IP or UDP fields of incoming frames. Firstly, when DS/TC match enable is set (bit 28), the DS (Differentiated Services) field of the received IPv4 header or TC field (traffic class) of IPv6 headers are matched against bits 11:4. Secondly, when UDP port match enable is set (bit 29), the UDP Destination Port of the received UDP frame is matched against bits 27:12. Both UDP and DS/TC matching can be enabled simultaneously or individually. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 1 screening registers is configured in the gem defines file. Up to 16 type 1 screening registers have been allocated APB address space between 0x500 and 0x53C. The bit mappings for these registers are as follows:"			
Offset	0x0500	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	reserved_31	"Reserved, read as 0, ignored on write."	RO	0
30	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
29	udp_port_match_enable	"UDP port match enable"	RW	0
28	dstc_enable	"DS/TC Enable"	RW	0
27:12	udp_port_match	"UDP Port Match"	RW	0x0000
11:4	dstc_match	"DS/TC Match"	RW	0x00
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-278: screening_type_1_register_1

Register Information			
Description	"Screening type 1 registers are used to allocate up to 16 priority queues to received frames based on certain IP or UDP fields of incoming frames. Firstly, when DS/TC match enable is set (bit 28), the DS (Differentiated Services) field of the received IPv4 header or TC field (traffic class) of IPv6 headers are matched against bits 11:4. Secondly, when UDP port match enable is set (bit 29), the UDP Destination Port of the received UDP frame is matched against bits 27:12. Both UDP and DS/TC matching can be enabled simultaneously or individually. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 1 screening registers is configured in the gem defines file. Up to 16 type 1 screening registers have been allocated APB address space between 0x500 and 0x53C. The bit mappings for these registers are as follows:"		
Offset	0x0504	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	reserved_31	"Reserved, read as 0, ignored on write."	RO	0
30	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
29	udp_port_match_enable	"UDP port match enable"	RW	0
28	dstc_enable	"DS/TC Enable"	RW	0
27:12	udp_port_match	"UDP Port Match"	RW	0x0000
11:4	dstc_match	"DS/TC Match"	RW	0x00
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-279: screening_type_1_register_2

Register Information			
Description	"Screening type 1 registers are used to allocate up to 16 priority queues to received frames based on certain IP or UDP fields of incoming frames. Firstly, when DS/TC match enable is set (bit 28), the DS (Differentiated Services) field of the received IPv4 header or TC field (traffic class) of IPv6 headers are matched against bits 11:4. Secondly, when UDP port match enable is set (bit 29), the UDP Destination Port of the received UDP frame is matched against bits 27:12. Both UDP and DS/TC matching can be enabled simultaneously or individually. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 1 screening registers is configured in the gem defines file. Up to 16 type 1 screening registers have been allocated APB address space between 0x500 and 0x53C. The bit mappings for these registers are as follows:"		
Offset	0x0508	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	reserved_31	"Reserved, read as 0, ignored on write."	RO	0
30	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
29	udp_port_match_enable	"UDP port match enable"	RW	0
28	dstc_enable	"DS/TC Enable"	RW	0
27:12	udp_port_match	"UDP Port Match"	RW	0x0000
11:4	dstc_match	"DS/TC Match"	RW	0x00
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-280: screening_type_1_register_3

Register Information			
Description	"Screening type 1 registers are used to allocate up to 16 priority queues to received frames based on certain IP or UDP fields of incoming frames. Firstly, when DS/TC match enable is set (bit 28), the DS (Differentiated Services) field of the received IPv4 header or TC field (traffic class) of IPv6 headers are matched against bits 11:4. Secondly, when UDP port match enable is set (bit 29), the UDP Destination Port of the received UDP frame is matched against bits 27:12. Both UDP and DS/TC matching can be enabled simultaneously or individually. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 1 screening registers is configured in the gem defines file. Up to 16 type 1 screening registers have been allocated APB address space between 0x500 and 0x53C. The bit mappings for these registers are as follows:"		
Offset	0x050C	**Type:**	RW

Bits	Name	Description	Access	Reset
31	reserved_31	"Reserved, read as 0, ignored on write."	RO	0
30	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
29	udp_port_match_enable	"UDP port match enable"	RW	0
28	dstc_enable	"DS/TC Enable"	RW	0
27:12	udp_port_match	"UDP Port Match"	RW	0x0000
11:4	dstc_match	"DS/TC Match"	RW	0x00
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-281: screening_type_1_register_4

Register Information			
Description	"Screening type 1 registers are used to allocate up to 16 priority queues to received frames based on certain IP or UDP fields of incoming frames. Firstly, when DS/TC match enable is set (bit 28), the DS (Differentiated Services) field of the received IPv4 header or TC field (traffic class) of IPv6 headers are matched against bits 11:4. Secondly, when UDP port match enable is set (bit 29), the UDP Destination Port of the received UDP frame is matched against bits 27:12. Both UDP and DS/TC matching can be enabled simultaneously or individually. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 1 screening registers is configured in the gem defines file. Up to 16 type 1 screening registers have been allocated APB address space between 0x500 and 0x53C. The bit mappings for these registers are as follows:"		
Offset	0x0510	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	reserved_31	"Reserved, read as 0, ignored on write."	RO	0
30	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
29	udp_port_match_enable	"UDP port match enable"	RW	0
28	dstc_enable	"DS/TC Enable"	RW	0
27:12	udp_port_match	"UDP Port Match"	RW	0x0000
11:4	dstc_match	"DS/TC Match"	RW	0x00
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-282: screening_type_1_register_5

Register Information			
Description	"Screening type 1 registers are used to allocate up to 16 priority queues to received frames based on certain IP or UDP fields of incoming frames. Firstly, when DS/TC match enable is set (bit 28), the DS (Differentiated Services) field of the received IPv4 header or TC field (traffic class) of IPv6 headers are matched against bits 11:4. Secondly, when UDP port match enable is set (bit 29), the UDP Destination Port of the received UDP frame is matched against bits 27:12. Both UDP and DS/TC matching can be enabled simultaneously or individually. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 1 screening registers is configured in the gem defines file. Up to 16 type 1 screening registers have been allocated APB address space between 0x500 and 0x53C. The bit mappings for these registers are as follows:"		
Offset	0x0514	**Type:**	RW

Bitfield Details

Bits	Name	Description	Access	Reset
31	reserved_31	"Reserved, read as 0, ignored on write."	RO	0
30	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
29	udp_port_match_enable	"UDP port match enable"	RW	0
28	dstc_enable	"DS/TC Enable"	RW	0
27:12	udp_port_match	"UDP Port Match"	RW	0x0000
11:4	dstc_match	"DS/TC Match"	RW	0x00
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-283: screening_type_1_register_6

Register Information			
Description	"Screening type 1 registers are used to allocate up to 16 priority queues to received frames based on certain IP or UDP fields of incoming frames. Firstly, when DS/TC match enable is set (bit 28), the DS (Differentiated Services) field of the received IPv4 header or TC field (traffic class) of IPv6 headers are matched against bits 11:4. Secondly, when UDP port match enable is set (bit 29), the UDP Destination Port of the received UDP frame is matched against bits 27:12. Both UDP and DS/TC matching can be enabled simultaneously or individually. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 1 screening registers is configured in the gem defines file. Up to 16 type 1 screening registers have been allocated APB address space between 0x500 and 0x53C. The bit mappings for these registers are as follows:"		
Offset	0x0518	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	reserved_31	"Reserved, read as 0, ignored on write."	RO	0
30	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
29	udp_port_match_enable	"UDP port match enable"	RW	0
28	dstc_enable	"DS/TC Enable"	RW	0
27:12	udp_port_match	"UDP Port Match"	RW	0x0000
11:4	dstc_match	"DS/TC Match"	RW	0x00
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-284: screening_type_1_register_7

Register Information			
Description	"Screening type 1 registers are used to allocate up to 16 priority queues to received frames based on certain IP or UDP fields of incoming frames. Firstly, when DS/TC match enable is set (bit 28), the DS (Differentiated Services) field of the received IPv4 header or TC field (traffic class) of IPv6 headers are matched against bits 11:4. Secondly, when UDP port match enable is set (bit 29), the UDP Destination Port of the received UDP frame is matched against bits 27:12. Both UDP and DS/TC matching can be enabled simultaneously or individually. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 1 screening registers is configured in the gem defines file. Up to 16 type 1 screening registers have been allocated APB address space between 0x500 and 0x53C. The bit mappings for these registers are as follows:"		
Offset	0x051C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	reserved_31	"Reserved, read as 0, ignored on write."	RO	0
30	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
29	udp_port_match_enable	"UDP port match enable"	RW	0
28	dstc_enable	"DS/TC Enable"	RW	0
27:12	udp_port_match	"UDP Port Match"	RW	0x0000
11:4	dstc_match	"DS/TC Match"	RW	0x00
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-285: screening_type_1_register_8

Register Information			
Description	"Screening type 1 registers are used to allocate up to 16 priority queues to received frames based on certain IP or UDP fields of incoming frames. Firstly, when DS/TC match enable is set (bit 28), the DS (Differentiated Services) field of the received IPv4 header or TC field (traffic class) of IPv6 headers are matched against bits 11:4. Secondly, when UDP port match enable is set (bit 29), the UDP Destination Port of the received UDP frame is matched against bits 27:12. Both UDP and DS/TC matching can be enabled simultaneously or individually. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 1 screening registers is configured in the gem defines file. Up to 16 type 1 screening registers have been allocated APB address space between 0x500 and 0x53C. The bit mappings for these registers are as follows:"		
Offset	0x0520	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	reserved_31	"Reserved, read as 0, ignored on write."	RO	0
30	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
29	udp_port_match_enable	"UDP port match enable"	RW	0
28	dstc_enable	"DS/TC Enable"	RW	0
27:12	udp_port_match	"UDP Port Match"	RW	0x0000
11:4	dstc_match	"DS/TC Match"	RW	0x00
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-286: screening_type_1_register_9

Register Information			
Description	"Screening type 1 registers are used to allocate up to 16 priority queues to received frames based on certain IP or UDP fields of incoming frames. Firstly, when DS/TC match enable is set (bit 28), the DS (Differentiated Services) field of the received IPv4 header or TC field (traffic class) of IPv6 headers are matched against bits 11:4. Secondly, when UDP port match enable is set (bit 29), the UDP Destination Port of the received UDP frame is matched against bits 27:12. Both UDP and DS/TC matching can be enabled simultaneously or individually. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 1 screening registers is configured in the gem defines file. Up to 16 type 1 screening registers have been allocated APB address space between 0x500 and 0x53C. The bit mappings for these registers are as follows:"		
Offset	0x0524	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	reserved_31	"Reserved, read as 0, ignored on write."	RO	0
30	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
29	udp_port_match_enable	"UDP port match enable"	RW	0
28	dstc_enable	"DS/TC Enable"	RW	0
27:12	udp_port_match	"UDP Port Match"	RW	0x0000
11:4	dstc_match	"DS/TC Match"	RW	0x00
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-287: screening_type_1_register_10

Register Information			
Description	"Screening type 1 registers are used to allocate up to 16 priority queues to received frames based on certain IP or UDP fields of incoming frames. Firstly, when DS/TC match enable is set (bit 28), the DS (Differentiated Services) field of the received IPv4 header or TC field (traffic class) of IPv6 headers are matched against bits 11:4. Secondly, when UDP port match enable is set (bit 29), the UDP Destination Port of the received UDP frame is matched against bits 27:12. Both UDP and DS/TC matching can be enabled simultaneously or individually. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 1 screening registers is configured in the gem defines file. Up to 16 type 1 screening registers have been allocated APB address space between 0x500 and 0x53C. The bit mappings for these registers are as follows:"		
Offset	0x0528	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	reserved_31	"Reserved, read as 0, ignored on write."	RO	0
30	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
29	udp_port_match_enable	"UDP port match enable"	RW	0
28	dstc_enable	"DS/TC Enable"	RW	0
27:12	udp_port_match	"UDP Port Match"	RW	0x0000
11:4	dstc_match	"DS/TC Match"	RW	0x00
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-288: screening_type_1_register_11

Register Information			
Description	"Screening type 1 registers are used to allocate up to 16 priority queues to received frames based on certain IP or UDP fields of incoming frames. Firstly, when DS/TC match enable is set (bit 28), the DS (Differentiated Services) field of the received IPv4 header or TC field (traffic class) of IPv6 headers are matched against bits 11:4. Secondly, when UDP port match enable is set (bit 29), the UDP Destination Port of the received UDP frame is matched against bits 27:12. Both UDP and DS/TC matching can be enabled simultaneously or individually. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 1 screening registers is configured in the gem defines file. Up to 16 type 1 screening registers have been allocated APB address space between 0x500 and 0x53C. The bit mappings for these registers are as follows:"		
Offset	0x052C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	reserved_31	"Reserved, read as 0, ignored on write."	RO	0
30	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
29	udp_port_match_enable	"UDP port match enable"	RW	0
28	dstc_enable	"DS/TC Enable"	RW	0
27:12	udp_port_match	"UDP Port Match"	RW	0x0000
11:4	dstc_match	"DS/TC Match"	RW	0x00
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-289: screening_type_1_register_12

Register Information			
Description	"Screening type 1 registers are used to allocate up to 16 priority queues to received frames based on certain IP or UDP fields of incoming frames. Firstly, when DS/TC match enable is set (bit 28), the DS (Differentiated Services) field of the received IPv4 header or TC field (traffic class) of IPv6 headers are matched against bits 11:4. Secondly, when UDP port match enable is set (bit 29), the UDP Destination Port of the received UDP frame is matched against bits 27:12. Both UDP and DS/TC matching can be enabled simultaneously or individually. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 1 screening registers is configured in the gem defines file. Up to 16 type 1 screening registers have been allocated APB address space between 0x500 and 0x53C. The bit mappings for these registers are as follows:"		
Offset	0x0530	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	reserved_31	"Reserved, read as 0, ignored on write."	RO	0
30	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
29	udp_port_match_enable	"UDP port match enable"	RW	0
28	dstc_enable	"DS/TC Enable"	RW	0
27:12	udp_port_match	"UDP Port Match"	RW	0x0000
11:4	dstc_match	"DS/TC Match"	RW	0x00
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-290: screening_type_1_register_13

Register Information			
Description	"Screening type 1 registers are used to allocate up to 16 priority queues to received frames based on certain IP or UDP fields of incoming frames. Firstly, when DS/TC match enable is set (bit 28), the DS (Differentiated Services) field of the received IPv4 header or TC field (traffic class) of IPv6 headers are matched against bits 11:4. Secondly, when UDP port match enable is set (bit 29), the UDP Destination Port of the received UDP frame is matched against bits 27:12. Both UDP and DS/TC matching can be enabled simultaneously or individually. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 1 screening registers is configured in the gem defines file. Up to 16 type 1 screening registers have been allocated APB address space between 0x500 and 0x53C. The bit mappings for these registers are as follows:"		
Offset	0x0534	**Type:**	RW

Bits	Name	Description	Access	Reset
31	reserved_31	"Reserved, read as 0, ignored on write."	RO	0
30	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
29	udp_port_match_enable	"UDP port match enable"	RW	0
28	dstc_enable	"DS/TC Enable"	RW	0
27:12	udp_port_match	"UDP Port Match"	RW	0x0000
11:4	dstc_match	"DS/TC Match"	RW	0x00
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-291: screening_type_1_register_14

Register Information			
Description	"Screening type 1 registers are used to allocate up to 16 priority queues to received frames based on certain IP or UDP fields of incoming frames. Firstly, when DS/TC match enable is set (bit 28), the DS (Differentiated Services) field of the received IPv4 header or TC field (traffic class) of IPv6 headers are matched against bits 11:4. Secondly, when UDP port match enable is set (bit 29), the UDP Destination Port of the received UDP frame is matched against bits 27:12. Both UDP and DS/TC matching can be enabled simultaneously or individually. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 1 screening registers is configured in the gem defines file. Up to 16 type 1 screening registers have been allocated APB address space between 0x500 and 0x53C. The bit mappings for these registers are as follows:"		
Offset	0x0538	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	reserved_31	"Reserved, read as 0, ignored on write."	RO	0
30	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
29	udp_port_match_enable	"UDP port match enable"	RW	0
28	dstc_enable	"DS/TC Enable"	RW	0
27:12	udp_port_match	"UDP Port Match"	RW	0x0000
11:4	dstc_match	"DS/TC Match"	RW	0x00
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-292: screening_type_1_register_15

Register Information			
Description	"Screening type 1 registers are used to allocate up to 16 priority queues to received frames based on certain IP or UDP fields of incoming frames. Firstly, when DS/TC match enable is set (bit 28), the DS (Differentiated Services) field of the received IPv4 header or TC field (traffic class) of IPv6 headers are matched against bits 11:4. Secondly, when UDP port match enable is set (bit 29), the UDP Destination Port of the received UDP frame is matched against bits 27:12. Both UDP and DS/TC matching can be enabled simultaneously or individually. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 1 screening registers is configured in the gem defines file. Up to 16 type 1 screening registers have been allocated APB address space between 0x500 and 0x53C. The bit mappings for these registers are as follows:"		
Offset	0x053C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	reserved_31	"Reserved, read as 0, ignored on write."	RO	0
30	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
29	udp_port_match_enable	"UDP port match enable"	RW	0
28	dstc_enable	"DS/TC Enable"	RW	0
27:12	udp_port_match	"UDP Port Match"	RW	0x0000
11:4	dstc_match	"DS/TC Match"	RW	0x00
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-293: screening_type_2_register_0

Register Information			
Description	"Screener Type 2 match registers allow a screen to be configured that is the combination of all or any of the following comparisons: 1) An enabled VLAN Priority. A VLAN Priority match will be performed if the VLAN priority enable is set. The extracted priority field in the VLAN header is compared against 3 bits within the screener type 2 register itself. 2) An enabled EtherType. 3) An enabled Field Compare A. 4) An enabled Field Compare B. 5) An enabled Field Compare C. All enabled comparisons are ANDed together to form the overall type 2 screener match. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 2 screening registers is configured in the gem defines file. Up to 16 type 2 screening registers have been allocated APB address space between 0x540 and 0x57C. The bit mappings for these registers are as follows:"		
Offset	0x0540	**Type:**	RW

Bitfield Details

Bits	Name	Description	Access	Reset
31	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
30	compare_c_enable	"Compare C Enable"	RW	0
29:25	compare_c	"Compare C - Index to screener type 2 Compare register"	RW	0x00
24	compare_b_enable	"Compare B Enable"	RW	0
23:19	compare_b	"Compare B - Index to screener type 2 Compare register"	RW	0x00
18	compare_a_enable	"Compare A Enable"	RW	0
17:13	compare_a	"Compare A - Index to screener type 2 Compare register"	RW	0x00
12	ethertype_enable	"EtherType Enable"	RW	0
11:9	ethertype_reg_index	"Index to screener type 2 EtherType register"	RW	0x0
8	vlan_enable	"VLAN Enable"	RW	0
7	reserved_7	"Reserved and implemented as RW"	RW	0
6:4	vlan_priority	"VLAN Priority"	RW	0x0
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-294: screening_type_2_register_1

Register Information			
Description	"Screener Type 2 match registers allow a screen to be configured that is the combination of all or any of the following comparisons: 1) An enabled VLAN Priority. A VLAN Priority match will be performed if the VLAN priority enable is set. The extracted priority field in the VLAN header is compared against 3 bits within the screener type 2 register itself. 2) An enabled EtherType. 3) An enabled Field Compare A. 4) An enabled Field Compare B. 5) An enabled Field Compare C. All enabled comparisons are ANDed together to form the overall type 2 screener match. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 2 screening registers is configured in the gem defines file. Up to 16 type 2 screening registers have been allocated APB address space between 0x540 and 0x57C. The bit mappings for these registers are as follows:"		
Offset	0x0544	**Type:**	RW

Bitfield Details

Bits	Name	Description	Access	Reset
31	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
30	compare_c_enable	"Compare C Enable"	RW	0
29:25	compare_c	"Compare C - Index to screener type 2 Compare register"	RW	0x00
24	compare_b_enable	"Compare B Enable"	RW	0
23:19	compare_b	"Compare B - Index to screener type 2 Compare register"	RW	0x00
18	compare_a_enable	"Compare A Enable"	RW	0
17:13	compare_a	"Compare A - Index to screener type 2 Compare register"	RW	0x00
12	ethertype_enable	"EtherType Enable"	RW	0
11:9	ethertype_reg_index	"Index to screener type 2 EtherType register"	RW	0x0
8	vlan_enable	"VLAN Enable"	RW	0
7	reserved_7	"Reserved and implemented as RW"	RW	0
6:4	vlan_priority	"VLAN Priority"	RW	0x0
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-295: screening_type_2_register_2

Register Information			
Description	"Screener Type 2 match registers allow a screen to be configured that is the combination of all or any of the following comparisons: 1) An enabled VLAN Priority. A VLAN Priority match will be performed if the VLAN priority enable is set. The extracted priority field in the VLAN header is compared against 3 bits within the screener type 2 register itself. 2) An enabled EtherType. 3) An enabled Field Compare A. 4) An enabled Field Compare B. 5) An enabled Field Compare C. All enabled comparisons are ANDed together to form the overall type 2 screener match. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 2 screening registers is configured in the gem defines file. Up to 16 type 2 screening registers have been allocated APB address space between 0x540 and 0x57C. The bit mappings for these registers are as follows:"		
Offset	0x0548	**Type:**	RW

Bitfield Details

Bits	Name	Description	Access	Reset
31	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
30	compare_c_enable	"Compare C Enable"	RW	0
29:25	compare_c	"Compare C - Index to screener type 2 Compare register"	RW	0x00
24	compare_b_enable	"Compare B Enable"	RW	0
23:19	compare_b	"Compare B - Index to screener type 2 Compare register"	RW	0x00
18	compare_a_enable	"Compare A Enable"	RW	0
17:13	compare_a	"Compare A - Index to screener type 2 Compare register"	RW	0x00
12	ethertype_enable	"EtherType Enable"	RW	0
11:9	ethertype_reg_index	"Index to screener type 2 EtherType register"	RW	0x0
8	vlan_enable	"VLAN Enable"	RW	0
7	reserved_7	"Reserved and implemented as RW"	RW	0
6:4	vlan_priority	"VLAN Priority"	RW	0x0
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-296: screening_type_2_register_3

Register Information			
Description	"Screener Type 2 match registers allow a screen to be configured that is the combination of all or any of the following comparisons: 1) An enabled VLAN Priority. A VLAN Priority match will be performed if the VLAN priority enable is set. The extracted priority field in the VLAN header is compared against 3 bits within the screener type 2 register itself. 2) An enabled EtherType. 3) An enabled Field Compare A. 4) An enabled Field Compare B. 5) An enabled Field Compare C. All enabled comparisons are ANDed together to form the overall type 2 screener match. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 2 screening registers is configured in the gem defines file. Up to 16 type 2 screening registers have been allocated APB address space between 0x540 and 0x57C. The bit mappings for these registers are as follows:"		
Offset	0x054C	**Type:**	RW

Bitfield Details

Bits	Name	Description	Access	Reset
31	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
30	compare_c_enable	"Compare C Enable"	RW	0
29:25	compare_c	"Compare C - Index to screener type 2 Compare register"	RW	0x00
24	compare_b_enable	"Compare B Enable"	RW	0
23:19	compare_b	"Compare B - Index to screener type 2 Compare register"	RW	0x00
18	compare_a_enable	"Compare A Enable"	RW	0
17:13	compare_a	"Compare A - Index to screener type 2 Compare register"	RW	0x00
12	ethertype_enable	"EtherType Enable"	RW	0
11:9	ethertype_reg_index	"Index to screener type 2 EtherType register"	RW	0x0
8	vlan_enable	"VLAN Enable"	RW	0
7	reserved_7	"Reserved and implemented as RW"	RW	0
6:4	vlan_priority	"VLAN Priority"	RW	0x0
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-297: screening_type_2_register_4

Register Information			
Description	"Screener Type 2 match registers allow a screen to be configured that is the combination of all or any of the following comparisons: 1) An enabled VLAN Priority. A VLAN Priority match will be performed if the VLAN priority enable is set. The extracted priority field in the VLAN header is compared against 3 bits within the screener type 2 register itself. 2) An enabled EtherType. 3) An enabled Field Compare A. 4) An enabled Field Compare B. 5) An enabled Field Compare C. All enabled comparisons are ANDed together to form the overall type 2 screener match. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 2 screening registers is configured in the gem defines file. Up to 16 type 2 screening registers have been allocated APB address space between 0x540 and 0x57C. The bit mappings for these registers are as follows:"		
Offset	0x0550	**Type:**	RW

Bitfield Details

Bits	Name	Description	Access	Reset
31	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
30	compare_c_enable	"Compare C Enable"	RW	0
29:25	compare_c	"Compare C - Index to screener type 2 Compare register"	RW	0x00
24	compare_b_enable	"Compare B Enable"	RW	0
23:19	compare_b	"Compare B - Index to screener type 2 Compare register"	RW	0x00
18	compare_a_enable	"Compare A Enable"	RW	0
17:13	compare_a	"Compare A - Index to screener type 2 Compare register"	RW	0x00
12	ethertype_enable	"EtherType Enable"	RW	0
11:9	ethertype_reg_index	"Index to screener type 2 EtherType register"	RW	0x0
8	vlan_enable	"VLAN Enable"	RW	0
7	reserved_7	"Reserved and implemented as RW"	RW	0
6:4	vlan_priority	"VLAN Priority"	RW	0x0
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-298: screening_type_2_register_5

Register Information			
Description	"Screener Type 2 match registers allow a screen to be configured that is the combination of all or any of the following comparisons: 1) An enabled VLAN Priority. A VLAN Priority match will be performed if the VLAN priority enable is set. The extracted priority field in the VLAN header is compared against 3 bits within the screener type 2 register itself. 2) An enabled EtherType. 3) An enabled Field Compare A. 4) An enabled Field Compare B. 5) An enabled Field Compare C. All enabled comparisons are ANDed together to form the overall type 2 screener match. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 2 screening registers is configured in the gem defines file. Up to 16 type 2 screening registers have been allocated APB address space between 0x540 and 0x57C. The bit mappings for these registers are as follows:"		
Offset	0x0554	**Type:**	RW

Bitfield Details

Bits	Name	Description	Access	Reset
31	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
30	compare_c_enable	"Compare C Enable"	RW	0
29:25	compare_c	"Compare C - Index to screener type 2 Compare register"	RW	0x00
24	compare_b_enable	"Compare B Enable"	RW	0
23:19	compare_b	"Compare B - Index to screener type 2 Compare register"	RW	0x00
18	compare_a_enable	"Compare A Enable"	RW	0
17:13	compare_a	"Compare A - Index to screener type 2 Compare register"	RW	0x00
12	ethertype_enable	"EtherType Enable"	RW	0
11:9	ethertype_reg_index	"Index to screener type 2 EtherType register"	RW	0x0
8	vlan_enable	"VLAN Enable"	RW	0
7	reserved_7	"Reserved and implemented as RW"	RW	0
6:4	vlan_priority	"VLAN Priority"	RW	0x0
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-299: screening_type_2_register_6

Register Information			
Description	"Screener Type 2 match registers allow a screen to be configured that is the combination of all or any of the following comparisons: 1) An enabled VLAN Priority. A VLAN Priority match will be performed if the VLAN priority enable is set. The extracted priority field in the VLAN header is compared against 3 bits within the screener type 2 register itself. 2) An enabled EtherType. 3) An enabled Field Compare A. 4) An enabled Field Compare B. 5) An enabled Field Compare C. All enabled comparisons are ANDed together to form the overall type 2 screener match. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 2 screening registers is configured in the gem defines file. Up to 16 type 2 screening registers have been allocated APB address space between 0x540 and 0x57C. The bit mappings for these registers are as follows:"		
Offset	0x0558	**Type:**	RW

Bitfield Details

Bits	Name	Description	Access	Reset
31	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
30	compare_c_enable	"Compare C Enable"	RW	0
29:25	compare_c	"Compare C - Index to screener type 2 Compare register"	RW	0x00
24	compare_b_enable	"Compare B Enable"	RW	0
23:19	compare_b	"Compare B - Index to screener type 2 Compare register"	RW	0x00
18	compare_a_enable	"Compare A Enable"	RW	0
17:13	compare_a	"Compare A - Index to screener type 2 Compare register"	RW	0x00
12	ethertype_enable	"EtherType Enable"	RW	0
11:9	ethertype_reg_index	"Index to screener type 2 EtherType register"	RW	0x0
8	vlan_enable	"VLAN Enable"	RW	0
7	reserved_7	"Reserved and implemented as RW"	RW	0
6:4	vlan_priority	"VLAN Priority"	RW	0x0
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-300: screening_type_2_register_7

Register Information			
Description	"Screener Type 2 match registers allow a screen to be configured that is the combination of all or any of the following comparisons: 1) An enabled VLAN Priority. A VLAN Priority match will be performed if the VLAN priority enable is set. The extracted priority field in the VLAN header is compared against 3 bits within the screener type 2 register itself. 2) An enabled EtherType. 3) An enabled Field Compare A. 4) An enabled Field Compare B. 5) An enabled Field Compare C. All enabled comparisons are ANDed together to form the overall type 2 screener match. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 2 screening registers is configured in the gem defines file. Up to 16 type 2 screening registers have been allocated APB address space between 0x540 and 0x57C. The bit mappings for these registers are as follows:"		
Offset	0x055C	**Type:**	RW

Bitfield Details

Bits	Name	Description	Access	Reset
31	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
30	compare_c_enable	"Compare C Enable"	RW	0
29:25	compare_c	"Compare C - Index to screener type 2 Compare register"	RW	0x00
24	compare_b_enable	"Compare B Enable"	RW	0
23:19	compare_b	"Compare B - Index to screener type 2 Compare register"	RW	0x00
18	compare_a_enable	"Compare A Enable"	RW	0
17:13	compare_a	"Compare A - Index to screener type 2 Compare register"	RW	0x00
12	ethertype_enable	"EtherType Enable"	RW	0
11:9	ethertype_reg_index	"Index to screener type 2 EtherType register"	RW	0x0
8	vlan_enable	"VLAN Enable"	RW	0
7	reserved_7	"Reserved and implemented as RW"	RW	0
6:4	vlan_priority	"VLAN Priority"	RW	0x0
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-301: screening_type_2_register_8

Register Information			
Description	"Screener Type 2 match registers allow a screen to be configured that is the combination of all or any of the following comparisons: 1) An enabled VLAN Priority. A VLAN Priority match will be performed if the VLAN priority enable is set. The extracted priority field in the VLAN header is compared against 3 bits within the screener type 2 register itself. 2) An enabled EtherType. 3) An enabled Field Compare A. 4) An enabled Field Compare B. 5) An enabled Field Compare C. All enabled comparisons are ANDed together to form the overall type 2 screener match. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 2 screening registers is configured in the gem defines file. Up to 16 type 2 screening registers have been allocated APB address space between 0x540 and 0x57C. The bit mappings for these registers are as follows:"		
Offset	0x0560	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
30	compare_c_enable	"Compare C Enable"	RW	0
29:25	compare_c	"Compare C - Index to screener type 2 Compare register"	RW	0x00
24	compare_b_enable	"Compare B Enable"	RW	0
23:19	compare_b	"Compare B - Index to screener type 2 Compare register"	RW	0x00
18	compare_a_enable	"Compare A Enable"	RW	0
17:13	compare_a	"Compare A - Index to screener type 2 Compare register"	RW	0x00
12	ethertype_enable	"EtherType Enable"	RW	0
11:9	ethertype_reg_index	"Index to screener type 2 EtherType register"	RW	0x0
8	vlan_enable	"VLAN Enable"	RW	0
7	reserved_7	"Reserved and implemented as RW"	RW	0
6:4	vlan_priority	"VLAN Priority"	RW	0x0
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-302: screening_type_2_register_9

Register Information			
Description	"Screener Type 2 match registers allow a screen to be configured that is the combination of all or any of the following comparisons: 1) An enabled VLAN Priority. A VLAN Priority match will be performed if the VLAN priority enable is set. The extracted priority field in the VLAN header is compared against 3 bits within the screener type 2 register itself. 2) An enabled EtherType. 3) An enabled Field Compare A. 4) An enabled Field Compare B. 5) An enabled Field Compare C. All enabled comparisons are ANDed together to form the overall type 2 screener match. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 2 screening registers is configured in the gem defines file. Up to 16 type 2 screening registers have been allocated APB address space between 0x540 and 0x57C. The bit mappings for these registers are as follows:"		
Offset	0x0564	**Type:**	RW

Bitfield Details

Bits	Name	Description	Access	Reset
31	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
30	compare_c_enable	"Compare C Enable"	RW	0
29:25	compare_c	"Compare C - Index to screener type 2 Compare register"	RW	0x00
24	compare_b_enable	"Compare B Enable"	RW	0
23:19	compare_b	"Compare B - Index to screener type 2 Compare register"	RW	0x00
18	compare_a_enable	"Compare A Enable"	RW	0
17:13	compare_a	"Compare A - Index to screener type 2 Compare register"	RW	0x00
12	ethertype_enable	"EtherType Enable"	RW	0
11:9	ethertype_reg_index	"Index to screener type 2 EtherType register"	RW	0x0
8	vlan_enable	"VLAN Enable"	RW	0
7	reserved_7	"Reserved and implemented as RW"	RW	0
6:4	vlan_priority	"VLAN Priority"	RW	0x0
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-303: screening_type_2_register_10

Register Information			
Description	"Screener Type 2 match registers allow a screen to be configured that is the combination of all or any of the following comparisons: 1) An enabled VLAN Priority. A VLAN Priority match will be performed if the VLAN priority enable is set. The extracted priority field in the VLAN header is compared against 3 bits within the screener type 2 register itself. 2) An enabled EtherType. 3) An enabled Field Compare A. 4) An enabled Field Compare B. 5) An enabled Field Compare C. All enabled comparisons are ANDed together to form the overall type 2 screener match. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 2 screening registers is configured in the gem defines file. Up to 16 type 2 screening registers have been allocated APB address space between 0x540 and 0x57C. The bit mappings for these registers are as follows:"		
Offset	0x0568	**Type:**	RW

Bitfield Details

Bits	Name	Description	Access	Reset
31	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
30	compare_c_enable	"Compare C Enable"	RW	0
29:25	compare_c	"Compare C - Index to screener type 2 Compare register"	RW	0x00
24	compare_b_enable	"Compare B Enable"	RW	0
23:19	compare_b	"Compare B - Index to screener type 2 Compare register"	RW	0x00
18	compare_a_enable	"Compare A Enable"	RW	0
17:13	compare_a	"Compare A - Index to screener type 2 Compare register"	RW	0x00
12	ethertype_enable	"EtherType Enable"	RW	0
11:9	ethertype_reg_index	"Index to screener type 2 EtherType register"	RW	0x0
8	vlan_enable	"VLAN Enable"	RW	0
7	reserved_7	"Reserved and implemented as RW"	RW	0
6:4	vlan_priority	"VLAN Priority"	RW	0x0
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-304: screening_type_2_register_11

Register Information			
Description	"Screener Type 2 match registers allow a screen to be configured that is the combination of all or any of the following comparisons: 1) An enabled VLAN Priority. A VLAN Priority match will be performed if the VLAN priority enable is set. The extracted priority field in the VLAN header is compared against 3 bits within the screener type 2 register itself. 2) An enabled EtherType. 3) An enabled Field Compare A. 4) An enabled Field Compare B. 5) An enabled Field Compare C. All enabled comparisons are ANDed together to form the overall type 2 screener match. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 2 screening registers is configured in the gem defines file. Up to 16 type 2 screening registers have been allocated APB address space between 0x540 and 0x57C. The bit mappings for these registers are as follows:"		
Offset	0x056C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
30	compare_c_enable	"Compare C Enable"	RW	0
29:25	compare_c	"Compare C - Index to screener type 2 Compare register"	RW	0x00
24	compare_b_enable	"Compare B Enable"	RW	0
23:19	compare_b	"Compare B - Index to screener type 2 Compare register"	RW	0x00
18	compare_a_enable	"Compare A Enable"	RW	0
17:13	compare_a	"Compare A - Index to screener type 2 Compare register"	RW	0x00
12	ethertype_enable	"EtherType Enable"	RW	0
11:9	ethertype_reg_index	"Index to screener type 2 EtherType register"	RW	0x0
8	vlan_enable	"VLAN Enable"	RW	0
7	reserved_7	"Reserved and implemented as RW"	RW	0
6:4	vlan_priority	"VLAN Priority"	RW	0x0
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-305: screening_type_2_register_12

Register Information			
Description	"Screener Type 2 match registers allow a screen to be configured that is the combination of all or any of the following comparisons: 1) An enabled VLAN Priority. A VLAN Priority match will be performed if the VLAN priority enable is set. The extracted priority field in the VLAN header is compared against 3 bits within the screener type 2 register itself. 2) An enabled EtherType. 3) An enabled Field Compare A. 4) An enabled Field Compare B. 5) An enabled Field Compare C. All enabled comparisons are ANDed together to form the overall type 2 screener match. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 2 screening registers is configured in the gem defines file. Up to 16 type 2 screening registers have been allocated APB address space between 0x540 and 0x57C. The bit mappings for these registers are as follows:"		
Offset	0x0570	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
30	compare_c_enable	"Compare C Enable"	RW	0
29:25	compare_c	"Compare C - Index to screener type 2 Compare register"	RW	0x00
24	compare_b_enable	"Compare B Enable"	RW	0
23:19	compare_b	"Compare B - Index to screener type 2 Compare register"	RW	0x00
18	compare_a_enable	"Compare A Enable"	RW	0
17:13	compare_a	"Compare A - Index to screener type 2 Compare register"	RW	0x00
12	ethertype_enable	"EtherType Enable"	RW	0
11:9	ethertype_reg_index	"Index to screener type 2 EtherType register"	RW	0x0
8	vlan_enable	"VLAN Enable"	RW	0
7	reserved_7	"Reserved and implemented as RW"	RW	0
6:4	vlan_priority	"VLAN Priority"	RW	0x0
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-306: screening_type_2_register_13

Register Information			
Description	"Screener Type 2 match registers allow a screen to be configured that is the combination of all or any of the following comparisons: 1) An enabled VLAN Priority. A VLAN Priority match will be performed if the VLAN priority enable is set. The extracted priority field in the VLAN header is compared against 3 bits within the screener type 2 register itself. 2) An enabled EtherType. 3) An enabled Field Compare A. 4) An enabled Field Compare B. 5) An enabled Field Compare C. All enabled comparisons are ANDed together to form the overall type 2 screener match. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 2 screening registers is configured in the gem defines file. Up to 16 type 2 screening registers have been allocated APB address space between 0x540 and 0x57C. The bit mappings for these registers are as follows:"		
Offset	0x0574	**Type:**	RW

Bitfield Details

Bits	Name	Description	Access	Reset
31	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
30	compare_c_enable	"Compare C Enable"	RW	0
29:25	compare_c	"Compare C - Index to screener type 2 Compare register"	RW	0x00
24	compare_b_enable	"Compare B Enable"	RW	0
23:19	compare_b	"Compare B - Index to screener type 2 Compare register"	RW	0x00
18	compare_a_enable	"Compare A Enable"	RW	0
17:13	compare_a	"Compare A - Index to screener type 2 Compare register"	RW	0x00
12	ethertype_enable	"EtherType Enable"	RW	0
11:9	ethertype_reg_index	"Index to screener type 2 EtherType register"	RW	0x0
8	vlan_enable	"VLAN Enable"	RW	0
7	reserved_7	"Reserved and implemented as RW"	RW	0
6:4	vlan_priority	"VLAN Priority"	RW	0x0
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-307: screening_type_2_register_14

Register Information			
Description	"Screener Type 2 match registers allow a screen to be configured that is the combination of all or any of the following comparisons: 1) An enabled VLAN Priority. A VLAN Priority match will be performed if the VLAN priority enable is set. The extracted priority field in the VLAN header is compared against 3 bits within the screener type 2 register itself. 2) An enabled EtherType. 3) An enabled Field Compare A. 4) An enabled Field Compare B. 5) An enabled Field Compare C. All enabled comparisons are ANDed together to form the overall type 2 screener match. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 2 screening registers is configured in the gem defines file. Up to 16 type 2 screening registers have been allocated APB address space between 0x540 and 0x57C. The bit mappings for these registers are as follows:"		
Offset	0x0578	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
30	compare_c_enable	"Compare C Enable"	RW	0
29:25	compare_c	"Compare C - Index to screener type 2 Compare register"	RW	0x00
24	compare_b_enable	"Compare B Enable"	RW	0
23:19	compare_b	"Compare B - Index to screener type 2 Compare register"	RW	0x00
18	compare_a_enable	"Compare A Enable"	RW	0
17:13	compare_a	"Compare A - Index to screener type 2 Compare register"	RW	0x00
12	ethertype_enable	"EtherType Enable"	RW	0
11:9	ethertype_reg_index	"Index to screener type 2 EtherType register"	RW	0x0
8	vlan_enable	"VLAN Enable"	RW	0
7	reserved_7	"Reserved and implemented as RW"	RW	0
6:4	vlan_priority	"VLAN Priority"	RW	0x0
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-308: screening_type_2_register_15

Register Information			
Description	"Screener Type 2 match registers allow a screen to be configured that is the combination of all or any of the following comparisons: 1) An enabled VLAN Priority. A VLAN Priority match will be performed if the VLAN priority enable is set. The extracted priority field in the VLAN header is compared against 3 bits within the screener type 2 register itself. 2) An enabled EtherType. 3) An enabled Field Compare A. 4) An enabled Field Compare B. 5) An enabled Field Compare C. All enabled comparisons are ANDed together to form the overall type 2 screener match. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 2 screening registers is configured in the gem defines file. Up to 16 type 2 screening registers have been allocated APB address space between 0x540 and 0x57C. The bit mappings for these registers are as follows:"		
Offset	0x057C	**Type:**	RW

Bits	Name	Description	Access	Reset
31	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
30	compare_c_enable	"Compare C Enable"	RW	0
29:25	compare_c	"Compare C - Index to screener type 2 Compare register"	RW	0x00
24	compare_b_enable	"Compare B Enable"	RW	0
23:19	compare_b	"Compare B - Index to screener type 2 Compare register"	RW	0x00
18	compare_a_enable	"Compare A Enable"	RW	0
17:13	compare_a	"Compare A - Index to screener type 2 Compare register"	RW	0x00
12	ethertype_enable	"EtherType Enable"	RW	0
11:9	ethertype_reg_index	"Index to screener type 2 EtherType register"	RW	0x0
8	vlan_enable	"VLAN Enable"	RW	0
7	reserved_7	"Reserved and implemented as RW"	RW	0
6:4	vlan_priority	"VLAN Priority"	RW	0x0
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-309: tx_sched_ctrl

Register Information					
Description	"This register controls the transmit scheduling algorithm the user can select for each active transmit queue. By default all queues are initialized to fixed priority, with the top indexed queue having overall priority."				
Offset	0x0580		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:30	tx_sched_q15	"Queue 15 selection. 00 : Fixed Priority 01 : CBS Enabled only valid for top two enabled queues and if CBS capability selected. 10 : DWRR Enabled 11 : ETS Enabled"		RW	0x0
29:28	tx_sched_q14	"Queue 14 selection. 00 : Fixed Priority 01 : CBS Enabled only valid for top two enabled queues and if CBS capability selected. 10 : DWRR Enabled 11 : ETS Enabled"		RW	0x0
27:26	tx_sched_q13	"Queue 13 selection. 00 : Fixed Priority 01 : CBS Enabled only valid for top two enabled queues and if CBS capability selected. 10 : DWRR Enabled 11 : ETS Enabled"		RW	0x0
25:24	tx_sched_q12	"Queue 12 selection. 00 : Fixed Priority 01 : CBS Enabled only valid for top two enabled queues and if CBS capability selected. 10 : DWRR Enabled 11 : ETS Enabled"		RW	0x0
23:22	tx_sched_q11	"Queue 11 selection. 00 : Fixed Priority 01 : CBS Enabled only valid for top two enabled queues and if CBS capability selected. 10 : DWRR Enabled 11 : ETS Enabled"		RW	0x0
21:20	tx_sched_q10	"Queue 10 selection. 00 : Fixed Priority 01 : CBS Enabled only valid for top two enabled queues and if CBS capability selected. 10 : DWRR Enabled 11 : ETS Enabled"		RW	0x0
19:18	tx_sched_q9	"Queue 9 selection. 00 : Fixed Priority 01 : CBS Enabled only valid for top two enabled queues and if CBS capability selected. 10 : DWRR Enabled 11 : ETS Enabled"		RW	0x0
17:16	tx_sched_q8	"Queue 8 selection. 00 : Fixed Priority 01 : CBS Enabled only valid for top two enabled queues and if CBS capability selected. 10 : DWRR Enabled 11 : ETS Enabled"		RW	0x0

15:14	tx_sched_q7	"Queue 7 selection. 00 : Fixed Priority 01 : CBS Enabled only valid for top two enabled queues and if CBS capability selected. 10 : DWRR Enabled 11 : ETS Enabled"	RW	0x0
13:12	tx_sched_q6	"Queue 6 selection. 00 : Fixed Priority 01 : CBS Enabled only valid for top two enabled queues and if CBS capability selected. 10 : DWRR Enabled 11 : ETS Enabled"	RW	0x0
11:10	tx_sched_q5	"Queue 5 selection. 00 : Fixed Priority 01 : CBS Enabled only valid for top two enabled queues and if CBS capability selected. 10 : DWRR Enabled 11 : ETS Enabled"	RW	0x0
9:8	tx_sched_q4	"Queue 4 selection. 00 : Fixed Priority 01 : CBS Enabled only valid for top two enabled queues and if CBS capability selected. 10 : DWRR Enabled 11 : ETS Enabled"	RW	0x0
7:6	tx_sched_q3	"Queue 3 selection. 00 : Fixed Priority 01 : CBS Enabled only valid for top two enabled queues and if CBS capability selected. 10 : DWRR Enabled 11 : ETS Enabled"	RW	0x0
5:4	tx_sched_q2	"Queue 2 selection. 00 : Fixed Priority 01 : CBS Enabled only valid for top two enabled queues and if CBS capability selected. 10 : DWRR Enabled 11 : ETS Enabled"	RW	0x0
3:2	tx_sched_q1	"Queue 1 selection. 00 : Fixed Priority 01 : CBS Enabled only valid for top two enabled queues and if CBS capability selected. 10 : DWRR Enabled 11 : ETS Enabled"	RW	0x0
1:0	tx_sched_q0	"Queue 0 selection. 00 : Fixed Priority 01 : CBS Enabled only valid for top two enabled queues and if CBS capability selected. 10 : DWRR Enabled 11 : ETS Enabled"	RW	0x0

Table 11-310: bw_rate_limit_q0to3

Register Information				
Description	"This register holds the DWRR weighting value or the ETS bandwidth percentage value used by the transmit scheduler for queues 0 to 3."			
Offset	0x0590	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	dwrr_ets_weight_q3	"DWRR Weighting / ETS Bandwidth Allocation for queue 3"	RW	0x00
23:16	dwrr_ets_weight_q2	"DWRR Weighting / ETS Bandwidth Allocation for queue 2"	RW	0x00
15:8	dwrr_ets_weight_q1	"DWRR Weighting / ETS Bandwidth Allocation for queue 1"	RW	0x00
7:0	dwrr_ets_weight_q0	"DWRR Weighting / ETS Bandwidth Allocation for queue 0"	RW	0x00

Table 11-311: bw_rate_limit_q4to7

Register Information				
Description	"This register holds the DWRR weighting value or the ETS bandwidth percentage value used by the transmit scheduler for queues 4 to 7."			
Offset	0x0594	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	dwrr_ets_weight_q7	"DWRR Weighting / ETS Bandwidth Allocation for queue 7"	RW	0x00
23:16	dwrr_ets_weight_q6	"DWRR Weighting / ETS Bandwidth Allocation for queue 6"	RW	0x00
15:8	dwrr_ets_weight_q5	"DWRR Weighting / ETS Bandwidth Allocation for queue 5"	RW	0x00
7:0	dwrr_ets_weight_q4	"DWRR Weighting / ETS Bandwidth Allocation for queue 4"	RW	0x00

Table 11-312: bw_rate_limit_q8to11

Register Information				
Description	"This register holds the DWRR weighting value or the ETS bandwidth percentage value used by the transmit scheduler for queues 8 to 11."			
Offset	0x0598	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	dwrr_ets_weight_q 11	"DWRR Weighting / ETS Bandwidth Allocation for queue 11"	RW	0x00
23:16	dwrr_ets_weight_q 10	"DWRR Weighting / ETS Bandwidth Allocation for queue 10"	RW	0x00
15:8	dwrr_ets_weight_q 9	"DWRR Weighting / ETS Bandwidth Allocation for queue 9"	RW	0x00
7:0	dwrr_ets_weight_q 8	"DWRR Weighting / ETS Bandwidth Allocation for queue 8"	RW	0x00

Table 11-313: bw_rate_limit_q12to15

Register Information				
Description	"This register holds the DWRR weighting value or the ETS bandwidth percentage value used by the transmit scheduler for queues 12 to 15."			
Offset	0x059C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	dwrr_ets_weight_q 15	"DWRR Weighting / ETS Bandwidth Allocation for queue 15"	RW	0x00
23:16	dwrr_ets_weight_q 14	"DWRR Weighting / ETS Bandwidth Allocation for queue 14"	RW	0x00
15:8	dwrr_ets_weight_q 13	"DWRR Weighting / ETS Bandwidth Allocation for queue 13"	RW	0x00
7:0	dwrr_ets_weight_q 12	"DWRR Weighting / ETS Bandwidth Allocation for queue 12"	RW	0x00

Table 11-314: tx_q_seg_alloc_q_lower

Register Information			
Description	"This register allows the user to distribute the Transmit SRAM used by the DMA across the priority queues, for queues 0 to 7. The SRAM itself is split into a number of evenly sized segments (this is defined in the verilog configuration defs file - for the configuration used to generate this register description, the total number of segments was set to '16'). Those segments can then be freely distributed across the active queues, in powers of 2. I.e. a value of 0 would mean 1 segment has been allocated to the queue. A value of 1 would mean 2 segments, a value of 2 means 4 segments and so on. The reset values of these registers are defined in the configuration defs file."		
Offset	0x05A0	**Type:**	RW

Bitfield Details

Bits	Name	Description	Access	Reset
31	reserved_31_31	"Reserved, read as 0, ignored on write."	RO	0
30:28	segment_alloc_q7	"Number of segments allocated to q7. This should be entered as a log 2, for example entering a value of 2 would grant 4 segments. A maximum of 16 segments can be granted."	RW	0x0
27	reserved_27_27	"Reserved, read as 0, ignored on write."	RO	0
26:24	segment_alloc_q6	"Number of segments allocated to q6. This should be entered as a log 2, for example entering a value of 2 would grant 4 segments. A maximum of 16 segments can be granted."	RW	0x0
23	reserved_23_23	"Reserved, read as 0, ignored on write."	RO	0
22:20	segment_alloc_q5	"Number of segments allocated to q5. This should be entered as a log 2, for example entering a value of 2 would grant 4 segments. A maximum of 16 segments can be granted."	RW	0x0
19	reserved_19_19	"Reserved, read as 0, ignored on write."	RO	0
18:16	segment_alloc_q4	"Number of segments allocated to q4. This should be entered as a log 2, for example entering a value of 2 would grant 4 segments. A maximum of 16 segments can be granted."	RW	0x0
15	reserved_15_15	"Reserved, read as 0, ignored on write."	RO	0
14:12	segment_alloc_q3	"Number of segments allocated to q3. This should be entered as a log 2, for example entering a value of 2 would grant 4 segments. A maximum of 16 segments can be granted."	RW	0x0
11	reserved_11_11	"Reserved, read as 0, ignored on write."	RO	0
10:8	segment_alloc_q2	"Number of segments allocated to q2. This should be entered as a log 2, for example entering a value of 2 would grant 4 segments. A maximum of 16 segments can be granted."	RW	0x0
7	reserved_7_7	"Reserved, read as 0, ignored on write."	RO	0
6:4	segment_alloc_q1	"Number of segments allocated to q1. This should be entered as a log 2, for example entering a value of 2 would grant 4 segments. A maximum of 16 segments can be granted."	RW	0x0
3	reserved_3_3	"Reserved, read as 0, ignored on write."	RO	0

2:0	segment_alloc_q0	"Number of segments allocated to q0. This should be entered as a log 2, for example entering a value of 2 would grant 4 segments. A maximum of 16 segments can be granted."	RW	0x0

Table 11-315: tx_q_seg_alloc_q_upper

Register Information			
Description	"This register allows the user to distribute the Transmit SRAM used by the DMA across the priority queues, for queues 8 to 15. The SRAM itself is split into a number of evenly sized segments (this is defined in the verilog configuration defs file - for the configuration used to generate this register description, the total number of segments was set to '16'). Those segments can then be freely distributed across the active queues, in powers of 2. I.e. a value of 0 would mean 1 segment has been allocated to the queue. A value of 1 would mean 2 segments, a value of 2 means 4 segments and so on. The reset values of these registers are defined in the configuration defs file."		
Offset	0x05A4	**Type:**	RW

Bitfield Details

Bits	Name	Description	Access	Reset
31	reserved_31_31	"Reserved, read as 0, ignored on write."	RO	0
30:28	segment_alloc_q15	"Number of segments allocated to q15. This should be entered as a log 2, for example entering a value of 2 would grant 4 segments. A maximum of 16 segments can be granted."	RW	0x0
27	reserved_27_27	"Reserved, read as 0, ignored on write."	RO	0
26:24	segment_alloc_q14	"Number of segments allocated to q14. This should be entered as a log 2, for example entering a value of 2 would grant 4 segments. A maximum of 16 segments can be granted."	RW	0x0
23	reserved_23_23	"Reserved, read as 0, ignored on write."	RO	0
22:20	segment_alloc_q13	"Number of segments allocated to q13. This should be entered as a log 2, for example entering a value of 2 would grant 4 segments. A maximum of 16 segments can be granted."	RW	0x0
19	reserved_19_19	"Reserved, read as 0, ignored on write."	RO	0
18:16	segment_alloc_q12	"Number of segments allocated to q12. This should be entered as a log 2, for example entering a value of 2 would grant 4 segments. A maximum of 16 segments can be granted."	RW	0x0
15	reserved_15_15	"Reserved, read as 0, ignored on write."	RO	0
14:12	segment_alloc_q11	"Number of segments allocated to q11. This should be entered as a log 2, for example entering a value of 2 would grant 4 segments. A maximum of 16 segments can be granted."	RW	0x0
11	reserved_11_11	"Reserved, read as 0, ignored on write."	RO	0
10:8	segment_alloc_q10	"Number of segments allocated to q10. This should be entered as a log 2, for example entering a value of 2 would grant 4 segments. A maximum of 16 segments can be granted."	RW	0x0
7	reserved_7_7	"Reserved, read as 0, ignored on write."	RO	0
6:4	segment_alloc_q9	"Number of segments allocated to q9. This should be entered as a log 2, for example entering a value of 2 would grant 4 segments. A maximum of 16 segments can be granted."	RW	0x0
3	reserved_3_3	"Reserved, read as 0, ignored on write."	RO	0

2:0	segment_alloc_q8	"Number of segments allocated to q8. This should be entered as a log 2, for example entering a value of 2 would grant 4 segments. A maximum of 16 segments can be granted."	RW	0x0

Table 11-316: receive_q8_ptr

Register Information				
Description	"This register holds the start address of the receive buffer queue (receive buffers descriptor list). The receive buffer queue base address must be initialized before receive is enabled through bit 2 of the network control register. Once reception is enabled, any write to the receive buffer queue base address register is ignored. Reading this register returns the location of the descriptor currently being accessed. This value increments as buffers are used. Software should not use this register for determining where to remove received frames from the queue as it constantly changes as new frames are received. Software should instead work its way through the buffer descriptor queue checking the used bits. In terms of AMBA (AHB/AXI) operation, the receive descriptors are read from memory using a single 32bit AHB/AXI access. When the datapath is configured at 64bit or 128bit, the receive descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is written to using a single 64bit AHB/AXI access. For 32bit datapaths, the receive descriptors should be aligned at 32-bit boundaries and are written to using two individual non sequential 32-bit accesses."			
Offset	0x05C0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:2	dma_rx_q_ptr	"Receive buffer queue base address - written with the address of the start of the receive queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_rx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while receive is not enabled."	RW	0

Table 11-317: receive_q9_ptr

Register Information			
Description	"This register holds the start address of the receive buffer queue (receive buffers descriptor list). The receive buffer queue base address must be initialized before receive is enabled through bit 2 of the network control register. Once reception is enabled, any write to the receive buffer queue base address register is ignored. Reading this register returns the location of the descriptor currently being accessed. This value increments as buffers are used. Software should not use this register for determining where to remove received frames from the queue as it constantly changes as new frames are received. Software should instead work its way through the buffer descriptor queue checking the used bits. In terms of AMBA (AHB/AXI) operation, the receive descriptors are read from memory using a single 32bit AHB/AXI access. When the datapath is configured at 64bit or 128bit, the receive descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is written to using a single 64bit AHB/AXI access. For 32bit datapaths, the receive descriptors should be aligned at 32-bit boundaries and are written to using two individual non sequential 32-bit accesses."		
Offset	0x05C4	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:2	dma_rx_q_ptr	"Receive buffer queue base address - written with the address of the start of the receive queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_rx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while receive is not enabled."	RW	0

Table 11-318: receive_q10_ptr

Register Information			
Description	"This register holds the start address of the receive buffer queue (receive buffers descriptor list). The receive buffer queue base address must be initialized before receive is enabled through bit 2 of the network control register. Once reception is enabled, any write to the receive buffer queue base address register is ignored. Reading this register returns the location of the descriptor currently being accessed. This value increments as buffers are used. Software should not use this register for determining where to remove received frames from the queue as it constantly changes as new frames are received. Software should instead work its way through the buffer descriptor queue checking the used bits. In terms of AMBA (AHB/AXI) operation, the receive descriptors are read from memory using a single 32bit AHB/AXI access. When the datapath is configured at 64bit or 128bit, the receive descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is written to using a single 64bit AHB/AXI access. For 32bit datapaths, the receive descriptors should be aligned at 32-bit boundaries and are written to using two individual non sequential 32-bit accesses."		
Offset	0x05C8	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:2	dma_rx_q_ptr	"Receive buffer queue base address - written with the address of the start of the receive queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_rx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while receive is not enabled."	RW	0

Table 11-319: receive_q11_ptr

Register Information					
Description	"This register holds the start address of the receive buffer queue (receive buffers descriptor list). The receive buffer queue base address must be initialized before receive is enabled through bit 2 of the network control register. Once reception is enabled, any write to the receive buffer queue base address register is ignored. Reading this register returns the location of the descriptor currently being accessed. This value increments as buffers are used. Software should not use this register for determining where to remove received frames from the queue as it constantly changes as new frames are received. Software should instead work its way through the buffer descriptor queue checking the used bits. In terms of AMBA (AHB/AXI) operation, the receive descriptors are read from memory using a single 32bit AHB/AXI access. When the datapath is configured at 64bit or 128bit, the receive descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is written to using a single 64bit AHB/AXI access. For 32bit datapaths, the receive descriptors should be aligned at 32-bit boundaries and are written to using two individual non sequential 32-bit accesses."				
Offset	0x05CC		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:2	dma_rx_q_ptr	"Receive buffer queue base address - written with the address of the start of the receive queue."		RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."		RO	0
0	dma_rx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while receive is not enabled."		RW	0

Table 11-320: receive_q12_ptr

Register Information					
Description	"This register holds the start address of the receive buffer queue (receive buffers descriptor list). The receive buffer queue base address must be initialized before receive is enabled through bit 2 of the network control register. Once reception is enabled, any write to the receive buffer queue base address register is ignored. Reading this register returns the location of the descriptor currently being accessed. This value increments as buffers are used. Software should not use this register for determining where to remove received frames from the queue as it constantly changes as new frames are received. Software should instead work its way through the buffer descriptor queue checking the used bits. In terms of AMBA (AHB/AXI) operation, the receive descriptors are read from memory using a single 32bit AHB/AXI access. When the datapath is configured at 64bit or 128bit, the receive descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is written to using a single 64bit AHB/AXI access. For 32bit datapaths, the receive descriptors should be aligned at 32-bit boundaries and are written to using two individual non sequential 32-bit accesses."				
Offset	0x05D0		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:2	dma_rx_q_ptr	"Receive buffer queue base address - written with the address of the start of the receive queue."		RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."		RO	0
0	dma_rx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while receive is not enabled."		RW	0

Table 11-321: receive_q13_ptr

Register Information			
Description	"This register holds the start address of the receive buffer queue (receive buffers descriptor list). The receive buffer queue base address must be initialized before receive is enabled through bit 2 of the network control register. Once reception is enabled, any write to the receive buffer queue base address register is ignored. Reading this register returns the location of the descriptor currently being accessed. This value increments as buffers are used. Software should not use this register for determining where to remove received frames from the queue as it constantly changes as new frames are received. Software should instead work its way through the buffer descriptor queue checking the used bits. In terms of AMBA (AHB/AXI) operation, the receive descriptors are read from memory using a single 32bit AHB/AXI access. When the datapath is configured at 64bit or 128bit, the receive descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is written to using a single 64bit AHB/AXI access. For 32bit datapaths, the receive descriptors should be aligned at 32-bit boundaries and are written to using two individual non sequential 32-bit accesses."		
Offset	0x05D4	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:2	dma_rx_q_ptr	"Receive buffer queue base address - written with the address of the start of the receive queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_rx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while receive is not enabled."	RW	0

Table 11-322: receive_q14_ptr

Register Information			
Description	"This register holds the start address of the receive buffer queue (receive buffers descriptor list). The receive buffer queue base address must be initialized before receive is enabled through bit 2 of the network control register. Once reception is enabled, any write to the receive buffer queue base address register is ignored. Reading this register returns the location of the descriptor currently being accessed. This value increments as buffers are used. Software should not use this register for determining where to remove received frames from the queue as it constantly changes as new frames are received. Software should instead work its way through the buffer descriptor queue checking the used bits. In terms of AMBA (AHB/AXI) operation, the receive descriptors are read from memory using a single 32bit AHB/AXI access. When the datapath is configured at 64bit or 128bit, the receive descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is written to using a single 64bit AHB/AXI access. For 32bit datapaths, the receive descriptors should be aligned at 32-bit boundaries and are written to using two individual non sequential 32-bit accesses."		
Offset	0x05D8	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:2	dma_rx_q_ptr	"Receive buffer queue base address - written with the address of the start of the receive queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_rx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while receive is not enabled."	RW	0

Table 11-323: receive_q15_ptr

Register Information			
Description	"This register holds the start address of the receive buffer queue (receive buffers descriptor list). The receive buffer queue base address must be initialized before receive is enabled through bit 2 of the network control register. Once reception is enabled, any write to the receive buffer queue base address register is ignored. Reading this register returns the location of the descriptor currently being accessed. This value increments as buffers are used. Software should not use this register for determining where to remove received frames from the queue as it constantly changes as new frames are received. Software should instead work its way through the buffer descriptor queue checking the used bits. In terms of AMBA (AHB/AXI) operation, the receive descriptors are read from memory using a single 32bit AHB/AXI access. When the datapath is configured at 64bit or 128bit, the receive descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is written to using a single 64bit AHB/AXI access. For 32bit datapaths, the receive descriptors should be aligned at 32-bit boundaries and are written to using two individual non sequential 32-bit accesses."		
Offset	0x05DC	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:2	dma_rx_q_ptr	"Receive buffer queue base address - written with the address of the start of the receive queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_rx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while receive is not enabled."	RW	0

Table 11-324: dma_rxbuf_size_q8

Register Information			
Description	"Receive Buffer Size"		
Offset	0x05E0	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:8	reserved_31_8	"Reserved, read as 0, ignored on write."	RO	0x00 0000
7:0	dma_rx_q_buf_size	"DMA receive buffer size in system memory. The value defined by these bits determines the size of buffer to use in main system memory when writing received data. The value is defined in multiples of 64 bytes. 0x01 corresponds to buffers of 64 bytes. 0x02 corresponds to 128 bytes etc. For example: 0x02: 128 byte 0x18: 1536 byte (1*max length frame/buffer) 0xA0: 10240 byte (1*10K jumbo frame/buffer) 0xFF: 16320 byte Note that this value should never be written as zero. Note. The reset value of this field is equal to the gem_rx_buffer_length_def define, which is user configurable."	RW	0x02

Table 11-325: dma_rxbuf_size_q9

Register Information				
Description	"Receive Buffer Size"			
Offset	0x05E4	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:8	reserved_31_8	"Reserved, read as 0, ignored on write."	RO	0x00 0000
7:0	dma_rx_q_buf_size	"DMA receive buffer size in system memory. The value defined by these bits determines the size of buffer to use in main system memory when writing received data. The value is defined in multiples of 64 bytes. 0x01 corresponds to buffers of 64 bytes. 0x02 corresponds to 128 bytes etc. For example: 0x02: 128 byte 0x18: 1536 byte (1*max length frame/buffer) 0xA0: 10240 byte (1*10K jumbo frame/buffer) 0xFF: 16320 byte Note that this value should never be written as zero. Note. The reset value of this field is equal to the gem_rx_buffer_length_def define, which is user configurable."	RW	0x02

Table 11-326: dma_rxbuf_size_q10

Register Information				
Description	"Receive Buffer Size"			
Offset	0x05E8	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:8	reserved_31_8	"Reserved, read as 0, ignored on write."	RO	0x00 0000
7:0	dma_rx_q_buf_size	"DMA receive buffer size in system memory. The value defined by these bits determines the size of buffer to use in main system memory when writing received data. The value is defined in multiples of 64 bytes. 0x01 corresponds to buffers of 64 bytes. 0x02 corresponds to 128 bytes etc. For example: 0x02: 128 byte 0x18: 1536 byte (1*max length frame/buffer) 0xA0: 10240 byte (1*10K jumbo frame/buffer) 0xFF: 16320 byte Note that this value should never be written as zero. Note. The reset value of this field is equal to the gem_rx_buffer_length_def define, which is user configurable."	RW	0x02

Table 11-327: dma_rxbuf_size_q11

Register Information				
Description	"Receive Buffer Size"			
Offset	0x05EC	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:8	reserved_31_8	"Reserved, read as 0, ignored on write."	RO	0x00 0000
7:0	dma_rx_q_buf_size	"DMA receive buffer size in system memory. The value defined by these bits determines the size of buffer to use in main system memory when writing received data. The value is defined in multiples of 64 bytes. 0x01 corresponds to buffers of 64 bytes. 0x02 corresponds to 128 bytes etc. For example: 0x02: 128 byte 0x18: 1536 byte (1*max length frame/buffer) 0xA0: 10240 byte (1*10K jumbo frame/buffer) 0xFF: 16320 byte Note that this value should never be written as zero. Note. The reset value of this field is equal to the gem_rx_buffer_length_def define, which is user configurable."	RW	0x02

Table 11-328: dma_rxbuf_size_q12

Register Information				
Description	"Receive Buffer Size"			
Offset	0x05F0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:8	reserved_31_8	"Reserved, read as 0, ignored on write."	RO	0x00 0000
7:0	dma_rx_q_buf_size	"DMA receive buffer size in system memory. The value defined by these bits determines the size of buffer to use in main system memory when writing received data. The value is defined in multiples of 64 bytes. 0x01 corresponds to buffers of 64 bytes. 0x02 corresponds to 128 bytes etc. For example: 0x02: 128 byte 0x18: 1536 byte (1*max length frame/buffer) 0xA0: 10240 byte (1*10K jumbo frame/buffer) 0xFF: 16320 byte Note that this value should never be written as zero. Note. The reset value of this field is equal to the gem_rx_buffer_length_def define, which is user configurable."	RW	0x02

Table 11-329: dma_rxbuf_size_q13

<table>
<tr><th colspan="5">Register Information</th></tr>
<tr><td>Description</td><td colspan="4">"Receive Buffer Size"</td></tr>
<tr><td>Offset</td><td>0x05F4</td><td>Type:</td><td colspan="2">RW</td></tr>
<tr><th colspan="5">Bitfield Details</th></tr>
<tr><th>Bits</th><th>Name</th><th>Description</th><th>Access</th><th>Reset</th></tr>
<tr><td>31:8</td><td>reserved_31_8</td><td>"Reserved, read as 0, ignored on write."</td><td>RO</td><td>0x00 0000</td></tr>
<tr><td>7:0</td><td>dma_rx_q_buf_size</td><td>"DMA receive buffer size in system memory. The value defined by these bits determines the size of buffer to use in main system memory when writing received data. The value is defined in multiples of 64 bytes.
0x01 corresponds to buffers of 64 bytes.
0x02 corresponds to 128 bytes etc.
For example:
0x02: 128 byte
0x18: 1536 byte (1*max length frame/buffer)
0xA0: 10240 byte (1*10K jumbo frame/buffer)
0xFF: 16320 byte
Note that this value should never be written as zero.
Note. The reset value of this field is equal to the gem_rx_buffer_length_def define, which is user configurable."</td><td>RW</td><td>0x02</td></tr>
</table>

Table 11-330: dma_rxbuf_size_q14

<table>
<tr><th colspan="5">Register Information</th></tr>
<tr><td>Description</td><td colspan="4">"Receive Buffer Size"</td></tr>
<tr><td>Offset</td><td>0x05F8</td><td>Type:</td><td colspan="2">RW</td></tr>
<tr><th colspan="5">Bitfield Details</th></tr>
<tr><th>Bits</th><th>Name</th><th>Description</th><th>Access</th><th>Reset</th></tr>
<tr><td>31:8</td><td>reserved_31_8</td><td>"Reserved, read as 0, ignored on write."</td><td>RO</td><td>0x00 0000</td></tr>
<tr><td>7:0</td><td>dma_rx_q_buf_size</td><td>"DMA receive buffer size in system memory. The value defined by these bits determines the size of buffer to use in main system memory when writing received data. The value is defined in multiples of 64 bytes.
0x01 corresponds to buffers of 64 bytes.
0x02 corresponds to 128 bytes etc.
For example:
0x02: 128 byte
0x18: 1536 byte (1*max length frame/buffer)
0xA0: 10240 byte (1*10K jumbo frame/buffer)
0xFF: 16320 byte
Note that this value should never be written as zero.
Note. The reset value of this field is equal to the gem_rx_buffer_length_def define, which is user configurable."</td><td>RW</td><td>0x02</td></tr>
</table>

Table 11-331: dma_rxbuf_size_q15

Register Information				
Description	"Receive Buffer Size"			
Offset	0x05FC	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:8	reserved_31_8	"Reserved, read as 0, ignored on write."	RO	0x00 0000
7:0	dma_rx_q_buf_size	"DMA receive buffer size in system memory. The value defined by these bits determines the size of buffer to use in main system memory when writing received data. The value is defined in multiples of 64 bytes. 0x01 corresponds to buffers of 64 bytes. 0x02 corresponds to 128 bytes etc. For example: 0x02: 128 byte 0x18: 1536 byte (1*max length frame/buffer) 0xA0: 10240 byte (1*10K jumbo frame/buffer) 0xFF: 16320 byte Note that this value should never be written as zero. Note. The reset value of this field is equal to the gem_rx_buffer_length_def define, which is user configurable."	RW	0x02

Table 11-332: int_q1_enable

Register Information				
Description	"At reset all interrupts are disabled. Writing a one to the relevant bit location enables the required interrupt. This register is write only and when read will return zero."			
Offset	0x0600	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	enable_resp_not_ok_interrupt	"Enable bresp/hresp not OK interrupt"	WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	enable_transmit_complete_interrupt	"Enable Transmit complete interrupt"	WO	0
6	enable_transmit_frame_corruption_due_to_amba_error_interrupt	"Enable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	enable_retry_limit_exceeded_or_late_collision_interrupt	"Enable Retry limit exceeded or late collision interrupt"	WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	enable_rx_used_bit_read_interrupt	"Enable RX used bit read interrupt"	WO	0
1	enable_receive_complete_interrupt	"Enable Receive complete interrupt"	WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-333: int_q2_enable

Register Information			
Description	"At reset all interrupts are disabled. Writing a one to the relevant bit location enables the required interrupt. This register is write only and when read will return zero."		
Offset	0x0604	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	enable_resp_not_ok_interrupt	"Enable bresp/hresp not OK interrupt"	WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	enable_transmit_complete_interrupt	"Enable Transmit complete interrupt"	WO	0
6	enable_transmit_frame_corruption_due_to_amba_error_interrupt	"Enable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	enable_retry_limit_exceeded_or_late_collision_interrupt	"Enable Retry limit exceeded or late collision interrupt"	WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	enable_rx_used_bit_read_interrupt	"Enable RX used bit read interrupt"	WO	0
1	enable_receive_complete_interrupt	"Enable Receive complete interrupt"	WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-334: int_q3_enable

Register Information			
Description	"At reset all interrupts are disabled. Writing a one to the relevant bit location enables the required interrupt. This register is write only and when read will return zero."		
Offset	0x0608	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	enable_resp_not_ok_interrupt	"Enable bresp/hresp not OK interrupt"	WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	enable_transmit_complete_interrupt	"Enable Transmit complete interrupt"	WO	0
6	enable_transmit_frame_corruption_due_to_amba_error_interrupt	"Enable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	enable_retry_limit_exceeded_or_late_collision_interrupt	"Enable Retry limit exceeded or late collision interrupt"	WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	enable_rx_used_bit_read_interrupt	"Enable RX used bit read interrupt"	WO	0
1	enable_receive_complete_interrupt	"Enable Receive complete interrupt"	WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-335: int_q4_enable

Register Information			
Description	"At reset all interrupts are disabled. Writing a one to the relevant bit location enables the required interrupt. This register is write only and when read will return zero."		
Offset	0x060C	**Type:**	RW

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	enable_resp_not_ok_interrupt	"Enable bresp/hresp not OK interrupt"	WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	enable_transmit_complete_interrupt	"Enable Transmit complete interrupt"	WO	0
6	enable_transmit_frame_corruption_due_to_amba_error_interrupt	"Enable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	enable_retry_limit_exceeded_or_late_collision_interrupt	"Enable Retry limit exceeded or late collision interrupt"	WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	enable_rx_used_bit_read_interrupt	"Enable RX used bit read interrupt"	WO	0
1	enable_receive_complete_interrupt	"Enable Receive complete interrupt"	WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-336: int_q5_enable

Register Information			
Description	"At reset all interrupts are disabled. Writing a one to the relevant bit location enables the required interrupt. This register is write only and when read will return zero."		
Offset	0x0610	**Type:**	RW

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	enable_resp_not_ok_interrupt	"Enable bresp/hresp not OK interrupt"	WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	enable_transmit_complete_interrupt	"Enable Transmit complete interrupt"	WO	0
6	enable_transmit_frame_corruption_due_to_amba_error_interrupt	"Enable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	enable_retry_limit_exceeded_or_late_collision_interrupt	"Enable Retry limit exceeded or late collision interrupt"	WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	enable_rx_used_bit_read_interrupt	"Enable RX used bit read interrupt"	WO	0
1	enable_receive_complete_interrupt	"Enable Receive complete interrupt"	WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-337: int_q6_enable

Register Information			
Description	"At reset all interrupts are disabled. Writing a one to the relevant bit location enables the required interrupt. This register is write only and when read will return zero."		
Offset	0x0614	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	enable_resp_not_ok_interrupt	"Enable bresp/hresp not OK interrupt"	WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	enable_transmit_complete_interrupt	"Enable Transmit complete interrupt"	WO	0
6	enable_transmit_frame_corruption_due_to_amba_error_interrupt	"Enable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	enable_retry_limit_exceeded_or_late_collision_interrupt	"Enable Retry limit exceeded or late collision interrupt"	WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	enable_rx_used_bit_read_interrupt	"Enable RX used bit read interrupt"	WO	0
1	enable_receive_complete_interrupt	"Enable Receive complete interrupt"	WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-338: int_q7_enable

Register Information			
Description	"At reset all interrupts are disabled. Writing a one to the relevant bit location enables the required interrupt. This register is write only and when read will return zero."		
Offset	0x0618	**Type:**	RW

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	enable_resp_not_ok_interrupt	"Enable bresp/hresp not OK interrupt"	WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	enable_transmit_complete_interrupt	"Enable Transmit complete interrupt"	WO	0
6	enable_transmit_frame_corruption_due_to_amba_error_interrupt	"Enable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	enable_retry_limit_exceeded_or_late_collision_interrupt	"Enable Retry limit exceeded or late collision interrupt"	WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	enable_rx_used_bit_read_interrupt	"Enable RX used bit read interrupt"	WO	0
1	enable_receive_complete_interrupt	"Enable Receive complete interrupt"	WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-339: int_q1_disable

Register Information			
Description	"Writing a 1 to the relevant bit location disables that particular interrupt. This register is write only and when read will return zero."		
Offset	0x0620	**Type:**	RW

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	disable_resp_not_ok_interrupt	"Disable bresp/hresp not OK interrupt"	WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	disable_transmit_complete_interrupt	"Disable Transmit complete interrupt"	WO	0
6	disable_transmit_frame_corruption_due_to_amba_error_interrupt	"Disable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	disable_retry_limit_exceeded_or_late_collision_interrupt	"Disable Retry limit exceeded or late collision interrupt"	WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	disable_rx_used_bit_read_interrupt	"Disable RX used bit read interrupt"	WO	0
1	disable_receive_complete_interrupt	"Disable Receive complete interrupt"	WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-340: int_q2_disable

Register Information			
Description	"Writing a 1 to the relevant bit location disables that particular interrupt. This register is write only and when read will return zero."		
Offset	0x0624	**Type:**	RW

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	disable_resp_not_ok_interrupt	"Disable bresp/hresp not OK interrupt"	WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	disable_transmit_complete_interrupt	"Disable Transmit complete interrupt"	WO	0
6	disable_transmit_frame_corruption_due_to_amba_error_interrupt	"Disable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	disable_retry_limit_exceeded_or_late_collision_interrupt	"Disable Retry limit exceeded or late collision interrupt"	WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	disable_rx_used_bit_read_interrupt	"Disable RX used bit read interrupt"	WO	0
1	disable_receive_complete_interrupt	"Disable Receive complete interrupt"	WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-341: int_q3_disable

Register Information			
Description	"Writing a 1 to the relevant bit location disables that particular interrupt. This register is write only and when read will return zero."		
Offset	0x0628	**Type:**	RW

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	disable_resp_not_ok_interrupt	"Disable bresp/hresp not OK interrupt"	WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	disable_transmit_complete_interrupt	"Disable Transmit complete interrupt"	WO	0
6	disable_transmit_frame_corruption_due_to_amba_error_interrupt	"Disable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	disable_retry_limit_exceeded_or_late_collision_interrupt	"Disable Retry limit exceeded or late collision interrupt"	WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	disable_rx_used_bit_read_interrupt	"Disable RX used bit read interrupt"	WO	0
1	disable_receive_complete_interrupt	"Disable Receive complete interrupt"	WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-342: int_q4_disable

Register Information			
Description	"Writing a 1 to the relevant bit location disables that particular interrupt. This register is write only and when read will return zero."		
Offset	0x062C	**Type:**	RW

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	disable_resp_not_ok_interrupt	"Disable bresp/hresp not OK interrupt"	WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	disable_transmit_complete_interrupt	"Disable Transmit complete interrupt"	WO	0
6	disable_transmit_frame_corruption_due_to_amba_error_interrupt	"Disable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	disable_retry_limit_exceeded_or_late_collision_interrupt	"Disable Retry limit exceeded or late collision interrupt"	WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	disable_rx_used_bit_read_interrupt	"Disable RX used bit read interrupt"	WO	0
1	disable_receive_complete_interrupt	"Disable Receive complete interrupt"	WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-343: int_q5_disable

Register Information					
Description	"Writing a 1 to the relevant bit location disables that particular interrupt. This register is write only and when read will return zero."				
Offset	0x0630		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."		RO	0x0 0000
11	disable_resp_not_ok_interrupt	"Disable bresp/hresp not OK interrupt"		WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."		RO	0x0
7	disable_transmit_complete_interrupt	"Disable Transmit complete interrupt"		WO	0
6	disable_transmit_frame_corruption_due_to_amba_error_interrupt	"Disable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"		WO	0
5	disable_retry_limit_exceeded_or_late_collision_interrupt	"Disable Retry limit exceeded or late collision interrupt"		WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."		RO	0x0
2	disable_rx_used_bit_read_interrupt	"Disable RX used bit read interrupt"		WO	0
1	disable_receive_complete_interrupt	"Disable Receive complete interrupt"		WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."		RO	0

Table 11-344: int_q6_disable

Register Information			
Description	"Writing a 1 to the relevant bit location disables that particular interrupt. This register is write only and when read will return zero."		
Offset	0x0634	**Type:**	RW

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	disable_resp_not_ok_interrupt	"Disable bresp/hresp not OK interrupt"	WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	disable_transmit_complete_interrupt	"Disable Transmit complete interrupt"	WO	0
6	disable_transmit_frame_corruption_due_to_amba_error_interrupt	"Disable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	disable_retry_limit_exceeded_or_late_collision_interrupt	"Disable Retry limit exceeded or late collision interrupt"	WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	disable_rx_used_bit_read_interrupt	"Disable RX used bit read interrupt"	WO	0
1	disable_receive_complete_interrupt	"Disable Receive complete interrupt"	WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-345: int_q7_disable

Register Information			
Description	"Writing a 1 to the relevant bit location disables that particular interrupt. This register is write only and when read will return zero."		
Offset	0x0638	**Type:**	RW

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	disable_resp_not_ok_interrupt	"Disable bresp/hresp not OK interrupt"	WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	disable_transmit_complete_interrupt	"Disable Transmit complete interrupt"	WO	0
6	disable_transmit_frame_corruption_due_to_amba_error_interrupt	"Disable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	disable_retry_limit_exceeded_or_late_collision_interrupt	"Disable Retry limit exceeded or late collision interrupt"	WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	disable_rx_used_bit_read_interrupt	"Disable RX used bit read interrupt"	WO	0
1	disable_receive_complete_interrupt	"Disable Receive complete interrupt"	WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-346: int_q1_mask

Register Information			
Description	"The interrupt mask register is a read only register indicating which interrupts are masked. All bits are set at reset and can be reset individually by writing to the interrupt enable register or set individually by writing to the interrupt disable register. Having separate address locations for enable and disable saves the need for performing a read modify write when updating the interrupt mask register. For test purposes there is a write-only function to this register that allows the bits in the interrupt status register to be set or cleared, regardless of the state of the mask register."		
Offset	0x0640	**Type:**	RO

Bitfield Details

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok_interrupt_mask	"bresp/hresp not OK interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete_interrupt_mask	"Transmit complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
6	amba_error_interrupt_mask	"A read of this register returns the value of the AMBA (AHB/AXI) error interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
5	retry_limit_exceeded_or_late_collision_interrupt_mask	"retry limit exceeded or late collision interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_interrupt_mask	"A read of this register returns the value of the RX Used interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
1	receive_complete_interrupt_mask	"Receive complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-347: int_q2_mask

Register Information			
Description	"The interrupt mask register is a read only register indicating which interrupts are masked. All bits are set at reset and can be reset individually by writing to the interrupt enable register or set individually by writing to the interrupt disable register. Having separate address locations for enable and disable saves the need for performing a read modify write when updating the interrupt mask register. For test purposes there is a write-only function to this register that allows the bits in the interrupt status register to be set or cleared, regardless of the state of the mask register."		
Offset	0x0644	**Type:**	RO

Bitfield Details

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok_interrupt_mask	"bresp/hresp not OK interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete_interrupt_mask	"Transmit complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
6	amba_error_interrupt_mask	"A read of this register returns the value of the AMBA (AHB/AXI) error interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
5	retry_limit_exceeded_or_late_collision_interrupt_mask	"retry limit exceeded or late collision interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_interrupt_mask	"A read of this register returns the value of the RX Used interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
1	receive_complete_interrupt_mask	"Receive complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-348: int_q3_mask

Register Information			
Description	"The interrupt mask register is a read only register indicating which interrupts are masked. All bits are set at reset and can be reset individually by writing to the interrupt enable register or set individually by writing to the interrupt disable register. Having separate address locations for enable and disable saves the need for performing a read modify write when updating the interrupt mask register. For test purposes there is a write-only function to this register that allows the bits in the interrupt status register to be set or cleared, regardless of the state of the mask register."		
Offset	0x0648	**Type:**	RO

Bitfield Details

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok_interrupt_mask	"bresp/hresp not OK interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete_interrupt_mask	"Transmit complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
6	amba_error_interrupt_mask	"A read of this register returns the value of the AMBA (AHB/AXI) error interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
5	retry_limit_exceeded_or_late_collision_interrupt_mask	"retry limit exceeded or late collision interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_interrupt_mask	"A read of this register returns the value of the RX Used interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
1	receive_complete_interrupt_mask	"Receive complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-349: int_q4_mask

Register Information			
Description	"The interrupt mask register is a read only register indicating which interrupts are masked. All bits are set at reset and can be reset individually by writing to the interrupt enable register or set individually by writing to the interrupt disable register. Having separate address locations for enable and disable saves the need for performing a read modify write when updating the interrupt mask register. For test purposes there is a write-only function to this register that allows the bits in the interrupt status register to be set or cleared, regardless of the state of the mask register."		
Offset	0x064C	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok_interrupt_mask	"bresp/hresp not OK interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete_interrupt_mask	"Transmit complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
6	amba_error_interrupt_mask	"A read of this register returns the value of the AMBA (AHB/AXI) error interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
5	retry_limit_exceeded_or_late_collision_interrupt_mask	"retry limit exceeded or late collision interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_interrupt_mask	"A read of this register returns the value of the RX Used interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
1	receive_complete_interrupt_mask	"Receive complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-350: int_q5_mask

Register Information			
Description	"The interrupt mask register is a read only register indicating which interrupts are masked. All bits are set at reset and can be reset individually by writing to the interrupt enable register or set individually by writing to the interrupt disable register. Having separate address locations for enable and disable saves the need for performing a read modify write when updating the interrupt mask register. For test purposes there is a write-only function to this register that allows the bits in the interrupt status register to be set or cleared, regardless of the state of the mask register."		
Offset	0x0650	**Type:**	RO

Bitfield Details

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok_interrupt_mask	"bresp/hresp not OK interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete_interrupt_mask	"Transmit complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
6	amba_error_interrupt_mask	"A read of this register returns the value of the AMBA (AHB/AXI) error interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
5	retry_limit_exceeded_or_late_collision_interrupt_mask	"retry limit exceeded or late collision interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_interrupt_mask	"A read of this register returns the value of the RX Used interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
1	receive_complete_interrupt_mask	"Receive complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-351: int_q6_mask

Register Information			
Description	"The interrupt mask register is a read only register indicating which interrupts are masked. All bits are set at reset and can be reset individually by writing to the interrupt enable register or set individually by writing to the interrupt disable register. Having separate address locations for enable and disable saves the need for performing a read modify write when updating the interrupt mask register. For test purposes there is a write-only function to this register that allows the bits in the interrupt status register to be set or cleared, regardless of the state of the mask register."		
Offset	0x0654	**Type:**	RO

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok_interrupt_mask	"bresp/hresp not OK interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete_interrupt_mask	"Transmit complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
6	amba_error_interrupt_mask	"A read of this register returns the value of the AMBA (AHB/AXI) error interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
5	retry_limit_exceeded_or_late_collision_interrupt_mask	"retry limit exceeded or late collision interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_interrupt_mask	"A read of this register returns the value of the RX Used interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
1	receive_complete_interrupt_mask	"Receive complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-352: int_q7_mask

Register Information			
Description	"The interrupt mask register is a read only register indicating which interrupts are masked. All bits are set at reset and can be reset individually by writing to the interrupt enable register or set individually by writing to the interrupt disable register. Having separate address locations for enable and disable saves the need for performing a read modify write when updating the interrupt mask register. For test purposes there is a write-only function to this register that allows the bits in the interrupt status register to be set or cleared, regardless of the state of the mask register."		
Offset	0x0658	**Type:**	RO

Bitfield Details

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok_interrupt_mask	"bresp/hresp not OK interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete_interrupt_mask	"Transmit complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
6	amba_error_interrupt_mask	"A read of this register returns the value of the AMBA (AHB/AXI) error interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
5	retry_limit_exceeded_or_late_collision_interrupt_mask	"retry limit exceeded or late collision interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_interrupt_mask	"A read of this register returns the value of the RX Used interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
1	receive_complete_interrupt_mask	"Receive complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-353: int_q8_enable

Register Information			
Description	"At reset all interrupts are disabled. Writing a one to the relevant bit location enables the required interrupt. This register is write only and when read will return zero."		
Offset	0x0660	**Type:**	RW

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	enable_resp_not_ok_interrupt	"Enable bresp/hresp not OK interrupt"	WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	enable_transmit_complete_interrupt	"Enable Transmit complete interrupt"	WO	0
6	enable_transmit_frame_corruption_due_to_amba_error_interrupt	"Enable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	enable_retry_limit_exceeded_or_late_collision_interrupt	"Enable Retry limit exceeded or late collision interrupt"	WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	enable_rx_used_bit_read_interrupt	"Enable RX used bit read interrupt"	WO	0
1	enable_receive_complete_interrupt	"Enable Receive complete interrupt"	WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-354: int_q9_enable

Register Information				
Description	"At reset all interrupts are disabled. Writing a one to the relevant bit location enables the required interrupt. This register is write only and when read will return zero."			
Offset	0x0664	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	enable_resp_not_ok_interrupt	"Enable bresp/hresp not OK interrupt"	WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	enable_transmit_complete_interrupt	"Enable Transmit complete interrupt"	WO	0
6	enable_transmit_frame_corruption_due_to_amba_error_interrupt	"Enable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	enable_retry_limit_exceeded_or_late_collision_interrupt	"Enable Retry limit exceeded or late collision interrupt"	WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	enable_rx_used_bit_read_interrupt	"Enable RX used bit read interrupt"	WO	0
1	enable_receive_complete_interrupt	"Enable Receive complete interrupt"	WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-355: int_q10_enable

Register Information			
Description	"At reset all interrupts are disabled. Writing a one to the relevant bit location enables the required interrupt. This register is write only and when read will return zero."		
Offset	0x0668	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	enable_resp_not_ok_interrupt	"Enable bresp/hresp not OK interrupt"	WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	enable_transmit_complete_interrupt	"Enable Transmit complete interrupt"	WO	0
6	enable_transmit_frame_corruption_due_to_amba_error_interrupt	"Enable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	enable_retry_limit_exceeded_or_late_collision_interrupt	"Enable Retry limit exceeded or late collision interrupt"	WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	enable_rx_used_bit_read_interrupt	"Enable RX used bit read interrupt"	WO	0
1	enable_receive_complete_interrupt	"Enable Receive complete interrupt"	WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-356: int_q11_enable

Register Information			
Description	"At reset all interrupts are disabled. Writing a one to the relevant bit location enables the required interrupt. This register is write only and when read will return zero."		
Offset	0x066C	**Type:**	RW

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	enable_resp_not_ok_interrupt	"Enable bresp/hresp not OK interrupt"	WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	enable_transmit_complete_interrupt	"Enable Transmit complete interrupt"	WO	0
6	enable_transmit_frame_corruption_due_to_amba_error_interrupt	"Enable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	enable_retry_limit_exceeded_or_late_collision_interrupt	"Enable Retry limit exceeded or late collision interrupt"	WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	enable_rx_used_bit_read_interrupt	"Enable RX used bit read interrupt"	WO	0
1	enable_receive_complete_interrupt	"Enable Receive complete interrupt"	WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-357: int_q12_enable

Register Information			
Description	"At reset all interrupts are disabled. Writing a one to the relevant bit location enables the required interrupt. This register is write only and when read will return zero."		
Offset	0x0670	**Type:**	RW

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	enable_resp_not_ok_interrupt	"Enable bresp/hresp not OK interrupt"	WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	enable_transmit_complete_interrupt	"Enable Transmit complete interrupt"	WO	0
6	enable_transmit_frame_corruption_due_to_amba_error_interrupt	"Enable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	enable_retry_limit_exceeded_or_late_collision_interrupt	"Enable Retry limit exceeded or late collision interrupt"	WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	enable_rx_used_bit_read_interrupt	"Enable RX used bit read interrupt"	WO	0
1	enable_receive_complete_interrupt	"Enable Receive complete interrupt"	WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-358: int_q13_enable

Register Information			
Description	"At reset all interrupts are disabled. Writing a one to the relevant bit location enables the required interrupt. This register is write only and when read will return zero."		
Offset	0x0674	**Type:**	RW

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	enable_resp_not_ok_interrupt	"Enable bresp/hresp not OK interrupt"	WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	enable_transmit_complete_interrupt	"Enable Transmit complete interrupt"	WO	0
6	enable_transmit_frame_corruption_due_to_amba_error_interrupt	"Enable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	enable_retry_limit_exceeded_or_late_collision_interrupt	"Enable Retry limit exceeded or late collision interrupt"	WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	enable_rx_used_bit_read_interrupt	"Enable RX used bit read interrupt"	WO	0
1	enable_receive_complete_interrupt	"Enable Receive complete interrupt"	WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-359: int_q14_enable

Register Information			
Description	"At reset all interrupts are disabled. Writing a one to the relevant bit location enables the required interrupt. This register is write only and when read will return zero."		
Offset	0x0678	**Type:**	RW

Bitfield Details

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	enable_resp_not_ok_interrupt	"Enable bresp/hresp not OK interrupt"	WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	enable_transmit_complete_interrupt	"Enable Transmit complete interrupt"	WO	0
6	enable_transmit_frame_corruption_due_to_amba_error_interrupt	"Enable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	enable_retry_limit_exceeded_or_late_collision_interrupt	"Enable Retry limit exceeded or late collision interrupt"	WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	enable_rx_used_bit_read_interrupt	"Enable RX used bit read interrupt"	WO	0
1	enable_receive_complete_interrupt	"Enable Receive complete interrupt"	WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-360: int_q15_enable

Register Information			
Description	"At reset all interrupts are disabled. Writing a one to the relevant bit location enables the required interrupt. This register is write only and when read will return zero."		
Offset	0x067C	**Type:**	RW

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	enable_resp_not_ok_interrupt	"Enable bresp/hresp not OK interrupt"	WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	enable_transmit_complete_interrupt	"Enable Transmit complete interrupt"	WO	0
6	enable_transmit_frame_corruption_due_to_amba_error_interrupt	"Enable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	enable_retry_limit_exceeded_or_late_collision_interrupt	"Enable Retry limit exceeded or late collision interrupt"	WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	enable_rx_used_bit_read_interrupt	"Enable RX used bit read interrupt"	WO	0
1	enable_receive_complete_interrupt	"Enable Receive complete interrupt"	WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-361: int_q8_disable

Register Information			
Description	"Writing a 1 to the relevant bit location disables that particular interrupt. This register is write only and when read will return zero."		
Offset	0x0680	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	disable_resp_not_ok_interrupt	"Disable bresp/hresp not OK interrupt"	WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	disable_transmit_complete_interrupt	"Disable Transmit complete interrupt"	WO	0
6	disable_transmit_frame_corruption_due_to_amba_error_interrupt	"Disable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	disable_retry_limit_exceeded_or_late_collision_interrupt	"Disable Retry limit exceeded or late collision interrupt"	WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	disable_rx_used_bit_read_interrupt	"Disable RX used bit read interrupt"	WO	0
1	disable_receive_complete_interrupt	"Disable Receive complete interrupt"	WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-362: int_q9_disable

Register Information			
Description	"Writing a 1 to the relevant bit location disables that particular interrupt. This register is write only and when read will return zero."		
Offset	0x0684	**Type:**	RW

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	disable_resp_not_ok_interrupt	"Disable bresp/hresp not OK interrupt"	WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	disable_transmit_complete_interrupt	"Disable Transmit complete interrupt"	WO	0
6	disable_transmit_frame_corruption_due_to_amba_error_interrupt	"Disable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	disable_retry_limit_exceeded_or_late_collision_interrupt	"Disable Retry limit exceeded or late collision interrupt"	WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	disable_rx_used_bit_read_interrupt	"Disable RX used bit read interrupt"	WO	0
1	disable_receive_complete_interrupt	"Disable Receive complete interrupt"	WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-363: int_q10_disable

Register Information				
Description	"Writing a 1 to the relevant bit location disables that particular interrupt. This register is write only and when read will return zero."			
Offset	0x0688	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	disable_resp_not_ok_interrupt	"Disable bresp/hresp not OK interrupt"	WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	disable_transmit_complete_interrupt	"Disable Transmit complete interrupt"	WO	0
6	disable_transmit_frame_corruption_due_to_amba_error_interrupt	"Disable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	disable_retry_limit_exceeded_or_late_collision_interrupt	"Disable Retry limit exceeded or late collision interrupt"	WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	disable_rx_used_bit_read_interrupt	"Disable RX used bit read interrupt"	WO	0
1	disable_receive_complete_interrupt	"Disable Receive complete interrupt"	WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-364: int_q11_disable

Register Information			
Description	"Writing a 1 to the relevant bit location disables that particular interrupt. This register is write only and when read will return zero."		
Offset	0x068C	**Type:**	RW

Bitfield Details

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	disable_resp_not_ok_interrupt	"Disable bresp/hresp not OK interrupt"	WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	disable_transmit_complete_interrupt	"Disable Transmit complete interrupt"	WO	0
6	disable_transmit_frame_corruption_due_to_amba_error_interrupt	"Disable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	disable_retry_limit_exceeded_or_late_collision_interrupt	"Disable Retry limit exceeded or late collision interrupt"	WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	disable_rx_used_bit_read_interrupt	"Disable RX used bit read interrupt"	WO	0
1	disable_receive_complete_interrupt	"Disable Receive complete interrupt"	WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-365: int_q12_disable

Register Information			
Description	"Writing a 1 to the relevant bit location disables that particular interrupt. This register is write only and when read will return zero."		
Offset	0x0690	**Type:**	RW

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	disable_resp_not_ok_interrupt	"Disable bresp/hresp not OK interrupt"	WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	disable_transmit_complete_interrupt	"Disable Transmit complete interrupt"	WO	0
6	disable_transmit_frame_corruption_due_to_amba_error_interrupt	"Disable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	disable_retry_limit_exceeded_or_late_collision_interrupt	"Disable Retry limit exceeded or late collision interrupt"	WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	disable_rx_used_bit_read_interrupt	"Disable RX used bit read interrupt"	WO	0
1	disable_receive_complete_interrupt	"Disable Receive complete interrupt"	WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-366: int_q13_disable

Register Information			
Description	"Writing a 1 to the relevant bit location disables that particular interrupt. This register is write only and when read will return zero."		
Offset	0x0694	**Type:**	RW

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	disable_resp_not_ok_interrupt	"Disable bresp/hresp not OK interrupt"	WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	disable_transmit_complete_interrupt	"Disable Transmit complete interrupt"	WO	0
6	disable_transmit_frame_corruption_due_to_amba_error_interrupt	"Disable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	disable_retry_limit_exceeded_or_late_collision_interrupt	"Disable Retry limit exceeded or late collision interrupt"	WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	disable_rx_used_bit_read_interrupt	"Disable RX used bit read interrupt"	WO	0
1	disable_receive_complete_interrupt	"Disable Receive complete interrupt"	WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-367: int_q14_disable

Register Information				
Description	"Writing a 1 to the relevant bit location disables that particular interrupt. This register is write only and when read will return zero."			
Offset	0x0698	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	disable_resp_not_ok_interrupt	"Disable bresp/hresp not OK interrupt"	WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	disable_transmit_complete_interrupt	"Disable Transmit complete interrupt"	WO	0
6	disable_transmit_frame_corruption_due_to_amba_error_interrupt	"Disable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	disable_retry_limit_exceeded_or_late_collision_interrupt	"Disable Retry limit exceeded or late collision interrupt"	WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	disable_rx_used_bit_read_interrupt	"Disable RX used bit read interrupt"	WO	0
1	disable_receive_complete_interrupt	"Disable Receive complete interrupt"	WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-368: int_q15_disable

Register Information					
Description	"Writing a 1 to the relevant bit location disables that particular interrupt. This register is write only and when read will return zero."				
Offset	0x069C		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."		RO	0x0 0000
11	disable_resp_not_ok_interrupt	"Disable bresp/hresp not OK interrupt"		WO	0
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."		RO	0x0
7	disable_transmit_complete_interrupt	"Disable Transmit complete interrupt"		WO	0
6	disable_transmit_frame_corruption_due_to_amba_error_interrupt	"Disable Transmit frame corruption due to AMBA (AHB/AXI) error interrupt"		WO	0
5	disable_retry_limit_exceeded_or_late_collision_interrupt	"Disable Retry limit exceeded or late collision interrupt"		WO	0
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."		RO	0x0
2	disable_rx_used_bit_read_interrupt	"Disable RX used bit read interrupt"		WO	0
1	disable_receive_complete_interrupt	"Disable Receive complete interrupt"		WO	0
0	reserved_0	"Reserved, read as 0, ignored on write."		RO	0

Table 11-369: int_q8_mask

Register Information			
Description	"The interrupt mask register is a read only register indicating which interrupts are masked. All bits are set at reset and can be reset individually by writing to the interrupt enable register or set individually by writing to the interrupt disable register. Having separate address locations for enable and disable saves the need for performing a read modify write when updating the interrupt mask register. For test purposes there is a write-only function to this register that allows the bits in the interrupt status register to be set or cleared, regardless of the state of the mask register."		
Offset	0x06A0	**Type:**	RO

Bitfield Details

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok_interrupt_mask	"bresp/hresp not OK interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete_interrupt_mask	"Transmit complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
6	amba_error_interrupt_mask	"A read of this register returns the value of the AMBA (AHB/AXI) error interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
5	retry_limit_exceeded_or_late_collision_interrupt_mask	"retry limit exceeded or late collision interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_interrupt_mask	"A read of this register returns the value of the RX Used interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
1	receive_complete_interrupt_mask	"Receive complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-370: int_q9_mask

Register Information			
Description	"The interrupt mask register is a read only register indicating which interrupts are masked. All bits are set at reset and can be reset individually by writing to the interrupt enable register or set individually by writing to the interrupt disable register. Having separate address locations for enable and disable saves the need for performing a read modify write when updating the interrupt mask register. For test purposes there is a write-only function to this register that allows the bits in the interrupt status register to be set or cleared, regardless of the state of the mask register."		
Offset	0x06A4	**Type:**	RO

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok_interrupt_mask	"bresp/hresp not OK interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete_interrupt_mask	"Transmit complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
6	amba_error_interrupt_mask	"A read of this register returns the value of the AMBA (AHB/AXI) error interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
5	retry_limit_exceeded_or_late_collision_interrupt_mask	"retry limit exceeded or late collision interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_interrupt_mask	"A read of this register returns the value of the RX Used interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
1	receive_complete_interrupt_mask	"Receive complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-371: int_q10_mask

Register Information			
Description	"The interrupt mask register is a read only register indicating which interrupts are masked. All bits are set at reset and can be reset individually by writing to the interrupt enable register or set individually by writing to the interrupt disable register. Having separate address locations for enable and disable saves the need for performing a read modify write when updating the interrupt mask register. For test purposes there is a write-only function to this register that allows the bits in the interrupt status register to be set or cleared, regardless of the state of the mask register."		
Offset	0x06A8	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok_interrupt_mask	"bresp/hresp not OK interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete_interrupt_mask	"Transmit complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
6	amba_error_interrupt_mask	"A read of this register returns the value of the AMBA (AHB/AXI) error interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
5	retry_limit_exceeded_or_late_collision_interrupt_mask	"retry limit exceeded or late collision interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_interrupt_mask	"A read of this register returns the value of the RX Used interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
1	receive_complete_interrupt_mask	"Receive complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-372: int_q11_mask

Register Information			
Description	"The interrupt mask register is a read only register indicating which interrupts are masked. All bits are set at reset and can be reset individually by writing to the interrupt enable register or set individually by writing to the interrupt disable register. Having separate address locations for enable and disable saves the need for performing a read modify write when updating the interrupt mask register. For test purposes there is a write-only function to this register that allows the bits in the interrupt status register to be set or cleared, regardless of the state of the mask register."		
Offset	0x06AC	**Type:**	RO

Bitfield Details

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok_interrupt_mask	"bresp/hresp not OK interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete_interrupt_mask	"Transmit complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
6	amba_error_interrupt_mask	"A read of this register returns the value of the AMBA (AHB/AXI) error interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
5	retry_limit_exceeded_or_late_collision_interrupt_mask	"retry limit exceeded or late collision interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_interrupt_mask	"A read of this register returns the value of the RX Used interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
1	receive_complete_interrupt_mask	"Receive complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-373: int_q12_mask

Register Information			
Description	"The interrupt mask register is a read only register indicating which interrupts are masked. All bits are set at reset and can be reset individually by writing to the interrupt enable register or set individually by writing to the interrupt disable register. Having separate address locations for enable and disable saves the need for performing a read modify write when updating the interrupt mask register. For test purposes there is a write-only function to this register that allows the bits in the interrupt status register to be set or cleared, regardless of the state of the mask register."		
Offset	0x06B0	**Type:**	RO

Bitfield Details

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok_interrupt_mask	"bresp/hresp not OK interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete_interrupt_mask	"Transmit complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
6	amba_error_interrupt_mask	"A read of this register returns the value of the AMBA (AHB/AXI) error interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
5	retry_limit_exceeded_or_late_collision_interrupt_mask	"retry limit exceeded or late collision interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_interrupt_mask	"A read of this register returns the value of the RX Used interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
1	receive_complete_interrupt_mask	"Receive complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-374: int_q13_mask

Register Information			
Description	"The interrupt mask register is a read only register indicating which interrupts are masked. All bits are set at reset and can be reset individually by writing to the interrupt enable register or set individually by writing to the interrupt disable register. Having separate address locations for enable and disable saves the need for performing a read modify write when updating the interrupt mask register. For test purposes there is a write-only function to this register that allows the bits in the interrupt status register to be set or cleared, regardless of the state of the mask register."		
Offset	0x06B4	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok_interrupt_mask	"bresp/hresp not OK interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete_interrupt_mask	"Transmit complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
6	amba_error_interrupt_mask	"A read of this register returns the value of the AMBA (AHB/AXI) error interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
5	retry_limit_exceeded_or_late_collision_interrupt_mask	"retry limit exceeded or late collision interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_interrupt_mask	"A read of this register returns the value of the RX Used interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
1	receive_complete_interrupt_mask	"Receive complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-375: int_q14_mask

Register Information			
Description	"The interrupt mask register is a read only register indicating which interrupts are masked. All bits are set at reset and can be reset individually by writing to the interrupt enable register or set individually by writing to the interrupt disable register. Having separate address locations for enable and disable saves the need for performing a read modify write when updating the interrupt mask register. For test purposes there is a write-only function to this register that allows the bits in the interrupt status register to be set or cleared, regardless of the state of the mask register."		
Offset	0x06B8	**Type:**	RO

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok_interrupt_mask	"bresp/hresp not OK interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete_interrupt_mask	"Transmit complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
6	amba_error_interrupt_mask	"A read of this register returns the value of the AMBA (AHB/AXI) error interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
5	retry_limit_exceeded_or_late_collision_interrupt_mask	"retry limit exceeded or late collision interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_interrupt_mask	"A read of this register returns the value of the RX Used interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
1	receive_complete_interrupt_mask	"Receive complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-376: int_q15_mask

Register Information			
Description	"The interrupt mask register is a read only register indicating which interrupts are masked. All bits are set at reset and can be reset individually by writing to the interrupt enable register or set individually by writing to the interrupt disable register. Having separate address locations for enable and disable saves the need for performing a read modify write when updating the interrupt mask register. For test purposes there is a write-only function to this register that allows the bits in the interrupt status register to be set or cleared, regardless of the state of the mask register."		
Offset	0x06BC	**Type:**	RO

Bitfield Details

Bits	Name	Description	Access	Reset
31:12	reserved_31_12	"Reserved, read as 0, ignored on write."	RO	0x0 0000
11	resp_not_ok_interrupt_mask	"bresp/hresp not OK interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
10:8	reserved_10_8	"Reserved, read as 0, ignored on write."	RO	0x0
7	transmit_complete_interrupt_mask	"Transmit complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
6	amba_error_interrupt_mask	"A read of this register returns the value of the AMBA (AHB/AXI) error interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
5	retry_limit_exceeded_or_late_collision_interrupt_mask	"retry limit exceeded or late collision interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
4:3	reserved_4_3	"Reserved, read as 0, ignored on write."	RO	0x0
2	rx_used_interrupt_mask	"A read of this register returns the value of the RX Used interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
1	receive_complete_interrupt_mask	"Receive complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
0	reserved_0	"Reserved, read as 0, ignored on write."	RO	0

Table 11-377: screening_type_2_ethertype_reg_0

Register Information				
Description	"EtherType Register"			
Offset	0x06E0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write"	RO	0x0000
15:0	compare_value	"EtherType compare value"	RW	0x0000

Table 11-378: screening_type_2_ethertype_reg_1

Register Information				
Description	"EtherType Register"			
Offset	0x06E4	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write"	RO	0x0000
15:0	compare_value	"EtherType compare value"	RW	0x0000

Table 11-379: screening_type_2_ethertype_reg_2

Register Information				
Description	"EtherType Register"			
Offset	0x06E8	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write"	RO	0x0000
15:0	compare_value	"EtherType compare value"	RW	0x0000

Table 11-380: screening_type_2_ethertype_reg_3

Register Information				
Description	"EtherType Register"			
Offset	0x06EC	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write"	RO	0x0000
15:0	compare_value	"EtherType compare value"	RW	0x0000

Table 11-381: screening_type_2_ethertype_reg_4

Register Information					
Description	"EtherType Register"				
Offset	0x06F0		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:16	reserved_31_16	"Reserved, read as 0, ignored on write"	RO	0x0000	
15:0	compare_value	"EtherType compare value"	RW	0x0000	

Table 11-382: screening_type_2_ethertype_reg_5

Register Information					
Description	"EtherType Register"				
Offset	0x06F4		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:16	reserved_31_16	"Reserved, read as 0, ignored on write"	RO	0x0000	
15:0	compare_value	"EtherType compare value"	RW	0x0000	

Table 11-383: screening_type_2_ethertype_reg_6

Register Information					
Description	"EtherType Register"				
Offset	0x06F8		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:16	reserved_31_16	"Reserved, read as 0, ignored on write"	RO	0x0000	
15:0	compare_value	"EtherType compare value"	RW	0x0000	

Table 11-384: screening_type_2_ethertype_reg_7

Register Information					
Description	"EtherType Register"				
Offset	0x06FC		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:16	reserved_31_16	"Reserved, read as 0, ignored on write"	RO	0x0000	
15:0	compare_value	"EtherType compare value"	RW	0x0000	

Table 11-385: type2_compare_0_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x0700	**Type:**	RW

Bits	Name	Description	Access	Reset
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-386: type2_compare_0_word_1

Register Information				
Description	"Type2 Compare Word 1"			
Offset	0x0704	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-387: type2_compare_1_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x0708	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-388: type2_compare_1_word_1

Register Information			
Description	"Type2 Compare Word 1"		
Offset	0x070C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-389: type2_compare_2_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x0710	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-390: type2_compare_2_word_1

Register Information			
Description	"Type2 Compare Word 1"		
Offset	0x0714	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-391: type2_compare_3_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x0718	**Type:**	RW

Bits	Name	Description	Access	Reset
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-392: type2_compare_3_word_1

Register Information			
Description	"Type2 Compare Word 1"		
Offset	0x071C	**Type:**	RW

Bits	Name	Description	Access	Reset
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-393: type2_compare_4_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x0720	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-394: type2_compare_4_word_1

Register Information			
Description	"Type2 Compare Word 1"		
Offset	0x0724	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-395: type2_compare_5_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x0728	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-396: type2_compare_5_word_1

Register Information				
Description	"Type2 Compare Word 1"			
Offset	0x072C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-397: type2_compare_6_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x0730	**Type:**	RW

Bits	Name	Description	Access	Reset
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-398: type2_compare_6_word_1

Register Information			
Description	"Type2 Compare Word 1"		
Offset	0x0734	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-399: type2_compare_7_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x0738	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-400: type2_compare_7_word_1

Register Information			
Description	"Type2 Compare Word 1"		
Offset	0x073C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-401: type2_compare_8_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x0740	**Type:**	RW

Bits	Name	Description	Access	Reset
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-402: type2_compare_8_word_1

Register Information				
Description	"Type2 Compare Word 1"			
Offset	0x0744	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-403: type2_compare_9_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x0748	**Type:**	RW

Bits	Name	Description	Access	Reset
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-404: type2_compare_9_word_1

Register Information			
Description	"Type2 Compare Word 1"		
Offset	0x074C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-405: type2_compare_10_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x0750	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-406: type2_compare_10_word_1

Register Information				
Description	"Type2 Compare Word 1"			
Offset	0x0754	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-407: type2_compare_11_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x0758	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-408: type2_compare_11_word_1

Register Information			
Description	"Type2 Compare Word 1"		
Offset	0x075C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-409: type2_compare_12_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x0760	**Type:**	RW

Bits	Name	Description	Access	Reset
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-410: type2_compare_12_word_1

Register Information			
Description	"Type2 Compare Word 1"		
Offset	0x0764	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-411: type2_compare_13_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x0768	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-412: type2_compare_13_word_1

Register Information			
Description	"Type2 Compare Word 1"		
Offset	0x076C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-413: type2_compare_14_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x0770	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-414: type2_compare_14_word_1

Register Information			
Description	"Type2 Compare Word 1"		
Offset	0x0774	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-415: type2_compare_15_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x0778	**Type:**	RW

Bits	Name	Description	Access	Reset
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-416: type2_compare_15_word_1

Register Information			
Description	"Type2 Compare Word 1"		
Offset	0x077C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-417: type2_compare_16_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x0780	**Type:**	RW

Bits	Name	Description	Access	Reset
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-418: type2_compare_16_word_1

Register Information				
Description	"Type2 Compare Word 1"			
Offset	0x0784	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-419: type2_compare_17_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x0788	**Type:**	RW

Bits	Name	Description	Access	Reset
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-420: type2_compare_17_word_1

Register Information				
Description	"Type2 Compare Word 1"			
Offset	0x078C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-421: type2_compare_18_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x0790	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-422: type2_compare_18_word_1

Register Information				
Description	"Type2 Compare Word 1"			
Offset	0x0794	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-423: type2_compare_19_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x0798	**Type:**	RW

Bits	Name	Description	Access	Reset
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-424: type2_compare_19_word_1

Register Information				
Description	"Type2 Compare Word 1"			
Offset	0x079C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-425: type2_compare_20_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x07A0	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-426: type2_compare_20_word_1

Register Information			
Description	"Type2 Compare Word 1"		
Offset	0x07A4	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-427: type2_compare_21_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x07A8	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-428: type2_compare_21_word_1

Register Information				
Description	"Type2 Compare Word 1"			
Offset	0x07AC	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-429: type2_compare_22_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x07B0	**Type:**	RW

Bits	Name	Description	Access	Reset
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-430: type2_compare_22_word_1

Register Information			
Description	"Type2 Compare Word 1"		
Offset	0x07B4	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-431: type2_compare_23_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x07B8	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-432: type2_compare_23_word_1

Register Information				
Description	"Type2 Compare Word 1"			
Offset	0x07BC	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-433: type2_compare_24_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x07C0	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-434: type2_compare_24_word_1

Register Information			
Description	"Type2 Compare Word 1"		
Offset	0x07C4	**Type:**	RW

Bits	Name	Description	Access	Reset
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-435: type2_compare_25_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x07C8	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-436: type2_compare_25_word_1

Register Information			
Description	"Type2 Compare Word 1"		
Offset	0x07CC	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-437: type2_compare_26_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x07D0	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-438: type2_compare_26_word_1

Register Information				
Description	"Type2 Compare Word 1"			
Offset	0x07D4	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-439: type2_compare_27_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x07D8	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-440: type2_compare_27_word_1

Register Information			
Description	"Type2 Compare Word 1"		
Offset	0x07DC	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-441: type2_compare_28_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x07E0	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-442: type2_compare_28_word_1

Register Information				
Description	"Type2 Compare Word 1"			
Offset	0x07E4	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-443: type2_compare_29_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x07E8	**Type:**	RW

Bits	Name	Description	Access	Reset
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-444: type2_compare_29_word_1

Register Information			
Description	"Type2 Compare Word 1"		
Offset	0x07EC	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-445: type2_compare_30_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x07F0	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-446: type2_compare_30_word_1

Register Information			
Description	"Type2 Compare Word 1"		
Offset	0x07F4	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-447: type2_compare_31_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x07F8	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-448: type2_compare_31_word_1

Register Information				
Description	"Type2 Compare Word 1"			
Offset	0x07FC	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-449: enst_start_time_q8

Register Information				
Description	"This register sets the absolute time at which queue q8 starts EnST transmit scheduling"			
Offset	0x0800	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	start_time_sec	"Bits 31:30 of the start time (seconds field), for q8"	RW	0x0
29:0	start_time_nsec	"Bits 29:0 of the start time (nanoseconds field), for q8"	RW	0x0000 0000

Table 11-450: enst_start_time_q9

Register Information					
Description	"This register sets the absolute time at which queue q9 starts EnST transmit scheduling"				
Offset	0x0804		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:30	start_time_sec	"Bits 31:30 of the start time (seconds field), for q9"	RW	0x0	
29:0	start_time_nsec	"Bits 29:0 of the start time (nanoseconds field), for q9"	RW	0x0000 0000	

Table 11-451: enst_start_time_q10

Register Information					
Description	"This register sets the absolute time at which queue q10 starts EnST transmit scheduling"				
Offset	0x0808		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:30	start_time_sec	"Bits 31:30 of the start time (seconds field), for q10"	RW	0x0	
29:0	start_time_nsec	"Bits 29:0 of the start time (nanoseconds field), for q10"	RW	0x0000 0000	

Table 11-452: enst_start_time_q11

Register Information					
Description	"This register sets the absolute time at which queue q11 starts EnST transmit scheduling"				
Offset	0x080C		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:30	start_time_sec	"Bits 31:30 of the start time (seconds field), for q11"	RW	0x0	
29:0	start_time_nsec	"Bits 29:0 of the start time (nanoseconds field), for q11"	RW	0x0000 0000	

Table 11-453: enst_start_time_q12

Register Information					
Description	"This register sets the absolute time at which queue q12 starts EnST transmit scheduling"				
Offset	0x0810		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:30	start_time_sec	"Bits 31:30 of the start time (seconds field), for q12"	RW	0x0	
29:0	start_time_nsec	"Bits 29:0 of the start time (nanoseconds field), for q12"	RW	0x0000 0000	

Table 11-454: enst_start_time_q13

Register Information				
Description	"This register sets the absolute time at which queue q13 starts EnST transmit scheduling"			
Offset	0x0814	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	start_time_sec	"Bits 31:30 of the start time (seconds field), for q13"	RW	0x0
29:0	start_time_nsec	"Bits 29:0 of the start time (nanoseconds field), for q13"	RW	0x0000 0000

Table 11-455: enst_start_time_q14

Register Information				
Description	"This register sets the absolute time at which queue q14 starts EnST transmit scheduling"			
Offset	0x0818	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	start_time_sec	"Bits 31:30 of the start time (seconds field), for q14"	RW	0x0
29:0	start_time_nsec	"Bits 29:0 of the start time (nanoseconds field), for q14"	RW	0x0000 0000

Table 11-456: enst_start_time_q15

Register Information				
Description	"This register sets the absolute time at which queue q15 starts EnST transmit scheduling"			
Offset	0x081C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	start_time_sec	"Bits 31:30 of the start time (seconds field), for q15"	RW	0x0
29:0	start_time_nsec	"Bits 29:0 of the start time (nanoseconds field), for q15"	RW	0x0000 0000

Table 11-457: enst_on_time_q8

Register Information				
Description	"This register sets the time period for which queue q8 is to be open, (on_time_nsec), this register has to be written according to the speed mode as: ||1Gbps or 2.5Gbps|| : on_time = on_time_nsec/8 ||100Mbps|| : on_time = on_time_nsec/80 ||10Mbps|| : on_time = on_time_nsec/800"			
Offset	0x0820	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:17	reserved_31_17	"Reserved, read as 0, ignored on write."	RO	0x0000
16:0	on_time	"on_time, for q8"	RW	0x1 FFFF

Table 11-458: enst_on_time_q9

Register Information			
Description	"This register sets the time period for which queue q9 is to be open, (on_time_nsec), this register has to be written according to the speed mode as: ||1Gbps or 2.5Gbps|| : on_time = on_time_nsec/8 ||100Mbps|| : on_time = on_time_nsec/80 ||10Mbps|| : on_time = on_time_nsec/800"		
Offset	0x0824	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:17	reserved_31_17	"Reserved, read as 0, ignored on write."	RO	0x0000
16:0	on_time	"on_time, for q9"	RW	0x1 FFFF

Table 11-459: enst_on_time_q10

Register Information			
Description	"This register sets the time period for which queue q10 is to be open, (on_time_nsec), this register has to be written according to the speed mode as: ||1Gbps or 2.5Gbps|| : on_time = on_time_nsec/8 ||100Mbps|| : on_time = on_time_nsec/80 ||10Mbps|| : on_time = on_time_nsec/800"		
Offset	0x0828	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:17	reserved_31_17	"Reserved, read as 0, ignored on write."	RO	0x0000
16:0	on_time	"on_time, for q10"	RW	0x1 FFFF

Table 11-460: enst_on_time_q11

Register Information			
Description	"This register sets the time period for which queue q11 is to be open, (on_time_nsec), this register has to be written according to the speed mode as: ||1Gbps or 2.5Gbps|| : on_time = on_time_nsec/8 ||100Mbps|| : on_time = on_time_nsec/80 ||10Mbps|| : on_time = on_time_nsec/800"		
Offset	0x082C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:17	reserved_31_17	"Reserved, read as 0, ignored on write."	RO	0x0000
16:0	on_time	"on_time, for q11"	RW	0x1 FFFF

Table 11-461: enst_on_time_q12

Register Information			
Description	"This register sets the time period for which queue q12 is to be open, (on_time_nsec), this register has to be written according to the speed mode as: ||1Gbps or 2.5Gbps|| : on_time = on_time_nsec/8 ||100Mbps|| : on_time = on_time_nsec/80 ||10Mbps|| : on_time = on_time_nsec/800"		
Offset	0x0830	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:17	reserved_31_17	"Reserved, read as 0, ignored on write."	RO	0x0000
16:0	on_time	"on_time, for q12"	RW	0x1 FFFF

Table 11-462: enst_on_time_q13

Register Information			
Description	"This register sets the time period for which queue q13 is to be open, (on_time_nsec), this register has to be written according to the speed mode as: ||1Gbps or 2.5Gbps|| : on_time = on_time_nsec/8 ||100Mbps|| : on_time = on_time_nsec/80 ||10Mbps|| : on_time = on_time_nsec/800"		
Offset	0x0834	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:17	reserved_31_17	"Reserved, read as 0, ignored on write."	RO	0x0000
16:0	on_time	"on_time, for q13"	RW	0x1 FFFF

Table 11-463: enst_on_time_q14

Register Information			
Description	"This register sets the time period for which queue q14 is to be open, (on_time_nsec), this register has to be written according to the speed mode as: ||1Gbps or 2.5Gbps|| : on_time = on_time_nsec/8 ||100Mbps|| : on_time = on_time_nsec/80 ||10Mbps|| : on_time = on_time_nsec/800"		
Offset	0x0838	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:17	reserved_31_17	"Reserved, read as 0, ignored on write."	RO	0x0000
16:0	on_time	"on_time, for q14"	RW	0x1 FFFF

Table 11-464: enst_on_time_q15

Register Information			
Description	"This register sets the time period for which queue q15 is to be open, (on_time_nsec), this register has to be written according to the speed mode as: ||1Gbps or 2.5Gbps|| : on_time = on_time_nsec/8 ||100Mbps|| : on_time = on_time_nsec/80 ||10Mbps|| : on_time = on_time_nsec/800"		
Offset	0x083C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:17	reserved_31_17	"Reserved, read as 0, ignored on write."	RO	0x0000
16:0	on_time	"on_time, for q15"	RW	0x1 FFFF

Table 11-465: enst_off_time_q8

Register Information			
Description	"This register sets the time period for which queue q8 is to be blocked, (off_time_nsec), this register has to be written according to the speed mode as: ||1Gbps or 2.5Gbps|| => off_time = off_time_nsec/8 ||100Mbps|| => off_time = off_time_nsec/80 ||10Mbps|| => off_time = off_time_nsec/800"		
Offset	0x0840	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:17	reserved_31_17	"Reserved, read as 0, ignored on write."	RO	0x0000
16:0	off_time	"Off_time, for q8"	RW	0x0 0000

Table 11-466: enst_off_time_q9

Register Information			
Description	"This register sets the time period for which queue q9 is to be blocked, (off_time_nsec), this register has to be written according to the speed mode as: ||1Gbps or 2.5Gbps|| => off_time = off_time_nsec/8 ||100Mbps|| => off_time = off_time_nsec/80 ||10Mbps|| => off_time = off_time_nsec/800"		
Offset	0x0844	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:17	reserved_31_17	"Reserved, read as 0, ignored on write."	RO	0x0000
16:0	off_time	"Off_time, for q9"	RW	0x0 0000

Table 11-467: enst_off_time_q10

Register Information					
Description	"This register sets the time period for which queue q10 is to be blocked, (off_time_nsec), this register has to be written according to the speed mode as: ||1Gbps or 2.5Gbps|| => off_time = off_time_nsec/8 ||100Mbps|| => off_time = off_time_nsec/80 ||10Mbps|| => off_time = off_time_nsec/800"				
Offset	0x0848		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:17	reserved_31_17	"Reserved, read as 0, ignored on write."		RO	0x0000
16:0	off_time	"Off_time, for q10"		RW	0x0 0000

Table 11-468: enst_off_time_q11

Register Information					
Description	"This register sets the time period for which queue q11 is to be blocked, (off_time_nsec), this register has to be written according to the speed mode as: ||1Gbps or 2.5Gbps|| => off_time = off_time_nsec/8 ||100Mbps|| => off_time = off_time_nsec/80 ||10Mbps|| => off_time = off_time_nsec/800"				
Offset	0x084C		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:17	reserved_31_17	"Reserved, read as 0, ignored on write."		RO	0x0000
16:0	off_time	"Off_time, for q11"		RW	0x0 0000

Table 11-469: enst_off_time_q12

Register Information					
Description	"This register sets the time period for which queue q12 is to be blocked, (off_time_nsec), this register has to be written according to the speed mode as: ||1Gbps or 2.5Gbps|| => off_time = off_time_nsec/8 ||100Mbps|| => off_time = off_time_nsec/80 ||10Mbps|| => off_time = off_time_nsec/800"				
Offset	0x0850		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:17	reserved_31_17	"Reserved, read as 0, ignored on write."		RO	0x0000
16:0	off_time	"Off_time, for q12"		RW	0x0 0000

Table 11-470: enst_off_time_q13

Register Information					
Description	"This register sets the time period for which queue q13 is to be blocked, (off_time_nsec), this register has to be written according to the speed mode as: ||1Gbps or 2.5Gbps|| => off_time = off_time_nsec/8 ||100Mbps|| => off_time = off_time_nsec/80 ||10Mbps|| => off_time = off_time_nsec/800"				
Offset	0x0854		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:17	reserved_31_17	"Reserved, read as 0, ignored on write."		RO	0x0000
16:0	off_time	"Off_time, for q13"		RW	0x0 0000

Table 11-471: enst_off_time_q14

Register Information					
Description	"This register sets the time period for which queue q14 is to be blocked, (off_time_nsec), this register has to be written according to the speed mode as: ||1Gbps or 2.5Gbps|| => off_time = off_time_nsec/8 ||100Mbps|| => off_time = off_time_nsec/80 ||10Mbps|| => off_time = off_time_nsec/800"				
Offset	0x0858		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:17	reserved_31_17	"Reserved, read as 0, ignored on write."		RO	0x0000
16:0	off_time	"Off_time, for q14"		RW	0x0 0000

Table 11-472: enst_off_time_q15

Register Information					
Description	"This register sets the time period for which queue q15 is to be blocked, (off_time_nsec), this register has to be written according to the speed mode as: ||1Gbps or 2.5Gbps|| => off_time = off_time_nsec/8 ||100Mbps|| => off_time = off_time_nsec/80 ||10Mbps|| => off_time = off_time_nsec/800"				
Offset	0x085C		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:17	reserved_31_17	"Reserved, read as 0, ignored on write."		RO	0x0000
16:0	off_time	"Off_time, for q15"		RW	0x0 0000

Table 11-473: enst_control

Register Information				
Description	"Enhancement for Scheduled Traffic control register. EnST scheduling can only be applied to a maximum number of 8 queues. If 802.3br operation has been configured and both an eMAC and pMAC are present then the eMAC only supports a single queue and EnST is enabled on the eMAC by writing to bit 0 of emac_enst_control."			
Offset	0x0880	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:8	reserved_31_8	"Reserved, read as 0, ignored on write."	RO	0x00 0000
7	enst_enable_q15	"Set to enable enhanced scheduler traffic (ENST), for q15"	RW	0
6	enst_enable_q14	"Set to enable enhanced scheduler traffic (ENST), for q14"	RW	0
5	enst_enable_q13	"Set to enable enhanced scheduler traffic (ENST), for q13"	RW	0
4	enst_enable_q12	"Set to enable enhanced scheduler traffic (ENST), for q12"	RW	0
3	enst_enable_q11	"Set to enable enhanced scheduler traffic (ENST), for q11"	RW	0
2	enst_enable_q10	"Set to enable enhanced scheduler traffic (ENST), for q10"	RW	0
1	enst_enable_q9	"Set to enable enhanced scheduler traffic (ENST), for q9"	RW	0
0	enst_enable_q8	"Set to enable enhanced scheduler traffic (ENST), for q8"	RW	0

Table 11-474: frer_timeout

Register Information				
Description	"FRER timeout register. This determines when the sequence recovery reset timers are reset and is used by all of the configured CB streams. It is programmed as a count of 8192 rx_clk periods. 8192 rx_clk periods is 65.536 microseconds at gigabit speeds and 327.68 microseconds at 100M speeds, this allows a max timeout value of about 4 seconds at gigabit speeds - for test purposes if bit 12 (retry_test) is set in the network configuration register at 0x004 then the timeout value just becomes a count of rx_clk periods (ie a speed-up of 8192 in the time-out process)."			
Offset	0x08A0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	timeout	"Count of 8192 rx_clk periods."	RW	0x0000

Table 11-475: frer_red_tag

Register Information			
Description	"FRER redundancy tag register. This defines the Ethertype used to identify the R-TAG (redundancy tag) and contains a control bit to enable stripping of the R-TAG from received frames. IEEE 802.1CB has defined the value of this Ethertype to be 0xF1C1. Whether or not a particular stream uses the redundancy tag to locate the sequence number is determined by bit 28 in that streams control register."		
Offset	0x08A4	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	strip_r_tag	"If set the redundancy tag is stripped from received frames. If this bit is set then the receive octet counters reflect post deletion frame size so the FRER functionality is transparent to higher level management if the stripping function is enabled. If the statistics counters need to reflect the actual number of octets received, then the stripping functionality should be disabled."	RW	0
30	six_byte_tag	"Draft 2.4 and earlier drafts of the 802.1CB standard defined a four-byte redundancy tag. Drafts 2.5 and later specified a six-byte tag. Releases 1p10 and 1p11 were implemented in accordance with draft 2.4 of the standard. Set this bit to zero to inter-operate with implementations that use a four-byte tag."	RW	1
29:16	reserved_29_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	redundancy_tag	"Ethertype value used to identify the redundancy tag (R-TAG)."	RW	0xF1C1

Table 11-476: frer_control_1_a

Register Information			
Description	"FRER control register A"		
Offset	0x08C0	**Type:**	RW

Bits	Name	Description	Access	Reset
31	en_elimination	"Enable 802.1CB elimination of received frames"	RW	0
30	en_vector_rec_alg	"Enable 802.1CB vector recovery algorithm 0 enables the match recovery algorithm 1 enables the vector recovery algorithm This bit should only be changed when bit 31 en_elimination is low"	RW	0
29	en_seqrecrst_timer	"Enable 802.1CB sequence recovery reset timer - this bit may be changed when bit 31 en_elimination is low"	RW	0
28	use_r_tag	"Set to one to use redundancy tag to identify sequence number otherwise use offset value to identify bottom of sequence number - this bit should only be changed when bit 31 en_elimination is low"	RW	0
27:17	reserved_27_17	"Reserved, read as 0, ignored on write."	RO	0x000
16:8	offset_value	"Offset value in bytes from the start packet delimiter to the most significant byte of the 802.1CB sequence number - 9 bits allow a maximum value of 511 - allowing a TCP sequence number to be used with IPv6 - this bit should only be changed when en_elimination is low"	RW	0x000
7:4	member_stream_2	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0
3:0	member_stream_1	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0

Table 11-477: frer_control_1_b

Register Information			
Description	"FRER control register B. This register has default values where the sequence number length defaults to 16 and the vector recovery window defaults to the size of the history vector. It is not expected that the user will need to change these default values."		
Offset	0x08C4	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:13	reserved_31_13	"Reserved, read as 0, ignored on write."	RO	0x0 0000
12:8	seq_num_length	"Number of significant bits of the 802.1CB sequence number. The value 0x00 or numbers greater than 16 bits are equivalent to 16, but the number written will still be read back if it is greater than 16. If a value of less than 16 is written, then the sequence recovery algorithm will only consider that number of LSBs of the sequence number. The minimum size of seq_num_length has to be such that 2^seq_num_len is at least 2xgem_seq_history_len otherwise the vector recovery algorithm will not work correctly - this bit should only be changed when en_elimination is low"	RW	0x00
7:6	reserved_7_6	"Reserved, read as 0, ignored on write."	RO	0x0
5:0	seq_rec_window	"Vector recovery window, defines the window size used by the vector recovery algorithm to determine whether to reject a packet. Six bits allow a window size of 63, for effective operation of FRER users should not write a value greater than the gem_seq_history_len configuration define, a value of zero means the entire history vector is used - this bit should only be changed when en_elimination is low."	RW	0x00

Table 11-478: frer_statistics_1_a

Register Information			
Description	"FRER statistics register A. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x08C8	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:26	reserved_31_26	"Reserved, read as 0, ignored on write."	RO	0x00
25:16	vec_rec_rogue	"Count of number of frames dropped by the vector recovery algorithm for being out of range"	RO RtoClr	0x000
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	latent_errors	"Count of sequence numbers seen without a duplicate. The latent error count is updated when a frame is dropped from the history vector. So the update only happens after a new frame is received."	RO RtoClr	0x000

Table 11-479: frer_statistics_1_b

Register Information			
Description	"FRER statistics register B. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x08CC	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	reserved_31_24	"Reserved, read as 0, ignored on write."	RO	0x00
23:16	seqrst_count	"Count of number of times the sequence recovery reset timer decrements to zero."	RO RtoClr	0x00
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	out_of_order	"Count of out of order sequence numbers received. Incremented when a frame is accepted but the sequence number is not +1 of the highest stored value."	RO RtoClr	0x000

Table 11-480: frer_control_2_a

Register Information				
Description	"FRER control register A"			
Offset	0x08D0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	en_elimination	"Enable 802.1CB elimination of received frames"	RW	0
30	en_vector_rec_alg	"Enable 802.1CB vector recovery algorithm 0 enables the match recovery algorithm 1 enables the vector recovery algorithm This bit should only be changed when bit 31 en_elimination is low"	RW	0
29	en_seqrecrst_timer	"Enable 802.1CB sequence recovery reset timer - this bit may be changed when bit 31 en_elimination is low"	RW	0
28	use_r_tag	"Set to one to use redundancy tag to identify sequence number otherwise use offset value to identify bottom of sequence number - this bit should only be changed when bit 31 en_elimination is low"	RW	0
27:17	reserved_27_17	"Reserved, read as 0, ignored on write."	RO	0x000
16:8	offset_value	"Offset value in bytes from the start packet delimiter to the most significant byte of the 802.1CB sequence number - 9 bits allow a maximum value of 511 - allowing a TCP sequence number to be used with IPv6 - this bit should only be changed when en_elimination is low"	RW	0x000
7:4	member_stream_2	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0
3:0	member_stream_1	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0

Table 11-481: frer_control_2_b

Register Information			
Description	"FRER control register B. This register has default values where the sequence number length defaults to 16 and the vector recovery window defaults to the size of the history vector. It is not expected that the user will need to change these default values."		
Offset	0x08D4	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:13	reserved_31_13	"Reserved, read as 0, ignored on write."	RO	0x0 0000
12:8	seq_num_length	"Number of significant bits of the 802.1CB sequence number. The value 0x00 or numbers greater than 16 bits are equivalent to 16, but the number written will still be read back if it is greater than 16. If a value of less than 16 is written, then the sequence recovery algorithm will only consider that number of LSBs of the sequence number. The minimum size of seq_num_length has to be such that 2^seq_num_len is at least 2xgem_seq_history_len otherwise the vector recovery algorithm will not work correctly - this bit should only be changed when en_elimination is low"	RW	0x00
7:6	reserved_7_6	"Reserved, read as 0, ignored on write."	RO	0x0
5:0	seq_rec_window	"Vector recovery window, defines the window size used by the vector recovery algorithm to determine whether to reject a packet. Six bits allow a window size of 63, for effective operation of FRER users should not write a value greater than the gem_seq_history_len configuration define, a value of zero means the entire history vector is used - this bit should only be changed when en_elimination is low."	RW	0x00

Table 11-482: frer_statistics_2_a

Register Information			
Description	"FRER statistics register A. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x08D8	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:26	reserved_31_26	"Reserved, read as 0, ignored on write."	RO	0x00
25:16	vec_rec_rogue	"Count of number of frames dropped by the vector recovery algorithm for being out of range"	RO RtoClr	0x000
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	latent_errors	"Count of sequence numbers seen without a duplicate. The latent error count is updated when a frame is dropped from the history vector. So the update only happens after a new frame is received."	RO RtoClr	0x000

Table 11-483: frer_statistics_2_b

Register Information			
Description	"FRER statistics register B. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x08DC	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	reserved_31_24	"Reserved, read as 0, ignored on write."	RO	0x00
23:16	seqrst_count	"Count of number of times the sequence recovery reset timer decrements to zero."	RO RtoClr	0x00
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	out_of_order	"Count of out of order sequence numbers received. Incremented when a frame is accepted but the sequence number is not +1 of the highest stored value."	RO RtoClr	0x000

Table 11-484: frer_control_3_a

Register Information			
Description	"FRER control register A"		
Offset	0x08E0	**Type:**	RW

Bits	Name	Description	Access	Reset
31	en_elimination	"Enable 802.1CB elimination of received frames"	RW	0
30	en_vector_rec_alg	"Enable 802.1CB vector recovery algorithm 0 enables the match recovery algorithm 1 enables the vector recovery algorithm This bit should only be changed when bit 31 en_elimination is low"	RW	0
29	en_seqrecrst_timer	"Enable 802.1CB sequence recovery reset timer - this bit may be changed when bit 31 en_elimination is low"	RW	0
28	use_r_tag	"Set to one to use redundancy tag to identify sequence number otherwise use offset value to identify bottom of sequence number - this bit should only be changed when bit 31 en_elimination is low"	RW	0
27:17	reserved_27_17	"Reserved, read as 0, ignored on write."	RO	0x000
16:8	offset_value	"Offset value in bytes from the start packet delimiter to the most significant byte of the 802.1CB sequence number - 9 bits allow a maximum value of 511 - allowing a TCP sequence number to be used with IPv6 - this bit should only be changed when en_elimination is low"	RW	0x000
7:4	member_stream_2	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0
3:0	member_stream_1	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0

Table 11-485: frer_control_3_b

Register Information			
Description	"FRER control register B. This register has default values where the sequence number length defaults to 16 and the vector recovery window defaults to the size of the history vector. It is not expected that the user will need to change these default values."		
Offset	0x08E4	**Type:**	RW

Bits	Name	Description	Access	Reset
31:13	reserved_31_13	"Reserved, read as 0, ignored on write."	RO	0x0 0000
12:8	seq_num_length	"Number of significant bits of the 802.1CB sequence number. The value 0x00 or numbers greater than 16 bits are equivalent to 16, but the number written will still be read back if it is greater than 16. If a value of less than 16 is written, then the sequence recovery algorithm will only consider that number of LSBs of the sequence number. The minimum size of seq_num_length has to be such that 2^seq_num_len is at least 2xgem_seq_history_len otherwise the vector recovery algorithm will not work correctly - this bit should only be changed when en_elimination is low"	RW	0x00
7:6	reserved_7_6	"Reserved, read as 0, ignored on write."	RO	0x0
5:0	seq_rec_window	"Vector recovery window, defines the window size used by the vector recovery algorithm to determine whether to reject a packet. Six bits allow a window size of 63, for effective operation of FRER users should not write a value greater than the gem_seq_history_len configuration define, a value of zero means the entire history vector is used - this bit should only be changed when en_elimination is low."	RW	0x00

Table 11-486: frer_statistics_3_a

Register Information			
Description	"FRER statistics register A. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x08E8	**Type:**	RO

Bits	Name	Description	Access	Reset
31:26	reserved_31_26	"Reserved, read as 0, ignored on write."	RO	0x00
25:16	vec_rec_rogue	"Count of number of frames dropped by the vector recovery algorithm for being out of range"	RO RtoClr	0x000
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	latent_errors	"Count of sequence numbers seen without a duplicate. The latent error count is updated when a frame is dropped from the history vector. So the update only happens after a new frame is received."	RO RtoClr	0x000

Table 11-487: frer_statistics_3_b

Register Information			
Description	"FRER statistics register B. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x08EC	**Type:**	RO

Bits	Name	Description	Access	Reset
31:24	reserved_31_24	"Reserved, read as 0, ignored on write."	RO	0x00
23:16	seqrst_count	"Count of number of times the sequence recovery reset timer decrements to zero."	RO RtoClr	0x00
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	out_of_order	"Count of out of order sequence numbers received. Incremented when a frame is accepted but the sequence number is not +1 of the highest stored value."	RO RtoClr	0x000

Table 11-488: frer_control_4_a

Register Information			
Description	"FRER control register A"		
Offset	0x08F0	**Type:**	RW

Bits	Name	Description	Access	Reset
31	en_elimination	"Enable 802.1CB elimination of received frames"	RW	0
30	en_vector_rec_alg	"Enable 802.1CB vector recovery algorithm 0 enables the match recovery algorithm 1 enables the vector recovery algorithm This bit should only be changed when bit 31 en_elimination is low"	RW	0
29	en_seqrecrst_timer	"Enable 802.1CB sequence recovery reset timer - this bit may be changed when bit 31 en_elimination is low"	RW	0
28	use_r_tag	"Set to one to use redundancy tag to identify sequence number otherwise use offset value to identify bottom of sequence number - this bit should only be changed when bit 31 en_elimination is low"	RW	0
27:17	reserved_27_17	"Reserved, read as 0, ignored on write."	RO	0x000
16:8	offset_value	"Offset value in bytes from the start packet delimiter to the most significant byte of the 802.1CB sequence number - 9 bits allow a maximum value of 511 - allowing a TCP sequence number to be used with IPv6 - this bit should only be changed when en_elimination is low"	RW	0x000
7:4	member_stream_2	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0
3:0	member_stream_1	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0

Table 11-489: frer_control_4_b

Register Information			
Description	"FRER control register B. This register has default values where the sequence number length defaults to 16 and the vector recovery window defaults to the size of the history vector. It is not expected that the user will need to change these default values."		
Offset	0x08F4	**Type:**	RW

Bits	Name	Description	Access	Reset
31:13	reserved_31_13	"Reserved, read as 0, ignored on write."	RO	0x0 0000
12:8	seq_num_length	"Number of significant bits of the 802.1CB sequence number. The value 0x00 or numbers greater than 16 bits are equivalent to 16, but the number written will still be read back if it is greater than 16. If a value of less than 16 is written, then the sequence recovery algorithm will only consider that number of LSBs of the sequence number. The minimum size of seq_num_length has to be such that 2^seq_num_len is at least 2xgem_seq_history_len otherwise the vector recovery algorithm will not work correctly - this bit should only be changed when en_elimination is low"	RW	0x00
7:6	reserved_7_6	"Reserved, read as 0, ignored on write."	RO	0x0
5:0	seq_rec_window	"Vector recovery window, defines the window size used by the vector recovery algorithm to determine whether to reject a packet. Six bits allow a window size of 63, for effective operation of FRER users should not write a value greater than the gem_seq_history_len configuration define, a value of zero means the entire history vector is used - this bit should only be changed when en_elimination is low."	RW	0x00

Table 11-490: frer_statistics_4_a

Register Information			
Description	"FRER statistics register A. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x08F8	**Type:**	RO

Bits	Name	Description	Access	Reset
31:26	reserved_31_26	"Reserved, read as 0, ignored on write."	RO	0x00
25:16	vec_rec_rogue	"Count of number of frames dropped by the vector recovery algorithm for being out of range"	RO RtoClr	0x000
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	latent_errors	"Count of sequence numbers seen without a duplicate. The latent error count is updated when a frame is dropped from the history vector. So the update only happens after a new frame is received."	RO RtoClr	0x000

Table 11-491: frer_statistics_4_b

Register Information			
Description	"FRER statistics register B. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x08FC	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	reserved_31_24	"Reserved, read as 0, ignored on write."	RO	0x00
23:16	seqrst_count	"Count of number of times the sequence recovery reset timer decrements to zero."	RO RtoClr	0x00
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	out_of_order	"Count of out of order sequence numbers received. Incremented when a frame is accepted but the sequence number is not +1 of the highest stored value."	RO RtoClr	0x000

Table 11-492: frer_control_5_a

Register Information				
Description	"FRER control register A"			
Offset	0x0900	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	en_elimination	"Enable 802.1CB elimination of received frames"	RW	0
30	en_vector_rec_alg	"Enable 802.1CB vector recovery algorithm 0 enables the match recovery algorithm 1 enables the vector recovery algorithm This bit should only be changed when bit 31 en_elimination is low"	RW	0
29	en_seqrecrst_timer	"Enable 802.1CB sequence recovery reset timer - this bit may be changed when bit 31 en_elimination is low"	RW	0
28	use_r_tag	"Set to one to use redundancy tag to identify sequence number otherwise use offset value to identify bottom of sequence number - this bit should only be changed when bit 31 en_elimination is low"	RW	0
27:17	reserved_27_17	"Reserved, read as 0, ignored on write."	RO	0x000
16:8	offset_value	"Offset value in bytes from the start packet delimiter to the most significant byte of the 802.1CB sequence number - 9 bits allow a maximum value of 511 - allowing a TCP sequence number to be used with IPv6 - this bit should only be changed when en_elimination is low"	RW	0x000
7:4	member_stream_2	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0
3:0	member_stream_1	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0

Table 11-493: frer_control_5_b

Register Information				
Description	"FRER control register B. This register has default values where the sequence number length defaults to 16 and the vector recovery window defaults to the size of the history vector. It is not expected that the user will need to change these default values."			
Offset	0x0904	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:13	reserved_31_13	"Reserved, read as 0, ignored on write."	RO	0x0 0000
12:8	seq_num_length	"Number of significant bits of the 802.1CB sequence number. The value 0x00 or numbers greater than 16 bits are equivalent to 16, but the number written will still be read back if it is greater than 16. If a value of less than 16 is written, then the sequence recovery algorithm will only consider that number of LSBs of the sequence number. The minimum size of seq_num_length has to be such that 2^seq_num_len is at least 2xgem_seq_history_len otherwise the vector recovery algorithm will not work correctly - this bit should only be changed when en_elimination is low"	RW	0x00
7:6	reserved_7_6	"Reserved, read as 0, ignored on write."	RO	0x0
5:0	seq_rec_window	"Vector recovery window, defines the window size used by the vector recovery algorithm to determine whether to reject a packet. Six bits allow a window size of 63, for effective operation of FRER users should not write a value greater than the gem_seq_history_len configuration define, a value of zero means the entire history vector is used - this bit should only be changed when en_elimination is low."	RW	0x00

Table 11-494: frer_statistics_5_a

Register Information				
Description	"FRER statistics register A. This register is cleared on a read and does not roll over if its maximum value is reached."			
Offset	0x0908	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:26	reserved_31_26	"Reserved, read as 0, ignored on write."	RO	0x00
25:16	vec_rec_rogue	"Count of number of frames dropped by the vector recovery algorithm for being out of range"	RO RtoClr	0x000
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	latent_errors	"Count of sequence numbers seen without a duplicate. The latent error count is updated when a frame is dropped from the history vector. So the update only happens after a new frame is received."	RO RtoClr	0x000

Table 11-495: frer_statistics_5_b

Register Information			
Description	"FRER statistics register B. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x090C	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	reserved_31_24	"Reserved, read as 0, ignored on write."	RO	0x00
23:16	seqrst_count	"Count of number of times the sequence recovery reset timer decrements to zero."	RO RtoClr	0x00
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	out_of_order	"Count of out of order sequence numbers received. Incremented when a frame is accepted but the sequence number is not +1 of the highest stored value."	RO RtoClr	0x000

Table 11-496: frer_control_6_a

Register Information			
Description	"FRER control register A"		
Offset	0x0910	**Type:**	RW

Bits	Name	Description	Access	Reset
31	en_elimination	"Enable 802.1CB elimination of received frames"	RW	0
30	en_vector_rec_alg	"Enable 802.1CB vector recovery algorithm 0 enables the match recovery algorithm 1 enables the vector recovery algorithm This bit should only be changed when bit 31 en_elimination is low"	RW	0
29	en_seqrecrst_timer	"Enable 802.1CB sequence recovery reset timer - this bit may be changed when bit 31 en_elimination is low"	RW	0
28	use_r_tag	"Set to one to use redundancy tag to identify sequence number otherwise use offset value to identify bottom of sequence number - this bit should only be changed when bit 31 en_elimination is low"	RW	0
27:17	reserved_27_17	"Reserved, read as 0, ignored on write."	RO	0x000
16:8	offset_value	"Offset value in bytes from the start packet delimiter to the most significant byte of the 802.1CB sequence number - 9 bits allow a maximum value of 511 - allowing a TCP sequence number to be used with IPv6 - this bit should only be changed when en_elimination is low"	RW	0x000
7:4	member_stream_2	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0
3:0	member_stream_1	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0

Table 11-497: frer_control_6_b

Register Information			
Description	"FRER control register B. This register has default values where the sequence number length defaults to 16 and the vector recovery window defaults to the size of the history vector. It is not expected that the user will need to change these default values."		
Offset	0x0914	**Type:**	RW

Bits	Name	Description	Access	Reset
31:13	reserved_31_13	"Reserved, read as 0, ignored on write."	RO	0x0 0000
12:8	seq_num_length	"Number of significant bits of the 802.1CB sequence number. The value 0x00 or numbers greater than 16 bits are equivalent to 16, but the number written will still be read back if it is greater than 16. If a value of less than 16 is written, then the sequence recovery algorithm will only consider that number of LSBs of the sequence number. The minimum size of seq_num_length has to be such that 2^seq_num_len is at least 2xgem_seq_history_len otherwise the vector recovery algorithm will not work correctly - this bit should only be changed when en_elimination is low"	RW	0x00
7:6	reserved_7_6	"Reserved, read as 0, ignored on write."	RO	0x0
5:0	seq_rec_window	"Vector recovery window, defines the window size used by the vector recovery algorithm to determine whether to reject a packet. Six bits allow a window size of 63, for effective operation of FRER users should not write a value greater than the gem_seq_history_len configuration define, a value of zero means the entire history vector is used - this bit should only be changed when en_elimination is low."	RW	0x00

Table 11-498: frer_statistics_6_a

Register Information			
Description	"FRER statistics register A. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x0918	**Type:**	RO

Bits	Name	Description	Access	Reset
31:26	reserved_31_26	"Reserved, read as 0, ignored on write."	RO	0x00
25:16	vec_rec_rogue	"Count of number of frames dropped by the vector recovery algorithm for being out of range"	RO RtoClr	0x000
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	latent_errors	"Count of sequence numbers seen without a duplicate. The latent error count is updated when a frame is dropped from the history vector. So the update only happens after a new frame is received."	RO RtoClr	0x000

Table 11-499: frer_statistics_6_b

Register Information			
Description	"FRER statistics register B. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x091C	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	reserved_31_24	"Reserved, read as 0, ignored on write."	RO	0x00
23:16	seqrst_count	"Count of number of times the sequence recovery reset timer decrements to zero."	RO RtoClr	0x00
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	out_of_order	"Count of out of order sequence numbers received. Incremented when a frame is accepted but the sequence number is not +1 of the highest stored value."	RO RtoClr	0x000

Table 11-500: frer_control_7_a

Register Information			
Description	"FRER control register A"		
Offset	0x0920	**Type:**	RW

Bits	Name	Description	Access	Reset
31	en_elimination	"Enable 802.1CB elimination of received frames"	RW	0
30	en_vector_rec_alg	"Enable 802.1CB vector recovery algorithm 0 enables the match recovery algorithm 1 enables the vector recovery algorithm This bit should only be changed when bit 31 en_elimination is low"	RW	0
29	en_seqrecrst_timer	"Enable 802.1CB sequence recovery reset timer - this bit may be changed when bit 31 en_elimination is low"	RW	0
28	use_r_tag	"Set to one to use redundancy tag to identify sequence number otherwise use offset value to identify bottom of sequence number - this bit should only be changed when bit 31 en_elimination is low"	RW	0
27:17	reserved_27_17	"Reserved, read as 0, ignored on write."	RO	0x000
16:8	offset_value	"Offset value in bytes from the start packet delimiter to the most significant byte of the 802.1CB sequence number - 9 bits allow a maximum value of 511 - allowing a TCP sequence number to be used with IPv6 - this bit should only be changed when en_elimination is low"	RW	0x000
7:4	member_stream_2	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0
3:0	member_stream_1	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0

Table 11-501: frer_control_7_b

Register Information				
Description	"FRER control register B. This register has default values where the sequence number length defaults to 16 and the vector recovery window defaults to the size of the history vector. It is not expected that the user will need to change these default values."			
Offset	0x0924	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:13	reserved_31_13	"Reserved, read as 0, ignored on write."	RO	0x0 0000
12:8	seq_num_length	"Number of significant bits of the 802.1CB sequence number. The value 0x00 or numbers greater than 16 bits are equivalent to 16, but the number written will still be read back if it is greater than 16. If a value of less than 16 is written, then the sequence recovery algorithm will only consider that number of LSBs of the sequence number. The minimum size of seq_num_length has to be such that 2^seq_num_len is at least 2xgem_seq_history_len otherwise the vector recovery algorithm will not work correctly - this bit should only be changed when en_elimination is low"	RW	0x00
7:6	reserved_7_6	"Reserved, read as 0, ignored on write."	RO	0x0
5:0	seq_rec_window	"Vector recovery window, defines the window size used by the vector recovery algorithm to determine whether to reject a packet. Six bits allow a window size of 63, for effective operation of FRER users should not write a value greater than the gem_seq_history_len configuration define, a value of zero means the entire history vector is used - this bit should only be changed when en_elimination is low."	RW	0x00

Table 11-502: frer_statistics_7_a

Register Information				
Description	"FRER statistics register A. This register is cleared on a read and does not roll over if its maximum value is reached."			
Offset	0x0928	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:26	reserved_31_26	"Reserved, read as 0, ignored on write."	RO	0x00
25:16	vec_rec_rogue	"Count of number of frames dropped by the vector recovery algorithm for being out of range"	RO RtoClr	0x000
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	latent_errors	"Count of sequence numbers seen without a duplicate. The latent error count is updated when a frame is dropped from the history vector. So the update only happens after a new frame is received."	RO RtoClr	0x000

Table 11-503: frer_statistics_7_b

Register Information			
Description	"FRER statistics register B. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x092C	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	reserved_31_24	"Reserved, read as 0, ignored on write."	RO	0x00
23:16	seqrst_count	"Count of number of times the sequence recovery reset timer decrements to zero."	RO RtoClr	0x00
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	out_of_order	"Count of out of order sequence numbers received. Incremented when a frame is accepted but the sequence number is not +1 of the highest stored value."	RO RtoClr	0x000

Table 11-504: frer_control_8_a

Register Information				
Description	"FRER control register A"			
Offset	0x0930	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	en_elimination	"Enable 802.1CB elimination of received frames"	RW	0
30	en_vector_rec_alg	"Enable 802.1CB vector recovery algorithm 0 enables the match recovery algorithm 1 enables the vector recovery algorithm This bit should only be changed when bit 31 en_elimination is low"	RW	0
29	en_seqrecrst_timer	"Enable 802.1CB sequence recovery reset timer - this bit may be changed when bit 31 en_elimination is low"	RW	0
28	use_r_tag	"Set to one to use redundancy tag to identify sequence number otherwise use offset value to identify bottom of sequence number - this bit should only be changed when bit 31 en_elimination is low"	RW	0
27:17	reserved_27_17	"Reserved, read as 0, ignored on write."	RO	0x000
16:8	offset_value	"Offset value in bytes from the start packet delimiter to the most significant byte of the 802.1CB sequence number - 9 bits allow a maximum value of 511 - allowing a TCP sequence number to be used with IPv6 - this bit should only be changed when en_elimination is low"	RW	0x000
7:4	member_stream_2	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0
3:0	member_stream_1	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0

Table 11-505: frer_control_8_b

Register Information			
Description	"FRER control register B. This register has default values where the sequence number length defaults to 16 and the vector recovery window defaults to the size of the history vector. It is not expected that the user will need to change these default values."		
Offset	0x0934	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:13	reserved_31_13	"Reserved, read as 0, ignored on write."	RO	0x0 0000
12:8	seq_num_length	"Number of significant bits of the 802.1CB sequence number. The value 0x00 or numbers greater than 16 bits are equivalent to 16, but the number written will still be read back if it is greater than 16. If a value of less than 16 is written, then the sequence recovery algorithm will only consider that number of LSBs of the sequence number. The minimum size of seq_num_length has to be such that 2^seq_num_len is at least 2xgem_seq_history_len otherwise the vector recovery algorithm will not work correctly - this bit should only be changed when en_elimination is low"	RW	0x00
7:6	reserved_7_6	"Reserved, read as 0, ignored on write."	RO	0x0
5:0	seq_rec_window	"Vector recovery window, defines the window size used by the vector recovery algorithm to determine whether to reject a packet. Six bits allow a window size of 63, for effective operation of FRER users should not write a value greater than the gem_seq_history_len configuration define, a value of zero means the entire history vector is used - this bit should only be changed when en_elimination is low."	RW	0x00

Table 11-506: frer_statistics_8_a

Register Information			
Description	"FRER statistics register A. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x0938	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:26	reserved_31_26	"Reserved, read as 0, ignored on write."	RO	0x00
25:16	vec_rec_rogue	"Count of number of frames dropped by the vector recovery algorithm for being out of range"	RO RtoClr	0x000
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	latent_errors	"Count of sequence numbers seen without a duplicate. The latent error count is updated when a frame is dropped from the history vector. So the update only happens after a new frame is received."	RO RtoClr	0x000

Table 11-507: frer_statistics_8_b

Register Information			
Description	"FRER statistics register B. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x093C	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	reserved_31_24	"Reserved, read as 0, ignored on write."	RO	0x00
23:16	seqrst_count	"Count of number of times the sequence recovery reset timer decrements to zero."	RO RtoClr	0x00
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	out_of_order	"Count of out of order sequence numbers received. Incremented when a frame is accepted but the sequence number is not +1 of the highest stored value."	RO RtoClr	0x000

Table 11-508: frer_control_9_a

Register Information					
Description	"FRER control register A"				
Offset	0x0940		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31	en_elimination	"Enable 802.1CB elimination of received frames"		RW	0
30	en_vector_rec_alg	"Enable 802.1CB vector recovery algorithm 0 enables the match recovery algorithm 1 enables the vector recovery algorithm This bit should only be changed when bit 31 en_elimination is low"		RW	0
29	en_seqrecrst_timer	"Enable 802.1CB sequence recovery reset timer - this bit may be changed when bit 31 en_elimination is low"		RW	0
28	use_r_tag	"Set to one to use redundancy tag to identify sequence number otherwise use offset value to identify bottom of sequence number - this bit should only be changed when bit 31 en_elimination is low"		RW	0
27:17	reserved_27_17	"Reserved, read as 0, ignored on write."		RO	0x000
16:8	offset_value	"Offset value in bytes from the start packet delimiter to the most significant byte of the 802.1CB sequence number - 9 bits allow a maximum value of 511 - allowing a TCP sequence number to be used with IPv6 - this bit should only be changed when en_elimination is low"		RW	0x000
7:4	member_stream_2	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"		RW	0x0
3:0	member_stream_1	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"		RW	0x0

Table 11-509: frer_control_9_b

Register Information			
Description	"FRER control register B. This register has default values where the sequence number length defaults to 16 and the vector recovery window defaults to the size of the history vector. It is not expected that the user will need to change these default values."		
Offset	0x0944	**Type:**	RW

Bits	Name	Description	Access	Reset
31:13	reserved_31_13	"Reserved, read as 0, ignored on write."	RO	0x0 0000
12:8	seq_num_length	"Number of significant bits of the 802.1CB sequence number. The value 0x00 or numbers greater than 16 bits are equivalent to 16, but the number written will still be read back if it is greater than 16. If a value of less than 16 is written, then the sequence recovery algorithm will only consider that number of LSBs of the sequence number. The minimum size of seq_num_length has to be such that 2^seq_num_len is at least 2xgem_seq_history_len otherwise the vector recovery algorithm will not work correctly - this bit should only be changed when en_elimination is low"	RW	0x00
7:6	reserved_7_6	"Reserved, read as 0, ignored on write."	RO	0x0
5:0	seq_rec_window	"Vector recovery window, defines the window size used by the vector recovery algorithm to determine whether to reject a packet. Six bits allow a window size of 63, for effective operation of FRER users should not write a value greater than the gem_seq_history_len configuration define, a value of zero means the entire history vector is used - this bit should only be changed when en_elimination is low."	RW	0x00

Table 11-510: frer_statistics_9_a

Register Information			
Description	"FRER statistics register A. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x0948	**Type:**	RO

Bits	Name	Description	Access	Reset
31:26	reserved_31_26	"Reserved, read as 0, ignored on write."	RO	0x00
25:16	vec_rec_rogue	"Count of number of frames dropped by the vector recovery algorithm for being out of range"	RO RtoClr	0x000
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	latent_errors	"Count of sequence numbers seen without a duplicate. The latent error count is updated when a frame is dropped from the history vector. So the update only happens after a new frame is received."	RO RtoClr	0x000

Table 11-511: frer_statistics_9_b

Register Information			
Description	"FRER statistics register B. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x094C	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	reserved_31_24	"Reserved, read as 0, ignored on write."	RO	0x00
23:16	seqrst_count	"Count of number of times the sequence recovery reset timer decrements to zero."	RO RtoClr	0x00
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	out_of_order	"Count of out of order sequence numbers received. Incremented when a frame is accepted but the sequence number is not +1 of the highest stored value."	RO RtoClr	0x000

Table 11-512: frer_control_10_a

Register Information				
Description	"FRER control register A"			
Offset	0x0950	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	en_elimination	"Enable 802.1CB elimination of received frames"	RW	0
30	en_vector_rec_alg	"Enable 802.1CB vector recovery algorithm 0 enables the match recovery algorithm 1 enables the vector recovery algorithm This bit should only be changed when bit 31 en_elimination is low"	RW	0
29	en_seqrecrst_timer	"Enable 802.1CB sequence recovery reset timer - this bit may be changed when bit 31 en_elimination is low"	RW	0
28	use_r_tag	"Set to one to use redundancy tag to identify sequence number otherwise use offset value to identify bottom of sequence number - this bit should only be changed when bit 31 en_elimination is low"	RW	0
27:17	reserved_27_17	"Reserved, read as 0, ignored on write."	RO	0x000
16:8	offset_value	"Offset value in bytes from the start packet delimiter to the most significant byte of the 802.1CB sequence number - 9 bits allow a maximum value of 511 - allowing a TCP sequence number to be used with IPv6 - this bit should only be changed when en_elimination is low"	RW	0x000
7:4	member_stream_2	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0
3:0	member_stream_1	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0

Table 11-513: frer_control_10_b

Register Information					
Description	"FRER control register B. This register has default values where the sequence number length defaults to 16 and the vector recovery window defaults to the size of the history vector. It is not expected that the user will need to change these default values."				
Offset	0x0954		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:13	reserved_31_13	"Reserved, read as 0, ignored on write."	RO	0x0 0000	
12:8	seq_num_length	"Number of significant bits of the 802.1CB sequence number. The value 0x00 or numbers greater than 16 bits are equivalent to 16, but the number written will still be read back if it is greater than 16. If a value of less than 16 is written, then the sequence recovery algorithm will only consider that number of LSBs of the sequence number. The minimum size of seq_num_length has to be such that 2^seq_num_len is at least 2xgem_seq_history_len otherwise the vector recovery algorithm will not work correctly - this bit should only be changed when en_elimination is low"	RW	0x00	
7:6	reserved_7_6	"Reserved, read as 0, ignored on write."	RO	0x0	
5:0	seq_rec_window	"Vector recovery window, defines the window size used by the vector recovery algorithm to determine whether to reject a packet. Six bits allow a window size of 63, for effective operation of FRER users should not write a value greater than the gem_seq_history_len configuration define, a value of zero means the entire history vector is used - this bit should only be changed when en_elimination is low."	RW	0x00	

Table 11-514: frer_statistics_10_a

Register Information					
Description	"FRER statistics register A. This register is cleared on a read and does not roll over if its maximum value is reached."				
Offset	0x0958		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:26	reserved_31_26	"Reserved, read as 0, ignored on write."	RO	0x00	
25:16	vec_rec_rogue	"Count of number of frames dropped by the vector recovery algorithm for being out of range"	RO RtoClr	0x000	
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00	
9:0	latent_errors	"Count of sequence numbers seen without a duplicate. The latent error count is updated when a frame is dropped from the history vector. So the update only happens after a new frame is received."	RO RtoClr	0x000	

Table 11-515: frer_statistics_10_b

Register Information			
Description	"FRER statistics register B. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x095C	**Type:**	RO

Bitfield Details

Bits	Name	Description	Access	Reset
31:24	reserved_31_24	"Reserved, read as 0, ignored on write."	RO	0x00
23:16	seqrst_count	"Count of number of times the sequence recovery reset timer decrements to zero."	RO RtoClr	0x00
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	out_of_order	"Count of out of order sequence numbers received. Incremented when a frame is accepted but the sequence number is not +1 of the highest stored value."	RO RtoClr	0x000

Table 11-516: frer_control_11_a

Register Information				
Description	"FRER control register A"			
Offset	0x0960	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	en_elimination	"Enable 802.1CB elimination of received frames"	RW	0
30	en_vector_rec_alg	"Enable 802.1CB vector recovery algorithm 0 enables the match recovery algorithm 1 enables the vector recovery algorithm This bit should only be changed when bit 31 en_elimination is low"	RW	0
29	en_seqrecrst_timer	"Enable 802.1CB sequence recovery reset timer - this bit may be changed when bit 31 en_elimination is low"	RW	0
28	use_r_tag	"Set to one to use redundancy tag to identify sequence number otherwise use offset value to identify bottom of sequence number - this bit should only be changed when bit 31 en_elimination is low"	RW	0
27:17	reserved_27_17	"Reserved, read as 0, ignored on write."	RO	0x000
16:8	offset_value	"Offset value in bytes from the start packet delimiter to the most significant byte of the 802.1CB sequence number - 9 bits allow a maximum value of 511 - allowing a TCP sequence number to be used with IPv6 - this bit should only be changed when en_elimination is low"	RW	0x000
7:4	member_stream_2	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0
3:0	member_stream_1	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0

Table 11-517: frer_control_11_b

Register Information			
Description	"FRER control register B. This register has default values where the sequence number length defaults to 16 and the vector recovery window defaults to the size of the history vector. It is not expected that the user will need to change these default values."		
Offset	0x0964	**Type:**	RW

Bits	Name	Description	Access	Reset
31:13	reserved_31_13	"Reserved, read as 0, ignored on write."	RO	0x0 0000
12:8	seq_num_length	"Number of significant bits of the 802.1CB sequence number. The value 0x00 or numbers greater than 16 bits are equivalent to 16, but the number written will still be read back if it is greater than 16. If a value of less than 16 is written, then the sequence recovery algorithm will only consider that number of LSBs of the sequence number. The minimum size of seq_num_length has to be such that 2^seq_num_len is at least 2xgem_seq_history_len otherwise the vector recovery algorithm will not work correctly - this bit should only be changed when en_elimination is low"	RW	0x00
7:6	reserved_7_6	"Reserved, read as 0, ignored on write."	RO	0x0
5:0	seq_rec_window	"Vector recovery window, defines the window size used by the vector recovery algorithm to determine whether to reject a packet. Six bits allow a window size of 63, for effective operation of FRER users should not write a value greater than the gem_seq_history_len configuration define, a value of zero means the entire history vector is used - this bit should only be changed when en_elimination is low."	RW	0x00

Table 11-518: frer_statistics_11_a

Register Information			
Description	"FRER statistics register A. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x0968	**Type:**	RO

Bits	Name	Description	Access	Reset
31:26	reserved_31_26	"Reserved, read as 0, ignored on write."	RO	0x00
25:16	vec_rec_rogue	"Count of number of frames dropped by the vector recovery algorithm for being out of range"	RO RtoClr	0x000
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	latent_errors	"Count of sequence numbers seen without a duplicate. The latent error count is updated when a frame is dropped from the history vector. So the update only happens after a new frame is received."	RO RtoClr	0x000

Table 11-519: frer_statistics_11_b

Register Information			
Description	"FRER statistics register B. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x096C	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	reserved_31_24	"Reserved, read as 0, ignored on write."	RO	0x00
23:16	seqrst_count	"Count of number of times the sequence recovery reset timer decrements to zero."	RO RtoClr	0x00
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	out_of_order	"Count of out of order sequence numbers received. Incremented when a frame is accepted but the sequence number is not +1 of the highest stored value."	RO RtoClr	0x000

Table 11-520: frer_control_12_a

Register Information			
Description	"FRER control register A"		
Offset	0x0970	**Type:**	RW

Bits	Name	Description	Access	Reset
31	en_elimination	"Enable 802.1CB elimination of received frames"	RW	0
30	en_vector_rec_alg	"Enable 802.1CB vector recovery algorithm 0 enables the match recovery algorithm 1 enables the vector recovery algorithm This bit should only be changed when bit 31 en_elimination is low"	RW	0
29	en_seqrecrst_timer	"Enable 802.1CB sequence recovery reset timer - this bit may be changed when bit 31 en_elimination is low"	RW	0
28	use_r_tag	"Set to one to use redundancy tag to identify sequence number otherwise use offset value to identify bottom of sequence number - this bit should only be changed when bit 31 en_elimination is low"	RW	0
27:17	reserved_27_17	"Reserved, read as 0, ignored on write."	RO	0x000
16:8	offset_value	"Offset value in bytes from the start packet delimiter to the most significant byte of the 802.1CB sequence number - 9 bits allow a maximum value of 511 - allowing a TCP sequence number to be used with IPv6 - this bit should only be changed when en_elimination is low"	RW	0x000
7:4	member_stream_2	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0
3:0	member_stream_1	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0

Table 11-521: frer_control_12_b

Register Information			
Description	"FRER control register B. This register has default values where the sequence number length defaults to 16 and the vector recovery window defaults to the size of the history vector. It is not expected that the user will need to change these default values."		
Offset	0x0974	**Type:**	RW

Bits	Name	Description	Access	Reset
31:13	reserved_31_13	"Reserved, read as 0, ignored on write."	RO	0x0 0000
12:8	seq_num_length	"Number of significant bits of the 802.1CB sequence number. The value 0x00 or numbers greater than 16 bits are equivalent to 16, but the number written will still be read back if it is greater than 16. If a value of less than 16 is written, then the sequence recovery algorithm will only consider that number of LSBs of the sequence number. The minimum size of seq_num_length has to be such that 2^seq_num_len is at least 2xgem_seq_history_len otherwise the vector recovery algorithm will not work correctly - this bit should only be changed when en_elimination is low"	RW	0x00
7:6	reserved_7_6	"Reserved, read as 0, ignored on write."	RO	0x0
5:0	seq_rec_window	"Vector recovery window, defines the window size used by the vector recovery algorithm to determine whether to reject a packet. Six bits allow a window size of 63, for effective operation of FRER users should not write a value greater than the gem_seq_history_len configuration define, a value of zero means the entire history vector is used - this bit should only be changed when en_elimination is low."	RW	0x00

Table 11-522: frer_statistics_12_a

Register Information			
Description	"FRER statistics register A. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x0978	**Type:**	RO

Bits	Name	Description	Access	Reset
31:26	reserved_31_26	"Reserved, read as 0, ignored on write."	RO	0x00
25:16	vec_rec_rogue	"Count of number of frames dropped by the vector recovery algorithm for being out of range"	RO RtoClr	0x000
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	latent_errors	"Count of sequence numbers seen without a duplicate. The latent error count is updated when a frame is dropped from the history vector. So the update only happens after a new frame is received."	RO RtoClr	0x000

Table 11-523: frer_statistics_12_b

Register Information			
Description	"FRER statistics register B. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x097C	**Type:**	RO

Bitfield Details

Bits	Name	Description	Access	Reset
31:24	reserved_31_24	"Reserved, read as 0, ignored on write."	RO	0x00
23:16	seqrst_count	"Count of number of times the sequence recovery reset timer decrements to zero."	RO RtoClr	0x00
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	out_of_order	"Count of out of order sequence numbers received. Incremented when a frame is accepted but the sequence number is not +1 of the highest stored value."	RO RtoClr	0x000

Table 11-524: frer_control_13_a

Register Information			
Description	"FRER control register A"		
Offset	0x0980	**Type:**	RW

Bits	Name	Description	Access	Reset
31	en_elimination	"Enable 802.1CB elimination of received frames"	RW	0
30	en_vector_rec_alg	"Enable 802.1CB vector recovery algorithm 0 enables the match recovery algorithm 1 enables the vector recovery algorithm This bit should only be changed when bit 31 en_elimination is low"	RW	0
29	en_seqrecrst_timer	"Enable 802.1CB sequence recovery reset timer - this bit may be changed when bit 31 en_elimination is low"	RW	0
28	use_r_tag	"Set to one to use redundancy tag to identify sequence number otherwise use offset value to identify bottom of sequence number - this bit should only be changed when bit 31 en_elimination is low"	RW	0
27:17	reserved_27_17	"Reserved, read as 0, ignored on write."	RO	0x000
16:8	offset_value	"Offset value in bytes from the start packet delimiter to the most significant byte of the 802.1CB sequence number - 9 bits allow a maximum value of 511 - allowing a TCP sequence number to be used with IPv6 - this bit should only be changed when en_elimination is low"	RW	0x000
7:4	member_stream_2	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0
3:0	member_stream_1	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0

Table 11-525: frer_control_13_b

Register Information			
Description	"FRER control register B. This register has default values where the sequence number length defaults to 16 and the vector recovery window defaults to the size of the history vector. It is not expected that the user will need to change these default values."		
Offset	0x0984	**Type:**	RW

Bits	Name	Description	Access	Reset
31:13	reserved_31_13	"Reserved, read as 0, ignored on write."	RO	0x0 0000
12:8	seq_num_length	"Number of significant bits of the 802.1CB sequence number. The value 0x00 or numbers greater than 16 bits are equivalent to 16, but the number written will still be read back if it is greater than 16. If a value of less than 16 is written, then the sequence recovery algorithm will only consider that number of LSBs of the sequence number. The minimum size of seq_num_length has to be such that 2^seq_num_len is at least 2xgem_seq_history_len otherwise the vector recovery algorithm will not work correctly - this bit should only be changed when en_elimination is low"	RW	0x00
7:6	reserved_7_6	"Reserved, read as 0, ignored on write."	RO	0x0
5:0	seq_rec_window	"Vector recovery window, defines the window size used by the vector recovery algorithm to determine whether to reject a packet. Six bits allow a window size of 63, for effective operation of FRER users should not write a value greater than the gem_seq_history_len configuration define, a value of zero means the entire history vector is used - this bit should only be changed when en_elimination is low."	RW	0x00

Table 11-526: frer_statistics_13_a

Register Information			
Description	"FRER statistics register A. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x0988	**Type:**	RO

Bits	Name	Description	Access	Reset
31:26	reserved_31_26	"Reserved, read as 0, ignored on write."	RO	0x00
25:16	vec_rec_rogue	"Count of number of frames dropped by the vector recovery algorithm for being out of range"	RO RtoClr	0x000
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	latent_errors	"Count of sequence numbers seen without a duplicate. The latent error count is updated when a frame is dropped from the history vector. So the update only happens after a new frame is received."	RO RtoClr	0x000

Table 11-527: frer_statistics_13_b

Register Information			
Description	"FRER statistics register B. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x098C	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	reserved_31_24	"Reserved, read as 0, ignored on write."	RO	0x00
23:16	seqrst_count	"Count of number of times the sequence recovery reset timer decrements to zero."	RO RtoClr	0x00
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	out_of_order	"Count of out of order sequence numbers received. Incremented when a frame is accepted but the sequence number is not +1 of the highest stored value."	RO RtoClr	0x000

Table 11-528: frer_control_14_a

Register Information				
Description	"FRER control register A"			
Offset	0x0990	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	en_elimination	"Enable 802.1CB elimination of received frames"	RW	0
30	en_vector_rec_alg	"Enable 802.1CB vector recovery algorithm 0 enables the match recovery algorithm 1 enables the vector recovery algorithm This bit should only be changed when bit 31 en_elimination is low"	RW	0
29	en_seqrecrst_timer	"Enable 802.1CB sequence recovery reset timer - this bit may be changed when bit 31 en_elimination is low"	RW	0
28	use_r_tag	"Set to one to use redundancy tag to identify sequence number otherwise use offset value to identify bottom of sequence number - this bit should only be changed when bit 31 en_elimination is low"	RW	0
27:17	reserved_27_17	"Reserved, read as 0, ignored on write."	RO	0x000
16:8	offset_value	"Offset value in bytes from the start packet delimiter to the most significant byte of the 802.1CB sequence number - 9 bits allow a maximum value of 511 - allowing a TCP sequence number to be used with IPv6 - this bit should only be changed when en_elimination is low"	RW	0x000
7:4	member_stream_2	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0
3:0	member_stream_1	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0

Table 11-529: frer_control_14_b

Register Information			
Description	"FRER control register B. This register has default values where the sequence number length defaults to 16 and the vector recovery window defaults to the size of the history vector. It is not expected that the user will need to change these default values."		
Offset	0x0994	**Type:**	RW

Bits	Name	Description	Access	Reset
31:13	reserved_31_13	"Reserved, read as 0, ignored on write."	RO	0x0 0000
12:8	seq_num_length	"Number of significant bits of the 802.1CB sequence number. The value 0x00 or numbers greater than 16 bits are equivalent to 16, but the number written will still be read back if it is greater than 16. If a value of less than 16 is written, then the sequence recovery algorithm will only consider that number of LSBs of the sequence number. The minimum size of seq_num_length has to be such that 2^seq_num_len is at least 2xgem_seq_history_len otherwise the vector recovery algorithm will not work correctly - this bit should only be changed when en_elimination is low"	RW	0x00
7:6	reserved_7_6	"Reserved, read as 0, ignored on write."	RO	0x0
5:0	seq_rec_window	"Vector recovery window, defines the window size used by the vector recovery algorithm to determine whether to reject a packet. Six bits allow a window size of 63, for effective operation of FRER users should not write a value greater than the gem_seq_history_len configuration define, a value of zero means the entire history vector is used - this bit should only be changed when en_elimination is low."	RW	0x00

Table 11-530: frer_statistics_14_a

Register Information			
Description	"FRER statistics register A. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x0998	**Type:**	RO

Bits	Name	Description	Access	Reset
31:26	reserved_31_26	"Reserved, read as 0, ignored on write."	RO	0x00
25:16	vec_rec_rogue	"Count of number of frames dropped by the vector recovery algorithm for being out of range"	RO RtoClr	0x000
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	latent_errors	"Count of sequence numbers seen without a duplicate. The latent error count is updated when a frame is dropped from the history vector. So the update only happens after a new frame is received."	RO RtoClr	0x000

Table 11-531: frer_statistics_14_b

Register Information			
Description	"FRER statistics register B. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x099C	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	reserved_31_24	"Reserved, read as 0, ignored on write."	RO	0x00
23:16	seqrst_count	"Count of number of times the sequence recovery reset timer decrements to zero."	RO RtoClr	0x00
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	out_of_order	"Count of out of order sequence numbers received. Incremented when a frame is accepted but the sequence number is not +1 of the highest stored value."	RO RtoClr	0x000

Table 11-532: frer_control_15_a

Register Information			
Description	"FRER control register A"		
Offset	0x09A0	**Type:**	RW

Bits	Name	Description	Access	Reset
31	en_elimination	"Enable 802.1CB elimination of received frames"	RW	0
30	en_vector_rec_alg	"Enable 802.1CB vector recovery algorithm 0 enables the match recovery algorithm 1 enables the vector recovery algorithm This bit should only be changed when bit 31 en_elimination is low"	RW	0
29	en_seqrecrst_timer	"Enable 802.1CB sequence recovery reset timer - this bit may be changed when bit 31 en_elimination is low"	RW	0
28	use_r_tag	"Set to one to use redundancy tag to identify sequence number otherwise use offset value to identify bottom of sequence number - this bit should only be changed when bit 31 en_elimination is low"	RW	0
27:17	reserved_27_17	"Reserved, read as 0, ignored on write."	RO	0x000
16:8	offset_value	"Offset value in bytes from the start packet delimiter to the most significant byte of the 802.1CB sequence number - 9 bits allow a maximum value of 511 - allowing a TCP sequence number to be used with IPv6 - this bit should only be changed when en_elimination is low"	RW	0x000
7:4	member_stream_2	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0
3:0	member_stream_1	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0

Table 11-533: frer_control_15_b

Register Information			
Description	"FRER control register B. This register has default values where the sequence number length defaults to 16 and the vector recovery window defaults to the size of the history vector. It is not expected that the user will need to change these default values."		
Offset	0x09A4	**Type:**	RW

Bits	Name	Description	Access	Reset
31:13	reserved_31_13	"Reserved, read as 0, ignored on write."	RO	0x0 0000
12:8	seq_num_length	"Number of significant bits of the 802.1CB sequence number. The value 0x00 or numbers greater than 16 bits are equivalent to 16, but the number written will still be read back if it is greater than 16. If a value of less than 16 is written, then the sequence recovery algorithm will only consider that number of LSBs of the sequence number. The minimum size of seq_num_length has to be such that 2^seq_num_len is at least 2xgem_seq_history_len otherwise the vector recovery algorithm will not work correctly - this bit should only be changed when en_elimination is low"	RW	0x00
7:6	reserved_7_6	"Reserved, read as 0, ignored on write."	RO	0x0
5:0	seq_rec_window	"Vector recovery window, defines the window size used by the vector recovery algorithm to determine whether to reject a packet. Six bits allow a window size of 63, for effective operation of FRER users should not write a value greater than the gem_seq_history_len configuration define, a value of zero means the entire history vector is used - this bit should only be changed when en_elimination is low."	RW	0x00

Table 11-534: frer_statistics_15_a

Register Information			
Description	"FRER statistics register A. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x09A8	**Type:**	RO

Bits	Name	Description	Access	Reset
31:26	reserved_31_26	"Reserved, read as 0, ignored on write."	RO	0x00
25:16	vec_rec_rogue	"Count of number of frames dropped by the vector recovery algorithm for being out of range"	RO RtoClr	0x000
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	latent_errors	"Count of sequence numbers seen without a duplicate. The latent error count is updated when a frame is dropped from the history vector. So the update only happens after a new frame is received."	RO RtoClr	0x000

Table 11-535: frer_statistics_15_b

Register Information			
Description	"FRER statistics register B. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x09AC	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	reserved_31_24	"Reserved, read as 0, ignored on write."	RO	0x00
23:16	seqrst_count	"Count of number of times the sequence recovery reset timer decrements to zero."	RO RtoClr	0x00
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	out_of_order	"Count of out of order sequence numbers received. Incremented when a frame is accepted but the sequence number is not +1 of the highest stored value."	RO RtoClr	0x000

Table 11-536: frer_control_16_a

Register Information				
Description	"FRER control register A"			
Offset	0x09B0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	en_elimination	"Enable 802.1CB elimination of received frames"	RW	0
30	en_vector_rec_alg	"Enable 802.1CB vector recovery algorithm 0 enables the match recovery algorithm 1 enables the vector recovery algorithm This bit should only be changed when bit 31 en_elimination is low"	RW	0
29	en_seqrecrst_timer	"Enable 802.1CB sequence recovery reset timer - this bit may be changed when bit 31 en_elimination is low"	RW	0
28	use_r_tag	"Set to one to use redundancy tag to identify sequence number otherwise use offset value to identify bottom of sequence number - this bit should only be changed when bit 31 en_elimination is low"	RW	0
27:17	reserved_27_17	"Reserved, read as 0, ignored on write."	RO	0x000
16:8	offset_value	"Offset value in bytes from the start packet delimiter to the most significant byte of the 802.1CB sequence number - 9 bits allow a maximum value of 511 - allowing a TCP sequence number to be used with IPv6 - this bit should only be changed when en_elimination is low"	RW	0x000
7:4	member_stream_2	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0
3:0	member_stream_1	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0

Table 11-537: frer_control_16_b

Register Information			
Description	"FRER control register B. This register has default values where the sequence number length defaults to 16 and the vector recovery window defaults to the size of the history vector. It is not expected that the user will need to change these default values."		
Offset	0x09B4	**Type:**	RW

Bits	Name	Description	Access	Reset
31:13	reserved_31_13	"Reserved, read as 0, ignored on write."	RO	0x0 0000
12:8	seq_num_length	"Number of significant bits of the 802.1CB sequence number. The value 0x00 or numbers greater than 16 bits are equivalent to 16, but the number written will still be read back if it is greater than 16. If a value of less than 16 is written, then the sequence recovery algorithm will only consider that number of LSBs of the sequence number. The minimum size of seq_num_length has to be such that 2^seq_num_len is at least 2xgem_seq_history_len otherwise the vector recovery algorithm will not work correctly - this bit should only be changed when en_elimination is low"	RW	0x00
7:6	reserved_7_6	"Reserved, read as 0, ignored on write."	RO	0x0
5:0	seq_rec_window	"Vector recovery window, defines the window size used by the vector recovery algorithm to determine whether to reject a packet. Six bits allow a window size of 63, for effective operation of FRER users should not write a value greater than the gem_seq_history_len configuration define, a value of zero means the entire history vector is used - this bit should only be changed when en_elimination is low."	RW	0x00

Table 11-538: frer_statistics_16_a

Register Information			
Description	"FRER statistics register A. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x09B8	**Type:**	RO

Bits	Name	Description	Access	Reset
31:26	reserved_31_26	"Reserved, read as 0, ignored on write."	RO	0x00
25:16	vec_rec_rogue	"Count of number of frames dropped by the vector recovery algorithm for being out of range"	RO RtoClr	0x000
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	latent_errors	"Count of sequence numbers seen without a duplicate. The latent error count is updated when a frame is dropped from the history vector. So the update only happens after a new frame is received."	RO RtoClr	0x000

Table 11-539: frer_statistics_16_b

Register Information			
Description	"FRER statistics register B. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x09BC	**Type:**	RO

Bits	Name	Description	Access	Reset
31:24	reserved_31_24	"Reserved, read as 0, ignored on write."	RO	0x00
23:16	seqrst_count	"Count of number of times the sequence recovery reset timer decrements to zero."	RO RtoClr	0x00
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	out_of_order	"Count of out of order sequence numbers received. Incremented when a frame is accepted but the sequence number is not +1 of the highest stored value."	RO RtoClr	0x000

Table 11-540: rx_q0_flush

Register Information			
Description	"Receive Queue flush register. This register defines the traffic policing mode of operation. Each mode can be set simultaneously with the exception of bits 2 and 3, which are exclusive. If bits 2 and 3 are both set then only bit 3 is treated as active."		
Offset	0x0B00	**Type:**	RW

Bits	Name	Description	Access	Reset
31:16	max_val	"This 16 bit vector is used when either bit 2 or 3 of this register is asserted. Refer to the description in those bits for details."	RW	0x0000
15:4	reserved_15_4	"Reserved, read as 0, ignored on write."	RO	0x000
3	limit_frame_size	"When set max_val (bits 31:16) indicates the maximum frame-length in bytes that may be received. Frames exceeding this length will be dropped. This traffic policing function is relevant to the 802.1Qci standard which specifies stream filtering based on a maximum SDU (service data unit) size."	RW	0
2	limit_num_bytes	"When set, the number of 128 byte chunks of data received for this queue and already stored in the SRAM awaiting DMA memory writes cannot exceed max_val (bits 31:16)."	RW	0
1	drop_on_resource_err	"When set, if a free DMA descriptor for this queue cannot be obtained (also referred to as lack of descriptor resource and occurs when the software either cannot free up descriptors quickly enough to meet the receive traffic rate or has deliberately decided not to free any descriptors), all new frames received on this queue will be automatically discarded."	RW	0
0	drop_all_frames	"When set, all frames on this queue will be dropped."	RW	0

Table 11-541: rx_q1_flush

Register Information			
Description	"Receive Queue flush register. This register defines the traffic policing mode of operation. Each mode can be set simultaneously with the exception of bits 2 and 3, which are exclusive. If bits 2 and 3 are both set then only bit 3 is treated as active."		
Offset	0x0B04	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	max_val	"This 16 bit vector is used when either bit 2 or 3 of this register is asserted. Refer to the description in those bits for details."	RW	0x0000
15:4	reserved_15_4	"Reserved, read as 0, ignored on write."	RO	0x000
3	limit_frame_size	"When set max_val (bits 31:16) indicates the maximum frame-length in bytes that may be received. Frames exceeding this length will be dropped. This traffic policing function is relevant to the 802.1Qci standard which specifies stream filtering based on a maximum SDU (service data unit) size."	RW	0
2	limit_num_bytes	"When set, the number of 128 byte chunks of data received for this queue and already stored in the SRAM awaiting DMA memory writes cannot exceed max_val (bits 31:16)."	RW	0
1	drop_on_resource_err	"When set, if a free DMA descriptor for this queue cannot be obtained (also referred to as lack of descriptor resource and occurs when the software either cannot free up descriptors quickly enough to meet the receive traffic rate or has deliberately decided not to free any descriptors), all new frames received on this queue will be automatically discarded."	RW	0
0	drop_all_frames	"When set, all frames on this queue will be dropped."	RW	0

Table 11-542: rx_q2_flush

Register Information			
Description	"Receive Queue flush register. This register defines the traffic policing mode of operation. Each mode can be set simultaneously with the exception of bits 2 and 3, which are exclusive. If bits 2 and 3 are both set then only bit 3 is treated as active."		
Offset	0x0B08	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	max_val	"This 16 bit vector is used when either bit 2 or 3 of this register is asserted. Refer to the description in those bits for details."	RW	0x0000
15:4	reserved_15_4	"Reserved, read as 0, ignored on write."	RO	0x000
3	limit_frame_size	"When set max_val (bits 31:16) indicates the maximum frame-length in bytes that may be received. Frames exceeding this length will be dropped. This traffic policing function is relevant to the 802.1Qci standard which specifies stream filtering based on a maximum SDU (service data unit) size."	RW	0
2	limit_num_bytes	"When set, the number of 128 byte chunks of data received for this queue and already stored in the SRAM awaiting DMA memory writes cannot exceed max_val (bits 31:16)."	RW	0
1	drop_on_resource_err	"When set, if a free DMA descriptor for this queue cannot be obtained (also referred to as lack of descriptor resource and occurs when the software either cannot free up descriptors quickly enough to meet the receive traffic rate or has deliberately decided not to free any descriptors), all new frames received on this queue will be automatically discarded."	RW	0
0	drop_all_frames	"When set, all frames on this queue will be dropped."	RW	0

Table 11-543: rx_q3_flush

Register Information			
Description	"Receive Queue flush register. This register defines the traffic policing mode of operation. Each mode can be set simultaneously with the exception of bits 2 and 3, which are exclusive. If bits 2 and 3 are both set then only bit 3 is treated as active."		
Offset	0x0B0C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	max_val	"This 16 bit vector is used when either bit 2 or 3 of this register is asserted. Refer to the description in those bits for details."	RW	0x0000
15:4	reserved_15_4	"Reserved, read as 0, ignored on write."	RO	0x000
3	limit_frame_size	"When set max_val (bits 31:16) indicates the maximum frame-length in bytes that may be received. Frames exceeding this length will be dropped. This traffic policing function is relevant to the 802.1Qci standard which specifies stream filtering based on a maximum SDU (service data unit) size."	RW	0
2	limit_num_bytes	"When set, the number of 128 byte chunks of data received for this queue and already stored in the SRAM awaiting DMA memory writes cannot exceed max_val (bits 31:16)."	RW	0
1	drop_on_resource_err	"When set, if a free DMA descriptor for this queue cannot be obtained (also referred to as lack of descriptor resource and occurs when the software either cannot free up descriptors quickly enough to meet the receive traffic rate or has deliberately decided not to free any descriptors), all new frames received on this queue will be automatically discarded."	RW	0
0	drop_all_frames	"When set, all frames on this queue will be dropped."	RW	0

Table 11-544: rx_q4_flush

Register Information			
Description	"Receive Queue flush register. This register defines the traffic policing mode of operation. Each mode can be set simultaneously with the exception of bits 2 and 3, which are exclusive. If bits 2 and 3 are both set then only bit 3 is treated as active."		
Offset	0x0B10	**Type:**	RW

Bits	Name	Description	Access	Reset
31:16	max_val	"This 16 bit vector is used when either bit 2 or 3 of this register is asserted. Refer to the description in those bits for details."	RW	0x0000
15:4	reserved_15_4	"Reserved, read as 0, ignored on write."	RO	0x000
3	limit_frame_size	"When set max_val (bits 31:16) indicates the maximum frame-length in bytes that may be received. Frames exceeding this length will be dropped. This traffic policing function is relevant to the 802.1Qci standard which specifies stream filtering based on a maximum SDU (service data unit) size."	RW	0
2	limit_num_bytes	"When set, the number of 128 byte chunks of data received for this queue and already stored in the SRAM awaiting DMA memory writes cannot exceed max_val (bits 31:16)."	RW	0
1	drop_on_resource_err	"When set, if a free DMA descriptor for this queue cannot be obtained (also referred to as lack of descriptor resource and occurs when the software either cannot free up descriptors quickly enough to meet the receive traffic rate or has deliberately decided not to free any descriptors), all new frames received on this queue will be automatically discarded."	RW	0
0	drop_all_frames	"When set, all frames on this queue will be dropped."	RW	0

Table 11-545: rx_q5_flush

Register Information			
Description	"Receive Queue flush register. This register defines the traffic policing mode of operation. Each mode can be set simultaneously with the exception of bits 2 and 3, which are exclusive. If bits 2 and 3 are both set then only bit 3 is treated as active."		
Offset	0x0B14	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	max_val	"This 16 bit vector is used when either bit 2 or 3 of this register is asserted. Refer to the description in those bits for details."	RW	0x0000
15:4	reserved_15_4	"Reserved, read as 0, ignored on write."	RO	0x000
3	limit_frame_size	"When set max_val (bits 31:16) indicates the maximum frame-length in bytes that may be received. Frames exceeding this length will be dropped. This traffic policing function is relevant to the 802.1Qci standard which specifies stream filtering based on a maximum SDU (service data unit) size."	RW	0
2	limit_num_bytes	"When set, the number of 128 byte chunks of data received for this queue and already stored in the SRAM awaiting DMA memory writes cannot exceed max_val (bits 31:16)."	RW	0
1	drop_on_resource_err	"When set, if a free DMA descriptor for this queue cannot be obtained (also referred to as lack of descriptor resource and occurs when the software either cannot free up descriptors quickly enough to meet the receive traffic rate or has deliberately decided not to free any descriptors), all new frames received on this queue will be automatically discarded."	RW	0
0	drop_all_frames	"When set, all frames on this queue will be dropped."	RW	0

Table 11-546: rx_q6_flush

Register Information			
Description	"Receive Queue flush register. This register defines the traffic policing mode of operation. Each mode can be set simultaneously with the exception of bits 2 and 3, which are exclusive. If bits 2 and 3 are both set then only bit 3 is treated as active."		
Offset	0x0B18	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	max_val	"This 16 bit vector is used when either bit 2 or 3 of this register is asserted. Refer to the description in those bits for details."	RW	0x0000
15:4	reserved_15_4	"Reserved, read as 0, ignored on write."	RO	0x000
3	limit_frame_size	"When set max_val (bits 31:16) indicates the maximum frame-length in bytes that may be received. Frames exceeding this length will be dropped. This traffic policing function is relevant to the 802.1Qci standard which specifies stream filtering based on a maximum SDU (service data unit) size."	RW	0
2	limit_num_bytes	"When set, the number of 128 byte chunks of data received for this queue and already stored in the SRAM awaiting DMA memory writes cannot exceed max_val (bits 31:16)."	RW	0
1	drop_on_resource_err	"When set, if a free DMA descriptor for this queue cannot be obtained (also referred to as lack of descriptor resource and occurs when the software either cannot free up descriptors quickly enough to meet the receive traffic rate or has deliberately decided not to free any descriptors), all new frames received on this queue will be automatically discarded."	RW	0
0	drop_all_frames	"When set, all frames on this queue will be dropped."	RW	0

Table 11-547: rx_q7_flush

Register Information			
Description	"Receive Queue flush register. This register defines the traffic policing mode of operation. Each mode can be set simultaneously with the exception of bits 2 and 3, which are exclusive. If bits 2 and 3 are both set then only bit 3 is treated as active."		
Offset	0x0B1C	**Type:**	RW

Bits	Name	Description	Access	Reset
31:16	max_val	"This 16 bit vector is used when either bit 2 or 3 of this register is asserted. Refer to the description in those bits for details."	RW	0x0000
15:4	reserved_15_4	"Reserved, read as 0, ignored on write."	RO	0x000
3	limit_frame_size	"When set max_val (bits 31:16) indicates the maximum frame-length in bytes that may be received. Frames exceeding this length will be dropped. This traffic policing function is relevant to the 802.1Qci standard which specifies stream filtering based on a maximum SDU (service data unit) size."	RW	0
2	limit_num_bytes	"When set, the number of 128 byte chunks of data received for this queue and already stored in the SRAM awaiting DMA memory writes cannot exceed max_val (bits 31:16)."	RW	0
1	drop_on_resource_err	"When set, if a free DMA descriptor for this queue cannot be obtained (also referred to as lack of descriptor resource and occurs when the software either cannot free up descriptors quickly enough to meet the receive traffic rate or has deliberately decided not to free any descriptors), all new frames received on this queue will be automatically discarded."	RW	0
0	drop_all_frames	"When set, all frames on this queue will be dropped."	RW	0

Table 11-548: rx_q8_flush

Register Information			
Description	"Receive Queue flush register. This register defines the traffic policing mode of operation. Each mode can be set simultaneously with the exception of bits 2 and 3, which are exclusive. If bits 2 and 3 are both set then only bit 3 is treated as active."		
Offset	0x0B20	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	max_val	"This 16 bit vector is used when either bit 2 or 3 of this register is asserted. Refer to the description in those bits for details."	RW	0x0000
15:4	reserved_15_4	"Reserved, read as 0, ignored on write."	RO	0x000
3	limit_frame_size	"When set max_val (bits 31:16) indicates the maximum frame-length in bytes that may be received. Frames exceeding this length will be dropped. This traffic policing function is relevant to the 802.1Qci standard which specifies stream filtering based on a maximum SDU (service data unit) size."	RW	0
2	limit_num_bytes	"When set, the number of 128 byte chunks of data received for this queue and already stored in the SRAM awaiting DMA memory writes cannot exceed max_val (bits 31:16)."	RW	0
1	drop_on_resource_err	"When set, if a free DMA descriptor for this queue cannot be obtained (also referred to as lack of descriptor resource and occurs when the software either cannot free up descriptors quickly enough to meet the receive traffic rate or has deliberately decided not to free any descriptors), all new frames received on this queue will be automatically discarded."	RW	0
0	drop_all_frames	"When set, all frames on this queue will be dropped."	RW	0

Table 11-549: rx_q9_flush

Register Information			
Description	"Receive Queue flush register. This register defines the traffic policing mode of operation. Each mode can be set simultaneously with the exception of bits 2 and 3, which are exclusive. If bits 2 and 3 are both set then only bit 3 is treated as active."		
Offset	0x0B24	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	max_val	"This 16 bit vector is used when either bit 2 or 3 of this register is asserted. Refer to the description in those bits for details."	RW	0x0000
15:4	reserved_15_4	"Reserved, read as 0, ignored on write."	RO	0x000
3	limit_frame_size	"When set max_val (bits 31:16) indicates the maximum frame-length in bytes that may be received. Frames exceeding this length will be dropped. This traffic policing function is relevant to the 802.1Qci standard which specifies stream filtering based on a maximum SDU (service data unit) size."	RW	0
2	limit_num_bytes	"When set, the number of 128 byte chunks of data received for this queue and already stored in the SRAM awaiting DMA memory writes cannot exceed max_val (bits 31:16)."	RW	0
1	drop_on_resource_err	"When set, if a free DMA descriptor for this queue cannot be obtained (also referred to as lack of descriptor resource and occurs when the software either cannot free up descriptors quickly enough to meet the receive traffic rate or has deliberately decided not to free any descriptors), all new frames received on this queue will be automatically discarded."	RW	0
0	drop_all_frames	"When set, all frames on this queue will be dropped."	RW	0

Table 11-550: rx_q10_flush

Register Information			
Description	"Receive Queue flush register. This register defines the traffic policing mode of operation. Each mode can be set simultaneously with the exception of bits 2 and 3, which are exclusive. If bits 2 and 3 are both set then only bit 3 is treated as active."		
Offset	0x0B28	**Type:**	RW

Bits	Name	Description	Access	Reset
31:16	max_val	"This 16 bit vector is used when either bit 2 or 3 of this register is asserted. Refer to the description in those bits for details."	RW	0x0000
15:4	reserved_15_4	"Reserved, read as 0, ignored on write."	RO	0x000
3	limit_frame_size	"When set max_val (bits 31:16) indicates the maximum frame-length in bytes that may be received. Frames exceeding this length will be dropped. This traffic policing function is relevant to the 802.1Qci standard which specifies stream filtering based on a maximum SDU (service data unit) size."	RW	0
2	limit_num_bytes	"When set, the number of 128 byte chunks of data received for this queue and already stored in the SRAM awaiting DMA memory writes cannot exceed max_val (bits 31:16)."	RW	0
1	drop_on_resource_err	"When set, if a free DMA descriptor for this queue cannot be obtained (also referred to as lack of descriptor resource and occurs when the software either cannot free up descriptors quickly enough to meet the receive traffic rate or has deliberately decided not to free any descriptors), all new frames received on this queue will be automatically discarded."	RW	0
0	drop_all_frames	"When set, all frames on this queue will be dropped."	RW	0

Table 11-551: rx_q11_flush

Register Information			
Description	"Receive Queue flush register. This register defines the traffic policing mode of operation. Each mode can be set simultaneously with the exception of bits 2 and 3, which are exclusive. If bits 2 and 3 are both set then only bit 3 is treated as active."		
Offset	0x0B2C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	max_val	"This 16 bit vector is used when either bit 2 or 3 of this register is asserted. Refer to the description in those bits for details."	RW	0x0000
15:4	reserved_15_4	"Reserved, read as 0, ignored on write."	RO	0x000
3	limit_frame_size	"When set max_val (bits 31:16) indicates the maximum frame-length in bytes that may be received. Frames exceeding this length will be dropped. This traffic policing function is relevant to the 802.1Qci standard which specifies stream filtering based on a maximum SDU (service data unit) size."	RW	0
2	limit_num_bytes	"When set, the number of 128 byte chunks of data received for this queue and already stored in the SRAM awaiting DMA memory writes cannot exceed max_val (bits 31:16)."	RW	0
1	drop_on_resource_err	"When set, if a free DMA descriptor for this queue cannot be obtained (also referred to as lack of descriptor resource and occurs when the software either cannot free up descriptors quickly enough to meet the receive traffic rate or has deliberately decided not to free any descriptors), all new frames received on this queue will be automatically discarded."	RW	0
0	drop_all_frames	"When set, all frames on this queue will be dropped."	RW	0

Table 11-552: rx_q12_flush

Register Information			
Description	"Receive Queue flush register. This register defines the traffic policing mode of operation. Each mode can be set simultaneously with the exception of bits 2 and 3, which are exclusive. If bits 2 and 3 are both set then only bit 3 is treated as active."		
Offset	0x0B30	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	max_val	"This 16 bit vector is used when either bit 2 or 3 of this register is asserted. Refer to the description in those bits for details."	RW	0x0000
15:4	reserved_15_4	"Reserved, read as 0, ignored on write."	RO	0x000
3	limit_frame_size	"When set max_val (bits 31:16) indicates the maximum frame-length in bytes that may be received. Frames exceeding this length will be dropped. This traffic policing function is relevant to the 802.1Qci standard which specifies stream filtering based on a maximum SDU (service data unit) size."	RW	0
2	limit_num_bytes	"When set, the number of 128 byte chunks of data received for this queue and already stored in the SRAM awaiting DMA memory writes cannot exceed max_val (bits 31:16)."	RW	0
1	drop_on_resource_err	"When set, if a free DMA descriptor for this queue cannot be obtained (also referred to as lack of descriptor resource and occurs when the software either cannot free up descriptors quickly enough to meet the receive traffic rate or has deliberately decided not to free any descriptors), all new frames received on this queue will be automatically discarded."	RW	0
0	drop_all_frames	"When set, all frames on this queue will be dropped."	RW	0

Table 11-553: rx_q13_flush

Register Information			
Description	"Receive Queue flush register. This register defines the traffic policing mode of operation. Each mode can be set simultaneously with the exception of bits 2 and 3, which are exclusive. If bits 2 and 3 are both set then only bit 3 is treated as active."		
Offset	0x0B34	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	max_val	"This 16 bit vector is used when either bit 2 or 3 of this register is asserted. Refer to the description in those bits for details."	RW	0x0000
15:4	reserved_15_4	"Reserved, read as 0, ignored on write."	RO	0x000
3	limit_frame_size	"When set max_val (bits 31:16) indicates the maximum frame-length in bytes that may be received. Frames exceeding this length will be dropped. This traffic policing function is relevant to the 802.1Qci standard which specifies stream filtering based on a maximum SDU (service data unit) size."	RW	0
2	limit_num_bytes	"When set, the number of 128 byte chunks of data received for this queue and already stored in the SRAM awaiting DMA memory writes cannot exceed max_val (bits 31:16)."	RW	0
1	drop_on_resource_err	"When set, if a free DMA descriptor for this queue cannot be obtained (also referred to as lack of descriptor resource and occurs when the software either cannot free up descriptors quickly enough to meet the receive traffic rate or has deliberately decided not to free any descriptors), all new frames received on this queue will be automatically discarded."	RW	0
0	drop_all_frames	"When set, all frames on this queue will be dropped."	RW	0

Table 11-554: rx_q14_flush

Register Information			
Description	"Receive Queue flush register. This register defines the traffic policing mode of operation. Each mode can be set simultaneously with the exception of bits 2 and 3, which are exclusive. If bits 2 and 3 are both set then only bit 3 is treated as active."		
Offset	0x0B38	**Type:**	RW

Bits	Name	Description	Access	Reset
31:16	max_val	"This 16 bit vector is used when either bit 2 or 3 of this register is asserted. Refer to the description in those bits for details."	RW	0x0000
15:4	reserved_15_4	"Reserved, read as 0, ignored on write."	RO	0x000
3	limit_frame_size	"When set max_val (bits 31:16) indicates the maximum frame-length in bytes that may be received. Frames exceeding this length will be dropped. This traffic policing function is relevant to the 802.1Qci standard which specifies stream filtering based on a maximum SDU (service data unit) size."	RW	0
2	limit_num_bytes	"When set, the number of 128 byte chunks of data received for this queue and already stored in the SRAM awaiting DMA memory writes cannot exceed max_val (bits 31:16)."	RW	0
1	drop_on_resource_err	"When set, if a free DMA descriptor for this queue cannot be obtained (also referred to as lack of descriptor resource and occurs when the software either cannot free up descriptors quickly enough to meet the receive traffic rate or has deliberately decided not to free any descriptors), all new frames received on this queue will be automatically discarded."	RW	0
0	drop_all_frames	"When set, all frames on this queue will be dropped."	RW	0

Table 11-555: rx_q15_flush

Register Information				
Description	"Receive Queue flush register. This register defines the traffic policing mode of operation. Each mode can be set simultaneously with the exception of bits 2 and 3, which are exclusive. If bits 2 and 3 are both set then only bit 3 is treated as active."			
Offset	0x0B3C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	max_val	"This 16 bit vector is used when either bit 2 or 3 of this register is asserted. Refer to the description in those bits for details."	RW	0x0000
15:4	reserved_15_4	"Reserved, read as 0, ignored on write."	RO	0x000
3	limit_frame_size	"When set max_val (bits 31:16) indicates the maximum frame-length in bytes that may be received. Frames exceeding this length will be dropped. This traffic policing function is relevant to the 802.1Qci standard which specifies stream filtering based on a maximum SDU (service data unit) size."	RW	0
2	limit_num_bytes	"When set, the number of 128 byte chunks of data received for this queue and already stored in the SRAM awaiting DMA memory writes cannot exceed max_val (bits 31:16)."	RW	0
1	drop_on_resource_err	"When set, if a free DMA descriptor for this queue cannot be obtained (also referred to as lack of descriptor resource and occurs when the software either cannot free up descriptors quickly enough to meet the receive traffic rate or has deliberately decided not to free any descriptors), all new frames received on this queue will be automatically discarded."	RW	0
0	drop_all_frames	"When set, all frames on this queue will be dropped."	RW	0

Table 11-556: scr2_reg0_rate_limit

Register Information				
Description	"Screener type 2 maximum rate register. This register defines the maximum receive rate for the corresponding type 2 screener. This is a traffic policing function and is relevant to the 802.1Qci standard. If the maximum rate is exceeded all frames matched by the corresponding type 2 screener will be discarded until the receive rate falls below the specified value."			
Offset	0x0B40	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	max_rate_val	"This is the maximum number of bytes this screener is permitted to match in the programmed interval time."	RW	0x0000
15:0	interval_time	"If during the interval time the total number of bytes of received frames matched by the screener exceeds max_rate_val then the current frame and frames subsequently matched will dropped until an interval time passes where max_rate_val is not exceeded. If this value is set to zero, then no rate limiting will be performed. The interval time is specified in units of 64 rx_clk periods."	RW	0x0000

Table 11-557: scr2_reg1_rate_limit

Register Information					
Description	"Screener type 2 maximum rate register. This register defines the maximum receive rate for the corresponding type 2 screener. This is a traffic policing function and is relevant to the 802.1Qci standard. If the maximum rate is exceeded all frames matched by the corresponding type 2 screener will be discarded until the receive rate falls below the specified value."				
Offset	0x0B44	**Type:**	RW		
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:16	max_rate_val	"This is the maximum number of bytes this screener is permitted to match in the programmed interval time."	RW	0x0000	
15:0	interval_time	"If during the interval time the total number of bytes of received frames matched by the screener exceeds max_rate_val then the current frame and frames subsequently matched will dropped until an interval time passes where max_rate_val is not exceeded. If this value is set to zero, then no rate limiting will be performed. The interval time is specified in units of 64 rx_clk periods."	RW	0x0000	

Table 11-558: scr2_reg2_rate_limit

Register Information					
Description	"Screener type 2 maximum rate register. This register defines the maximum receive rate for the corresponding type 2 screener. This is a traffic policing function and is relevant to the 802.1Qci standard. If the maximum rate is exceeded all frames matched by the corresponding type 2 screener will be discarded until the receive rate falls below the specified value."				
Offset	0x0B48	**Type:**	RW		
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:16	max_rate_val	"This is the maximum number of bytes this screener is permitted to match in the programmed interval time."	RW	0x0000	
15:0	interval_time	"If during the interval time the total number of bytes of received frames matched by the screener exceeds max_rate_val then the current frame and frames subsequently matched will dropped until an interval time passes where max_rate_val is not exceeded. If this value is set to zero, then no rate limiting will be performed. The interval time is specified in units of 64 rx_clk periods."	RW	0x0000	

Table 11-559: scr2_reg3_rate_limit

Register Information			
Description	"Screener type 2 maximum rate register. This register defines the maximum receive rate for the corresponding type 2 screener. This is a traffic policing function and is relevant to the 802.1Qci standard. If the maximum rate is exceeded all frames matched by the corresponding type 2 screener will be discarded until the receive rate falls below the specified value."		
Offset	0x0B4C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	max_rate_val	"This is the maximum number of bytes this screener is permitted to match in the programmed interval time."	RW	0x0000
15:0	interval_time	"If during the interval time the total number of bytes of received frames matched by the screener exceeds max_rate_val then the current frame and frames subsequently matched will dropped until an interval time passes where max_rate_val is not exceeded. If this value is set to zero, then no rate limiting will be performed. The interval time is specified in units of 64 rx_clk periods."	RW	0x0000

Table 11-560: scr2_reg4_rate_limit

Register Information			
Description	"Screener type 2 maximum rate register. This register defines the maximum receive rate for the corresponding type 2 screener. This is a traffic policing function and is relevant to the 802.1Qci standard. If the maximum rate is exceeded all frames matched by the corresponding type 2 screener will be discarded until the receive rate falls below the specified value."		
Offset	0x0B50	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	max_rate_val	"This is the maximum number of bytes this screener is permitted to match in the programmed interval time."	RW	0x0000
15:0	interval_time	"If during the interval time the total number of bytes of received frames matched by the screener exceeds max_rate_val then the current frame and frames subsequently matched will dropped until an interval time passes where max_rate_val is not exceeded. If this value is set to zero, then no rate limiting will be performed. The interval time is specified in units of 64 rx_clk periods."	RW	0x0000

Table 11-561: scr2_reg5_rate_limit

Register Information					
Description	"Screener type 2 maximum rate register. This register defines the maximum receive rate for the corresponding type 2 screener. This is a traffic policing function and is relevant to the 802.1Qci standard. If the maximum rate is exceeded all frames matched by the corresponding type 2 screener will be discarded until the receive rate falls below the specified value."				
Offset	0x0B54		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:16	max_rate_val	"This is the maximum number of bytes this screener is permitted to match in the programmed interval time."		RW	0x0000
15:0	interval_time	"If during the interval time the total number of bytes of received frames matched by the screener exceeds max_rate_val then the current frame and frames subsequently matched will dropped until an interval time passes where max_rate_val is not exceeded. If this value is set to zero, then no rate limiting will be performed. The interval time is specified in units of 64 rx_clk periods."		RW	0x0000

Table 11-562: scr2_reg6_rate_limit

Register Information					
Description	"Screener type 2 maximum rate register. This register defines the maximum receive rate for the corresponding type 2 screener. This is a traffic policing function and is relevant to the 802.1Qci standard. If the maximum rate is exceeded all frames matched by the corresponding type 2 screener will be discarded until the receive rate falls below the specified value."				
Offset	0x0B58		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:16	max_rate_val	"This is the maximum number of bytes this screener is permitted to match in the programmed interval time."		RW	0x0000
15:0	interval_time	"If during the interval time the total number of bytes of received frames matched by the screener exceeds max_rate_val then the current frame and frames subsequently matched will dropped until an interval time passes where max_rate_val is not exceeded. If this value is set to zero, then no rate limiting will be performed. The interval time is specified in units of 64 rx_clk periods."		RW	0x0000

Table 11-563: scr2_reg7_rate_limit

Register Information					
Description	"Screener type 2 maximum rate register. This register defines the maximum receive rate for the corresponding type 2 screener. This is a traffic policing function and is relevant to the 802.1Qci standard. If the maximum rate is exceeded all frames matched by the corresponding type 2 screener will be discarded until the receive rate falls below the specified value."				
Offset	0x0B5C		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:16	max_rate_val	"This is the maximum number of bytes this screener is permitted to match in the programmed interval time."		RW	0x0000
15:0	interval_time	"If during the interval time the total number of bytes of received frames matched by the screener exceeds max_rate_val then the current frame and frames subsequently matched will dropped until an interval time passes where max_rate_val is not exceeded. If this value is set to zero, then no rate limiting will be performed. The interval time is specified in units of 64 rx_clk periods."		RW	0x0000

Table 11-564: scr2_reg8_rate_limit

Register Information					
Description	"Screener type 2 maximum rate register. This register defines the maximum receive rate for the corresponding type 2 screener. This is a traffic policing function and is relevant to the 802.1Qci standard. If the maximum rate is exceeded all frames matched by the corresponding type 2 screener will be discarded until the receive rate falls below the specified value."				
Offset	0x0B60		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:16	max_rate_val	"This is the maximum number of bytes this screener is permitted to match in the programmed interval time."		RW	0x0000
15:0	interval_time	"If during the interval time the total number of bytes of received frames matched by the screener exceeds max_rate_val then the current frame and frames subsequently matched will dropped until an interval time passes where max_rate_val is not exceeded. If this value is set to zero, then no rate limiting will be performed. The interval time is specified in units of 64 rx_clk periods."		RW	0x0000

Table 11-565: scr2_reg9_rate_limit

Register Information					
Description	"Screener type 2 maximum rate register. This register defines the maximum receive rate for the corresponding type 2 screener. This is a traffic policing function and is relevant to the 802.1Qci standard. If the maximum rate is exceeded all frames matched by the corresponding type 2 screener will be discarded until the receive rate falls below the specified value."				
Offset	0x0B64		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:16	max_rate_val	"This is the maximum number of bytes this screener is permitted to match in the programmed interval time."		RW	0x0000
15:0	interval_time	"If during the interval time the total number of bytes of received frames matched by the screener exceeds max_rate_val then the current frame and frames subsequently matched will dropped until an interval time passes where max_rate_val is not exceeded. If this value is set to zero, then no rate limiting will be performed. The interval time is specified in units of 64 rx_clk periods."		RW	0x0000

Table 11-566: scr2_reg10_rate_limit

Register Information					
Description	"Screener type 2 maximum rate register. This register defines the maximum receive rate for the corresponding type 2 screener. This is a traffic policing function and is relevant to the 802.1Qci standard. If the maximum rate is exceeded all frames matched by the corresponding type 2 screener will be discarded until the receive rate falls below the specified value."				
Offset	0x0B68		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:16	max_rate_val	"This is the maximum number of bytes this screener is permitted to match in the programmed interval time."		RW	0x0000
15:0	interval_time	"If during the interval time the total number of bytes of received frames matched by the screener exceeds max_rate_val then the current frame and frames subsequently matched will dropped until an interval time passes where max_rate_val is not exceeded. If this value is set to zero, then no rate limiting will be performed. The interval time is specified in units of 64 rx_clk periods."		RW	0x0000

Table 11-567: scr2_reg11_rate_limit

Register Information					
Description	"Screener type 2 maximum rate register. This register defines the maximum receive rate for the corresponding type 2 screener. This is a traffic policing function and is relevant to the 802.1Qci standard. If the maximum rate is exceeded all frames matched by the corresponding type 2 screener will be discarded until the receive rate falls below the specified value."				
Offset	0x0B6C	**Type:**	RW		
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:16	max_rate_val	"This is the maximum number of bytes this screener is permitted to match in the programmed interval time."	RW	0x0000	
15:0	interval_time	"If during the interval time the total number of bytes of received frames matched by the screener exceeds max_rate_val then the current frame and frames subsequently matched will dropped until an interval time passes where max_rate_val is not exceeded. If this value is set to zero, then no rate limiting will be performed. The interval time is specified in units of 64 rx_clk periods."	RW	0x0000	

Table 11-568: scr2_reg12_rate_limit

Register Information					
Description	"Screener type 2 maximum rate register. This register defines the maximum receive rate for the corresponding type 2 screener. This is a traffic policing function and is relevant to the 802.1Qci standard. If the maximum rate is exceeded all frames matched by the corresponding type 2 screener will be discarded until the receive rate falls below the specified value."				
Offset	0x0B70	**Type:**	RW		
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:16	max_rate_val	"This is the maximum number of bytes this screener is permitted to match in the programmed interval time."	RW	0x0000	
15:0	interval_time	"If during the interval time the total number of bytes of received frames matched by the screener exceeds max_rate_val then the current frame and frames subsequently matched will dropped until an interval time passes where max_rate_val is not exceeded. If this value is set to zero, then no rate limiting will be performed. The interval time is specified in units of 64 rx_clk periods."	RW	0x0000	

Table 11-569: scr2_reg13_rate_limit

Register Information			
Description	"Screener type 2 maximum rate register. This register defines the maximum receive rate for the corresponding type 2 screener. This is a traffic policing function and is relevant to the 802.1Qci standard. If the maximum rate is exceeded all frames matched by the corresponding type 2 screener will be discarded until the receive rate falls below the specified value."		
Offset	0x0B74	**Type:**	RW

Bits	Name	Description	Access	Reset
31:16	max_rate_val	"This is the maximum number of bytes this screener is permitted to match in the programmed interval time."	RW	0x0000
15:0	interval_time	"If during the interval time the total number of bytes of received frames matched by the screener exceeds max_rate_val then the current frame and frames subsequently matched will dropped until an interval time passes where max_rate_val is not exceeded. If this value is set to zero, then no rate limiting will be performed. The interval time is specified in units of 64 rx_clk periods."	RW	0x0000

Table 11-570: scr2_reg14_rate_limit

Register Information			
Description	"Screener type 2 maximum rate register. This register defines the maximum receive rate for the corresponding type 2 screener. This is a traffic policing function and is relevant to the 802.1Qci standard. If the maximum rate is exceeded all frames matched by the corresponding type 2 screener will be discarded until the receive rate falls below the specified value."		
Offset	0x0B78	**Type:**	RW

Bits	Name	Description	Access	Reset
31:16	max_rate_val	"This is the maximum number of bytes this screener is permitted to match in the programmed interval time."	RW	0x0000
15:0	interval_time	"If during the interval time the total number of bytes of received frames matched by the screener exceeds max_rate_val then the current frame and frames subsequently matched will dropped until an interval time passes where max_rate_val is not exceeded. If this value is set to zero, then no rate limiting will be performed. The interval time is specified in units of 64 rx_clk periods."	RW	0x0000

Table 11-571: scr2_reg15_rate_limit

Register Information			
Description	"Screener type 2 maximum rate register. This register defines the maximum receive rate for the corresponding type 2 screener. This is a traffic policing function and is relevant to the 802.1Qci standard. If the maximum rate is exceeded all frames matched by the corresponding type 2 screener will be discarded until the receive rate falls below the specified value."		
Offset	0x0B7C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	max_rate_val	"This is the maximum number of bytes this screener is permitted to match in the programmed interval time."	RW	0x0000
15:0	interval_time	"If during the interval time the total number of bytes of received frames matched by the screener exceeds max_rate_val then the current frame and frames subsequently matched will dropped until an interval time passes where max_rate_val is not exceeded. If this value is set to zero, then no rate limiting will be performed. The interval time is specified in units of 64 rx_clk periods."	RW	0x0000

Table 11-572: scr2_rate_status

Register Information			
Description	"Screener rate limit exceeded status register. For each screener type 2 register configured a status bit will be set and cleared on read if the maximum receive rate is exceeded."		
Offset	0x0B80	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15	scr2_15_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
14	scr2_14_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
13	scr2_13_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
12	scr2_12_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
11	scr2_11_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
10	scr2_10_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
9	scr2_9_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
8	scr2_8_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
7	scr2_7_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
6	scr2_6_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
5	scr2_5_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
4	scr2_4_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
3	scr2_3_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
2	scr2_2_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
1	scr2_1_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
0	scr2_0_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0

Table 11-573: asf_int_status

Register Information				
Description	"ASF Interrupt Status Register. This register indicates the source of ASF interrupts. The corresponding bit in the mask register must be clear for a bit to be set. If any bit is set in this register the asf_fatal or asf_nonfatal signal will be asserted. Writing to either raw or masked status registers, clear both registers. For test purposes, trigger signal interrupt event by writing to the ASF interrupt status test register."			
Offset	0x0E00	**Type:**	RW	

Bitfield Details

Bits	Name	Description	Access	Reset
31:7	reserved_31_7	"Reserved, read as 0, ignored on write."	RO	0x000 0000
6	asf_integrity_err	"Integrity error interrupt"	RW W1toClr	0
5	asf_protocol_err	"Protocol error interrupt"	RW W1toClr	0
4	asf_trans_to_err	"Transaction timeouts error interrupt"	RW W1toClr	0
3	asf_csr_err	"Configuration and status registers error interrupt"	RW W1toClr	0
2	asf_dap_err	"Data and address paths parity error interrupt"	RW W1toClr	0
1	asf_sram_uncorr_err	"SRAM uncorrectable error interrupt"	RW W1toClr	0
0	asf_sram_corr_err	"SRAM correctable error interrupt"	RW W1toClr	0

Table 11-574: asf_int_raw_status

Register Information			
Description	"ASF Interrupt Raw Status Register. A bit set in this raw register indicates a source of ASF fault in the corresponding feature. Writing to either raw or masked status registers, clear both registers. For test purposes, trigger signal interrupt event by writing to the ASF interrupt status test register."		
Offset	0x0E04	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:7	reserved_31_7	"Reserved, read as 0, ignored on write."	RO	0x000 0000
6	asf_integrity_err	"Integrity error interrupt"	RW W1toClr	0
5	asf_protocol_err	"Protocol error interrupt"	RW W1toClr	0
4	asf_trans_to_err	"Transaction timeouts error interrupt"	RW W1toClr	0
3	asf_csr_err	"Configuration and status registers error interrupt"	RW W1toClr	0
2	asf_dap_err	"Data and address paths parity error interrupt"	RW W1toClr	0
1	asf_sram_uncorr_err	"SRAM uncorrectable error interrupt"	RW W1toClr	0
0	asf_sram_corr_err	"SRAM correctable error interrupt"	RW W1toClr	0

Table 11-575: asf_int_mask

Register Information			
Description	"The ASF interrupt mask register indicating which interrupt bits in the ASF interrupt status register are masked. All bits are set at reset. Clear the individual bit to enable the corresponding interrupt."		
Offset	0x0E08	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:7	reserved_31_7	"Reserved, read as 0, ignored on write."	RO	0x000 0000
6	asf_integrity_err_mask	"Mask bit for integrity error interrupt"	RW	1
5	asf_protocol_err_mask	"Mask bit for protocol error interrupt."	RW	1
4	asf_trans_to_err_mask	"Mask bit for transaction timeouts error interrupt."	RW	1
3	asf_csr_err_mask	"Mask bit for configuration and status registers error interrupt."	RW	1
2	asf_dap_err_mask	"Mask bit for data and address paths parity error interrupt."	RW	1
1	asf_sram_uncorr_err_mask	"Mask bit for SRAM uncorrectable error interrupt."	RW	1
0	asf_sram_corr_err_mask	"Mask bit for SRAM correctable error interrupt."	RW	1

Table 11-576: asf_int_test

Register Information			
Description	"The ASF interrupt test register emulate hardware even. Write one to individual bit to trigger single event in (masked and raw) status registers according to mask and will generate interrupt accordingly."		
Offset	0x0E0C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:7	reserved_31_7	"Reserved, read as 0, ignored on write."	RO	0x000 0000
6	asf_integrity_err_test	"Test bit for integrity error interrupt"	WO	0
5	asf_protocol_err_test	"Test bit for protocol error interrupt."	WO	0
4	asf_trans_to_err_test	"Test bit for transaction timeouts error interrupt."	WO	0
3	asf_csr_err_test	"Test bit for configuration and status registers error interrupt."	WO	0
2	asf_dap_err_test	"Test bit for data and address paths parity error interrupt."	WO	0
1	asf_sram_uncorr_err_test	"Test bit for SRAM uncorrectable error interrupt."	WO	0
0	asf_sram_corr_err_test	"Test bit for SRAM correctable error interrupt."	WO	0

Table 11-577: asf_fatal_nonfatal_select

Register Information			
Description	"The fatal or non-fatal interrupt register selects whether a fatal (asf_int_fatal) or non-fatal (asf_int_nonfatal) interrupt is triggered. If the bit of the event will be set to one then fatal interrupt (asf_int_fatal) will be triggered. Otherwise the non-fatal interrupt (asf_int_nonfatal) will be triggered."		
Offset	0x0E10	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:7	reserved_31_7	"Reserved, read as 0, ignored on write."	RO	0x000 0000
6	asf_integrity_err	"Enable integrity error interrupt as fatal"	RW	1
5	asf_protocol_err	"Enable protocol error interrupt as fatal."	RW	1
4	asf_trans_to_err	"Enable transaction timeouts error interrupt as fatal."	RW	1
3	asf_csr_err	"Enable configuration and status registers error interrupt as fatal."	RW	1
2	asf_dap_err	"Enable data and address paths parity error interrupt as fatal."	RW	1
1	asf_sram_uncorr_err	"Enable SRAM uncorrectable error interrupt as fatal."	RW	1
0	asf_sram_corr_err	"Enable SRAM correctable error interrupt as fatal."	RW	1

Table 11-578: asf_sram_corr_fault_status

Register Information			
Description	"Status register for SRAM correctable fault. These fields are updated whenever asf_sram_corr_fault input is active."		
Offset	0x0E20	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	asf_sram_corr_fault_inst	"Last SRAM instance that generated fault."	RO	0x00
23:0	asf_sram_corr_fault_addr	"Last SRAM address that generated fault."	RO	0x00 0000

Table 11-579: asf_sram_uncorr_fault_status

Register Information			
Description	"Status register for SRAM uncorrectable fault. These fields are updated whenever asf_sram_uncorr_fault input is active."		
Offset	0x0E24	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	asf_sram_uncorr_fault_inst	"Last SRAM instance that generated fault."	RO	0x00
23:0	asf_sram_uncorr_fault_addr	"Last SRAM address that generated fault."	RO	0x00 0000

Table 11-580: asf_sram_fault_stats

Register Information				
Description	"Statistics register for SRAM faults. Note that this register clears when software writes to any field."			
Offset	0x0E28	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	asf_sram_fault_corr_stats	"Count of number of correctable errors if implemented. Count value will saturate at 0xffff."	RW W1toClr	0x0000

Table 11-581: asf_trans_to_ctrl

Register Information				
Description	"Control register to configure the ASF transaction timeout monitors."			
Offset	0x0E30	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	asf_trans_to_en	"Enable transaction timeout monitoring."	RW	0
30:16	reserved_30_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	asf_trans_to_ctrl	"Timer value to use for transaction timeout monitor."	RW	0x0000

Table 11-582: asf_trans_to_fault_mask

Register Information				
Description	"Control register to mask out ASF transaction timeout faults from triggering interrupts. On reset, all bits are set to mask out all sources. Clear the corresponding bit to enable the interrupt source."			
Offset	0x0E34	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:5	reserved_31_5	"Reserved, read as 0, ignored on write."	RO	0x000 0000
4	host_trans_to_mask	"Mask register for AMBA host transaction timeouts."	RW	1
3	dma_rx_to_mask	"Mask register for DMA RX lockup detection."	RW	1
2	dma_tx_to_mask	"Mask register for DMA TX lockup detection."	RW	1
1	mac_rx_to_mask	"Mask register for MAC RX lockup detection."	RW	1
0	mac_tx_to_mask	"Mask register for MAC TX lockup detection."	RW	1

Table 11-583: asf_trans_to_fault_status

Register Information			
Description	"Status register for transaction timeouts fault. If a fault occurs the revelant status bit will be set to 1. Each bit can be cleared by software writing 1 to each bit."		
Offset	0x0E38	**Type:**	RW

Bits	Name	Description	Access	Reset
31:5	reserved_31_5	"Reserved, read as 0, ignored on write."	RO	0x000 0000
4	host_trans_to_status	"Status bit for AMBA host transaction timeouts."	RW W1toClr	0
3	dma_rx_to_status	"Status bit for DMA RX lockup detection."	RW W1toClr	0
2	dma_tx_to_status	"Status bit for DMA TX lockup detection."	RW W1toClr	0
1	mac_rx_to_status	"Status bit for MAC RX lockup detection."	RW W1toClr	0
0	mac_tx_to_status	"Status bit for MAC TX lockup detection."	RW W1toClr	0

Table 11-584: asf_protocol_fault_mask

Register Information			
Description	"Control register to mask out ASF Protocol faults from triggering interrupts. On reset, all bits are set to mask out all sources. Clear the corresponding bit to enable the interrupt source."		
Offset	0x0E40	**Type:**	RW

Bits	Name	Description	Access	Reset
31:22	reserved_31_22	"Reserved, read as 0, ignored on write."	RO	0x000
21	rx_dma_pkt_flush_mask	"Mask bit for RX packet was flushed from the DMA."	RW	1
20	rx_overflow_mask	"Mask bit for DMA buffer overflow and packet was dropped."	RW	1
19	rx_hresp_err_mask	"Mask bit for DMA hresp error (AHB only)."	RW	1
18	tx_hresp_err_mask	"Mask bit for DMA hresp error (AHB only)."	RW	1
17	tx_buff_ex_mid_mask	"Mask bit for Transmit buffers exhausted before end of packet."	RW	1
16	tx_underrun_mask	"Mask bit for Transmit DMA underrun occurred."	RW	1
15:9	reserved_15_9	"Reserved, read as 0, ignored on write."	RO	0x00
8	tx_too_many_retries_mask	"Mask bit for too many retry attempts after collision error (half duplex only)."	RW	1
7	rx_udp_ck_err_mask	"Mask bit for RX packet UDP checksum error."	RW	1
6	rx_tcp_ck_err_mask	"Mask bit for RX packet TCP checksum error."	RW	1
5	rx_ip_ck_err_mask	"Mask bit for RX packet IP checksum error."	RW	1
4	rx_length_err_mask	"Mask bit for RX packet with length field error."	RW	1
3	rx_symbol_err_mask	"Mask bit for RX packet with symbol errors."	RW	1
2	rx_long_err_mask	"Mask bit for RX packet too long."	RW	1
1	rx_short_err_mask	"Mask bit for RX packet too short."	RW	1
0	rx_crc_err_mask	"Mask bit for RX packet with bad CRC."	RW	1

Table 11-585: asf_protocol_fault_status

Register Information			
Description	"Status register for protocol faults. If a fault occurs the revelant status bit will be set to 1. Each bit can be cleared by software writing 1 to each bit"		
Offset	0x0E44	**Type:**	RW

Bits	Name	Description	Access	Reset
31:22	reserved_31_22	"Reserved, read as 0, ignored on write."	RO	0x000
21	rx_dma_pkt_flush_status	"Status bit for RX packet was flushed from the DMA."	RW W1toClr:emac_asf_protocol_fault_status.rx_dma_pkt_flush_status	0
20	rx_overflow_status	"Status bit for DMA buffer overflow and packet was dropped."	RW W1toClr:emac_asf_protocol_fault_status.rx_overflow_status	0
19	rx_hresp_err_status	"Status bit for DMA hresp error (AHB only)."	RW W1toClr:emac_asf_protocol_fault_status.rx_hresp_err_status	0
18	tx_hresp_err_status	"Status bit for DMA hresp error (AHB only)."	RW W1toClr:emac_asf_protocol_fault_status.tx_hresp_err_status	0
17	tx_buff_ex_mid_status	"Status bit for Transmit buffers exhausted before end of packet."	RW W1toClr:emac_asf_protocol_fault_status.tx_buff_ex_mid_status	0
16	tx_underrun_status	"Status bit for Transmit DMA underrun occurred."	RW W1toClr:emac_asf_protocol_fault_status.tx_underrun_status	0
15:9	reserved_15_9	"Reserved, read as 0, ignored on write."	RO	0x00
8	tx_too_many_retries_status	"Status bit for too many retry attempts after collision error (half duplex only)."	RW W1toClr:emac_asf_protocol_fault_status.tx_too_many_retries_status	0
7	rx_udp_ck_err_status	"Status bit for RX packet UDP checksum error."	RW W1toClr:emac_asf_protocol_fault_status.rx_udp_ck_err_status	0
6	rx_tcp_ck_err_status	"Status bit for RX packet TCP checksum error."	RW W1toClr:emac_asf_protocol_fault_status.rx_tcp_ck_err_status	0
5	rx_ip_ck_err_status	"Status bit for RX packet IP checksum error."	RW W1toClr:emac_asf_protocol_fault_status.rx_ip_ck_err_status	0
4	rx_length_err_status	"Status bit for RX packet with length field error."	RW W1toClr:emac_asf_protocol_fault_status.rx_length_err_status	0
3	rx_symbol_err_status	"Status bit for RX packet with symbol errors."	RW W1toClr:emac_asf_protocol_fault_status.rx_symbol_err_status	0
2	rx_long_err_status	"Status bit for RX packet too long."	RW W1toClr:emac_asf_protocol_fault_status.rx_long_err_status	0

1	rx_short_err_status	"Status bit for RX packet too short."	RW W1toClr:emac_asf_protocol_fault_status.rx_short_err_status	0
0	rx_crc_err_status	"Status bit for RX packet with bad CRC."	RW W1toClr:emac_asf_protocol_fault_status.rx_crc_err_status	0

Table 11-586: mmsl_control

Register Information			
Description	"MMSL Control Register. This register contains the control bits for the 802.3br MAC Merge Sublayer (MMSL)"		
Offset	0x0F00	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:8	reserved_31_8	"Reserved, read as 0, ignored on write."	RO	0x00 0000
7	invert_mcrc	"When set to 1 the generated mCRC will be inverted before transmission and the receive path will require the received mCRC to be inverted. Keep this bit 0 for normal operation."	RW	0
6	mmsl_debug_mode	"This bit is used for test purposes only. When set, the verify_timer counter is sped up by a factor of 256. The main effect of this is that about 40 us are spent waiting for a response packet rather than 10 ms."	RW	0
5	route_rx_to_pmac	"When set, the MMSL routes all received traffic to the pMAC rather than the eMAC when pre_active is 0 in the MMSL status register."	RW	1
4	restart_ver	"Write a one to this bit to initiate the verification procedure. Writing a one to this bit has no effect if any of the following are true: pre_enable is zero, verify_disable is one or the verification procedure is already in progress. If pre-emption is active, route_rx_to_pmac should be set to zero before writing to the restart_ver bit otherwise express MAC frames will be sent to the pMAC and will be lost if they are interspersed between pMAC frame fragments. This bit always returns zero when read."	WO	0
3	pre_enable	"This bit is for enabling/disabling the pre-emption operation. If set to zero pre-emption will not occur and verify mpackets will not be responded to or sent. If verify_disable is zero the verification process starts when this bit is written with one and pre-emption can occur once the verification process completes. If verify_disable is high pre-emption can occur as soon as this bit is set. If this bit is reset to zero from one and pre-emption is in progress, then pre-emption will complete and no more pre-emption will occur."	RW	0
2	verify_disable	"This bit is for enabling/disabling the verification procedure that determine whether the link partner can support 802.3br. If disabled the link partner will not be verified and pre-emption is enabled as soon as pre_enable is set to one. Verify mpackets are responded to regardless of whether this bit is set. This bit is static and must be valid before pre_enable is set."	RW	0

1:0	add_frag_size	"This bit field determines the minimum number of bytes which the pMAC sends before pre-emption is allowed. The vector is encoded as follows: 0: 64 bytes 1: 128 bytes 2: 192 bytes 3: 256 bytes These bits are static and must be valid before pre_enable is set."	RW	0x0

Table 11-587: mmsl_status

Register Information				
Description	"MMSL Status Register. This register contains the MMSL status bits."			
Offset	0x0F04	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	smd_error	"This bit is set to 1 if an illegal SMD is received, that is if the SMD was not an Express, Verify, Response, Start Preemptible or Continuation Preemptible SMD."	RO RtoClr	0
9	frer_count_err	"This bit is set to 1 if a frame count error occurs, that is if the SMD-C received indicates a different frame count than expected (i.e. the fragment belongs to another frame and not to the Start packet already received before this) or if a Fragment error happened, which means the field following the SMD-C received was encoding a different fragment count than it was supposed to be"	RO RtoClr	0
8	smdc_error	"This bit is set to 1 if an SMD-C is received when waiting for an SMD-S"	RO RtoClr	0
7	smds_error	"This is set to 1 if an SMD-S is received when waiting for an SMD-C"	RO RtoClr	0
6	rcv_v_error	"This bit is set to 1 if the verification m-packet received is incorrect"	RO RtoClr	0
5	rcv_r_error	"This bit is set to 1 if the response m-packet received is incorrect. If this occurs the verification process fails."	RO RtoClr	0
4:2	verify_status	"These bits indicate the state of the verification state machine: 3'b000 - INIT_VERIFICATION 3'b001 - VERIFICATION_IDLE 3'b010 - SEND_VERIFY 3'b011 - WAIT_FOR_RESPONSE 3'b100 - VERIFIED 3'b101 - VERIFY_FAIL"	RO	0x0
1	respond_status	"This bit indicates the state of the respond state machine: 0 - R_IDLE 1 - SEND_RESPOND"	RO	0
0	pre_active	"This bit is set to one if pre-emption capability is active. It is set to one after the verification process completes, or if verify_disable is one when pre_enable is set to one."	RO	0

Table 11-588: mmsl_err_stats

Register Information				
Description	"MMSL error statistics register. This register contains error counters for the MMSL. It is cleared on a read and does not roll if its maximum value is reached."			
Offset	0x0F08	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	reserved_31_24	"Reserved, read as 0, ignored on write."	RO	0x00
23:16	smd_err_count	"A count of received MAC frames/MAC frame fragments rejected due to unknown SMD value or arriving with an SMD-C when no frame is in progress."	RO RtoClr	0x00
15:8	reserved_15_8	"Reserved, read as 0, ignored on write."	RO	0x00
7:0	ass_error_count	"A count of MAC frames with reassembly errors. The counter is incremented by one every time the ASSEMBLY_ERROR state in the Receive Processing State Diagram is entered."	RO RtoClr	0x00

Table 11-589: mmsl_ass_ok_count

Register Information				
Description	"MMSL frames reassembled OK register. This register contains a count of MAC frames that were successfully reassembled and delivered to MAC. It is cleared on a read and does not roll if its maximum value is reached."			
Offset	0x0F0C	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:17	reserved_31_17	"Reserved, read as 0, ignored on write."	RO	0x0000
16:0	ass_ok_count	"A count of MAC frames that were successfully reassembled and delivered to the MAC."	RO RtoClr	0x0 0000

Table 11-590: mmsl_frag_count_rx

Register Information				
Description	"MMSL fragments received register. This register contains a count of the number of additional mPackets received due to preemption. It is cleared on a read and does not roll if its maximum value is reached."			
Offset	0x0F10	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:17	reserved_31_17	"Reserved, read as 0, ignored on write."	RO	0x0000
16:0	frag_count_rx	"A count of the number of additional mPackets received due to preemption."	RO RtoClr	0x0 0000

Table 11-591: mmsl_frag_count_tx

Register Information					
Description	"MMSL fragments transmitted register. This register contains a count of the number of additional mPackets transmitted due to preemption. It is cleared on a read and does not roll if its maximum value is reached."				
Offset	0x0F14		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:17	reserved_31_17	"Reserved, read as 0, ignored on write."		RO	0x0000
16:0	frag_count_tx	"A count of the number of additional mPackets transmitted due to preemption."		RO RtoClr	0x0 0000

Table 11-592: mmsl_int_status

Register Information					
Description	"MMSL Interrupt Status Register. This register indicates the source of an MMSL interrupt. The corresponding bit in the MMSL interrupt mask register must be clear for a bit to be set. If any bit is set in this register the ethernet_int signal will be asserted. For test purposes each bit can be set or reset by writing to the interrupt mask register. All interrupt status bits are reset to zero on read if the `gem_irq_read_clear define is set otherwise a one is written to the appropriate bit in order to clear it. In this mode reading has no effect on the status of the bit."				
Offset	0x0F18		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:6	reserved_31_6	"Reserved, read as 0, ignored on write."		RO	0x000 0000
5	smd_err	"Indicates an illegal SMD was received, that is the SMD was not an Express, Verify, Response, Start Pre-emptible or Continuation Pre-emptible SMD."		RW RtoClr	0
4	fr_count_err	"Indicates a Frame Count error happened, that is the SMD-C received indicates a different frame count than expected (i.e. the fragment belongs to another frame and not to the Start packet already received before this) or if a Fragment error happened, which means the field following the SMD-C received was encoding a different fragment count than it was supposed to be"		RW RtoClr	0
3	smdc_err	"Indicates an SMD-C was received when waiting for an SMD-S"		RW RtoClr	0
2	smds_err	"Indicates an SMD-S was received when waiting for an SMD-C"		RW RtoClr	0
1	rcv_v_err	"Indicates an incorrect verification m-packet was received"		RW RtoClr	0
0	rcv_r_err	"Indicates an incorrect response m-packet was received. If this occurs the verification process fails."		RW RtoClr	0

Table 11-593: mmsl_int_enable

Register Information				
Description	"MMSL Interrupt Enable Register. At reset all interrupts are disabled. Writing a one to the relevant bit location enables the required interrupt. This register is write only and when read will return zero."			
Offset	0x0F1C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:6	reserved_31_6	"Reserved, read as 0, ignored on write."	RO	0x000 0000
5	smd_error_int_en	"Enable SMD error interrupt."	WO	0
4	fr_count_error_int_en	"Enable frame count error interrupt."	WO	0
3	smdc_error_int_en	"Enable SMD-C error interrupt."	WO	0
2	smds_error_int_en	"Enable SMD-S error interrupt."	WO	0
1	rcv_v_error_int_en	"Enable verify packet received error interrupt."	WO	0
0	rcv_r_error_int_en	"Enable response packet received error interrupt."	WO	0

Table 11-594: mmsl_int_disable

Register Information				
Description	"MMSL Interrupt Disable Register. Writing a 1 to the relevant bit location disables that particular interrupt. This register is write only and when read will return zero."			
Offset	0x0F20	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:6	reserved_31_6	"Reserved, read as 0, ignored on write."	RO	0x000 0000
5	smd_error_int_dis	"Disable SMD error interrupt."	WO	0
4	fr_count_error_int_dis	"Disable frame count error interrupt."	WO	0
3	smdc_error_int_dis	"Disable SMD-C error interrupt."	WO	0
2	smds_error_int_dis	"Disable SMD-S error interrupt."	WO	0
1	rcv_v_error_int_dis	"Disable verify packet received error interrupt."	WO	0
0	rcv_r_error_int_dis	"Disable response packet received error interrupt."	WO	0

Table 11-595: mmsl_int_mask

Register Information			
Description	"MMSL Interrupt Mask Register. The interrupt mask register is a read only register indicating which interrupts are masked. All bits are set at reset and can be reset individually by writing to the interrupt enable register or set individually by writing to the interrupt disable register. Having separate address locations for enable and disable saves the need for performing a read modify write when updating the interrupt mask register. For test purposes there is a write-only function to this register that allows the bits in the interrupt status register to be set or cleared, regardless of the state of the mask register."		
Offset	0x0F24	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:6	reserved_31_6	"Reserved, read as 0, ignored on write."	RO	0x000 0000
5	smd_error_mask	"Mask bit for SMD error interrupt."	RO	1
4	fr_count_error_mask	"Mask bit for frame count error interrupt."	RO	1
3	smdc_error_mask	"Mask bit for SMD-C error interrupt."	RO	1
2	smds_error_mask	"Mask bit for SMD-S error interrupt."	RO	1
1	rcv_v_error_mask	"Mask bit for verify packet received error interrupt."	RO	1
0	rcv_r_error_mask	"Mask bit for response packet received error interrupt."	RO	1

Table 11-596: emac_network_control

Register Information			
Description	"The network control register contains general MAC control functions for both receiver and transmitter."		
Offset	0x1000	**Type:**	RW

Bits	Name	Description	Access	Reset
31	reserved_31	"Reserved, read as 0, ignored on write."	RO	0
30	ifg_eats_qav_credit	"Setting this bit high modifies the CBS algorithm so the IFG/IPG associated with a transmit frame counts towards its 802.1Qav credit."	RW	0
29	reserved_29	"Reserved"	RW	0
28	sel_mii_on_rgmii	"If the RGMII interface being used set this bit high to configure the interface for MII operation (in 802.3br configurations this bit has no effect for the eMAC)."	RW	0
27	oss_correction_field	"1588 One Step Correction Field Update. Set this bit high to enable updating the correction field of PTP 1588 version 2 sync frames by adding current TSU timer value."	RW	0
26	ext_rxq_sel_en	"Enable external selection of receive queue. When this bit is high the ext_match1, ext_match2, ext_match3 and ext_match4 inputs will determine which receive queue a frame is routed to. This will be the case regardless of the state of the external address match enable bit 9 of the network config register. Note that receive frames will be dropped unless they are matched by the internal frame filtering functionality. If the external address match enable bit 9 in the network config register is set frames may be matched by an external address match filter as long as one of the ext_match1, ext_match2, ext_match3 and ext_match4 inputs is asserted early enough. When set ext_rxq_sel_en takes precedence over the existing screener functionality. This bit is only relevant if priority queuing is configured."	RW	0
25	pfc_ctrl	"Enable multiple PFC pause quantums, one per pause priority."	RW	0
24	one_step_sync_mode	"1588 One Step Sync Mode. Write 1 to enable. Replace timestamp field in the 1588 header for TX Sync Frames with current TSU timer value."	RW	0
23	ext_tsu_port_enable	"External TSU timer port enable (1 = enable) - for the eMAC this should be enabled."	RW	0
22	store_udp_offset	"Store UDP / TCP offset to memory. Setting this bit to one will cause the upper 16-bits of the CRC of every received frame to be replaced with the offset from start of frame to the beginning of the UDP or TCP header. The lower 16-bits of the CRC are replaced with zero and reserved for future use. The offset is measured in units of 2 bytes. Set to zero for normal operation."	RW	0
21	alt_sgmii_mode	"Alternative sgmii mode. If asserted with sgmii_mode in the network control register the ACK bit is driven before ability detect during transfer of status information from the PHY to the MAC."	RW	0
20	ptp_unicast_ena	"Enable detection of unicast PTP unicast frames."	RW	0

19	tx_lpi_en	"LPI is supported by the pMAC not the eMAC. Do not set this bit in the eMAC."	RW	0
18	flush_rx_pkt_pclk	"Flush the next packet from the external RX DPRAM. This bit flushes the frame that is the next in line to be pushed out to the AXI interface. Writing one to this bit will only have an effect if the DMA is not currently writing a packet already stored in the DPRAM to memory."	WO	0
17	transmit_pfc_priority_based_pause_frame	"Write a one to transmit PFC priority based pause frame. Takes the values stored in the Transmit PFC Pause Register."	WO	0
16	pfc_enable	"Enable PFC Priority Based Pause Reception capabilities. Setting this bit will enable PFC negotiation and recognition of priority based pause frames."	RW	0
15	store_rx_ts	"Store receive time stamp to memory. Setting this bit to one will cause the CRC of every received frame to be replaced with the value of the nanoseconds field of the 1588 timer that was captured as the receive frame passed the message time stamp point (for releases prior to 1p10 bits 31 and 30 of the value written to the CRC field were hardwired to zero, from release 1p10 onwards these bits represent the two least significant seconds bit of the captured timer value). Set to zero for normal operation."	RW	0
14	stats_read_snap	"Read snapshot - writing a one means that the snapshot value of the statistics register will be read back, otherwise the raw statistic register will be read. The default GEM configuration does not support this function. See Parameterization section under Implementation Application Notes for more details."	RW	0
13	stats_take_snap	"Take snapshot - writing a one will record the current value of all statistics registers in the snapshot registers and clear the statistics registers. The default GEM configuration does not support this function."	WO	0
12	tx_pause_frame_zero	"Transmit zero quantum pause frame - writing one to this bit causes a pause frame with zero quantum to be transmitted."	WO	0
11	tx_pause_frame_req	"Transmit pause frame - writing one to this bit causes a pause frame to be transmitted."	WO	0
10	transmit_halt	"Transmit halt - writing one to this bit resets the tx_go variable and halts the dma from reading more transmit frames into the transmit SRAM buffer. Any frames already read into the transmit SRAM will still be transmitted."	WO	0
9	transmit_start	"Start transmission - writing one to this bit starts transmission."	WO	0
8	back_pressure	"Back pressure if set in 10M or 100M half-duplex mode will force collisions on all received frames. Ignored in gigabit half-duplex mode."	RW	0
7	stats_write_en	"Write enable for statistics registers - setting this bit to one means the statistics registers can be written for functional test purposes."	RW	0
6	inc_all_stats_regs	"Incremental statistics registers - this bit is write only. Writing a one increments all the statistics registers by one for test purposes."	WO	0

5	clear_all_stats_regs	"Clear statistics registers - this bit is write only. Writing a one clears the statistics registers."	WO	0
4	man_port_en	"Management port enable - set to one to enable the management port. When zero forces mdio to high impedance state and mdc low."	RW	0
3	enable_transmit	"Transmit enable - when set, it enables the GEM transmitter to send data. When reset transmission will stop immediately, the transmit pipeline and control registers will be cleared and the transmit queue pointer register will reset to point to the start of the transmit descriptor list. This bit needs to be set before bit 9 the start transmission bit. (Setting this bit low resets the transmit queue pointer however reading the transmit queue pointer register through the APB interface may still return an old value until transmission is restarted.)"	RW	0
2	enable_receive	"Receive enable - when set, it enables the GEM to receive data. When reset frame reception will stop immediately and the receive pipeline will be cleared. The receive queue pointer register is unaffected."	RW	0
1	loopback_local	"Loopback local - asserts the loopback_local signal to the system clock generator. Also connects txd to rxd, tx_en to rx_dv and forces full duplex mode. If the GEM configuration contains the PCS then bit 11 of the network configuration register must be set low to disable TBI mode when in internal loopback. rx_clk and tx_clk may malfunction as the GEM is switched into and out of internal loopback. It is important that receive and transmit circuits have already been disabled when making the switch into and out of internal loopback. Local loopback functionality is optional and may not be supported by some instantiations of the GEM."	RW	0
0	loopback	"Loopback - controls the loopback output pin."	RW	0

Table 11-597: emac_network_config

Register Information			
Description	"The network configuration register contains functions for setting the mode of operation for the Gigabit Ethernet MAC."		
Offset	0x1004	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	uni_direction_enable	"Uni-direction-enable. When low the PCS will transmit idle symbols if the link goes down. When high the PCS can transmit frame data when the link is down."	RW	0
30	ignore_ipg_rx_er	"Ignore IPG rx_er. When set rx_er has no effect on the GEMs operation when rx_dv is low. Set this when using the RGMII wrapper in half-duplex mode."	RW	0
29	nsp_change	"Receive bad preamble. When set frames with non-standard preamble are not rejected (the first byte of preamble needs to be 55 and the last D5, bytes other than 55 or D5 may be inserted in between)."	RW	0
28	ipg_stretch_enable	"IPG stretch enable - when set the transmit IPG can be increased above 96 bit times depending on the previous frame length using the IPG stretch register."	RW	0
27	sgmii_mode_enable	"SGMII mode enable - changes behaviour of the auto-negotiation advertisement and link partner ability registers to meet the requirements of SGMII and reduces the duration of the link timer from 10 ms to 1.6 ms."	RW	0
26	ignore_rx_fcs	"Ignore RX FCS - when set frames with FCS/CRC errors will not be rejected. FCS error statistics will still be collected for frames with bad FCS and FCS status will be recorded in frame's DMA descriptor. For normal operation this bit must be set to zero."	RW	0
25	en_half_duplex_rx	"Enable frames to be received in half-duplex mode while transmitting."	RW	0
24	receive_checksum_offload_enable	"Receive checksum offload enable - when set, the receive checksum engine is enabled. Frames with bad IP, TCP or UDP checksums are discarded."	RW	0
23	disable_copy_of_pause_frames	"Disable copy of pause frames - set to one to prevent pause frames being copied to memory. When set, neither control frames with type id 8808, nor pause frames with destination address 010000c28001 are copied to memory, this functionality was enhanced in release 1p09. Note that valid pause frames received will still increment pause statistics and pause the transmission of frames as required."	RW	0

22:21	data_bus_width	"Data bus width - set according to AMBA (AHB/AXI) or external FIFO data bus width. The reset value for this can be changed by defining a new value for gem_dma_bus_width_def in gem_defs. Only valid bus widths may be written if the system is configured to a maximum width less than 128-bits. 00: 32 bit data bus width 01: 64 bit AMBA (AHB/AXI) data bus width 10: 128 bit AMBA (AHB/AXI) data bus width 11: invalid Note. The reset value of this field is equal to the gem_dma_bus_width_def define, which is user configurable."	RW	0x0
20:18	mdc_clock_division	"MDC clock division - set according to pclk speed. These three bits determine the number pclk will be divided by to generate MDC. For conformance with the 802.3 specification, MDC must not exceed 2.5 MHz (MDC is only active during MDIO read and write operations). The reset value for this can be changed by defining a new value for gem_mdc_clock_div in gem_defs.v 000: divide pclk by 8 (pclk up to 20 MHz) 001: divide pclk by 16 (pclk up to 40 MHz) 010: divide pclk by 32 (pclk up to 80 MHz) 011: divide pclk by 48 (pclk up to 120MHz) 100: divide pclk by 64 (pclk up to 160 MHz) 101: divide pclk by 96 (pclk up to 240 MHz) 110: divide pclk by 128 (pclk up to 320 MHz) 111: divide pclk by 224 (pclk up to 540 MHz). Note. The reset value of this field is equal to the gem_mdc_clock_div define, which is user configurable."	RW	0x2
17	fcs_remove	"FCS remove - setting this bit will cause received frames to be written to memory without their frame check sequence (last 4 bytes). The frame length indicated will be reduced by four bytes in this mode."	RW	0
16	length_field_error_frame_discard	"Length field error frame discard - setting this bit causes frames with a measured length shorter than the extracted length field (as indicated by bytes 13 and 14 in a non-VLAN tagged frame) to be discarded. This only applies to frames with a length field less than 0x0600."	RW	0
15:14	receive_buffer_offset	"Receive buffer offset - indicates the number of bytes by which the received data is offset from the start of the receive buffer. Note that when the define gem_pbuf_rsc has been set then these bits cannot be used."	RW	0x0
13	pause_enable	"Pause enable - when set, transmission will pause if a non-zero 802.3 classic pause frame is received and PFC has not been negotiated."	RW	0
12	retry_test	"Retry test - must be set to zero for normal operation. If set to one the backoff between collisions will always be one slot time. Setting this bit to one helps test the too many retries condition. Also used in the pause frame tests to reduce the pause counter's decrement time from 512 bit times, to every rx_clk cycle."	RW	0

11	pcs_select	"PCS select - selects between MII/GMII and TBI. Must be set for SGMII operation. 0: GMII/MII interface enabled, TBI disabled 1: TBI enabled, GMII/MII disabled (in 802.3br configurations this bit must be set identically in the eMAC and pMAC)"	RW	0
10	gigabit_mode_enable	"Gigabit mode enable - setting this bit configures the GEM for 1000 Mbps operation. 0: 10/100 operation using MII or TBI interface 1: Gigabit operation using GMII or TBI interface (in 802.3br configurations this bit must be set identically in the eMAC and pMAC)"	RW	0
9	external_address_match_enable	"External address match enable - when set the external address match interface can be used to copy frames to memory."	RW	0
8	receive_1536_byte_frames	"Receive 1536 byte frames - setting this bit means the GEM will accept frames up to 1536 bytes in length. Normally the GEM would reject any frame above 1518 bytes."	RW	0
7	unicast_hash_enable	"Unicast hash enable - when set, unicast frames will be accepted when the 6 bit hash function of the destination address points to a bit that is set in the hash register."	RW	0
6	multicast_hash_enable	"Multicast hash enable - when set, multicast frames will be accepted when the 6 bit hash function of the destination address points to a bit that is set in the hash register."	RW	0
5	no_broadcast	"No broadcast - when set to logic one, frames addressed to the broadcast address of all ones will not be accepted."	RW	0
4	copy_all_frames	"Copy all frames - when set to logic one, all valid frames will be accepted."	RW	0
3	jumbo_frames	"Jumbo frames - set to one to enable jumbo frames up to `gem_jumbo_max_length bytes to be accepted. The default length is 10,240 bytes."	RW	0
2	discard_non_vlan_frames	"Discard non-VLAN frames - when set only VLAN tagged frames will be passed to the address matching logic."	RW	0
1	full_duplex	"Full duplex - if set to logic one, the transmit block ignores the state of collision and carrier sense and allows receive while transmitting. Also controls the half_duplex pin."	RW	0
0	speed	"Speed - set to logic one to indicate 100Mbps operation, logic zero for 10Mbps. The value of this pin is reflected on the speed_mode[0] output pin (in 802.3br configurations this bit has no effect for the eMAC)."	RW	0

Table 11-598: emac_network_status

Register Information				
Description	"The network status register returns status information with respect to the PHY management MDIO interface, the PCS, priority flow control, LPI and other status."			
Offset	0x1008	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10:9	reserved_10_9	"Reserved, read as 0, ignored on write."	RO	0x0
8	axi_xaction_outstanding	"Outstanding AXI Transactions - This status bit is set when one or more AXI read or write transactions have been issued by the DUT but the responses have not yet been fully collected."	RO	0
7	lpi_indicate_pclk	"LPI Indication - Low power idle has been detected on receive. This bit is set when LPI is detected and reset when normal idle is detected. An interrupt is generated when the state of this bit changes."	RO	0
6	pfc_negotiate_pclk	"Set when PFC Priority Based Pause has been negotiated."	RO	0
5	mac_pause_tx_en	"PCS auto-negotiation pause transmit resolution."	RO	0
4	mac_pause_rx_en	"PCS auto-negotiation pause receive resolution."	RO	0
3	mac_full_duplex	"PCS auto-negotiation duplex resolution. Set to one if the resolution function determines that both devices are capable of full duplex operation. If zero half-duplex operation is possible as long as bit 0 (PCS link state) is also one."	RO	0
2	man_done	"The PHY management logic is idle (i.e. has completed)."	RO	1
1	mdio_in	"Returns status of the mdio_in pin."	RO	0
0	pcs_link_state	"Returns status of PCS link state. If auto-negotiation is disabled this returns the synchronisation status. If auto-negotiation is enabled it is set in the LINK_OK state as long as a compatible duplex mode is resolved, it is always set in the LINK_OK state in SGMII mode."	RO	0

Table 11-599: emac_dma_config

Register Information			
Description	"DMA Configuration Register"		
Offset	0x1010	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	reserved_31	"Reserved, read as 0, ignored on write."	RO	0
30	dma_addr_bus_width_1	"DMA address bus width. 0 = 32b, 1 = 64b."	RW	0
29	tx_bd_extended_mode_en	"Enable TX extended BD mode. See TX BD control register definition for description of feature."	RW	0
28	rx_bd_extended_mode_en	"Enable RX extended BD mode. See RX BD control register definition for description of feature."	RW	0
27	reserved_27	"Reserved, read as 0, ignored on write."	RO	0
26	force_max_amba_burst_tx	"Force max length bursts on TX. Force the TX DMA to always issue max length bursts on EOP (end of packet) or EOB (end of buffer) transfers as defined by bits 4:0 of this register, even when there is less than max burst data bytes to read. Residual data read is ignored. AHB only - does not apply on AXI bursts (prior to release 1p10) or bursts that break 1k boundary rule - supported on AXI from release 1p10."	RW	0
25	force_max_amba_burst_rx	"Force max length bursts on RX. Force the RX DMA to always issue max length bursts on EOP (end of packet) or EOB (end of buffer) transfers, even if there is less than max burst real packet data required to write. Any extra bytes of pad data is set to 0x00. AHB only - does not apply on AXI bursts (prior to release 1p10) or bursts that break 1k boundary rule - supported on AXI from release 1p10."	RW	0
24	force_discard_on_err	"Auto Discard RX frames during lack of resource. When set, the GEM DMA will automatically discard the next frame (that is the oldest frame) from the receiver packet buffer memory when a receive buffer descriptor is read with its used bit set. When low, then received frames will remain to be stored in the SRAM based packet buffer until AMBA (AHB/AXI) buffer resource next becomes available. In this case if the SRAM memory fills up then there will be a receive overflow condition and the most recently received frame (that is the newest) will be discarded. A write to this bit is ignored if the DMA is not configured in the packet buffer full store and forward mode."	RW	0

23:16	rx_buf_size	"DMA receive buffer size in external AMBA (AHB/AXI) system memory. The value defined by these bits determines the size of buffer to use in main system memory when writing received data. The value is defined in multiples of 64 bytes. 0x01 corresponds to buffers of 64 bytes 0x02 corresponds to 128 bytes etc. For example: 0x02: 128 byte. 0x18: 1536 byte (1*max length frame/buffer) 0xA0: 10240 byte (1*10K jumbo frame/buffer) 0xFF: 16320 byte Note that this value should never be written as zero. Note. The reset value of this field is equal to the gem_rx_buffer_length_def define, which is user configurable."	RW	0x02
15:14	reserved_15_14	"Reserved, read as 0, ignored on write."	RO	0x0
13	crc_error_report	"When the bit is set, bit 16 of the receive buffer descriptor will represent FCS/CRC error (only if frames with FCS are copied to memory as enabled by bit 26 in the network config register). When this bit is clear, bit 16 of the receive buffer descriptor will represent the Canonical format indicator (CFI) bit as extracted from the receive frame (if the receive buffer descriptor is pointing to the last data buffer of the receive frame and the received frame was VLAN tagged)."	RW	0
12	infinite_last_dbuf_size_en	"Forces the receive DMA to consider the data buffer pointed to by last descriptor in the descriptor list to be of elastic size (the last descriptor is the one with its wrap bit set). This means the first buffer pointed to in the list will always contain the beginning of a frame (this helps if there is a desire to build custom logic that interfaces with the receive buffer directly without software intervention). When set the rx_buf_size bits 23:16 in the dma configuration register are ignored for the last receive buffer in the descriptor list and data will be written into the buffer sequentially until the frame is completely received and the buffer descriptor status will be updated with the frame length as normal."	RW	0
11	tx_pbuf_tcp_en	"Transmitter IP, TCP and UDP checksum generation offload enable (not supported when in TX Partial Store and Forward mode). When set, the transmitter checksum generation engine is enabled, to calculate and substitute checksums for transmit frames. When clear, frame data is unaffected. If the GEM is not configured to use the DMA packet buffer, this bit is not implemented and will be treated as reserved, read as 0, ignored on write."	RW	0

10	tx_pbuf_size	"Transmitter packet buffer memory size select. Having this bit at zero halves the amount of memory used for the transmit packet buffer. This reduces the amount of memory used by the GEM. It is important to set this bit to one if the full configured physical memory is available. The value in brackets below represents the size that would result for the default maximum configured memory size of 4 Kbytes. 1: Use full configured addressable space (4 Kb) 0: Do not use top address bit (2 Kb) If the GEM is not configured to use the DMA packet buffer, this bit is not implemented and will be treated as reserved, read as 0, ignored on write. Note. The reset value of this field is equal to the gem_tx_pbuf_size_def define, which is user configurable."	RW	1
9:8	rx_pbuf_size	"Receiver packet buffer memory size select. Having these bits at less than 11 reduces the amount of memory used for the receive packet buffer. This reduces the amount of memory used by the GEM. It is important to set these bits both to one if the full configured physical memory is available. The value in brackets below represents the size that would result for the default maximum configured memory size of 8 Kbytes. 11: Use full configured addressable space (8 Kb) 10: Do not use top address bit (4 Kb) 01: Do not use top two address bits (2 Kb) 00: Do not use top three address bits (1 Kb) If the GEM is not configured to use the DMA packet buffer, these bits are not implemented and will be treated as reserved, read as 0, ignored on write. Note. The reset value of this field is equal to the gem_rx_pbuf_size_def define, which is user configurable."	RW	0x3
7	endian_swap_packet	"endian swap mode enable for packet data accesses. When set, selects swapped endianism for AMBA (AHB/AXI) transfers. When clear, selects little endian mode."	RW	1
6	endian_swap_management	"endian swap mode enable for management descriptor accesses. When set, selects swapped endianism for AMBA (AHB/AXI) transfers. When clear, selects little endian mode."	RW	1
5	hdr_data_splitting_en	"Enable header data Splitting. When set, receive frames will be forwarded to main memory using a minimum of two DMA data buffers. The first X data buffers will contain the frame header, consisting of the Ethernet,VLAN,(IPv4 or IPv6),(TCP or UDP). X= (frame header size divided by rx_buf_size as defined in bits 23:16 of this register). The last Y data buffers will contain the frame payload. Y= (frame payload size divided by rx_buf_size). Note that for non VLAN/IP/TCP/UDP frames, the header will always be 14 bytes. When this feature is disabled, the frame is forwarded to main memory in blocks of rx_buf_size."	RW	0

4:0	amba_burst_length	"Selects the burst length to use on the AMBA (AHB/AXI) when transferring frame data. Not used for DMA management operations and only used where space and data size allow and respecting AXI/AHB burst boundary rules. One-hot priority encoding enforced automatically on register writes as follows, where x represents don't care: 1xxxx: Attempt to use bursts of up to 16. 01xxx: Attempt to use bursts of up to 8. 001xx: Attempt to use bursts of up to 4. 0001x: Always use SINGLE bursts. 00001: Always use SINGLE bursts. 00000: Attempt to use bursts of up to 256."	RW	0x04

Table 11-600: emac_transmit_status

Register Information			
Description	"This register, when read, provides details of the status of the transmit path. Once read, individual bits may be cleared by writing 1 to them. It is not possible to set a bit to 1 by writing to the register."		
Offset	0x1014	**Type:**	RW

Bits	Name	Description	Access	Reset
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	tx_dma_lockup_detected	"TX DMA lockup detected - set when lockup has been detected the transmit DMA lockup monitor."	RW W1toClr	0
9	tx_mac_lockup_detected	"TX MAC lockup detected - set when lockup has been detected by the lockup monitor on the MAC transmit path."	RW W1toClr	0
8	resp_not_ok	"bresp/hresp not OK - set when the DMA block sees bresp/hresp not OK. Cleared by writing a one to this bit."	RW W1toClr	0
7	late_collision_occurred	"Late collision occurred - only set if the condition occurs in gigabit mode, as retry is not attempted. Cleared by writing a one to this bit."	RW W1toClr	0
6	transmit_under_run	"Transmit under run - this bit is set if the transmitter was forced to terminate a frame that it had already began transmitting due to further data being unavailable. This bit is set if a transmitter status write back has not completed when another status write back is attempted. When using the DMA interface configured for internal FIFO mode, this bit is also set when the transmit DMA has written the SOP data into the FIFO and either the AHB bus was not granted in time for further data, or because an AHB not OK response was returned, or because a used bit was read. When using the DMA interface configured for packet buffer mode, this bit will never be set. When using the external FIFO interface, this bit is also set when the tx_r_underflow input is asserted during a frame transfer. Cleared by writing a 1."	RW W1toClr	0
5	transmit_complete	"Transmit complete - set when a frame has been transmitted. Cleared by writing a one to this bit."	RW W1toClr	0
4	amba_error	"Transmit frame corruption due to AMBA (AHB/AXI) errors. Set if an error occurs whilst midway through reading transmit frame from external memory including HRESP(AHB), RRESP or BRESP(AXI) errors and buffers exhausted mid frame (if the buffers run out during transmission of a frame then transmission stops, FCS shall be bad and tx_er asserted). Also set in DMA packet buffer mode if single frame is too large for configured packet buffer memory size. Cleared by writing a one to this bit."	RW W1toClr	0
3	transmit_go	"Transmit go - if high transmit is active. When using the exposed FIFO interface, this bit represents bit 3 of the network control register. When using the DMA interface this bit represents the tx_go variable as specified in the transmit buffer description."	RO	0
2	retry_limit_exceeded	"Retry limit exceeded - cleared by writing a one to this bit."	RW W1toClr	0

1	collision_occurred	"Collision occurred - set by the assertion of collision. Cleared by writing a one to this bit. When operating in 10/100 mode, this status indicates either a collision or a late collision. In gigabit mode, this status is not set for a late collision."	RW W1toClr	0
0	used_bit_read	"Used bit read - set when a transmit buffer descriptor is read with its used bit set. Cleared by writing a one to this bit."	RW W1toClr	0

Table 11-601: emac_receive_q_ptr

Register Information				
Description	"This register holds the start address of the receive buffer queue (receive buffers descriptor list). The receive buffer queue base address must be initialized before receive is enabled through bit 2 of the network control register. Once reception is enabled, any write to the receive buffer queue base address register is ignored. Reading this register returns the location of the descriptor currently being accessed. This value increments as buffers are used. Software should not use this register for determining where to remove received frames from the queue as it constantly changes as new frames are received. Software should instead work its way through the buffer descriptor queue checking the used bits. In terms of AMBA (AHB/AXI) operation, the receive descriptors are read from memory using a single 32bit AHB/AXI access. When the datapath is configured at 64bit or 128bit, the receive descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is written to using a single 64bit AHB/AXI access. For 32bit datapaths, the receive descriptors should be aligned at 32-bit boundaries and are written to using two individual non sequential 32-bit accesses."			
Offset	0x1018	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:2	dma_rx_q_ptr	"Receive buffer queue base address - written with the address of the start of the receive queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_rx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while receive is not enabled."	RW	0

Table 11-602: emac_transmit_q_ptr

Register Information			
Description	"This register holds the start address of the transmit buffer queue (transmit buffers descriptor list). The transmit buffer queue base address register must be initialized before transmit is started through bit 9 of the network control register. Once transmission has started, any write to the transmit buffer queue base address register is illegal and therefore ignored. Note that due to clock boundary synchronization, it takes a maximum of four pclk cycles from the writing of the transmit start bit before the transmitter is active. Writing to the transmit buffer queue base address register during this time may produce unpredictable results. Reading this register returns the location of the descriptor currently being accessed. Because the DMA can store data for multiple frames at once, this may not necessarily be pointing to the current frame being transmitted. In terms of AMBA AHB/AXI operation, the transmit descriptors are written to memory using a single 32bit AHB access. When the datapath is configured as 64bit or 128bit, the transmit descriptors should be aligned at 64-bit boundaries and each pair of 32-bit descriptors is read from memory using a single AHB/AXI access. For 32bit datapaths, the descriptors should be aligned at 32-bit boundaries and the descriptors are read from memory using two individual 32-bit non sequential accesses."		
Offset	0x101C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:2	dma_tx_q_ptr	"Transmit buffer queue base address - written with the address of the start of the transmit queue."	RW	0x0000 0000
1	reserved_1	"Reserved, read as 0, ignored on write."	RO	0
0	dma_tx_dis_q	"Disable queue if set to 1. This can be used to reduce the number of active queues and should only be changed while transmit is not enabled."	RW	0

Table 11-603: emac_receive_status

Register Information			
Description	"This register, when read provides details of the status of the receive path. Once read, individual bits may be cleared by writing 1 to them. It is not possible to set a bit to 1 by writing to the register."		
Offset	0x1020	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:6	reserved_31_6	"Reserved, read as 0, ignored on write."	RO	0x000 0000
5	rx_dma_lockup_detected	"RX DMA lockup detected - set when lockup has been detected the receive DMA lockup monitor."	RW W1toClr	0
4	rx_mac_lockup_detected	"RX MAC lockup detected - set when lockup has been detected by the lockup monitor on the MAC receive path."	RW W1toClr	0
3	resp_not_ok	"bresp/hresp not OK - set when the DMA block sees bresp/hresp not OK. Cleared by writing a one to this bit."	RW W1toClr	0
2	receive_overrun	"Receive overrun - this bit is set if either the gem_dma RX FIFO or external RX FIFO were unable to store the receive frame due to a FIFO overflow, or if the receive status, reported by the gem_rx module to the gem_dma was not taken at end of frame. This bit is also set in DMA packet buffer mode if the packet buffer overflows. For DMA operation the buffer will be recovered if an overrun occurs. This bit is cleared by writing a one to it."	RW W1toClr	0
1	frame_received	"Frame received - one or more frames have been received and placed in memory. Cleared by writing a one to this bit."	RW W1toClr	0
0	buffer_not_available	"Buffer not available - an attempt was made to get a new buffer and the pointer indicated that it was owned by the processor. The DMA will reread the pointer each time an end of frame is received until a valid pointer is found. This bit is set following each descriptor read attempt that fails, even if consecutive pointers are unsuccessful and software has in the meantime cleared the status flag. Cleared by writing a one to this bit."	RW W1toClr	0

Table 11-604: emac_int_status

Register Information			
Description	"If not configured for priority queuing, the GEM generates a single interrupt. This register indicates the source of this interrupt. The corresponding bit in the mask register must be clear for a bit to be set. If any bit is set in this register the ethernet_int signal will be asserted. For test purposes each bit can be set or reset by writing to the interrupt mask register. The default configuration is shown below whereby all bits are reset to zero on read. Changing the validity of the `gem_irq_read_clear define will instead require a one to be written to the appropriate bit in order to clear it. In this mode reading has no effect on the status of the bit."		
Offset	0x1024	**Type:**	RW

Bitfield Details

Bits	Name	Description	Access	Reset
31	tx_lockup_detected	"TX lockup detection - set when lockup has been detected by any of the lockup monitors on the transmit datapath."	RW RtoClr	0
30	rx_lockup_detected	"RX lockup detection - set when lockup has been detected by any of the lockup monitors on the receive datapath."	RW RtoClr	0
29	tsu_timer_comparison_interrupt	"TSU timer comparison interrupt. Indicates when TSU timer count value is equal to programmed value."	RW RtoClr	0
28	wol_interrupt	"WOL interrupt. Indicates a WOL event has been received."	RW RtoClr	0
27	receive_lpi_indication_status_bit_change	"Receive LPI indication status bit change"	RW RtoClr	0
26	tsu_seconds_register_increment	"TSU seconds register increment indicates the register has incremented. Cleared on read."	RW RtoClr	0
25	ptp_pdelay_resp_frame_transmitted	"PTP pdelay_resp frame transmitted indicates a PTP pdelay_resp frame has been transmitted. Cleared on read."	RW RtoClr	0
24	ptp_pdelay_req_frame_transmitted	"PTP pdelay_req frame transmitted indicates a PTP pdelay_req frame has been transmitted. Cleared on read."	RW RtoClr	0
23	ptp_pdelay_resp_frame_received	"PTP pdelay_resp frame received indicates a PTP pdelay_resp frame has been received. Cleared on read."	RW RtoClr	0
22	ptp_pdelay_req_frame_received	"PTP pdelay_req frame received indicates a PTP pdelay_req frame has been received. Cleared on read."	RW RtoClr	0
21	ptp_sync_frame_transmitted	"PTP sync frame transmitted indicates a PTP sync frame has been transmitted. Cleared on read."	RW RtoClr	0
20	ptp_delay_req_frame_transmitted	"PTP delay_req frame transmitted indicates a PTP delay_req frame has been transmitted. Cleared on read."	RW RtoClr	0
19	ptp_sync_frame_received	"PTP sync frame received indicates a PTP sync frame has been received. Cleared on read."	RW RtoClr	0
18	ptp_delay_req_frame_received	"PTP delay_req frame received indicates a PTP delay_req frame has been received. Cleared on read."	RW RtoClr	0

17	pcs_link_partner_page_received	"PCS link partner page received - set when a new base page or next page is received from the link partner. The first time this interrupt is received, it will indicate base page received and subsequent reads will indicate next pages. The next page and base page registers should only be read when this interrupt is signalled. For next pages, the link partner next page register should be read first to avoid the register being over written. This interrupt also indicates when the host should write a new page into the next page register. If further next page exchange is only required by the link partner, this register should be written with a null message page (0x2001). Cleared on read."	RW RtoClr	0
16	pcs_auto_negotiation_complete	"PCS auto-negotiation complete - set once the internal PCS layer has completed auto-negotiation. Cleared on read."	RW RtoClr	0
15	external_interrupt	"External interrupt - set when a rising edge has been detected on the ext_interrupt_in input pin. Cleared on read."	RW RtoClr	0
14	pause_frame_transmitted	"Pause frame transmitted - indicates a pause frame has been successfully transmitted after being initiated from the network control register or from the tx_pause control pin. Cleared on read."	RW RtoClr	0
13	pause_time_elapsed	"Pause Time elapsed - set when either the pause time register at address 0x38 decrements to zero, or when a valid pause frame is received with a zero pause quantum field. Not set for PFC pause frames. Cleared on read."	RW RtoClr	0
12	pause_frame_with_non_zero_pause_quantum_received	"Pause frame with non-zero pause quantum received - indicates a valid legacy pause frame with a non-zero pause quantum field has been received or any valid PFC pause frame has been received. Cleared on read."	RW RtoClr	0
11	resp_not_ok	"bresp/hresp not OK - set when the DMA block sees bresp/hresp not OK. Cleared on read."	RW RtoClr	0
10	receive_overrun	"Receive overrun - set when the receive overrun status bit gets set. Cleared on read."	RW RtoClr	0
9	link_change	"Link change - set when the PCS link status changes. If auto-negotiation is enabled then link status goes high when it completes. If AN is disabled link status goes high when synchronization is achieved. If 802.3cb link fault signalling is enabled a link change interrupt is triggered when the link fault indication as reported in the network status register changes. Cleared on read."	RW RtoClr	0
8	reserved_8	"Reserved, read as 0, ignored on write."	RO	0
7	transmit_complete	"Transmit complete - set when a frame has been transmitted. Cleared on read."	RW RtoClr	0
6	amba_error	"Transmit frame corruption due to AMBA (AHB/AXI) error. Set if an error occurs whilst midway through reading transmit frame from external system memory, including HRESP errors(AHB), RRESP or BRESP(AXI) errors and buffers exhausted mid frame (if the buffers run out during transmission of a frame then transmission stops, FCS shall be bad and tx_er asserted). Also set in DMA packet buffer mode if single frame is too large for configured packet buffer memory size. Cleared on a read."	RW RtoClr	0

5	retry_limit_exceeded_or_late_collision	"Retry limit exceeded or late collision - transmit error. Late collision will only cause this status bit to be set in gigabit mode (as a retry is not attempted). Cleared on read."	RW RtoClr	0
4	transmit_under_run	"Transmit under run - this interrupt is set if the transmitter was forced to terminate a frame that it has already began transmitting due to further data being unavailable. If an under run occurs, the transmitter will force bad crc and tx_er high. This interrupt is set if a transmitter status write back has not completed when another status write back is attempted. When using the DMA interface configured for internal FIFO mode, this interrupt is also set when the transmit DMA has written the SOP data into the FIFO and either the AHB bus was not granted in time for further data, or because an AHB/AXI error response was returned by the connected slave, or because a used bit was read. When using the DMA interface configured for packet buffer mode, this bit will never be set. When using the external FIFO interface, this interrupt is also set when the tx_r_underflow input was asserted during a frame transfer. Cleared on read."	RW RtoClr	0
3	tx_used_bit_read	"TX used bit read - set when a transmit buffer descriptor is read with its used bit set. Cleared on read."	RW RtoClr	0
2	rx_used_bit_read	"RX used bit read - set when a receive buffer descriptor is read with its used bit set. Cleared on read."	RW RtoClr	0
1	receive_complete	"Receive complete - a frame has been stored in memory. Cleared on read."	RW RtoClr	0
0	management_frame_sent	"Management frame sent - the PHY management register has completed its operation. Cleared on read."	RW RtoClr	0

Table 11-605: emac_int_enable

Register Information			
Description	"At reset all interrupts are disabled. Writing a one to the relevant bit location enables the required interrupt. This register is write only and when read will return zero."		
Offset	0x1028	**Type:**	RW

Bits	Name	Description	Access	Reset
31	enable_tx_lockup_detected_interrupt	"Enable TX lockup detection interrupt."	WO	0
30	enable_rx_lockup_detected_interrupt	"Enable RX lockup detection interrupt."	WO	0
29	enable_tsu_timer_comparison_interrupt	"Enable TSU timer comparison interrupt."	WO	0
28	enable_wol_event_received_interrupt	"Enable WOL event received interrupt"	WO	0
27	enable_rx_lpi_indication_interrupt	"Enable RX LPI indication interrupt"	WO	0
26	enable_tsu_seconds_register_increment	"Enable TSU seconds register increment"	WO	0
25	enable_ptp_pdelay_resp_frame_transmitted	"Enable PTP pdelay_resp frame transmitted"	WO	0
24	enable_ptp_pdelay_req_frame_transmitted	"Enable PTP pdelay_req frame transmitted"	WO	0
23	enable_ptp_pdelay_resp_frame_received	"Enable PTP pdelay_resp frame received"	WO	0
22	enable_ptp_pdelay_req_frame_received	"Enable PTP pdelay_req frame received"	WO	0
21	enable_ptp_sync_frame_transmitted	"Enable PTP sync frame transmitted"	WO	0
20	enable_ptp_delay_req_frame_transmitted	"Enable PTP delay_req frame transmitted"	WO	0
19	enable_ptp_sync_frame_received	"Enable PTP sync frame received"	WO	0
18	enable_ptp_delay_req_frame_received	"Enable PTP delay_req frame received"	WO	0
17	enable_pcs_link_partner_page_received	"Enable PCS link partner page received"	WO	0
16	enable_pcs_auto_negotiation_complete_interrupt	"Enable PCS auto-negotiation complete interrupt"	WO	0
15	enable_external_interrupt	"Enable external interrupt"	WO	0

14	enable_pause_frame_transmitted_interrupt	"Enable pause frame transmitted interrupt"	WO	0
13	enable_pause_time_zero_interrupt	"Enable pause time zero interrupt"	WO	0
12	enable_pause_frame_with_non_zero_pause_quantum_interrupt	"Enable pause frame with non-zero pause quantum interrupt"	WO	0
11	enable_resp_not_ok_interrupt	"Enable bresp/hresp not OK interrupt"	WO	0
10	enable_receive_overrun_interrupt	"Enable receive overrun interrupt"	WO	0
9	enable_link_change_interrupt	"Enable link change interrupt"	WO	0
8	reserved_8	"Reserved, read as 0, ignored on write."	RO	0
7	enable_transmit_complete_interrupt	"Enable transmit complete interrupt"	WO	0
6	enable_transmit_frame_corruption_due_to_amba_error_interrupt	"Enable transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	enable_retry_limit_exceeded_or_late_collision_interrupt	"Enable retry limit exceeded or late collision interrupt"	WO	0
4	enable_transmit_buffer_under_run_interrupt	"Enable transmit buffer under run interrupt"	WO	0
3	enable_transmit_used_bit_read_interrupt	"Enable transmit used bit read interrupt"	WO	0
2	enable_receive_used_bit_read_interrupt	"Enable receive used bit read interrupt"	WO	0
1	enable_receive_complete_interrupt	"Enable receive complete interrupt"	WO	0
0	enable_management_done_interrupt	"Enable management done interrupt"	WO	0

Table 11-606: emac_int_disable

Register Information			
Description	"Writing a 1 to the relevant bit location disables that particular interrupt. This register is write only and when read will return zero."		
Offset	0x102C	**Type:**	RW

Bits	Name	Description	Access	Reset
31	disable_tx_lockup_detected_interrupt	"Disable TX lockup detection interrupt."	WO	0
30	disable_rx_lockup_detected_interrupt	"Disable RX lockup detection interrupt."	WO	0
29	disable_tsu_timer_comparison_interrupt	"Disable TSU timer comparison interrupt."	WO	0
28	disable_wol_event_received_interrupt	"Disable WOL event received interrupt"	WO	0
27	disable_rx_lpi_indication_interrupt	"Disable RX LPI indication interrupt"	WO	0
26	disable_tsu_seconds_register_increment	"Disable TSU seconds register increment"	WO	0
25	disable_ptp_pdelay_resp_frame_transmitted	"Disable PTP pdelay_resp frame transmitted"	WO	0
24	disable_ptp_pdelay_req_frame_transmitted	"Disable PTP pdelay_req frame transmitted"	WO	0
23	disable_ptp_pdelay_resp_frame_received	"Disable PTP pdelay_resp frame received"	WO	0
22	disable_ptp_pdelay_req_frame_received	"Disable PTP pdelay_req frame received"	WO	0
21	disable_ptp_sync_frame_transmitted	"Disable PTP sync frame transmitted"	WO	0
20	disable_ptp_delay_req_frame_transmitted	"Disable PTP delay_req frame transmitted"	WO	0
19	disable_ptp_sync_frame_received	"Disable PTP sync frame received"	WO	0
18	disable_ptp_delay_req_frame_received	"Disable PTP delay_req frame received"	WO	0
17	disable_pcs_link_partner_page_received	"Disable PCS link partner page received"	WO	0
16	disable_pcs_auto_negotiation_complete_interrupt	"Disable PCS auto-negotiation complete interrupt"	WO	0
15	disable_external_interrupt	"Disable external interrupt"	WO	0

14	disable_pause_frame_transmitted_interrupt	"Disable pause frame transmitted interrupt"	WO	0
13	disable_pause_time_zero_interrupt	"Disable pause time zero interrupt"	WO	0
12	disable_pause_frame_with_non_zero_pause_quantum_interrupt	"Disable pause frame with non-zero pause quantum interrupt"	WO	0
11	disable_resp_not_ok_interrupt	"Disable bresp/hresp not OK interrupt"	WO	0
10	disable_receive_overrun_interrupt	"Disable receive overrun interrupt"	WO	0
9	disable_link_change_interrupt	"Disable link change interrupt"	WO	0
8	reserved_8	"Reserved, read as 0, ignored on write."	RO	0
7	disable_transmit_complete_interrupt	"Disable transmit complete interrupt"	WO	0
6	disable_transmit_frame_corruption_due_to_amba_error_interrupt	"Disable transmit frame corruption due to AMBA (AHB/AXI) error interrupt"	WO	0
5	disable_retry_limit_exceeded_or_late_collision_interrupt	"Disable retry limit exceeded or late collision interrupt"	WO	0
4	disable_transmit_buffer_under_run_interrupt	"Disable transmit buffer under run interrupt"	WO	0
3	disable_transmit_used_bit_read_interrupt	"Disable transmit used bit read interrupt"	WO	0
2	disable_receive_used_bit_read_interrupt	"Disable receive used bit read interrupt"	WO	0
1	disable_receive_complete_interrupt	"Disable receive complete interrupt"	WO	0
0	disable_management_done_interrupt	"Disable management done interrupt"	WO	0

Table 11-607: emac_int_mask

Register Information			
Description	"The interrupt mask register is a read only register indicating which interrupts are masked. All bits are set at reset and can be reset individually by writing to the interrupt enable register or set individually by writing to the interrupt disable register. Having separate address locations for enable and disable saves the need for performing a read modify write when updating the interrupt mask register. For test purposes there is a write-only function to this register that allows the bits in the interrupt status register to be set or cleared, regardless of the state of the mask register."		
Offset	0x1030	**Type:**	RO

Bits	Name	Description	Access	Reset
31	tx_lockup_detected_mask	"TX lockup interrupt mask."	RO	1
30	rx_lockup_detected_mask	"RX lockup interrupt mask."	RO	1
29	tsu_timer_comparison_mask	"Enable TSU timer comparison interrupt mask."	RO	1
28	wol_event_received_mask	"A read of this register returns the value of the WOL event received mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
27	rx_lpi_indication_mask	"A read of this register returns the value of the RX LPI indication mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written"	RO	1
26	tsu_seconds_register_increment_mask	"A read of this register returns the value of the TSU seconds register increment mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
25	ptp_pdelay_resp_frame_transmitted_mask	"A read of this register returns the value of the PTP pdelay_resp frame transmitted mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
24	ptp_pdelay_req_frame_transmitted_mask	"A read of this register returns the value of the PTP pdelay_req frame transmitted mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1

23	ptp_pdelay_resp_frame_received_mask	"A read of this register returns the value of the PTP pdelay_resp frame received mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
22	ptp_pdelay_req_frame_received_mask	"A read of this register returns the value of the PTP pdelay_req frame received mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
21	ptp_sync_frame_transmitted_mask	"A read of this register returns the value of the PTP sync frame transmitted mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
20	ptp_delay_req_frame_transmitted_mask	"A read of this register returns the value of the PTP delay_req frame transmitted mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
19	ptp_sync_frame_received_mask	"A read of this register returns the value of the PTP sync frame received mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
18	ptp_delay_req_frame_received_mask	"A read of this register returns the value of the PTP delay_req frame received mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
17	pcs_link_partner_page_mask	"A read of this register returns the value of the PCS link partner page mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
16	pcs_auto_negotiation_complete_interrupt_mask	"A read of this register returns the value of the PCS auto-negotiation complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1

15	external_interrupt_mask	"External interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
14	pause_frame_transmitted_interrupt_mask	"Pause frame transmitted interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
13	pause_time_zero_interrupt_mask	"Pause time zero interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
12	pause_frame_with_non_zero_pause_quantum_interrupt_mask	"Pause frame with non-zero pause quantum interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
11	resp_not_ok_interrupt_mask	"bresp/hresp not OK interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
10	receive_overrun_interrupt_mask	"Receive overrun interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
9	link_change_interrupt_mask	"A read of this register returns the value of the link change interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
8	reserved_8	"Reserved, read as 0, ignored on write."	RO	0
7	transmit_complete_interrupt_mask	"Transmit complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
6	amba_error_interrupt_mask	"Transmit frame corruption due to AMBA (AHB/AXI) error interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1

5	retry_limit_exceeded_or_late_collision_mask	"A read of this register returns the value of the retry limit exceeded or late collision (gigabit mode only) interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
4	transmit_buffer_under_run_interrupt_mask	"Transmit buffer under run interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
3	transmit_used_bit_read_interrupt_mask	"Transmit used bit read interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
2	receive_used_bit_read_interrupt_mask	"Receive used bit read interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
1	receive_complete_interrupt_mask	"Receive complete interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1
0	management_done_interrupt_mask	"Management done interrupt mask. 0: Interrupt is enabled. 1: Interrupt is disabled. A write to this register directly affects the state of the corresponding bit in the interrupt status register, causing an interrupt to be generated if a 1 is written."	RO	1

Table 11-608: emac_phy_management

Register Information				
Description	"Do not use this register. The eMAC MDIO port is unconnected."			
Offset	0x1034	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	write0	"Must be written with 0."	RW	0
30	write1	"Must be written to 1 for a valid Clause 22 frame and to 0 for a valid Clause 45 frame."	RW	0
29:28	operation	"Operation. For a Clause 45 frame: 00 is an addr, 01 is a write, 10 is a post read increment, 11 is a read frame. For a Clause 22 frame: 10 is a read, 01 is a write."	RW	0x0
27:23	phy_address	"PHY address."	RW	0x00
22:18	register_address	"Register address - specifies the register in the PHY to access."	RW	0x00
17:16	write10	"Must be written with 10."	RW	0x0
15:0	phy_write_read_data	"For a write operation this is written with the data to be written to the PHY. After a read operation this contains the data read from the PHY."	RW	0x0000

Table 11-609: emac_pause_time

Register Information				
Description	"Received Pause Quantum Register"			
Offset	0x1038	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	quantum	"Received pause quantum - stores the current value of the received pause quantum register which is decremented every 512 bit times."	RO	0x0000

Table 11-610: emac_tx_pause_quantum

Register Information				
Description	"Transmit Pause Quantum Register"			
Offset	0x103C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	quantum_p1	"Transmit pause quantum - written with the pause quantum value for pause frame transmission of priority 1."	RW	0xFFFF
15:0	quantum	"Transmit pause quantum - written with the pause quantum value for pause frame transmission."	RW	0xFFFF

Table 11-611: emac_pbuf_txcutthru

Register Information				
Description	"Transmit Partial Store and Forward Register. Partial store and forward is only applicable when using the DMA configured in SRAM based packet buffer mode. It is also not available when using multi buffer frames. This register contains the enable bit and watermark value for transmit cut-through operation."			
Offset	0x1040	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	dma_tx_cutthru	"Enable TX partial store and forward operation"	RW	0
30:11	reserved	"Reserved, read as 0, ignored on write."	RO	0x0 0000
10:0	dma_tx_cutthru_threshold	"Watermark value corresponding to number of SRAM locations. This value must be >= 0x14. The actual number of bytes for the watermark is obtained by multiplying the watermark value by the value of the gem_tx_pbuf_data define divided by 8. The reset value depends on the value of the configuration option `gem_emac_tx_pbuf_addr, which is defined in the verilog defs configuration file."	RW	0x7FF

Table 11-612: emac_pbuf_rxcutthru

Register Information				
Description	"Receive Partial Store and Forward Register. Partial store and forward is only applicable when using the DMA configured in SRAM based packet buffer mode. It is also not available when using multi buffer frames. This register contains the enable bit and watermark value for receive cut-through operation."			
Offset	0x1044	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	dma_rx_cutthru	"Enable RX partial store and forward operation"	RW	0
30:11	reserved	"Reserved, read as 0, ignored on write."	RO	0x0 0000
10:0	dma_rx_cutthru_threshold	"Watermark value corresponding to number of SRAM locations. The actual number of bytes for the watermark is obtained by multiplying the watermark value by the value of the gem_rx_pbuf_data define divided by 8. The reset value depends on the value of the configuration option `gem_emac_rx_pbuf_addr, which is defined in the verilog defs configuration file."	RW	0x7FF

Table 11-613: emac_jumbo_max_length

Register Information				
Description	"Maximum Jumbo Frame Size."			
Offset	0x1048	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:14	reserved_31_14	"Reserved, read as 0, ignored on write."	RO	0x0 0000
13:0	jumbo_max_length	"Maximum Jumbo Frame Size - resets to the gem_jumbo_max_length define value."	RW	0x2800

Table 11-614: emac_external_fifo_interface

Register Information				
Description	"External FIFO Interface Enable (only valid when gem_host_if_soft_select is defined)"			
Offset	0x104C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:1	reserved_31_1	"Reserved, read as 0, ignored on write."	RO	0x0000 0000
0	external_fifo_interface	"Enable external fifo interface. 1: Enabled. 0: Disabled."	RW	0

Table 11-615: emac_axi_max_pipeline

Register Information				
Description	"Used to set the maximum amount of outstanding transactions on the AXI bus between AR / R channels and AW / W channels. Cannot be more than the depth of the configured AXI pipeline FIFO (defined in verilog defs.v)"			
Offset	0x1054	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:17	reserved	"Reserved, read as 0, ignored on write."	RO	0x0000
16	use_aw2b_fill	"For the write issuing capability as defined in bits 15:8 of this register, select whether the max number of transactions operates between the AW to W AXI channel or the AW to B channel. Set to 0 to operate between the AW and W channels. Set to 1 to operate between the AW and B channels."	RW	0
15:8	aw2w_max_pipeline	"Defines the maximum number of outstanding AXI write requests that can be issued by the DMA via the AW channel. This is effectively the write issuing capability."	RW	0x01
7:0	ar2r_max_pipeline	"Defines the maximum number of outstanding AXI read requests that can be issued by the DMA via the AR channel. This is effectively the read issuing capability."	RW	0x01

Table 11-616: emac_int_moderation

Register Information			
Description	"Used to moderate the number of transmit and receive complete interrupts issued. With interrupt moderation enabled receive and transmit interrupts are not generated immediately a frame is transmitted or received. Instead when a receive or transmit event occurs a timer is started and the interrupt is asserted after it times out. This limits the frequency with which the CPU receives interrupts. When interrupt moderation is enabled interrupt status bit one is always used for receive and bit 7 is always used for transmit even when priority queuing is enabled. With interrupt moderation 800ns periods are counted. GEM determines what constitutes an 800ns period by looking at the tbi (bit 11), gigabit bit (10) and speed (bit 0) bits in the network configuration register and counting tx_clk cycles. Bit 0 needs to be set to 1 for 100M operation. From release 1p11 onwards a frame threshold value may also be used to moderate interrupts. If both time based and frame threshold moderation is enabled the interrupt will be asserted as soon as the first moderation method expires."		
Offset	0x105C	**Type:**	RW

Bits	Name	Description	Access	Reset
31:24	tx_int_mod_thresh	"Count of transmitted frames before bit 7 is set in the interrupt status register. A non-zero value indicates transmit interrupt moderation will be performed."	RW	0x00
23:16	tx_int_moderation	"Count of 800ns periods before bit 7 is set in the interrupt status register after a frame is transmitted. A non-zero value indicates transmit interrupt moderation will be performed."	RW	0x00
15:8	rx_int_mod_thresh	"Count of received frames before bit 1 is set in the interrupt status register. A non-zero value indicates receive interrupt moderation will be performed."	RW	0x00
7:0	rx_int_moderation	"Count of 800ns periods before bit 1 is set in the interrupt status register after a frame is received. A non-zero value indicates receive interrupt moderation will be performed."	RW	0x00

Table 11-617: emac_sys_wake_time

Register Information			
Description	"Used to pause transmission after deassertion of tx_lpi_en. Each unit in this register corresponds to 25.6ns in 2.5G mode, 64ns in gigabit mode, 320ns in 100M mode and 3200ns at 10M. After tx_lpi_en is deasserted transmission will pause for the set time."		
Offset	0x1060	**Type:**	RW

Bits	Name	Description	Access	Reset
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	sys_wake_time	"Count of 25.6ns, 64ns, 320ns or 3200ns intervals before transmission starts after deassertion of tx_lpi_en (each interval is equivalent to eight tx_clk periods and so varies with data rate)."	RW	0x0000

Table 11-618: emac_lockup_config

Register Information			
Description	"The lockup detection and recovery configuration register."		
Offset	0x1068	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	tx_dma_lockup_mon_en	"Enable the TX DMA lockup detector. Enable the monitor that detects lockups in the transmit DMA."	RW	0
30	tx_mac_lockup_mon_en	"Enable the TX MAC lockup detector. Enable the monitor that detects lockups in the MAC transmit path."	RW	0
29	rx_dma_lockup_mon_en	"Enable the RX DMA lockup detector. Enable the monitor that detects lockups in the receive DMA."	RW	0
28	rx_mac_lockup_mon_en	"Enable the RX MAC lockup detector. Enable the monitor that detects lockups in the MAC receive path."	RW	0
27	lockup_recovery_en	"Lockup recovery. If this bit is set then the module will go into a soft reset state if a lockup is detected on the transmit or receive data paths."	RW	0
26:16	dma_lockup_time	"Prior to release 1p11 this field was reserved. From release 1p11 onwards this field defines the timeout value for transmit and receive DMA lockup detection defined as a multiple of the prescaler value stored in bits 15:0 of this register."	RW	0x7FF
15:0	prescaler_value	"For release 1p10 this field defined the time (measured in units of 1024 tx_clk periods) after which the lockup detection monitors would trigger, except for the receive MAC lock up detector which had its lockup time register. From release 1p11 onwards this field defines a prescaler value which is the number of tx_clk periods used to scale the timeout registers."	RW	0xFFFF

Table 11-619: emac_mac_lockup_time

Register Information				
Description	"MAC lockup detection time register. For receive this register specifies the maximum time between received frames. If no valid EOP is seen at the receive FIFO interface within the timer period then a lockup is considered to have occurred in the receive MAC. For transmit the MAC lockup time is simply the time it takes for data to be seen on the MII output pins after entering on the MAC FIFO interface."			
Offset	0x106C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:27	reserved_31_27	"Reserved, read as 0, ignored on write."	RO	0x00
26:16	tx_mac_lockup_time	"Prior to release 1p11 this field was reserved. From release 1p11 onwards this field defines the timeout value for transmit MAC lockup detection defined as a multiple of the prescaler value stored in bits 15:0 of the lockup config register."	RW	0x7FF
15:0	rx_mac_lockup_time	"The time after which the receive MAC lockup detection monitor will trigger. For release 1p10 this was measured in units of 1024 tx_clk periods. From release 1p11 onwards it is used in conjunction with the prescaler stored in bits 15:0 of the lockup config register."	RW	0xFFFF

Table 11-620: emac_lockup_config3

Register Information				
Description	"eMAC DMA TX Lockup Enable Control Register. This register enables the lockup timer for queue 0."			
Offset	0x1070	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:1	reserved_31_1	"Reserved, read as 0, ignored on write."	RO	0x0000 0000
0	dma_tx_lockup_en_q_0	"Enable DMA TX lockup timer for queue 0."	RW	0

Table 11-621: emac_rx_water_mark

Register Information					
Description	"rx_water_mark - Receive water mark register. This register contains the high and low water-marks for automatic transmission of pause frames. The water-marks are compared against the internal signal rx_dpram_fill_lvl which is readable in the dpram_fill_dbg register; rx_dpram_fill_lvl indicates the number of words used in the receive SRAM (word length is configuration dependent and may be 4, 8 and 16 bytes). A value of zero in a field disables the corresponding functionality."				
Offset	0x107C		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:16	rx_low_watermark	"If this field is non-zero and the last pause frame transmitted was non-zero then a zero length pause frame is transmitted when the receive SRAM fill level falls below this value."	RW	0x0000	
15:0	rx_high_watermark	"If this field is non-zero and the receive SRAM fill level exceeds this value then a pause frame is transmitted"	RW	0x0000	

Table 11-622: emac_hash_bottom

Register Information					
Description	"The unicast hash enable and the multicast hash enable bits in the network configuration register enable the reception of hash matched frames. Hash Register Bottom 31:0."				
Offset	0x1080		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:0	address	"The first 32 bits of the hash address register."	RW	0x0000 0000	

Table 11-623: emac_hash_top

Register Information					
Description	"Hash Register Top 63:32"				
Offset	0x1084		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:0	address	"The remaining 32 bits of the hash address register."	RW	0x0000 0000	

Table 11-624: emac_spec_add1_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x1088	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-625: emac_spec_add1_top

Register Information				
Description	"Specific Address Top"			
Offset	0x108C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:17	reserved_31_17	"Reserved, read as 0, ignored on write."	RO	0x0000
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. Octets 5 and 4 of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-626: emac_spec_add2_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x1090	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-627: emac_spec_add2_top

Register Information				
Description	"Specific Address Top"			
Offset	0x1094	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-628: emac_spec_add3_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x1098	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-629: emac_spec_add3_top

Register Information				
Description	"Specific Address Top"			
Offset	0x109C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-630: emac_spec_add4_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x10A0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-631: emac_spec_add4_top

Register Information				
Description	"Specific Address Top"			
Offset	0x10A4	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-632: emac_spec_type1

Register Information				
Description	"Type ID Match 1"			
Offset	0x10A8	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	enable_copy	"Enable copying of type ID match 1 matched frames."	RW	0
30:16	reserved_30_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	match	"Type ID match 1. For use in comparisons with received frames type ID/length field."	RW	0x0000

Table 11-633: emac_spec_type2

Register Information				
Description	"Type ID Match 2"			
Offset	0x10AC	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	enable_copy	"Enable copying of type ID match 2 matched frames."	RW	0
30:16	reserved_30_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	match	"Type ID match 2. For use in comparisons with received frames type ID/length field."	RW	0x0000

Table 11-634: emac_spec_type3

Register Information				
Description	"Type ID Match 3"			
Offset	0x10B0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	enable_copy	"Enable copying of type ID match 3 matched frames."	RW	0
30:16	reserved_30_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	match	"Type ID match 3. For use in comparisons with received frames type ID/length field."	RW	0x0000

Table 11-635: emac_spec_type4

Register Information				
Description	"Type ID Match 4"			
Offset	0x10B4	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	enable_copy	"Enable copying of type ID match 4 matched frames."	RW	0
30:16	reserved_30_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	match	"Type ID match 4. For use in comparisons with received frames type ID/length field."	RW	0x0000

Table 11-636: emac_wol_register

Register Information			
Description	"Wake on LAN Register"		
Offset	0x10B8	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:20	reserved_31_20	"Reserved - read 0, ignored when written."	RO	0x000
19	wol_mask_3	"Wake on LAN multicast hash event enable. When set multicast hash events will cause the wol output to be asserted."	RW	0
18	wol_mask_2	"Wake on LAN specific address register 1 event enable. When set specific address 1 events will cause the wol output to be asserted."	RW	0
17	wol_mask_1	"Wake on LAN ARP request event enable. When set ARP request events will cause the wol output to be asserted."	RW	0
16	wol_mask_0	"Wake on LAN magic packet event enable. When set magic packet events will cause the wol output to be asserted."	RW	0
15:0	addr	"Wake on LAN ARP request IP address. Written to define the least significant 16 bits of the target IP address that is matched to generate a Wake on LAN event. A value of zero will not generate an event, even if this is matched by the received frame."	RW	0x0000

Table 11-637: emac_stretch_ratio

Register Information			
Description	"IPG stretch register"		
Offset	0x10BC	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	ipg_stretch	"IPG Stretch. Bits 7:0 are multiplied with the previously transmitted frame length (including preamble) bits 15:8 +1 divide the frame length. If the resulting number is greater than 96 and bit 28 is set in the network configuration register then the resulting number is used for the transmit inter-packet-gap. 1 is added to bits 15:8 to prevent a divide by zero."	RW	0x0000

Table 11-638: emac_stacked_vlan

Register Information					
Description	"Stacked VLAN Register"				
Offset	0x10C0		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31	enable_processing	"Enable stacked VLAN processing mode"		RW	0
30:16	reserved_30_16	"Reserved, read as 0, ignored on write."		RO	0x0000
15:0	match	"User defined VLAN_TYPE field. When Stacked VLAN is enabled, the first VLAN tag in a received frame will only be accepted if the VLAN type field is equal to this user defined VLAN_TYPE OR equal to the standard VLAN type (0x8100). Note that the second VLAN tag of a Stacked VLAN packet will only be matched correctly if its VLAN_TYPE field equals 0x8100."		RW	0x0000

Table 11-639: emac_tx_pfc_pause

Register Information					
Description	"Transmit PFC Pause Register"				
Offset	0x10C4		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."		RO	0x0000
15:8	vector	"Priority Vector Pause Size. If bit 17 of the network control register is written with a one then for each entry equal to zero in the Transmit PFC Pause Register[15:8], the PFC pause frame's pause quantum field associated with that entry will be taken from the transmit pause quantum register. For each entry equal to one in the Transmit PFC Pause Register [15:8], the pause quantum associated with that entry will be zero."		RW	0x00
7:0	vector_enable	"Priority Vector Enable. If bit 17 of the network control register is written with a one then the priority enable vector of the PFC priority based pause frame will be set equal to the value stored in this register [7:0]."		RW	0x00

Table 11-640: emac_mask_add1_bottom

Register Information					
Description	"Specific Address Mask 1 Bottom 31:0"				
Offset	0x10C8		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	address_mask	"Specific Address Mask. Setting a bit to one masks the corresponding bit in the specific address 1 register"		RW	0x0000 0000

Table 11-641: emac_mask_add1_top

Register Information				
Description	"Specific Address Mask 1 Top 47:32"			
Offset	0x10CC	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	address_mask	"Specific Address Mask. Setting a bit to one masks the corresponding bit in the specific address 1 register"	RW	0x0000

Table 11-642: emac_dma_addr_or_mask

Register Information				
Description	"Receive DMA Data Buffer Address Mask - only applies to AHB operation"			
Offset	0x10D0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:28	mask_value	"Data Buffer Address Mask Value. Values used to force bits 31:28 of the receive data buffer AHB address to a particular value when the associated enable bits stored in this register [3:0] are set. Any changes to this register will be ignored while the DMA is currently processing a receive packet. It will only affect the next full packet to be written to external system memory."	RW	0x0
27:4	reserved_27_4	"Reserved, read as 0, ignored on write."	RO	0x00 0000
3:0	mask_enable	"Data Buffer Address Mask Enable. These bits are associated directly with bits[31:28].When bit 0 is set, the AHB address bit 28 used for accessing the receive data buffers will be forced to the value stored in bit 28 of this register. When bit 1 is set, the AHB address bit 29 used for accessing the receive data buffers will be forced to the value stored in bit 29 of this register. When bit 2 is set, the AHB address bit 30 used for accessing the receive data buffers will be forced to the value stored in bit 30 of this register. When bit 3 is set, the AHB address bit 31 used for accessing the receive data buffers will be forced to the value stored in bit 31 of this register. When these bits are clear, the associated value stored in bits 31:28 have no effect on the AHB address used for receive data buffer accesses. Any changes to this register will be ignored while the DMA is currently processing a receive packet. It will only affect the next full packet to be written to external memory."	RW	0x0

Table 11-643: emac_rx_ptp_unicast

Register Information					
Description	"PTP RX unicast IP destination address"				
Offset	0x10D4		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:0	address	"Unicast IP destination address. Used for detection of PTP frames on receive path."	RW	0x0000 0000	

Table 11-644: emac_tx_ptp_unicast

Register Information					
Description	"PTP TX unicast IP destination address"				
Offset	0x10D8		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:0	address	"Unicast IP destination address. Used for detection of PTP frames on transmit path."	RW	0x0000 0000	

Table 11-645: emac_tsu_nsec_cmp

Register Information					
Description	"TSU timer comparison value nanoseconds"				
Offset	0x10DC		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:22	reserved_31_22	"Reserved, read as 0, ignored on write."	RO	0x000	
21:0	comparison_value	"TSU timer comparison value (ns). Value is compared to the bits[45:24] of the TSU timer count value (upper 22 bits of nanosecond value). The output tsu_timer_cmp_val is driven high when the timer comparison values match the TSU count value. From release 1p11 onwards this comparison only occurs once the nanosecond compare register has been written and when the match occurs tsu_timer_cmp_val will only be driven high for a single tsu_clk period."	RW	0x00 0000	

Table 11-646: emac_tsu_sec_cmp

Register Information				
Description	"TSU timer comparison value seconds 31:0"			
Offset	0x10E0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	comparison_value	"TSU timer comparison value (s). Value is compared to seconds value bits [31:0] of the TSU timer count value."	RW	0x0000 0000

Table 11-647: emac_tsu_msb_sec_cmp

Register Information				
Description	"TSU timer comparison value seconds 47:32"			
Offset	0x10E4	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	comparison_value	"TSU timer comparison value (s). Value is compared to the top 16 bits (most significant 16-bits [47:32] of seconds value) of the TSU timer count value."	RW	0x0000

Table 11-648: emac_tsu_ptp_tx_msb_sec

Register Information				
Description	"PTP Event Frame Transmitted Seconds Register 47:32"			
Offset	0x10E8	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	timer_seconds	"PTP Event Frame TX Seconds. The register is updated with the value that the 1588 timer seconds register held when the SFD of a PTP transmit primary event crosses the MII interface. The actual update occurs when the GEM recognizes the frame as a PTP sync or delay_req frame. An interrupt is issued when the register is updated."	RO	0x0000

Table 11-649: emac_tsu_ptp_rx_msb_sec

Register Information				
Description	"PTP Event Frame Received Seconds Register 47:32"			
Offset	0x10EC	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	timer_seconds	"PTP Event Frame TX Seconds. The register is updated with the value that the 1588 timer seconds register held when the SFD of a PTP receive primary event crosses the MII interface. The actual update occurs when the GEM recognizes the frame as a PTP sync or delay_req frame. An interrupt is issued when the register is updated."	RO	0x0000

Table 11-650: emac_tsu_peer_tx_msb_sec

Register Information				
Description	"PTP Peer Event Frame Transmitted Seconds Register 47:32"			
Offset	0x10F0	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	timer_seconds	"PTP Peer Event Frame TX Seconds. The register is updated with the value that the 1588 timer seconds register held when the SFD of a PTP transmit peer event crosses the MII interface. The actual update occurs when the GEM recognizes the frame as a PTP pdelay_req or pdelay_resp frame. An interrupt is issued when the register is updated."	RO	0x0000

Table 11-651: emac_tsu_peer_rx_msb_sec

Register Information				
Description	"PTP Peer Event Frame Received Seconds Register 47:32"			
Offset	0x10F4	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	timer_seconds	"PTP Peer Event Frame RX Seconds. The register is updated with the value that the 1588 timer seconds register held when the SFD of a PTP receive peer event crosses the MII interface. The actual update occurs when the GEM recognizes the frame as a PTP pdelay_req or pdelay_resp frame. An interrupt is issued when the register is updated."	RO	0x0000

Table 11-652: emac_dpram_fill_dbg

Register Information				
Description	"The fill levels for the TX and RX packet buffer SRAMs can be read using this register, including the fill level for each queue in the TX direction. The fill level is reported as the number of word locations used. The number of bytes will depend on the SRAM data width."			
Offset	0x10F8	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	dma_tx_rx_fill_level	"Fill Level - TX or RX packet buffer fill level, selected by the tx_q_fill_level_select and tx_rx_fill_level_select registers. Read this register to determine the fill level."	RO	0x0000
15:8	reserved_15_8	"Reserved, read as 0, ignored on write."	RO	0x00
7:4	dma_tx_q_fill_level_select	"TX queue fill level select - select what TX queue to report fill levels for."	RW	0x0
3:1	reserved_3_1	"Reserved, read as 0, ignored on write."	RO	0x0
0	dma_tx_rx_fill_level_select	"TX/RX Fill Level select - report the fill level for the TX or RX packet buffer. Set 1 for transmit and 0 for receive."	RW	0

Table 11-653: emac_revision_reg

Register Information				
Description	"This register indicates a Cadence module identification number and module revision. The value of this register is read only as defined by `gem_revision_reg_value"			
Offset	0x10FC	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:28	fix_number	"Fix number - incremented for fix releases."	RO	0x0
27:16	module_identification_number	"Module identification number - for the GEM, this value is fixed."	RO	0x007
15:0	module_revision	"Module revision - fixed value specific to the revision of the design which is incremented for each non-fix release of the IP."	RO	0x0200

Table 11-654: emac_octets_txed_bottom

Register Information			
Description	"These registers reset to zero on a read and stick at all ones when they count to their maximum value. They should be read frequently enough to prevent loss of data. In order to reduce overall design area, the statistics registers may be optionally removed in the configuration file if they are deemed unnecessary for a particular design. The receive statistics registers are only incremented when the receive enable bit is set in the network control register. The statistics registers optionally have a snapshot capability which, when exercised, will simultaneously store and clear the current values of all the statistics registers into a snapshot register set in order to allow a consistent set of statistics to be read by the processor. The snapshot is controlled using bit 13 of the network control register. The read snapshot control indicated by bit 14 of the network control register determines whether the processor reads the snapshot registers (logic 1) or the incrementing registers (logic 0). The default GEM configuration does not support the snapshot capability. See Parameterization section under Implementation Application Notes for an explanation of how to enable this function. All the statistics registers are read only. For test purposes they may be written by setting bit 7 (Write Enable) in the network control register. Setting bit 6 (increment statistics) in the network control register causes all the statistics registers to increment by one, again for test purposes. Once a statistics register has been read, it is automatically cleared. When reading the octets transmitted and octets received registers, bits 31:0 should be read prior to bits 47:32 to ensure reliable operation. The statistics register block contains the following registers. Octets Transmitted [31:0]"		
Offset	0x1100	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	count	"Transmitted octets in frame without errors [31:0]. The number of octets transmitted in valid frames of any type. This counter is 48-bits, and is read through two registers. This count does not include octets from automatically generated pause frames."	RO RtoClr	0x0000 0000

Table 11-655: emac_octets_txed_top

Register Information			
Description	"Octets Transmitted 47:32"		
Offset	0x1104	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	count	"Transmitted octets in frame without errors [47:32]. The number of octets transmitted in valid frames of any type. This counter is 48-bits, and is read through two registers. This count does not include octets from automatically generated pause frames."	RO RtoClr	0x0000

Table 11-656: emac_frames_txed_ok

Register Information					
Description	"Frames Transmitted"				
Offset	0x1108		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:0	count	"Frames transmitted without error. A 32 bit register counting the number of frames successfully transmitted, i.e. no under run and not too many retries. Excludes pause frames."	RO RtoClr	0x0000 0000	

Table 11-657: emac_broadcast_txed

Register Information					
Description	"Broadcast Frames Transmitted"				
Offset	0x110C		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:0	count	"Broadcast frames transmitted without error. A 32 bit register counting the number of broadcast frames successfully transmitted without error, i.e. no under run and not too many retries. Excludes pause frames."	RO RtoClr	0x0000 0000	

Table 11-658: emac_multicast_txed

Register Information					
Description	"Multicast Frames Transmitted"				
Offset	0x1110		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:0	count	"Multicast frames transmitted without error. A 32 bit register counting the number of multicast frames successfully transmitted without error, i.e. no under run and not too many retries. Excludes pause frames."	RO RtoClr	0x0000 0000	

Table 11-659: emac_pause_frames_txed

Register Information					
Description	"Pause Frames Transmitted"				
Offset	0x1114		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."		RO	0x0000
15:0	count	"Transmitted pause frames - a 16 bit register counting the number of pause frames transmitted. Only pause frames triggered by the register interface or through the external pause pins or automatically when rx_water_mark is reached are counted as pause frames. Pause frames received through the external FIFO interface are counted in the frames transmitted counter."		RO RtoClr	0x0000

Table 11-660: emac_frames_txed_64

Register Information					
Description	"64 Byte Frames Transmitted"				
Offset	0x1118		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	count	"64 byte frames transmitted without error. A 32 bit register counting the number of 64 byte frames successfully transmitted without error, i.e. no under run and not too many retries. Excludes pause frames."		RO RtoClr	0x0000 0000

Table 11-661: emac_frames_txed_65

Register Information					
Description	"65 to 127 Byte Frames Transmitted"				
Offset	0x111C		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	count	"65 to127 byte frames transmitted without error. A 32 bit register counting the number of 65 to127 byte frames successfully transmitted without error, i.e. no under run and not too many retries."		RO RtoClr	0x0000 0000

Table 11-662: emac_frames_txed_128

Register Information					
Description	"128 to 255 Byte Frames Transmitted"				
Offset	0x1120		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	count	"128 to 255 byte frames transmitted without error. A 32 bit register counting the number of 128 to 255 byte frames successfully transmitted without error, i.e. no under run and not too many retries."		RO RtoClr	0x0000 0000

Table 11-663: emac_frames_txed_256

Register Information					
Description	"256 to 511 Byte Frames Transmitted"				
Offset	0x1124		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	count	"256 to 511 byte frames transmitted without error. A 32 bit register counting the number of 256 to 511 byte frames successfully transmitted without error, i.e. no under run and not too many retries."		RO RtoClr	0x0000 0000

Table 11-664: emac_frames_txed_512

Register Information					
Description	"512 to 1023 Byte Frames Transmitted"				
Offset	0x1128		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	count	"512 to 1023 byte frames transmitted without error. A 32 bit register counting the number of 512 to 1023 byte frames successfully transmitted without error, i.e. no under run and not too many retries."		RO RtoClr	0x0000 0000

Table 11-665: emac_frames_txed_1024

Register Information					
Description	"1024 to 1518 Byte Frames Transmitted"				
Offset	0x112C		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:0	count	"1024 to 1518 byte frames transmitted without error. A 32 bit register counting the number of 1024 to 1518 byte frames successfully transmitted without error, i.e. no under run and not too many retries."	RO RtoClr	0x0000 0000	

Table 11-666: emac_frames_txed_1519

Register Information					
Description	"Greater Than 1518 Byte Frames Transmitted"				
Offset	0x1130		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:0	count	"Greater than 1518 byte frames transmitted without error. A 32 bit register counting the number of 1518 or above byte frames successfully transmitted without error, i.e. no under run and not too many retries."	RO RtoClr	0x0000 0000	

Table 11-667: emac_tx_underruns

Register Information					
Description	"Transmit Under Runs"				
Offset	0x1134		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:10	reserved_31_10	"Reserved, read as 0, ignored on write."	RO	0x00 0000	
9:0	count	"Transmit under runs - a 10 bit register counting the number of frames not transmitted due to a transmit under run. If this register is incremented then no other statistics register is incremented."	RO RtoClr	0x000	

Table 11-668: emac_single_collisions

Register Information				
Description	"Single Collision Frames"			
Offset	0x1138	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:18	reserved_31_18	"Reserved, read as 0, ignored on write."	RO	0x0000
17:0	count	"Single collision frames - an 18 bit register counting the number of frames experiencing a single collision before being successfully transmitted, i.e. no under run."	RO RtoClr	0x0 0000

Table 11-669: emac_multiple_collisions

Register Information				
Description	"Multiple Collision Frames"			
Offset	0x113C	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:18	reserved_31_18	"Reserved, read as 0, ignored on write."	RO	0x0000
17:0	count	"Multiple collision frames - an 18 bit register counting the number of frames experiencing between two and fifteen collisions prior to being successfully transmitted, i.e. no under run and not too many retries."	RO RtoClr	0x0 0000

Table 11-670: emac_excessive_collisions

Register Information				
Description	"Excessive Collisions"			
Offset	0x1140	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:10	reserved_31_10	"Reserved, read as 0, ignored on write."	RO	0x00 0000
9:0	count	"Excessive collisions - a 10 bit register counting the number of frames that failed to be transmitted because they experienced 16 collisions."	RO RtoClr	0x000

Table 11-671: emac_late_collisions

Register Information					
Description	"Late Collisions"				
Offset	0x1144		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:10	reserved_31_10	"Reserved, read as 0, ignored on write."		RO	0x00 0000
9:0	count	"Late collisions - a 10 bit register counting the number of late collision occurring after the slot time (512 bits) has expired. In 10/100 mode, late collisions are counted twice i.e. both as a collision and a late collision. In gigabit mode, a late collision causes the transmission to be aborted, thus the single and multi collision registers are not updated."		RO RtoClr	0x000

Table 11-672: emac_deferred_frames

Register Information					
Description	"Deferred Transmission Frames"				
Offset	0x1148		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:18	reserved_31_18	"Reserved, read as 0, ignored on write."		RO	0x0000
17:0	count	"Deferred transmission frames - an 18 bit register counting the number of frames experiencing deferral due to carrier sense being active on their first attempt at transmission. Frames involved in any collision are not counted nor are frames that experienced a transmit under run."		RO RtoClr	0x0 0000

Table 11-673: emac_crs_errors

Register Information					
Description	"Carrier Sense Errors"				
Offset	0x114C		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:10	reserved_31_10	"Reserved, read as 0, ignored on write."	RO	0x00 0000	
9:0	count	"Carrier sense errors - a 10 bit register counting the number of frames transmitted where carrier sense was not seen during transmission or where carrier sense was deasserted after being asserted in a transmit frame without collision (no under run). Only incremented in half-duplex mode. The only effect of a carrier sense error is to increment this register. The behaviour of the other statistics registers is unaffected by the detection of a carrier sense error."	RO RtoClr	0x000	

Table 11-674: emac_octets_rxed_bottom

Register Information					
Description	"Octets Received 31:0"				
Offset	0x1150		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:0	count	"Received octets in frame without errors [31:0]. The number of octets received in valid frames of any type. This counter is 48-bits and is read through two registers. This count does not include octets from pause frames, and is only incremented if the frame is successfully filtered."	RO RtoClr	0x0000 0000	

Table 11-675: emac_octets_rxed_top

Register Information					
Description	"Octets Received 47:32"				
Offset	0x1154		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000	
15:0	count	"Received octets in frame without errors [47:32]. The number of octets received in valid frames of any type. This counter is 48-bits and is read through two registers. This count does not include octets from pause frames, and is only incremented if the frame is successfully filtered."	RO RtoClr	0x0000	

Table 11-676: emac_frames_rxed_ok

Register Information					
Description	"Frames Received"				
Offset	0x1158		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:0	count	"Frames received without error. A 32 bit register counting the number of frames successfully received. Excludes pause frames, and is only incremented if the frame is successfully filtered."	RO RtoClr	0x0000 0000	

Table 11-677: emac_broadcast_rxed

Register Information					
Description	"Broadcast Frames Received"				
Offset	0x115C		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:0	count	"Broadcast frames received without error. A 32 bit register counting the number of broadcast frames successfully received without error. Excludes pause frames, and is only incremented if the frame is successfully filtered."	RO RtoClr	0x0000 0000	

Table 11-678: emac_multicast_rxed

Register Information					
Description	"Multicast Frames Received"				
Offset	0x1160		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:0	count	"Multicast frames received without error. A 32 bit register counting the number of multicast frames successfully received without error. Excludes pause frames, and is only incremented if the frame is successfully filtered."	RO RtoClr	0x0000 0000	

Table 11-679: emac_pause_frames_rxed

Register Information					
Description	"Pause Frames Received"				
Offset	0x1164		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."		RO	0x0000
15:0	count	"Received pause frames - a 16 bit register counting the number of pause frames received without error."		RO RtoClr	0x0000

Table 11-680: emac_frames_rxed_64

Register Information					
Description	"64 Byte Frames Received"				
Offset	0x1168		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	count	"64 byte frames received without error. A 32 bit register counting the number of 64 byte frames successfully received without error. Excludes pause frames, and is only incremented if the frame is successfully filtered."		RO RtoClr	0x0000 0000

Table 11-681: emac_frames_rxed_65

Register Information					
Description	"65 to 127 Byte Frames Received"				
Offset	0x116C		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	count	"65 to 127 byte frames received without error. A 32 bit register counting the number of 65 to 127 byte frames successfully received without error. Excludes pause frames, and is only incremented if the frame is successfully filtered."		RO RtoClr	0x0000 0000

Table 11-682: emac_frames_rxed_128

Register Information				
Description	"128 to 255 Byte Frames Received"			
Offset	0x1170	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	count	"128 to 255 byte frames received without error. A 32 bit register counting the number of 128 to 255 byte frames successfully received without error. Excludes pause frames, and is only incremented if the frame is successfully filtered."	RO RtoClr	0x0000 0000

Table 11-683: emac_frames_rxed_256

Register Information				
Description	"256 to 511 Byte Frames Received"			
Offset	0x1174	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	count	"256 to 511 byte frames received without error. A 32 bit register counting the number of 256 to 511 byte frames successfully transmitted without error. Excludes pause frames, and is only incremented if the frame is successfully filtered."	RO RtoClr	0x0000 0000

Table 11-684: emac_frames_rxed_512

Register Information				
Description	"512 to 1023 Byte Frames Received"			
Offset	0x1178	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	count	"512 to 1023 byte frames received without error. A 32 bit register counting the number of 512 to 1023 byte frames successfully received without error. Excludes pause frames, and is only incremented if the frame is successfully filtered."	RO RtoClr	0x0000 0000

Table 11-685: emac_frames_rxed_1024

Register Information					
Description	"1024 to 1518 Byte Frames Received"				
Offset	0x117C		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	count	"1024 to 1518 byte frames received without error. A 32 bit register counting the number of 1024 to 1518 byte frames successfully received without error. Excludes pause frames, and is only incremented if the frame is successfully filtered."		RO RtoClr	0x0000 0000

Table 11-686: emac_frames_rxed_1519

Register Information					
Description	"1519 to maximum Byte Frames Received"				
Offset	0x1180		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	count	"1519 to maximum byte frames received without error. A 32 bit register counting the number of 1519 byte or above frames successfully received without error. Maximum frame size is determined by the network configuration register bit 8 (1536 maximum frame size) or bit 3 (jumbo frame size). Excludes pause frames, and is only incremented if the frame is successfully filtered."		RO RtoClr	0x0000 0000

Table 11-687: emac_undersize_frames

Register Information					
Description	"Undersized Frames Received"				
Offset	0x1184		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:10	reserved_31_10	"Reserved, read as 0, ignored on write."		RO	0x00 0000
9:0	count	"Undersize frames received - a 10 bit register counting the number of frames received less than 64 bytes in length (10/100 mode or gigabit mode, full duplex) that do not have either a CRC error or an alignment error. In gigabit mode, half-duplex, this register counts either frames not conforming to the minimum slot time of 512 bytes or frames not conforming to the minimum frame size once bursting is active."		RO RtoClr	0x000

Table 11-688: emac_excessive_rx_length

Register Information					
Description	"Oversize Frames Received"				
Offset	0x1188		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:10	reserved_31_10	"Reserved read 0, ignored on write."		RO	0x00 0000
9:0	count	"Oversize frames received - a 10 bit register counting the number of frames received exceeding 1518 bytes (1536 bytes if bit 8 is set in network configuration register, 10,240 bytes if bit 3 is set in the network configuration register) in length but do not have either a CRC error, an alignment error nor a receive symbol error."		RO RtoClr	0x000

Table 11-689: emac_rx_jabbers

Register Information					
Description	"Jabbers Received"				
Offset	0x118C		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:10	reserved_31_10	"Reserved, read as 0, ignored on write."		RO	0x00 0000
9:0	count	"Jabbers received - a 10 bit register counting the number of frames received exceeding 1518 bytes (1536 if bit 8 set in network configuration register, 10,240 bytes if bit 3 is set in the network configuration register) in length and have either a CRC error, an alignment error or a receive symbol error."		RO RtoClr	0x000

Table 11-690: emac_fcs_errors

Register Information					
Description	"Frame Check Sequence Errors"				
Offset	0x1190		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:10	reserved_31_10	"Reserved, read as 0, ignored on write."		RO	0x00 0000
9:0	count	"Frame check sequence errors - a 10 bit register counting frames that are an integral number of bytes, have bad CRC and are between 64 and 1518 bytes in length (1536 if bit 8 set in network configuration register, 10,240 bytes if bit 3 is set in the network configuration register). This register is also incremented if a symbol error is detected and the frame is of valid length and has an integral number of bytes. This register is incremented for a frame with bad FCS, regardless of whether it is copied to memory due to ignore FCS mode being enabled in bit 26 of the network configuration register."		RO RtoClr	0x000

Table 11-691: emac_rx_length_errors

Register Information					
Description	"Length Field Frame Errors"				
Offset	0x1194		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:10	reserved_31_10	"Reserved, read as 0, ignored on write."		RO	0x00 0000
9:0	count	"Length field frame errors - this 10-bit register counts the number of frames received that have a measured length shorter than that extracted from the length field (bytes 13 and 14). This condition is only counted if the value of the length field is less than 0x0600, the frame is not of excessive length and checking is enabled through bit 16 of the network configuration register."		RO RtoClr	0x000

Table 11-692: emac_rx_symbol_errors

Register Information					
Description	"Receive Symbol Errors"				
Offset	0x1198		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:10	reserved_31_10	"Reserved, read as 0, ignored on write."		RO	0x00 0000
9:0	count	"Receive symbol errors - a 10-bit register counting the number of frames that had rx_er asserted during reception. For 10/100 mode symbol errors are counted regardless of frame length checks. For gigabit mode the frame must satisfy slot time requirements in order to count a symbol error. Additionally, in gigabit half-duplex mode, carrier extension errors are also recorded. Receive symbol errors will also be counted as an FCS or alignment error if the frame is between 64 and 1518 bytes (1536 bytes if bit 8 is set in the network configuration register, 10240 bytes if bit 3 is set in the network configuration register). If the frame is larger it will be recorded as a jabber error."		RO RtoClr	0x000

Table 11-693: emac_alignment_errors

Register Information					
Description	"Alignment Errors"				
Offset	0x119C		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:10	reserved_31_10	"Reserved, read as 0, ignored on write."		RO	0x00 0000
9:0	count	"Alignment errors - a 10 bit register counting frames that are not an integral number of bytes long and have bad CRC when their length is truncated to an integral number of bytes and are between 64 and 1518 bytes in length (1536 if bit 8 set in network configuration register, 10,240 bytes if bit 3 is set in the network configuration register). This register is also incremented if a symbol error is detected and the frame is of valid length and does not have an integral number of bytes."		RO RtoClr	0x000

Table 11-694: emac_rx_resource_errors

Register Information					
Description	"Receive Resource Errors"				
Offset	0x11A0		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:18	reserved_31_18	"Reserved, read as 0, ignored on write."		RO	0x0000
17:0	count	"Receive resource errors - an 18 bit register counting the number of times a receive buffer descriptor was read with its used bit set. This can be used as a means of checking the efficiency of the software in processing and releasing receive buffers. In an ideally configured system, this counter would not increase."		RO RtoClr	0x0 0000

Table 11-695: emac_rx_overruns

Register Information					
Description	"Receive Overruns"				
Offset	0x11A4		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:10	reserved_31_10	"Reserved, read as 0, ignored on write."		RO	0x00 0000
9:0	count	"Receive overruns - a 10 bit register counting the number of frames that are address recognized but were not copied to memory due to a receive overrun."		RO RtoClr	0x000

Table 11-696: emac_rx_ip_ck_errors

Register Information					
Description	"IP Header Checksum Errors"				
Offset	0x11A8		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:8	reserved_31_8	"Reserved, read as 0, ignored on write."		RO	0x00 0000
7:0	count	"IP header checksum errors - an 8-bit register counting the number of frames discarded due to an incorrect IP header checksum, but are between 64 and 1518 bytes (1536 bytes if bit 8 is set in the network configuration register or 10240 bytes if bit 3 is in the network configuration register) and do not have a CRC error, an alignment error, nor a symbol error."		RO RtoClr	0x00

Table 11-697: emac_rx_tcp_ck_errors

Register Information					
Description	"TCP Checksum Errors"				
Offset	0x11AC		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:8	reserved_31_8	"Reserved, read as 0, ignored on write."		RO	0x00 0000
7:0	count	"TCP checksum errors - an 8-bit register counting the number of frames discarded due to an incorrect TCP checksum, but are between 64 and 1518 bytes (1536 bytes if bit 8 is set in the network configuration register or 10240 bytes if bit 3 is in the network configuration register) and do not have a CRC error, an alignment error, nor a symbol error."		RO RtoClr	0x00

Table 11-698: emac_rx_udp_ck_errors

Register Information					
Description	"UDP Checksum Errors"				
Offset	0x11B0		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:8	reserved_31_8	"Reserved, read as 0, ignored on write."		RO	0x00 0000
7:0	count	"UDP checksum errors - an 8-bit register counting the number of frames discarded due to an incorrect UDP checksum, but are between 64 and 1518 bytes (1536 bytes if bit 8 is set in the network configuration register or 10240 bytes if bit 3 is in the network configuration register) and do not have a CRC error, an alignment error, nor a symbol error."		RO RtoClr	0x00

Table 11-699: emac_auto_flushed_pkts

Register Information					
Description	"Receive DMA Flushed Packets"				
Offset	0x11B4		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."		RO	0x0000
15:0	count	"Flushed RX packets counter. A 16 bit register counting the number of frames that have been flushed from the receive SRAM based packet buffer due to one of the following reasons: 1. When either partial store and forward mode, bit 24 of the DMA configuration register, or the drop_on_resource_err mode of the traffic policing feature is active, and a packet is received while there is no AMBA (AHB/AXI) resource (no free descriptors for the DUT to use). 2. When partial store and forward mode is enabled and an AMBA (AHB/AXI) error is encountered while writing the packet data to external memory. 3. When bit 18 of the network control register (software action to flush a packet from the head of the PBUF queue) is pulsed and the GEM DMA is not currently busy. 4. When a frame is dropped due to the policing actions defined in the rx_qX_flush registers located at 0x0b00-0x0b3c."		RO RtoClr	0x0000

Table 11-700: emac_tsu_timer_incr_sub_nsec

Register Information					
Description	"1588 Timer Increment Register sub nsec. From release 1p08f1 onwards this register must be written before the tsu_timer_incr register and the value written will not take effect until the tsu_timer_incr register is written to."				
Offset	0x11BC		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:24	sub_ns_incr_lsb	"These are the least significant bits [7:0] of the sub-ns value by which the 1588 timer will be incremented each clock cycle."		RW	0x00
23:16	reserved_23_16	"Reserved, read as 0, ignored on write."		RO	0x00
15:0	sub_ns_incr	"These are the most significant bits [23:8] of the sub-ns value by which the 1588 timer will be incremented each clock cycle. 24 bits of sub nanosecond precision gives resolution of approximately 5.86E-17 seconds (16 bits gives 15.2 femtoseconds)."		RW	0x0000

Table 11-701: emac_tsu_timer_msb_sec

Register Information				
Description	"1588 Timer Seconds Register 47:32"			
Offset	0x11C0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	timer	"TSU timer value (s). Most significant 16 bits of seconds timer count. The register is writeable. The 48-bit counter increments by one when the 1588 nanoseconds counter counts to one second. It may also be incremented or decremented when the timer adjust register is written (if decremented from zero the 48-bit combined count would roll back to 0xFFFFFFFFFFFF). Note: The value of this register is used only when the lower 32 bit register is written to. This is to ensure a single update of the 48 bit seconds value"	RW	0x0000

Table 11-702: emac_tsu_strobe_msb_sec

Register Information				
Description	"1588 Timer Sync Strobe Seconds Register 47:32"			
Offset	0x11C4	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	strobe	"1588 Timer Sync Strobe Seconds. The most significant 16-bit value of the Timer Seconds register captured when gem_tsu_ms and gem_tsu_inc_ctrl are zero."	RO	0x0000

Table 11-703: emac_tsu_strobe_sec

Register Information				
Description	"1588 Timer Sync Strobe Seconds Register 31:0"			
Offset	0x11C8	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	strobe	"1588 Timer Sync Strobe Seconds. The lowest significant 32-bit value of the Timer Seconds register captured when gem_tsu_ms and gem_tsu_inc_ctrl are zero."	RO	0x0000 0000

Table 11-704: emac_tsu_strobe_nsec

Register Information				
Description	"1588 Timer Sync Strobe Nanoseconds Register"			
Offset	0x11CC	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:0	strobe	"1588 Timer Sync Strobe Nanoseconds. The value of the Timer Nanoseconds register captured when gem_tsu_ms and gem_tsu_inc_ctrl are zero."	RO	0x0000 0000

Table 11-705: emac_tsu_timer_sec

Register Information				
Description	"1588 Timer Seconds Register 31:0"			
Offset	0x11D0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	timer	"1588 Timer Seconds Register. TSU timer value (s). Least significant 32 bits of seconds timer count. This register is writeable. The 48-bit counter increments by one when the 1588 nanoseconds counter counts to one second. It may also be incremented or decremented when the timer adjust register is written (if decremented from zero the 48-bit combined count would roll back to 0xFFFFFFFFFFFF)."	RW	0x0000 0000

Table 11-706: emac_tsu_timer_nsec

Register Information				
Description	"1588 Timer Nanoseconds Register"			
Offset	0x11D4	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:0	timer	"Timer count in nanoseconds. This register is writeable. It can also be adjusted by writes to the 1588 timer adjust register. It increments by the value of the 1588 timer increment register each clock cycle (if this register is close to zero and a write to the timer adjust register causes a decrement the seconds register will be decremented if necessary and the nanoseconds register will roll back to 9999999xx (decimal))."	RW	0x0000 0000

Table 11-707: emac_tsu_timer_adjust

Register Information					
Description	"This register is used to adjust the value of the timer in the TSU. It allows an integral number of nanoseconds to be added or subtracted from the timer in a one-off operation. This register returns all zeroes when read."				
Offset	0x11D8		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31	add_subtract	"Write as one to subtract from the 1588 timer. Write as zero to add to it."		WO	0
30	reserved_30	"Reserved, read as 0, ignored on write."		RO	0
29:0	increment_value	"Timer increment value. The number of nanoseconds to increment or decrement the 1588 timer nanoseconds register. If necessary the 1588 seconds register will be incremented or decremented."		WO	0x0000 0000

Table 11-708: emac_tsu_timer_incr

Register Information					
Description	"1588 Timer Increment Register. From release 1p08f1 onwards this register must be written after the tsu_timer_incr_sub_ns register and the write operation will cause the value written to the tsu_timer_incr_sub_ns register to take effect."				
Offset	0x11DC		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:24	reserved_31_24	"Reserved, read as 0, ignored on write."		RO	0x00
23:16	num_incs	"Number of incs before alt inc. The number of increments after which the alternative increment is used."		RW	0x00
15:8	alt_ns_incr	"Alternative nanoseconds count. Alternative count of nanoseconds by which the 1588 timer nanoseconds register will be incremented each clock cycle."		RW	0x00
7:0	ns_increment	"A count of nanoseconds by which the 1588 timer nanoseconds register will be incremented each clock cycle. These are the most significant 8 bits of the 32 bit timer_increment counter. The tsu_timer_incr_sub_nsec register holds the least significant 24 bits of the increment."		RW	0x00

Table 11-709: emac_tsu_ptp_tx_sec

Register Information				
Description	"PTP Event Frame Transmitted Seconds Register 31:0"			
Offset	0x11E0	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	timer	"PTP Event Frame Transmitted Seconds. The register is updated with the value that the 1588 timer seconds register held when the SFD of a PTP transmit primary event crosses the MII interface. The actual update occurs when the GEM recognizes the frame as a PTP sync or delay_req frame. An interrupt is issued when the register is updated."	RO	0x0000 0000

Table 11-710: emac_tsu_ptp_tx_nsec

Register Information				
Description	"PTP Event Frame Transmitted Nanoseconds Register"			
Offset	0x11E4	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:0	timer	"PTP Event Frame Transmitted Nanoseconds. The register is updated with the value that the 1588 timer nanoseconds register held when the SFD of a PTP transmit primary event crosses the MII interface. The actual update occurs when the GEM recognizes the frame as a PTP sync or delay_req frame. An interrupt is issued when the register is updated."	RO	0x0000 0000

Table 11-711: emac_tsu_ptp_rx_sec

Register Information				
Description	"PTP Event Frame Received Seconds Register 31:0"			
Offset	0x11E8	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	timer	"PTP Event Frame Received Seconds. The register is updated with the value that the 1588 timer seconds register held when the SFD of a PTP receive primary event crosses the MII interface. The actual update occurs when the GEM recognizes the frame as a PTP sync or delay_req frame. An interrupt is issued when the register is updated."	RO	0x0000 0000

Table 11-712: emac_tsu_ptp_rx_nsec

Register Information				
Description	"PTP Event Frame Received Nanoseconds Register"			
Offset	0x11EC	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:0	timer	"PTP Event Frame Received Nanoseconds. The register is updated with the value that the 1588 timer nanoseconds register held when the SFD of a PTP receive primary event crosses the MII interface. The actual update occurs when the GEM recognizes the frame as a PTP sync or delay_req frame. An interrupt is issued when the register is updated."	RO	0x0000 0000

Table 11-713: emac_tsu_peer_tx_sec

Register Information				
Description	"PTP Peer Event Frame Transmitted Seconds Register 31:0"			
Offset	0x11F0	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	timer	"PTP Peer Event Frame Received Seconds. The register is updated with the value that the 1588 timer seconds register held when the SFD of a PTP transmit peer event crosses the MII interface. The actual update occurs when the GEM recognizes the frame as a PTP pdelay_req or pdelay_resp frame. An interrupt is issued when the register is updated."	RO	0x0000 0000

Table 11-714: emac_tsu_peer_tx_nsec

Register Information				
Description	"PTP Peer Event Frame Transmitted Nanoseconds Register"			
Offset	0x11F4	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:0	timer	"PTP Peer Event Frame Transmitted Nanoseconds. The register is updated with the value that the 1588 timer nanoseconds register held when the SFD of a PTP transmit peer event crosses the MII interface. The actual update occurs when the GEM recognizes the frame as a PTP pdelay_req or pdelay_resp frame. An interrupt is issued when the register is updated."	RO	0x0000 0000

Table 11-715: emac_tsu_peer_rx_sec

Register Information				
Description	"PTP Peer Event Frame Received Seconds Register 31:0"			
Offset	0x11F8	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	timer	"PTP Peer Event Frame Received Seconds. The register is updated with the value that the 1588 timer seconds register held when the SFD of a PTP receive peer event crosses the MII interface. The actual update occurs when the GEM recognizes the frame as a PTP pdelay_req or pdelay_resp frame. An interrupt is issued when the register is updated."	RO	0x0000 0000

Table 11-716: emac_tsu_peer_rx_nsec

Register Information				
Description	"PTP Peer Event Frame Received Nanoseconds Register"			
Offset	0x11FC	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:0	timer	"PTP Peer Event Frame Received Nanoseconds. The register is updated with the value that the 1588 timer nanoseconds register held when the SFD of a PTP receive peer event crosses the MII interface. The actual update occurs when the GEM recognizes the frame as a PTP pdelay_req or pdelay_resp frame. An interrupt is issued when the register is updated."	RO	0x0000 0000

Table 11-717: emac_tx_pause_quantum1

Register Information				
Description	"Transmit Pause Quantum Register 1"			
Offset	0x1260	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	quantum_p3	"Transmit pause quantum - written with the pause quantum value for pause frame transmission of priority 3."	RW	0xFFFF
15:0	quantum_p2	"Transmit pause quantum - written with the pause quantum value for pause frame transmission of priority 2."	RW	0xFFFF

Table 11-718: emac_tx_pause_quantum2

Register Information				
Description	"Transmit Pause Quantum Register 2"			
Offset	0x1264	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	quantum_p5	"Transmit pause quantum - written with the pause quantum value for pause frame transmission of priority 5."	RW	0xFFFF
15:0	quantum_p4	"Transmit pause quantum - written with the pause quantum value for pause frame transmission of priority 4."	RW	0xFFFF

Table 11-719: emac_tx_pause_quantum3

Register Information				
Description	"Transmit Pause Quantum Register 3"			
Offset	0x1268	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	quantum_p7	"Transmit pause quantum - written with the pause quantum value for pause frame transmission of priority 7."	RW	0xFFFF
15:0	quantum_p6	"Transmit pause quantum - written with the pause quantum value for pause frame transmission of priority 6."	RW	0xFFFF

Table 11-720: emac_pfc_status

Register Information				
Description	"Priority Flow Control status register - indicates whether PFC has been negotiated and the current state of the PFC counters for each priority."			
Offset	0x126C	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:9	reserved_31_9	"Reserved, read as 0, ignored on write."	RO	0x00 0000
8	pfc_negotiate_pclk	"Set when PFC Priority Based Pause has been negotiated."	RO	0
7:0	rx_pfc_paused	"Reflects the state of the rx_pfc_paused signals. Each bit in the vector corresponds to a priority indicated within the received PFC priority based pause frame. A bit is set when a PFC priority based pause frame has been received, and the associated priority pause time quantum is non-zero. The bit is cleared when the associated pause time has elapsed."	RO	0x00

Table 11-721: emac_rx_lpi

Register Information					
Description	"Received LPI transitions"				
Offset	0x1270		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:16	reserved_31_16	"Unused, read zero"	RO	0x0000	
15:0	count	"Count of RX LPI transitions. A count of the number of times there is a transition from receiving normal idle to receiving low power idle. Cleared on read."	RO RtoClr	0x0000	

Table 11-722: emac_rx_lpi_time

Register Information					
Description	"Received LPI time"				
Offset	0x1274		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:24	reserved_31_24	"Unused, read zero"	RO	0x00	
23:0	lpi_time	"Time in LPI. This register increments once every 16 pclk cycles when the LPI indication bit 7 is set in the network status register. Cleared on read."	RO RtoClr	0x00 0000	

Table 11-723: emac_tx_lpi

Register Information					
Description	"Transmit LPI transitions"				
Offset	0x1278		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:16	reserved_31_16	"Unused, read zero"	RO	0x0000	
15:0	count	"Count of TX LPI transmissions. A count of the number of times the enable LPI transmission bit 19 goes from low to high in the network control register. Cleared on read."	RO RtoClr	0x0000	

Table 11-724: emac_tx_lpi_time

Register Information				
Description	"Transmit LPI time"			
Offset	0x127C	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	reserved_31_24	"Unused, read zero"	RO	0x00
23:0	lpi_time	"Time in LPI. This register increments once every 16 pclk cycles when the enable LPI transmission bit 19 is set in the network control register. Cleared on read."	RO RtoClr	0x00 0000

Table 11-725: emac_designcfg_debug1

Register Information				
Description	"Design Configuration Register 1 - The GEM has many parameterisation options to configure the IP during compilation stage. This is achieved using Verilog define compiler directives in an include file called gem_defs.v. This configuration is readable through APB addressable designcfg_debug registers."			
Offset	0x1280	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:28	axi_cache_value	"Takes the value of the `gem_axi_awcache_value DEFINE"	RO	0x0
27:25	dma_bus_width	"Takes the value of bits 7:5 of the `gem_dma_bus_width DEFINE. So if the define is set to decimal 64 this will return binary 010."	RO	0x2
24	exclude_cbs	"Takes the value of the `gem_exclude_cbs DEFINE."	RO	0
23	irq_read_clear	"Takes the value of the `gem_irq_read_clear DEFINE"	RO	1
22	no_snapshot	"Takes the value of the `gem_no_snapshot DEFINE"	RO	0
21	no_stats	"Takes the value of the `gem_no_stats DEFINE"	RO	0
20	reserved_20	"Reserved, read as 1, ignored on write."	RO	1
19:15	user_in_width	"Takes the value of the `gem_user_in_width DEFINE or 1 if user_io not defined"	RO	0x01
14:10	user_out_width	"Takes the value of the `gem_user_out_width DEFINE or 1 if user_io not defined"	RO	0x01
9	user_io	"user_io not defined for the eMAC"	RO	0
8	reserved_8	"Reserved, read as 1, ignored on write."	RO	1
7	reserved_7	"Reserved, read as 0, ignored on write."	RO	0
6	ext_fifo_interface	"Takes the value of the `gem_ext_fifo_interface DEFINE"	RO	0
5	reserved_5	"Reserved, read as 0, ignored on write."	RO	0
4	int_loopback	"Takes the value of the `gem_int_loopback DEFINE"	RO	1
3:2	reserved_3_2	"Reserved, read as 0, ignored on write."	RO	0x0
1	exclude_qbv	"Takes the value of the `gem_exclude_qbv DEFINE"	RO	0
0	no_pcs	"Takes the value of the `gem_no_pcs DEFINE"	RO	0

Table 11-726: emac_designcfg_debug2

Register Information				
Description	"Design Configuration Register 2"			
Offset	0x1284	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	spram	"Takes the value of the `gem_spram DEFINE"	RO	0
30	axi	"Takes the value of the `gem_axi DEFINE"	RO	1
29:26	tx_pbuf_addr	"Takes the value of the `gem_emac_tx_pbuf_addr DEFINE or zero if pbuf_address is decimal 16"	RO	0xB
25:22	rx_pbuf_addr	"Takes the value of the `gem_emac_rx_pbuf_addr DEFINE or zero if pbuf_address is decimal 16"	RO	0xB
21	tx_pkt_buffer	"Takes the value of the `gem_tx_pkt_buffer DEFINE"	RO	1
20	rx_pkt_buffer	"Takes the value of the `gem_rx_pkt_buffer DEFINE"	RO	1
19:16	hprot_value	"Takes the value of the `gem_hprot_value DEFINE"	RO	0x1
15:14	reserved_15_14	"Unused, read zero"	RO	0x0
13:0	jumbo_max_length	"Takes the value of the `gem_jumbo_max_length DEFINE"	RO	0x2800

Table 11-727: emac_designcfg_debug3

Register Information				
Description	"Design Configuration Register 3"			
Offset	0x1288	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Unused, read zero"	RO	0x0
29:24	num_spec_add_filters	"Takes the value of the `num_spec_add_filters DEFINE"	RO	0x24
23:21	reserved_23_21	"Unused, read zero"	RO	0x0
20:0	reserved_20_0	"Unused, read zero - reserved for rx_base2_fifo_size and rx_fifo_size defines in internal FIFO mode"	RO	0x00 0000

Table 11-728: emac_designcfg_debug4

Register Information				
Description	"Design Configuration Register 4"			
Offset	0x128C	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:20	reserved_31_20	"Unused, read zero"	RO	0x000
19:0	reserved_19_0	"Unused, read zero - reserved for tx_base2_fifo_size and tx_fifo_size defines in internal FIFO mode"	RO	0x0 0000

Table 11-729: emac_designcfg_debug5

Register Information				
Description	"Design Configuration Register 5"			
Offset	0x1290	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:29	axi_prot_value	"Takes the value of the edma_axi_prot_value parameter which is set by the `gem_axi_prot_value DEFINE but will be zero if the FIFO interface has been configured"	RO	0x2
28	tsu_clk	"Takes the value of the `gem_tsu_clk DEFINE"	RO	1
27:20	rx_buffer_length_def	"Takes the value of the `gem_rx_buffer_length_def DEFINE"	RO	0x02
19	tx_pbuf_size_def	"Takes the value of the edma_tx_pbuf_size_def parameter which is set by the `gem_tx_pbuf_size_def DEFINE but will be zero if the FIFO inteface has been configured"	RO	1
18:17	rx_pbuf_size_def	"Takes the value of the `gem_rx_pbuf_size_def DEFINE"	RO	0x3
16:15	endian_swap_def	"Takes the value of the `gem_endian_swap_def DEFINE"	RO	0x3
14:12	mdc_clock_div	"Takes the value of the `gem_mdc_clock_div DEFINE"	RO	0x2
11:10	dma_bus_width_def	"Takes the value of the `gem_dma_bus_width_def DEFINE"	RO	0x0
9	phy_ident	"Indicates whether the top and bottom PHY_ID registers are present in the address map"	RO	1
8	tsu	"Takes the value of the `gem_tsu DEFINE"	RO	1
7:4	tx_fifo_cnt_width	"Takes the value of the `gem_tx_fifo_cnt_width DEFINE"	RO	0x4
3:0	rx_fifo_cnt_width	"Takes the value of the `gem_rx_fifo_cnt_width DEFINE"	RO	0x5

Table 11-730: emac_designcfg_debug6

Register Information				
Description	"Design Configuration Register 6"			
Offset	0x1294	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:28	reserved_31_28	"Reserved. Set to zero."	RO	0x0
27	pbuf_lso	"Takes the value of the `gem_pbuf_lso DEFINE"	RO	0
26	reserved_26	"Reserved. Set to zero."	RO	0
25	pbuf_cutthru	"Takes the value of the `gem_pbuf_cutthru DEFINE"	RO	1
24	pfc_multi_quantum	"Takes the value of the `gem_pfc_multi_quantum DEFINE"	RO	1
23	dma_addr_width_is_64b	"Takes the value of the `gem_dma_addr_width_is_64b DEFINE"	RO	1
22	host_if_soft_select	"Takes the value of the `gem_host_if_soft_select DEFINE"	RO	1
21	tx_add_fifo_if	"Takes the value of the `gem_tx_add_fifo_if DEFINE"	RO	0
20	ext_tsu_timer	"Set to one to indicate external TSU"	RO	1
19:16	tx_pbuf_queue_segment_size	"Takes the value of the `gem_tx_pbuf_queue_segment_size DEFINE"	RO	0x4
15:0	reserved_15_0	"Reserved. Set to zero."	RO	0x0000

Table 11-731: emac_designcfg_debug7

Register Information				
Description	"Design Configuration Register 7"			
Offset	0x1298	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:28	tx_pbuf_num_segments_q7	"Takes the value of the `gem_tx_pbuf_num_segments_q7 DEFINE"	RO	0x0
27:24	tx_pbuf_num_segments_q6	"Takes the value of the `gem_tx_pbuf_num_segments_q6 DEFINE"	RO	0x0
23:20	tx_pbuf_num_segments_q5	"Takes the value of the `gem_tx_pbuf_num_segments_q5 DEFINE"	RO	0x0
19:16	tx_pbuf_num_segments_q4	"Takes the value of the `gem_tx_pbuf_num_segments_q4 DEFINE"	RO	0x0
15:12	tx_pbuf_num_segments_q3	"Takes the value of the `gem_tx_pbuf_num_segments_q3 DEFINE"	RO	0x0
11:8	tx_pbuf_num_segments_q2	"Takes the value of the `gem_tx_pbuf_num_segments_q2 DEFINE"	RO	0x0
7:4	tx_pbuf_num_segments_q1	"Takes the value of the `gem_tx_pbuf_num_segments_q1 DEFINE"	RO	0x0
3:0	tx_pbuf_num_segments_q0	"Takes the value of the edma_tx_pbuf_num_segments_q0 parameter which is set by the `gem_tx_pbuf_num_segments_q0 DEFINE or is zero if priority queuing is not configured."	RO	0x0

Table 11-732: emac_designcfg_debug8

Register Information				
Description	"Design Configuration Register 8"			
Offset	0x129C	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	num_type1_screeners	"Set to 1 for the eMAC (express MAC)"	RO	0x01
23:16	num_type2_screeners	"Set to 2 for the eMAC (express MAC)"	RO	0x02
15:8	num_scr2_ethtype_regs	"Set to 0 for the eMAC (express MAC)"	RO	0x00
7:0	num_scr2_compare_regs	"Set to 6 for the eMAC (express MAC)"	RO	0x06

Table 11-733: emac_designcfg_debug9

Register Information				
Description	"Design Configuration Register 9"			
Offset	0x12A0	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:28	tx_pbuf_num_segments_q15	"Takes the value of the `gem_tx_pbuf_num_segments_q15 DEFINE"	RO	0x0
27:24	tx_pbuf_num_segments_q14	"Takes the value of the `gem_tx_pbuf_num_segments_q14 DEFINE"	RO	0x0
23:20	tx_pbuf_num_segments_q13	"Takes the value of the `gem_tx_pbuf_num_segments_q13 DEFINE"	RO	0x0
19:16	tx_pbuf_num_segments_q12	"Takes the value of the `gem_tx_pbuf_num_segments_q12 DEFINE"	RO	0x0
15:12	tx_pbuf_num_segments_q11	"Takes the value of the `gem_tx_pbuf_num_segments_q11 DEFINE"	RO	0x0
11:8	tx_pbuf_num_segments_q10	"Takes the value of the `gem_tx_pbuf_num_segments_q10 DEFINE"	RO	0x0
7:4	tx_pbuf_num_segments_q9	"Takes the value of the `gem_tx_pbuf_num_segments_q9 DEFINE"	RO	0x0
3:0	tx_pbuf_num_segments_q8	"Takes the value of the `gem_tx_pbuf_num_segments_q8 DEFINE"	RO	0x0

Table 11-734: emac_designcfg_debug10

Register Information			
Description	"Design Configuration Register 10"		
Offset	0x12A4	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:28	emac_bus_width	"Takes the value of the `gem_emac_bus_width DEFINE. 1 - The MAC has a datawidth of 32bits. 2 - The MAC has a datawidth of 64bits. 4 - The MAC has a datawidth of 128bits"	RO	0x2
27:24	tx_pbuf_data	"Takes the value of the `gem_tx_pbuf_data DEFINE. 1 - The TX DPRAM has a datawidth of 32bits. 2 - The TX DPRAM has a datawidth of 64bits. 4 - The TX DPRAM has a datawidth of 128bits"	RO	0x2
23:20	rx_pbuf_data	"Takes the value of the `gem_rx_pbuf_data DEFINE. 1 - The RX DPRAM has a datawidth of 32bits. 2 - The RX DPRAM has a datawidth of 64bits. 4 - RX The DPRAM has a datawidth of 128bits"	RO	0x2
19:16	axi_access_pipeline_bits	"Takes the value of the edma_axi_access_pipeline_bits parameter which is set by the `gem_axi_access_pipeline_bits DEFINE or is zero if the FIFO interface is configured."	RO	0x4
15:12	axi_tx_descr_rd_buff_bits	"Takes the value of the edma_axi_tx_descr_rd_buff_bits parameter which is set by the `gem_axi_tx_descr_rd_buff_bits DEFINE or is zero if the FIFO interface is configured."	RO	0x4
11:8	axi_rx_descr_rd_buff_bits	"Takes the value of the edma_axi_rx_descr_rd_buff_bits parameter which is set by the `gem_axi_rx_descr_rd_buff_bits DEFINE or is zero if the FIFO interface is configured."	RO	0x4
7:4	axi_tx_descr_wr_buff_bits	"Takes the value of the edma_axi_tx_descr_wr_buff_bits parameter which is set by the `gem_axi_tx_descr_wr_buff_bits DEFINE or is zero if the FIFO interface is configured."	RO	0x4
3:0	axi_rx_descr_wr_buff_bits	"Takes the value of the edma_axi_rx_descr_wr_buff_bits parameter which is set by the `gem_axi_rx_descr_wr_buff_bits DEFINE or is zero if the FIFO interface is configured."	RO	0x4

Table 11-735: emac_designcfg_debug11

Register Information			
Description	"Design Configuration Register 11"		
Offset	0x12A8	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:8	reserved_31_8	"Reserved, read as 0, ignored on write."	RO	0x00 0000
7	asf_prot_tx_sched	"Takes the value of the edma_asf_asf_prot_tx_sched parameter which is set by the `gem_asf_prot_tx_sched DEFINE."	RO	1
6	asf_host_par	"Takes the value of the edma_asf_host_par parameter which is set by the `gem_asf_host_par DEFINE."	RO	1
5	asf_trans_to_prot	"Takes the value of the edma_asf_trans_to_prot parameter which is set by the `gem_asf_enable DEFINE."	RO	1
4	asf_integrity_prot	"Takes the value of the edma_asf_integrity_prot parameter which is set by the `gem_asf_enable DEFINE."	RO	1
3	protect_tsu	"akes the value of the edma_asf_prot_tsu parameter which is set by the `gem_asf_prot_tsu DEFINE or is zero if no TSU is present."	RO	1
2	csr_protection	"Takes the value of the edma_asf_csr_prot parameter which is set by the `gem_asf_enable DEFINE."	RO	1
1	dap_protection	"Takes the value of the edma_asf_dap_prot parameter which is set by the `gem_asf_enable DEFINE."	RO	1
0	ecc_sram	"Takes the value of the edma_asf_ecc_sram parameter which is set by the `gem_asf_ecc_sram DEFINE."	RO	1

Table 11-736: emac_designcfg_debug12

Register Information				
Description	"Design Configuration Register 12"			
Offset	0x12AC	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:26	reserved_31_26	"Reserved, read as 0, ignored on write."	RO	0x00
25	gem_has_802p3_br	"A value of one indicates that gem_has_802p3_br is defined. If defined the configuration contains both an express MAC (eMAC) and a pre-emptable MAC (pMAC) to allow 802.3br frame pre-emption."	RO	1
24:21	emac_tx_pbuf_addr	"Takes the value of the `gem_emac_tx_pbuf_addr DEFINE - this defines the size of the transmit SRAM for express MAC (eMAC) when 802.3br is configured - will be zero if emac_tx_pbuf_address is decimal 16"	RO	0xB
20:17	emac_rx_pbuf_addr	"Takes the value of the `gem_emac_rx_pbuf_addr DEFINE - this defines the size of the receive SRAM for express MAC (eMAC) when 802.3br is configured - will be zero if emac_tx_pbuf_address is decimal 16"	RO	0xB
16	gem_has_cb	"Indicates whether gem has 802.1CB/FRER configured"	RO	1
15:8	gem_cb_history_len	"Takes the value of the `gem_seq_history_len DEFINE or 0x01 if undefined."	RO	0x40
7:0	gem_num_cb_streams	"Set to 2 for the eMAC (express MAC)"	RO	0x02

Table 11-737: emac_axi_qos_cfg

Register Information				
Description	"eMAC AXI Quality of Service register. This register is only present if AXI is configured. This register contains 8-bits to control driving of the ARQOS and AWQOS outputs (4-bits each). The lower four bits are for data accesses and the upper four bits are for descriptor accesses."			
Offset	0x12E0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:8	reserved_31_8	"Reserved, read as 0, ignored on write."	RO	0x00 0000
7:4	emac_descr_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for eMAC descriptor accesses."	RW	0x0
3:0	emac_data_qos	"Determines what value to drive to the ARQOS and AWQOS outputs (4-bits each) for eMAC data accesses."	RW	0x0

Table 11-738: emac_spec_add5_bottom

Register Information					
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."				
Offset	0x1300		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."		RW	0x0000 0000

Table 11-739: emac_spec_add5_top

Register Information					
Description	"Specific Address Top"				
Offset	0x1304		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."		RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."		RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."		RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."		RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."		RW	0x0000

Table 11-740: emac_spec_add6_bottom

Register Information					
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."				
Offset	0x1308		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."		RW	0x0000 0000

Table 11-741: emac_spec_add6_top

Register Information					
Description	"Specific Address Top"				
Offset	0x130C		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."		RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."		RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."		RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."		RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."		RW	0x0000

Table 11-742: emac_spec_add7_bottom

Register Information			
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."		
Offset	0x1310	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-743: emac_spec_add7_top

Register Information			
Description	"Specific Address Top"		
Offset	0x1314	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-744: emac_spec_add8_bottom

Register Information					
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."				
Offset	0x1318		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000	

Table 11-745: emac_spec_add8_top

Register Information					
Description	"Specific Address Top"				
Offset	0x131C		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0	
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00	
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00	
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0	
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000	

Table 11-746: emac_spec_add9_bottom

Register Information			
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."		
Offset	0x1320	**Type:**	RW

Bits	Name	Description	Access	Reset
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-747: emac_spec_add9_top

Register Information			
Description	"Specific Address Top"		
Offset	0x1324	**Type:**	RW

Bits	Name	Description	Access	Reset
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-748: emac_spec_add10_bottom

Register Information			
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."		
Offset	0x1328	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-749: emac_spec_add10_top

Register Information			
Description	"Specific Address Top"		
Offset	0x132C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-750: emac_spec_add11_bottom

Register Information					
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."				
Offset	0x1330		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."		RW	0x0000 0000

Table 11-751: emac_spec_add11_top

Register Information					
Description	"Specific Address Top"				
Offset	0x1334		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."		RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."		RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."		RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."		RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."		RW	0x0000

Table 11-752: emac_spec_add12_bottom

Register Information					
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."				
Offset	0x1338		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."		RW	0x0000 0000

Table 11-753: emac_spec_add12_top

Register Information					
Description	"Specific Address Top"				
Offset	0x133C		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."		RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."		RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."		RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."		RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."		RW	0x0000

Table 11-754: emac_spec_add13_bottom

Register Information			
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."		
Offset	0x1340	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-755: emac_spec_add13_top

Register Information			
Description	"Specific Address Top"		
Offset	0x1344	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-756: emac_spec_add14_bottom

Register Information			
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."		
Offset	0x1348	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-757: emac_spec_add14_top

Register Information			
Description	"Specific Address Top"		
Offset	0x134C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-758: emac_spec_add15_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x1350	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-759: emac_spec_add15_top

Register Information				
Description	"Specific Address Top"			
Offset	0x1354	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-760: emac_spec_add16_bottom

Register Information			
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."		
Offset	0x1358	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-761: emac_spec_add16_top

Register Information			
Description	"Specific Address Top"		
Offset	0x135C	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-762: emac_spec_add17_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x1360	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-763: emac_spec_add17_top

Register Information				
Description	"Specific Address Top"			
Offset	0x1364	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-764: emac_spec_add18_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x1368	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-765: emac_spec_add18_top

Register Information				
Description	"Specific Address Top"			
Offset	0x136C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-766: emac_spec_add19_bottom

Register Information			
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."		
Offset	0x1370	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-767: emac_spec_add19_top

Register Information			
Description	"Specific Address Top"		
Offset	0x1374	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-768: emac_spec_add20_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x1378	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-769: emac_spec_add20_top

Register Information				
Description	"Specific Address Top"			
Offset	0x137C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-770: emac_spec_add21_bottom

Register Information					
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."				
Offset	0x1380		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."		RW	0x0000 0000

Table 11-771: emac_spec_add21_top

Register Information					
Description	"Specific Address Top"				
Offset	0x1384		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."		RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."		RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."		RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."		RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."		RW	0x0000

Table 11-772: emac_spec_add22_bottom

Register Information					
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."				
Offset	0x1388		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000	

Table 11-773: emac_spec_add22_top

Register Information					
Description	"Specific Address Top"				
Offset	0x138C		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0	
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00	
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00	
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0	
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000	

Table 11-774: emac_spec_add23_bottom

Register Information			
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."		
Offset	0x1390	**Type:**	RW

Bits	Name	Description	Access	Reset
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-775: emac_spec_add23_top

Register Information			
Description	"Specific Address Top"		
Offset	0x1394	**Type:**	RW

Bits	Name	Description	Access	Reset
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-776: emac_spec_add24_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x1398	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-777: emac_spec_add24_top

Register Information				
Description	"Specific Address Top"			
Offset	0x139C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-778: emac_spec_add25_bottom

Register Information					
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."				
Offset	0x13A0		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."		RW	0x0000 0000

Table 11-779: emac_spec_add25_top

Register Information					
Description	"Specific Address Top"				
Offset	0x13A4		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."		RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."		RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."		RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."		RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."		RW	0x0000

Table 11-780: emac_spec_add26_bottom

Register Information			
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."		
Offset	0x13A8	**Type:**	RW

Bits	Name	Description	Access	Reset
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-781: emac_spec_add26_top

Register Information			
Description	"Specific Address Top"		
Offset	0x13AC	**Type:**	RW

Bits	Name	Description	Access	Reset
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-782: emac_spec_add27_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x13B0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-783: emac_spec_add27_top

Register Information				
Description	"Specific Address Top"			
Offset	0x13B4	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-784: emac_spec_add28_bottom

Register Information					
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."				
Offset	0x13B8		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."		RW	0x0000 0000

Table 11-785: emac_spec_add28_top

Register Information					
Description	"Specific Address Top"				
Offset	0x13BC		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."		RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."		RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."		RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."		RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."		RW	0x0000

Table 11-786: emac_spec_add29_bottom

Register Information			
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."		
Offset	0x13C0	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-787: emac_spec_add29_top

Register Information			
Description	"Specific Address Top"		
Offset	0x13C4	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-788: emac_spec_add30_bottom

Register Information			
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."		
Offset	0x13C8	**Type:**	RW

Bits	Name	Description	Access	Reset
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-789: emac_spec_add30_top

Register Information			
Description	"Specific Address Top"		
Offset	0x13CC	**Type:**	RW

Bits	Name	Description	Access	Reset
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-790: emac_spec_add31_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x13D0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-791: emac_spec_add31_top

Register Information				
Description	"Specific Address Top"			
Offset	0x13D4	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-792: emac_spec_add32_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x13D8	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-793: emac_spec_add32_top

Register Information				
Description	"Specific Address Top"			
Offset	0x13DC	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-794: emac_spec_add33_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x13E0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-795: emac_spec_add33_top

Register Information				
Description	"Specific Address Top"			
Offset	0x13E4	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-796: emac_spec_add34_bottom

Register Information				
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."			
Offset	0x13E8	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-797: emac_spec_add34_top

Register Information				
Description	"Specific Address Top"			
Offset	0x13EC	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-798: emac_spec_add35_bottom

Register Information			
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."		
Offset	0x13F0	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."	RW	0x0000 0000

Table 11-799: emac_spec_add35_top

Register Information			
Description	"Specific Address Top"		
Offset	0x13F4	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."	RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."	RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."	RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."	RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."	RW	0x0000

Table 11-800: emac_spec_add36_bottom

Register Information					
Description	"The addresses stored in the specific address registers are deactivated at reset or when their corresponding specific address register bottom is written. They are activated when specific address register top is written."				
Offset	0x13F8		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:0	address	"Octets 3 to 0 of the destination address, that is bits 31:0. Bit zero indicates whether the address is multicast or unicast and corresponds to the least significant bit of the first byte received. See the MAC Filtering Block of the user guide for more information on how to program this register."		RW	0x0000 0000

Table 11-801: emac_spec_add36_top

Register Information					
Description	"Specific Address Top"				
Offset	0x13FC		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:30	reserved_31_30	"Reserved, read as 0, ignored on write."		RO	0x0
29:24	filter_byte_mask	"When high, the associated byte of the specific address will not be compared. Bit 24 controls whether the first byte received should be compared. Bit 29 controls whether the last byte received should be compared."		RW	0x00
23:17	reserved_23_17	"Reserved, read as 0, ignored on write."		RO	0x00
16	filter_type	"This control bit selects whether this filter should be comparing the MAC source address or the MAC destination address of the received Ethernet frame. When set to zero, the filter is a destination address filter. When set to one, the filter is a source address filter."		RW	0
15:0	address	"Specific address 1. The most significant bits of the destination/source address that is to be compared, that is bits 47:32. See MAC Filtering Block of the user guide for more information."		RW	0x0000

Table 11-802: emac_cbs_control

Register Information			
Description	"The IdleSlope value is defined as the rate of change of credit when a packet is waiting to be sent. This must not exceed the portTransmitRate which is dependent on the speed of operation, eg, portTransmitRate: 1Gb/s = 32'h07735940 (125 Mbytes/s), 100Mb/sec = 32'h017D7840 (25 Mnibbles/s), 10Mb/sec = 32'h002625A0 (2.5 Mnibbles/s). If 50% of bandwidth was to be allocated to a particular queue in 1Gb/sec mode then the IdleSlope value for that queue would be calculated as 32'h07735940/2. Note: Credit-Based Shaping should be disabled prior to updating the IdleSlope values. As another example, for a 1722 audio packet with a payload of 6 samples per channel, the packet size would be: 7 (preamble) + 1 (SFD) + 50 (packet header) + 6x4x2(payload) + 4 (CRC) = 110 bytes. For a rate of 8000 packets per second, the desired rate would 110 x 8000 bytes per second, so the programmed idleSlope value would be 880000 for a 1Gb/s connection, or 1760000 for a 100Mb/s or 10Mbs connection. See Figure 6.3 in the IEEE 1722 standard. In practice, the actual transmission rate will be vary slightly from that calculated. In this case, the idleSlope value should be recalibrated based on the variance between the measured and expected rate, and in this case very accurate transmission rates can be achieved. (The idleslope value is scaled by 2.5 in 2.5G operation and so the port transmit rate is the same for 1G and 2.5G.)"		
Offset	0x14BC	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:1	reserved_31_1	"Reserved, read as 0, ignored on write."	RO	0x0000 0000
0	cbs_enable_queue_a	"Enable Credit-Based Shaping on the highest priority queue (queue A). Write 1 to enable"	RW	0

Table 11-803: emac_cbs_idleslope_q_a

Register Information			
Description	"Queue A is the highest priority queue. This is the highest indexed active queue, e.g. For a system with Q0 to Q7, this would be Q7 if all queues were active."		
Offset	0x14C0	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	idleslope_a	"IdleSlope value for queue A in bytes/sec for gigabit operation and nibbles/sec for 10/100 operation"	RW	0x0000 0000

Table 11-804: emac_cbs_idleslope_q_b

Register Information			
Description	"Queue B is the 2nd highest priority queue. This is the second highest indexed active queue, e.g. For a system with Q0 to Q7, this would be Q6 if all queues were active."		
Offset	0x14C4	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	idleslope_b	"IdleSlope value for queue B in bytes/sec for gigabit operation and nibbles/sec for 10/100 operation"	RW	0x0000 0000

Table 11-805: emac_upper_tx_q_base_addr

Register Information			
Description	"Upper 32 bits of transmit buffer descriptor queue base address."		
Offset	0x14C8	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	upper_tx_q_base_addr	"Upper 32 bits of transmit buffer descriptor queue base address. Used when 64 bit addressing is enabled. (In releases earlier to 1p06f2 this register also affected the receive descriptor queue.)"	RW	0x0000 0000

Table 11-806: emac_tx_bd_control

Register Information			
Description	"Transmit buffer descriptor control register - this register determines which transmit frames, with time stamps reported in the buffer descriptor status field in extended buffer descriptor mode, set bit 23 in transmit buffer descriptor word 1."		
Offset	0x14CC	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:6	reserved_31_6	"Reserved, read as 0, ignored on write."	RO	0x000 0000
5:4	tx_bd_ts_mode	"Transmit Descriptor Timestamp Insertion mode - bit 23 in transmit buffer descriptor word 1 when extended buffer descriptor mode is enabled, 00: Bit 23 always zero, 01: Bit 23 high for PTP Event Frames only, 10: Bit 23 high for all PTP Frames only, 11: Bit 23 always high"	RW	0x0
3:0	reserved_3_0	"Reserved, read as 0, ignored on write."	RO	0x0

Table 11-807: emac_rx_bd_control

Register Information			
Description	"Receive buffer descriptor control register - this register determines which receive frames have time stamps reported in the buffer descriptor status field"		
Offset	0x14D0	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:6	reserved_31_6	"Reserved, read as 0, ignored on write."	RO	0x000 0000
5:4	rx_bd_ts_mode	"Receive Descriptor Timestamp Insertion mode, 00: TS insertion disable, 01: TS inserted for PTP Event Frames only, 10: TS inserted for All PTP Frames only, 11: TS insertion for All Frames"	RW	0x0
3:0	reserved_3_0	"Reserved, read as 0, ignored on write."	RO	0x0

Table 11-808: emac_upper_rx_q_base_addr

Register Information				
Description	"Upper 32 bits of receive buffer descriptor queue base address."			
Offset	0x14D4	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:0	upper_rx_q_base_addr	"Upper 32 bits of receive buffer descriptor queue base address. Used when 64 bit addressing is enabled."	RW	0x0000 0000

Table 11-809: emac_wd_counter

Register Information				
Description	"Hidden control register - do not alter"			
Offset	0x14EC	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:4	reserved_31_4	"Reserved, read as 0, ignored on write."	RO	0x000 0000
3:0	rx_bd_reread_timer	"Controls counter used to retry receive descriptor reads (retries may occur if there was a resource error and there are frames waiting in the internal SRAM buffer to be off-loaded to host memory)"	RW	0x7

Table 11-810: emac_wd_axi_tx_full_thresh0

Register Information				
Description	"Hidden control register - do not alter"			
Offset	0x14F8	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	reserved_31_24	"Reserved, read as 0, ignored on write."	RO	0x00
23:16	axi_tx_full_adj_0	"AXI single port SRAM fine tuning"	RW	0x06
15:8	reserved_15_8	"Reserved, read as 0, ignored on write."	RO	0x00
7:0	axi_tx_full_adj_1	"AXI single port SRAM fine tuning"	RW	0x08

Table 11-811: emac_wd_axi_tx_full_thresh1

Register Information				
Description	"Hidden control register - do not alter"			
Offset	0x14FC	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	reserved_31_24	"Reserved, read as 0, ignored on write."	RO	0x00
23:16	axi_tx_full_adj_2	"AXI single port SRAM fine tuning"	RW	0x00
15:8	reserved_15_8	"Reserved, read as 0, ignored on write."	RO	0x00
7:0	axi_tx_full_adj_3	"AXI single port SRAM fine tuning"	RW	0x00

Table 11-812: emac_screening_type_1_register

Register Information				
Description	"Screening type 1 registers are used to allocate up to 16 priority queues to received frames based on certain IP or UDP fields of incoming frames. Firstly, when DS/TC match enable is set (bit 28), the DS (Differentiated Services) field of the received IPv4 header or TC field (traffic class) of IPv6 headers are matched against bits 11:4. Secondly, when UDP port match enable is set (bit 29), the UDP Destination Port of the received UDP frame is matched against bits 27:12. Both UDP and DS/TC matching can be enabled simultaneously or individually. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 1 screening registers is configured in the gem defines file. Up to 16 type 1 screening registers have been allocated APB address space between 0x500 and 0x53C. The bit mappings for these registers are as follows:"			
Offset	0x1500	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	reserved_31	"Reserved, read as 0, ignored on write."	RO	0
30	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
29	udp_port_match_enable	"UDP port match enable"	RW	0
28	dstc_enable	"DS/TC Enable"	RW	0
27:12	udp_port_match	"UDP Port Match"	RW	0x0000
11:4	dstc_match	"DS/TC Match"	RW	0x00
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-813: emac_screening_type_2_register_0

Register Information			
Description	"Screener Type 2 match registers allow a screen to be configured that is the combination of all or any of the following comparisons: 1) An enabled VLAN Priority. A VLAN Priority match will be performed if the VLAN priority enable is set. The extracted priority field in the VLAN header is compared against 3 bits within the screener type 2 register itself. 2) An enabled EtherType. 3) An enabled Field Compare A. 4) An enabled Field Compare B. 5) An enabled Field Compare C. All enabled comparisons are ANDed together to form the overall type 2 screener match. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 2 screening registers is configured in the gem defines file. Up to 16 type 2 screening registers have been allocated APB address space between 0x540 and 0x57C. The bit mappings for these registers are as follows:"		
Offset	0x1540	**Type:**	RW

Bits	Name	Description	Access	Reset
31	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
30	compare_c_enable	"Compare C Enable"	RW	0
29:25	compare_c	"Compare C - Index to screener type 2 Compare register"	RW	0x00
24	compare_b_enable	"Compare B Enable"	RW	0
23:19	compare_b	"Compare B - Index to screener type 2 Compare register"	RW	0x00
18	compare_a_enable	"Compare A Enable"	RW	0
17:13	compare_a	"Compare A - Index to screener type 2 Compare register"	RW	0x00
12	ethertype_enable	"EtherType Enable"	RW	0
11:9	ethertype_reg_index	"Index to screener type 2 EtherType register"	RW	0x0
8	vlan_enable	"VLAN Enable"	RW	0
7	reserved_7	"Reserved and implemented as RW"	RW	0
6:4	vlan_priority	"VLAN Priority"	RW	0x0
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-814: emac_screening_type_2_register_1

Register Information			
Description	"Screener Type 2 match registers allow a screen to be configured that is the combination of all or any of the following comparisons: 1) An enabled VLAN Priority. A VLAN Priority match will be performed if the VLAN priority enable is set. The extracted priority field in the VLAN header is compared against 3 bits within the screener type 2 register itself. 2) An enabled EtherType. 3) An enabled Field Compare A. 4) An enabled Field Compare B. 5) An enabled Field Compare C. All enabled comparisons are ANDed together to form the overall type 2 screener match. If a match is successful, then the queue value programmed in bits 3:0 is allocated to the frame. The required number of Type 2 screening registers is configured in the gem defines file. Up to 16 type 2 screening registers have been allocated APB address space between 0x540 and 0x57C. The bit mappings for these registers are as follows:"		
Offset	0x1544	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	drop_on_match	"When set, if the frame matches it will be completely dropped. In this case the queue number is don't care."	RW	0
30	compare_c_enable	"Compare C Enable"	RW	0
29:25	compare_c	"Compare C - Index to screener type 2 Compare register"	RW	0x00
24	compare_b_enable	"Compare B Enable"	RW	0
23:19	compare_b	"Compare B - Index to screener type 2 Compare register"	RW	0x00
18	compare_a_enable	"Compare A Enable"	RW	0
17:13	compare_a	"Compare A - Index to screener type 2 Compare register"	RW	0x00
12	ethertype_enable	"EtherType Enable"	RW	0
11:9	ethertype_reg_index	"Index to screener type 2 EtherType register"	RW	0x0
8	vlan_enable	"VLAN Enable"	RW	0
7	reserved_7	"Reserved and implemented as RW"	RW	0
6:4	vlan_priority	"VLAN Priority"	RW	0x0
3:0	queue_number	"Queue Number (0 to 15)"	RW	0x0

Table 11-815: emac_tx_sched_ctrl

Register Information					
Description	"This register controls the transmit scheduling algorithm the user can select for the eMAC."				
Offset	0x1580		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:2	reserved_31_2	"Reserved, read as 0, ignored on write."		RO	0x0000 0000
1:0	tx_sched	"Transmit scheduling algorithm. 00 : Normal Priority 01 : CBS Enabled only valid if CBS capability selected. 10 : DWRR Enabled 11 : ETS Enabled"		RW	0x0

Table 11-816: emac_bw_rate_limit

Register Information					
Description	"This register holds the DWRR weighting value or the ETS bandwidth percentage value used by the transmit scheduler for the eMAC."				
Offset	0x1590		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:8	reserved_31_8	"Reserved, read as 0, ignored on write."		RO	0x00 0000
7:0	emac_dwrr_ets_weight	"DWRR Weighting / ETS Bandwidth Allocation for the eMAC"		RW	0x00

Table 11-817: emac_tx_q_seg_alloc_q_lower

Register Information					
Description	"There should be no need to change this register for the eMAC."				
Offset	0x15A0		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:3	reserved_31_3	"Reserved, read as 0, ignored on write."		RO	0x0000 0000
2:0	segment_alloc	"Number of segments allocated. This should be entered as a log 2, for example entering a value of 2 would grant 4 segments. A maximum of 16 segments can be granted."		RW	0x0

Table 11-818: emac_type2_compare_0_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x1700	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-819: emac_type2_compare_0_word_1

Register Information			
Description	"Type2 Compare Word 1"		
Offset	0x1704	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-820: emac_type2_compare_1_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x1708	**Type:**	RW

Bits	Name	Description	Access	Reset
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-821: emac_type2_compare_1_word_1

Register Information				
Description	"Type2 Compare Word 1"			
Offset	0x170C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-822: emac_type2_compare_2_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x1710	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-823: emac_type2_compare_2_word_1

Register Information				
Description	"Type2 Compare Word 1"			
Offset	0x1714	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-824: emac_type2_compare_3_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x1718	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-825: emac_type2_compare_3_word_1

Register Information				
Description	"Type2 Compare Word 1"			
Offset	0x171C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-826: emac_type2_compare_4_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x1720	**Type:**	RW

Bits	Name	Description	Access	Reset
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-827: emac_type2_compare_4_word_1

Register Information				
Description	"Type2 Compare Word 1"			
Offset	0x1724	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-828: emac_type2_compare_5_word_0

Register Information			
Description	"Compare A, B and C fields of the screener type 2 match register are pointers to a pool of up to 32 compare registers. If enabled the compare is true if the data at the OFFSET into the frame, ANDed with the MASK Value if the mask is enabled, is equal to the COMPARE Value. Either a 16 bit comparison or a 32 bit comparison is done. This selection is made via the associated compare word1 register bit 9. If a 16 bit comparison is selected, then a 16 bit mask is also available to the user to select which bits should be compared. If the 32 bit compare option is selected, then no mask is available. The byte at the OFFSET number of bytes from the index start is compared thru bits 7:0 of the configured VALUE. The byte at the OFFSET number of bytes + 1 from the index start is compared thru bits 15:8 of the configured VALUE and so on. The OFFSET can be configured to be from 0 to 127 bytes from either the start of the frame, the byte following the therType field (last EtherType in the header if the frame is VLAN tagged), the byte following the IP header (IPv4 or IPv6) or from the byte following the start of the TCP/UDP header. The required number of Type 2 screening registers up to a maximum of 32 is configurable in the gem defines file and have been allocated APB address space between 0x700 and 0x7fc. Note. when using RX Partial Store and Forward mode and priority queues, the frame offset must be less than the Partial Store and Forward watermark. If the offset is higher than the watermark value it's not possible to identify the priority queue before the frame is sent to the AMBA interface, and an incorrect priority queue may be used."		
Offset	0x1728	**Type:**	RW

Bits	Name	Description	Access	Reset
31:16	compare_value	"2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+2 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+3. If bit 9 of the associated compare_word1 register is clear, then the byte stored in bits [23:16] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [31:24] is compared against the byte in the received frame from the selected offset+1. "	RW	0x0000
15:0	mask_value	"These bits can be either a 2 byte mask field or an additional 2 byte Compare Value. If bit 9 of the associated compare_word1 register is set, then the byte stored in bits [7:0] is compared against the byte in the received frame from the selected offset+0 and the byte stored in bits [15:8] is compared against the byte in the received frame from the selected offset+1. If bit 9 of the associated compare_word1 register is clear, these bits become a direct 2-byte mask for the 2-byte compare register in bits [31:16]. A value of zero in a mask bit masks the corresponding data bit, a value of 1 enables the comparison."	RW	0x0000

Table 11-829: emac_type2_compare_5_word_1

Register Information				
Description	"Type2 Compare Word 1"			
Offset	0x172C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:11	reserved_31_11	"Reserved, read as 0, ignored on write."	RO	0x00 0000
10	compare_vlan_id	"If set bits 9, 8 and 6:0 must be 0 and bit 7 of compare_offset is used as follows: 0 Compare C-TAG VID (VLAN has EtherType of 0x8100) 1 Compare S-TAG VID comparison (VLAN has EtherType of the value in the stacked VLAN register at 0x00C0) Note the byte order is such that if 81 00 00 20 is received to indicate a C-TAG frame with VID 020 0x00200FFF would be written to the compare register. So in the special case of VLAN comparison the last byte received is the least significant byte in the compare register"	RW	0
9	disable_mask	"This bit is used to control whether the compare register word_0 contains a 4-byte compare value, or a 2-byte compare value with a 2-byte mask value. 1 - 4-byte compare value 0 - 2-byte compare, 2-byte mask"	RW	0
8:7	compare_offset	"Compare byte offset. 00 : Offset from beginning of the frame. 01 : Offset from byte after Ether Type. 10 : Offset from byte following end of IP header. 11 : Offset from byte following end of TCP/UDP header"	RW	0x0
6:0	offset_value	"Offset value in bytes"	RW	0x00

Table 11-830: emac_enst_start_time

Register Information				
Description	"This register sets the absolute time at which this queue starts EnST transmit scheduling"			
Offset	0x1800	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:30	start_time_sec	"Bits 31:30 of the start time (seconds field)"	RW	0x0
29:0	start_time_nsec	"Bits 29:0 of the start time (nanoseconds field)"	RW	0x0000 0000

Table 11-831: emac_enst_on_time

Register Information					
Description	"This register sets the time period for which queue zero is to be open, (on_time_nsec), this register has to be written according to the speed mode as: ||1Gbps or 2.5Gbps|| : on_time = on_time_nsec/8 ||100Mbps|| : on_time = on_time_nsec/80 ||10Mbps|| : on_time = on_time_nsec/800"				
Offset	0x1820		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:17	reserved_31_17	"Reserved, read as 0, ignored on write."	RO	0x0000	
16:0	on_time	"on_time"	RW	0x1 FFFF	

Table 11-832: emac_enst_off_time

Register Information					
Description	"This register sets the time period for which queue zero is to be blocked, (off_time_nsec), this register has to be written according to the speed mode as: ||1Gbps or 2.5Gbps|| => off_time = off_time_nsec/8 ||100Mbps|| => off_time = off_time_nsec/80 ||10Mbps|| => off_time = off_time_nsec/800"				
Offset	0x1840		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:17	reserved_31_17	"Reserved, read as 0, ignored on write."	RO	0x0000	
16:0	off_time	"Off_time"	RW	0x0 0000	

Table 11-833: emac_enst_control

Register Information				
Description	"Enhancement for Scheduled Traffic control register. EnST scheduling can only be applied to a maximum number of 8 queues. If 802.3br operation has been configured and both an eMAC and pMAC are present then the eMAC only supports a single queue and EnST is enabled on the eMAC by writing to bit 0 of emac_enst_control."			
Offset	0x1880	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:8	reserved_31_8	"Reserved, read as 0, ignored on write."	RO	0x00 0000
7	enst_enable_q15	"Set to enable enhanced scheduler traffic (ENST), for q15"	RW	0
6	enst_enable_q14	"Set to enable enhanced scheduler traffic (ENST), for q14"	RW	0
5	enst_enable_q13	"Set to enable enhanced scheduler traffic (ENST), for q13"	RW	0
4	enst_enable_q12	"Set to enable enhanced scheduler traffic (ENST), for q12"	RW	0
3	enst_enable_q11	"Set to enable enhanced scheduler traffic (ENST), for q11"	RW	0
2	enst_enable_q10	"Set to enable enhanced scheduler traffic (ENST), for q10"	RW	0
1	enst_enable_q9	"Set to enable enhanced scheduler traffic (ENST), for q9"	RW	0
0	enst_enable_q8	"Set to enable enhanced scheduler traffic (ENST), for q8"	RW	0

Table 11-834: emac_frer_timeout

Register Information				
Description	"FRER timeout register. This determines when the sequence recovery reset timers are reset and is used by all of the configured CB streams. It is programmed as a count of 8192 rx_clk periods. 8192 rx_clk periods is 65.536 microseconds at gigabit speeds and 327.68 microseconds at 100M speeds, this allows a max timeout value of about 4 seconds at gigabit speeds - for test purposes if bit 12 (retry_test) is set in the network configuration register at 0x004 then the timeout value just becomes a count of rx_clk periods (ie a speed-up of 8192 in the time-out process)."			
Offset	0x18A0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	timeout	"Count of 8192 rx_clk periods."	RW	0x0000

Table 11-835: emac_frer_red_tag

Register Information			
Description	"FRER redundancy tag register. This defines the Ethertype used to identify the R-TAG (redundancy tag) and contains a control bit to enable stripping of the R-TAG from received frames. IEEE 802.1CB has defined the value of this Ethertype to be 0xF1C1. Whether or not a particular stream uses the redundancy tag to locate the sequence number is determined by bit 28 in that streams control register."		
Offset	0x18A4	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	strip_r_tag	"If set the redundancy tag is stripped from received frames. If this bit is set then the receive octet counters reflect post deletion frame size so the FRER functionality is transparent to higher level management if the stripping function is enabled. If the statistics counters need to reflect the actual number of octets received, then the stripping functionality should be disabled."	RW	0
30	six_byte_tag	"Draft 2.4 and earlier drafts of the 802.1CB standard defined a four-byte redundancy tag. Drafts 2.5 and later specified a six-byte tag. Releases 1p10 and 1p11 were implemented in accordance with draft 2.4 of the standard. Set this bit to zero to inter-operate with implementations that use a four-byte tag."	RW	1
29:16	reserved_29_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15:0	redundancy_tag	"Ethertype value used to identify the redundancy tag (R-TAG)."	RW	0xF1C1

Table 11-836: emac_frer_control_1_a

Register Information				
Description	"FRER control register A"			
Offset	0x18C0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	en_elimination	"Enable 802.1CB elimination of received frames"	RW	0
30	en_vector_rec_alg	"Enable 802.1CB vector recovery algorithm 0 enables the match recovery algorithm 1 enables the vector recovery algorithm This bit should only be changed when bit 31 en_elimination is low"	RW	0
29	en_seqrecrst_timer	"Enable 802.1CB sequence recovery reset timer - this bit may be changed when bit 31 en_elimination is low"	RW	0
28	use_r_tag	"Set to one to use redundancy tag to identify sequence number otherwise use offset value to identify bottom of sequence number - this bit should only be changed when bit 31 en_elimination is low"	RW	0
27:17	reserved_27_17	"Reserved, read as 0, ignored on write."	RO	0x000
16:8	offset_value	"Offset value in bytes from the start packet delimiter to the most significant byte of the 802.1CB sequence number - 9 bits allow a maximum value of 511 - allowing a TCP sequence number to be used with IPv6 - this bit should only be changed when en_elimination is low"	RW	0x000
7:4	member_stream_2	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0
3:0	member_stream_1	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0

Table 11-837: emac_frer_control_1_b

Register Information					
Description	"FRER control register B. This register has default values where the sequence number length defaults to 16 and the vector recovery window defaults to the size of the history vector. It is not expected that the user will need to change these default values."				
Offset	0x18C4		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:13	reserved_31_13	"Reserved, read as 0, ignored on write."	RO	0x0 0000	
12:8	seq_num_length	"Number of significant bits of the 802.1CB sequence number. The value 0x00 or numbers greater than 16 bits are equivalent to 16, but the number written will still be read back if it is greater than 16. If a value of less than 16 is written, then the sequence recovery algorithm will only consider that number of LSBs of the sequence number. The minimum size of seq_num_length has to be such that 2^seq_num_len is at least 2xgem_seq_history_len otherwise the vector recovery algorithm will not work correctly - this bit should only be changed when en_elimination is low"	RW	0x00	
7:6	reserved_7_6	"Reserved, read as 0, ignored on write."	RO	0x0	
5:0	seq_rec_window	"Vector recovery window, defines the window size used by the vector recovery algorithm to determine whether to reject a packet. Six bits allow a window size of 63, for effective operation of FRER users should not write a value greater than the gem_seq_history_len configuration define, a value of zero means the entire history vector is used - this bit should only be changed when en_elimination is low."	RW	0x00	

Table 11-838: emac_frer_statistics_1_a

Register Information					
Description	"FRER statistics register A. This register is cleared on a read and does not roll over if its maximum value is reached."				
Offset	0x18C8		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:26	reserved_31_26	"Reserved, read as 0, ignored on write."	RO	0x00	
25:16	vec_rec_rogue	"Count of number of frames dropped by the vector recovery algorithm for being out of range"	RO RtoClr	0x000	
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00	
9:0	latent_errors	"Count of sequence numbers seen without a duplicate. The latent error count is updated when a frame is dropped from the history vector. So the update only happens after a new frame is received."	RO RtoClr	0x000	

Table 11-839: emac_frer_statistics_1_b

Register Information					
Description	"FRER statistics register B. This register is cleared on a read and does not roll over if its maximum value is reached."				
Offset	0x18CC		**Type:**	RO	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:24	reserved_31_24	"Reserved, read as 0, ignored on write."		RO	0x00
23:16	seqrst_count	"Count of number of times the sequence recovery reset timer decrements to zero."		RO RtoClr	0x00
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."		RO	0x00
9:0	out_of_order	"Count of out of order sequence numbers received. Incremented when a frame is accepted but the sequence number is not +1 of the highest stored value."		RO RtoClr	0x000

Table 11-840: emac_frer_control_2_a

Register Information				
Description	"FRER control register A"			
Offset	0x18D0	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31	en_elimination	"Enable 802.1CB elimination of received frames"	RW	0
30	en_vector_rec_alg	"Enable 802.1CB vector recovery algorithm 0 enables the match recovery algorithm 1 enables the vector recovery algorithm This bit should only be changed when bit 31 en_elimination is low"	RW	0
29	en_seqrecrst_timer	"Enable 802.1CB sequence recovery reset timer - this bit may be changed when bit 31 en_elimination is low"	RW	0
28	use_r_tag	"Set to one to use redundancy tag to identify sequence number otherwise use offset value to identify bottom of sequence number - this bit should only be changed when bit 31 en_elimination is low"	RW	0
27:17	reserved_27_17	"Reserved, read as 0, ignored on write."	RO	0x000
16:8	offset_value	"Offset value in bytes from the start packet delimiter to the most significant byte of the 802.1CB sequence number - 9 bits allow a maximum value of 511 - allowing a TCP sequence number to be used with IPv6 - this bit should only be changed when en_elimination is low"	RW	0x000
7:4	member_stream_2	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0
3:0	member_stream_1	"Pointer to screener type 2 register used for member stream identification. The member stream 1 and 2 values may be programmed to identical values. - this bit should only be changed when en_elimination is low"	RW	0x0

Table 11-841: emac_frer_control_2_b

Register Information			
Description	"FRER control register B. This register has default values where the sequence number length defaults to 16 and the vector recovery window defaults to the size of the history vector. It is not expected that the user will need to change these default values."		
Offset	0x18D4	**Type:**	RW

Bits	Name	Description	Access	Reset
31:13	reserved_31_13	"Reserved, read as 0, ignored on write."	RO	0x0 0000
12:8	seq_num_length	"Number of significant bits of the 802.1CB sequence number. The value 0x00 or numbers greater than 16 bits are equivalent to 16, but the number written will still be read back if it is greater than 16. If a value of less than 16 is written, then the sequence recovery algorithm will only consider that number of LSBs of the sequence number. The minimum size of seq_num_length has to be such that 2^seq_num_len is at least 2xgem_seq_history_len otherwise the vector recovery algorithm will not work correctly - this bit should only be changed when en_elimination is low"	RW	0x00
7:6	reserved_7_6	"Reserved, read as 0, ignored on write."	RO	0x0
5:0	seq_rec_window	"Vector recovery window, defines the window size used by the vector recovery algorithm to determine whether to reject a packet. Six bits allow a window size of 63, for effective operation of FRER users should not write a value greater than the gem_seq_history_len configuration define, a value of zero means the entire history vector is used - this bit should only be changed when en_elimination is low."	RW	0x00

Table 11-842: emac_frer_statistics_2_a

Register Information			
Description	"FRER statistics register A. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x18D8	**Type:**	RO

Bits	Name	Description	Access	Reset
31:26	reserved_31_26	"Reserved, read as 0, ignored on write."	RO	0x00
25:16	vec_rec_rogue	"Count of number of frames dropped by the vector recovery algorithm for being out of range"	RO RtoClr	0x000
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	latent_errors	"Count of sequence numbers seen without a duplicate. The latent error count is updated when a frame is dropped from the history vector. So the update only happens after a new frame is received."	RO RtoClr	0x000

Table 11-843: emac_frer_statistics_2_b

Register Information			
Description	"FRER statistics register B. This register is cleared on a read and does not roll over if its maximum value is reached."		
Offset	0x18DC	**Type:**	RO

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	reserved_31_24	"Reserved, read as 0, ignored on write."	RO	0x00
23:16	seqrst_count	"Count of number of times the sequence recovery reset timer decrements to zero."	RO RtoClr	0x00
15:10	reserved_15_10	"Reserved, read as 0, ignored on write."	RO	0x00
9:0	out_of_order	"Count of out of order sequence numbers received. Incremented when a frame is accepted but the sequence number is not +1 of the highest stored value."	RO RtoClr	0x000

Table 11-844: emac_rx_q0_flush

Register Information			
Description	"Receive Queue flush register. This register defines the traffic policing mode of operation. Each mode can be set simultaneously with the exception of bits 2 and 3, which are exclusive. If bits 2 and 3 are both set then only bit 3 is treated as active."		
Offset	0x1B00	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:16	max_val	"This 16 bit vector is used when either bit 2 or 3 of this register is asserted. Refer to the description in those bits for details."	RW	0x0000
15:4	reserved_15_4	"Reserved, read as 0, ignored on write."	RO	0x000
3	limit_frame_size	"When set max_val (bits 31:16) indicates the maximum frame-length in bytes that may be received. Frames exceeding this length will be dropped. This traffic policing function is relevant to the 802.1Qci standard which specifies stream filtering based on a maximum SDU (service data unit) size."	RW	0
2	limit_num_bytes	"When set, the number of 128 byte chunks of data received for this queue and already stored in the SRAM awaiting DMA memory writes cannot exceed max_val (bits 31:16)."	RW	0
1	drop_on_resource_err	"When set, if a free DMA descriptor for this queue cannot be obtained (also referred to as lack of descriptor resource and occurs when the software either cannot free up descriptors quickly enough to meet the receive traffic rate or has deliberately decided not to free any descriptors), all new frames received on this queue will be automatically discarded."	RW	0
0	drop_all_frames	"When set, all frames on this queue will be dropped."	RW	0

Table 11-845: emac_scr2_reg0_rate_limit

Register Information					
Description	"Screener type 2 maximum rate register. This register defines the maximum receive rate for the corresponding type 2 screener. This is a traffic policing function and is relevant to the 802.1Qci standard. If the maximum rate is exceeded all frames matched by the corresponding type 2 screener will be discarded until the receive rate falls below the specified value."				
Offset	0x1B40		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:16	max_rate_val	"This is the maximum number of bytes this screener is permitted to match in the programmed interval time."		RW	0x0000
15:0	interval_time	"If during the interval time the total number of bytes of received frames matched by the screener exceeds max_rate_val then the current frame and frames subsequently matched will dropped until an interval time passes where max_rate_val is not exceeded. If this value is set to zero, then no rate limiting will be performed. The interval time is specified in units of 64 rx_clk periods."		RW	0x0000

Table 11-846: emac_scr2_reg1_rate_limit

Register Information					
Description	"Screener type 2 maximum rate register. This register defines the maximum receive rate for the corresponding type 2 screener. This is a traffic policing function and is relevant to the 802.1Qci standard. If the maximum rate is exceeded all frames matched by the corresponding type 2 screener will be discarded until the receive rate falls below the specified value."				
Offset	0x1B44		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**		**Access**	**Reset**
31:16	max_rate_val	"This is the maximum number of bytes this screener is permitted to match in the programmed interval time."		RW	0x0000
15:0	interval_time	"If during the interval time the total number of bytes of received frames matched by the screener exceeds max_rate_val then the current frame and frames subsequently matched will dropped until an interval time passes where max_rate_val is not exceeded. If this value is set to zero, then no rate limiting will be performed. The interval time is specified in units of 64 rx_clk periods."		RW	0x0000

Table 11-847: emac_scr2_rate_status

Register Information			
Description	"Screener rate limit exceeded status register. For each screener type 2 register configured a status bit will be set and cleared on read if the maximum receive rate is exceeded."		
Offset	0x1B80	**Type:**	RO

Bits	Name	Description	Access	Reset
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000
15	scr2_15_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
14	scr2_14_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
13	scr2_13_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
12	scr2_12_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
11	scr2_11_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
10	scr2_10_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
9	scr2_9_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
8	scr2_8_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
7	scr2_7_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
6	scr2_6_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
5	scr2_5_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
4	scr2_4_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
3	scr2_3_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
2	scr2_2_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
1	scr2_1_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0
0	scr2_0_excess_rate	"Set to 1 if screeners rate limiting mechanism has been triggered"	RO RtoClr	0

Table 11-848: emac_asf_int_status

Register Information			
Description	"ASF Interrupt Status Register. This register indicates the source of ASF interrupts. The corresponding bit in the mask register must be clear for a bit to be set. If any bit is set in this register the asf_fatal or asf_nonfatal signal will be asserted. Writing to either raw or masked status registers, clear both registers. For test purposes, trigger signal interrupt event by writing to the ASF interrupt status test register."		
Offset	0x1E00	**Type:**	RW

Bitfield Details

Bits	Name	Description	Access	Reset
31:7	reserved_31_7	"Reserved, read as 0, ignored on write."	RO	0x000 0000
6	asf_integrity_err	"Integrity error interrupt"	RW W1toClr	0
5	asf_protocol_err	"Protocol error interrupt"	RW W1toClr	0
4	asf_trans_to_err	"Transaction timeouts error interrupt"	RW W1toClr	0
3	asf_csr_err	"Configuration and status registers error interrupt"	RW W1toClr	0
2	asf_dap_err	"Data and address paths parity error interrupt"	RW W1toClr	0
1	asf_sram_uncorr_err	"SRAM uncorrectable error interrupt"	RW W1toClr	0
0	asf_sram_corr_err	"SRAM correctable error interrupt"	RW W1toClr	0

Table 11-849: emac_asf_int_raw_status

Register Information			
Description	"ASF Interrupt Status Register. This register indicates the source of ASF interrupts. The corresponding bit in the mask register must be clear for a bit to be set. If any bit is set in this register the asf_fatal or asf_nonfatal signal will be asserted. Writing to either raw or masked status registers, clear both registers. For test purposes, trigger signal interrupt event by writing to the ASF interrupt status test register."		
Offset	0x1E04	**Type:**	RW

Bits	Name	Description	Access	Reset
31:7	reserved_31_7	"Reserved, read as 0, ignored on write."	RO	0x000 0000
6	asf_integrity_err	"Integrity error interrupt"	RW W1toClr	0
5	asf_protocol_err	"Protocol error interrupt"	RW W1toClr	0
4	asf_trans_to_err	"Transaction timeouts error interrupt"	RW W1toClr	0
3	asf_csr_err	"Configuration and status registers error interrupt"	RW W1toClr	0
2	asf_dap_err	"Data and address paths parity error interrupt"	RW W1toClr	0
1	asf_sram_uncorr_err	"SRAM uncorrectable error interrupt"	RW W1toClr	0
0	asf_sram_corr_err	"SRAM correctable error interrupt"	RW W1toClr	0

Table 11-850: emac_asf_int_mask

Register Information				
Description	"The ASF interrupt mask register indicating which interrupt bits in the ASF interrupt status register are masked. All bits are set at reset. Clear the individual bit to enable the corresponding interrupt."			
Offset	0x1E08	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:7	reserved_31_7	"Reserved, read as 0, ignored on write."	RO	0x000 0000
6	asf_integrity_err_mask	"Mask bit for integrity error interrupt"	RW	1
5	asf_protocol_err_mask	"Mask bit for protocol error interrupt."	RW	1
4	asf_trans_to_err_mask	"Mask bit for transaction timeouts error interrupt."	RW	1
3	asf_csr_err_mask	"Mask bit for configuration and status registers error interrupt."	RW	1
2	asf_dap_err_mask	"Mask bit for data and address paths parity error interrupt."	RW	1
1	asf_sram_uncorr_err_mask	"Mask bit for SRAM uncorrectable error interrupt."	RW	1
0	asf_sram_corr_err_mask	"Mask bit for SRAM correctable error interrupt."	RW	1

Table 11-851: emac_asf_int_test

Register Information				
Description	"The ASF interrupt test register emulate hardware even. Write one to individual bit to trigger single event in (masked and raw) status registers according to mask and will generate interrupt accordingly."			
Offset	0x1E0C	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:7	reserved_31_7	"Reserved, read as 0, ignored on write."	RO	0x000 0000
6	asf_integrity_err_test	"Test bit for integrity error interrupt"	WO	0
5	asf_protocol_err_test	"Test bit for protocol error interrupt."	WO	0
4	asf_trans_to_err_test	"Test bit for transaction timeouts error interrupt."	WO	0
3	asf_csr_err_test	"Test bit for configuration and status registers error interrupt."	WO	0
2	asf_dap_err_test	"Test bit for data and address paths parity error interrupt."	WO	0
1	asf_sram_uncorr_err_test	"Test bit for SRAM uncorrectable error interrupt."	WO	0
0	asf_sram_corr_err_test	"Test bit for SRAM correctable error interrupt."	WO	0

Table 11-852: emac_asf_fatal_nonfatal_select

Register Information				
Description	"The fatal or non-fatal interrupt register selects whether a fatal (asf_int_fatal) or non-fatal (asf_int_nonfatal) interrupt is triggered. If the bit of the event will be set to one then fatal interrupt (asf_int_fatal) will be triggered. Otherwise the non-fatal interrupt (asf_int_nonfatal) will be triggered."			
Offset	0x1E10	**Type:**	RW	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:7	reserved_31_7	"Reserved, read as 0, ignored on write."	RO	0x000 0000
6	asf_integrity_err	"Enable integrity error interrupt as fatal"	RW	1
5	asf_protocol_err	"Enable protocol error interrupt as fatal."	RW	1
4	asf_trans_to_err	"Enable transaction timeouts error interrupt as fatal."	RW	1
3	asf_csr_err	"Enable configuration and status registers error interrupt as fatal."	RW	1
2	asf_dap_err	"Enable data and address paths parity error interrupt as fatal."	RW	1
1	asf_sram_uncorr_err	"Enable SRAM uncorrectable error interrupt as fatal."	RW	1
0	asf_sram_corr_err	"Enable SRAM correctable error interrupt as fatal."	RW	1

Table 11-853: emac_asf_sram_corr_fault_status

Register Information				
Description	"Status register for SRAM correctable fault. These fields are updated whenever asf_sram_corr_fault input is active."			
Offset	0x1E20	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	asf_sram_corr_fault_inst	"Last SRAM instance that generated fault."	RO	0x00
23:0	asf_sram_corr_fault_addr	"Last SRAM address that generated fault."	RO	0x00 0000

Table 11-854: emac_asf_sram_uncorr_fault_status

Register Information				
Description	"Status register for SRAM uncorrectable fault. These fields are updated whenever asf_sram_uncorr_fault input is active."			
Offset	0x1E24	**Type:**	RO	
Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:24	asf_sram_uncorr_fault_inst	"Last SRAM instance that generated fault."	RO	0x00
23:0	asf_sram_uncorr_fault_addr	"Last SRAM address that generated fault."	RO	0x00 0000

Table 11-855: emac_asf_sram_fault_stats

Register Information					
Description	"Statistics register for SRAM faults. Note that this register clears when software writes to any field."				
Offset	0x1E28		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:16	reserved_31_16	"Reserved, read as 0, ignored on write."	RO	0x0000	
15:0	asf_sram_fault_corr_stats	"Count of number of correctable errors if implemented. Count value will saturate at 0xffff."	RW W1toClr	0x0000	

Table 11-856: emac_asf_trans_to_ctrl

Register Information					
Description	"Control register to configure the ASF transaction timeout monitors."				
Offset	0x1E30		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31	asf_trans_to_en	"Enable transaction timeout monitoring."	RW	0	
30:16	reserved_30_16	"Reserved, read as 0, ignored on write."	RO	0x0000	
15:0	asf_trans_to_ctrl	"Timer value to use for transaction timeout monitor."	RW	0x0000	

Table 11-857: emac_asf_trans_to_fault_mask

Register Information					
Description	"Control register to mask out ASF transaction timeout faults from triggering interrupts. On reset, all bits are set to mask out all sources. Clear the corresponding bit to enable the interrupt source."				
Offset	0x1E34		**Type:**	RW	
Bitfield Details					
Bits	**Name**	**Description**	**Access**	**Reset**	
31:5	reserved_31_5	"Reserved, read as 0, ignored on write."	RO	0x000 0000	
4	host_trans_to_mask	"Mask register for AMBA host transaction timeouts."	RW	1	
3	dma_rx_to_mask	"Mask register for DMA RX lockup detection."	RW	1	
2	dma_tx_to_mask	"Mask register for DMA TX lockup detection."	RW	1	
1	mac_rx_to_mask	"Mask register for MAC RX lockup detection."	RW	1	
0	mac_tx_to_mask	"Mask register for MAC TX lockup detection."	RW	1	

Table 11-858: emac_asf_trans_to_fault_status

Register Information			
Description	"Status register for transaction timeouts fault. If a fault occurs the revelant status bit will be set to 1. Each bit can be cleared by software writing 1 to each bit."		
Offset	0x1E38	**Type:**	RW

Bitfield Details				
Bits	**Name**	**Description**	**Access**	**Reset**
31:5	reserved_31_5	"Reserved, read as 0, ignored on write."	RO	0x000 0000
4	host_trans_to_status	"Status bit for AMBA host transaction timeouts."	RW W1toClr	0
3	dma_rx_to_status	"Status bit for DMA RX lockup detection."	RW W1toClr	0
2	dma_tx_to_status	"Status bit for DMA TX lockup detection."	RW W1toClr	0
1	mac_rx_to_status	"Status bit for MAC RX lockup detection."	RW W1toClr	0
0	mac_tx_to_status	"Status bit for MAC TX lockup detection."	RW W1toClr	0

Table 11-859: emac_asf_protocol_fault_mask

Register Information			
Description	"Control register to mask out ASF Protocol faults from triggering interrupts. On reset, all bits are set to mask out all sources. Clear the corresponding bit to enable the interrupt source."		
Offset	0x1E40	**Type:**	RW

Bits	Name	Description	Access	Reset
31:22	reserved_31_22	"Reserved, read as 0, ignored on write."	RO	0x000
21	rx_dma_pkt_flush_mask	"Mask bit for RX packet was flushed from the DMA."	RW	1
20	rx_overflow_mask	"Mask bit for DMA buffer overflow and packet was dropped."	RW	1
19	rx_hresp_err_mask	"Mask bit for DMA hresp error (AHB only)."	RW	1
18	tx_hresp_err_mask	"Mask bit for DMA hresp error (AHB only)."	RW	1
17	tx_buff_ex_mid_mask	"Mask bit for Transmit buffers exhausted before end of packet."	RW	1
16	tx_underrun_mask	"Mask bit for Transmit DMA underrun occurred."	RW	1
15:9	reserved_15_9	"Reserved, read as 0, ignored on write."	RO	0x00
8	tx_too_many_retries_mask	"Mask bit for too many retry attempts after collision error (half duplex only)."	RW	1
7	rx_udp_ck_err_mask	"Mask bit for RX packet UDP checksum error."	RW	1
6	rx_tcp_ck_err_mask	"Mask bit for RX packet TCP checksum error."	RW	1
5	rx_ip_ck_err_mask	"Mask bit for RX packet IP checksum error."	RW	1
4	rx_length_err_mask	"Mask bit for RX packet with length field error."	RW	1
3	rx_symbol_err_mask	"Mask bit for RX packet with symbol errors."	RW	1
2	rx_long_err_mask	"Mask bit for RX packet too long."	RW	1
1	rx_short_err_mask	"Mask bit for RX packet too short."	RW	1
0	rx_crc_err_mask	"Mask bit for RX packet with bad CRC."	RW	1

Table 11-860: emac_asf_protocol_fault_status

Register Information				
Description	"Status register for protocol faults. If a fault occurs the revelant status bit will be set to 1. Each bit can be cleared by software writing 1 to each bit"			
Offset	0x1E44	**Type:**	RW	

Bits	Name	Description	Access	Reset
31:22	reserved_31_22	"Reserved, read as 0, ignored on write."	RO	0x000
21	rx_dma_pkt_flush_status	"Status bit for RX packet was flushed from the DMA."	RW W1toClr	0
20	rx_overflow_status	"Status bit for DMA buffer overflow and packet was dropped."	RW W1toClr	0
19	rx_hresp_err_status	"Status bit for DMA hresp error (AHB only)."	RW W1toClr	0
18	tx_hresp_err_status	"Status bit for DMA hresp error (AHB only)."	RW W1toClr	0
17	tx_buff_ex_mid_status	"Status bit for Transmit buffers exhausted before end of packet."	RW W1toClr	0
16	tx_underrun_status	"Status bit for Transmit DMA underrun occurred."	RW W1toClr	0
15:9	reserved_15_9	"Reserved, read as 0, ignored on write."	RO	0x00
8	tx_too_many_retries_status	"Status bit for too many retry attempts after collision error (half duplex only)."	RW W1toClr	0
7	rx_udp_ck_err_status	"Status bit for RX packet UDP checksum error."	RW W1toClr	0
6	rx_tcp_ck_err_status	"Status bit for RX packet TCP checksum error."	RW W1toClr	0
5	rx_ip_ck_err_status	"Status bit for RX packet IP checksum error."	RW W1toClr	0
4	rx_length_err_status	"Status bit for RX packet with length field error."	RW W1toClr	0
3	rx_symbol_err_status	"Status bit for RX packet with symbol errors."	RW W1toClr	0
2	rx_long_err_status	"Status bit for RX packet too long."	RW W1toClr	0
1	rx_short_err_status	"Status bit for RX packet too short."	RW W1toClr	0
0	rx_crc_err_status	"Status bit for RX packet with bad CRC."	RW W1toClr	0

gem_ip7014a_ug has no common memories.

Change Log

Revision	Date	Description
1.00	November 6, 2017	• Initial version based on legacy UG doc # I-IPA01-0266-USR, revision 18 (22 June 2017) • Added new content for IEEE 802.1Qci: Receive (Ingress) Traffic Policing • Updated FRER Redundancy Tag and Sequence Number Identification